

IPv4 Addressing and Subnetting Workbook

Version 2.1

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Student Name:

IPv4 Address Classes

Class A	1 – 127	Leading bit pattern	0	00000000.00000000.00000000.00000000 Network . Host . Host . Host
Class B	128 – 191	Leading bit pattern	10	10000000.00000000.00000000.00000000 Network . Network . Host . Host
Class C	192 – 223	Leading bit pattern	110	11000000.00000000.00000000.00000000 Network . Network . Network . Host
Class D	224 – 239	(Reserved for multicast)		
Class E	240 – 255	(Reserved for experimental, used for research)		

Speciality Address Ranges

Loopback -	Only the single 127.0.0.1 address is used, addresses 127.0.0.0 to 127.255.255.255 are reserved. Any address within this block will loop back to the local host.
Link-Local Addresses -	IPv4 addresses in the address block 169.254.0.0 to 169.254.255.255 (169.254.0.0/16) are designated as link-local addresses.
TEST-NET Addresses -	The address block 192.0.2.0 to 192.0.2.255 (192.0.2.0/24) is set aside for teaching and learning purposes.
Experimental Addresses -	The addresses in the block 240.0.0.0 to 255.255.255.254 are listed as reserved for future use (RFC 3330).

Private Address Space

Class A	10.0.0.0 to 10.255.255.255
Class B	172.16.0.0 to 172.31.255.255
Class C	192.168.0.0 to 192.168.255.255

Default Subnet Masks

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

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and to everyone who has sent in suggestions to improve the series.

Inside Cover

Binary To Decimal Conversion

128	64	32	16	8	4	2	1	Answers	Scratch Area	
1	0	0	1	0	0	1	0	<u>146</u>	<i>128</i>	<i>64</i>
0	1	1	1	0	1	1	1	<u>119</u>	<i>16</i>	<i>32</i>
1	1	1	1	1	1	1	1		<i>2</i>	<i>16</i>
1	1	0	0	0	1	0	1		<i>146</i>	<i>4</i>
1	1	1	1	0	1	1	0			<i>2</i>
0	0	0	1	0	0	1	1			<i>1</i>
1	0	0	0	0	0	0	1			<u>119</u>
0	0	1	1	0	0	0	1			
0	1	1	1	1	0	0	0			
1	1	1	1	0	0	0	0			
0	0	1	1	1	0	1	1			
0	0	0	0	0	1	1	1			
								00011011		
								10101010		
								01101111		
								11111000		
								00100000		
								01010101		
								00111110		
								00000011		
								11101101		
								11000000		

Decimal To Binary Conversion

Use all 8 bits for each problem

128	64	32	16	8	4	2	1 =	255	Scratch Area	
1	1	1	0	1	1	1	0	238	238	34
									-128	-32
0	0	1	0	0	0	1	0	34	110	2
									-64	-2
								123	46	0
									-32	
								50	14	
									-8	
								255	6	
									-4	
								200	2	
									-2	
								10	0	
								138		
								1		
								13		
								250		
								107		
								224		
								114		
								192		
								172		
								100		
								119		
								57		
								98		
								179		
								2		

Address Class Identification

Address	Class
10.250.1.1	<u>A</u>
150.10.15.0	<u>B</u>
192.14.2.0	<u> </u>
148.17.9.1	<u> </u>
193.42.1.1	<u> </u>
126.8.156.0	<u> </u>
220.200.23.1	<u> </u>
230.230.45.58	<u> </u>
177.100.18.4	<u> </u>
119.18.45.0	<u> </u>
249.240.80.78	<u> </u>
199.155.77.56	<u> </u>
117.89.56.45	<u> </u>
215.45.45.0	<u> </u>
199.200.15.0	<u> </u>
95.0.21.90	<u> </u>
33.0.0.0	<u> </u>
158.98.80.0	<u> </u>
219.21.56.0	<u> </u>

Network & Host Identification

Circle the network portion
of these addresses:

177.100.18.4

119.18.45.0

209.240.80.78

199.155.77.56

117.89.56.45

215.45.45.0

192.200.15.0

95.0.21.90

33.0.0.0

158.98.80.0

217.21.56.0

10.250.1.1

150.10.15.0

192.14.2.0

148.17.9.1

193.42.1.1

126.8.156.0

220.200.23.1

Circle the host portion of
these addresses:

10.15.123.50

171.2.199.31

198.125.87.177

223.250.200.222

17.45.222.45

126.201.54.231

191.41.35.112

155.25.169.227

192.15.155.2

123.102.45.254

148.17.9.155

100.25.1.1

195.0.21.98

25.250.135.46

171.102.77.77

55.250.5.5

218.155.230.14

10.250.1.1

Network Addresses

Using the IP address and subnet mask shown write out the network address:

188.10.18.2	<i>188 . 10 . 0 . 0</i>
255.255.0.0	_____

10.10.48.80	<i>10 . 10 . 48 . 0</i>
255.255.255.0	_____

192.149.24.191	_____
255.255.255.0	

150.203.23.19	_____
255.255.0.0	

10.10.10.10	_____
255.0.0.0	

186.13.23.110	_____
255.255.255.0	

223.69.230.250	_____
255.255.0.0	

200.120.135.15	_____
255.255.255.0	

27.125.200.151	_____
255.0.0.0	

199.20.150.35	_____
255.255.255.0	

191.55.165.135	_____
255.255.255.0	

28.212.250.254	_____
255.255.0.0	

Host Addresses

Using the IP address and subnet mask shown write out the host address:

188.10.18.2
255.255.0.0

0 . 0 . 18 . 2

10.10.48.80
255.255.255.0

0 . 0 . 0 . 80

222.49.49.11
255.255.255.0

128.23.230.19
255.255.0.0

10.10.10.10
255.0.0.0

200.113.123.11
255.255.255.0

223.169.23.20
255.255.0.0

203.20.35.215
255.255.255.0

117.15.2.51
255.0.0.0

199.120.15.135
255.255.255.0

191.55.165.135
255.255.255.0

48.21.25.54
255.255.0.0

Default Subnet Masks

Write the correct default subnet mask for each of the following addresses:

177.100.18.4	<i>255 . 255 . 0 . 0</i> _____
119.18.45.0	<i>255 . 0 . 0 . 0</i> _____
191.249.234.191	_____
223.23.223.109	_____
10.10.250.1	_____
126.123.23.1	_____
223.69.230.250	_____
192.12.35.105	_____
77.251.200.51	_____
189.210.50.1	_____
88.45.65.35	_____
128.212.250.254	_____
193.100.77.83	_____
125.125.250.1	_____
1.1.10.50	_____
220.90.130.45	_____
134.125.34.9	_____
95.250.91.99	_____

ANDING with Default subnet masks

Every IP address must be accompanied by a subnet mask. By now you should be able to look at an IP address and tell what class it is. Unfortunately your computer doesn't think that way. For your computer to determine the network and subnet portion of an IP address it must "AND" the IP address with the subnet mask.

Default Subnet Masks:

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

ANDING Equations:

1 AND 1 = 1
 1 AND 0 = 0
 0 AND 1 = 0
 0 AND 0 = 0

Sample:

What you see...

IP Address: 192 . 100 . 10 . 33

What you can figure out in your head...

Address Class:	C
Network Portion:	<u>192 . 100 . 10</u> . 33
Host Portion:	192 . 100 . 10 . <u>33</u>

In order for your computer to get the same information it must AND the IP address with the subnet mask in binary.

	Network	Host	
IP Address:	1 1 0 0 0 0 0 0 . 0 1 1 0 0 1 0 0 . 0 0 0 0 1 0 1 0 .	0 0 1 0 0 0 0 1	(192 . 100 . 10 . 33)
Default Subnet Mask:	1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 .	0 0 0 0 0 0 0 0	(255 . 255 . 255 . 0)
AND:	1 1 0 0 0 0 0 0 . 0 1 1 0 0 1 0 0 . 0 0 0 0 1 0 1 0 .	0 0 0 0 0 0 0 0	(192 . 100 . 10 . 0)

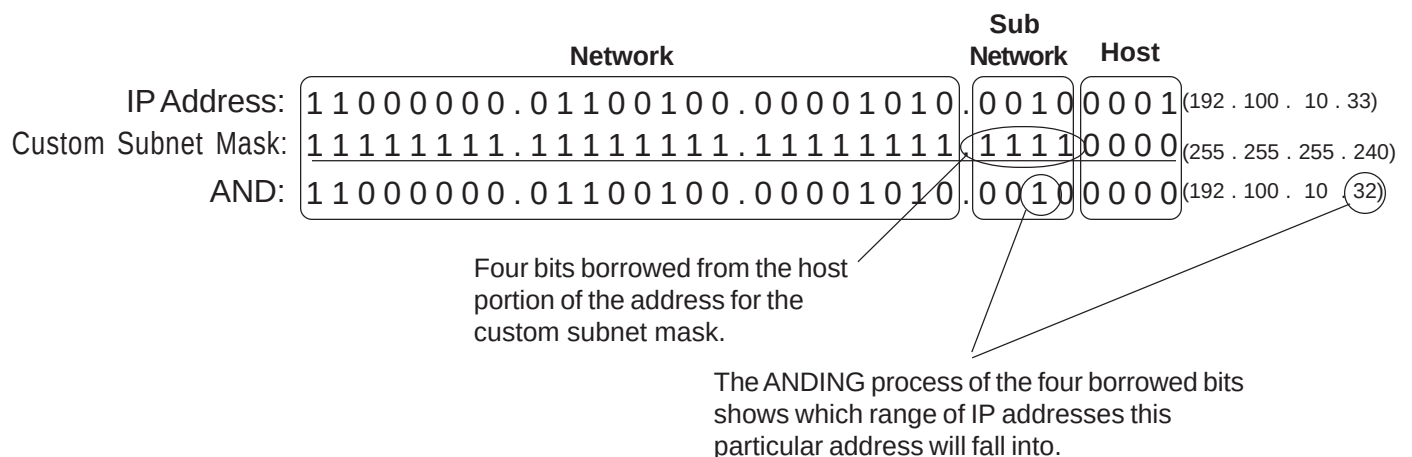
ANDING with the default subnet mask allows your computer to figure out the network portion of the address.

ANDING with Custom subnet masks

When you take a single network such as 192.100.10.0 and divide it into five smaller networks (192.100.10.16, 192.100.10.32, 192.100.10.48, 192.100.10.64, 192.100.10.80) the outside world still sees the network as 192.100.10.0, but the internal computers and routers see five smaller subnetworks. Each independent of the other. This can only be accomplished by using a custom subnet mask. A custom subnet mask borrows bits from the host portion of the address to create a subnetwork address between the network and host portions of an IP address. In this example each range has 14 usable addresses in it. The computer must still AND the IP address against the custom subnet mask to see what the network portion is and which subnetwork it belongs to.

IP Address: 192 . 100 . 10 . 0
Custom Subnet Mask: 255.255.255.240

Address Ranges: 192.10.10.0 to 192.100.10.15
 192.100.10.16 to 192.100.10.31
 192.100.10.32 to 192.100.10.47 (Range in the sample below)
 192.100.10.48 to 192.100.10.63
 192.100.10.64 to 192.100.10.79
 192.100.10.80 to 192.100.10.95
 192.100.10.96 to 192.100.10.111
 192.100.10.112 to 192.100.10.127
 192.100.10.128 to 192.100.10.143
 192.100.10.144 to 192.100.10.159
 192.100.10.160 to 192.100.10.175
 192.100.10.176 to 192.100.10.191
 192.100.10.192 to 192.100.10.207
 192.100.10.208 to 192.100.10.223
 192.100.10.224 to 192.100.10.239
 192.100.10.240 to 192.100.10.255



In the next set of problems you will determine the necessary information to determine the correct subnet mask for a variety of IP addresses.

How to determine the number of subnets and the number of hosts per subnet

The formula that can provide this basic information:

$$\text{Number of subnets} = 2^s$$

$$\text{Number of usable hosts per subnet} = 2^h - 2$$

This formula calculates the number of hosts or subnets based on the number of binary bits used. For example if you borrow three bits from the host portion of the address use the *number of subnets* formula to determine the total number of subnets gained by borrowing the three bits. This would be 2^3 or $2 \times 2 \times 2 = 8$ subnets

To determine the number of hosts per subnet you would take the number of binary bits used in the host portion and apply this to the *number of hosts per subnet* formula. If five bits are in the host portion of the address this would be 2^5 or $2 \times 2 \times 2 \times 2 \times 2 = 32$ hosts.

When dealing with the *number of hosts per subnet* you have to subtract two addresses from the range. The first address in every range is the subnet number. The last address in every range is the broadcast address. These two addresses cannot be assigned to any device in the network which is why you have to subtract two addresses to find the number of usable addresses in each range.

For example, if two bits are borrowed for the network portion of the address you can easily determine the number of subnets and hosts per subnets using the formula.

195. 223 . 50 . 0 0 | 0 0 0 0 0 0

The number of subnets created by borrowing 2 bits is 2^2 or $2 \times 2 = 4$ subnets.

The number of hosts created by leaving 6 bits is $2^6 - 2$ or $2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64 - 2 = 62$ usable hosts per subnet.

Custom Subnet Problems

Custom Subnet Masks

Problem 1

Number of needed subnets **14**
 Number of needed usable hosts **14**
 Network Address **192.10.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 240

Total number of subnets 16

Total number of host addresses 16

Number of usable addresses 14

Number of bits borrowed 4

Show your work for Problem 1 in the space below.

	256	128	64	32	16	8	4	2	1	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
192 . 10 . 10 . 0	0	0	0	0	0	0	0	0	0	

Add the binary value numbers to the left of the line to create the custom subnet mask.

128
64
32
+16
240

16	Observe the total number of hosts.
-2	
14	Subtract 2 for the number of usable hosts.

Custom Subnet Masks

Problem 2

Number of needed subnets **1000**

Number of needed usable hosts **60**

Network Address **165.100.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

Show your work for Problem 2 in the space below.

Number of Hosts -	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets -	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values -	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
	165	100	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Add the binary value numbers to the left of the line to create the custom subnet mask.

128	128
64	+64
32	192
16	
8	
4	
2	
+1	
255	

Observe the total number of hosts.

64

-2

62

Subtract 2 for the number of usable hosts.

Custom Subnet Masks

Problem 3

Network Address **148.75.0.0 /26**

/26 indicates the total number of bits used for the network and subnetwork portion of the address. All bits remaining belong to the host portion of the address.

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

Show your work for Problem 3 in the space below.

	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Hosts	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Number of Subnets	-	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768
Binary values	-	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2
		148	75	0	0	0	0	0	0	0	0	0	0	0	0	0

Add the binary value numbers to the left of the line to create the custom subnet mask.

128	128
64	+64
32	192
16	
8	
4	
2	
+1	
<u>255</u>	

64	Observe the total number of hosts.
-2	
<u>62</u>	Subtract 2 for the number of usable hosts.

1024	
-2	
<u>1,022</u>	Subtract 2 for the total number of subnets to get the usable number of subnets.

Custom Subnet Masks

Problem 4

Number of needed subnets **6**

Number of needed usable hosts **30**

Network Address **195.85.8.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 5 in the space below.

											Number of				
						256	128	64	32	16	8	4	2	-	Hosts
Number of Subnets	-	2	4	8		16	32	64	128	256					
		128	64	32		16	8	4	2	1	- Binary values				

Custom Subnet Masks

Problem 5

Number of needed subnets **6**

Number of needed usable hosts **30**

Network Address **210.100.56.0**

Address class _____

Default subnet mask _____

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses _____

Number of bits borrowed

Show your work for Problem 4 in the space below.

[illegible]

Custom Subnet Masks

Problem 6

Number of needed subnets **126**

Number of needed usable hosts **131,070**

Network Address **118.0.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 6 in the space below.

Number of Hosts	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576	2097152	4194304	8388608	16777216	33554432	67108864	134217728	268435456	536870912	1073741824	2147483648	4294967296	8589934592	17179869184	34359738368	68719476736	137438953472	274877906944	549755813888	1099511627776	2199023255552	4398046511104	8796093022208	17592186044416	35184372088832	70368744177664	140737488355328	281474976710656	562949953421312	1125899906842624	2251799813685248	4503599627370496	9007199254740992	18014398509481984	36028797018963968	72057594037927936	144115188075855872	288230376151711744	576460752303423488	1152921504606846976	2305843009213693952	4611686018427387904	9223372036854775808	18446744073709551616	36893488147419103232	73786976294838206464	147573952589676412928	295147905179352825856	590295810358705651712	1180591620717411303424	2361183241434822606848	4722366482869645213696	9444732965739290427392	18889465931478580854784	37778931862957161709568	75557863725914323419136	151115727451828646838272	302231454903657293676544	604462909807314587353088	1208925819614629174706176	2417851639229258349412352	4835703278458516698824704	9671406556917033397649408	19342813113834066795298816	38685626227668133590597632	77371252455336267181195264	154742504910672534362390528	309485009821345068724781056	618970019642690137449562112	1237940039285380274899124224	2475880078570760549798248448	4951760157141521099596496896	9903520314283042199192993792	19807040628566084398385987584	39614081257132168796771975168	79228162514264337593543950336	158456325028528675187087900672	316912650057057350374175801344	633825300114114700748351602688	1267650600228229401496703205376	2535301200456458802993406410752	5070602400912917605986812821504	10141204801825835211973625643008	20282409603651670423947251286016	40564819207303340847894502572032	81129638414606681695789005144064	162259276829213363391578010288128	324518553658426726783156020576256	649037107316853453566312041152512	1298074214633706907132624082305024	2596148429267413814265248164610048	5192296858534827628530496329220096	10384593717069655257060992658440192	20769187434139310514121985316880384	41538374868278621028243970633760768	83076749736557242056487941267521536	166153499473114484112975882535043072	332306998946228968225951765070086144	664613997892457936451903530140172288	1329227995784915872903807060280344576	2658455991569831745807614120560689152	5316911983139663491615228241121378304	10633823966279326983230456482242756608	21267647932558653966460912964485513216	42535295865117307932921825928971026432	85070591730234615865843651857942052864	170141183460469231731687303715884105728	340282366920938463463374607431768211456	680564733841876926926749214863536422912	1361129467683753853853498429727072845824	272225893536750770770699685945414569152	544451787073501541541399371890829138304	1088903574147003083082798743781658276608	2177807148294006166165597487563316553216	4355614296588012332331194975126633106432	8711228593176024664662389950253266212864	17422457186352049329324779900506524425728	34844914372704098658649559801013048851456	69689828745408197317299119602026097702912	139379657490816394634598239204052195405824	278759314981632789269196478408104390811648	557518629963265578538392956816208781623296	1115037259926531157076785913632417563246592	2230074519853062314153571827264835126493184	
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Custom Subnet Masks

Problem 7

Number of needed subnets **2000**
 Number of needed usable hosts **15**
 Network Address **178.100.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 7 in the space below.

		65,536	32,768	16,384	8,192	4,096	2,048	1,024	512		256	128	64	32	16	8	4	2
Number of Hosts	-	-----	-----	-----	-----	-----	-----	-----	-----									
		2	4	8	16	32	64	128	256		512	1024	2048	4096	8192	16384	32768	65536
Number of Subnets	-	-----	-----	-----	-----	-----	-----	-----	-----									
Binary values	-	128	64	32	16	8	4	2	1	.	128	64	32	16	8	4	2	1
		178	100	0	0	0	0	0	0	.	0	0	0	0	0	0	0	0

Custom Subnet Masks

Problem 8

Number of needed subnets **3**

Number of needed usable hosts **45**

Network Address **200.175.14.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 8 in the space below.

Custom Subnet Masks

Problem 9

Number of needed subnets **60**

Number of needed usable hosts **1,000**

Network Address **128.77.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 9 in the space below.

Custom Subnet Masks

Problem 10

Number of needed usable hosts **60**

Network Address **198.100.10.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 10 in the space below.

Custom Subnet Masks

Problem 11

Number of needed subnets **250**

Network Address **101.0.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 11 in the space below.

Custom Subnet Masks

Problem 12

Number of needed subnets **5**

Network Address **218.35.50.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 12 in the space below.

Custom Subnet Masks

Problem 13

Number of needed usable hosts **25**

Network Address **218.35.50.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 13 in the space below.

Custom Subnet Masks

Problem 14

Number of needed subnets **10**

Network Address **172.59.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 14 in the space below.

Custom Subnet Masks

Problem 15

Number of needed usable hosts **50**

Network Address **172.59.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 15 in the space below.

Custom Subnet Masks

Problem 16

Number of needed usable hosts **29**

Network Address **23.0.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

Show your work for Problem 16 in the space below.

Subnetting

Problem 1

Number of needed subnets **14**

Number of needed usable hosts **14**

Network Address **192.10.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 240

Total number of subnets 16

Total number of host addresses 16

Number of usable addresses 14

Number of bits borrowed 4

What is the 4th subnet range? 192.10.10.48 to 192.10.10.63

What is the subnet number for the 8th subnet? 192 . 10 . 10 . 112

What is the subnet broadcast address for the 13th subnet? 192 . 10 . 10 . 207

What are the assignable addresses for the 9th subnet? 192.10.10.129 to 192.10.10.142

Show your work for Problem 1 in the space below.

Number of Subnets	256 128 64 32				16 8 4 2				Number of Hosts	
	2	4	8	16	32	64	128	256		
	128 64 32 16				8 4 2 1				Binary values	
192.10.10.0	0	0	0	0	0	0	0	0		
(1)	0	0	0	0	192.10.10.0	to	192.10.10.15			
(2)	0	0	0	1	192.10.10.16	to	192.10.10.31			
(3)	0	0	1	0	192.10.10.32	to	192.10.10.47			
(4)	0	0	1	1	192.10.10.48	to	192.10.10.63			
(5)	0	1	0	0	192.10.10.64	to	192.10.10.79			
(6)	0	1	0	1	192.10.10.80	to	192.10.10.95			
(7)	0	1	1	0	192.10.10.96	to	192.10.10.111			
(8)	0	1	1	1	192.10.10.112	to	192.10.10.127			
(9)	1	0	0	0	192.10.10.128	to	192.10.10.143			
(10)	1	0	0	1	192.10.10.144	to	192.10.10.159			
(11)	1	0	1	0	192.10.10.160	to	192.10.10.175			
(12)	1	0	1	1	192.10.10.176	to	192.10.10.191			
(13)	1	1	0	0	192.10.10.192	to	192.10.10.207			
(14)	1	1	0	1	192.10.10.208	to	192.10.10.223			
(15)	1	1	1	0	192.10.10.224	to	192.10.10.239			
(16)	1	1	1	1	192.10.10.240	to	192.10.10.255			

$$\begin{array}{r}
 128 \\
 64 \\
 32 \\
 +16 \\
 \hline
 \text{Custom subnet mask } 240
 \end{array}$$

$$\begin{array}{r}
 16 \\
 -2 \\
 \hline
 \text{Usable subnets } 14
 \end{array}$$

$$\begin{array}{r}
 16 \\
 -2 \\
 \hline
 \text{Usable hosts } 14
 \end{array}$$

The binary value of the last bit borrowed is the range. In this problem the range is 16.

The first address in each subnet range is the subnet number.

The last address in each subnet range is the subnet broadcast address.

Subnetting

Problem 2

Number of needed subnets **1000**

Number of needed usable hosts **60**

Network Address **165.100.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

What is the 15th subnet range? 165.100.3.128 to 165.100.3.191

What is the subnet number for the 6th subnet? 165 . 100 . 1 . 64

What is the subnet broadcast address for the 6th subnet? 165 . 100 . 1 . 127

What are the assignable addresses for the 9th subnet? 165.100.2.1 to 165.100.0.62

Show your work for Problem 2 in the space below.

[illegible]

Subnetting

Problem 3

Hint: It is possible to borrow one bit to create two subnets.

Number of needed subnets **2**

Network Address **195.223.50.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 2nd
subnet range? _____

What is the subnet number
for the 2nd subnet? _____

What is the subnet
broadcast address for
the 1st subnet? _____

What are the assignable
addresses for the 1st
subnet? _____

Show your work for Problem 3 in the space below.

		256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256		
		128	64	32	16	8	4	2	1	-	Binary values
195. 223 . 50 . 0		0	0	0	0	0	0	0	0		

Subnetting

Problem 4

Number of needed subnets **750**

Network Address **190.35.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 15th
subnet range? _____

What is the subnet number
for the 13th subnet? _____

What is the subnet
broadcast address for
the 10th subnet? _____

What are the assignable
addresses for the 6th
subnet? _____

Show your work for Problem 4 in the space below.

Subnetting

Problem 5

Number of needed usable hosts **6**

Network Address **126.0.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 2nd
subnet range? _____

What is the subnet number
for the 5th subnet? _____

What is the subnet
broadcast address for
the 7th subnet? _____

What are the assignable
addresses for the 10th
subnet? _____

Show your work for Problem 5 in the space below.

Subnetting

Problem 6

Number of needed subnets **10**

Network Address **192.70.10.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 9th
subnet range? _____

What is the subnet number
for the 4th subnet? _____

What is the subnet
broadcast address for
the 12th subnet? _____

What are the assignable
addresses for the 10th
subnet? _____

Show your work for Problem 6 in the space below.

Subnetting

Problem 7

Network Address **10.0.0.0 /16**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 11th
subnet range? _____

What is the subnet number
for the 6th subnet? _____

What is the subnet
broadcast address for
the 2nd subnet? _____

What are the assignable
addresses for the 9th
subnet? _____

Show your work for Problem 7 in the space below.

Subnetting

Problem 8

Number of needed subnets **5**

Network Address **172.50.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 4th
subnet range? _____

What is the subnet number
for the 5th subnet? _____

What is the subnet
broadcast address for
the 6th subnet? _____

What are the assignable
addresses for the 3rd
subnet? _____

Show your work for Problem 8 in the space below.

Subnetting

Problem 9

Number of needed usable hosts **28**

Network Address **172.50.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 2nd
subnet range? _____

What is the subnet number
for the 10th subnet? _____

What is the subnet broadcast
address for
the 4th subnet? _____

What are the assignable
addresses for the 6th
subnet? _____

Show your work for Problem 9 in the space below.

Subnetting

Problem 10

Number of needed subnets **45**

Network Address **220.100.100.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 5th
subnet range? _____

What is the subnet number
for the 4th subnet? _____

What is the subnet
broadcast address for
the 13th subnet? _____

What are the assignable
addresses for the 12th
subnet? _____

Show your work for Problem 10 in the space below.

Subnetting

Problem 11

Number of needed usable hosts **8,000**

Network Address **135.70.0.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 6th
subnet range? _____

What is the subnet number
for the 7th subnet? _____

What is the subnet
broadcast address for
the 3rd subnet? _____

What are the assignable
addresses for the 5th
subnet? _____

Show your work for Problem 11 in the space below.

Subnetting

Problem 12

Number of needed usable hosts **45**

Network Address **198.125.50.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 2nd
subnet range? _____

What is the subnet number
for the 2nd subnet? _____

What is the subnet
broadcast address for
the 4th subnet? _____

What are the assignable
addresses for the 3rd
subnet? _____

Show your work for Problem 12 in the space below.

Subnetting

Problem 13

Network Address **165.200.0.0 /26**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 10th
subnet range? _____

What is the subnet number
for the 11th subnet? _____

What is the subnet
broadcast address for
the 1023rd subnet? _____

What are the assignable
addresses for the 1022nd
subnet? _____

Show your work for Problem 13 in the space below.

Subnetting

Problem 14

Number of needed usable hosts **16**

Network Address **200.10.10.0**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 7th
subnet range? _____

What is the subnet number
for the 5th subnet? _____

What is the subnet
broadcast address for
the 4th subnet? _____

What are the assignable
addresses for the 6th
subnet? _____

Show your work for Problem 14 in the space below.

Subnetting

Problem 15

Network Address **93.0.0.0 /19**

Address class _____

Default subnet mask _____

Custom subnet mask _____

Total number of subnets _____

Total number of host addresses _____

Number of usable addresses _____

Number of bits borrowed _____

What is the 15th
subnet range? _____

What is the subnet number
for the 9th subnet? _____

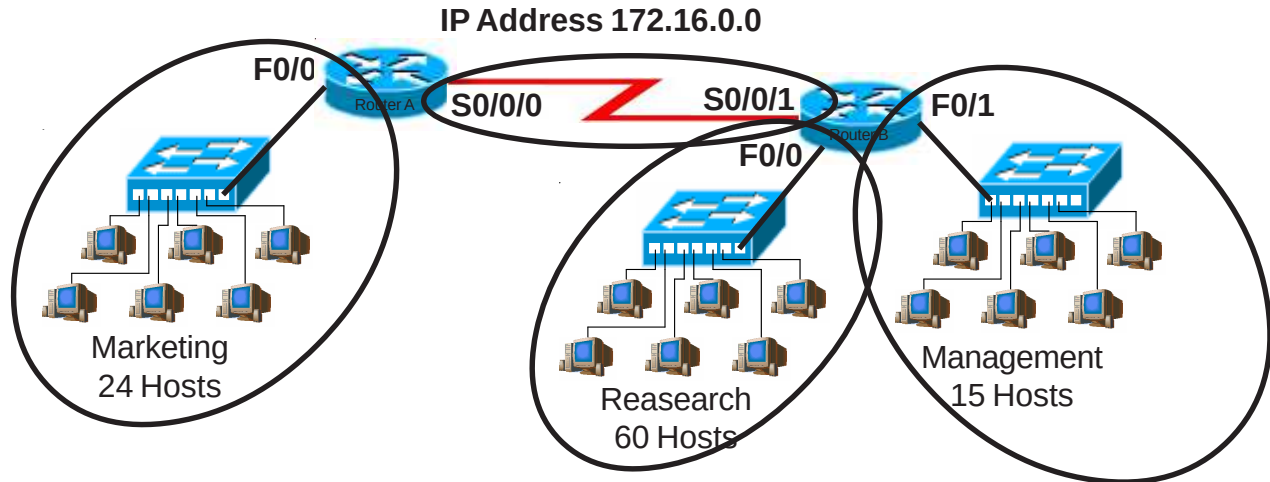
What is the subnet
broadcast address for
the 7th subnet? _____

What are the assignable
addresses for the 12th
subnet? _____

Show your work for Problem 15 in the space below.

Practical Subnetting 1

Based on the information in the graphic shown, design a network addressing scheme that will supply the minimum number of subnets, and allow enough extra subnets and hosts for 100% growth in both areas. Circle each subnet on the graphic and answer the questions below.



Address class	<u>B</u>
Custom subnet mask	<u>255.255.224.0</u>
Minimum number of subnets needed	<u>4</u>
Extra subnets required for 100% growth (Round up to the next whole number)	<u>+ 4</u>
Total number of subnets needed	<u>= 8</u>
Number of host addresses in the largest subnet group	<u>60</u>
Number of addresses needed for 100% growth in the largest subnet (Round up to the next whole number)	<u>+ 60</u>
Total number of address needed for the largest subnet	<u>= 120</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Research	<u>172.16.0.0 to 172.31.255</u>
IP address range for Marketing	<u>172.16.32.0 to 172.63.255</u>
IP address range for Management	<u>172.16.64.0 to 172.95.255</u>
IP address range for Router A to Router B serial connection	<u>172.16.96.0 to 172.127.255</u>

Show your work for Practical Subnetting 1 in the space below.

Number of Hosts	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072
Number of Subnets	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64
	172	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

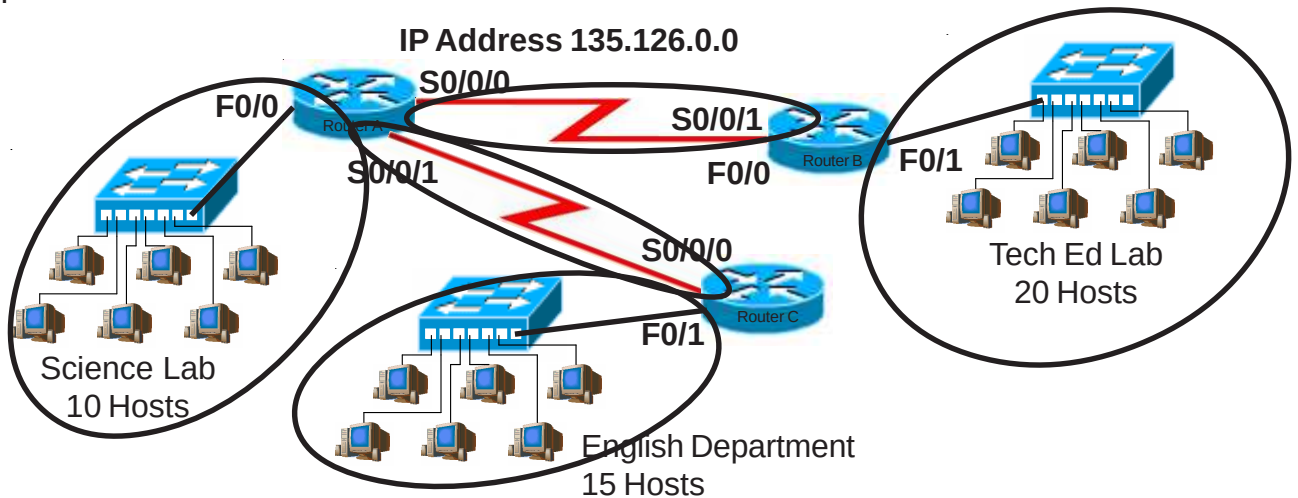
(1)			0	172.16.0.0	to	172.16.31.255
(2)			1	172.16.32.0	to	172.16.63.255
(3)		1	0	172.16.64.0	to	172.16.95.255
(4)		1	1	172.16.96.0	to	172.16.127.255
(5)	1	0	0	172.16.128.0	to	172.16.159.255
(6)	1	0	1	172.16.160.0	to	172.16.191.255
(7)	1	1	0	172.16.192.0	to	172.16.223.255
(8)	1	1	1	172.16.224.0	to	172.16.255.255

$$\begin{array}{r} 4 \\ \times 1.0 \\ \hline 4 \end{array}$$

$$\begin{array}{r} 60 \\ \times 1.0 \\ \hline 60 \end{array}$$

Practical Subnetting 2

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 30% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class	<u>B</u>
Custom subnet mask	<u>255.255.255.224</u>
Minimum number of subnets needed	<u>5</u>
Extra subnets required for 30% growth (Round up to the next whole number)	<u>+ 2</u>
Total number of subnets needed	<u>= 7</u>
Number of host addresses in the largest subnet group	<u>20</u>
Number of addresses needed for 30% growth in the largest subnet (Round up to the next whole number)	<u>+ 6</u>
Total number of address needed for the largest subnet	<u>= 26</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Tech Ed	<u>135.126.0.0 to 135.126.0.31</u>
IP address range for English	<u>135.126.0.32 to 135.126.0.63</u>
IP address range for Science	<u>135.126.0.64 to 135.126.0.95</u>
IP address range for Router A to Router B serial connection	<u>135.126.0.96 to 135.126.0.127</u>
IP address range for Router A to Router C serial connection	<u>135.126.0.128 to 135.126.0.159</u>

Show your work for Problem 2 in the space below.

Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
	135	126	0	0	0	0	0	0	0	0	0	0	0	0	0	0

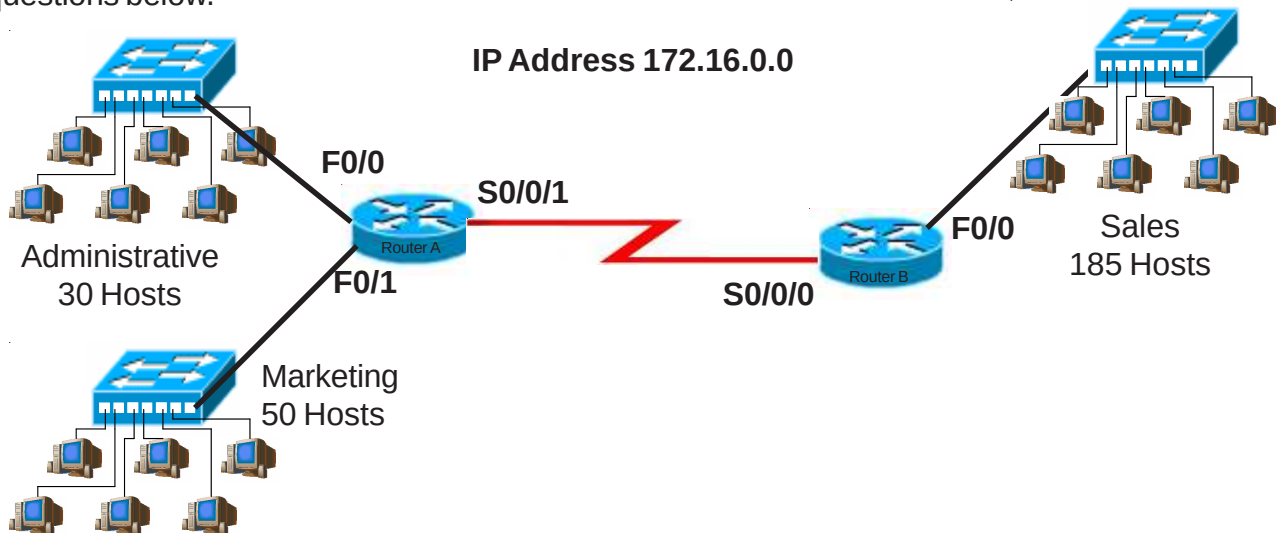
$$\begin{array}{r} 5 \\ \times 3 \\ \hline 15 \\ \text{(Round up to 2)} \end{array}$$

$$\begin{array}{r} 20 \\ \times 3 \\ \hline 6 \end{array}$$

(1)				0	135.126.0.0	to	135.126.0.31
(2)				1	135.126.0.32	to	135.126.0.63
(3)			1	0	135.126.0.64	to	135.126.0.95
(4)			1	1	135.126.0.96	to	135.126.0.127
(5)		1	0	0	135.126.0.128	to	135.126.0.159
(6)		1	0	1	135.126.0.160	to	135.126.0.191
(7)		1	1	0	135.126.0.192	to	135.126.0.223
(8)		1	1	1	135.126.0.224	to	135.126.0.255
(9)	1	0	0	0	135.126.1.0	to	135.126.1.31
(10)	1	0	0	1	135.126.1.32	to	135.126.1.63
(11)	1	0	1	0	135.126.1.64	to	135.126.1.95
(12)	1	0	1	1	135.126.1.96	to	135.126.1.127
(13)	1	1	0	0	135.126.1.128	to	135.126.1.159
(14)	1	1	0	1	135.126.1.160	to	135.126.1.191
(15)	1	1	1	0	135.126.1.192	to	135.126.1.223
(16)	1	1	1	1	135.126.1.224	to	135.126.1.255

Practical Subnetting 3

Based on the information in the graphic shown, design a classfull network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 25% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class _____

Custom subnet mask _____

Minimum number of subnets needed _____

Extra subnets required for 25% growth + _____
(Round up to the next whole number)

Total number of subnets needed = _____

Number of host addresses
in the largest subnet group _____

Number of addresses needed for
25% growth in the largest subnet + _____
(Round up to the next whole number)

Total number of address
needed for the largest subnet = _____

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Sales _____

IP address range for Marketing _____

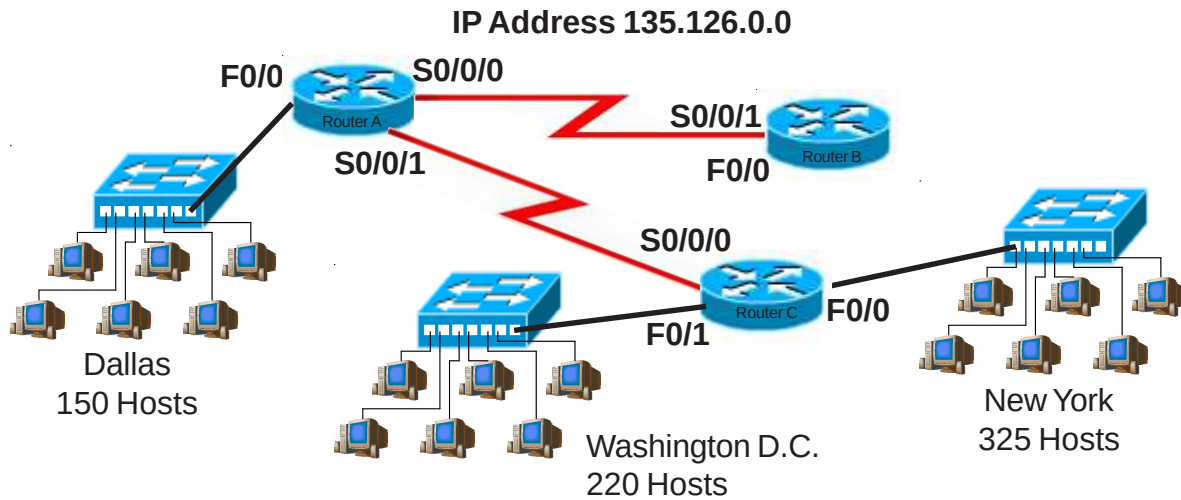
IP address range for Administrative _____

IP address range for Router A
to Router B serial connection _____

Show your work for Problem 3 in the space below.

Practical Subnetting 4

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of subnets**, and allow enough extra subnets and hosts for 70% growth in all areas. Circle each subnet on the graphic and answer the questions below. Circle each subnet on the graphic and answer the questions below.



Address class _____

Custom subnet mask _____

Minimum number of subnets needed _____

Extra subnets required for 70% growth $+$ _____
(Round up to the next whole number)

Total number of subnets needed $=$ _____

Number of host addresses
in the largest subnet group _____

Number of addresses needed for
70% growth in the largest subnet $+$ _____
(Round up to the next whole number)

Total number of address
needed for the largest subnet $=$ _____

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for New York _____

IP address range for Washington D. C. _____

IP address range for Dallas _____

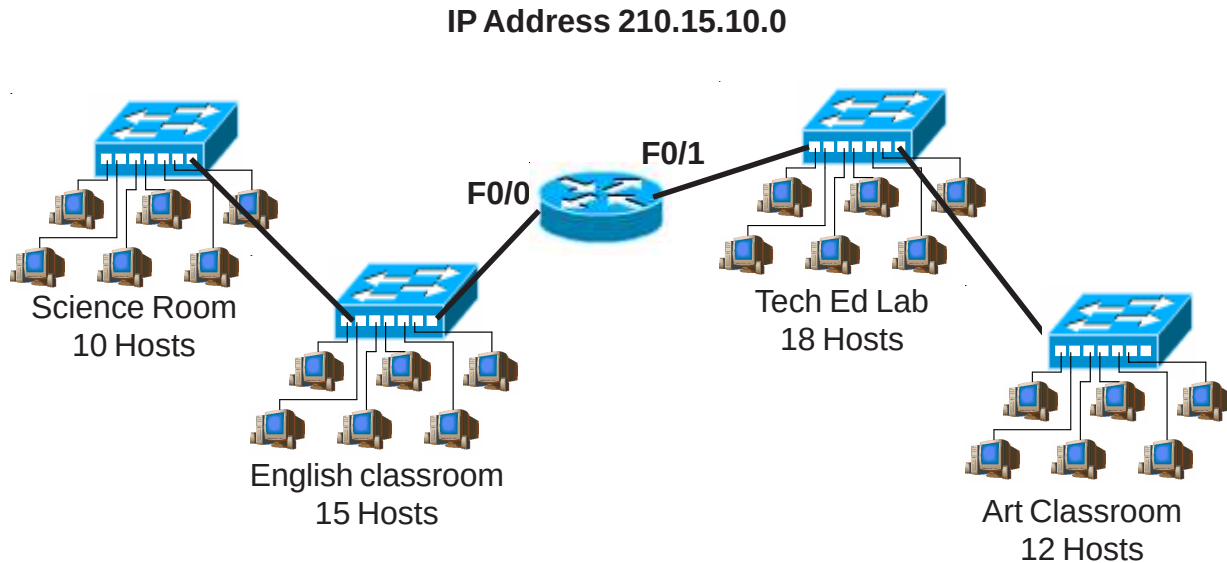
IP address range for Router A
to Router B serial connection _____

IP address range for Router A
to Router C serial connection _____

Show your work for Problem 4 in the space below.

Practical Subnetting 5

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 100% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class _____

Custom subnet mask _____

Minimum number of subnets needed _____

Extra subnets required for 100% growth $+$ _____
(Round up to the next whole number)

Total number of subnets needed $=$ _____

Number of host addresses
in the largest subnet group _____

Number of addresses needed for
100% growth in the largest subnet $+$ _____
(Round up to the next whole number)

Total number of address
needed for the largest subnet $=$ _____

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

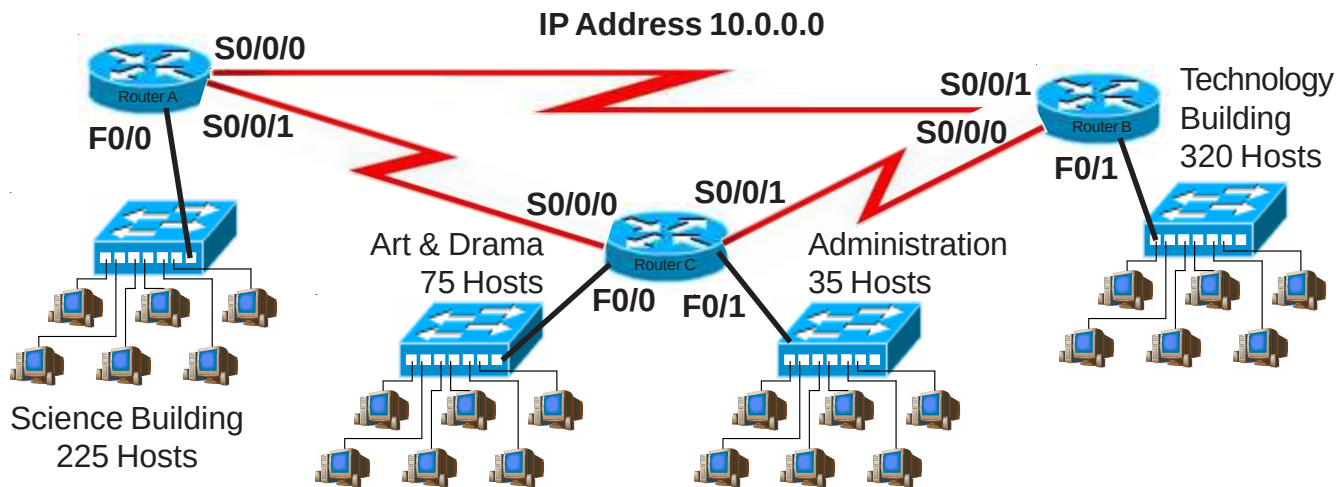
IP address range for Router F0/0 Port _____

IP address range for Router F0/1 Port _____

Show your work for Problem 5 in the space below.

Practical Subnetting 6

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of subnets**, and allow enough extra subnets and hosts for 20% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class _____

Custom subnet mask _____

Minimum number of subnets needed _____

Extra subnets required for 20% growth $+$ _____
(Round up to the next whole number)

Total number of subnets needed $=$ _____

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Technology _____

IP address range for Science _____

IP address range for Arts & Drama _____

IP Address range Administration _____

IP address range for Router A
to Router B serial connection _____

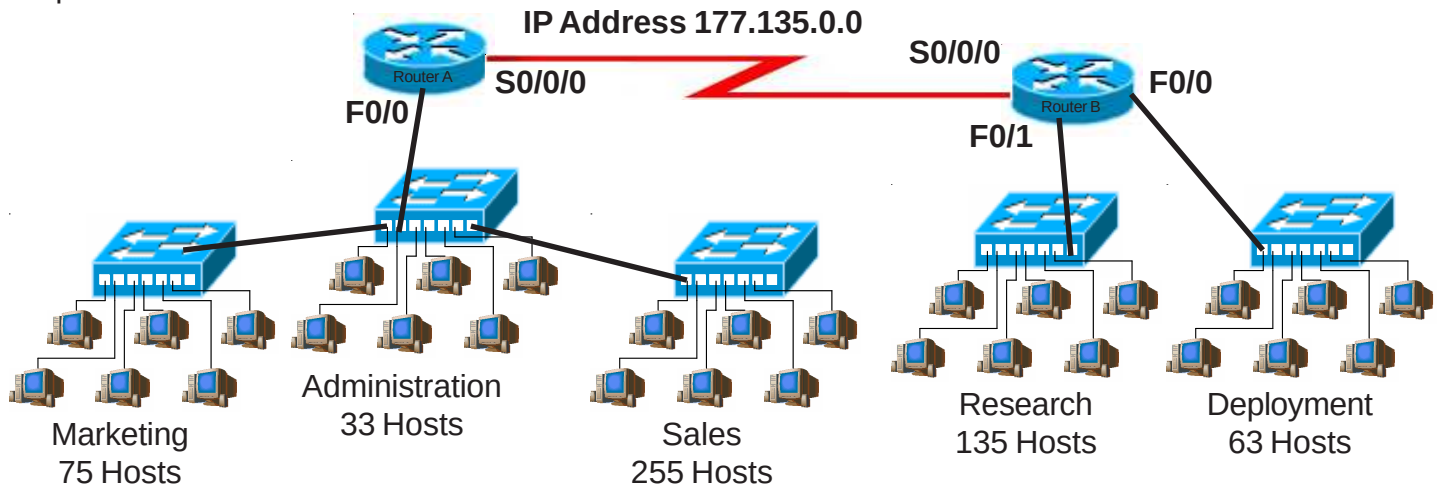
IP address range for Router A
to Router C serial connection _____

IP address range for Router B
to Router C serial connection _____

Show your work for Problem 6 in the space below.

Practical Subnetting 7

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 125% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class _____

Custom subnet mask _____

Minimum number of subnets needed _____

Extra subnets required for 125% growth + _____
(Round up to the next whole number)

Total number of subnets needed = _____

Number of host addresses in the largest subnet group _____

Number of addresses needed for 125% growth in the largest subnet + _____
(Round up to the next whole number)

Total number of address needed for the largest subnet = _____

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Router A Port F0/0 _____

IP address range for Research _____

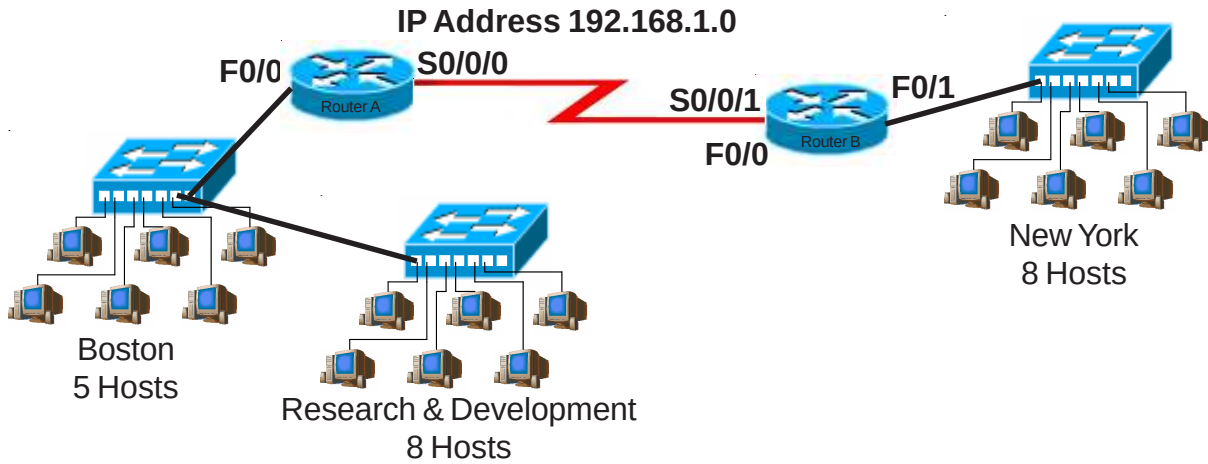
IP address range for Deployment _____

IP address range for Router A to Router B serial connection _____

Show your work for Problem 7 in the space below.

Practical Subnetting 8

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number subnets**, and allow enough extra subnets and hosts for 85% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class _____

Custom subnet mask _____

Minimum number of subnets needed _____

Extra subnets required for 85% growth $+$ _____
(Round up to the next whole number)

Total number of subnets needed $=$ _____

Number of host addresses
in the largest subnet group _____

Number of addresses needed for
85% growth in the largest subnet $+$ _____
(Round up to the next whole number)

Total number of address
needed for the largest subnet $=$ _____

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Router A F0/0 _____

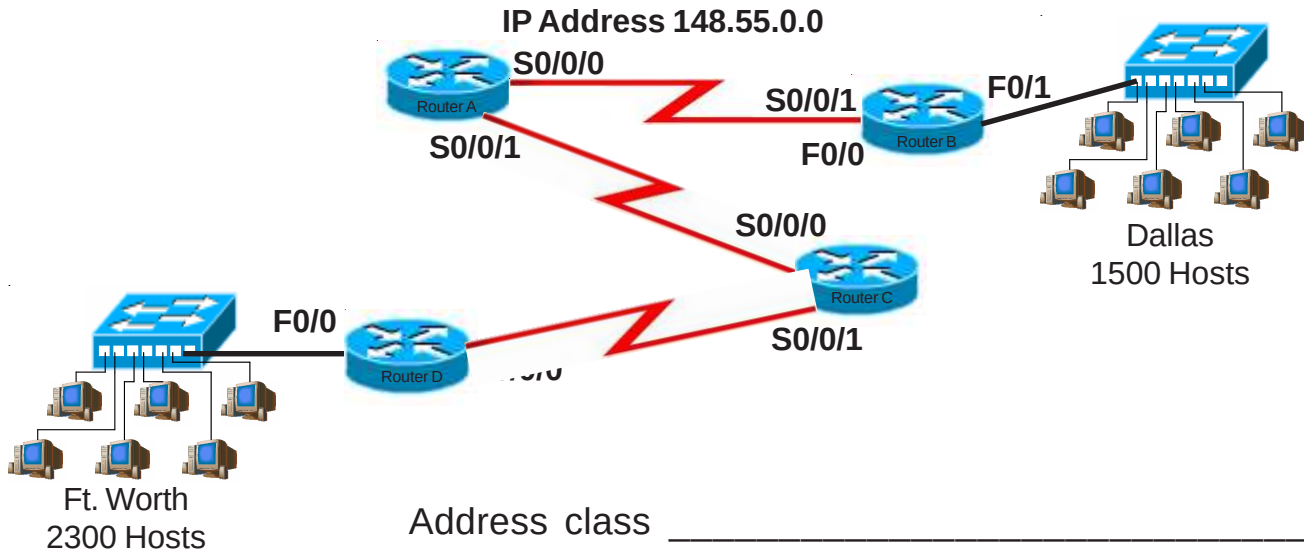
IP address range for New York _____

IP address range for Router A
to Router B serial connection _____

Show your work for Problem 8 in the space below.

Practical Subnetting 9

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 15% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class _____

Custom subnet mask _____

Minimum number of subnets needed _____

Extra subnets required for 15% growth + _____
(Round up to the next whole number)

Total number of subnets needed = _____

Number of host addresses in the largest subnet group _____

Number of addresses needed for 15% growth in the largest subnet + _____
(Round up to the next whole number)

Total number of address needed for the largest subnet = _____

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Ft. Worth _____

IP address range for Dallas _____

IP address range for Router A to Router B serial connection _____

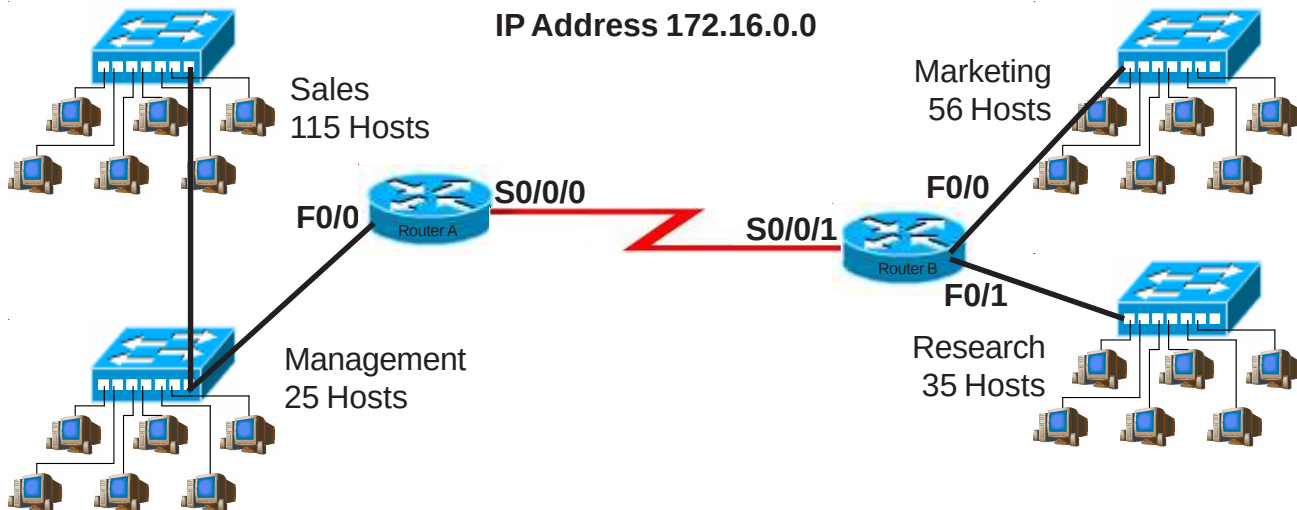
IP address range for Router A to Router C serial connection _____

IP address range for Router C to Router D serial connection _____

Show your work for Problem 9 in the space below.

Practical Subnetting 10

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of subnets**, and allow enough extra subnets and hosts for 110% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class _____

Custom subnet mask _____

Minimum number of subnets needed _____

Extra subnets required for 110% growth + _____
(Round up to the next whole number)

Total number of subnets needed = _____

Number of host addresses
in the largest subnet group _____

Number of addresses needed for
110% growth in the largest subnet + _____
(Round up to the next whole number)

Total number of address
needed for the largest subnet = _____

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Sales/Managemnt _____

IP address range for Marketing _____

IP address range for Research _____

IP address range for Router A
to Router B serial connection _____

Show your work for Problem 10 in the space below.

Valid and Non-Valid IP Addresses

Using the material in this workbook identify which of the addresses below are correct and usable. If they are not usable addresses explain why.

IP Address: 0.230.190.192

Subnet Mask: 255.0.0.0

Reference Page Inside Front Cover

The network ID cannot be 0.

IP Address: 192.10.10.1

Subnet Mask: 255.255.255.0

Reference Pages 28-29

OK

IP Address: 245.150.190.10

Subnet Mask: 255.255.255.0

Reference Page Inside Front Cover

IP Address: 135.70.191.255

Subnet Mask: 255.255.254.0

Reference Pages 48-49

IP Address: 127.100.100.10

Subnet Mask: 255.0.0.0

Reference Pages Inside Front Cover

IP Address: 93.0.128.1

Subnet Mask: 255.255.224.0

Reference Pages 56-57

IP Address: 200.10.10.128

Subnet Mask: 255.255.255.224

Reference Pages 54-55

IP Address: 165.100.255.189

Subnet Mask: 255.255.255.192

Reference Pages 30-31

IP Address: 190.35.0.10

Subnet Mask: 255.255.255.192

Reference Pages 34-35

IP Address: 218.35.50.195

Subnet Mask: 255.255.0.0

Reference Page Inside Front Cover

IP Address: 200.10.10.175 /22

Reference Pages 54-55 and/or Inside Front Cover

IP Address: 135.70.255.255

Subnet Mask: 255.255.224.0

Reference Pages 48-49

Reference Charts and Support Materials

Class A Addresses
VLSM Chart 8-15 Bits (2nd octet)

#	#	#10	#11	#12	#13	#14	#15
0-27.77% male	0-27.77% male	0-27.77% male	0-27.77% male	0-27.77% male	0-27.77% male	0-27.77% male	0-27.77% male
				0-51			
				6-75	7-77	7-77	
			0-51	16-23	16-23	16-23	
			16-23	24-31	24-31	24-31	
			32-47	32-47	32-47	32-47	
			48-55	48-55	48-55	48-55	
			56-63	56-63	56-63	56-63	
			64-71	64-71	64-71	64-71	
			72-79	72-79	72-79	72-79	
			80-87	80-87	80-87	80-87	
			88-95	88-95	88-95	88-95	
			96-101	96-101	96-101	96-101	
			102-117	102-117	102-117	102-117	
			118-129	118-129	118-129	118-129	
			130-143	130-143	130-143	130-143	
			144-159	144-159	144-159	144-159	
			160-175	160-175	160-175	160-175	
			176-188	176-188	176-188	176-188	
			189-195	189-195	189-195	189-195	
			196-201	196-201	196-201	196-201	
			202-207	202-207	202-207	202-207	
			208-215	208-215	208-215	208-215	
			216-221	216-221	216-221	216-221	
			222-228	222-228	222-228	222-228	
			229-236	229-236	229-236	229-236	
			237-243	237-243	237-243	237-243	
			244-250	244-250	244-250	244-250	
			251-257	251-257	251-257	251-257	
			258-265	258-265	258-265	258-265	

Class B Addresses
VLSM Chart 16-23 Bits (3rd octet)

[illegible]

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

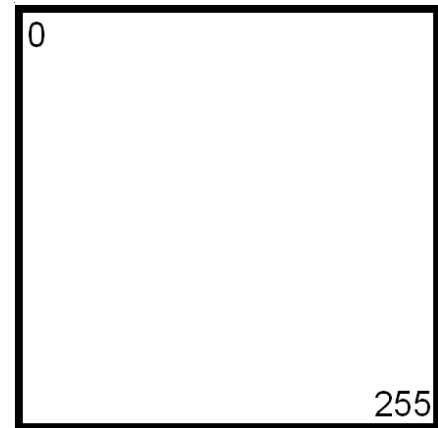
[illegible]

Visualizing Subnets Using The Box Method

The box method is the simplest way to visualize the breakdown of subnets and addresses into smaller sizes.

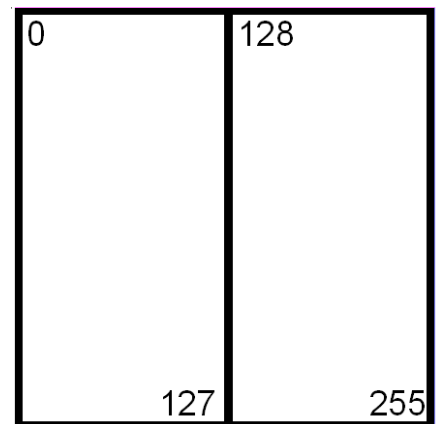
Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



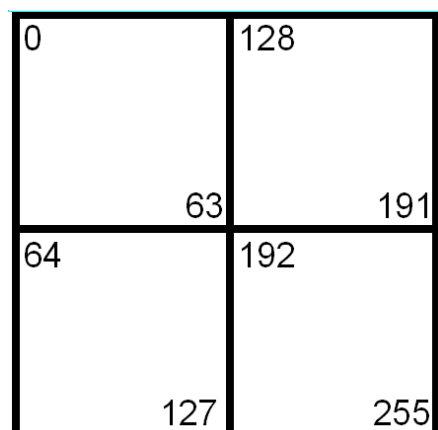
Split the box in half and you get two subnets with 128 addresses,

/25
255.255.255.128
128 Hosts
2 Subnets



Divide the box into quarters and you get four subnets with 64 addresses,

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you
get eight subnets with 32 addresses,

/27
255.255.255.224
32 Hosts
8 Subnets

0	32	128	160
	31	63	159
64	96	192	224
	95	127	255

Split the boxes in half again and you
get sixteen subnets with sixteen
addresses,

/28
255.255.255.240
16 Hosts
16 Subnets

0	32	128	160
	15	47	143
16	48	144	176
	31	63	159
64	96	192	224
	79	111	207
80	112	208	240
	95	127	255

The next split gives you thirty two
subnets with eight addresses,

/29
255.255.255.248
8 Hosts
32 Subnets

0	8	32	40	128	136	160	168
	7	15	39	47	135	143	175
16	24	48	56	144	152	176	184
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	71	79	103	111	199	207	239
80	88	112	120	208	216	240	248
	87	95	119	127	215	223	255

The last split gives sixty four subnets
with four addresses each,

/30
255.255.255.252
4 Hosts
64 Subnets

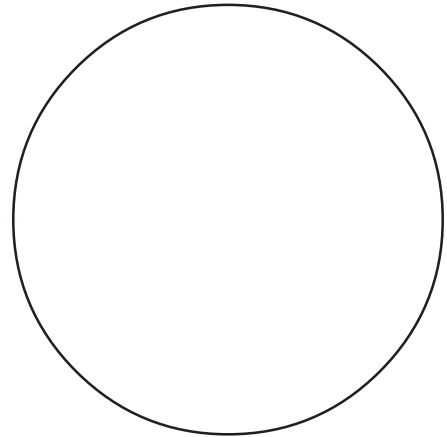
0	8	32	40	128	136	160	168
	3	11	35	43	131	139	171
4	12	36	44	132	140	164	172
	7	15	39	47	135	143	175
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	235
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	239
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	251
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	255

Visualizing Subnets Using The Circle Method

The circle method is another method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the different sections of the circle you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

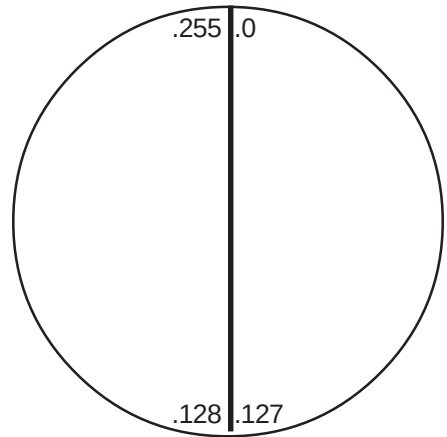
Start with a circle. The whole circle is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



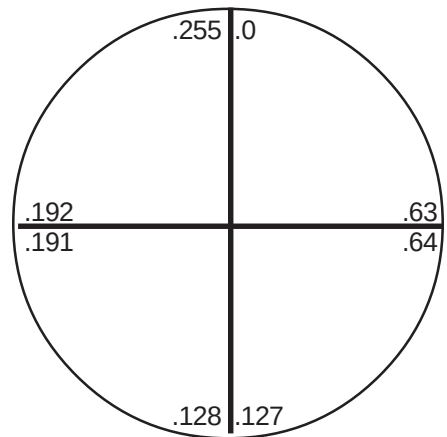
Split the circle in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



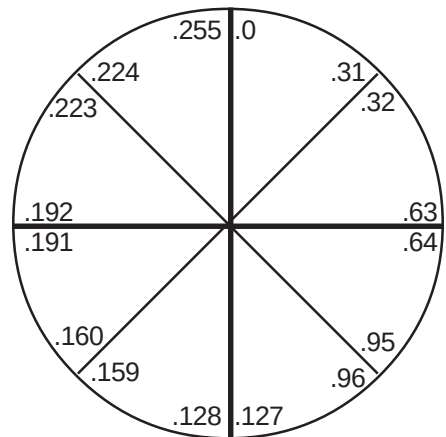
Divide the circle into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



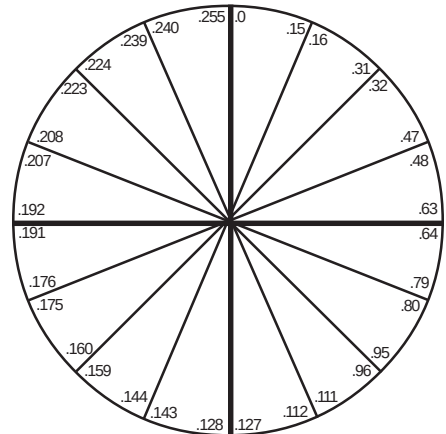
Split each quarter and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets



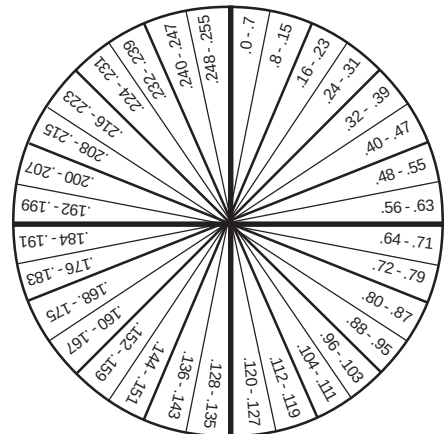
Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets



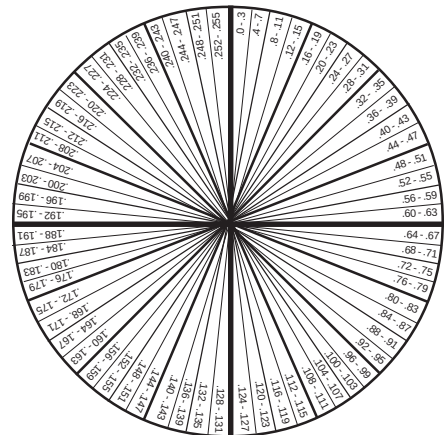
The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets



The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets



Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31	16-23	8-11	4-5
						12-15	6-7
					24-31	16-19	8-9
						20-23	10-11
		32-63	32-47	32-39	24-27	22-23	12-13
					28-31	24-25	14-15
				40-47	32-35	26-27	16-17
					36-39	28-29	18-19
			48-63	48-55	40-43	30-31	20-21
					44-47	32-33	22-23
				56-63	48-51	34-35	24-25
					52-55	36-37	26-27
		64-127	64-95	64-79	56-59	38-39	28-29
					60-63	40-41	30-31
				80-95	64-67	42-43	32-33
					68-71	44-45	34-35
			96-127	96-103	72-75	46-47	36-37
					76-79	48-49	38-39
				104-111	80-83	50-51	40-41
					84-87	52-53	42-43
			128-159	128-143	88-91	54-55	44-45
					92-95	56-57	46-47
				144-159	96-99	58-59	48-49
					100-103	60-61	50-51
		128-255	160-191	160-175	104-107	62-63	52-53
					108-111	64-65	54-55
				176-191	112-115	66-67	56-57
					116-119	68-69	58-59
		192-255	192-223	192-207	120-123	70-71	60-61
					124-127	72-73	62-63
				208-223	128-131	74-75	64-65
					132-135	76-77	66-67
			224-255	224-231	136-139	78-79	68-69
					140-143	80-81	70-71
				240-247	144-147	82-83	72-73
					148-151	84-85	74-75
			248-255	248-251	152-155	86-87	76-77
					156-159	88-89	78-79
				252-255	160-163	90-91	80-81
					164-167	92-93	82-83

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
			16-31	16-23		16-19	8-9
						20-23	10-11
					24-31	24-27	12-13
						28-31	14-15
			32-63	32-39		32-35	16-17
						36-39	18-19
				40-47		40-43	20-21
						44-47	22-23
				48-63		48-51	24-25
						52-55	26-27
					56-63	56-59	28-29
						60-63	30-31
		64-127	64-95	64-79	64-71	64-67	32-33
						68-71	34-35
				80-95	72-79	72-75	36-37
					80-87	80-83	38-39
			96-127	96-111	88-95	88-91	40-41
						92-95	42-43
				104-111	96-103	96-99	44-45
						100-103	46-47
				112-127	104-111	104-107	48-49
						108-111	50-51
					112-119	112-115	52-53
					120-127	116-119	54-55
	128-255	128-191	128-159	128-143		120-123	56-57
						124-127	58-59
				144-159	128-135	128-131	60-61
					136-143	132-135	62-63
			160-191	160-175	134-135	136-139	64-65
					140-143	140-141	66-67
				176-191	144-151	144-147	68-69
					152-159	148-151	70-71
		192-255	192-223	192-207	156-159	156-155	72-73
					160-163	160-163	74-75
				208-223	164-167	164-167	76-77
					168-175	168-171	78-79
			224-255	224-231	172-175	172-175	80-81
					176-179	176-179	82-83
				240-255	180-183	180-183	84-85
					184-187	184-187	86-87
			248-255	248-251	188-191	188-191	88-89
					192-195	192-195	90-91
				252-255	196-199	196-199	92-93
					200-203	200-203	94-95
					204-207	204-207	96-97
					208-211	208-211	98-99
					212-215	212-215	100-101
					216-219	216-219	102-103
					220-223	220-223	104-105
					224-227	224-227	106-107
					228-231	228-231	108-109
					232-235	232-235	110-111
					236-239	236-239	112-113
					240-243	240-243	114-115
					244-247	244-247	116-117
					248-251	248-251	118-119
					252-255	252-255	120-121
							122-123
							124-125
							126-127
							128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
					4-7	4-7
				16-31	8-15	8-11
					12-15	12-15
		64-127	32-63	32-47	16-23	16-19
					20-23	20-23
				48-63	24-31	24-27
					28-31	28-31
			64-95	64-79	32-39	32-35
					36-39	36-39
				80-95	40-47	40-43
					44-47	44-47
		96-127	96-111	96-103	48-55	48-51
					52-55	52-55
				104-111	56-59	56-59
					60-63	60-63
			112-127	112-119	64-71	64-67
					68-71	68-71
				120-127	72-79	72-75
					76-79	76-79
	128-255	128-191	128-159	128-143	80-87	80-83
					84-87	84-87
				144-159	88-95	88-91
					92-95	92-95
		160-191	160-175	160-167	96-99	96-99
					100-103	100-103
				176-191	104-107	104-107
					108-111	108-111
		192-255	192-223	192-207	112-115	112-115
					116-119	116-119
				208-223	120-123	120-123
					124-127	124-127
		224-255	224-239	224-231	128-131	128-131
					132-135	132-135
				232-239	136-143	136-139
					140-143	140-143
		240-255	240-247	240-247	144-147	144-147
					148-151	148-151
				248-255	152-155	152-155
					156-159	156-159
		252-255	252-255	252-255	160-163	160-163
					164-167	164-167
				252-255	168-171	168-171
					172-175	172-175
		252-255	252-255	252-255	176-179	176-179
					180-183	180-183
				252-255	184-187	184-187
					188-191	188-191
		252-255	252-255	252-255	192-195	192-195
					196-199	196-199
				252-255	200-203	200-203
					204-207	204-207
		252-255	252-255	252-255	208-211	208-211
					212-215	212-215
				252-255	216-219	216-219
					220-223	220-223
		252-255	252-255	252-255	224-227	224-227
					228-231	228-231
				252-255	232-235	232-235
					236-239	236-239
		252-255	252-255	252-255	240-243	240-243
					244-247	244-247
				252-255	248-251	248-251
					252-255	252-255

Class A Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/8	0	255.0.0.0	1	16,777,216	16,777,214
/9	1	255.128.0.0	2	8,388,608	8,388,606
/10	2	255.192.0.0	4	4,194,304	4,194,302
/11	3	255.224.0.0	8	2,097,152	2,097,150
/12	4	255.240.0.0	16	1,048,576	1,048,574
/13	5	255.248.0.0	32	524,288	524,286
/14	6	255.252.0.0	64	262,144	262,142
/15	7	255.254.0.0	128	131,072	131,070
/16	8	255.255.0.0	256	65,536	65,534
/17	9	255.255.128.0	512	32,768	32,766
/18	10	255.255.192.0	1,024	16,384	16,382
/19	11	255.255.224.0	2,048	8,192	8,190
/20	12	255.255.240.0	4,096	4,096	4,094
/21	13	255.255.248.0	8,192	2,048	2,046
/22	14	255.255.252.0	16,384	1,024	1,022
/23	15	255.255.254.0	32,768	512	510
/24	16	255.255.255.0	65,536	256	254
/25	17	255.255.255.128	131,072	128	126
/26	18	255.255.255.192	262,144	64	62
/27	19	255.255.255.224	524,288	32	30
/28	20	255.255.255.240	1,048,576	16	14
/29	21	255.255.255.248	2,097,152	8	6
/30	22	255.255.255.252	4,194,304	4	2

Class B Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/16	0	255.255.0.0	1	65,536	65,534
/17	1	255.255.128.0	2	32,768	32,766
/18	2	255.255.192.0	4	16,384	16,382
/19	3	255.255.224.0	8	8,192	8,190
/20	4	255.255.240.0	16	4,096	4,094
/21	5	255.255.248.0	32	2,048	2,046
/22	6	255.255.252.0	64	1,024	1,022
/23	7	255.255.254.0	128	512	510
/24	8	255.255.255.0	256	256	254
/25	9	255.255.255.128	512	128	126
/26	10	255.255.255.192	1,024	64	62
/27	11	255.255.255.224	2,048	32	30
/28	12	255.255.255.240	4,096	16	14
/29	13	255.255.255.248	8,192	8	6
/30	14	255.255.255.252	16,384	4	2

Class C Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/24	0	255.255.255.0	1	256	254
/25	1	255.255.255.128	2	128	126
/26	2	255.255.255.192	4	64	62
/27	3	255.255.255.224	8	32	30
/28	4	255.255.255.240	16	16	14
/29	5	255.255.255.248	32	8	6
/30	6	255.255.255.252	64	4	2

IPv6 Addressing and Subnetting Workbook

Version 1

Global Routing Prefix

Subnet ID

Interface ID

Hexadecimal

Student Name:

2001:0DB9

Types of IPv6 Addresses

Unspecified, Loopback, Embedded IPv4

Unspecified address is an all 0 address and cannot be assigned to an interface. It would be typed as ::. This is only used as a source address to indicate the absence of an actual address.

Loopback Address is all 0's except for the last bit, which is 1. It would be typed as ::1. It operates the same as the IPv4 127.0.0.1 loopback address.

IPv4 Embedded addresses are IPv6 addresses with an IPv4 address embedded in the low-order 32 bits. They are used to transition networks from IPv4 to IPv6.

Address Range:

0000:0000:0000:0000:0000:0000:0000:0000/8 to 00FF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/8

Global Unicast

Global Unicast addresses are used to uniquely identify a specific interface on a host and can be used as a public address on the internet.

Address Range:

2000:0000:0000:0000:0000:0000:0000:0000/3 to 3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/3

Unique local Unicast

Unique local Unicast addresses are roughly the same as IPv4 private addresses.

Address Range:

FC00:0000:0000:0000:0000:0000:0000:0000/7 to FDFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/7

Link-local Unicast

Link-local addresses are unicast addresses that are limited to a point to point connection within a local network. Routers will not forward packets with a link-local address.

Address Range:

FE80:0000:0000:0000:0000:0000:0000:0000/10 to FEBF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/10

Multicast

Multicast addresses are used to send a single packet to multiple destinations simultaneously.

Address Range:

FF00:0000:0000:0000:0000:0000:0000:0000/8 to FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/8

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Special Thanks to Melvin Baker and Jim Dorsch for taking the time to check this workbook for errors,
and to everyone who has sent in suggestions to improve the series.

A Brief History of TCP/IP Versions

TCP version 1 through TCP version 3 were developed as test versions and not widely used. Contrary to popular belief there was never an IPv1, IPv2, or IPv3. The version numbers were kept intact to avoid confusion when the TCP protocol was split into TCP and IP.

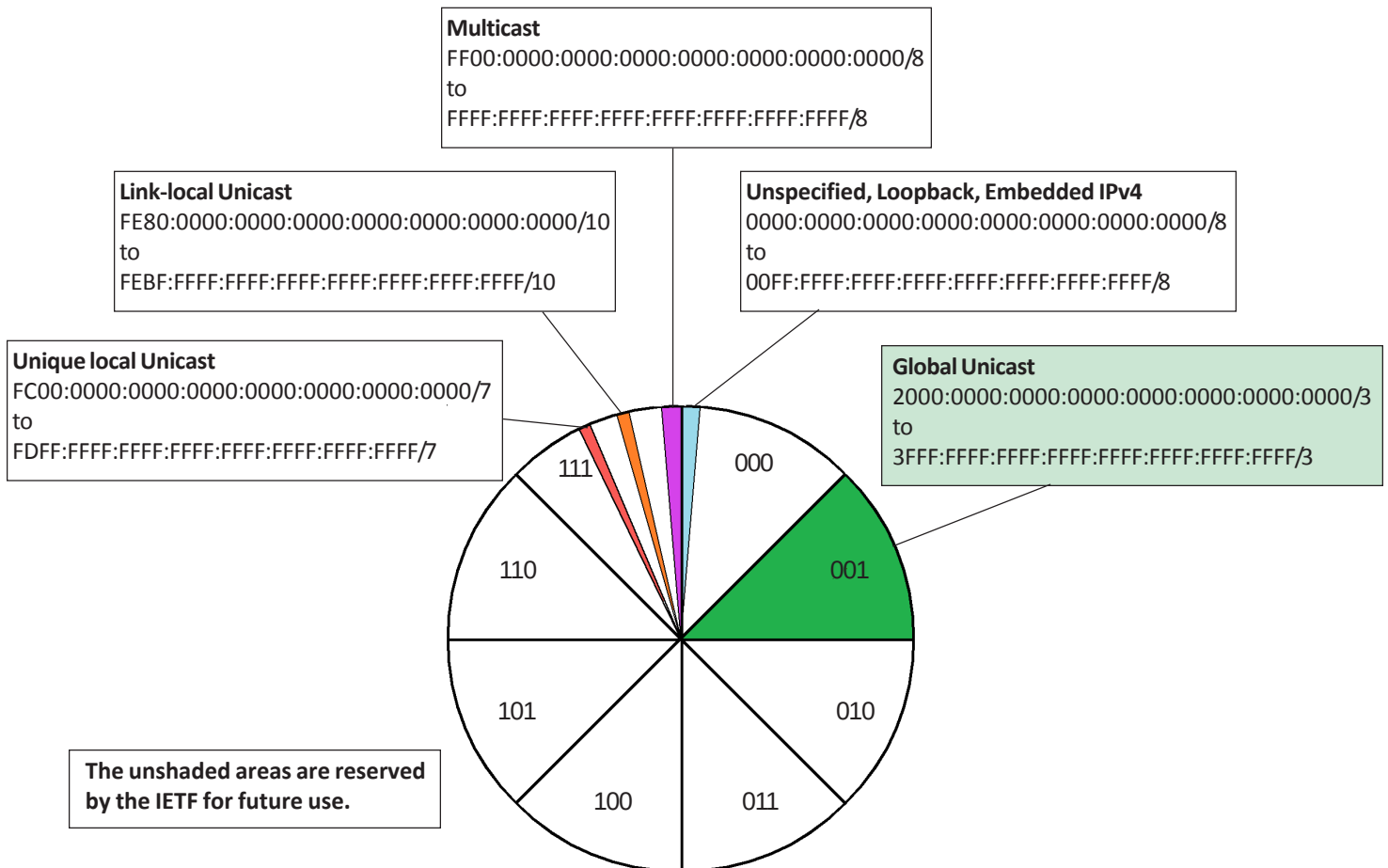
- 1973 - TCP version 1 was developed and documented in RFC 675. At this time IP was part of TCP.
- 1977 - TCP version 2 was developed and documented in March. In August of 1977 it was decided that the TCP protocol was going in the wrong direction.
- 1978 - TCP and IP were split into two separate protocols. Both TCP and IP were part of version 3.
- 1980 - Early development of IPv4 defined in RFC 760.
- 1981 - The current version of IPv4 is defined in RFC 791, 792 and 793. It was the first widely used version of the Internet Protocol.
- 1983 - On January 1, 1983, TCP/IP protocols became the only approved protocol on the ARPANET, replacing the earlier NCP protocol. This was known as flag day.
- 1984 - The number of hosts on the internet breaks 1000.
- 1987 - Hosts on the internet exceeds 10,000.
- 1989 - Host accessing the internet surpasses 100,000.
- 1990 - IPv5 relates to an experimental TCP/IP protocol called the Internet Stream Protocol, Version 2, originally defined in RFC 1190. This protocol was a peer of IPv4 and was designed to work with voice conversations and conferences with delay and bandwidth guarantees. These packets were assigned IP version 5 to differentiate them from "normal" IPv4 packets. This protocol was never introduced to the public, and was always considered experimental. To be sure there would be no confusion, version 5 was skipped over in favor of version 6.
- 1992 - The number of hosts on the internet breaks 1,000,000.
- 1995 - IPv6, introduced as IP Next Generation, was presented in RFC 1883.
- 1997 - The number of hosts using the internet exceeds 19,000,000.
- 1998 - The more fully developed IPv6 obsoletes RFC 1883 with the updated RFC 2460.

IPv4 has been well established for years. IPv6 is still in flux as it undergoes growing pains with changes and adjustments to the rules as it is being implemented.

IPv6

There are 340,282,366,920,938,463,463,374,607,431,768,211,456 possible IPv6 addresses.

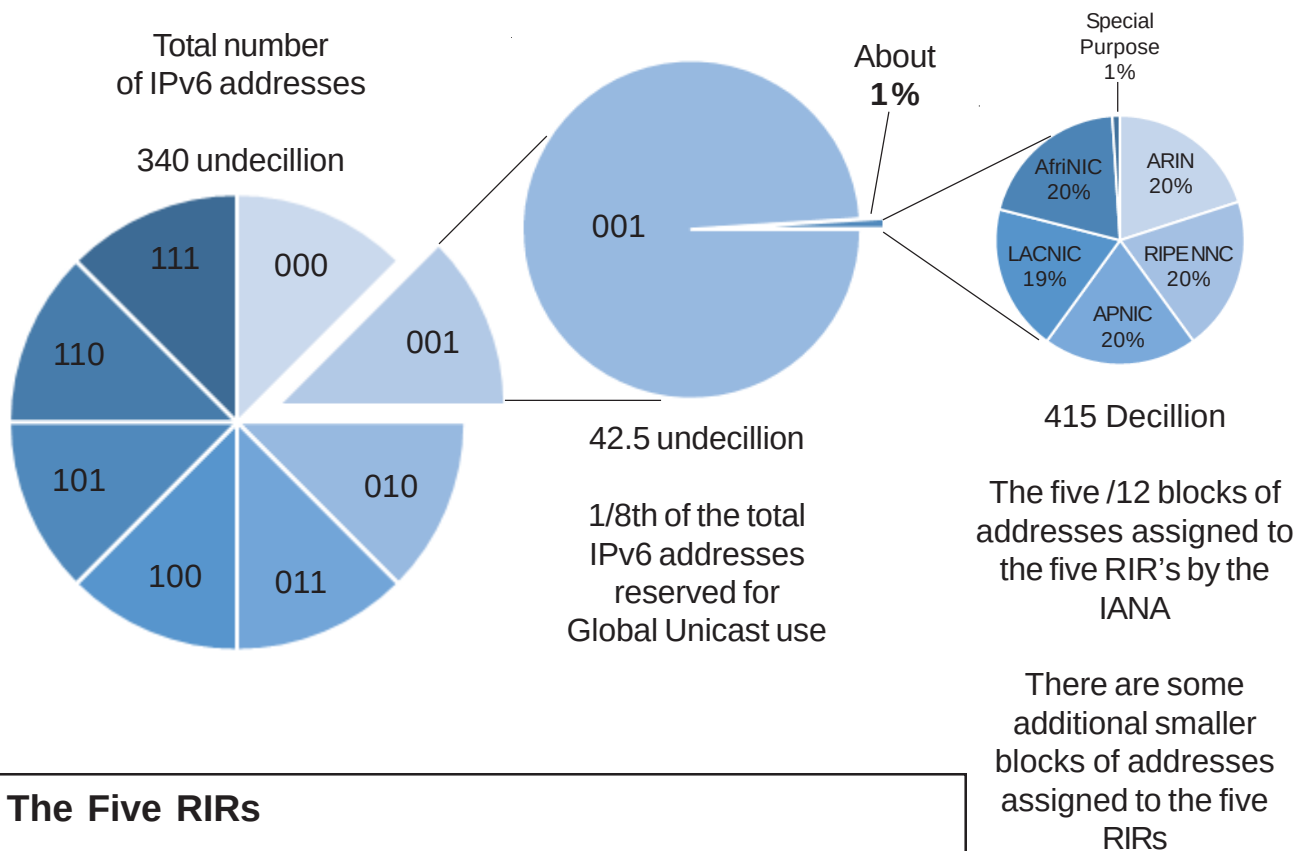
If you want to actually say the number it is three hundred and forty undecillion, two hundred and eighty-two decillion, three hundred and sixty-six nonillion, nine hundred and twenty octillion, nine hundred and thirty-eight septillion, four hundred and sixty-three sextillion, four hundred and sixty-three quintillion, three hundred and seventy-four quadrillion, six hundred and seven trillion, four hundred and thirty-one billion, seven hundred and sixty-eight million, two hundred and eleven thousand, four hundred and fifty-six. (or you can have Windows Narrator say it for you.)



The Internet Assigned Numbers Authority (IANA) divided the available IPv6 addresses into eight equal segments based on the three leading bits of the addresses (000, 001, 010, 011, 100, 101, 110, and 111). Only one eighth of the total available addresses have been reserved for use as global unicast addresses. Four smaller subgroups have been made available for unique local unicast, link-local unicast, multicast, and (unspecified, loopback, embedded IPv4).

IPv6 by the Numbers

340,282,366,920,938,463,463,374,607,431,768,211,456 Total number of IPv6 Addresses
 42,535,295,865,117,307,932,921,825,928,971,026,432 1/8 or the reserved Global Unicast addresses
 415,383,748,682,786,210,282,439,706,337,607,680 The five /12 ranges assigned to the RIRs
 7,119,157,000 Estimated world population
 58,347,322,398,253,923,924,200,534 Estimated number of IPv6 addresses per person
 (That's over 58 septillion addresses per person and doesn't include the additional smaller blocks of addresses assigned to the five RIRs by the IANA)



The Five RIRs

The **Regional Internet Registry** is an organization that manages the allocation and registration of internet number resources world wide. It has evolved over time to divide the world into five areas, or RIRs.

AfriNIC - African Network Information Centre

ARIN - American Registry for Internet Numbers

APNIC - Asia-Pacific Network Information Centre

LACNIC - Latin America and Caribbean Network Information Centre

RIPE NCC - Réseaux IP Européens Network Coordination Centre

There is a chart in the Reference Section that has all of these listed.

To make IPv6 addresses a little less imposing, two rules were developed to make them easier to work with. Rule 1: Omission of the Leading 0s, and Rule 2: Omission of the all-0 Hextets.

Rule 1: Omission of the Leading 0s

Rule 1 allows you to remove all the leading 0s in each individual hextet.

Sample 1 Unspecified address

Preferred Format: **0000:0000:0000:0000:0000:0000:0000:0000**
 Leading 0's removed: 0: 0: 0: 0: 0: 0: 0: 0
 or
 0:0:0:0:0:0:0:0

Sample 2 - Loopback Address

Preferred Format: **0000:0000:0000:0000:0000:0000:0000:0001**
 Leading 0's removed: 0: 0: 0: 0: 0: 0: 0: 1
 or
 0:0:0:0:0:0:0:1

Sample 3 – Global Unicast Address

Preferred Format: **2000:0000:0000:0000:0000:0000:0000:0001**
 Leading 0's removed: 2000: 0: 0: 0: 0: 0: 0: 1
 or
 2000:0:0:0:0:0:0:1

Sample 4 – Global Unicast Address

Preferred Format: **2001:00FE:ACAD:2013:0000:0000:00AA:0271**
 Leading 0's removed: 2001: FE:ACAD:2013: 0: 0: AA: 271
 or
 2001: FE:ACAD:2013:0:0:AA:271

Sample 5 – Unique local Unicast Address

Preferred Format: **FC80:0000:0000:ACAD:0000:0000:0000:0001**
 Leading 0's removed: FC80: 0: 0:ACAD: 0: 0: 0: 1
 or
 FC80:0:0:ACAD:0:0:0:1

Sample 6 – Link-local Address

Preferred Format: **FE80:ACAD:0000:0197:0000:0000:0000:FF01**
 Leading 0's removed: FE80:ACAD: 0: 197: 0: 0: 0:FF01
 or
 FE80:ACAD:0:197:0:0:0:FF01

Sample 7 – Multicast Address

Preferred Format: **FF00:0000:0000:ACAD:0000:0000:FE00:0721**
 Leading 0's removed: FF00: 0: 0:ACAD: 0: 0:FE00: 721
 or
 FF00:0:0:ACAD:0:0:FE00:721

Rule 1: Omission of the Leading 0s Problems

Using Rule 1 reduce the IPv6 addresses to their shortened form.

1. 0000:0000:0000:0000:0000:0000:0000:0000

2. 0000:0000:0000:0000:0000:0000:0000:0001

3. 2000:0000:0000:0000:0000:ABCD:0000:0025

4. 3F00:0090:0000:0000:0000:0098:0000:0001

5. 2001:3756:0005:0000:ACAD:0000:0000:0025

6. 3FFF:FF00:0000:0000:ACAD:0000:0000:0127

7. 2001:0000:0000:ABCD:FFFF:0000:0000:0001

8. 3ABC:0001:ACAD:0000:0000:1234:0000:0005

9. FC00:0000:0000:0000:3E00:1275:0000:0034

10. FE95:FC6C:C540:0000:0000:0000:0000:9800

11. FF00:ACAD:0000:0000:1234:0000:0000:0001

Rule 2: Omission of the All-0 Hextets

Rule 2 uses a double colon :: to represent a single contiguous set of all zero hextets. It can only be used once in any IPv6 address.

Sample 1 Unspecified address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0000
Contiguous 0's removed: ::

Sample 2 - Loopback Address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0001
Contiguous 0's removed: ::0001

Sample 3 – Global Unicast Address

Preferred Format: 2000:0000:0000:0000:0000:0000:0000:0001
Contiguous 0's removed: 2000: :0001
or
2000::0001

Sample 4 – Global Unicast Address

Preferred Format: 2001:00FE:ACAD:2013:0000:0000:00AA:0271
Contiguous 0's removed: 2001:00FE:ACAD:2013: :00AA:0271
or
2001:00FE:ACAD:2013::00AA:0271

Sample 5 – Unique local Unicast Address

Preferred Format: FC80:0000:0000:ACAD:0000:0000:0000:0001
Contiguous 0's removed: FC80:0000:0000:ACAD: :0001
or
FC80:0000:0000:ACAD::0001

Sample 6 – Link-local Address

Preferred Format: FE80:ACAD:0000:0197:0000:0000:0000:FF01
Contiguous 0's removed: FE80:ACAD:0000:0197: :FF01
or
FE80:ACAD:0000:0197::FF01

Sample 7 – Multicast Address

Preferred Format: FF00:0000:0000:ACAD:0000:0000:FE00:0721
Contiguous 0's removed: FF00: :ACAD:0000:0000:FE00:0721 (Option #1)
FF00:0000:0000:ACAD: :FE00:0721 (Option #2)
or
FF00::ACAD:0000:0000:FE00:0721 (Option #1)
FF00:0000:0000:ACAD::FE00:0721 (Option #2)

Rule 2: Omission of the All-0 Hextets Problems

Using Rule 2 reduce the IPv6 addresses to their shortened form.

1. 0000:0000:0000:0000:0000:0000:0000:0000

2. 0000:0000:0000:0000:0000:0000:0000:0001

3. 2000:0000:0000:0000:0000:ABCD:0000:0025

4. 3F00:0090:0000:0000:0000:0098:0000:0001

5. 2001:3756:0005:0000:ACAD:0000:0000:0025

6. 3FFF:FF00:0000:0000:ACAD:0025:0000:0127

7. 2001:ACAD:0000:ABCD:FFFF:0000:0000:0001

8. 3ABC:0001:ACAD:0000:0000:1234:0000:0005

9. FC00:0000:0000:0000:3E00:1275:0000:0034

10. FE95:FC6C:C540:0000:0000:0000:0000:9800

11. FF00:ACAD:0000:0000:1234:0000:0000:0001

Combining Rule 1 and Rule 2

To reduce the size of IPv6 address even more you can combine Rule 1 with Rule 2.

Sample 1 Unspecified address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0000
Combined reduction: ::

Sample 2 - Loopback Address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0001
Combined reduction: ::1

Sample 3 – Global Unicast Address

Preferred Format: 2000:0000:0000:0000:0000:0000:0000:0001
Combined reduction: 2000: :1
or
2000::1

Sample 4 – Global Unicast Address

Preferred Format: 2001:00FE:ACAD:2013:0000:0000:00AA:0271
Combined reduction: 2001: FE:ACAD:2013: : AA: 271
or
2001:FE:ACAD:2013::AA:271

Sample 5 – Unique local Unicast Address

Preferred Format: FC80:0000:0000:ACAD:0000:0000:0000:0001
Combined reduction: FC80: 0: 0:ACAD: : 1
or
FC80:0:0:ACAD::1

Sample 6 – Link-local Address

Preferred Format: FE80:ACAD:0000:0197:0000:0000:0000:FF01
Combined reduction: FE80:ACAD:0 : 197: :FF01
or
FE80:ACAD:0:197::FF01

Sample 7 – Multicast Address

Preferred Format: FF00:0000:0000:ACAD:0000:0000:FE00:0721
Combined reduction: FF00: :ACAD: 0: 0:FE00: 721 (Option #1)
FF00: 0: 0:ACAD: :FE00: 721 (Option #2)
or
FF00::ACAD:0:0:FE00:721 (Option #1)
FF00:0:0:ACAD::FE00:721 (Option #2)

Combining Rule 1 and Rule 2 Problems

Using Rule 1 and 2 reduce the IPv6 addresses to their shortest form.

1. 0000:0000:0000:0000:0000:0000:0000:0000

2. 0000:0000:0000:0000:0000:0000:0000:0001

3. 2000:0000:0000:0000:0000:ABCD:0000:0025

4. 3F00:0090:0000:0000:0000:0098:0000:0001

5. 2001:3756:0005:0000:ACAD:0000:0000:0025

6. 3FFF:FF00:0000:0000:ACAD:0025:0000:0127

7. 2001:ACAD:0000:ABCD:FFFF:0000:0000:0001

8. 3ABC:0001:ACAD:0000:0000:1234:0000:0005

9. FC00:0000:0000:0000:3E00:1275:0000:0034

10. FE95:FC6C:C540:0000:0000:0000:0000:9800

11. FF00:ACAD:0000:0000:1234:0000:0000:0001

Reverting Reduced Address Problems

The following addresses have been shorted using Rule1 and/or Rule 2. Expand them back to their preferred format.

Sample: FF00:ACAD:ABCD:0:1234::1

FF00:ACAD:ABCD:0000:1234:0000:0000:0001

1. 2000::1

2. ::1

3. 2001:0:0:0:0:ABCD:0:127

4. 3E80:0070::0098:0000:0001

5. 2FFF:38:5:0:ACAD::5

6. 3FFF::ACAD:25:0:100

7. 2002:ACAD:0:1BCD:FFFF::4

8. 3FAA:0025:ACAD::ABCD:0000:0005

9. FFFF::4E00:1235:0:34

10. 3E01:6C:40::9800

IPv6 Address Categories

All IPv6 addresses fall into one of three categories

Unicast -

Unicast addresses identify a unique interface on an IPv6 device.
It is a one to one connection between a source and destination.

Examples of IPv6 Unicast addresses include:

Global Unicast -

Similar to a public IPv4 address
Can be used as a public address on the internet
Globally unique
Can be static or dynamic

Link-Local -

Only used on a local network link to uniquely identify a host
Not routable on the public internet
No router will forward a link-local address
Every IPv6 enabled networked device is required to have a link-local address
Multiple interfaces on the same device can have the same link-local address

Loopback - (::1/128)

Used by a host to ping itself to test the TCP/IP stack
It cannot be assigned to a physical interface

Unspecified Address - (::/128)

Is only used as a source address to indicate the absence of an actual address

Unique Local -

Unique local addresses are roughly the same as IPv4 private addresses

Embedded IPv4 -

Used to transition IPv4 networks into IPv6

Multicast -

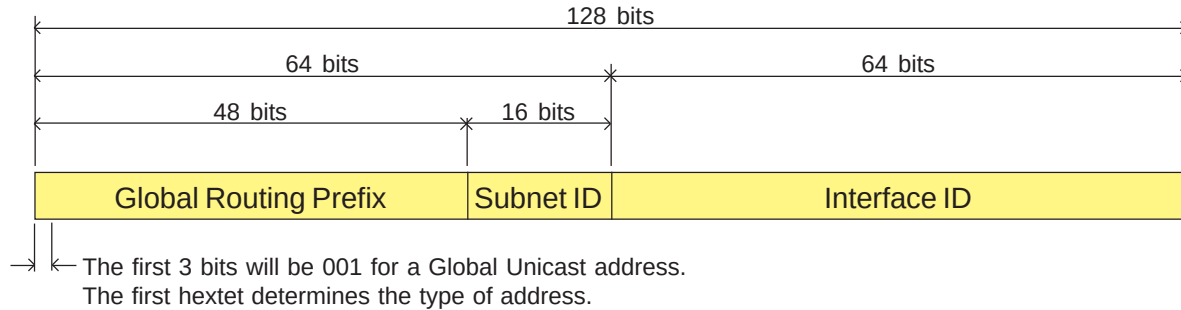
Multicast addresses are used to send a single packet to multiple destinations simultaneously.

Anycast -

Anycast addresses are described as a one-to-nearest or one-to-one-of-many packet delivery. For example, all routers in the same network will be assigned the same Anycast address. A packet sent to that address will only be delivered to the closest router with that address based on routing protocol metrics. Anycast addresses can be pulled from Global Unicast, Site-Local, or Link-Local address ranges. The first address and the last 128 addresses in a /64 Global Unicast range are reserved as the Subnet-Router Anycast Address.

There are no broadcast addresses in IPv6.

Global Unicast IPv6 Address



Global Routing Prefix -

This is assigned by the ISP to a customer or site.

The Global Routing Prefix is determined by the *prefix-length* notation. (example /48 or /64).

This is similar to the network portion of an IPv4 address.

Subnet ID -

This is similar to the subnet portion of an IPv4 address.

The difference is in IPv4 the subnet is borrowed from the host portion of the address.

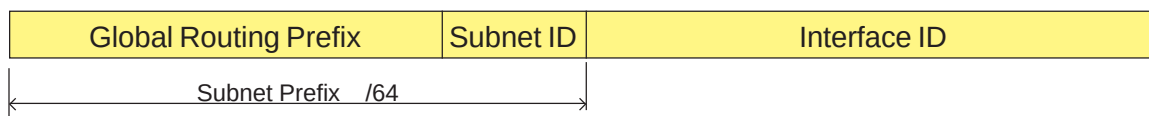
In IPv6 the Subnet ID is a separate field (/48 to /64) and not necessarily part of the Interface ID.

Interface ID -

The Interface ID uniquely identifies a interface on the local subnet.

Subnet Prefix

The Subnet Prefix is the address space used by the Global Routing Prefix and the Subnet ID, and can range from 0 to 128. The preferred Subnet Prefix length is /48 to /64 for customers or sites.



RFC 4291 recommends that the Interface-ID or host portion of your IPv6 address be 64 bits.

A minimum /64 prefix length is required to support Stateless Address Auto-configuration.

IPv6 Prefix Length vs IPv4 CIDR

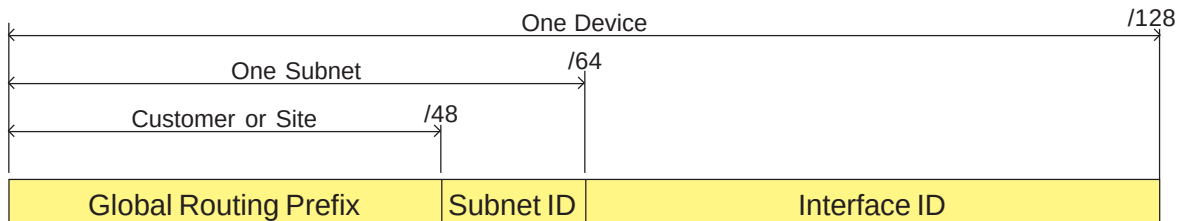
In IPv4 the network portion of the IP address is indicated with a dotted-decimal subnet mask; such as 255.255.255.0. It can also be identified with a CIDR (classless interdomain routing) notation; such as /24.

IPv6 does not use either of these terms to indicate the network portion of an IPv6 address. The network portion of the address is indicated with a **Prefix Length** at the end of the address. While a /48 or /56 looks like a CIDR notation it is not a classless interdomain routing notation. The prefix length indicates the number of nibbles or bits used in the network or subnetwork portion of an IPv6 address.

Global Unicast Prefix Allocations

In 2001, RFC 3177 was written to provide recommendations for how IPv6 Global Unicast addresses should be allocated to customers or Sites.

“In particular, it recommends the assignment of /48 in the general case, /64 when it is known that one and only one subnet is needed and /128 when it is absolutely known that one and only one device is connecting.”

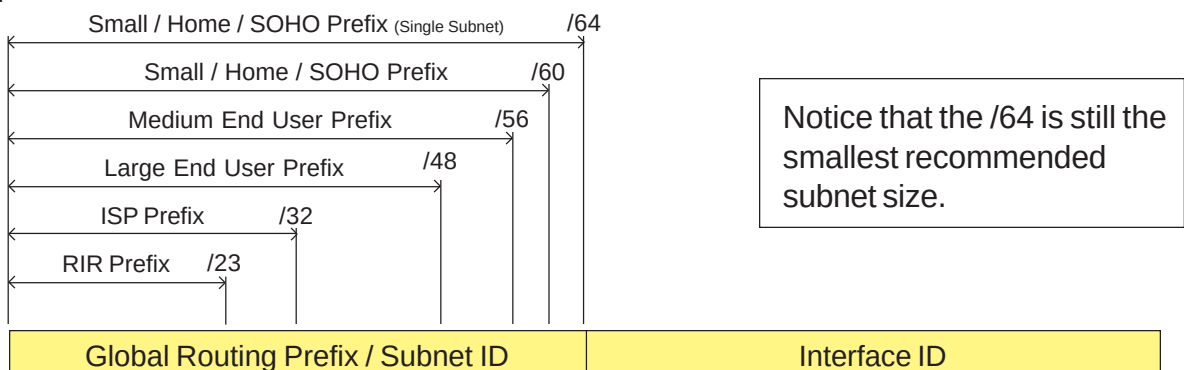


A /48 prefix allows each customer or Site to have 1,208,925,819,614,629,174,706,176 addresses. These can be used as a single subnet, or up to 65,536 subnets with a /64 prefix-length.

The Regional Internet Registries (RIRs) adopted RTC 3177 recommendation in 2002, but began reconsidering the policy in 2005. In March of 2011 RTC 6177 obsoleted RTC 3177 with a new recommendation.

“The exact choice of how much address space to assign end sites is an issue for the operational community.”

This gives local ISPs more options when assigning IPv6 addresses to their customers or Sites.



Typical IPv6 Prefix Assignments:

<u>Subscribers</u>	<u>Prefix- Length</u>	<u># of Subnets at /64</u>	<u>Total number of Possible Addresses</u>
RIR	/23	2,199,023,255,552	40,564,819,207,303,340,847,894,502,572,032
Service Provider (*LIR) -	/32	4,294,967,296	79,228,162,514,264,337,593,543,950,336
Large End User -	/48	65,536	1,208,925,819,614,629,174,706,176
Medium End User -	/56	256	4,722,366,482,869,645,213,696
Small / Home / SOHO -	/60	16	295,147,905,179,352,825,856
Small / Home / SOHO -	/64	1	18,446,744,073,709,551,616

Subnetting on the Nibble Boundary

One of the design themes driving IPv6 was to keep subnetting as simple as possible. The fourth hexet is reserved for subnetting, giving network administrators multiple options for developing a network plan. In order to keep IPv6 addressing as simple as possible it is a **Best Practice** to subnet on the nibble boundary.

Every IPv6 address is comprised of 128 bits, which is represented with 32 hexadecimal numbers.

2001:ACAD:1234:0000:0000:0000:0000:0000/48

Showing this address with 128 binary characters makes it difficult to read and almost impossible for most people to work with.

00100000000000001:1010110010101101:0001001000110100:0000000000000000:
0000000000000000:0000000000000000:0000000000000000:0000000000000000

Each hexadecimal number in an IPv6 address represents 4 bits or a Nibble. An IPv6 address is composed of 32 hexadecimal numbers or 32 Nibbles.

2001:ACAD:1234:0000:0000:0000:0000:0000/48

IPv6 can be subnetted just like IPv4 using individual binary bits. To keep subnetting as simple as possible it is a **Best Practice** to borrow 4 bits, or one Nibble at a time.

Nibble Boundary Subnets

(Subnetting on the Nibble Boundary)

Prefix	# of /64 subnets
/48	65,536
/52	4096
/56	256
/60	16
/64	1

It is a Best Practice to subnet on the Nibble Boundary.

Subnets Based on Individual Binary Bits

(Subnetting within a Nibble)

Prefix subnets	# of /64
/48	65,536
/49	32,768
/50	16,384
/51	8,192
/52	4,096
/53	2,048
/54	1,024
/55	512
/56	256
/57	128
/58	64
/59	32
/60	16
/61	8
/62	4
/63	2
/64	1

Site ID and Sub-Site IDs

Subnetting on the nibble boundary allows you to easily set up proper route aggregation and summarization to use for each subnet or location. It also allows for easier deployment of firewalls based on location or network users. In order to do this you need to assign Site ID's and Sub-Site ID's as necessary. The Site ID is the first address in the subnet you have assigned to a specific location or user group. Sub-Site ID's come into play if you subnet a location or user group into smaller subnets. The first address in each range becomes the Sub-Site ID.

Subnetting on the Nibble boundary gives you these subnetting options.

/48 No Nibbles	/52 1 Nibble	/56 2 Nibbles	/60 3 Nibbles	/64 4 Nibbles
/48 - 1 Subnet				
/52 - 16 Subnets	/52 - 1 Subnet			
/56 - 256 Subnets	/56 - 16 Subnets	/56 - 1 Subnets		
/60 - 4096 Subnets	/60 - 256 Subnets	/60 - 16 Subnets	/60 - 1 Subnets	
/64 - 65,536 Subnets	/64 - 4096 Subnets	/64 - 256 Subnets	/64 - 16 Subnets	/64 - 1 Subnet

As an example, your company has two offices, and within each office there are several groups you want on separate subnets. The groups include: Infrastructure, Management, Marketing, Finance, Research, Warehouse, and Sales.

Your ISP has assigned your company the IPv6 address 2001:ACAD:1234::/48. You will need one Site ID for each office. A /52 Subnet Prefix will give you 16 subnets, or you could use a /56 Subnet Prefix and have 256 subnets. For our purposes we'll use the /56 Subnet Prefix.

Our original IPv6 Range was: 2001:ACAD:1234::/48. Subnetting this address with a /56 Subnet Prefix will take two nibbles from the Subnet ID and give you the following address ranges:

2001:ACAD:1234:**0000**::/56 Save this range for over all infrastructure needs.

2001:ACAD:1234:**0100**::/56 This becomes the Site ID for Office 1.

2001:ACAD:1234:**0200**::/56 This becomes the Site ID for Office 2.

2001:ACAD:1234:**0300**::/56

(Lots of subnets omitted)

2001:ACAD:1234:**FF00**::/56

Office 1 needs subnets for Infrastructure, Management, and Sales. Subnet the Site ID for Office 1 with a /60 Subnet Prefix.

2001:ACAD:1234:01**00**::/60 This becomes the Sub-Site ID for Infrastructure needs.

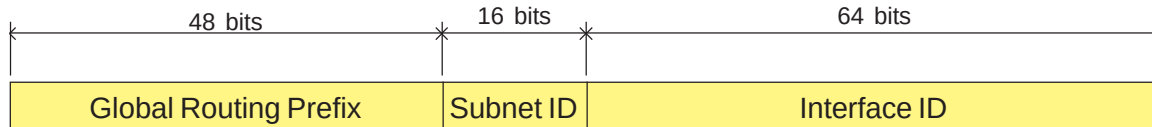
2001:ACAD:1234:01**10**::/60 This becomes the Sub-Site ID for Managment.

2001:ACAD:1234:01**20**::/60 This becomes the Sub-Site ID for Sales.

(Lots of subnets omitted)

2001:ACAD:1234:01**F0**::/60

Subnetting IPv6



Unlike IPv4 which requires you to borrow bits from the host portion, IPv6 has 16 bits or four hexadecimal numbers built into the address specifically allocated for creating subnets. A /48 address will allow you to have a single subnet or up to 65,536 subnets.

Lets say that your ISP has assigned you the 2001:ACAD:1234::/48 IPv6 address.

Global Routing Prefix	Subnet ID	Interface ID
2001:ACAD:1234:	0000	0000:0000:0000:0000

Basic subnetting in IPv6 was designed to make subnetting a very simple process. Start at the /64 bit and start counting up until you've used all the available bits in the subnet ID.

It really is that simple!

2001:ACAD:1234:0000:0000:0000:0000/64	1st subnet
2001:ACAD:1234:0001:0000:0000:0000:0000/64	2nd subnet
2001:ACAD:1234:0002:0000:0000:0000:0000/64	3rd subnet
2001:ACAD:1234:0003:0000:0000:0000:0000/64	4th subnet
2001:ACAD:1234:0004:0000:0000:0000:0000/64	5th subnet
2001:ACAD:1234:0005:0000:0000:0000:0000/64	6th subnet
2001:ACAD:1234:0006:0000:0000:0000:0000/64	7th subnet
2001:ACAD:1234:0007:0000:0000:0000:0000/64	8th subnet
2001:ACAD:1234:0008:0000:0000:0000:0000/64	9th subnet
2001:ACAD:1234:0009:0000:0000:0000:0000/64	10th subnet
2001:ACAD:1234:000A:0000:0000:0000:0000/64	11th subnet
2001:ACAD:1234:000B:0000:0000:0000:0000/64	12th subnet
2001:ACAD:1234:000C:0000:0000:0000:0000/64	13th subnet
2001:ACAD:1234:000D:0000:0000:0000:0000/64	14th subnet
2001:ACAD:1234:000E:0000:0000:0000:0000/64	15th subnet
2001:ACAD:1234:000F:0000:0000:0000:0000/64	16th subnet
2001:ACAD:1234:0010:0000:0000:0000:0000/64	17th subnet
2001:ACAD:1234:0011:0000:0000:0000:0000/64	18th subnet
2001:ACAD:1234:0012:0000:0000:0000:0000/64	19th subnet
2001:ACAD:1234:0013:0000:0000:0000:0000/64	20th subnet

Each subnet contains over 18 quintillion addresses.

(Lots of subnets omitted for space.)

2001:ACAD:1234:FFFC:0000:0000:0000:0000/64	65,533rd subnet
2001:ACAD:1234:FFFD:0000:0000:0000:0000/64	65,534th subnet
2001:ACAD:1234:FFFE:0000:0000:0000:0000/64	65,535th subnet
2001:ACAD:1234:FFFF:0000:0000:0000:0000/64	65,536th subnet

A Medium End user might receive a /56 IPv6 address from their ISP.

Global Routing Prefix	Subnet ID	Interface ID	
2001:ACAD:1234:12	00	0000:0000:0000:0000	
2001:ACAD:1234:12	00	0000:0000:0000:0000/64	1st subnet
2001:ACAD:1234:12	01	0000:0000:0000:0000/64	2nd subnet
2001:ACAD:1234:12	02	0000:0000:0000:0000/64	3rd subnet
2001:ACAD:1234:12	03	0000:0000:0000:0000/64	4th subnet
(Lots of subnets omitted for space.)			
2001:ACAD:1234:12	FD	0000:0000:0000:0000/64	254th subnet
2001:ACAD:1234:12	FE	0000:0000:0000:0000/64	255th subnet
2001:ACAD:1234:12	FF	0000:0000:0000:0000/64	256th subnet

Each subnet contains over 18 quintillion addresses.

A Small End user might receive a /60 IPv6 address from their ISP.

Global Routing Prefix	Subnet ID	Interface ID
2001:ACAD:1234:1230:0000:0000:0000:0000		
2001:ACAD:1234:1230:0000:0000:0000:0000/64	1st subnet	
2001:ACAD:1234:1231:0000:0000:0000:0000/64	2nd subnet	
2001:ACAD:1234:1232:0000:0000:0000:0000/64	3rd subnet	
2001:ACAD:1234:1233:0000:0000:0000:0000/64	4th subnet	
2001:ACAD:1234:1234:0000:0000:0000:0000/64	5th subnet	
2001:ACAD:1234:1235:0000:0000:0000:0000/64	6th subnet	
2001:ACAD:1234:1236:0000:0000:0000:0000/64	7th subnet	
2001:ACAD:1234:1237:0000:0000:0000:0000/64	8th subnet	
2001:ACAD:1234:1238:0000:0000:0000:0000/64	9th subnet	
2001:ACAD:1234:1239:0000:0000:0000:0000/64	10th subnet	
2001:ACAD:1234:123A:0000:0000:0000:0000/64	11th subnet	
2001:ACAD:1234:123B:0000:0000:0000:0000/64	12th subnet	
2001:ACAD:1234:123C:0000:0000:0000:0000/64	13th subnet	
2001:ACAD:1234:123D:0000:0000:0000:0000/64	14th subnet	
2001:ACAD:1234:123E:0000:0000:0000:0000/64	15th subnet	
2001:ACAD:1234:123F:0000:0000:0000:0000/64	16th subnet	

Each subnet contains over 18 quintillion addresses.

A Home or single Site might receive a /64 IPv6 address from their ISP.

Global Routing Prefix	Interface ID
2001:ACAD:1234:1234	0000:0000:0000:0000
2001:ACAD:1234:1234:0000:0000:0000:0000/64	1 subnet

Over 18 quintillion addresses

Common Prefix's and Number of Subnets

Using the Subnet ID, the common Subnet Prefix's available from your ISP are /48, /52, /56, /60, and /64. The ISP could assign a lower Prefix Length, but most business will not need more than 65,536 subnets per Site.

The commonly available /64 subnets are:

/48	65,536 subnets
/52	4096 subnets
/56	256 subnets
/60	16 subnets
/64	1 subnet

Notice that the Subnet ID always changes by four even though we're only using a single hexadecimal character.

Each hexadecimal character equals 4 binary numbers.

(/48, /52, /56, /60, /64)

Each /64 subnet contains over 18 quintillion addresses.

With IPv4 the main concern was using the fewest possible number of addresses through creative subnetting.

The primary concern with IPv6 is making sure you have a prefix length that will cover all the subnets your Site will require and allow for future growth.

Basic Subnetting Problems

Sample Problem

Your ISP has given you the IPv6 address 2001:FE12:A231::/48.

How many /64 subnets are available with this address? 65,536

What are the first six /64 subnets?

2001:FE12:A231::/64

2001:FE12:A231:1::/64

2001:FE12:A231:2::/64

2001:FE12:A231:3::/64

2001:FE12:A231:4::/64

2001:FE12:A231:5::/64

Problem 1

Your ISP has given you the IPv6 address 2000:ACAD:1234:6600::/56.

How many /64 subnets are available with this address? _____

What are the first four /64 subnets?

What are the last two /64 subnets in this range?

Problem 2

Your ISP has given you the IPv6 address 3FFF:5801:DEAF::/48.

How many /64 subnets are available with this address? _____

What are the first four /64 subnets?

What are the last two /64 subnets in this range?

Problem 3

Your ISP has given you the IPv6 address 2001:ACAD:5678:1840::/60.

How many /64 subnets are available with this address? _____

Complete the /64 subnets in this range. (The ISP's Global Routing Prefix is already printed for you.)

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

2001:ACAD:5678: _____

Problem 4

Your ISP has given you the IPv6 address 3100:6523:AD14:8000::/52.

How many /64 subnets are available with this address? _____

What are the first four /64 subnets? (The ISP's Global Routing Prefix is already printed for you.)

3100:6523:AD14: _____

3100:6523:AD14: _____

3100:6523:AD14: _____

3100:6523:AD14: _____

What are the last two /64 subnets in this range?

3100:6523:AD14: _____

3100:6523:AD14: _____

Problem 5

Your ISP has given you the IPv6 address 2100:89:4500::/48.

How many /64 subnets are available with this address? _____

What are the first four /64 subnets? (The ISP's Global Routing Prefix is already printed for you.)

2100:89:4500: _____

2100:89:4500: _____

2100:89:4500: _____

2100:89:4500: _____

What are the last two /64 subnets in this range?

2100:89:4500: _____

2100:89:4500: _____

IPv6 Subnetting Best Practices

IPv6 was designed to make subnetting as simple as possible by using ridiculously large blocks of addresses. Yes, it's wasteful! Most experts agree that IPv6 will last for 100 years and IP will be replaced before we run out of IPv6 address space.

So until the rules and/or the consensus of the experts change, these are the common subnets and best practices you will be working with. Information for subnets can be found in RFC & RTC documents, plus some additional resources listed in the Reference section of this workbook.

The common /64 subnets are:

You can negotiate with your ISP for a larger block

/48	65,536 subnets
/52	4096 subnets
/56	256 subnets
/60	16 subnets
/64	1 subnet

Specific Address Rules and Best Practices:

Point-To-Point Connections -

Best Practice: Use a /64 address range for these two addresses.

In rare cases this approach might present an issue with certain router setups where addresses are bounced back and forth between routers. Reducing the number of addresses in the range can also help avoid neighbor cache exhaustion attacks. (RFC 6164 - section 5)

Acceptable options include:

- /127 -is recommended in RFC 6164. You must disable the router's Subnet-Router Anycast option to avoid issues with the all routers anycast address.
- /126 -is discussed in RFC 2526. The /126 allows the all-zero reserved Anycast address to be avoided.
- /120 - allows all the reserved anycast address to be avoided.
- /112 - allows you to avoid the reserved anycast address, and gives you the entire four-digit hex value after the last colon.

Of all these options the /120 is probably the best choice since it avoids any issues with the Anycast addresses.

It is recommended that you reserve the entire /64 network with each of the above options.

/128 Single Address Subnets -

Best Practice: reserve the first subnet to use for infrastructure needs, such as loopback addresses, etc..

Acceptable option:

Allocate a full /64 range of addresses for each loopback address, but assign it a /128 subnet Prefix.

Anycast addresses -

Best Practice: Don't use the very first address or the last 128 addresses in any /64 network. These can only be assigned to an interface as an Anycast address.

Anycast addresses can be pulled from Global Unicast, Site-Local or Link-Local address ranges. Any address assigned to more than one interface on a subnet becomes an Anycast address. Anycast addresses can only be used by network devices, not a host. No anycast address can be used as the source address of an IPv6 packet.

The first address in every /64 subnet range is reserved for special use. The Interface ID is all zero's. (Example: 2001:ACAD:1234:5678:0000:0000:0000:0000/64) This address is the Subnet-Anycast Address. These addresses are typically used by different protocols such as Mobile IPv6. (RFC 4291)

The last eight bits in every /64 subnet range are reserved for Anycast addresses. These bits are 10000000 to 11111111. This means you can not assign any addresses if the last hexet falls between FF80 and FFFF unless it is a Anycast address. (RFC 2526)

If you try to apply the last 128 addresses to a router without setting them up as an Anycast address you will get the following error message.

```
Router(config-if)#ipv6 address 2001:ACAD:1234:5678:FFFF:FFFF:FFFF:FFFF/64
% 2001:DB8:1:1:FFFF:FFFF:FFFF:FFFF/64 should not be configured on FastEthernet0/0, a reserved anycast
```

Developing an Address Plan

(or IPv6 Subnetting in the Real World)

There is no one right way for developing an IPv6 addressing plan, but the recommended general guidelines include the following:

Step 1: Decide how you are going to divide your network:

- a. by location
- b. by user groups



Administration

Users Include:
Administration
Staff



Academic

Users Include:
Staff
Students



Dormitory

Users Include:
Staff
Students

Subnetting by Location:

To divide by location you would need four subnets. One for each building and one for the overall network infrastructure needs. You also need to hold several extra subnets in reserve for later growth.

Advantages:

This allows you to optimize your routing tables. All the networks within each location will aggregate to a single route.

Subnetting by User Groups:

To subnet the network by user groups you would need four subnets. One for Administration, Staff, and Students, plus one for overall network infrastructure needs. You also need to hold several extra subnets in reserve for later growth.

Advantages:

Subnetting by user groups makes it much easier to implement a security policy. Grouping by usage also helps track addresses for allocation and management.

Best Practice:

Subnetting by either location or user is acceptable. However, with the emphasis on network security, most networks are better served by subnetting user groups. It makes it much easier to maintain a higher level of security.

Step 2: Determine how many primary and secondary subnets your Site will need.

- Create the primary subnets first.
- Then create secondary subnets.

Subnetting by Location:

Primary Subnets: Quantity 4

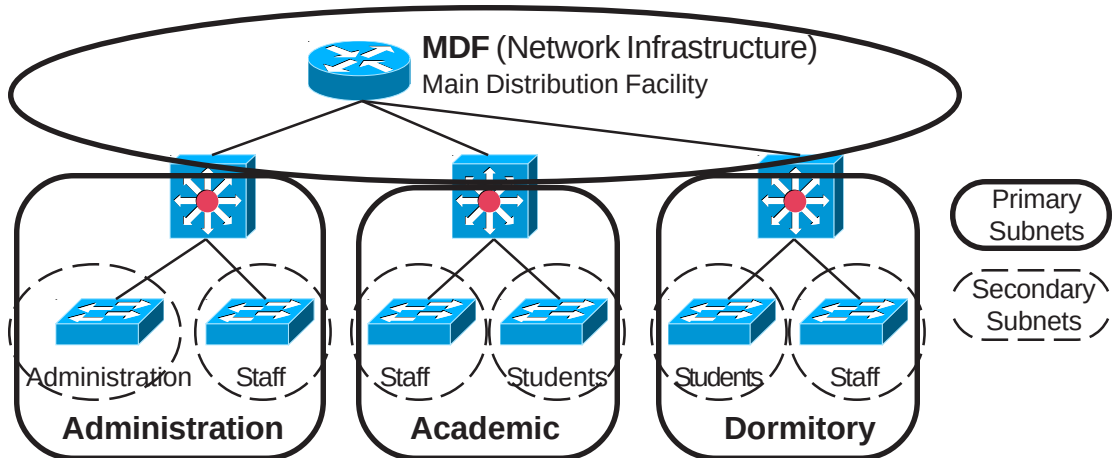
With three buildings you will need four primary subnets. One for each building and one for the overall infrastructure needs.

Secondary Subnets: Quantity 6

Administration will need two secondary subnets: Administration and Staff.

Academic will need two secondary subnets: Staff and Students.

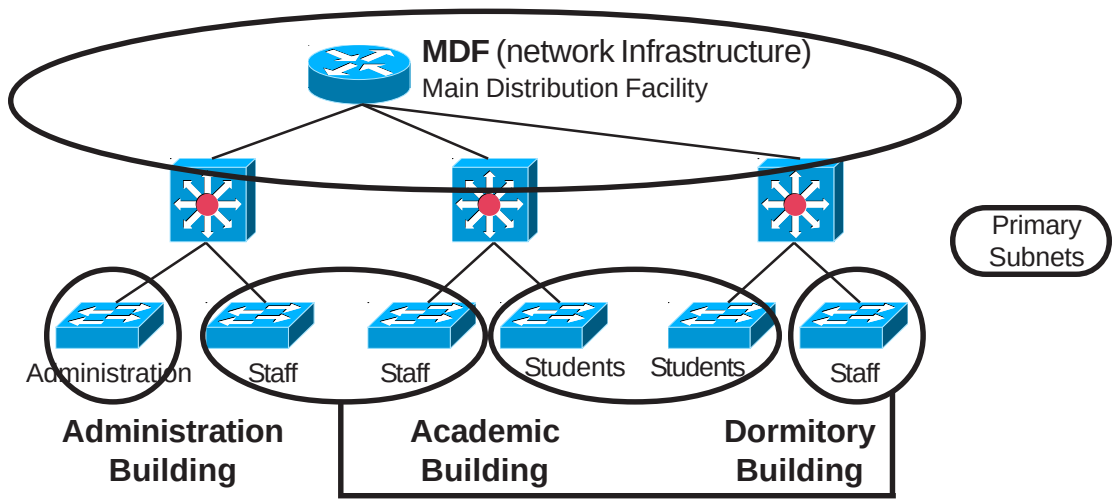
Dormitory will need two secondary subnets: Staff and Students.



Subnetting by User Groups:

Primary Subnets: Quantity 4

With three user groups you will need four subnets. One for each group and one for the overall infrastructure needs.



Step 3: Based on the number of primary and secondary subnets needed assign the address ranges. The ISP has assigned you 2001:ACAD:1234::/48.

Subnetting Options:

/48 No Nibbles	/52 1 Nibble	/56 2 Nibbles	/60 3 Nibbles	/64 4 Nibbles
/48 - 1 Subnet				
/52 - 16 Subnets	/52 - 1 Subnet			
/56 - 256 Subnets	/56 - 16 Subnets	/56 - 1 Subnets		
/60 - 4096 Subnets	/60 - 256 Subnets	/60 - 16 Subnets	/60 - 1 Subnets	
/64 - 65,536 Subnets	/64 - 4096 Subnets	/64 - 256 Subnets	/64 - 16 Subnets	/64 - 1 Subnet

Subnetting by Location:

Primary Subnets: Quantity 4

With three buildings you will need four primary subnets. One for each building and one for the overall infrastructure needs.

Secondary Subnets: Quantity 6

Administration will need two secondary subnets: Administration and Staff.

Academic will need two secondary subnets: Staff and Students.

Dormitory will need two secondary subnets: Staff and Students.

Take the addresses assigned to you by the ISP use one nibble and subnet it into 16 subnets using a /52 Subnet Prefix. This will give you the 4 primary subnets required with several to spare for future growth.

2001:ACAD:1234::/48 becomes:

- ▣ 2001:ACAD:1234::/52 Site ID for over all infrastructure needs.
- ▣ 2001:ACAD:1234:1000::/52 Site ID for the Administration Building.
- ▣ 2001:ACAD:1234:2000::/52 Site ID for the Academic Building.
- ▣ 2001:ACAD:1234:3000::/52 Site ID for the Dormitory.
- ▣ 2001:ACAD:1234:4000::/52
- (Subnets omitted for space.)
- ▣ 2001:ACAD:1234:F000::/52

Site IDs and Sub-Site IDs will be the addresses found in the routing tables.

Take the second, third, and forth ranges and subnet them again by using the next Nibble with a /56 Subnet Prefix. This will create 16 subnets for each location.

Administration Building Site ID 2001:ACAD:1234:1000::/52 becomes:

- ▣ 2001:ACAD:1234:1000::/52 Administration Building Site ID
 - ▣ 2001:ACAD:1234:1000::/56 Sub-Site ID for infrastructure needs.
 - ▣ 2001:ACAD:1234:1100::/56 Sub-Site ID for Administration
 - ▣ 2001:ACAD:1234:1200::/56 Sub-Site ID for Staff

Academic Building Site ID 2001:ACAD:1234:2000::/52 becomes:

- ▣ 2001:ACAD:1234:2000::/52 Academic Building Site ID
 - ▣ 2001:ACAD:1234:2000::/56 Sub-Site ID for infrastructure needs.
 - ▣ 2001:ACAD:1234:2100::/56 Sub-Site ID for Students
 - ▣ 2001:ACAD:1234:2200::/56 Sub-Site ID for Staff

Dormitory Building Site ID 2001:ACAD:1234:3000::/52 becomes:

- ▣ 2001:ACAD:1234:3000::/52 Dormitory Building Site ID
 - ▣ 2001:ACAD:1234:3000::/56 Sub-Site ID for infrastructure needs.
 - ▣ 2001:ACAD:1234:3100::/56 Sub-Site ID for Students
 - ▣ 2001:ACAD:1234:3200::/56 Sub-Site ID for Staff

Subnetting by User Group:

Primary Subnets: Quantity 4

With three user groups you will need four primary subnets. One for each group and one for the overall infrastructure needs. In this example no secondary subnets are required.

Take the address assigned to you by the ISP use one nibble and subnet it into 16 subnets using a /52 Subnet Prefix. This will give you the 4 primary subnets required with several to spare for future growth.

2001:ACAD:1234::/48 becomes:

- ▣ 2001:ACAD:1234::/52 Site ID for over all infrastructure needs.
- ▣ 2001:ACAD:1234:1000::/52 Site ID for the Administration employees.
- ▣ 2001:ACAD:1234:2000::/52 Site ID for the Staff.
- ▣ 2001:ACAD:1234:3000::/52 Site ID for the Students.
- ▣ 2001:ACAD:1234:4000::/52
- (Subnets omitted for space.)
- ▣ 2001:ACAD:1234:F000::/52

IPv6 Subnetting Problems

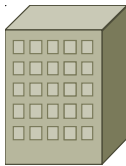
Subnetting on the Nibble Boundary

Sample Problem 1

Using the **minimum number of subnets required** for the primary and secondary sites, design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

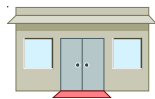
A coffee shop is opening three new stores and a central office/warehouse in your community and needs an IPv6 network plan developed. Each store will need secure network access for three groups: managers, the registers (which will monitor inventory and revenue), and wireless access for guests. The central office/warehouse will need secure network access for several departments: Managers, Finance, Human Relations.

The ISP has given the company 2000:FE23:0054::/48.



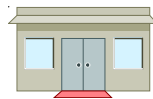
Central Office

Users Include:
Mangers
Finance
Human Relations



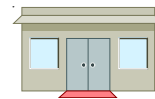
Store 1

Users Include:
Mangers
Registers
Wireless



Store 2

Users Include:
Mangers
Registers
Wireless



Store 3

Users Include:
Mangers
Registers
Wireless

Subnets Based on User Groups

ISP Address: 2000:FE23:0054::/48

- ❑ Infrastructure Site ID: 2000:FE23:0054::/52
- ❑ Managers Site ID: 2000:FE23:0054:1000::/52
- ❑ Finance Site ID: 2000:FE23:0054:2000::/52
- ❑ Human Relations Site ID: 2000:FE23:0054:3000::/52
- ❑ Registers Site ID: 2000:FE23:0054:4000::/52
- ❑ Wireless Site ID: 2000:FE23:0054:5000::/52

Subnets Based on Location

ISP Address: 2000:FE23:0054::/48

- ▣ Infrastructure Site ID: 2000:FE23:0054::/52
- ▣ Central Office Site ID: 2000:FE23:0054:1000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:1000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:1100::/56
- ▣ Finance Sub-Site ID: 2000:FE23:0054:1200::/56
- ▣ Human Relations Sub-Site ID: 2000:FE23:0054:1300::/56
- ▣ Store 1 Site ID: 2000:FE23:0054:2000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:2000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:2100::/56
- ▣ Registers Sub-Site ID: 2000:FE23:0054:2200::/56
- ▣ Wireless Sub-Site ID: 2000:FE23:0054:2300::/56
- ▣ Store 2 Site ID: 2000:FE23:0054:3000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:3000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:3100::/56
- ▣ Registers Sub-Site ID: 2000:FE23:0054:3200::/56
- ▣ Wireless Sub-Site ID: 2000:FE23:0054:3300::/56
- ▣ Store 3 Site ID: 2000:FE23:0054:4000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:4000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:4100::/56
- ▣ Registers Sub-Site ID: 2000:FE23:0054:4200::/56
- ▣ Wireless Sub-Site ID: 2000:FE23:0054:4300::/56

IPv6 Subnetting Problems

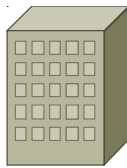
Subnetting on the Nibble Boundary

Sample Problem 2

Using the **minimum number of subnets required** for the primary and secondary sites, design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

A Medical Building is opening in your community and needs an IPv6 network plan developed.

The ISP has given the company 2001:5378:8801::/48.



Medical Building

First Floor Rooms

Patient Check-in
Emergency Room

Users Include:
Administrators
Staff
Guests

Second Floor Rooms

Nurses Station
Ward A

Users Include:
Staff
Guests

Subnets Based on User Groups

ISP Address: 2001:5378:8801::/48

- ▣ Infrastructure Site ID: 2001:5378:8801::/52
- ▣ Administrators Site ID: 2001:5378:8801:1000::/52
- ▣ Staff Site ID: 2001:5378:8801:2000::/52
- ▣ Guests Site ID: 2001:5378:8801:3000::/52

Subnets Based on Location

ISP Address: 2001:5378:8801::/48

- ▣ Infrastructure Site ID: 2001:5378:8801::/52
- ▣ First Floor Site ID: 2001:5378:8801:1000::/52
- ▣ Infrastructure Sub-Site ID: 2001:5378:8801:1000::/56
- ▣ Patient Check-in Sub-Site ID: 2001:5378:8801:1100::/56
- ▣ Administrators Sub-Site ID: 2001:5378:8801:1100::/60
- ▣ Staff Sub-Site ID: 2001:5378:8801:1110::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:1120::/60
- ▣ Emergency Room Sub-Site ID: 2001:5378:8801:1200::/56
- ▣ Administrators Sub-Site ID: 2001:5378:8801:1200::/60
- ▣ Staff Sub-Site ID: 2001:5378:8801:1210::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:1220::/60
- ▣ Second Floor Site ID: 2001:5378:8801:2000::/52
- ▣ Infrastructure Sub-Site ID: 2001:5378:8801:2000::/56
- ▣ Nurses Station Sub-Site ID: 2001:5378:8801:2100::/56
- ▣ Staff Sub-Site ID: 2001:5378:8801:2100::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:2110::/60
- ▣ Ward A Sub-Site ID: 2001:5378:8801:2200::/56
- ▣ Staff Sub-Site ID: 2001:5378:8801:2200::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:2210::/60

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 1

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

The XYZ Film company is setting up a new office and needs an IPv6 network plan developed. The company will have the following departments:

The ISP has given the company 2001:EE00:2575::/48.

Administration Building

Users Include:
Sales
Accounting
Distribution

Production Building

Users Include:
Casting
Editing

Subnets Based on User Groups

ISP Address: 2001:EE00:2575::/48

- ☐ Infrastructure Site ID: _____
- ☐ Sales Site ID: _____
- ☐ Accounting Site ID: _____
- ☐ Distribution Site ID: _____
- ☐ Casting Site ID: _____
- ☐ Editing Site ID: _____

Subnets Based on Location

ISP Address: 2001:EE00:2575::/48

▣ Infrastructure Site ID: _____

▣ Administration Site ID: _____

▣ Infrastructure Sub-Site ID: _____

▣ Sales Sub-Site ID: _____

▣ Accounting Sub-Site ID: _____

▣ Distribution Sub-Site ID: _____

▣ Production Site ID: _____

▣ Infrastructure Sub-Site ID: _____

▣ Casting Sub-Site ID: _____

▣ Editing Sub-Site ID: _____

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 2

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

A new medical supply company is opening and needs an IPv6 network plan developed. The company has three buildings and will include the following user groups:

The ISP has given the company 3F01:ABCD:8875::/48.

Building 1

Users Include:
Management
Sales

Building 2

Users Include:
Management
Human Resources

Building 3

Users Include:
Management
Warehouse

Subnets Based on User Groups

ISP Address: 3F01:ABCD:8875::/48

▣ Infrastructure Site ID: _____

▣ Management Site ID: _____

▣ Sales Site ID: _____

▣ Human Resources Site ID: _____

▣ Warehouse Site ID: _____

Subnets Based on Location

ISP Address: 3F01:ABCD:8875::/48

▣ Infrastructure Site ID: _____

▣ Building 1 Site ID: _____

▣ Infrastructure Sub-Site ID: _____

▣ Management Sub-Site ID: _____

▣ Sales Sub-Site ID: _____

▣ Building 2 Site ID: _____

▣ Infrastructure Sub-Site ID: _____

▣ Management Sub-Site ID: _____

▣ Human Resources Sub-Site ID: _____

▣ Building 2 Site ID: _____

▣ Infrastructure Sub-Site ID: _____

▣ Management Sub-Site ID: _____

▣ Warehouse Sub-Site ID: _____

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 3

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

A paper supply company needs an IPv6 network plan developed. The company has two buildings and will include the following user groups and sub-user groups:

The ISP has given the company 2001:CA21:9000::/48.

Building A

Management Groups

Human Resources

Sales:

Wholesale

Retail

Building B

Production Groups

Warehouse

Shipping:

Domestic

Worldwide

Subnets Based on User Groups

ISP Address: 2001:CA21:9000::/48 (The ISP's Global Routing Prefix is already printed for you.)

▣ Infrastructure Site ID: 2001:CA21:9000:

▣ Management Groups Site ID: 2001:CA21:9000:

▣ HR Sub-Site ID: 2001:CA21:9000:

▣ Sales Sub-Site ID: 2001:CA21:9000:

▣ Wholesale Sub- Site ID: 2001:CA21:9000:

▣ Retail Sub- Site ID: 2001:CA21:9000:

▣ Production Groups Site ID: 2001:CA21:9000:

▣ Warehouse Sub-Site ID: 2001:CA21:9000:

▣ Shipping Sub-Site ID: 2001:CA21:9000:

▣ Domestic Sub-Site ID: 2001:CA21:9000:

▣ Worldwide Sub-Site ID: 2001:CA21:9000:

Subnets Based on Location

ISP Address: 2001:CA21:9000::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2001:CA21:9000:

☐ Building A Site ID: 2001:CA21:9000:

☐ Infrastructure Sub-Site ID: 2001:CA21:9000:

☐ HR Sub-Site ID: 2001:CA21:9000:

☐ Sales Sub-Site ID: 2001:CA21:9000:

☐ Wholesale Sub-Site ID: 2001:CA21:9000:

☐ Retail Sub-Site ID: 2001:CA21:9000:

☐ Building B Site ID: 2001:CA21:9000:

☐ Infrastructure Sub-Site ID: 2001:CA21:9000:

☐ Warehouse Sub-Site ID: 2001:CA21:9000:

☐ Shipping Sub-Site ID: 2001:CA21:9000:

☐ Domestic Sub-Site ID: 2001:CA21:9000:

☐ Worldwide Sub-Site ID: 2001:CA21:9000:

IPv6 Subnetting Problems

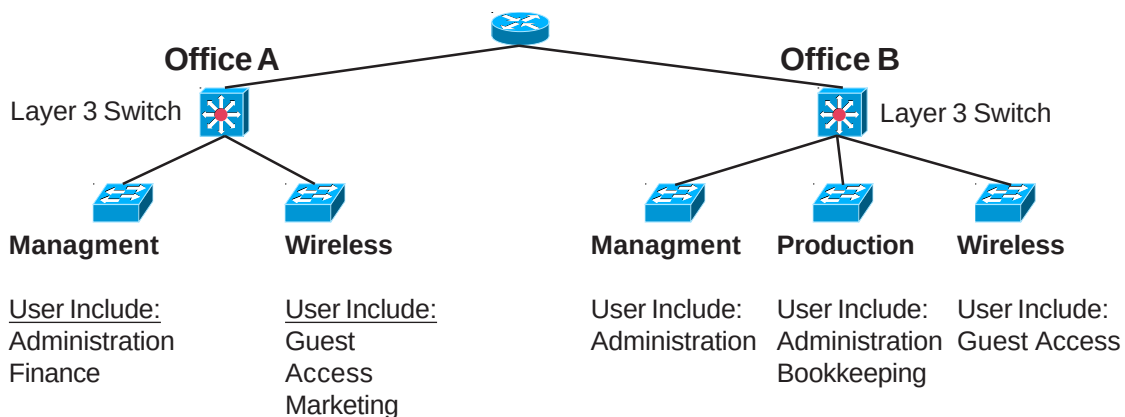
Subnetting on the Nibble Boundary

Problem 4

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

A company is setting up a new server farm and needs an IPv6 network plan developed. The Core Router and the direct connections with layer 3 switches will pull their IPv6 addresses from the Infrastructure Site ID range.

The ISP has given the company 2000:ACAD:1145::/48.



Subnets Based on User Groups

☐ ISP Address: 2000:ACAD:1145::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2000:ACAD:1145:

☐ Administration Site ID: 2000:ACAD:1145:

☐ Finance Site ID: 2000:ACAD:1145:

☐ Guest Access Site ID: 2000:ACAD:1145:

☐ Marketing Site ID: 2000:ACAD:1145:

☐ Bookkeeping Site ID: 2000:ACAD:1145:

Subnets Based on Location

ISP Address: 2000:ACAD:1145::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2000:ACAD:1145:

☐ Office A Site ID: 2000:ACAD:1145:

☐ Infrastructure Sub-Site ID: 2000:ACAD:1145:

☐ Management Sub-Site ID: 2000:ACAD:1145:

☐ Administration Sub-Site ID: 2000:ACAD:1145:

☐ Finance Sub-Site ID: 2000:ACAD:1145:

☐ Wireless Access Sub-Site ID: 2000:ACAD:1145:

☐ Guest Access Sub-Site ID: 2000:ACAD:1145:

☐ Marketing Sub-Site ID: 2000:ACAD:1145:

☐ Office B Site ID: 2000:ACAD:1145:

☐ Infrastructure Sub-Site ID: 2000:ACAD:1145:

☐ Management Sub-Site ID: 2000:ACAD:1145:

☐ Administration Sub-Site ID: 2000:ACAD:1145:

☐ Production Sub-Site ID: 2000:ACAD:1145:

☐ Administration Sub-Site ID: 2000:ACAD:1145:

☐ Bookkeeping Sub-Site ID: 2000:ACAD:1145:

☐ Wireless Access Sub-Site ID: 2000:ACAD:1145:

☐ Guest Access Sub-Site ID: 2000:ACAD:1145:

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 5

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

The company has multiple floors in a high rise building and will include the following user groups and sub-user groups:

The ISP has given the company 3F01:AA07:3907::/48.

33rd Floor

Manufacturing Groups

Marketing

Inventory

Shipping

34th Floor

Admin Groups

Human Resources (HR):

Hiring

Benifits

Financial:

Purchasing

Sales

Subnets Based on User Groups

☐ ISP Address: 3F01:AA07:3907::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 3F01:AA07:3907:

☐ Manufacturing Groups Site ID: 3F01:AA07:3907:1

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:

☐ Marketing Sub-Site ID: 3F01:AA07:3907:

☐ Inventory Sub-Site ID: 3F01:AA07:3907:

☐ Shipping Sub- Site ID: 3F01:AA07:3907:

☐ Admin Groups Site ID: 3F01:AA07:3907:

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:

☐ HR Sub-Site ID: 3F01:AA07:3907:

☐ Hiring Sub-Site ID: 3F01:AA07:3907:

☐ Benfits Sub-Site ID: 3F01:AA07:3907:

☐ Financial Sub-Site ID: 3F01:AA07:3907:

☐ Purchasing Sub-Site ID: 3F01:AA07:3907:

☐ Sales Sub-Site ID: 3F01:AA07:3907:

Subnets Based on Location

ISP Address: 3F01:AA07:3907::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 3F01:AA07:3907:

☐ 33rd Floor Site ID: 3F01:AA07:3907:

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:

☐ Manufacturing Groups Sub- Site ID: 3F01:AA07:3907:

☐ Marketing Sub-Site ID: 3F01:AA07:3907:

☐ Inventory Sub-Site ID: 3F01:AA07:3907:

☐ Shipping Sub- Site ID: 3F01:AA07:3907:

☐ 34th Floor Site ID: 3F01:AA07:3907:

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:

☐ Admin Groups Sub-Site ID: 3F01:AA07:3907:

☐ HR Sub-Site ID: 3F01:AA07:3907:

☐ Hiring Sub-Site ID: 3F01:AA07:3907:

☐ Benefits Sub-Site ID: 3F01:AA07:3907:

☐ Financial Sub-Site ID: 3F01:AA07:3907:

☐ Purchasing Sub-Site ID: 3F01:AA07:3907:

☐ Sales Sub-SiteID: 3F01:AA07:3907:

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 6

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

This medium sized company will include the following user groups and sub-user groups:

The ISP has given the company 2001:0:17::/52

Management

Users Include:
Laboratory
Administrators
Staff

Finance

Users Include:
Staff

Marketing Department

Users Include:
Advertising:
Radio
TV
Web
Sales:
Retail
Wholesale

Subnets Based on User Groups

☐ ISP Address: 2001:0:17::/52 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2001:0:17:

☐ Administrators Site ID: 2001:0:17:

☐ Staff Site ID: 2001:0:17:

☐ Advertising Site ID: 2001:0:17:

☐ Infrastructure Sub-Site ID: 2001:0:17:

☐ Radio Sub-Site ID: 2001:0:17:

☐ TV Sub- Site ID: 2001:0:17:

☐ Web Sub- Site ID: 2001:0:17:

☐ Sales Site ID: 2001:0:17:

☐ Infrastructure Sub-Site ID: 2001:0:17:

☐ Retail Sub-Site ID: 2001:0:17:

☐ Wholesale Sub-Site ID: 2001:0:17:

Subnets Based on Location

☐ ISP Address: 2001:0:17::/52 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2001:0:17:

☐ Management Site ID: 2001:0:17:

☐ Infrastructure Sub-Site ID: 2001:0:17:

☐ Administrators Sub-Site ID: 2001:0:17:

☐ Staff Sub- Site ID: 2001:0:17:

☐ Finance Site ID: 2001:0:17:

☐ Infrastructure Sub-Site ID: 2001:0:17:

☐ Staff Sub-Site ID: 2001:0:17:

☐ Marketing Dept Site ID: 2001:0:17:

☐ Infrastructure Sub-Site ID: 2001:0:17:

☐ Advertising Sub-Site ID: 2001:0:17:

☐ Radio Sub-Site ID: 2001:0:17:

☐ TV Sub-Site ID: 2001:0:17:

☐ Web Sub-Site ID: 2001:0:17:

☐ Sales Sub-Site ID: 2001:0:17:

☐ Retail Sub-Site ID: 2001:0:17:

☐ Wholesale Sub-Site ID: 2001:0:17:

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 7

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

This medium sized company is setting up a new IPv6 addressing plan which will include the following user groups and sub-user groups:

The ISP has given the company 3F00:3589:0:5000::/52

Office A

Users Include:
Management
Human Relations (HR):
 Record Keeping
 Insurance

Office B

Users Include:
Management
Finance:
 Sales

Office C

Users Include:
Management
Purchasing:
 Inventory
 Distribution

Subnets Based on User Groups

ISP Address: 3F00:3589:0:5000::/52 (The ISP's Global Routing Prefix is already printed for you.)

- ☐ Infrastructure Site ID: 3F00:3589:0:
- ☐ Management Site ID: 3F00:3589:0:
- ☐ HR Site ID: 3F00:3589:0:
- ☐ Infrastructure Sub-Site ID: 3F00:3589:0:
- ☐ Record Keeping Sub-Site ID: 3F00:3589:0:
- ☐ Insurance Site ID: 3F00:3589:0:
- ☐ Finance Site ID: 3F00:3589:0:
- ☐ Infrastructure Sub-Site ID: 3F00:3589:0:
- ☐ Sales Sub-Site ID: 3F00:3589:0:
- ☐ Purchasing Sub-Site ID: 3F00:3589:0:
- ☐ Inventory Sub-Site ID: 3F00:3589:0:
- ☐ Distribution Sub-Site ID: 3F00:3589:0:

Subnets Based on Location

ISP Address: 3F00:3589:0:5000::/52 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 3F00:3589:0:

☐ Office A Site ID: 3F00:3589:0:

☐ Infrastructure Sub-Site ID: 3F00:3589:0:

☐ Management Sub-Site ID: 3F00:3589:0:

☐ HR Sub- Site ID: 3F00:3589:0:

☐ Record Keeping Sub-Site ID: 3F00:3589:0:

☐ Insurance Sub-Site ID: 3F00:3589:0:

☐ Office B Site ID: 3F00:3589:0:

☐ Infrastructure Sub-Site ID: 3F00:3589:0:

☐ Management Sub-Site ID: 3F00:3589:0:

☐ Finance Sub-Site ID: 3F00:3589:0:

☐ Sales Sub-Site ID: 3F00:3589:0:

☐ Office C Site ID: 3F00:3589:0:

☐ Infrastructure Sub-Site ID: 3F00:3589:0:

☐ Management Sub-Site ID: 3F00:3589:0:

☐ Purchasing Sub-Site ID: 3F00:3589:0:

☐ Inventory Sub-Site ID: 3F00:3589:0:

☐ Distribution Sub-Site ID: 3F00:3589:0:

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 8

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

A Health Care facility is upgrading their network to IPv6 and will include the following user groups and sub-user groups:

The ISP has given the company 2000:2531:FE00::/48.

Emergency

Users Include:
Nurses/Staff
Laboratory
Obstetrics
Pediactric

Admissions

Users Include:
Nurses/Staff
Records

Patient Wards

Users Include:
Ward A:
Nurses/Staff
Guest WIFI
Ward B:
Nurses/Staff
Guest WIFI

Subnets Based on User Groups

ISP Address: 2000:2531:FE00::/48 (The ISP's Global Routing Prefix is already printed for you.)

- ▣ Infrastructure Site ID: 2000:2531:FE00:
- ▣ Nurses/Staff Site ID: 2000:2531:FE00:
- ▣ Laboratory Site ID: 2000:2531:FE00:
- ▣ Obstetrics Site ID: 2000:2531:FE00:
- ▣ Pediatric Site ID: 2000:2531:FE00:
- ▣ Records Site ID: 2000:2531:FE00:
- ▣ Guest WIFI Site ID: 2000:2531:FE00:

Subnets Based on Location

ISP Address: 2000:2531:FE00::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2000:2531:FE00:

☐ Emergency Site ID: 2000:2531:FE00:

☐ Infrastructure Sub-Site ID: 2000:2531:FE00:

☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:

☐ Laboratory Sub-Site ID: 2000:2531:FE00:

☐ Obstetrics Sub-Site ID: 2000:2531:FE00:

☐ Pediatric Sub-Site ID: 2000:2531:FE00:

☐ Admissions Site ID: 2000:2531:FE00:

☐ Infrastructure Sub-Site ID: 2000:2531:FE00:

☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:

☐ Records Sub-Site ID: 2000:2531:FE00:

☐ Patient Wards Site ID: 2000:2531:FE00:

☐ Infrastructure Sub-Site ID: 2000:2531:FE00:

☐ Ward A Sub-Site ID: 2000:2531:FE00:

☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:

☐ Guest WIFI Sub-Site ID: 2000:2531:FE00:

☐ Ward B Sub-Site ID: 2000:2531:FE00:

☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:

☐ Guest WIFI Sub-Site ID: 2000:2531:FE00:

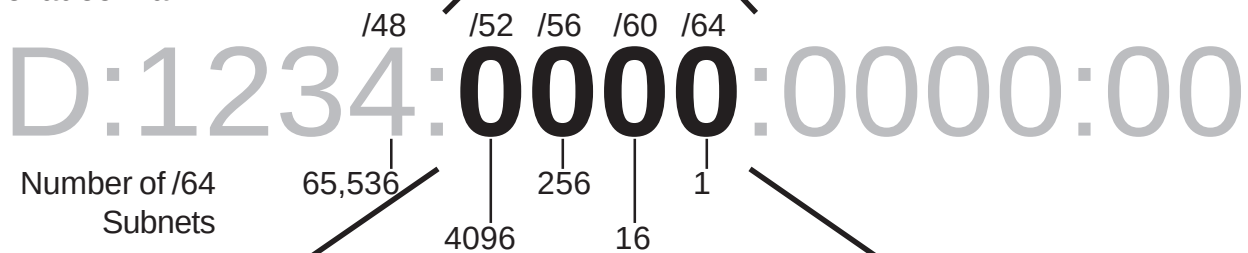
Subnetting Within a Nibble

It is a Best Practice to subnet on the Nibble Boundary. However, subnetting within a Nibble is an acceptable practice. It tends to make subnetting, implementation, and troubleshooting more difficult.

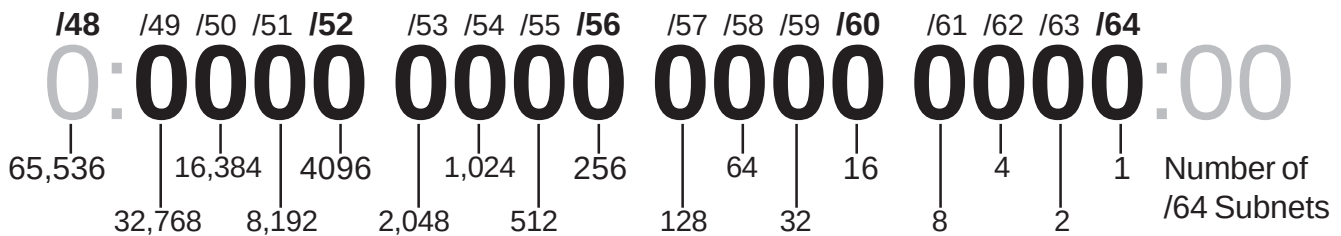
Subnet ID Break Down

2001:ACAD:1234:0000:0000:0000:0000:0000/48

Hexadecimal



Binary

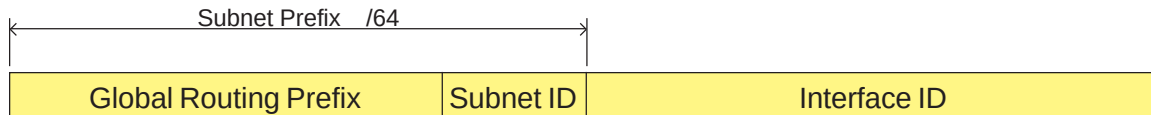


Subnetting Beyond the /64 Boundary

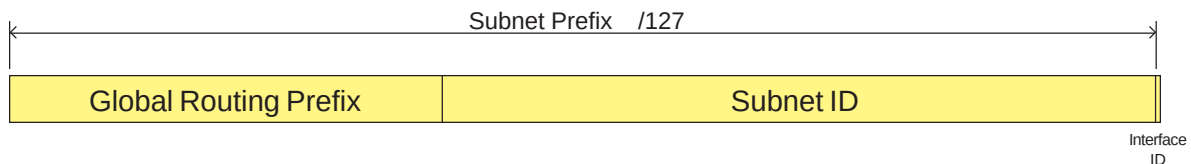
What happens if you need more subnets than the 16 bit Subnet Prefix will allow? Or your IPv4 address conservatism kicks in and you decide that using a /64 Global Routing Prefix with over 18 quintillion addresses for a single point-to-point serial connection is more than you can handle; there are some options.

IPv6 was designed to be very flexible. The Subnet Prefix is the address space used by the Global Routing Prefix and the Subnet ID, and can range from 0 to 128.

Standard /48 Subnet Prefix



Point-To-Point /127 Subnet Prefix



Just as you could borrow host bits in IPv4, you can borrow Interface ID bits in IPv6. This allows you to create more subnets with fewer addresses. Before you get too excited, there are a few rules and best practices you need to take into account.

Borrowing bits from the Interface ID should only be done on network infrastructure links. Loopback addresses, point-to-point links, addresses that are usually statically assigned.

Any subnet that includes end devices needs to stay on a /64 or lower prefix. This would be computers, tablets, smart phones, servers, printers, anything that might be on a subnet that connects to the internet.

According to RFC 5375, a /64 prefix is required to support a number of benefits offered by IPv6; such as:

- Stateless Address Autoconfiguration
- Neighbor Discovery (ND)
- Secure Neighborhood Discovery (SEND)
- privacy extensions
- parts of Mobile IPv6
- PIM-SM with Embedded-RP
- SHIM6 [SHIM6]

Plus a number of other features currently in development, or being proposed, which will rely on a /64 prefix.

The bottom line is that IPv6 was designed to waste an unbelievable amount of addresses and it's OK. So while it is possible to subnet beyond the /64 subnet prefix it is not recommended.

Define the Following IPv6 Terms:

(Use the definitions/explanations from this workbook)

1. Global Routing Prefix -

2. Subnet ID -

3. Interface ID -

4. Subnet Prefix -

5. Nibble -

6. Unspecified address -

7. Loopback address -

8. Ipv4 Embedded address -

9. Global Unicast address -

10. Unique Local address -

11. Multicast address -

12. Unicast address -

13. Anycast address -

14. Site ID -

15. Sub-Site ID -

16. Prefix Length -

What Type of IPv6 Address is this?

Base on the information supplied on the inside front cover of this workbook identify the following IPv6 addresses as either: Unspecified, Loopback, Embedded IPv4, Global Unicast, Unique local Unicast, Link-local Unicast, or Multicast.

1. 2000:ACAD:1234::/48 Global Unicast
2. 0000:0000:0000:0000:0000:0000:0000:0000 Unspecified
3. FE80:ACAD:1234::/48 _____
4. FDFF:8771:3321::/48 _____
5. FFCD:984:1::/48 _____
6. 3F98::/48 _____
7. ::1 _____
9. 3000::0001/64 _____
10. FEA1:8934:3021:8945:1234:ACAD:FE23:0001/64 _____
11. 00AB:2307:4829::/56 _____
13. FF45:6543:ACAD::/60 _____
14. 2ABC:ACAD:AAAA:0000:0000:0000:0000:00001/64 _____
15. FC12:0000:ACAD:1234:5678:9101:1121:3141/48 _____
16. 2345:FE66:FECD:9999:2365::1/52 _____
17. :: _____
18. 0000:0000:0000:0000:0000:0000:0000:0001 _____
19. FFF8:0000:00001::0023/64 _____
20. 0023:5935:F441::/48 _____
22. 2001:ABCD:1234:FFFF:ACAD::45/60 _____
23. 3211:FCAB:EEEE::/48 _____
24. FCCC:25:1::/48 _____

Reference Section

IANA IPv6 Global Unicast Address Allocations to the RIRs

Prefix	Designation	Date	Whois	Status
2001:0000::/23	IANA	7/1/1999	whois.iana.org	ALLOCATED
2001:0200::/23	APNIC	7/1/1999	whois.apnic.net	ALLOCATED
2001:0400::/23	ARIN	7/1/1999	whois.arin.net	ALLOCATED
2001:0600::/23	RIPE NCC	7/1/1999	whois.ripe.net	ALLOCATED
2001:0800::/23	RIPE NCC	5/2/2002	whois.ripe.net	ALLOCATED
2001:0a00::/23	RIPE NCC	11/2/2002	whois.ripe.net	ALLOCATED
2001:0c00::/23	APNIC	5/2/2002	whois.apnic.net	ALLOCATED
2001:0e00::/23	APNIC	1/1/2003	whois.apnic.net	ALLOCATED
2001:1200::/23	LACNIC	11/1/2002	whois.lacnic.net	ALLOCATED
2001:1400::/23	RIPE NCC	2/1/2003	whois.ripe.net	ALLOCATED
2001:1600::/23	RIPE NCC	7/1/2003	whois.ripe.net	ALLOCATED
2001:1800::/23	ARIN	4/1/2003	whois.arin.net	ALLOCATED
2001:1a00::/23	RIPE NCC	1/1/2004	whois.ripe.net	ALLOCATED
2001:1c00::/22	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:2000::/20	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:3000::/21	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:3800::/22	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:3c00::/22	IANA			RESERVED
2001:4000::/23	RIPE NCC	6/11/2004	whois.ripe.net	ALLOCATED
2001:4200::/23	AFRINIC	6/1/2004	whois.afrinic.net	ALLOCATED
2001:4400::/23	APNIC	6/11/2004	whois.apnic.net	ALLOCATED
2001:4600::/23	RIPE NCC	8/17/2004	whois.ripe.net	ALLOCATED
2001:4800::/23	ARIN	8/24/2004	whois.arin.net	ALLOCATED
2001:4a00::/23	RIPE NCC	10/15/2004	whois.ripe.net	ALLOCATED
2001:4c00::/23	RIPE NCC	12/17/2004	whois.ripe.net	ALLOCATED
2001:5000::/20	RIPE NCC	9/10/2004	whois.ripe.net	ALLOCATED
2001:8000::/19	APNIC	11/30/2004	whois.apnic.net	ALLOCATED
2001:a000::/20	APNIC	11/30/2004	whois.apnic.net	ALLOCATED
2001:b000::/20	APNIC	3/8/2006	whois.apnic.net	ALLOCATED
2002:0000::/16	6to4	2/1/2001		ALLOCATED
2003:0000::/18	RIPE NCC	1/12/2005	whois.ripe.net	ALLOCATED
2400:0000::/12	APNIC	10/3/2006	whois.apnic.net	ALLOCATED
2600:0000::/12	ARIN	10/3/2006	whois.arin.net	ALLOCATED
2610:0000::/23	ARIN	11/17/2005	whois.arin.net	ALLOCATED
2620:0000::/23	ARIN	9/12/2006	whois.arin.net	ALLOCATED
2800:0000::/12	LACNIC	10/3/2006	whois.lacnic.net	ALLOCATED
2a00:0000::/12	RIPE NCC	10/3/2006	whois.ripe.net	ALLOCATED
2c00:0000::/12	AFRINIC	10/3/2006	whois.afrinic.net	ALLOCATED
2d00:0000::/8	IANA	7/1/1999		RESERVED
2e00:0000::/7	IANA	7/1/1999		RESERVED
3000:0000::/4	IANA	7/1/1999		RESERVED
3ffe::/16	IANA	2008-04		RESERVED
5f00::/8	IANA	2008-04		RESERVED

Prefix-Length	Number of IPs		
/128	1	/68	1,152,921,504,606,840,000
/127	2	/67	2,305,843,009,213,690,000
/126	4	/66	4,611,686,018,427,380,000
/125	8	/65	9,223,372,036,854,770,000
/124	16	/64 - Residential	18,446,744,073,709,500,000
/123	32	/63	36,893,488,147,419,100,000
/122	64	/62	73,786,976,294,838,200,000
/121	128	/61	147,573,952,589,676,000,000
/120	256	/60 - Residential	295,147,905,179,352,000,000
/119	512	/59	590,295,810,358,705,000,000
/118	1,024	/58	1,180,591,620,717,410,000,000
/117	2,048	/57	2,361,183,241,434,820,000,000
/116	4,096	/56 - Medium	4,722,366,482,869,640,000,000
/115	8,192	/55	9,444,732,965,739,290,000,000
/114	16,384	/54	18,889,465,931,478,500,000,000
/113	32,768	/53	37,778,931,862,957,100,000,000
/112	65,536	/52	75,557,863,725,914,300,000,000
/111	131,072	/51	151,115,727,451,828,000,000,000
/110	262,144	/50	302,231,454,903,657,000,000,000
/109	524,288	/49	604,462,909,807,314,000,000,000
/108	1,048,576	/48 - Large	1,208,925,819,614,620,000,000,000
/107	2,097,152	/47	2,417,851,639,229,250,000,000,000
/106	4,194,304	/46	4,835,703,278,458,510,000,000,000
/105	8,388,608	/45	9,671,406,556,917,030,000,000,000
/104	16,777,216	/44	19,342,813,113,834,000,000,000,000
/103	33,554,432	/43	38,685,626,227,668,100,000,000,000
/102	67,108,864	/42	77,371,252,455,336,200,000,000,000
/101	134,217,728	/41	154,742,504,910,672,000,000,000,000
/100	268,435,456	/40	309,485,009,821,345,000,000,000,000
/99	536,870,912	/39	618,970,019,642,690,000,000,000,000
/98	1,073,741,824	/38	1,237,940,039,285,380,000,000,000,000
/97	2,147,483,648	/37	2,475,880,078,570,760,000,000,000,000
/96	4,294,967,296	/36	4,951,760,157,141,520,000,000,000,000
/95	8,589,934,592	/35	9,903,520,314,283,040,000,000,000,000
/94	17,179,869,184	/34	19,807,040,628,566,000,000,000,000,000
/93	34,359,738,368	/33	39,614,081,257,132,100,000,000,000,000
/92	68,719,476,736	/32 - Service LIR	79,228,162,514,264,300,000,000,000,000
/91	137,438,953,472	/31	158,456,325,028,528,000,000,000,000,000
/90	274,877,906,944	/30	316,912,650,057,057,000,000,000,000,000
/89	549,755,813,888	/29	633,825,300,114,114,000,000,000,000,000
/88	1,099,511,627,776	/28	1,267,650,600,228,220,000,000,000,000,000
/87	2,199,023,255,552	/27	2,535,301,200,456,450,000,000,000,000,000
/86	4,398,046,511,104	/26	5,070,602,400,912,910,000,000,000,000,000
/85	8,796,093,022,208	/25	10,141,204,801,825,800,000,000,000,000,000
/84	17,592,186,044,416	/24	20,282,409,603,651,600,000,000,000,000,000
/83	35,184,372,088,832	/23 - ISP	40,564,819,207,303,300,000,000,000,000,000
/82	70,368,744,177,664	/22	81,129,638,414,606,600,000,000,000,000,000
/81	140,737,488,355,328	/21	162,259,276,829,213,000,000,000,000,000,000
/80	281,474,976,710,656	/20	324,518,553,658,426,000,000,000,000,000,000
/79	562,949,953,421,312	/19	649,037,107,316,853,000,000,000,000,000,000
/78	1,125,899,906,842,620	/18	1,298,074,214,633,700,000,000,000,000,000,000
/77	2,251,799,813,685,240	/17	2,596,148,429,267,410,000,000,000,000,000,000
/76	4,503,599,627,370,490	/16	5,192,296,858,534,820,000,000,000,000,000,000
/75	9,007,199,254,740,990	/15	10,384,593,717,069,600,000,000,000,000,000,000
/74	18,014,398,509,481,900	/14	20,769,187,434,139,300,000,000,000,000,000,000
/73	36,028,797,018,963,900	/13	41,538,374,868,278,600,000,000,000,000,000,000
/72	72,057,594,037,927,900	/12	83,076,749,736,557,200,000,000,000,000,000,000
/71	144,115,188,075,855,000	/11	166,153,499,473,114,000,000,000,000,000,000,000
/70	288,230,376,151,711,000	/10	332,306,998,946,228,000,000,000,000,000,000,000
/69	576,460,752,303,423,000	/9	664,613,997,892,457,000,000,000,000,000,000,000
		/8	1,329,227,995,784,910,000,000,000,000,000,000,000

IPv6 Resources

Web Sites:

ARIN IPv6 Wiki

<http://www.getipv6.info/display/IPv6/IPv6+Info+Home>

Cisco Support Community IPv6 Subnetting - Overview and Case Study

<https://supportforums.cisco.com/docs/DOC-17232>

Videos:

IPv6 for CCNAs with Anthony Sequeira

Video Series by the Cisco Learning Network - Parts 1, 2 & 3

<https://learningnetwork.cisco.com/docs/DOC-20357>

PDF Resources:

Preparing An IPv6 Address Plan, Version 2, 18 September 2013

https://www.ripe.net/lir-services/training/material/IPv6-for-LIRs-Training-Course/IPv6_addr_plan4.pdf

Best Current Operational Practices - IPv6 Subnetting (v1)

http://www.ipbcop.org/wp-content/uploads/2012/02/BCOP-IPv6_Subnetting.pdf

6net An IPv6 Deployment Guide by The European 6NET Consortium

<http://www.6net.org/book/deployment-guide.pdf>

IPv6 Addressing At-A-Glance By Cisco

http://www.cisco.com/en/US/technologies/tk648/tk872/technologies_white_paper0900aecd8026003d.pdf

IPv6 Implementation Guide, Cisco IOS Release 15.2M&T

<http://www.cisco.com/en/US/docs/ios-xml/ios/ipv6/configuration/15-2mt/ipv6-15-2mt-book.pdf>

Printed Books:

IPv6 Fundamentals A Straightforward Approach to Understanding IPv6

By Rick Graziani

ISBN-13: 978-1-58714-313-7

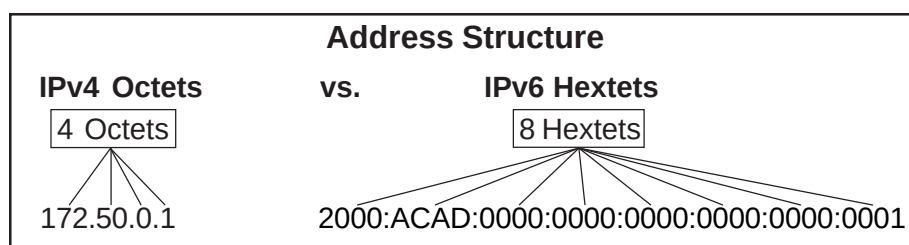
Understanding IPv6 Third Edition

By Joseph Davies

ISBN: 978-0-7356-5914-8

Available subnets within the Subnet ID using the Nibble boundary.

/48 No Nibbles	/52 1 Nibble	/56 2 Nibbles	/60 3 Nibbles	/64 4 Nibbles
/48 - 1 Subnet				
/52 - 16 Subnets	/52 - 1 Subnet			
/56 - 256 Subnets	/56 - 16 Subnets	/56 - 1 Subnets		
/60 - 4096 Subnets	/60 - 256 Subnets	/60 - 16 Subnets	/60 - 1 Subnets	
/64 - 65,536 Subnets	/64 - 4096 Subnets	/64 - 256 Subnets	/64 - 16 Subnets	/64 - 1 Subnet



What is a Site?

A Site = one building
A home, apartment, or house = a Site
A campus with 10 buildings = 10 Sites
A single building with 15 businesses = 15 Sites

Conversion Chart

Decimal (Base 10)	Hexadecimal (Base 16)	Binary (Base 2)
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

VLSM

Variable-Length Subnet Mask

IPv4

Workbook

Version 2.3

192.168.

192.168.10.96

192.168.10.126

172.31.15.0

10.250.1.0

Student Name:

IPv4 Address Classes

Class A	1 – 127	Leading bit pattern	0	00000000.00000000.00000000.00000000 Network . Host . Host . Host
Class B	128 – 191	Leading bit pattern	10	10000000.00000000.00000000.00000000 Network . Network . Host . Host
Class C	192 – 223	Leading bit pattern	110	11000000.00000000.00000000.00000000 Network . Network . Network . Host
Class D	224 – 239	(Reserved for multicast)		
Class E	240 – 255	(Reserved for experimental, used for research)		

Speciality Address Ranges

Loopback -	Only the single 127.0.0.1 address is used, addresses 127.0.0.0 to 127.255.255.255 are reserved. Any address within this block will loop back to the local host.
Link-Local Addresses -	IPv4 addresses in the address block 169.254.0.0 to 169.254.255.255 (169.254.0.0/16) are designated as link-local addresses.
TEST-NET Addresses -	The address block 192.0.2.0 to 192.0.2.255 (192.0.2.0/24) is set aside for teaching and learning purposes.
Experimental Addresses -	The addresses in the block 240.0.0.0 to 255.255.255.254 are listed as reserved for future use (RFC 3330).

Private Address Space

Class A	10.0.0.0 to 10.255.255.255
Class B	172.16.0.0 to 172.31.255.255
Class C	192.168.0.0 to 192.168.255.255

Default Subnet Masks

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

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Special Thanks to Melvin Baker and Jim Dorsch for taking the time to check this workbook for errors,
and to everyone who has sent in suggestions to improve the series.

Classful vs. Classless Subnetting

When you're subnetting an IP address for a network you have two options: classful and classless. Classful subnetting is the simplest method. It tends to be the most wasteful because it uses more addresses than are necessary. In classful subnetting you use the same subnet mask for each subnet, and all the subnets have the same number of addresses in them.

Classless addressing allows you to use different subnet masks and create subnets tailored to the number of users in each group. This technique is referred to as VLSM, Variable Length Subnet Masks.

What is VLSM

Variable Length Subnet Masks allow you a much tighter control over your addressing scheme. If you use a class C address with a default subnet mask you end up with one subnet containing 256 addresses. By using VLSM you can adjust the number of subnets and number of addresses depending on the specific needs of your network. The same rules apply to a class A or B addresses.

VLSM is supported by the following protocols: RIP version 2, OSPF, EIGRP, Dual IS-IS, and BGP. You need to configure your router for Variable Length Subnet Masks by setting up one of these protocols. Then configure the subnet masks of the various interfaces in the IP address interface sub-command.

Benefits of VLSM

- Allows efficient use of address space
- Allows the use of multiple subnet mask lengths
- Breaks up an address block into smaller custom blocks
- Allows for route summarization
- Provides more flexibility in network design
- Supports hierarchical enterprise networks

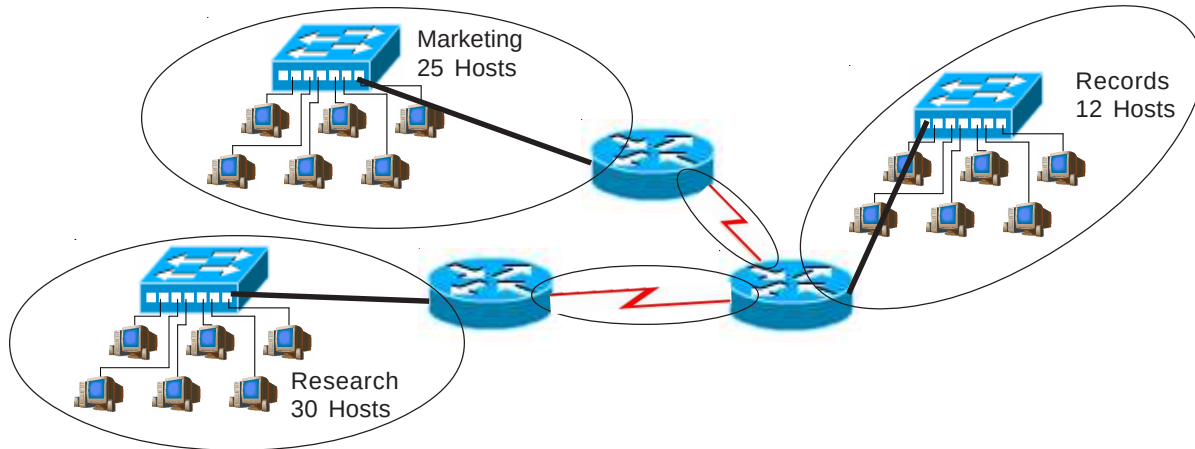
This workbook explores three different methods to figure out sub-subnets: the box method, the circle method, and a VLSM chart.

Classful Subnetting Example

When you're subnetting an IP address for a network you have two options: classful and classless. Classful subnetting is the simplest method. It also tends to be the most wasteful because it uses more addresses than are necessary. In classful subnetting you use the same subnet mask for each subnet, and all the subnets have the same number of addresses in them.

In this example you need five subnets, each one containing 30 hosts. The serial connections only require two address each so you are wasting 28 usable addresses in each of the serial subnet ranges.

IP Address: 192.168.1.0



The Box Method for visualizing subnets

Classful Subnet Ranges

```

192.168.1.0      to 192.168.1.31  /27
192.168.1.32    to 192.168.1.63  /27
192.168.1.64    to 192.168.1.95  /27
192.168.1.96    to 192.168.1.127 /27
192.168.1.128   to 192.168.1.159 /27
192.168.1.160   to 192.168.1.191 /27
192.168.1.192   to 192.168.1.223 /27
192.168.1.224   to 192.168.1.255 /27

                                     /27
                                     255.255.255.224
                                     32 Hosts
                                     8 Subnets

```

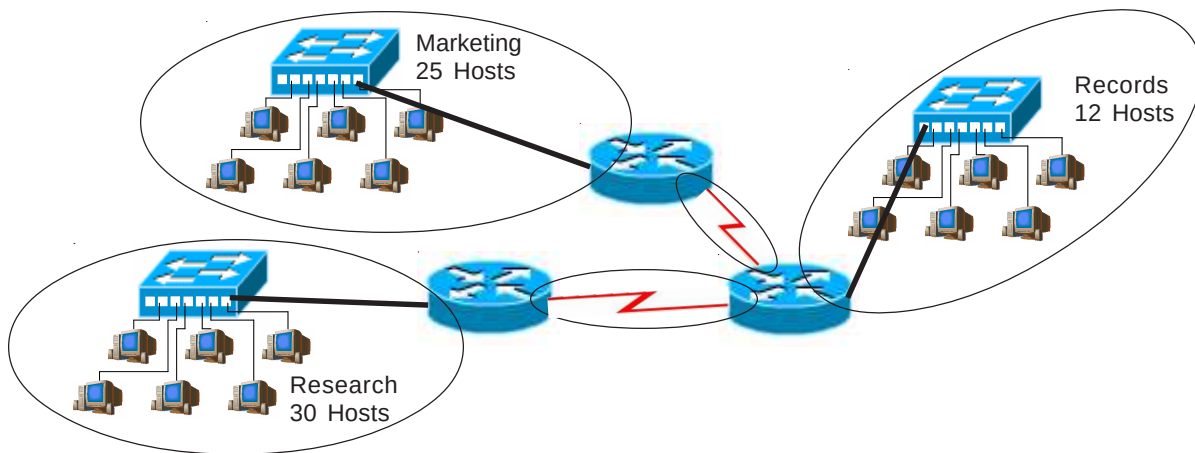
0 31	32 63	128 159	160 191
64 95	96 127	192 223	224 255

Classless Subnetting Example

Classless addressing allows you to use different subnet masks and create subnets tailored to the number of users in each subnetwork. There are fewer wasted IP addresses using smaller subnets.

In this example you need at total of five subnets, two containing 30 hosts, one containing 12 hosts, and two serial connections that only require two usable addresses each.

IP Address: 192.168.1.0

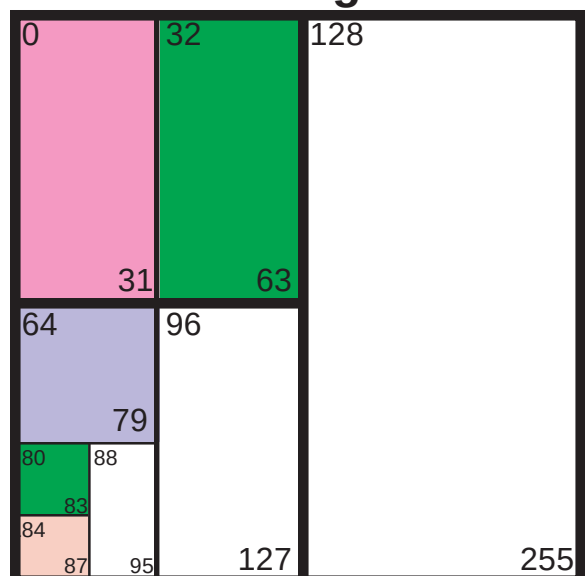


By adjusting the subnet masks you can cut your address usage by almost half in this example. This type of subnetting requires a network protocol which will support it such as: RIP version 2, EIGRP, OSPF, or BGP.

The Box Method for visualizing subnets

Classless Subnet Ranges

192.168.1.0	to	192.168.1.31	/27
192.168.1.32	to	192.168.1.63	/27
192.168.1.64	to	192.168.1.79	/28
192.168.1.80	to	192.168.1.87	/30
192.168.1.88	to	192.168.1.95	/29
192.168.1.96	to	192.168.1.127	/27
192.168.1.128	to	192.168.1.255	/25

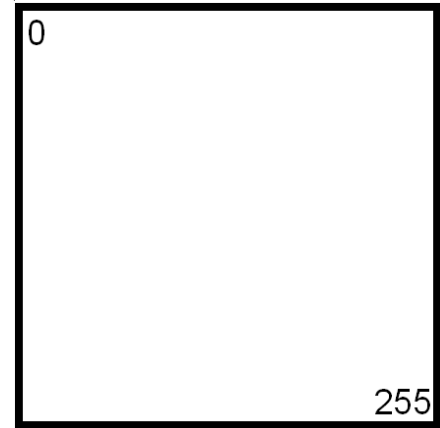


Visualizing Subnets Using The Box Method

The box method is a simple way to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

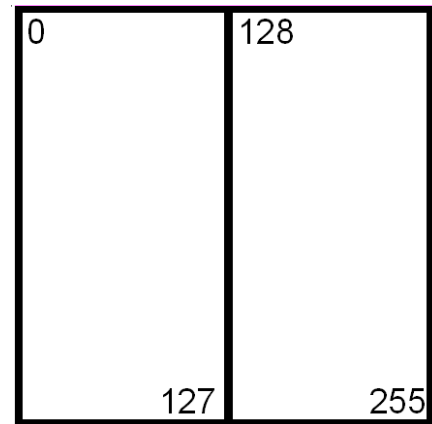
Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



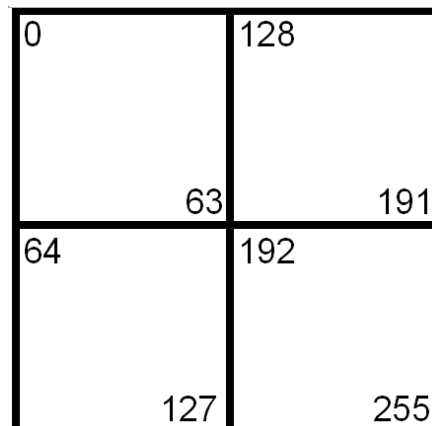
Split the box in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



Divide the box into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets

0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets

0	32	128	160
15	47	143	175
16	48	144	176
31	63	159	191
64	96	192	224
79	111	207	239
80	112	208	240
95	127	223	255

The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets

0	8	32	40	128	136	160	168
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
87	95	119	127	215	223	247	255

The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets

0	8	32	40	128	136	160	168
3	11	35	43	131	139	163	171
4	12	36	44	132	140	164	172
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
19	27	51	59	147	155	179	187
20	28	52	60	148	156	180	188
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
67	75	99	107	195	203	227	235
68	76	100	108	196	204	228	236
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
83	91	115	123	211	219	243	251
84	92	116	124	212	220	244	252
87	95	119	127	215	223	247	255

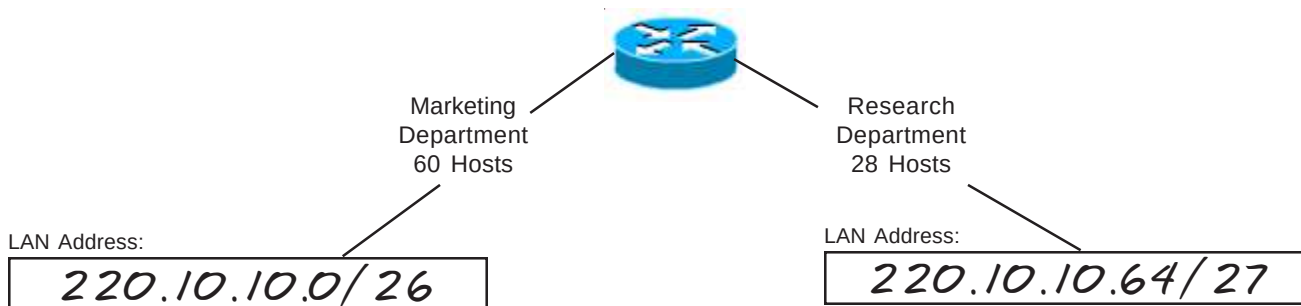
VLSM Addressing

Box Method

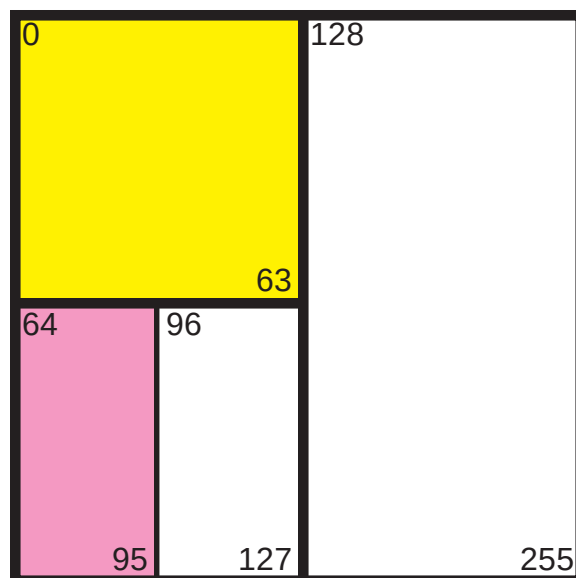
(Sample)

Problem 1

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This business will be using the class C address 220.10.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.



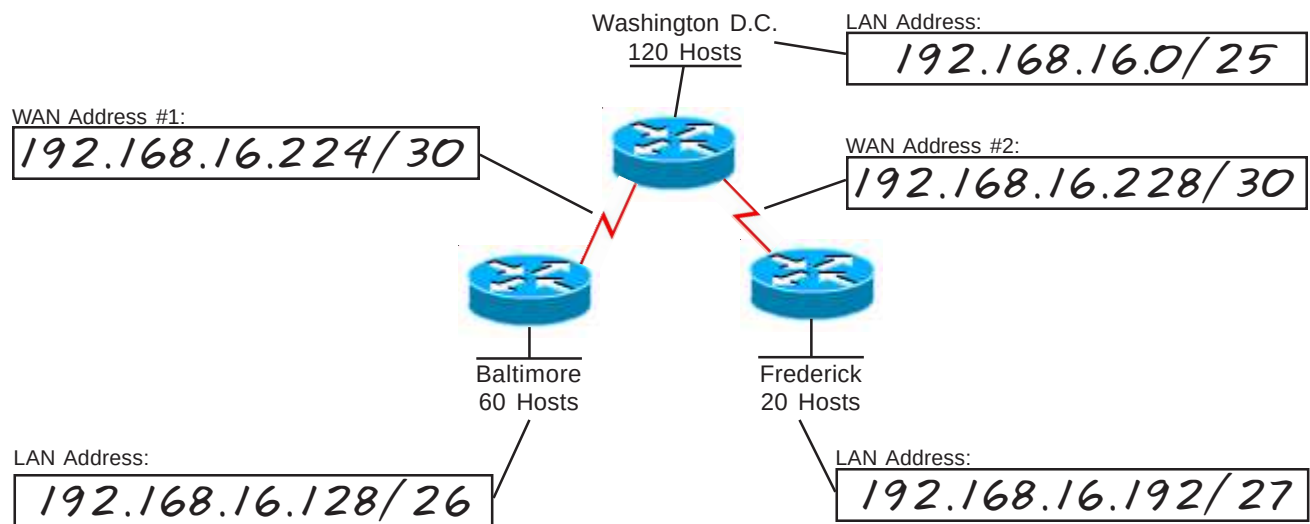
VLSM Addressing

Box Method

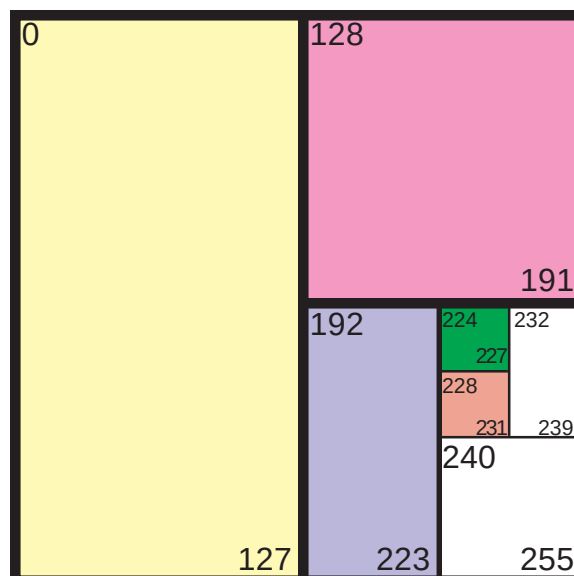
(Sample)

Problem 2

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.16.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

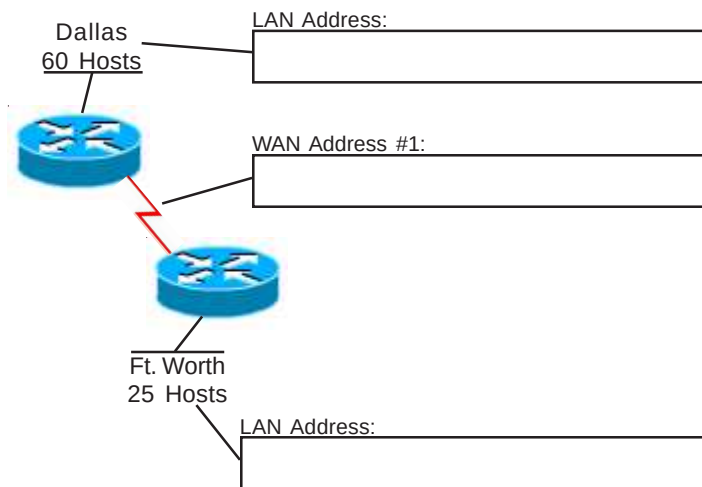


VLSM Addressing

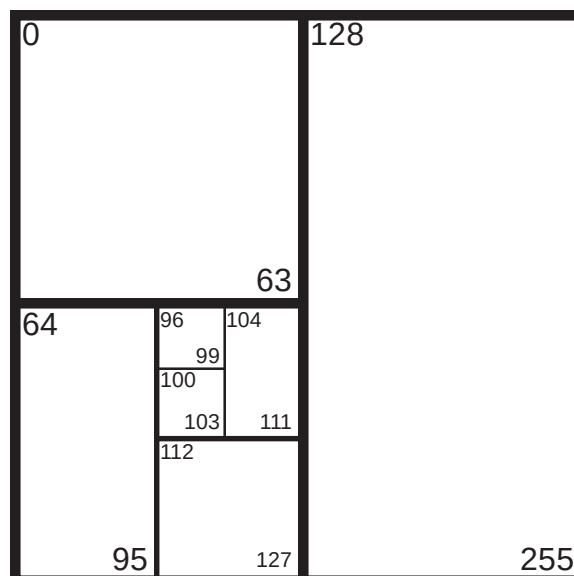
Box Method

Problem 3

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 190.10.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

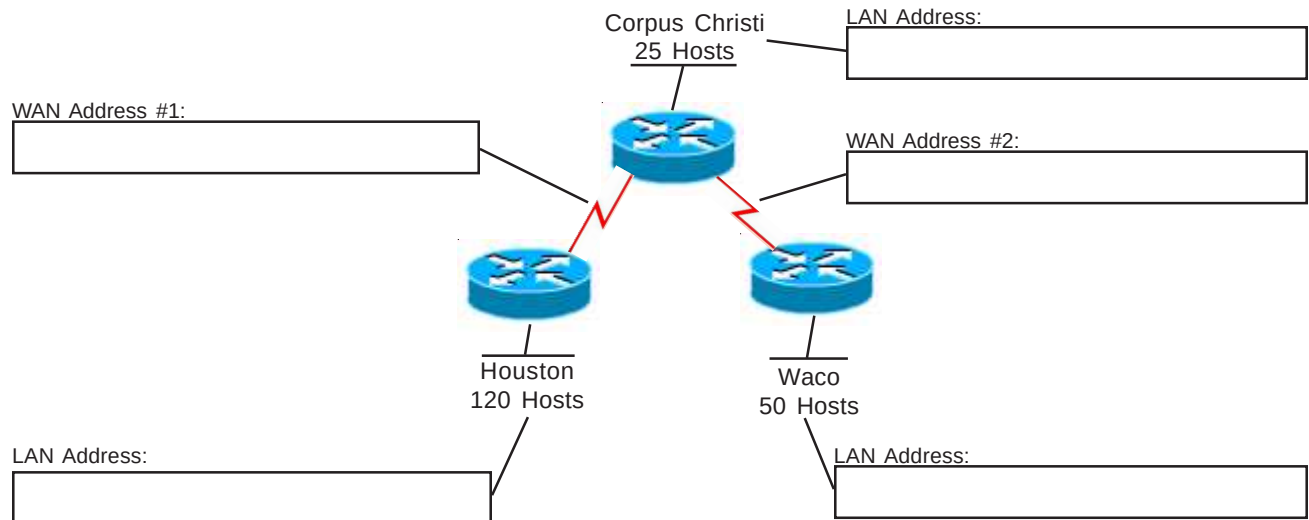


VLSM Addressing

Box Method

Problem 4

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 220.108.38.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

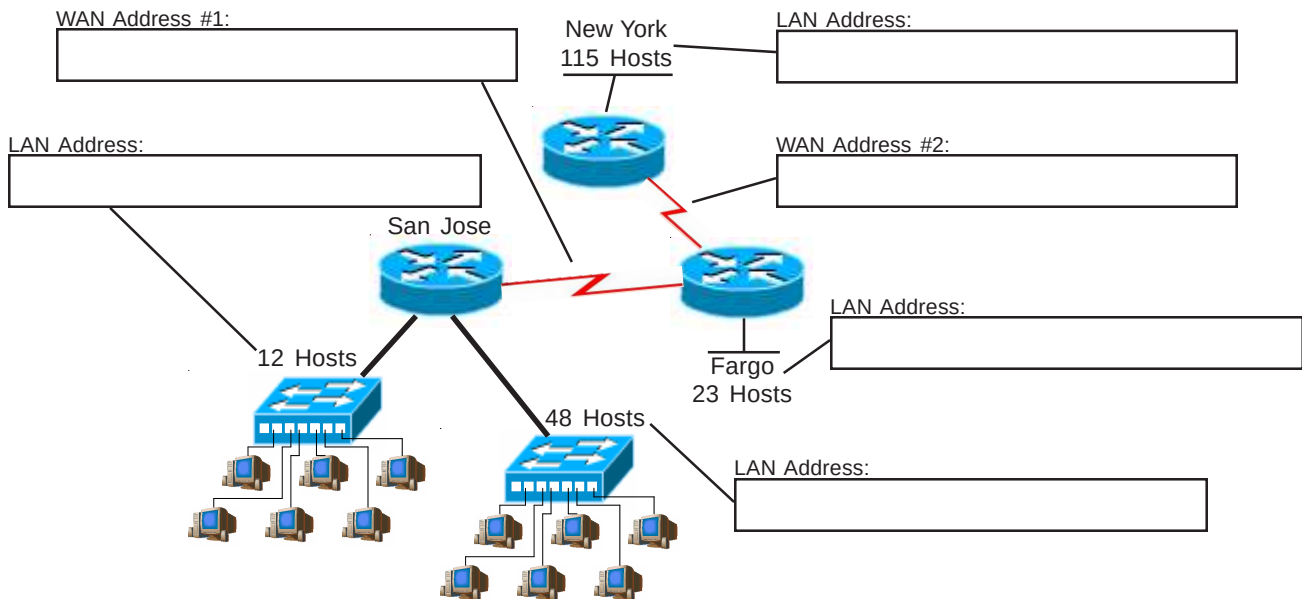
0	8	32	40	128	136	160	168
	3	11	35	43	131	139	163
4	12	36	44	132	140	164	172
	7	15	39	47	135	143	167
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247

VLSM Addressing

Box Method

Problem 5

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

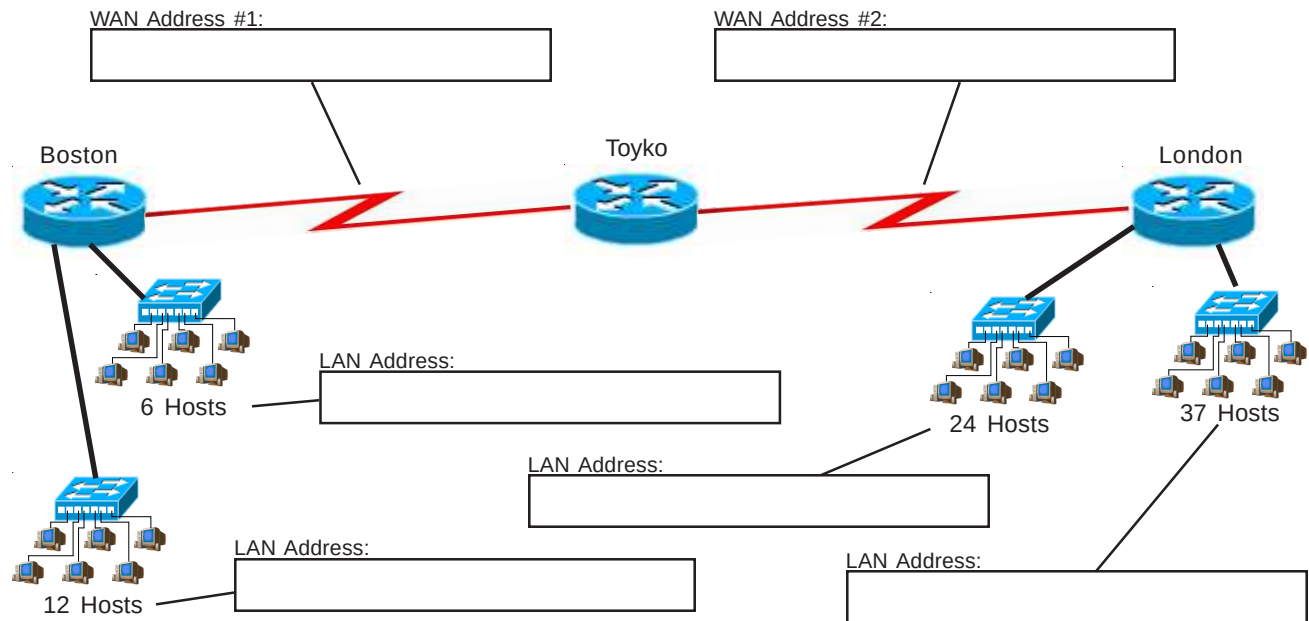
0	8	32	40	128	136	160	168
	3	11	35	43	131	139	171
4	12	36	44	132	140	164	172
	7	15	39	47	135	143	175
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247

VLSM Addressing

Box Method

Problem 6

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 222.10.150.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

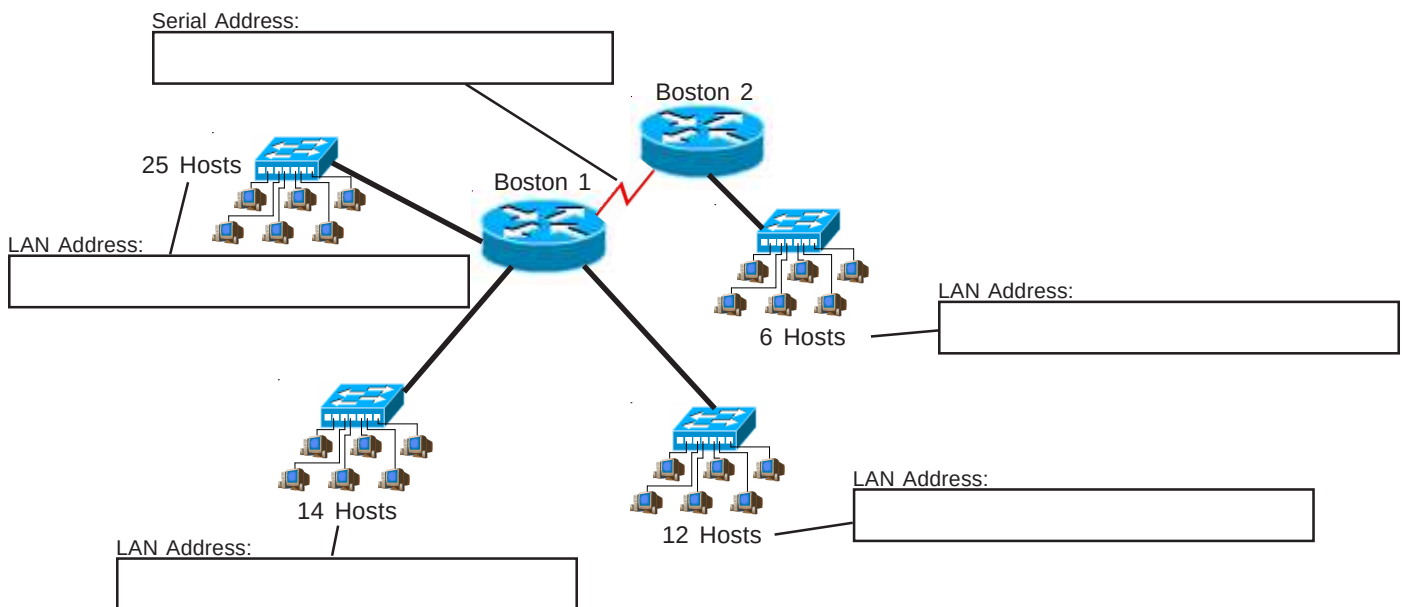
0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

VLSM Addressing

Box Method

Problem 7

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and subnet mask in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 200.150.70.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

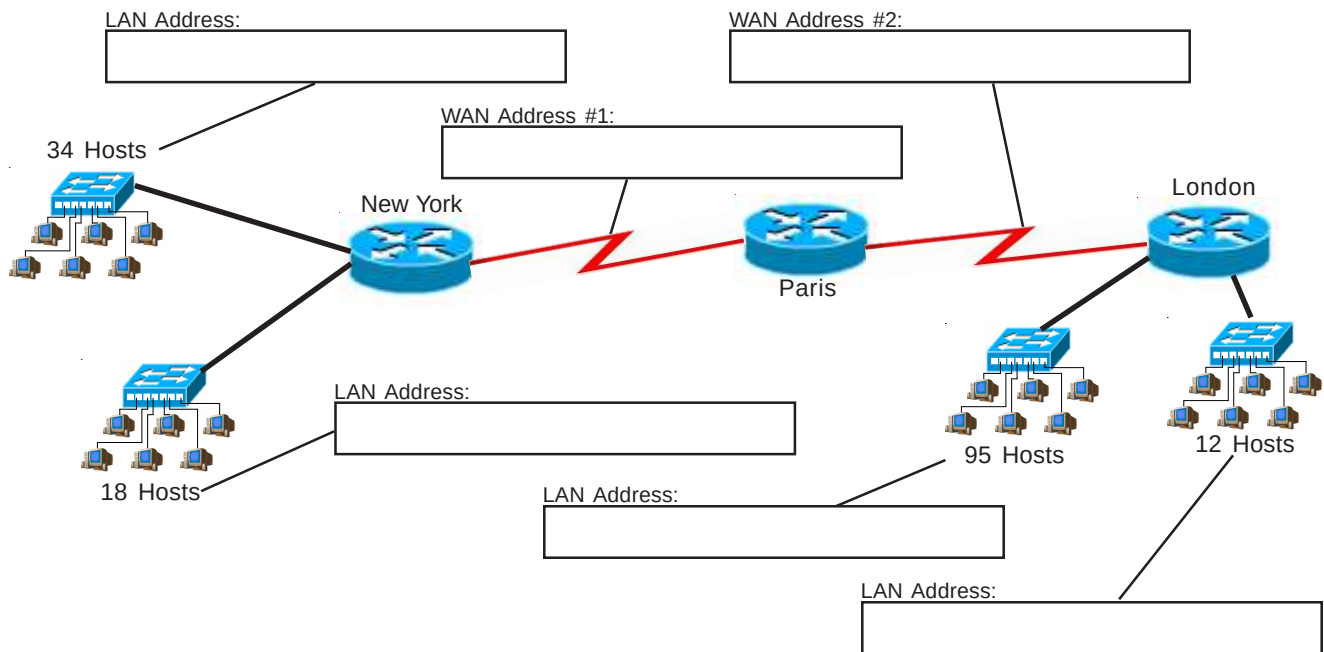
0	128
63	191
64	192
127	255

VLSM Addressing

Box Method

Problem 8

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and subnet mask in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.24.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

0
255

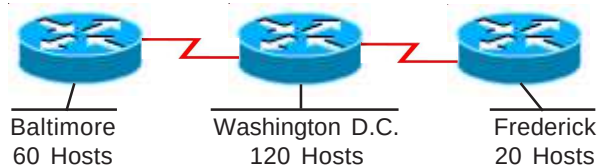
Configuring Subnets Using The Numeric Method

The Numeric method extends the technique from the IP Addressing and Subnetting workbook to allow for classless subnetting. Start by listing the needed hosts per subnets from largest to smallest.

Step 1

In this sample the needed host groups are:

- 120 Washington D. C.
- 60 Baltimore
- 20 Frederick
- 2 Serial #1
- 2 Serial #2



Step 2

Lay out the IP address the same way you would in the IP Addressing and Subnetting workbook showing the binary values of the bits, the number of subnets, and the number of hosts.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
192.10.10.0	0	0	0	0	0	0	0	0	0	

Step 3

Draw a line allowing for 120 hosts. Using the value of the last bit borrowed or magic number, write in your first range. Then add the subnet address to start the second range.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
192.10.10.0	0	0	0	0	0	0	0	0	0	

	Address Ranges		
Hosts	192.168.16.0	to	192.168.16.127
120	192.168.16.128	to	
			CIDR /25

Draw a second line for the 60 hosts. Use the value of the last bit borrowed or magic number to lay out the second largest subnet. Add the subnet address which will begin the third range.

Step 5

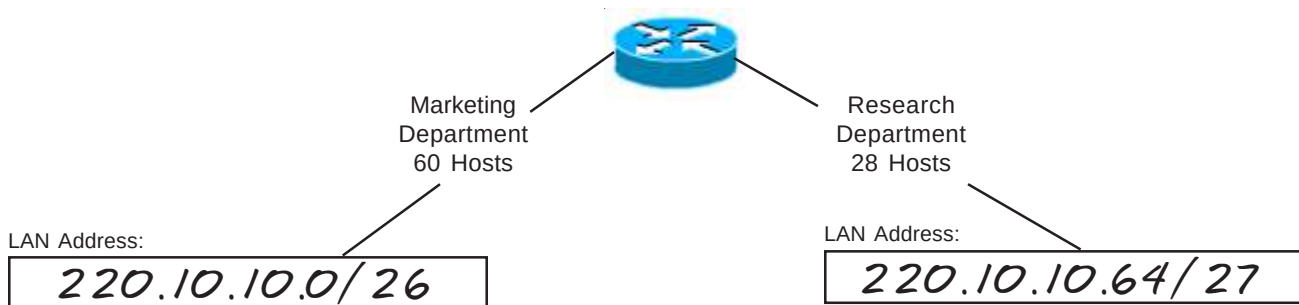
Continue the same process until all the subnets are accounted for. The CIDR is the total number of bits borrowed for the network and subnetwork portion of the address.

	120 Hosts	60 Hosts	20 Hosts	2 Hosts						
Number of Subnets	256	128	64	32	16	8	4	2	-	Number of Hosts
	2	4	8	16	32	64	128	256		
	128	64	32	16	8	4	2	1	-	Binary values
192.10.10.0	0	0	0	0	0	0	0	0		

Hosts	Address Ranges	CIDR
120	192.168.16.0 to 192.168.16.127	/25
60	192.168.16.128 to 192.168.16.191	/26
20	192.168.16.192 to 192.168.16.223	/27
2	192.168.16.224 to 192.168.16.227	/30
2	192.168.16.228 to 192.168.16.231	/30

(Sample)

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 220.10.10.0. Start with your largest groups first.

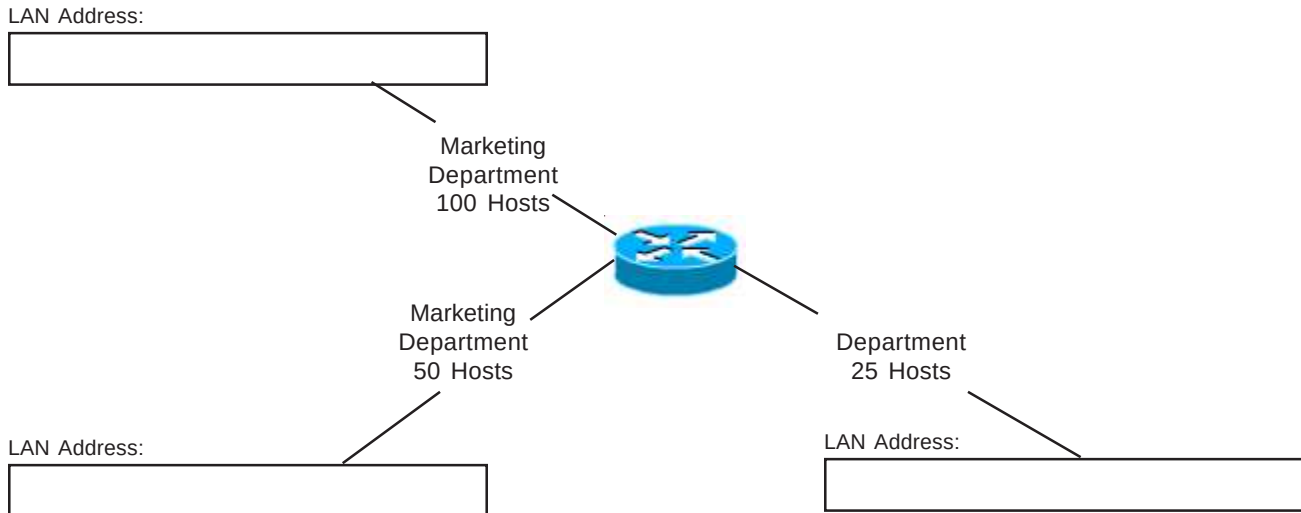
[illegible]

<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>
60	220.10.10.0 to 220.10.10.63	/26
28	220.10.10.64 to 220.10.10.95	/27

Numeric Method

Problem 11

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 200.20.20.0. Start with your largest groups first.



Show your work for **Problem 11** in the space below.

	100 Hosts	50 Hosts	25 Hosts								
Number of Subnets	256	128	64	32	16	8	4	2	-	Number of Hosts	
	2	4	8	16	32	64	128	256			
	128	64	32	16	8	4	2	1	-	Binary values	
200.20.20.0	0	0	0	0	0	0	0	0			

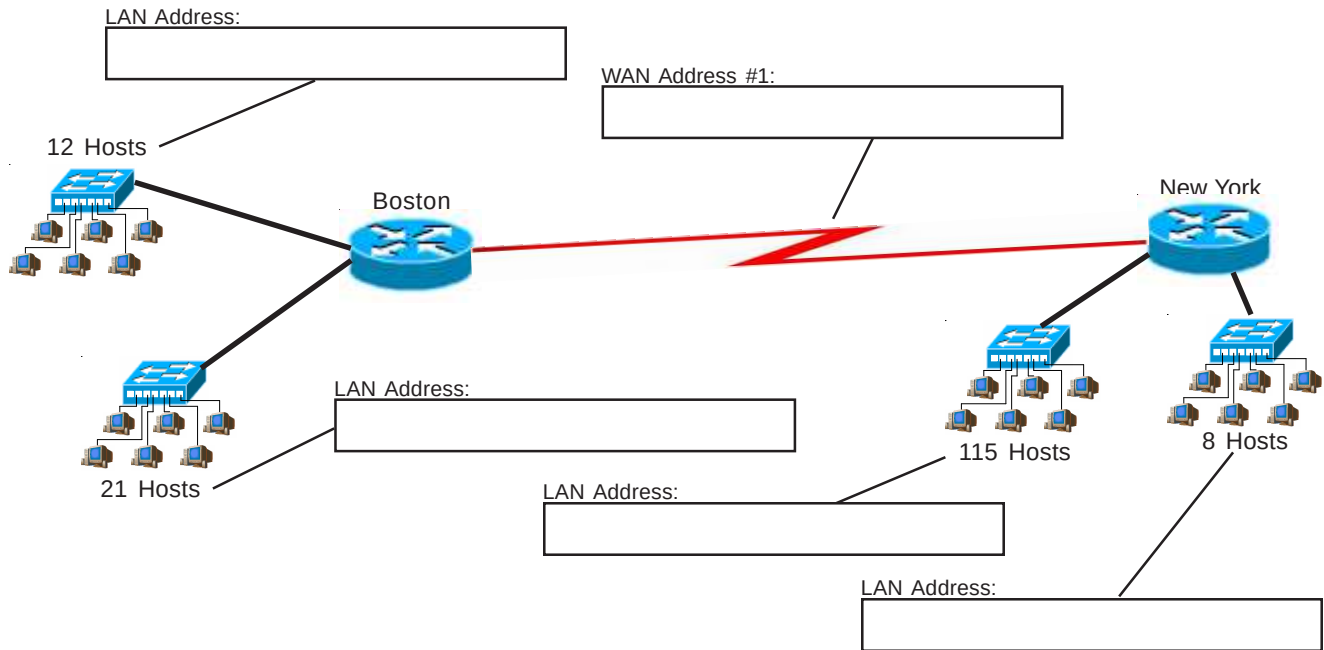
<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>
100		
50		
25		

VLSM Addressing

Numeric Method

Problem 12

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 199.55.70.0. Start with your largest groups first.



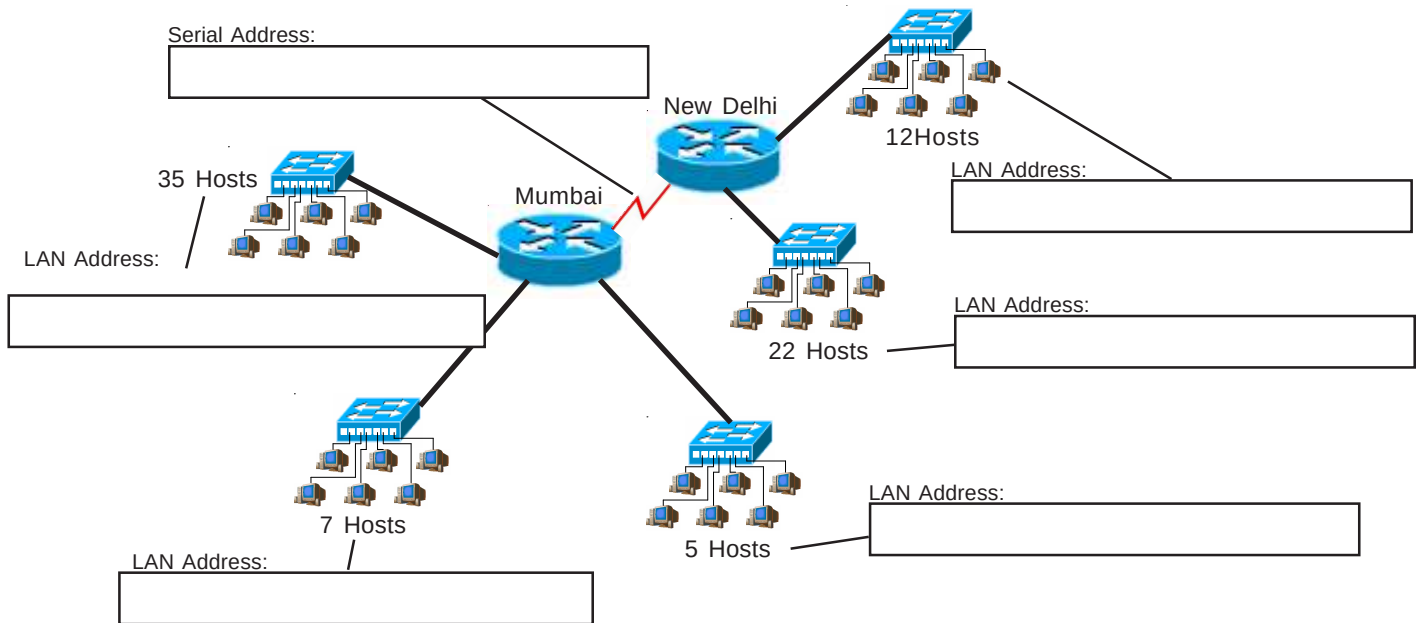
Show your work for Problem 12 in the space below.

	Hosts	Hosts	Hosts	Hosts	Hosts	Hosts	Hosts	Hosts	Hosts	Number of Hosts
Number of Subnets	256	128	64	32	16	8	4	2	1	
	2	4	8	16	32	64	128	256		
	128	64	32	16	8	4	2	1		Binary values
199.55.70.0	0	0	0	0	0	0	0	0	0	
Hosts	Address Ranges									CIDR
115										
21										
12										
8										
2										

Numeric Method

Problem 13

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 200.150.70.0. Start with your largest groups first.



Show your work for Problem 13 in the space below.

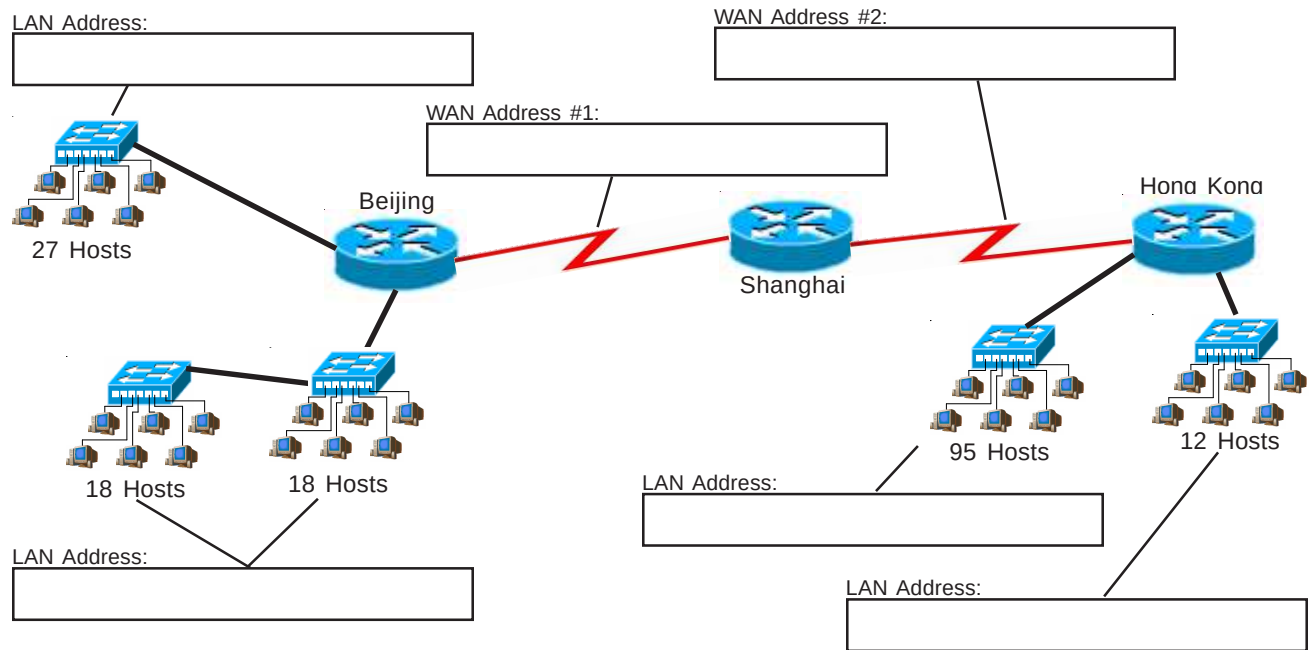
		256		128		64		32		16		8		4		2		Number of Hosts	
Number of Subnets	-	2	4	8	16	32	64	128	256										
		128	64	32	16	8	4	2	1	Binary values									
200.150.70.0	0	0	0	0	0	0	0	0	0										
<u>Hosts</u>		<u>Address Ranges</u>														<u>CIDR</u>			

VLSM Addressing

Numeric Method

Problem 14

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 210.10.10.0. Start with your largest groups first.



Show your work for Problem 14 in the space below.

210.10.10.00000000

Hosts

Address Ranges

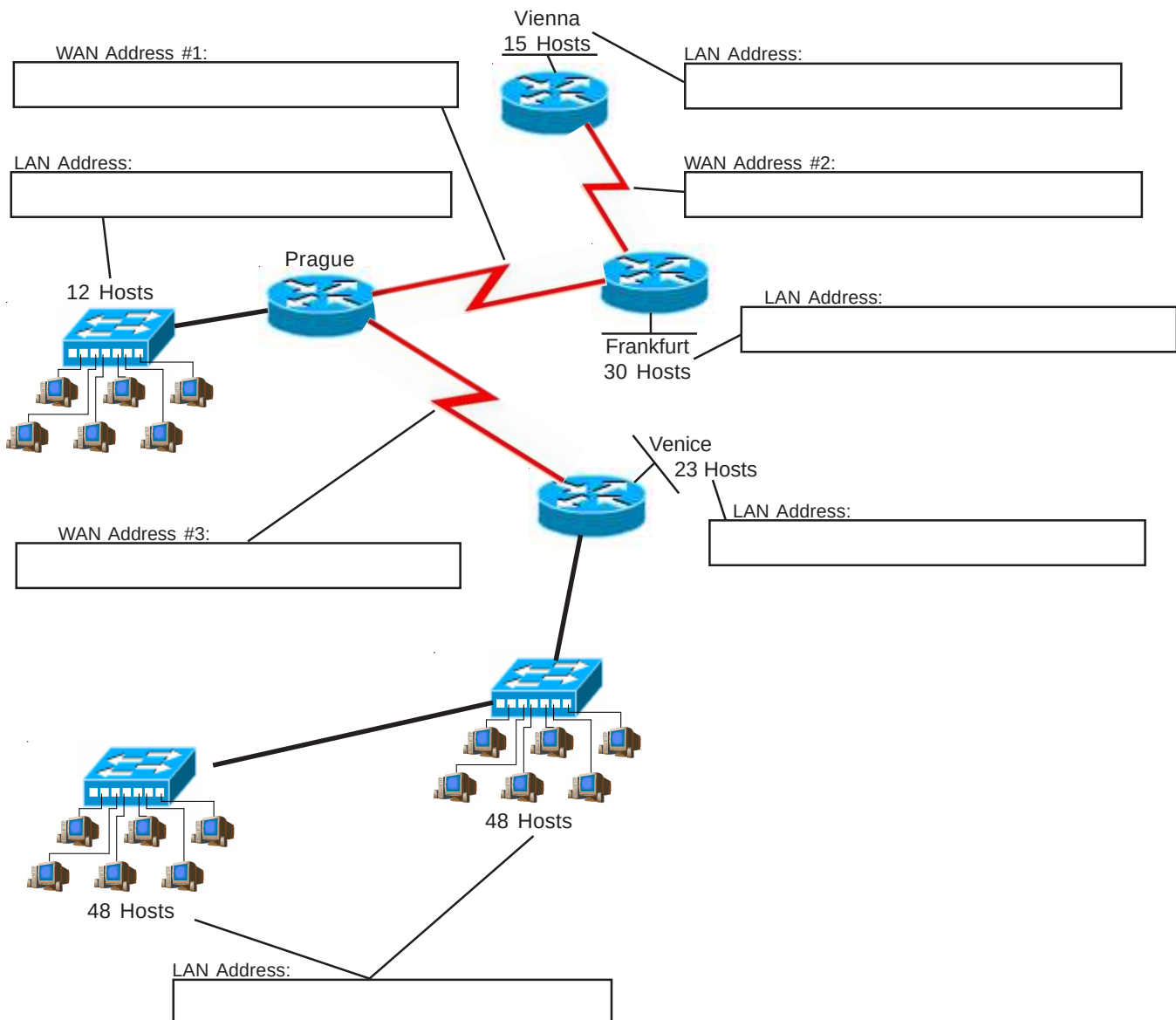
CIDR

VLSM Addressing

Numeric Method

Problem 15

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 192.168.150.0. Start with your largest groups first.



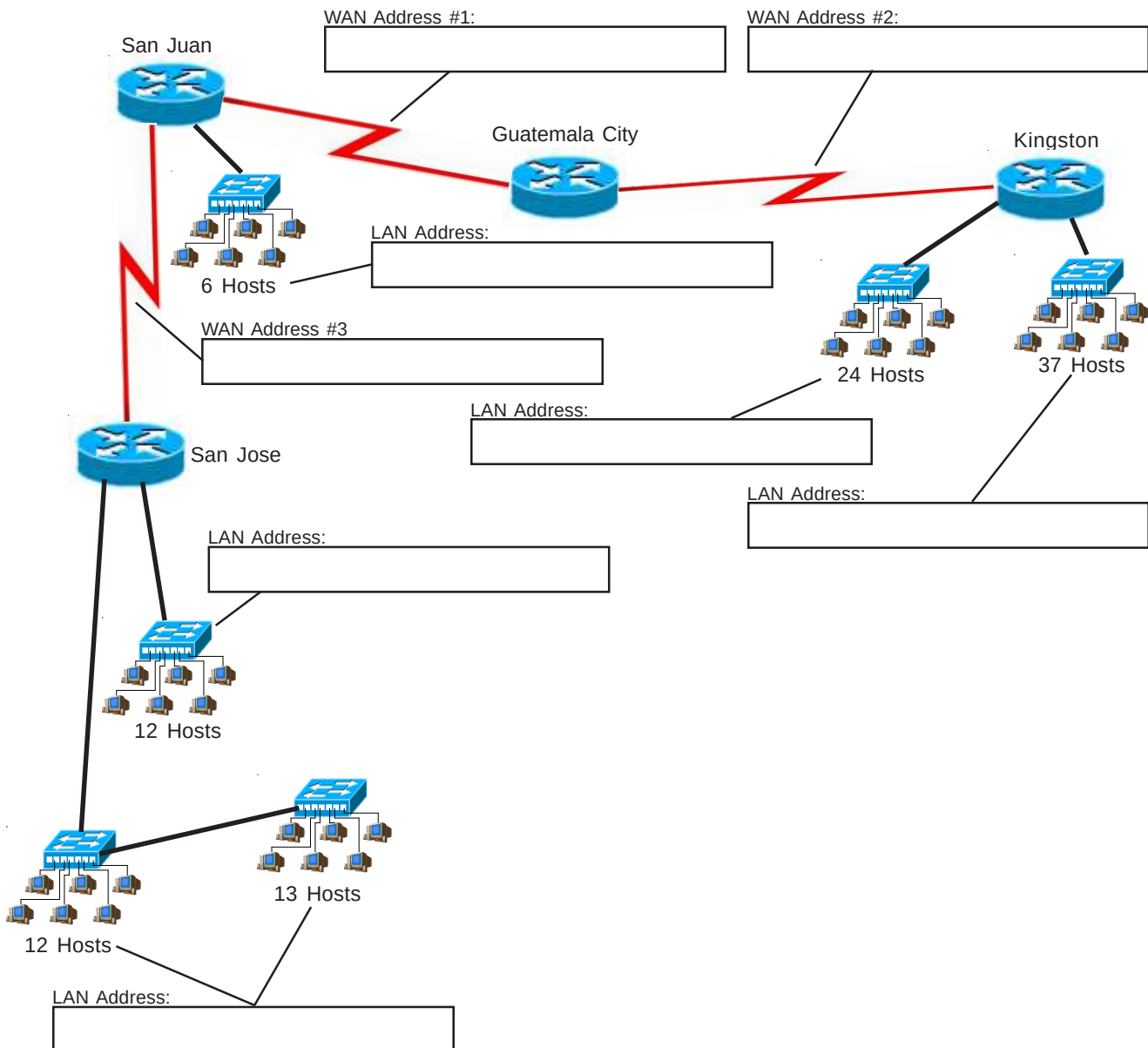
Show your work for Problem 15 in the space below.

VLSM Addressing

Numeric Method

Problem 16

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 195.75.23.0. Start with your largest groups first.



Show your work for Problem 16 in the space below.

Visualizing Subnets Using a VLSM Chart

The VLSM chart is the third method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You can adjust each sub-subnet to the correct size needed.

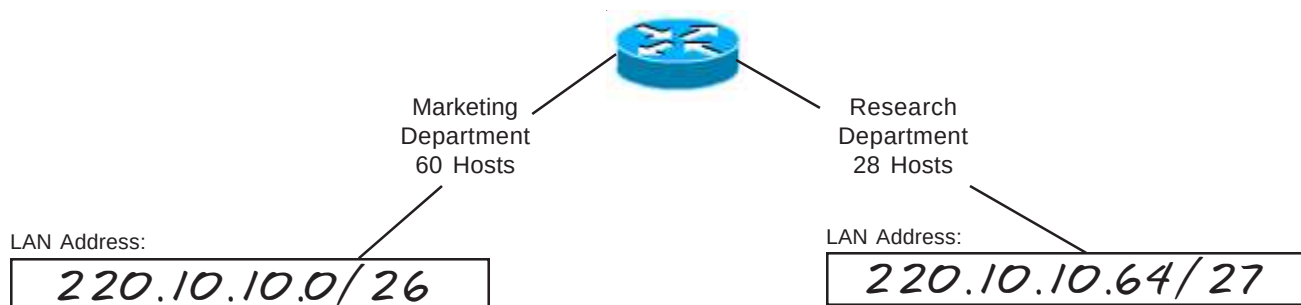
VLSM Addressing

VLSM Chart Method

(Sample)

Problem 17

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This business will be using the class C address 220.10.10.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

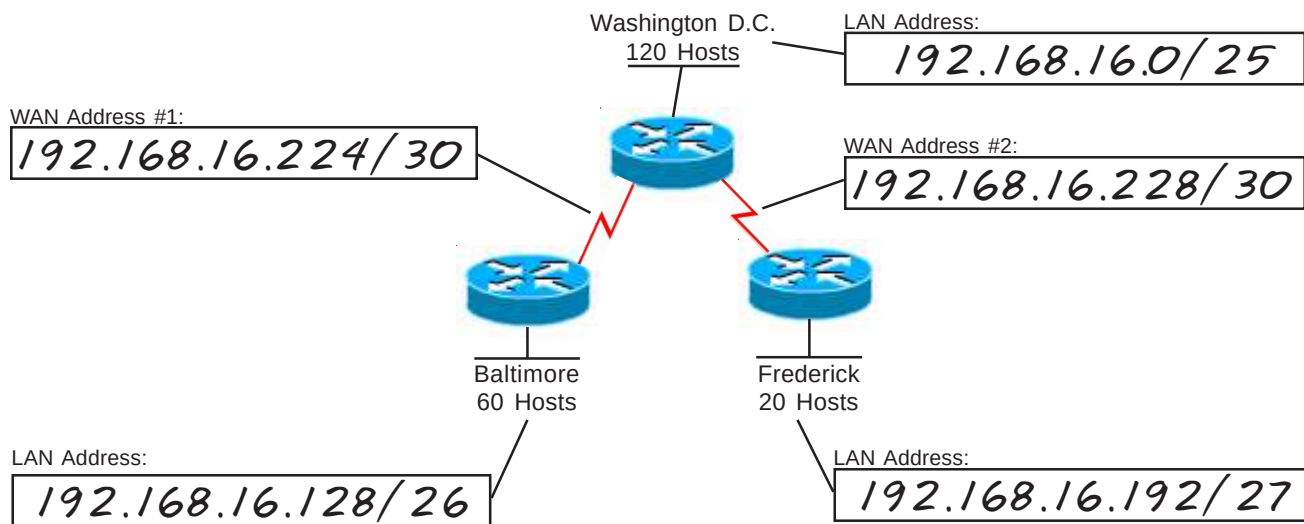
VLSM Addressing

VLSM Chart Method

(Sample)

Problem 18

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 192.168.16.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

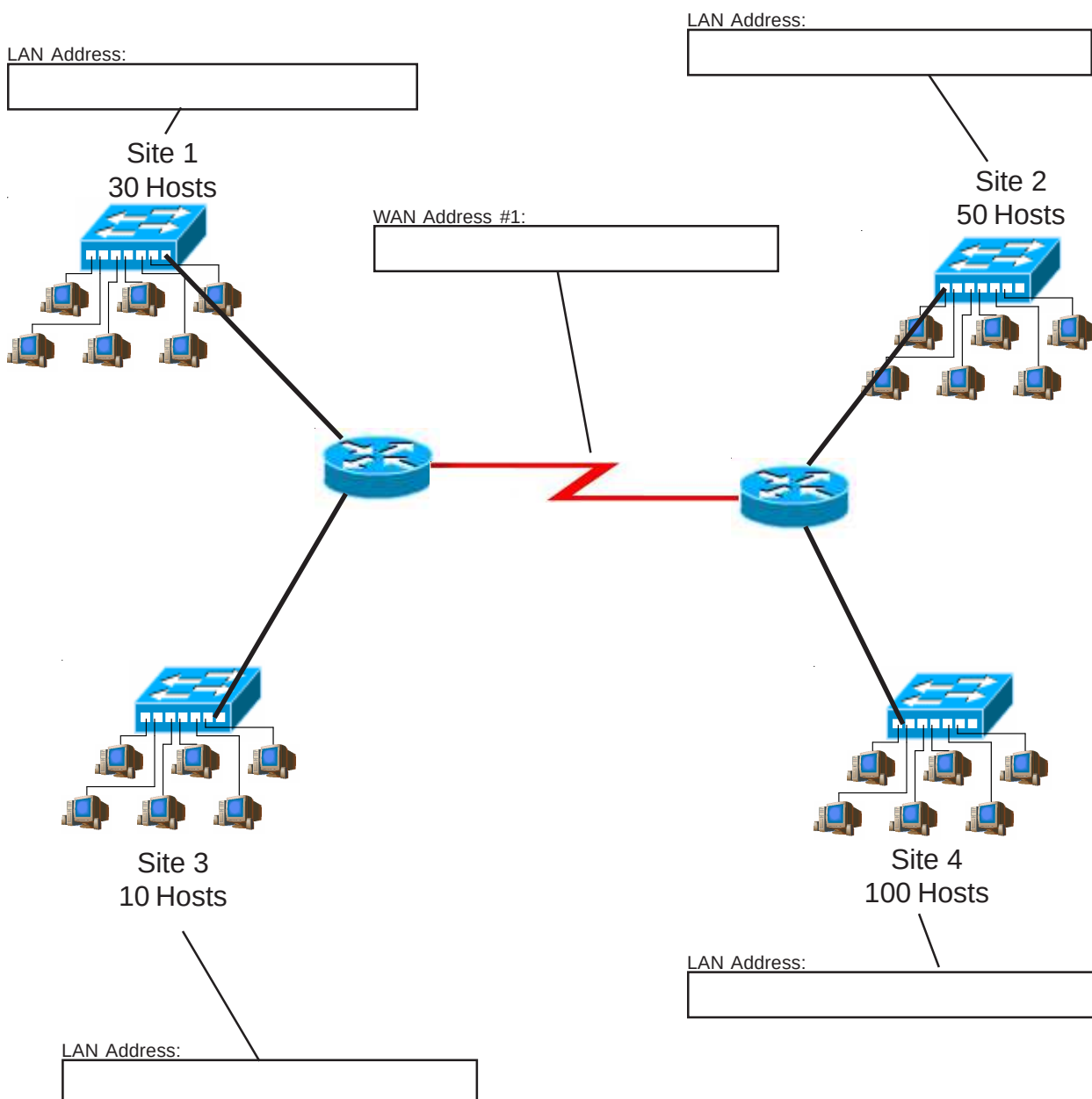
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 19

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 199.55.78.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

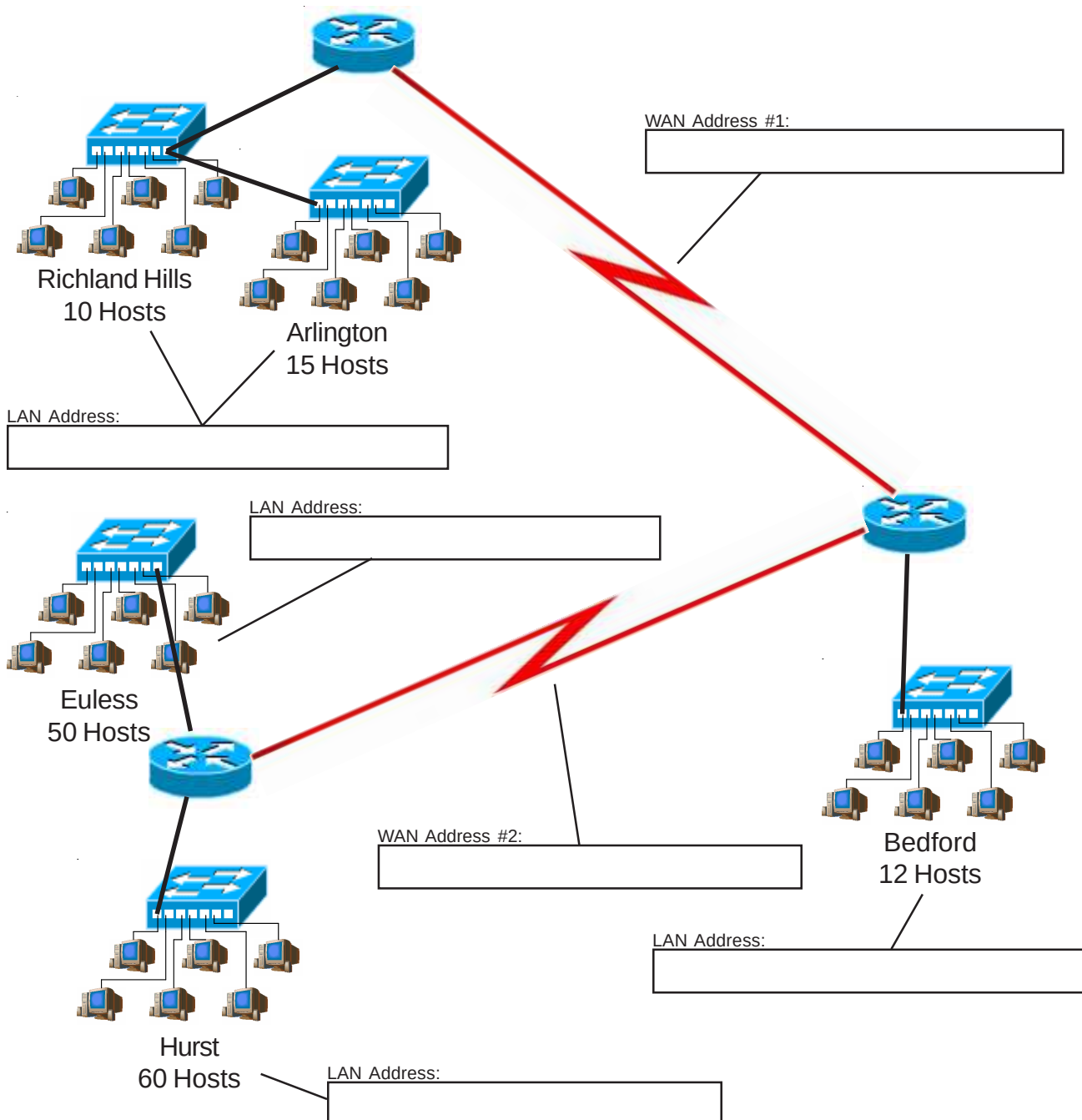
/24	/25	/26	/27	/28	/29	/30							
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts							
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7 8-11 12-15 16-19 20-23 24-27 28-31							
					16-31	8-15	32-35 36-39 40-43 44-47						
						32-47	16-23	48-51 52-55 56-59 60-63					
							48-63	24-31	64-67 68-71 72-75 76-79				
				64-127				64-95	64-79	32-39 40-47	80-83 84-87 88-91 92-95		
					80-95					48-55 56-63	96-99 100-103 104-107 108-111		
						96-127			96-111	64-71 72-79	112-115 116-119 120-123 124-127		
					112-127		80-87 88-95			128-131 132-135 136-139 140-143			
			128-255				128-191	128-159	128-143	128-135 136-143	144-147 148-151 152-155 156-159		
					144-159					144-151 152-159	160-163 164-167 168-171 172-175		
						160-191				160-175	16-167 168-175	176-179 180-183 184-187 188-191	
											176-191	176-183 184-191	192-195 196-199 200-203 204-207
				192-223					192-207			208-215 216-223	224-227 228-231 232-235 236-239
					208-223							224-231 232-239	240-243 244-247 248-251 252-255
										224-255		224-239	240-247 248-255
											240-255		

VLSM Addressing

VLSM Chart Method

Problem 20

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 223.150.50.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

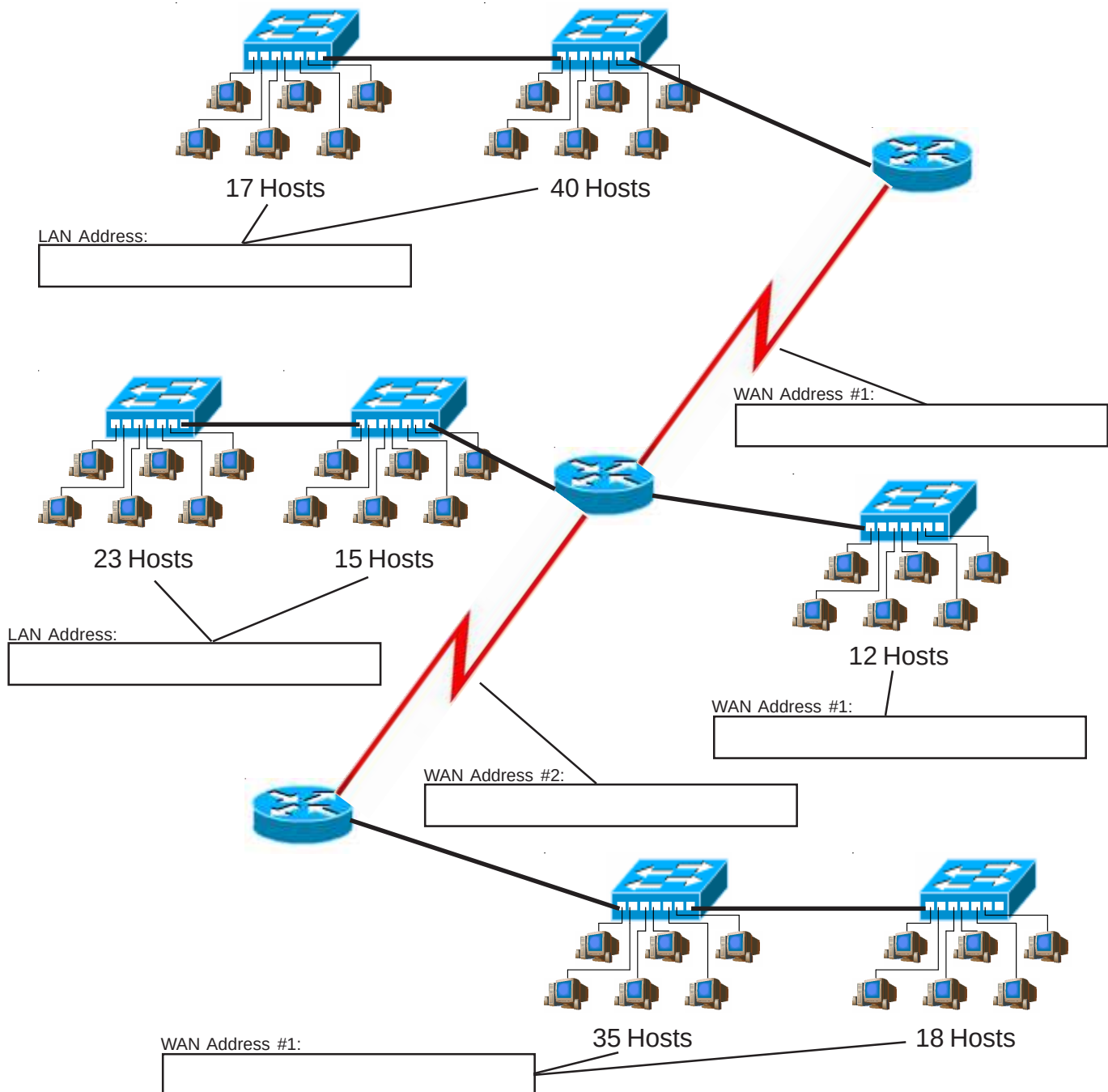
/24	/25	/26	/27	/28	/29	/30				
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts				
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7				
					8-15	8-11 12-15				
				16-31	16-23	16-19 20-23				
					24-31	24-27 28-31				
					32-63	32-47	32-39	32-35 36-39		
							40-47	40-43 44-47		
				48-63			48-55	48-51 52-55		
							56-63	56-59 60-63		
			64-127			64-95	64-79	64-71	64-67 68-71	
								72-79	72-75 76-79	
				80-95				80-87	80-83 84-87	
								88-95	88-91 92-95	
					96-127		96-111	96-103	96-99 100-103	
								104-111	104-107 108-111	
				112-127				112-119	112-115 116-119	
								120-127	120-123 124-127	
		128-255				128-191	128-159	128-143	128-135	128-131 132-135
									136-143	136-139 140-143
				144-159					144-151	144-147 148-151
									152-159	152-155 156-159
					160-191			160-175	16-167	160-163 164-167
									168-175	168-171 172-175
				176-191					176-183	176-179 180-183
									184-191	184-187 188-191
			192-255				192-223	192-207	192-199	192-195 196-199
									200-207	200-203 204-207
				208-223					208-215	208-211 212-215
									216-223	216-219 220-223
					224-255			224-239	224-231	224-227 228-231
									232-239	232-235 236-239
				240-255					240-247	240-243 244-247
									248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 21

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 222.22.2.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

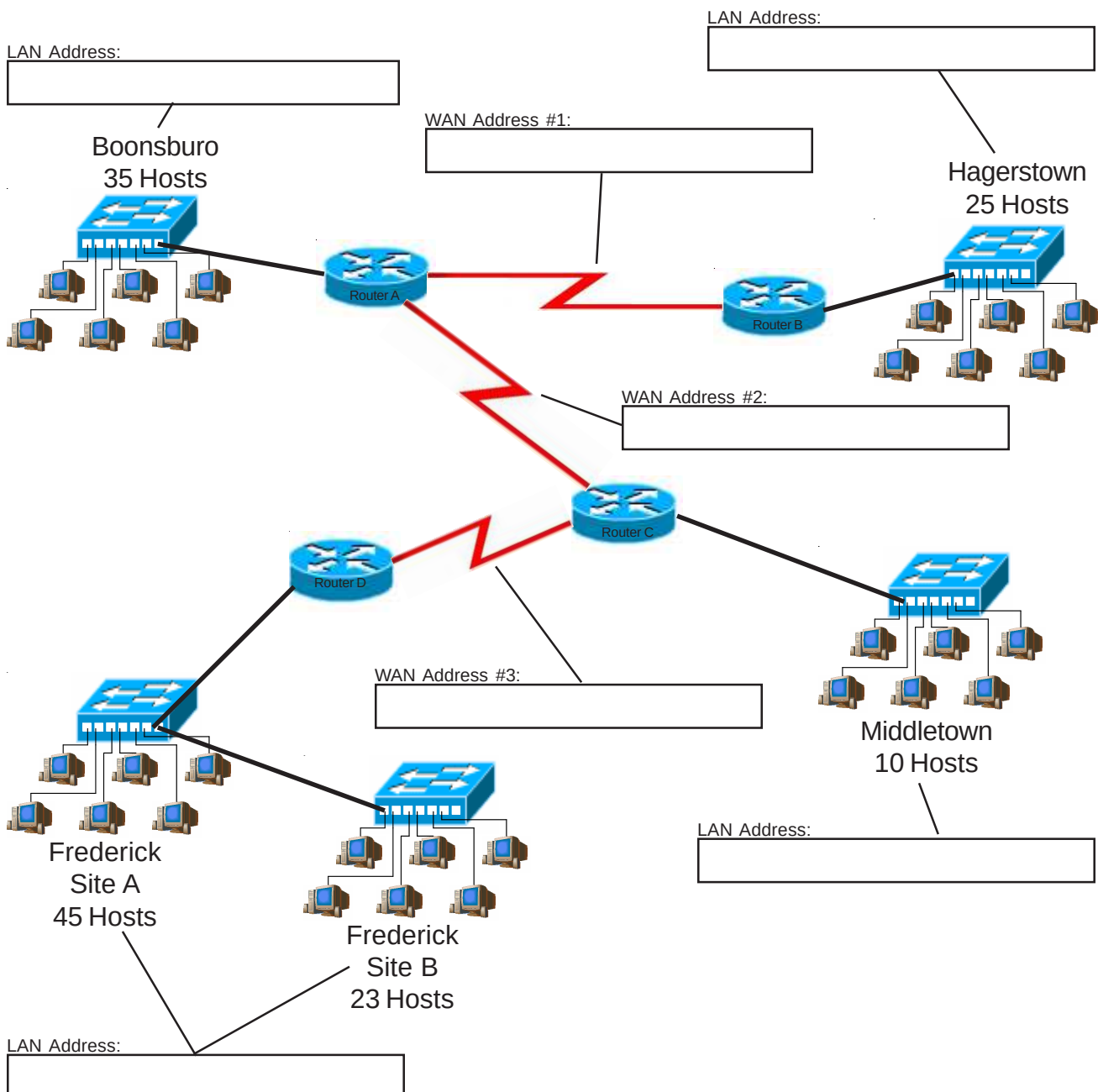
/24	/25	/26	/27	/28	/29	/30		
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts		
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7 8-11 12-15 16-19 20-23 24-27 28-31		
					16-31	8-15	32-35 36-39 40-43 44-47 48-51 52-55 56-59 60-63	
						32-47	16-23	64-67 68-71 72-75 76-79 80-83 84-87 88-91 92-95 96-99
							48-63	24-31
				64-95				32-39
					80-95			40-47
						96-111		48-55
							112-127	56-63
			64-127					64-71
					80-95			72-79
						96-111		80-87
							112-127	88-95
				128-191				96-103
					144-159			104-111
						160-175		112-119
							176-191	120-127
		192-223	128-135					
			208-223		136-143			
					224-239	144-151		
						240-255	152-159	
				224-255			16-167	
			248-255				168-175	
					252-255		176-183	
						256-259	184-191	

VLSM Addressing

VLSM Chart Method

Problem 22

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 200.20.2.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

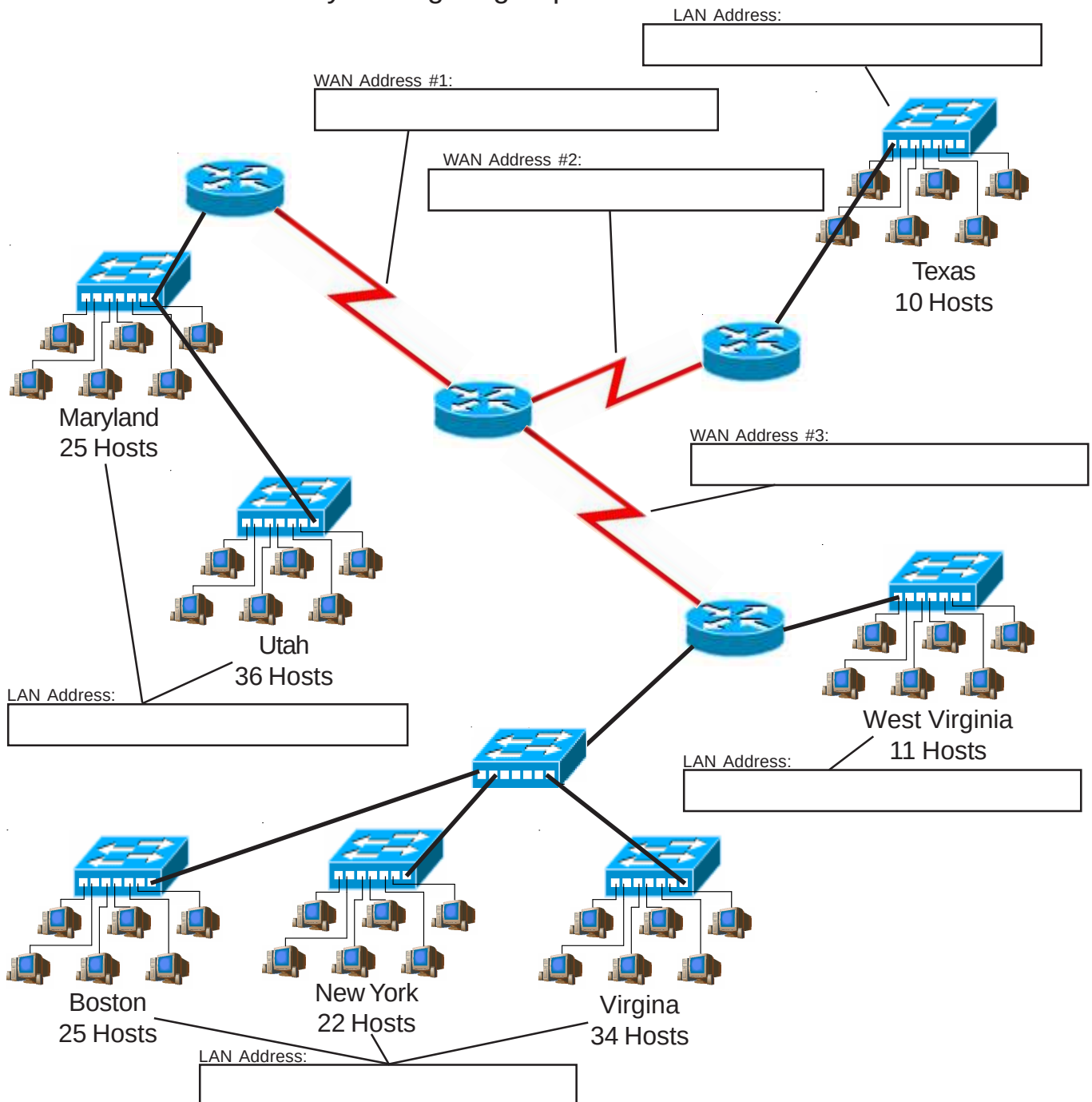
/24	/25	/26	/27	/28	/29	/30						
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts						
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7 8-11 12-15 16-19 20-23 24-27 28-31						
					16-31	8-15	32-35 36-39 40-43 44-47					
						32-47	16-23	48-51 52-55 56-59 60-63				
							48-63	24-31	64-67 68-71 72-75 76-79			
				64-127				64-95	64-79	80-83 84-87 88-91 92-95		
					80-95					96-99 100-103 104-107 108-111		
						96-127			96-111	112-115 116-119 120-123 124-127		
					112-127		128-131 132-135 136-139 140-143 144-147 148-151 152-155 156-159					
			128-255				128-191	128-159	128-143	160-163 164-167 168-171 172-175 176-179 180-183 184-187 188-191		
					144-159					192-195 196-199 200-203 204-207 208-211 212-215 216-219 220-223 224-227		
						160-191				160-175	228-231 232-235 236-239 240-243 244-247 248-251 252-255	
											176-191	240-247 248-255
				192-255					192-223			192-207
					208-223							
										224-255		224-239
					240-255						240-247 248-255	

VLSM Addressing

VLSM Chart Method

Problem 23

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 190.150.23.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

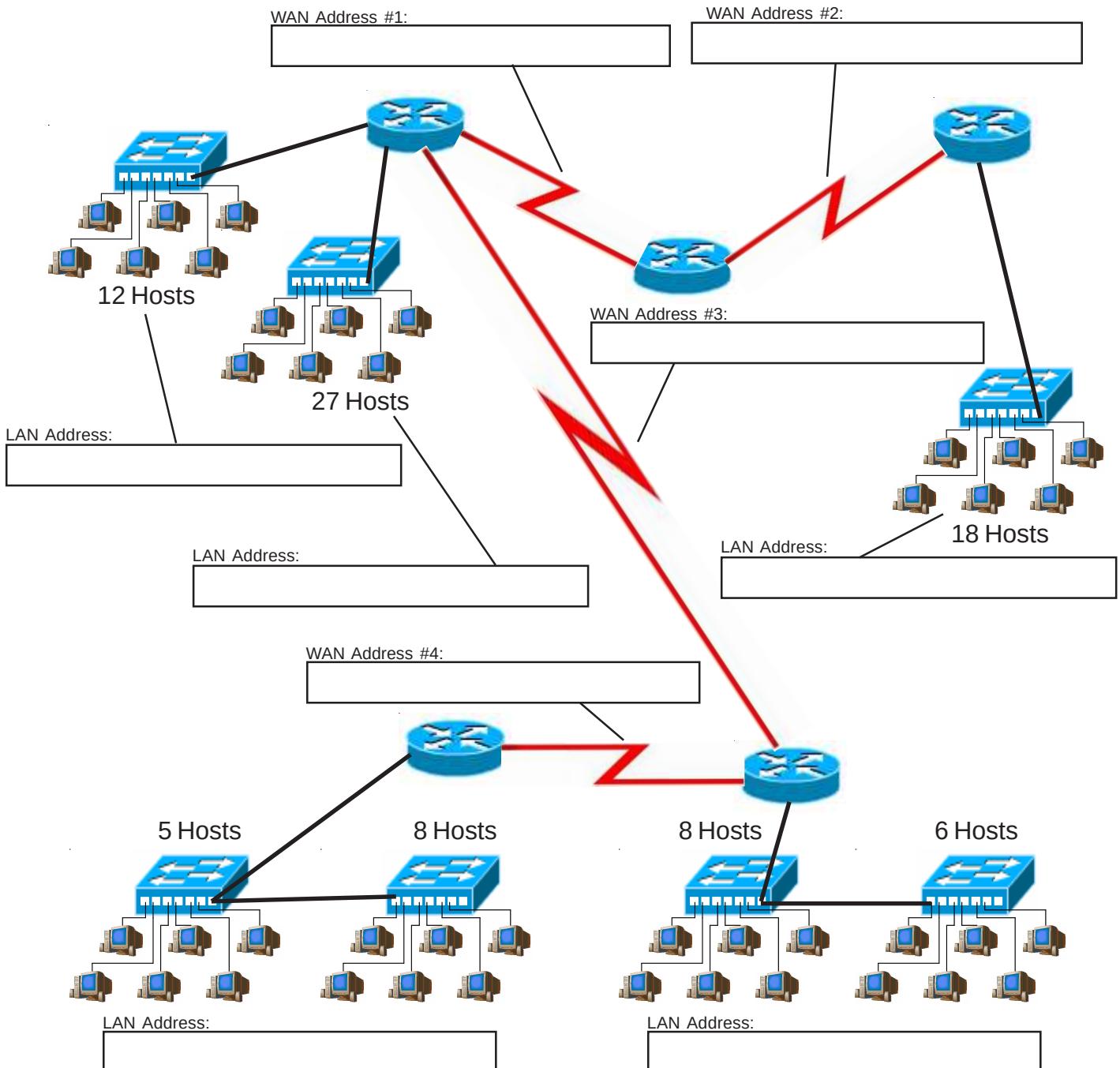
/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	16-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

VLSM Addressing

VLSM Chart Method

Problem 24

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 192.168.1.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

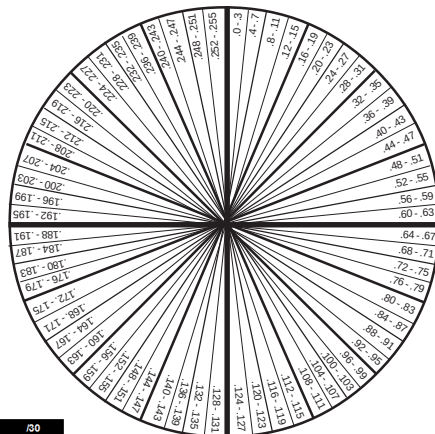
/24 255.255.255.0 256 Hosts		/25 255.255.255.128 128 Hosts		/26 255.255.255.192 64 Hosts						/29 255.255.255.248 8 Hosts		/30 255.255.255.252 4 Hosts				
0 - 255	0-127	0-63	0-31	0-15	0-7		0-3									
					4-7		4-7									
					8-15	8-11		8-11								
						12-15		12-15								
				16-31	16-23		16-19									
					20-23		20-23									
					24-31	24-27		24-27								
						28-31		28-31								
			32-63	32-47	32-39		32-35									
					36-39		36-39									
					40-47	40-43		40-43								
						44-47		44-47								
				48-63	48-55		48-51									
					52-55		52-55									
					56-63	56-59		56-59								
						60-63		60-63								
		64-127	64-95	64-79	64-71		64-67									
					68-71		68-71									
					72-79	72-75		72-75								
						76-79		76-79								
				80-95	80-87		80-83									
					84-87		84-87									
					88-95	88-91		88-91								
						92-95		92-95								
				96-127	96-111	96-103		96-99								
						100-103		100-103								
						104-111	104-107		104-107							
							108-111		108-111							
			112-127		112-115		112-115									
					116-119		116-119									
					120-127	120-123		120-123								
						124-127		124-127								
			128-255		128-191	128-159	128-143	128-135		128-131						
								132-135		132-135						
								136-143	136-139		136-139					
									140-143		140-143					
				144-159			144-151		144-147							
							148-151		148-151							
							152-159	152-155		152-155						
								156-159		156-159						
	160-191	160-175		16-167		160-163										
				164-167		164-167										
				168-175		168-171		168-171								
						172-175		172-175								
		176-191		176-183		176-179										
				180-183		180-183										
				184-191		184-187		184-187								
						188-191		188-191								
	192-255	192-223		192-207	192-199		192-195									
					196-199		196-199									
					200-207	200-203		200-203								
						204-207		204-207								
				208-223	208-215		208-211									
					212-215		212-215									
					216-223	216-219		216-219								
						220-223		220-223								
	224-255	224-239	224-231		224-227											
			228-231		228-231											
			232-239	232-235		232-235										
				236-239		236-239										
		240-255	240-247		240-243											
			244-247		244-247											
			248-255	248-251		248-251										
				252-255		252-255										

Practical VLSM Problems

Use the VLSM method of your choice to complete the following problems.

(Reference pages are in the back of the workbook.)

0	8	32	40	128	136	160	168
	3	11	35	43	131	139	163
4	12	36	44		132	140	164
	7	15	39	47	135	143	167
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247
							255



VLSM Chart 24-30 Bits (4th octet)

/24 256 Hosts	/25 128 Hosts	/26 64 Hosts	/27 32 Hosts	/28 16 Hosts	/29 8 Hosts	/30 4 Hosts
0-255	0-127	0-63	0-31	0-15	0-7	0-3
				16-31	8-15	4-7
				32-47	16-23	8-15
				48-63	24-31	16-23
				64-79	32-39	24-31
				80-95	40-47	32-39
				96-111	48-55	40-47
				112-127	56-63	48-55
				128-143	64-71	56-63
				144-159	72-79	64-71
				160-175	80-87	72-79
				176-191	88-95	80-87
				192-207	96-103	88-95
				208-223	104-111	96-103
				224-239	112-119	104-111
				240-255	120-127	112-119
					128-135	120-127
					136-143	128-135
					144-151	136-143
					152-159	144-151
					160-167	152-159
					168-175	160-167
					176-183	168-175
					184-191	176-183
					192-199	184-191
					200-207	192-199
					208-215	200-207
					216-223	208-215
					224-231	216-223
					232-239	224-231
					240-247	232-239
					248-255	240-247

VLSM Addressing

(Sample)

Problem 25

You are developing a school network with the class C address 192.168.2.0/24. There will be three computer labs with 30 computers each that need to be on different sub-subnets. Forty eight classrooms with one computer each that will comprise a single sub-subnet. The administrative office and guidance office contain a total of seven computers which will need to be grouped together. Plan for four more mini labs with six computers to each sub-subnetwork. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.2.0	/26	192.168.2.1	192.168.2.62	192.168.2.63
2	192.168.2.64	/27	192.168.2.65	192.168.2.94	192.168.2.95
3	192.168.2.96	/27	192.168.2.97	192.168.2.126	192.168.2.127
4	192.168.2.128	/27	192.168.2.129	192.168.2.158	192.168.2.159
5	192.168.2.160	/28	192.168.2.161	192.168.2.174	192.168.2.175
6	192.168.2.176	/29	192.168.2.177	192.168.2.182	192.168.2.183
7	192.168.2.184	/29	192.168.2.185	192.168.2.190	192.168.2.191
8	192.168.2.192	/29	192.168.2.193	192.168.2.198	192.168.2.199
9	192.168.2.200	/29	192.168.2.201	192.168.2.206	192.168.2.207
10					
11					
12					
13					
14					

VLSM Addressing

(Sample)

Problem 26

You are setting up a small business network with the class C address 220.55.80.0/24. The marketing division will need 12 computers. Research and development needs 27 computers. The reception area will need two computers. Management requires 19 computers. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	220.55.80.0	/27	220.55.80.1	220.55.80.30	220.55.80.31
2	220.55.80.32	/27	220.55.80.	220.55.80.62	220.55.80.63
3	220.55.80.64	/28	220.55.80.65	220.55.80.78	220.55.80.79
4	220.55.80.80	/30	220.55.80.81	220.55.80.82	220.55.80.83
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 27

You are setting up a medium sized network with the class C address 222.37.34.0/24. Marketing needs 29 computers. Research and development needs 110 computers. Bookkeeping will use 12 computers. The reception area will need three computers. Management requires 60 computers. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 28

A shipping company needs to set up its network across several locations. The Denver office needs six computers. The Waco office needs 22 computers. The Fargo office will need five computers. The WAN links between all three locations need to be included in the solution. Using the IP address 192.168.10.0/24 divide the network using VLSM. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 29

A new school is being built in the local school district. It will have three computer labs with 28 computers each. There will be 58 classrooms with 2 computers each that need to be on one sub-subnet. The office staff and administrators will need 7 computers. The guidance and attendance office will have 5 computers. The school has been given the address 223.145.75.0/24. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 30

A local college is setting up a campus wide network. The technology wing will be on its own network address of 192.168.250.0/24. The office wing will include 15 computers. There are 2 labs of 20 computers each, 2 labs of 30 computers each and one lab of 35 computers. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 31

You are setting up a network for a company in four locations. Location A has 8 computers. Location B has 122 computers. Location C has 4 computers. Location D has 55 computers. There is a WAN connection between all four locations. Complete the information required below using the class C address 192.168.10.0. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 32

A college dormitory is being remodeled. A new network is being installed. There are 50 dorm rooms with two drops each that will be on one sub-subnet. The offices will have 5 drops. The reception desk will have three drops. A small study hall will include 30 drops. Using the IP address 192.168.12.0/24 complete the information required below using VLSM. Work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 33

You are setting up a business network with the class C address 219.75.160.0/24. The marketing division will need 19 computers. Research and development needs 40 computers. The reception area will need four computers. Management requires 12 computers. Divide the network using variable length subnet information. On the opposite page draw a detailed map of this network. Include the name and sub-subnet IP addresses for each branch of the network with the subnet mask. One router with four ethernet ports will be used for this network.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

Problem 33 - Detailed Map

Draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network, and the subnet mask.

VLSM Addressing

Problem 34

A small company needs to set up its network across several locations. The New York branch office needs 15 computers. The San Jose office needs 66 computers. The Trinidad office will need 18 computers. You will need two WAN links between the routers. Using the IP address 195.20.5.0/24 divide the network using VLSM. On the opposite page draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network. Label the WAN links with the same information. Complete the information required below. Work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

Problem 34 - Detailed Map

Draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network.

Class A and B VLSM Problems

10 . 0 . 0 . 0

Class A Addresses
VLSM Chart 8-15 Bits (2nd octet)

10	11	12	13	14	15
255.0.0.0 8,192,384 hosts	255.128.0.0 4,096,384 hosts	255.192.0.0 2,048,384 hosts	255.240.0.0 1,024,384 hosts	255.288.0.0 512,384 hosts	255.336.0.0 256,384 hosts
0-255	0-127	0-63	0-5	6-11	12-15
			16-23	24-31	32-39
			40-47	48-55	56-63
		64-127	64-71	72-79	80-87
			88-95	96-103	104-111
			112-119	120-127	
	128-255	128-191	128-135	136-143	144-151
			152-159	160-167	168-175
			176-183	184-191	192-199
			200-207	208-215	216-223
		192-255	224-231	232-239	240-247
			248-255		

Class B Addresses
VLSM Chart 16-23 Bits (3rd octet)

16	17	18	19	20	21	22	23
255.255.0.0 65,536 hosts	255.255.128.0 32,768 hosts	255.255.192.0 16,384 hosts	255.255.224.0 8,192 hosts	255.255.240.0 4,096 hosts	255.255.248.0 2,048 hosts	255.255.252.0 1,024 hosts	255.255.254.0 512 hosts
0-255	0-127	0-63	0-5	6-11	12-15	16-23	24-31
			32-39	40-47	48-55	56-63	64-71
			72-79	80-87	88-95	96-103	104-111
		64-127	112-119	120-127			
	128-255	128-191	128-135	136-143	144-151	152-159	160-167
			168-175	176-183	184-191	192-199	200-207
			208-215	216-223	224-231	232-239	240-247
			248-255				
		192-255					

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

24	25	26	27	28	29	30
255.255.255.0 256 hosts	255.255.255.128 128 hosts	255.255.255.192 64 hosts	255.255.255.224 32 hosts	255.255.255.240 16 hosts	255.255.255.248 8 hosts	255.255.255.252 4 hosts
0-255	0-127	0-63	0-5	6-11	12-15	16-23
			24-31	32-39	40-47	48-55
			56-63	64-71	72-79	80-87
		64-127	88-95	96-103	104-111	112-119
			120-127			
	128-255	128-191	128-135	136-143	144-151	152-159
			160-167	168-175	176-183	184-191
			192-199	200-207	208-215	216-223
			224-231	232-239	240-247	248-255
		192-255				

VLSM

with

Class A and B Addresses

We've gone over the practical applications of using VLSM on class C addresses. The same approach works with class A and B addresses. For example an ISP may have a class A address which it needs to subnet between its customers. Each customer may need to take their addresses and subnet them again in order to use them more effectively. The real trick to this is to remember which octet of the IP address you are working with.

Sample Problem 35

Part 1 of 3

Use the **Class A** address chart to break down the address for different ISP customers. At this stage of the problem you are creating subnets using the second octet of the IP address.

ISP Addresses 15.0.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Customer #1	8 million	<i>15.0.0.0 to 15.127.255.255</i>	<i>/9</i>
Customer #2	2 million	<i>15.128.0.0 to 15.159.255.255</i>	<i>/11</i>
Customer #3	2,000,000	<i>15.160.0.0 to 15.191.255.255</i>	<i>/11</i>
Customer #4	1,000,000	<i>15.192.0.0 to 15.207.255.255</i>	<i>/12</i>
Customer #5	500,000	<i>15.208.0.0 to 15.215.255.255</i>	<i>/13</i>
Customer #6	450,000	<i>15.216.0.0 to 15.223.255.255</i>	<i>/13</i>
Customer #7	200,000	<i>15.224.0.0 to 15.227.255.255</i>	<i>/14</i>
Customer #8	130,000	<i>15.228.0.0 to 15.229.255.255</i>	<i>/15</i>
Customer #9	100,000	<i>15.230.0.0 to 15.231.255.255</i>	<i>/15</i>

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
				16-31		16-19	8-9
					16-23	20-23	10-11
						24-27	12-13
					24-31	28-31	14-15
			32-63	32-47		32-35	16-17
					32-39	36-39	18-19
						40-43	20-21
					40-47	44-47	22-23
				48-63		48-51	24-25
					48-55	52-55	26-27
						56-59	28-29
					56-63	60-63	30-31
		64-127	64-95	64-79	64-71	64-67	32-33
						68-71	34-35
					72-79	72-75	36-37
						76-79	38-39
			80-95	80-87	80-83	84-87	40-41
						88-91	42-43
					88-95	92-95	44-45
						96-99	46-47
			96-127	96-103	96-103	100-103	48-49
						104-107	50-51
				104-111	104-111	108-111	52-53
						112-115	54-55
				112-127	112-119	116-119	56-57
						120-123	58-59
					120-127	124-127	60-61
						128-131	62-63
	128-255	128-191	128-159	128-143	128-135	128-131	64-65
						132-135	66-67
					136-143	136-139	68-69
						140-143	70-71
			144-159	144-151	144-147	148-151	72-73
						152-155	74-75
				152-159	152-155	156-159	76-77
						160-163	78-79
			160-191	160-175	160-167	164-167	80-81
						168-171	82-83
				168-175	168-171	172-175	84-85
						176-179	86-87
				176-191	176-183	180-183	88-89
						184-187	90-91
				184-191	184-187	188-191	92-93
						192-195	94-95
		192-255	192-223	192-207	192-199	196-199	96-97
						200-203	98-99
				200-207	200-207	204-207	100-101
						208-211	102-103
			208-223	208-215	208-211	212-215	104-105
						216-219	106-107
				216-223	216-219	220-223	108-109
						224-227	110-111
			224-255	224-231	224-231	228-231	112-113
						232-235	114-115
				232-239	232-235	236-239	116-117
						240-243	118-119
				240-247	240-247	244-247	120-121
						248-251	122-123
				248-255	248-251	252-255	124-125
							126-127

VLSM

with

Class A and B Addresses

Sample Problem 35

Part 2 of 3

Customer #5 has a total of 524,288 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for their different clients. At this stage of the problem you are creating sub-subnets with the third octet of the IP address.

ISP Addresses 15.208.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Client #1	7,500	<i>15.208.0.0 to 15.208.31.255</i>	<i>/19</i>
Client #2	5,000	<i>15.208.32.0 to 15.208.63.255</i>	<i>/19</i>
Client #3	4,500	<i>15.208.64.0 to 15.208.95.255</i>	<i>/19</i>
Client #4	2,000	<i>15.208.96.0 to 15.208.103.255</i>	<i>/21</i>
Client #5	1,450	<i>15.208.104.0 to 15.208.111.255</i>	<i>/21</i>
Client #6	1,150	<i>15.208.112.0 to 15.208.119.255</i>	<i>/21</i>
Client #7	900	<i>15.208.120.0 to 15.208.123.255</i>	<i>/22</i>
Client #8	750	<i>15.208.124.0 to 15.208.127.255</i>	<i>/22</i>
Client #9	450	<i>15.208.128.0 to 15.208.129.255</i>	<i>/23</i>

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
						8-11	4-5
						12-15	6-7
						16-19	8-9
				16-31	16-23	20-23	10-11
					24-31	28-31	12-13
						32-35	14-15
						36-39	16-17
						40-43	18-19
		64-127	32-63	32-47	40-47	44-47	20-21
					48-55	52-55	22-23
					56-63	60-63	24-25
						64-67	26-27
						68-71	28-29
			64-95	64-79	72-79	76-79	30-31
					80-87	84-87	32-33
					88-95	92-95	34-35
						96-99	36-37
						100-103	38-39
			96-127	96-111	104-111	108-111	40-41
					112-119	116-119	42-43
				112-127	120-127	124-127	44-45
						128-129	46-47
						130-131	48-49
		128-255	128-191	128-143	128-135	132-135	50-51
					136-143	140-143	52-53
				144-159	144-151	148-151	54-55
					152-159	156-159	56-57
				160-175	160-167	164-167	58-59
					168-175	172-175	60-61
				176-191	176-183	180-183	62-63
					184-191	188-191	64-65
			192-255	192-207	192-199	196-199	66-67
					200-207	204-207	68-69
				208-223	208-215	212-215	70-71
					216-223	220-223	72-73
				224-239	224-231	228-231	74-75
					232-239	236-239	76-77
				240-255	240-247	244-247	78-79
					248-255	252-255	80-81
							82-83
							84-85
							86-87
							88-89
							90-91
							92-93
							94-95
							96-97
							98-99
							100-101
							102-103
							104-105
							106-107
							108-109
							110-111
							112-113
							114-115
							116-117
							118-119
							120-121
							122-123
							124-125
							126-127
							128-129
							130-131
							132-133
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							136-137
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							140-141
							142-143
							144-145
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							152-153
							154-155
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							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
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							192-193
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							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Sample Problem 35

Part 3 of 3

Client #8 has a total of 1,024 addresses. Use the **Class C** address chart to break down the sub-subnetwork addresses for their different branch offices. At this stage of the problem you are creating sub-subnets with the fourth octet of the IP address.

ISP Addresses 15.208.124.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Branch #1	100	<i>15.208.124.0 to 15.208.124.127</i>	<i>/ 25</i>
Branch #2	55	<i>15.208.124.128 to 15.208.124.191</i>	<i>/ 26</i>
Branch #3	25	<i>15.208.124.192 to 15.208.124.223</i>	<i>/ 27</i>
Branch #4	6	<i>15.208.124.224 to 15.208.124.231</i>	<i>/ 29</i>
Branch #5	4	<i>15.208.124.232 to 15.208.124.239</i>	<i>/ 29</i>
Branch #6	2	<i>15.208.124.240 to 15.208.124.243</i>	<i>/ 30</i>
Branch #7	2	<i>15.208.124.244 to 15.208.124.247</i>	<i>/ 30</i>
Branch #8	2	<i>15.208.124.248 to 15.208.124.251</i>	<i>/ 30</i>
Branch #9	2	<i>15.208.124.252 to 15.208.124.255</i>	<i>/ 30</i>

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30		
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts		
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3		
					4-7	4-7		
				16-31	8-15	8-11		
					12-15	12-15		
			32-63	32-47	16-23	16-19		
					20-23	20-23		
				48-63	24-31	24-27		
					28-31	28-31		
				64-127		64-79	32-39	32-35
							36-39	36-39
		64-95	80-95	40-47	40-43			
				44-47	44-47			
			96-111	48-55	48-51			
				52-55	52-55			
		96-127	112-127	56-63	56-59			
				60-63	60-63			
			128-255		128-143	64-71	64-67	
						68-71	68-71	
		128-159			72-79	72-75	72-75	
						76-79	76-79	
					144-159	80-87	80-83	
						84-87	84-87	
		160-191			176-191	88-95	88-91	
						92-95	92-95	
					192-255	192-207	96-103	96-99
							100-103	100-103
	224-255	208-223				104-111	104-107	
						108-111	108-111	
		240-255	112-119	112-115				
			116-119	116-119				
	192-255		192-207	120-123	120-123			
				124-127	124-127			
			192-223	208-215	128-131	128-131		
					132-135	132-135		
				224-239	136-143	136-139		
					140-143	140-143		
			240-255	248-255	144-147	144-147		
					148-151	148-151		
				252-255	152-155	152-155		
					156-159	156-159		
			128-255		160-175	160-167	160-163	
						164-167	164-167	
	176-191	168-175			168-171			
		172-175			172-175			
		192-207			176-179	176-179		
					180-183	180-183		
	224-239	208-215			184-187	184-187		
					188-191	188-191		
		240-255			192-199	192-195		
					196-199	196-199		
	192-255				192-207	200-207	200-203	
						204-207	204-207	
			208-223	216-223	208-211	208-211		
					212-215	212-215		
				224-239	216-219	216-219		
					220-223	220-223		
			240-255	224-231	224-227	224-227		
					228-231	228-231		
				248-255	232-235	232-235		
					236-239	236-239		
			128-255		224-239	240-247	240-243	
						244-247	244-247	
	240-255	248-255			248-251	248-251		
					252-255	252-255		

VLSM

with

Class A and B Addresses

Problem 36

Part 1 of 3

The company you are working for is using the IP address 110.0.0.0 sub-subneted for multiple offices around the world. Use the **Class A** address chart to break down the sub-subnetwork addresses for the different offices.

At this stage of the problem you are creating sub-subnets with the third octet of the IP address. Remember which octet of the IP address you are working in.

Company Address 110.0.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Moskva	3,050,000		
New York	1,540,000		
St. Petersburg	1,075,000		
London	975,000		
Ekaterinoburg	525,000		
Munchen	450,000		
Napoli	150,000		
Birmingham	130,000		
Rotterdam	95,000		

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0 - 127	0 - 63	0 - 31	0 - 15	0 - 7	0 - 3	0 - 1
							2 - 3
							4 - 5
							6 - 7
							8 - 9
							10 - 11
							12 - 13
							14 - 15
							16 - 17
							18 - 19
							20 - 21
							22 - 23
							24 - 25
							26 - 27
							28 - 29
							30 - 31
							32 - 33
							34 - 35
							36 - 37
							38 - 39
							40 - 41
							42 - 43
							44 - 45
							46 - 47
							48 - 49
							50 - 51
							52 - 53
							54 - 55
							56 - 57
							58 - 59
							60 - 61
							62 - 63
							64 - 65
							66 - 67
							68 - 69
							70 - 71
							72 - 73
							74 - 75
							76 - 77
							78 - 79
							80 - 81
							82 - 83
							84 - 85
							86 - 87
							88 - 89
							90 - 91
							92 - 93
							94 - 95
							96 - 97
							98 - 99
							100 - 101
							102 - 103
							104 - 105
							106 - 107
							108 - 109
							110 - 111
							112 - 113
							114 - 115
							116 - 117
							118 - 119
							120 - 121
							122 - 123
							124 - 125
							126 - 127
							128 - 129
							130 - 131
							132 - 133
							134 - 135
							136 - 137
							138 - 139
							140 - 141
							142 - 143
							144 - 145
							146 - 147
							148 - 149
							150 - 151
							152 - 153
							154 - 155
							156 - 157
							158 - 159
							160 - 161
							162 - 163
							164 - 165
							166 - 167
							168 - 169
							170 - 171
							172 - 173
							174 - 175
							176 - 177
							178 - 179
							180 - 181
							182 - 183
							184 - 185
							186 - 187
							188 - 189
							190 - 191
							192 - 193
							194 - 195
							196 - 197
							198 - 199
							200 - 201
							202 - 203
							204 - 205
							206 - 207
							208 - 209
							210 - 211
							212 - 213
							214 - 215
							216 - 217
							218 - 219
							220 - 221
							222 - 223
							224 - 225
							226 - 227
							228 - 229
							230 - 231
							232 - 233
							234 - 235
							236 - 237
							238 - 239
							240 - 241
							242 - 243
							244 - 245
							246 - 247
							248 - 249
							250 - 251
							252 - 253
							254 - 255

VLSM

with

Class A and B Addresses

Problem 36

Part 2 of 3

London in part 1 of this problem has been given 1,048,576 hosts, with the address range of 110.128.0.0 to 110.143.255.255 /12 (255.240.0.0).

Based on the information below supply the required address ranges and subnet masks for each office. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different areas of the network.

London
Address Range 110.128.0.0 to 110.143.255.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Office #1	6,450		
Office #2	3,780		
Office #3	2,750		
Office #4	2,000		
Office #5	1,000		
Office #6	845		
Office #7	500		
Office #8	450		
Office #9	300		

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16	/17	/18	/19		/23				
255.255.0.0 65,536 Hosts	255.255.128.0 32,768 Hosts	255.255.192.0 16,384 Hosts	255.255.224.0 8,192 Hosts			255.255.254.0 512 Hosts			
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1		
						4-7	2-3		
				16-31	8-15	8-11	4-5		
					16-23	16-19	12-15	6-7	
						20-23	18-19	8-9	
						24-27	20-21	10-11	
			32-63			32-39	24-27	22-23	12-13
				28-31	24-25		14-15		
				32-35	26-27		16-17		
				40-47	36-39	30-31	18-19		
					40-43	32-33	20-21		
					44-47	34-35	22-23		
					48-55	48-51	36-37	24-25	
						52-55	38-39	26-27	
						56-59	40-41	28-29	
		64-127	64-79	64-71	60-63	42-43	30-31		
					64-67	44-45	32-33		
					68-71	46-47	34-35		
				72-79	72-75	48-49	36-37		
					76-79	50-51	38-39		
					78-79	52-53	40-41		
			80-95	80-87	80-83	54-55	42-43		
					84-87	56-57	44-45		
					88-91	58-59	46-47		
				96-111	92-95	60-61	48-49		
					96-99	62-63	50-51		
			96-127		96-103	100-103	64-65	52-53	
						104-107	66-67	54-55	
				104-111	108-111	68-69	56-57		
					112-127	112-119	112-115	70-71	58-59
							116-119	72-73	60-61
	120-127	120-123	74-75	62-63					
		124-127	76-77	64-65					
		128-255	128-191	128-159		128-143	128-135	128-131	66-67
	136-143				132-135		68-69		
	144-159				140-143	136-139	70-71		
					144-147	140-141	72-73		
					148-151	142-143	74-75		
					152-155	144-145	76-77		
					156-159	146-147	78-79		
					160-161	148-149	80-81		
	160-191			160-175	160-163	150-151	82-83		
					164-167	152-153	84-85		
					168-171	154-155	86-87		
					172-175	156-157	88-89		
				176-191	176-179	158-159	90-91		
					180-183	160-161	92-93		
					184-187	162-163	94-95		
					188-191	164-165	96-97		
	192-255			192-223	192-207	192-195	166-167	98-99	
						196-199	168-169	100-101	
						200-203	170-171	102-103	
						204-207	172-173	104-105	
					208-223	208-211	174-175	106-107	
						212-215	176-177	108-109	
						216-219	178-179	110-111	
						220-223	180-181	112-113	
				224-255	224-239	224-227	182-183	114-115	
						228-231	184-185	116-117	
						232-235	186-187	118-119	
						236-239	188-189	120-121	
						240-243	190-191	122-123	
						244-247	192-193	124-125	
240-255					240-247	248-251	194-195	126-127	
						252-255	196-197	128-129	
						198-199	130-131		
						200-201	132-133		
			248-255		248-249	202-203	134-135		
					250-251	204-205	136-137		
					252-253	206-207	138-139		
					254-255	208-209	140-141		
						210-211	142-143		

VLSM

with

Class A and B Addresses

Problem 36

Part 3 of 3

Office #7 in part 2 of this problem has been given 512 hosts, with the address range of 110.128.80.0 / 23 (255.255.254.0).

Based on the information below supply the required address ranges and subnet masks for each school area. Use the **Class C** address chart to break down the sub-subnetwork addresses for the different areas of the network. **Hint:** Another way to look at this problem is to see that with the third octet range of 80 to 81 you have access to 2 groups of 255 addresses or two Class C VLSM charts.

Office #7
Address Range 110.128.80.0 to 110.128.81.255

Customer Name	Number of Addresses	Address Range	CIDR
1st Floor	125		
2nd Floor	75		
5th Floor	50		
8th Floor	45		
4th Floor	30		
Basement	14		
7th Floor	12		
3rd Floor	6		
6th Floor	4		

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-103	96-99	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

VLSM

with

Class A and B Addresses

Problem 37

Part 1 of 3

The school system you are working for is using the private address of 172.32.0.0 to subnet the entire district. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different schools and offices.

At this stage of the problem you are creating sub-subnets with the third octet of the IP address. Remember which octet of the IP address you are working in.

School System Address 172.32.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
North High	2,400		
South High	2,000		
North Middle	1,200		
South Middle	1,000		
Central Elem.	550		
Southern Elem.	475		
Eastern Elem.	450		
Central Office	400		
Western Elem.	300		

Numeric Method
Show your work for Problem 36 in the space below.

Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1

Hosts

Address Ranges

CIDR

VLSM

with

Class A and B Addresses

Problem 37

Part 2 of 3

Eastern Elementary has been given 512 hosts, with the address range of 172.32.42.0 / 21 (255.255.248.0).

Based on the information below supply the required address ranges and subnet masks for each school area.

Hint:

Another way to look at this problem is to see that with the third octet range of 42 to 43 you have access to 2 groups of 255 addresses (172.32.42.0 and 172.32.43.0). Think in terms of having two Class C address ranges.

Eastern Elementary School
Address Range 172.32.42.0 to 172.32.43.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Students	250		
Printers	45		
Staff	40		
Network Devices	25		
Administrative	12		

Numeric Method

Show your work for Problem 36 in the space below.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
		128	64	32	16	8	4	2	1	- Binary values

172. 32. **42** . 0 0 0 0 0 0 0 0 0

Hosts

Address Ranges

CIDR

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
		128	64	32	16	8	4	2	1	- Binary values

172. 32. **43** . 0 0 0 0 0 0 0 0 0

Hosts

Address Ranges

CIDR

VLSM

with

Class A and B Addresses

Problem 37

Part 3 of 3

South High in part 1 of this problem has been given 2,048 hosts, with the address range of 172.32.16.0 / 21 (255.255.248.0).

Based on the information below supply the required address ranges and subnet masks for each school area.

South High School
Address Range 172.32.16.0 to 172.32.23.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Students	1,000		
Network Devices	250		
Printers	200		
Staff	150		
Administrative	50		

Numeric Method

Show your work for Problem 36 in the space below.

Reminder - Keep within the South High School address range.

VLSM

with

Class A and B Addresses

Problem 38

Part 1 of 4

Use the **Class A** address chart to break down the address for different business customers by country. At this stage of this problem you are creating subnets in the second octet of the IP address.

Addresses 75.0.0.0

Customer Name	Number of Addresses	Address Range	CIDR
United States	6.5 million		
China	4 million		
Japan	1 million		
Germany	500,000		
Russia	455,000		
Australia	450,000		
Brazil	125,000		
Canda	90,000		
Denmark	88,000		

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31	16-23	8-11	4-5
						12-15	6-7
					24-31	16-19	8-9
						20-23	10-11
			32-63	32-47	32-39	24-27	12-13
						28-31	14-15
					40-47	32-35	16-17
						36-39	18-19
				48-63	48-55	40-43	20-21
						44-47	22-23
					56-63	48-51	24-25
						52-55	26-27
		64-127	64-95	64-79	64-71	56-59	28-29
						60-63	30-31
					72-79	64-67	32-33
						68-71	34-35
				80-95	80-87	72-75	36-37
						76-79	38-39
					88-95	80-83	40-41
						84-87	42-43
			96-127	96-111	96-103	88-91	44-45
						92-95	46-47
					104-111	96-99	48-49
						100-103	50-51
				112-127	112-119	104-107	52-53
						108-111	54-55
					120-127	112-115	56-57
						116-119	58-59
	128-255	128-191	128-159	128-143	128-135	120-123	60-61
						124-127	62-63
				136-143	136-139	128-131	64-65
						132-135	66-67
			144-159	144-151	144-147	136-137	68-69
						140-143	70-71
				152-159	152-155	144-145	72-73
						148-151	74-75
			160-191	160-175	160-163	152-153	76-77
						156-157	78-79
				168-175	168-171	160-161	80-81
						164-167	82-83
				176-183	176-179	168-169	84-85
						172-175	86-87
			176-191	184-191	184-187	176-177	88-89
						180-183	90-91
		192-255	192-207	192-199	192-195	188-191	92-93
						192-199	94-95
			200-207	200-203	200-203	196-199	96-97
						204-207	98-99
			208-215	208-211	208-211	204-207	100-101
						208-211	102-103
			216-223	216-219	216-219	212-215	104-105
						216-219	106-107
			224-231	224-227	224-227	220-223	108-109
						224-227	110-111
			228-231	228-231	228-231	228-231	112-113
						232-235	114-115
			232-239	232-235	232-235	236-239	116-117
						240-243	118-119
			240-247	240-243	240-243	244-247	120-121
						248-251	122-123
			248-255	248-251	248-251	252-255	124-125
						252-255	126-127
				252-255	252-255	252-255	128-129
						252-255	130-131
						252-255	132-133
						252-255	134-135
						252-255	136-137
						252-255	138-139
						252-255	140-141
						252-255	142-143
						252-255	144-145
						252-255	146-147
						252-255	148-149
						252-255	150-151
						252-255	152-153
						252-255	154-155
						252-255	156-157
						252-255	158-159
						252-255	160-161
						252-255	162-163
						252-255	164-165
						252-255	166-167
						252-255	168-169
						252-255	170-171
						252-255	172-173
						252-255	174-175
						252-255	176-177
						252-255	178-179
						252-255	180-181
						252-255	182-183
						252-255	184-185
						252-255	186-187
						252-255	188-189
						252-255	190-191
						252-255	192-193
						252-255	194-195
						252-255	196-197
						252-255	198-199
						252-255	200-201
						252-255	202-203
						252-255	204-205
						252-255	206-207
						252-255	208-209
						252-255	210-211
						252-255	212-213
						252-255	214-215
						252-255	216-217
						252-255	218-219
						252-255	220-221
						252-255	222-223
						252-255	224-225
						252-255	226-227
						252-255	228-229
						252-255	230-231
						252-255	232-233
						252-255	234-235
						252-255	236-237
						252-255	238-239
						252-255	240-241
						252-255	242-243
						252-255	244-245
						252-255	246-247
						252-255	248-249
						252-255	250-251
						252-255	252-253
						252-255	254-255

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 2 of 4

The United States customers have a total of 8,388,608 addresses. Use the **Class A** address chart to break down the sub-subnetwork addresses for their different areas. At this stage of this problem you are creating sub-subnets in the second octet of the IP address.

Addresses Range: 75.0.0.0 to 75.127.255.255

Customer Name	Number of Addresses	Address Range	CIDR
Client #1	1,950,000		
Client #2	1,000,000		
Client #3	950,000		
Client #4	700,000		
Client #5	550,000		
Client #6	500,000		
Client #7	450,000		

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					4-7	4-7	2-3
				16-31	8-15	8-11	4-5
					16-23	16-19	6-7
			32-63	24-31	20-23	20-23	8-9
					24-27	24-27	10-11
				32-47	28-31	28-31	12-13
					32-35	32-35	14-15
					36-39	36-39	16-17
					40-43	40-43	18-19
			48-63	40-47	44-47	44-47	20-21
					48-51	48-51	22-23
				56-63	52-55	52-55	24-25
					56-59	56-59	26-27
		64-127	64-95	64-79	60-63	60-63	28-29
					64-67	64-67	30-31
				80-95	68-71	68-71	32-33
					72-75	72-75	34-35
			96-127	88-95	76-79	76-79	36-37
					80-83	80-83	38-39
				96-111	84-87	84-87	40-41
					88-91	88-91	42-43
			128-159	128-143	92-95	92-95	44-45
					96-99	96-99	46-47
				144-159	100-103	100-103	48-49
					104-107	104-107	50-51
		128-255	160-191	160-175	108-111	108-111	52-53
					112-115	112-115	54-55
				176-191	116-119	116-119	56-57
					120-123	120-123	58-59
			192-223	192-207	124-127	124-127	60-61
					128-131	128-131	62-63
				208-223	132-135	132-135	64-65
					136-139	136-139	66-67
			224-255	224-239	140-143	140-143	68-69
					144-147	144-147	70-71
				240-255	148-151	148-151	72-73
					152-155	152-155	74-75
			248-255	248-251	156-159	156-159	76-77
					160-163	160-163	78-79
				252-255	164-167	164-167	80-81
					168-171	168-171	82-83
					172-175	172-175	84-85
					176-179	176-179	86-87
					180-183	180-183	88-89
					184-187	184-187	90-91
					188-191	188-191	92-93
					192-195	192-195	94-95
					196-199	196-199	96-97
					200-203	200-203	98-99
					204-207	204-207	100-101
					208-211	208-211	102-103
					212-215	212-215	104-105
					216-219	216-219	106-107
					220-223	220-223	108-109
					224-227	224-227	110-111
					228-231	228-231	112-113
					232-235	232-235	114-115
					236-239	236-239	116-117
					240-243	240-243	118-119
					244-247	244-247	120-121
					248-251	248-251	122-123
					252-255	252-255	124-125
							126-127
							128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
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							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 3 of 4

Client #7 has a total of 524,288 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for their different clients. At this stage of this problem you are creating sub-subnets in the third or forth octet of the IP address.

Hint: Another way to look at this problem is to see that with the second octet range of 104 to 111 you have access to 8 groups of 65,536 addresses or 8 Class B VLSM charts.

ISP Addresses 75.104.0.0 to 75.111.255.255

Customer Name	Number of Addresses	Address Range	CIDR
Office #1	60,000		
Office #2	45,000		
Office #3	30,000		
Office #4	24,000		
Office #5	15,000		
Office #6	10,000		
Office #7	8,000		
Office #8	2,000		
Office #9	1,000		

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
							2-3
							4-5
							6-7
							8-9
							10-11
							12-13
							14-15
							16-17
							18-19
							20-21
							22-23
							24-25
							26-27
							28-29
							30-31
							32-33
							34-35
							36-37
							38-39
							40-41
							42-43
							44-45
							46-47
							48-49
							50-51
							52-53
							54-55
							56-57
							58-59
							60-61
							62-63
							64-65
							66-67
							68-69
							70-71
							72-73
							74-75
							76-77
							78-79
							80-81
							82-83
							84-85
							86-87
							88-89
							90-91
							92-93
							94-95
							96-97
							98-99
							100-101
							102-103
							104-105
							106-107
							108-109
							110-111
							112-113
							114-115
							116-117
							118-119
							120-121
							122-123
							124-125
							126-127
							128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 4 of 4

Office #7 from part 3 of 4 has a total of 8,192 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different branch offices. At this stage of this problem you are creating sub-subnets in the third octet of the IP address.

Hint: Remember that the range of this problem is between 128 and 159 in the third octet. Your subnetting will start in the middle of the chart not at the top for this range.

ISP Addresses 75.107.128.0 to 75.107.159.255

Customer Name	Number of Addresses	Address Range	CIDR
Branch #1	4,000		
Branch #2	2,000		
Branch #3	1,000		
Branch #4	500		
Branch #5	450		

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
			16-31	16-31	16-23	16-19	8-9
						20-23	10-11
					24-31	24-27	12-13
						28-31	14-15
			32-63	32-47	32-39	32-35	16-17
						36-39	18-19
					40-47	40-43	20-21
						44-47	22-23
				48-63	48-55	48-51	24-25
						52-55	26-27
					56-63	56-59	28-29
						60-63	30-31
		64-127	64-95	64-79	64-71	64-67	32-33
						68-71	34-35
					72-79	72-75	36-37
						76-79	38-39
			80-95	80-87	80-83	80-81	40-41
						84-87	42-43
					88-95	88-91	44-45
						92-95	46-47
			96-127	96-111	96-103	96-99	48-49
						100-103	50-51
					104-111	104-107	52-53
						108-111	54-55
				112-127	112-119	112-115	56-57
						116-119	58-59
					120-127	120-123	60-61
						124-127	62-63
	128-255	128-191	See Hint 128-159	128-143	128-135	128-131	64-65
						132-135	66-67
					136-143	136-139	68-69
						140-143	70-71
				144-159	144-151	144-147	72-73
						148-151	74-75
					152-159	152-155	76-77
						156-159	78-79
			160-191	160-175	160-167	160-163	80-81
						164-167	82-83
					168-175	168-171	84-85
						172-175	86-87
				176-191	176-183	176-179	88-89
						180-183	90-91
					184-191	184-187	92-93
						188-191	94-95
		192-255	192-223	192-207	192-199	192-195	96-97
						196-199	98-99
					200-207	200-203	100-101
						204-207	102-103
				208-223	208-215	208-211	104-105
						212-215	106-107
					216-223	216-219	108-109
						220-223	110-111
			224-255	224-239	224-231	224-227	112-113
						228-231	114-115
					232-239	232-235	116-117
						236-239	118-119
				240-255	240-247	240-243	120-121
						244-247	122-123
					248-255	248-251	124-125
						252-255	126-127

Reference Charts and Support Materials

Class A Addresses
VLSM Chart 8-15 Bits (2nd octet)

#	#	#10	#11	#12	#13	#14	#15
0-27.77% male	0-27.77% male	0-27.77% male	0-27.77% male	0-27.77% male	0-27.77% male	0-27.77% male	0-27.77% male
				0-51			
				6-75	7-77	7-77	
			0-51	18-23	21-23	21-23	
			18-21	24-31	24-31	24-31	
			32-47	32-47	32-47	32-47	
			32-49	48-55	48-55	48-55	
			48-55	56-63	56-63	56-63	
			64-71	72-79	72-79	72-79	
			80-87	88-95	88-95	88-95	
			96-101	102-109	102-109	102-109	
			110-117	118-125	118-125	118-125	
			126-133	134-141	134-141	134-141	
			142-149	150-157	150-157	150-157	
			158-165	166-173	166-173	166-173	
			174-181	182-189	182-189	182-189	
			190-197	198-205	198-205	198-205	
			206-213	214-221	214-221	214-221	
			222-229	230-237	230-237	230-237	
			238-245	246-253	246-253	246-253	
			254-261	262-269	262-269	262-269	
			270-277	278-285	278-285	278-285	
			286-293	294-301	294-301	294-301	
			302-309	310-317	310-317	310-317	
			318-325	326-333	326-333	326-333	
			334-341	342-349	342-349	342-349	
			350-357	358-365	358-365	358-365	
			366-373	374-381	374-381	374-381	
			382-389	390-397	390-397	390-397	
			398-405	406-413	406-413	406-413	
			414-421	422-429	422-429	422-429	
			430-437	438-445	438-445	438-445	
			446-453	454-461	454-461	454-461	
			462-469	470-477	470-477	470-477	
			478-485	486-493	486-493	486-493	
			494-501	502-509	502-509	502-509	
			510-517	518-525	518-525	518-525	
			526-533	534-541	534-541	534-541	
			542-549	550-557	550-557	550-557	
			558-565	566-573	566-573	566-573	
			574-581	582-589	582-589	582-589	
			590-597	598-605	598-605	598-605	
			606-613	614-621	614-621	614-621	
			622-629	630-637	630-637	630-637	
			638-645	646-653	646-653	646-653	
			654-661	662-669	662-669	662-669	
			670-677	678-685	678-685	678-685	
			686-693	694-701	694-701	694-701	
			702-709	710-717	710-717	710-717	
			718-725	726-733	726-733	726-733	
			734-741	742-749	742-749	742-749	
			750-757	758-765	758-765	758-765	
			766-773	774-781	774-781	774-781	
			782-789	790-797	790-797	790-797	
			798-805	806-813	806-813	806-813	
			814-821	822-829	822-829	822-829	
			830-837	838-845	838-845	838-845	
			846-853	854-861	854-861	854-861	
			862-869	870-877	870-877	870-877	
			878-885	886-893	886-893	886-893	
			894-901	902-909	902-909	902-909	
			910-917	918-925	918-925	918-925	
			926-933	934-941	934-941	934-941	
			942-949	950-957	950-957	950-957	
			958-965	966-973	966-973	966-973	
			974-981	982-989	982-989	982-989	
			990-997	998-1005	998-1005	998-1005	
			1006-1013	1014-1021	1014-1021	1014-1021	
			1022-1029	1030-1037	1030-1037	1030-1037	
			1038-1045	1046-1053	1046-1053	1046-1053	
			1054-1061	1062-1069	1062-1069	1062-1069	
			1070-1077	1078-1085	1078-1085	1078-1085	
			1086-1093	1094-1101	1094-1101	1094-1101	
			1102-1109	1110-1117	1110-1117	1110-1117	
			1118-1125	1126-1133	1126-1133	1126-1133	
			1134-1141	1142-1149	1142-1149	1142-1149	
			1150-1157	1158-1165	1158-1165	1158-1165	
			1166-1173	1174-1181	1174-1181	1174-1181	
			1182-1189	1190-1197	1190-1197	1190-1197	
			1198-1205	1206-1213	1206-1213	1206-1213	
			1214-1221	1222-1229	1222-1229	1222-1229	
			1230-1237	1238-1245	1238-1245	1238-1245	
			1246-1253	1254-1261	1254-1261	1254-1261	
			1262-1269	1270-1277	1270-1277	1270-1277	
			1278-1285	1286-1293	1286-1293	1286-1293	
			1294-1301	1302-1309	1302-1309	1302-1309	
			1310-1317	1318-1325	1318-1325	1318-1325	
			1326-1333	1334-1341	1334-1341	1334-1341	
			1342-1349	1350-1357	1350-1357	1350-1357	
			1358-1365	1366-1373	1366-1373	1366-1373	
			1374-1381	1382-1389	1382-1389	1382-1389	
			1390-1397	1398-1405	1398-1405	1398-1405	
			1406-1413	1414-1421	1414-1421	1414-1421	
			1422-1429	1430-1437	1430-1437	1430-1437	
			1438-1445	1446-1453	1446-1453	1446-1453	
			1454-1461	1462-1469	1462-1469	1462-1469	
			1470-1477	1478-1485	1478-1485	1478-1485	
			1486-1493	1494-1501	1494-1501	1494-1501	
			1502-1509	1510-1517	1510-1517	1510-1517	
			1518-1525	1526-1533	1526-1533	1526-1533	
			1534-1541	1542-1549	1542-1549	1542-1549	
			1550-1557	1558-1565	1558-1565	1558-1565	
			1566-1573	1574-1581	1574-1581	1574-1581	
			1582-1589	1590-1597	1590-1597	1590-1597	
			1598-1605	1606-1613	1606-1613	1606-1613	
			1614-1621	1622-1629	1622-1629	1622-1629	
			1630-1637	1638-1645	1638-1645	1638-1645	
			1646-1653	1654-1661	1654-1661	1654-1661	
			1662-1669	1670-1677	1670-1677	1670-1677	
			1678-1685	1686-1693	1686-1693	1686-1693	
			1694-1701	1702-1709	1702-1709	1702-1709	
			1710-1717	1718-1725	1718-1725	1718-1725	
			1726-1733	1734-1741	1734-1741	1734-1741	
			1742-1749	1750-1757	1750-1757	1750-1757	
			1758-1765	1766-1773	1766-1773	1766-1773	
			1774-1781	1782-1789	1782-1789	1782-1789	
			1790-1797	1798-1805	1798-1805	1798-1805	
			1806-1813	1814-1821	1814-1821	1814-1821	
			1822-1829	1830-1837	1830-1837	1830-1837	
			1838-1845	1846-1853	1846-1853	1846-1853	
			1854-1861	1862-1869	1862-1869	1862-1869	
			1870-1877	1878-1885	1878-1885	1878-1885	
			1886-1893	1894-1901	1894-1901	1894-1901	
			1902-1909	1910-1917	1910-1917	1910-1917	
			1918-1925	1926-1933	1926-1933	1926-1933	
			1934-1941	1942-1949	1942-1949	1942-1949	
			1950-1957	1958-1965	1958-1965	1958-1965	
			1966-1973	1974-1981	1974-1981	1974-1981	
			1982-1989	1990-1997	1990-1997	1990-1997	
			1998-2005	2006-2013	2006-2013	2006-2013	
			2014-2021	2022-2029	2022-2029	2022-2029	
			2030-2037	2038-2045	2038-2045	2038-2045	
			2046-2053	2054-2061	2054-2061	2054-2061	
			2062-2069	2070-2077	2070-2077	2070-2077	
			2078-2085	2086-2093	2086-2093	2086-2093	
			2094-2101	2102-2109	2102-2109	2102-2109	
			2110-2117	2118-2125	2118-2125	2118-2125	
			2126-2133	2134-2141	2134-2141	2134-2141	
			2142-2149	2150-2157	2150-2157	2150-2157	
			2158-2165	2166-2173	2166-2173	2166-2173	
			2174-2181	2182-2189	2182-2189	2182-2189	
			2190-2197	2198-2205	2198-2205	2198-2205	
			2206-2213	2214-2221	2214-2221	2214-2221	
			2222-2229	2230-2237	2230-2237	2230-2237	
			2238-2245	2246-2253	2246-2253	2246-2253	
			2254-2261	2262-2269	2262-2269	2262-2269	
			2270-2277	2278-2285	2278-2285	2278-2285	
			2286-2293	2294-2301	2294-2301	2294-2301	
			2302-2309	2310-2317	2310-2317	2310-2317	
			2318-2325	2326-2333	2326-2333	2326-2333	
			2334-2341	2342-2349	2342-2349	2342-2349	
			2350-2357	2358-2365	2358-2365	2358-2365	
			2366-2373	2374-2381	2374-2381	2374-2381	
			2382-2389	2390-2397	2390-2397	2390-2397	
			2398-2405	2406-2413	2406-2413	2406-2413	
			2414-2421	2422-2429	2422-2429	2422-2429	
			2430-2437	2438-2445	2438-2445	2438-2445	
			2446-2453	2454-2461	2454-2461	2454-2461	
			2462-2469	2470-2477	2470-2477	2470-2477	
			2478-2485	2486-2493	2486-2493	2486-2493	
			2494-2501	2502-2509	2502-2509	2502-2509	
			2510-2517	2518-2525	2518-2525	2518-2525	
			2526-2533	2534-2541	2534-2541	2534-2541	
			2542-2549	2550-2557	2550-2557	2550-2557	
			2558-2565	2566-2573	2566-2573	2566-2573	
			2574-2581	2582-2589	2582-2589	2582-2589	
			2590-2597	2598-2605	2598-2605	2598-2605	
			2606-2613	2614-2621	2614-2621	2614-2621	
			2622-2629	2630-2637	2630-2637	2630-2637	
			2638-2645	2646-2653	2646-2653	2646-2653	

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

[illegible]

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

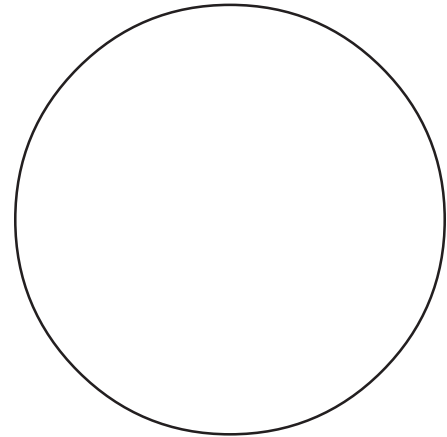
[illegible]

Visualizing Subnets Using The Circle Method

The circle method is another method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the different sections of the circle you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

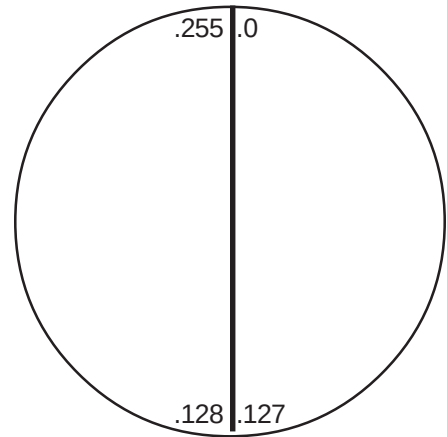
Start with a circle. The whole circle is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



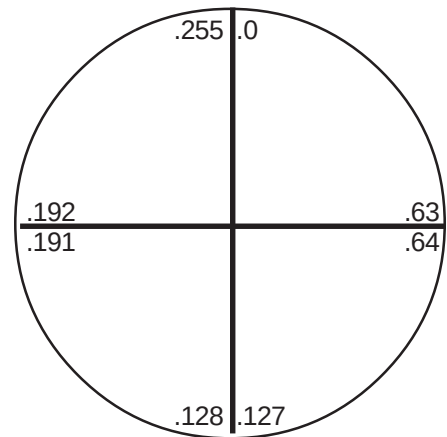
Split the circle in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



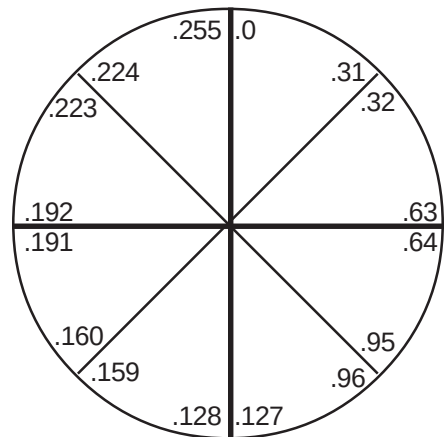
Divide the circle into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



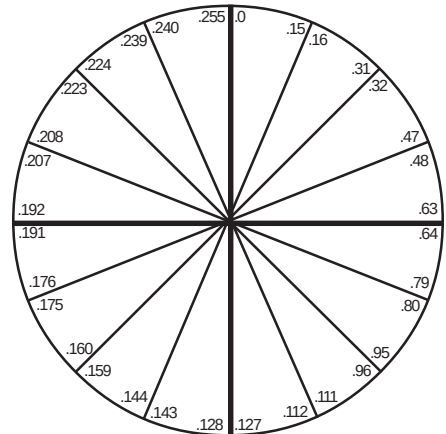
Split each quarter and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets



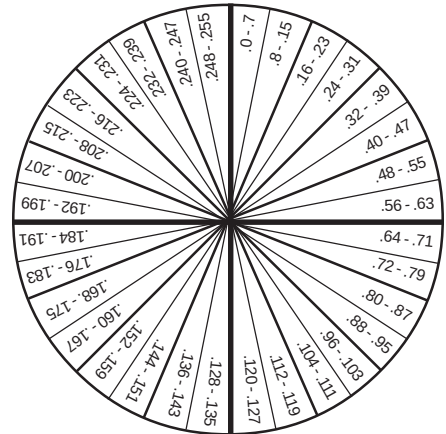
Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets



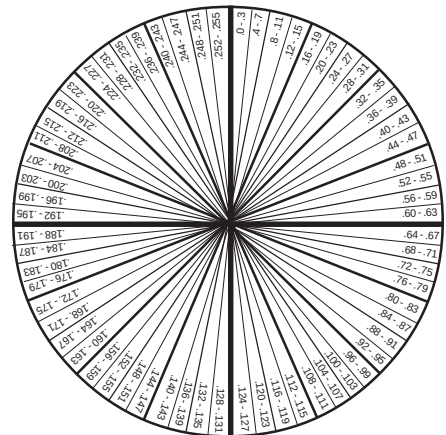
The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets



The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets

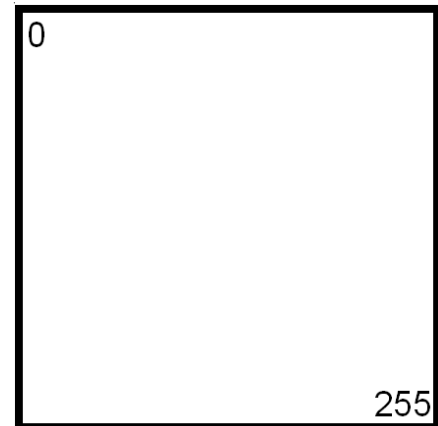


Visualizing Subnets Using The Box Method

The box method is a simple way to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

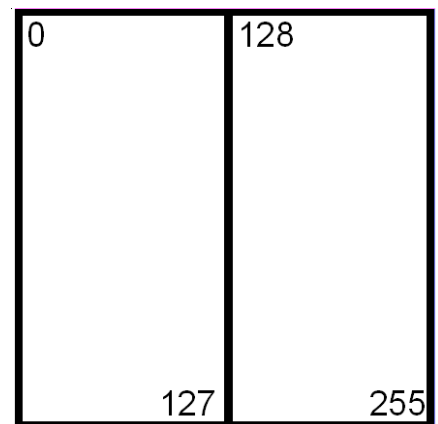
Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



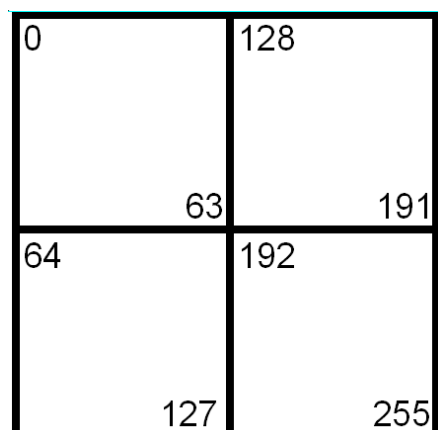
Split the box in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



Divide the box into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets

0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets

0	32	128	160
15	47	143	175
16	48	144	176
31	63	159	191
64	96	192	224
79	111	207	239
80	112	208	240
95	127	223	255

The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets

0	8	32	40	128	136	160	168
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
71	79	103	111	199	207	321	239
80	88	112	120	208	216	240	248
87	95	119	127	215	223	247	255

The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets

0	8	32	40	128	136	160	168
3	11	35	43	131	139	163	171
4	12	36	44	132	140	164	172
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
19	27	51	59	147	155	179	187
20	28	52	60	148	156	180	188
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
67	75	99	107	195	203	227	235
68	76	100	108	196	204	228	236
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
83	91	115	123	211	219	243	251
84	92	116	124	212	220	244	252
87	95	119	127	215	223	247	255

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
						16-19	8-9
						20-23	10-11
						24-27	12-13
						28-31	14-15
						32-35	16-17
						36-39	18-19
						40-43	20-21
						44-47	22-23
						48-51	24-25
						52-55	26-27
						56-59	28-29
						60-63	30-31
		64-127	32-63	32-47	32-39	32-35	32-33
						40-43	34-35
					48-55	48-51	36-37
						52-55	38-39
				48-63	56-63	56-59	40-41
						60-63	42-43
			64-95	64-79	64-71	64-67	44-45
						68-71	46-47
					72-79	72-75	48-49
						76-79	50-51
				80-95	80-87	80-83	52-53
						84-87	54-55
				96-111	88-95	88-91	56-57
						92-95	58-59
					96-103	96-99	60-61
						100-103	62-63
			96-127	104-111	104-107	104-107	64-65
						108-111	66-67
				112-127	112-119	112-115	68-69
						116-119	70-71
				120-127	120-123	120-123	72-73
						124-127	74-75
		128-255	128-159	128-143	128-135	128-131	76-77
						132-135	78-79
					136-143	136-139	80-81
						140-143	82-83
				144-159	144-151	144-147	84-85
						148-151	86-87
				160-175	160-167	160-167	88-89
					168-175	168-171	90-91
			160-191	176-191	176-183	176-179	92-93
						180-183	94-95
				184-191	184-187	184-187	96-97
						188-191	98-99
			192-255	192-207	192-199	192-195	100-101
						196-199	102-103
				200-207	200-203	200-203	104-105
						204-207	106-107
				208-215	208-211	208-211	108-109
						212-215	110-111
				216-223	216-219	216-219	112-113
						220-223	114-115
				224-231	224-227	224-227	116-117
						228-231	118-119
				232-239	232-235	232-235	120-121
						236-239	122-123
			224-255	240-247	240-243	240-243	124-125
						244-247	126-127
				248-255	248-251	248-251	128-129
						252-255	130-131

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
						8-11	4-5
						12-15	6-7
						16-19	8-9
						20-23	10-11
			16-31	16-31	16-23	16-19	12-13
					24-31	20-23	14-15
						24-27	16-17
						28-31	18-19
						32-35	20-21
						36-39	22-23
			32-63	32-47	32-39	32-35	24-25
					40-47	40-43	26-27
						44-47	28-29
						48-51	30-31
						52-55	32-33
						56-59	34-35
			48-63	48-55	56-63	56-59	36-37
						60-63	38-39
						64-67	40-41
						68-71	42-43
						72-75	44-45
						76-79	46-47
		64-127	64-95	64-79	64-71	64-67	48-49
					72-79	68-71	50-51
						72-75	52-53
						80-83	54-55
						84-87	56-57
						88-91	58-59
			80-95	80-87	88-95	88-91	60-61
						92-95	62-63
						96-99	64-65
						100-103	66-67
						104-107	68-69
						108-111	70-71
		96-127	96-111	96-103	100-103	100-103	72-73
					104-111	104-107	74-75
						108-111	76-77
						112-115	78-79
						116-119	80-81
						120-123	82-83
			112-127	112-119	120-127	124-127	84-85
						128-131	86-87
						132-135	88-89
						136-139	90-91
						140-143	92-93
						144-147	94-95
	128-255	128-191	128-159	128-143	128-135	128-131	96-97
					136-143	132-135	98-99
						136-139	100-101
						140-143	102-103
						144-147	104-105
						148-151	106-107
			144-159	144-151	148-151	148-149	108-109
					152-155	152-153	110-111
					156-159	156-157	112-113
						160-163	114-115
						164-167	116-117
						168-171	118-119
		160-191	160-175	160-167	164-167	164-165	120-121
					168-171	166-167	122-123
					172-175	172-173	124-125
					176-179	176-177	126-127
					180-183	180-181	128-129
					184-187	184-185	130-131
			176-191	176-183	188-191	188-189	132-133
					192-195	192-193	134-135
					196-199	196-197	136-137
					200-203	200-201	138-139
					204-207	204-205	140-141
					208-211	208-209	142-143
		192-255	192-223	192-207	204-207	204-207	144-145
					208-211	208-211	146-147
					212-215	212-213	148-149
					216-219	216-217	150-151
					220-223	220-221	152-153
					224-227	224-225	154-155
			224-255	224-231	228-231	228-229	156-157
					232-235	232-233	158-159
					236-239	236-237	160-161
					240-243	240-241	162-163
					244-247	244-245	164-165
					248-251	248-249	166-167
			240-255	240-247	252-255	252-253	168-169
						254-255	170-171
							172-173
							174-175
							176-177
							178-179

Class C Addresses

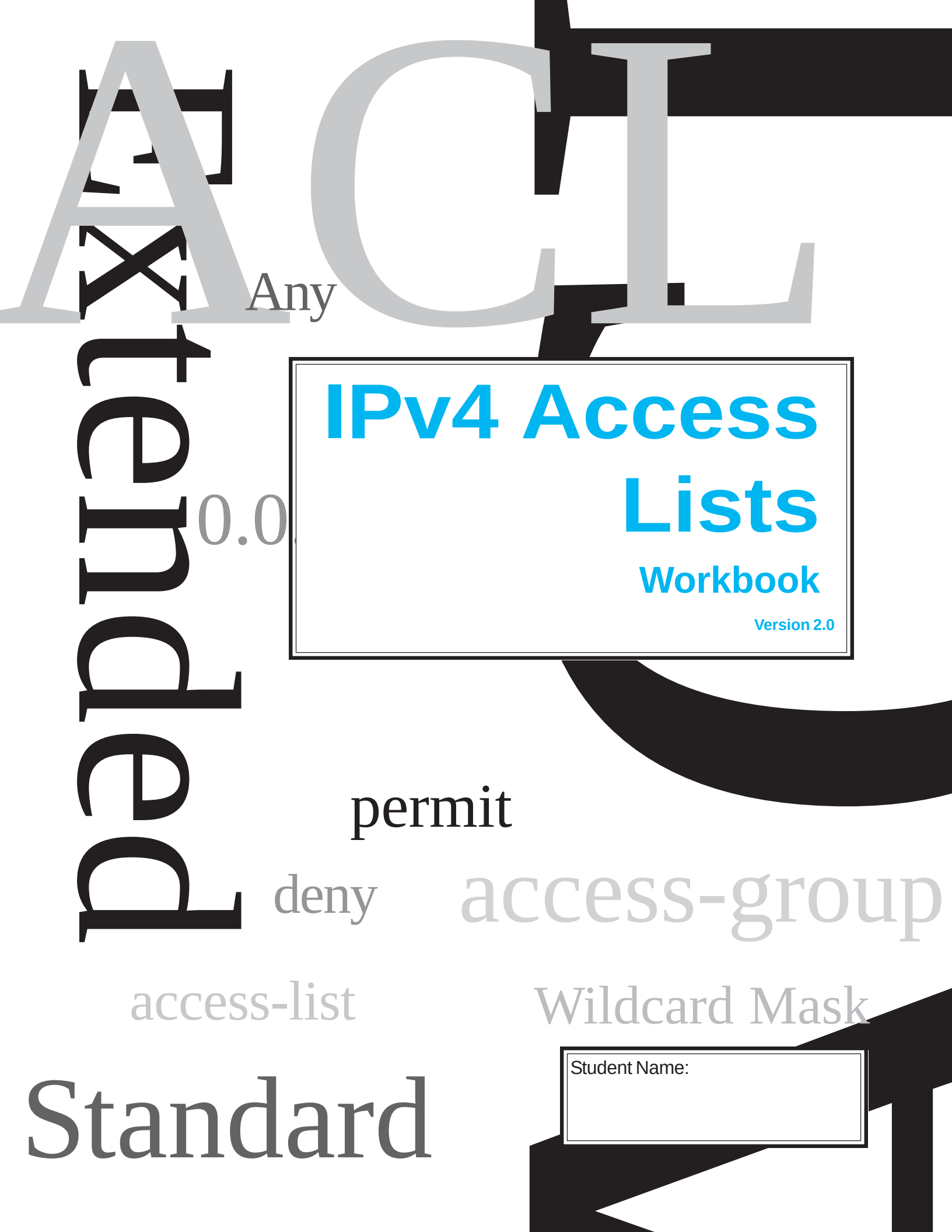
VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

Class A Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/8	0	255.0.0.0	1	16,777,216	16,777,214
/9	1	255.128.0.0	2	8,388,608	8,388,606
/10	2	255.192.0.0	4	4,194,304	4,194,302
/11	3	255.224.0.0	8	2,097,152	2,097,150
/12	4	255.240.0.0	16	1,048,576	1,048,574
/13	5	255.248.0.0	32	524,288	524,286
/14	6	255.252.0.0	64	262,144	262,142
/15	7	255.254.0.0	128	131,072	131,070
/16	8	255.255.0.0	256	65,536	65,534
/17	9	255.255.128.0	512	32,768	32,766
/18	10	255.255.192.0	1,024	16,384	16,382
/19	11	255.255.224.0	2,048	8,192	8,190
/20	12	255.255.240.0	4,096	4,096	4,094
/21	13	255.255.248.0	8,192	2,048	2,046
/22	14	255.255.252.0	16,384	1,024	1,022
/23	15	255.255.254.0	32,768	512	510
/24	16	255.255.255.0	65,536	256	254
/25	17	255.255.255.128	131,072	128	126
/26	18	255.255.255.192	262,144	64	62
/27	19	255.255.255.224	524,288	32	30
/28	20	255.255.255.240	1,048,576	16	14
/29	21	255.255.255.248	2,097,152	8	6
/30	22	255.255.255.252	4,194,304	4	2

Class B Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/16	0	255.255.0.0	1	65,536	65,534
/17	1	255.255.128.0	2	32,768	32,766
/18	2	255.255.192.0	4	16,384	16,382
/19	3	255.255.224.0	8	8,192	8,190
/20	4	255.255.240.0	16	4,096	4,094
/21	5	255.255.248.0	32	2,048	2,046
/22	6	255.255.252.0	64	1,024	1,022
/23	7	255.255.254.0	128	512	510
/24	8	255.255.255.0	256	256	254
/25	9	255.255.255.128	512	128	126
/26	10	255.255.255.192	1,024	64	62
/27	11	255.255.255.224	2,048	32	30
/28	12	255.255.255.240	4,096	16	14
/29	13	255.255.255.248	8,192	8	6
/30	14	255.255.255.252	16,384	4	2

Class C Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/24	0	255.255.255.0	1	256	254
/25	1	255.255.255.128	2	128	126
/26	2	255.255.255.192	4	64	62
/27	3	255.255.255.224	8	32	30
/28	4	255.255.255.240	16	16	14
/29	5	255.255.255.248	32	8	6
/30	6	255.255.255.252	64	4	2



IPv4 Access Lists Workbook

Version 2.0

Student Name:

Access-List Numbers

IP Standard	1	to	99
IP Extended	100	to	199
Ethernet Type Code	200	to	299
Ethernet Address	700	to	799
DECnet and Extended DECnet	300	to	399
XNS	400	to	499
Extended XNS	500	to	599
Appletalk	600	to	699
48-bit MAC Addresses	700	to	799
IPX Standard	800	to	899
IPX Extended	900	to	999
IPX SAP (service advertisement protocol)	1000	to	1099
IPX SAP SPX	1000	to	1099
Extended 48-bit MAC Addresses	1100	to	1199
IPX NLSP	1200	to	1299
IP Standard, expanded range	1300	to	1999
IP Extended, expanded range	2000	to	2699
SS7 (voice)	2700	to	2999
Standard Vines	1	to	100
Extended Vines	101	to	200
Simple Vines	201	to	300
Transparent bridging (protocol type)	200	to	299
Transparent bridging (vendor type)	700	to	799
Extended Transparent bridging	1100	to	1199
Source-route bridging (protocol type)	200	to	299
Source-route bridging (vendor type)	700	to	799

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Special Thanks to Melvin Baker, Jim Dorsch, and Brent Sieling
for taking the time to check this workbook for errors, and making suggestions for improvements.

Instructors (and anyone else for that matter) please do not post the Instructors version on public websites.
When you do this you are giving everyone else worldwide the answers. Yes, students look for answers this way.
It also discourages others; myself included, from posting high quality materials.

What are Access Control Lists?

ACLs...

...are a sequential list of instructions that tell a router which packets to permit or deny.

General Access Lists Information

Access Lists...

- ...are read sequentially.
- ...are set up so that as soon as the packet matches a statement it stops comparing and permits or denies the packet.
- ...need to be written to take care of the most abundant traffic first.
- ...must be configured on your router before you can deny packets.
- ...can be written for all supported routed protocols; but each routed protocol must have a different ACL for each interface.
- ...must be applied to an interface to work.

How routers use Access Lists

(Outbound Port - Default)

- ❑ The router checks to see if the packet is routable. If it is it looks up the route in its routing table.
- ❑ The router then checks for an ACL on that outbound interface.
- ❑ If there is no ACL the router switches the packet out that interface to its destination.
- ❑ If there is an ACL the router checks the packet against the access list statements sequentially. Then permits or denies each packet as it is matched.
- ❑ If the packet does not match any statement written in the ACL it is denied because there is an implicit “deny any” statement at the end of every ACL.

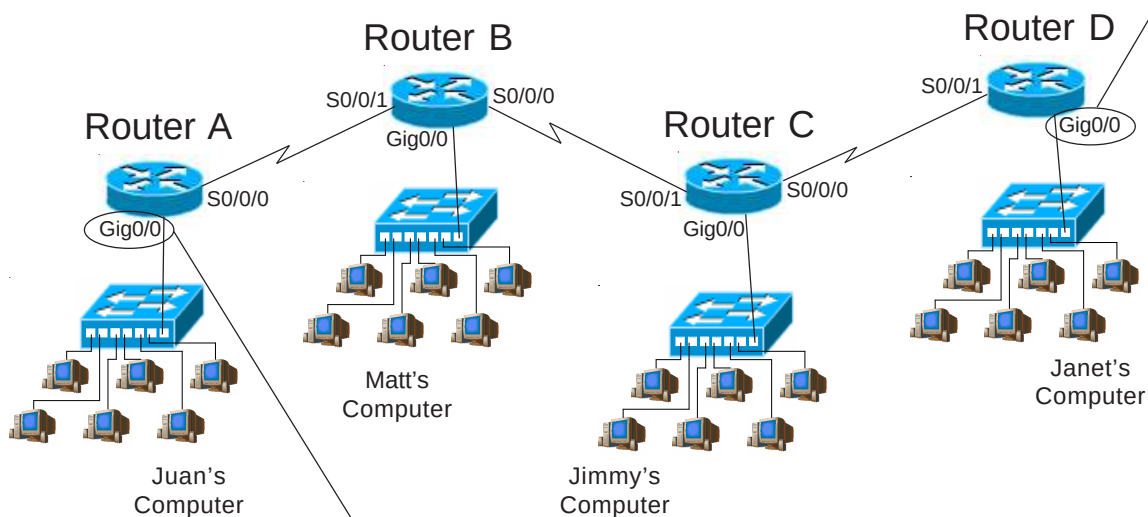
Standard Access Lists

Standard Access Lists...

- ...are numbered from 1 to 99 or 1300 to 1999.
- ...filter (permit or deny) only source addresses.
- ...do not have any destination information so it must be placed as close to the destination as possible.
- ...work at layer 3 of the OSI model.

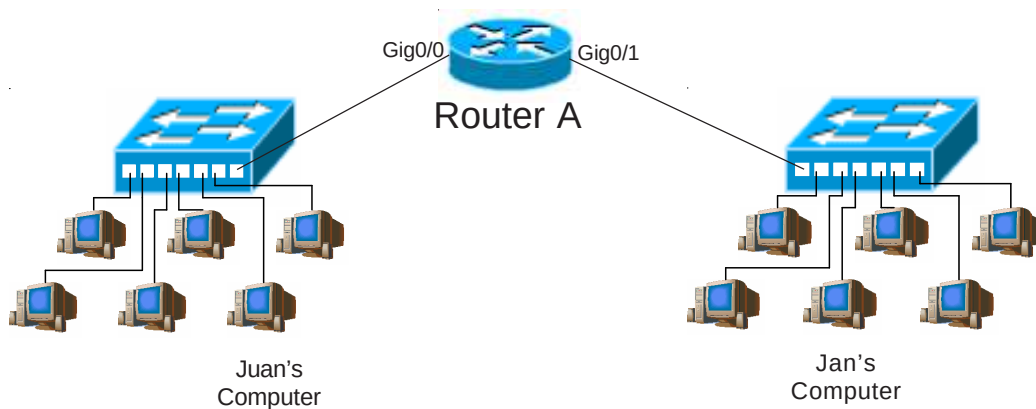
Why standard ACLs are placed close to the destination.

If you want to block traffic from Juan's computer from reaching Janet's computer with a standard access list you would place the ACL close to the destination on Router D, interface Gig0/0. Since it is using only the source address to permit or deny packets the ACL here will not effect packets reaching Routers B, or C.

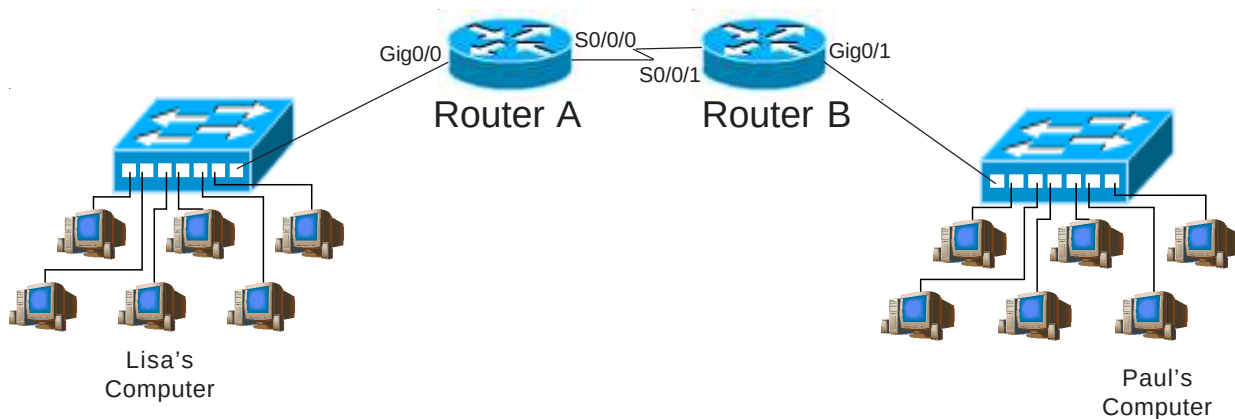


If you place the ACL on router A to block traffic to Router D it will also block all packets going to Routers B, and C; because all the packets will have the same source address.

Standard Access List Placement Sample Problems



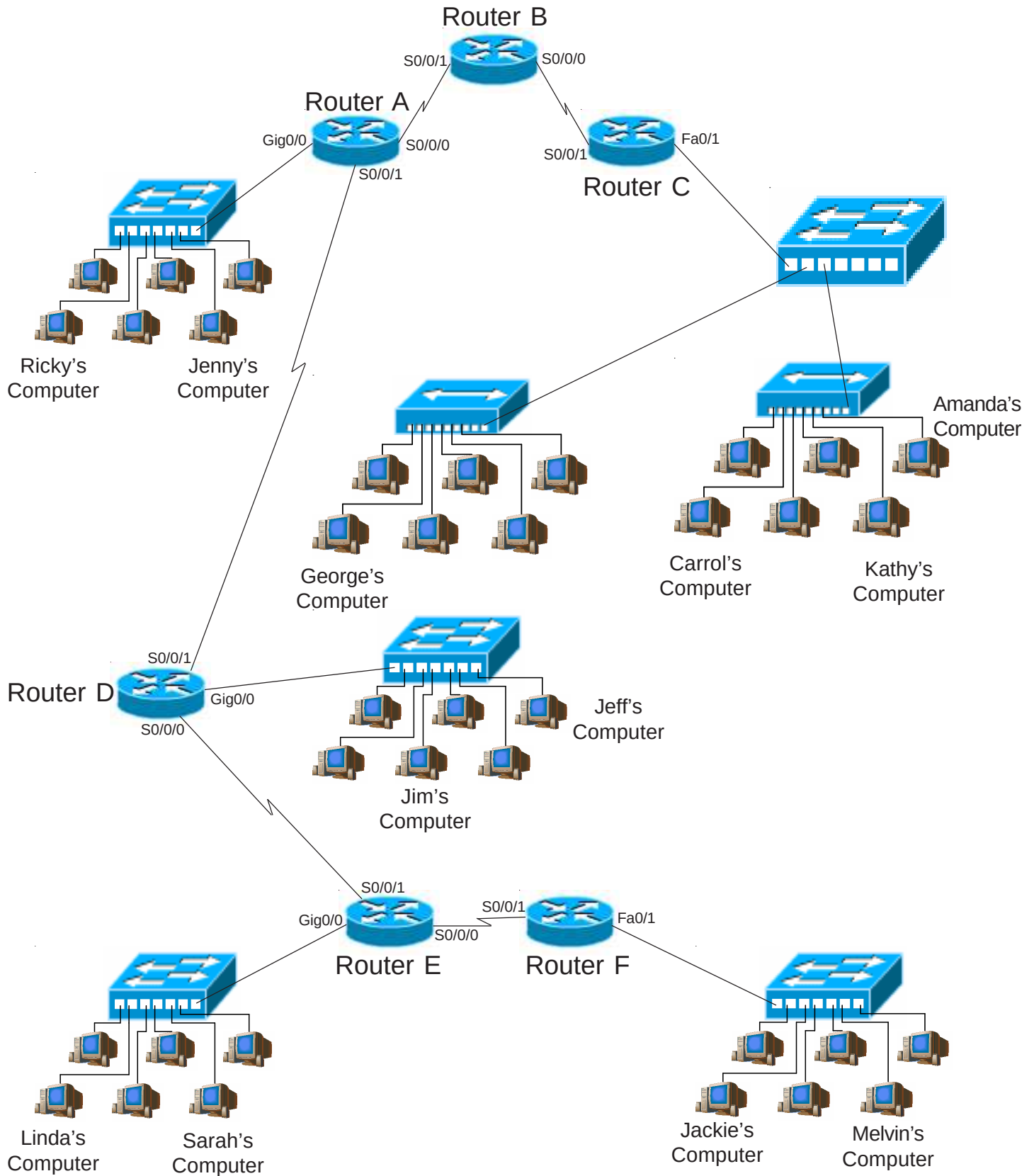
In order to permit packets from Juan's computer to arrive at Jan's computer you would place the standard access list at router interface Gig0/1.



Lisa has been sending unnecessary information to Paul. Where would you place the standard ACL to deny all traffic from Lisa to Paul?
Router Name Router B Interface Gig0/1

Where would you place the standard ACL to deny traffic from Paul to Lisa?
Router Name Router A Interface Gig0/0

Standard Access List Placement



Standard Access List Placement

1. Where would you place a standard access list to permit traffic from Ricky's computer to reach Jeff's computer?

Router Name Router D
Interface Gig0/0

2. Where would you place a standard access list to deny traffic from Melvin's computer from reaching Jenny's computer?

Router Name Router A
Interface Gig0/0

3. Where would you place a standard access list to deny traffic to Carrol's computer from Sarah's computer?

Router Name _____
Interface _____

4. Where would you place a standard access list to permit traffic to Ricky's computer from Jeff's computer?

Router Name _____
Interface _____

5. Where would you place a standard access list to deny traffic from Amanda's computer from reaching Jeff and Jim's computer?

Router Name _____
Interface _____

6. Where would you place a standard access list to permit traffic from Jackie's computer to reach Linda's computer?

Router Name _____
Interface _____

7. Where would you place a standard access list to permit traffic from Ricky's computer to reach Carrol and Amanda's computer?

Router Name _____
Interface _____

8. Where would you place a standard access list to deny traffic to Jenny's computer from Jackie's computer?

Router Name _____
Interface _____

9. Where would you place a standard access list to permit traffic from George's computer to reach Linda and Sarah's computer?

Router Name _____
Interface _____

10. Where would you place an ACL to deny traffic from Jeff's computer from reaching George's computer?

Router Name _____
Interface _____

11. Where would you place a standard access list to deny traffic to Sarah's computer from Ricky's computer?

Router Name _____
Interface _____

12. Where would you place an ACL to deny traffic from Linda's computer from reaching Jackie's computer?

Router Name _____
Interface _____

Extended Access Lists

Extended Access Lists...

...are numbered from 100 to 199 or 2000 to 2699.

...filter (permit or deny) based on the:

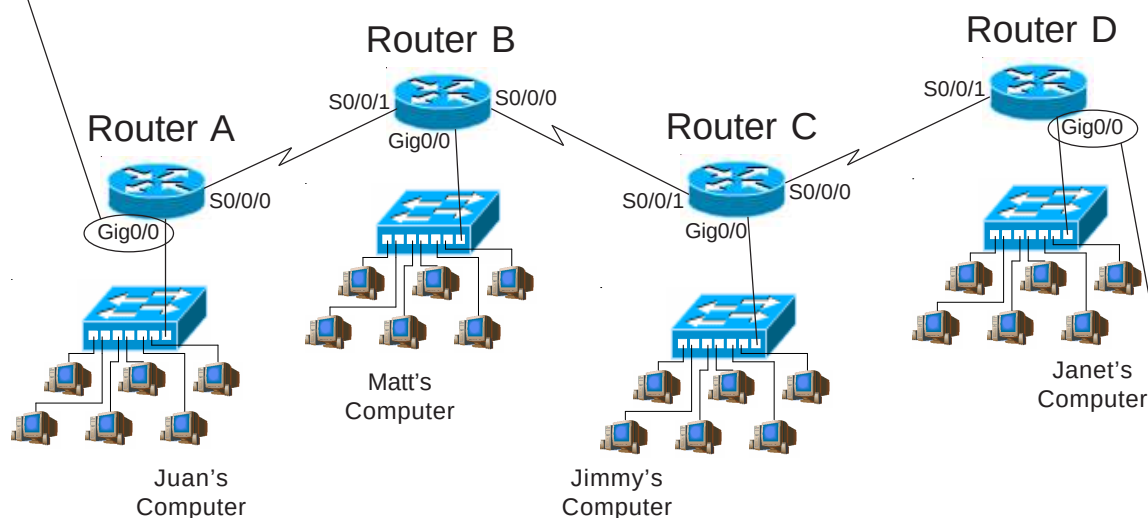
- source address
- destination address
- protocol
- application / port number

... are placed close to the source.

...work at both layer 3 and 4 of the OSI model.

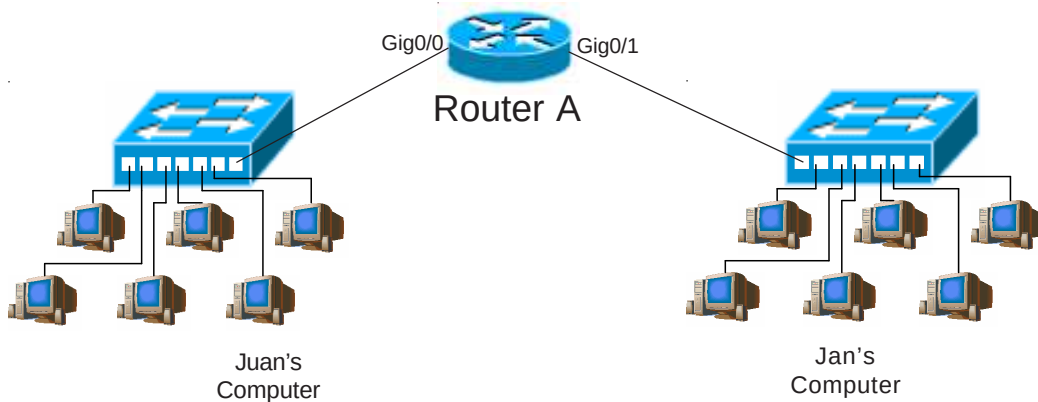
Why extended ACLs are placed close to the source.

If you want to deny traffic from Juan's computer from reaching Janet's computer with an extended access list you would place the ACL **close to the source** on Router A, interface Gig0/0. Since it can permit or deny based on the destination address it can reduce backbone overhead and not affect traffic to Routers B or C.

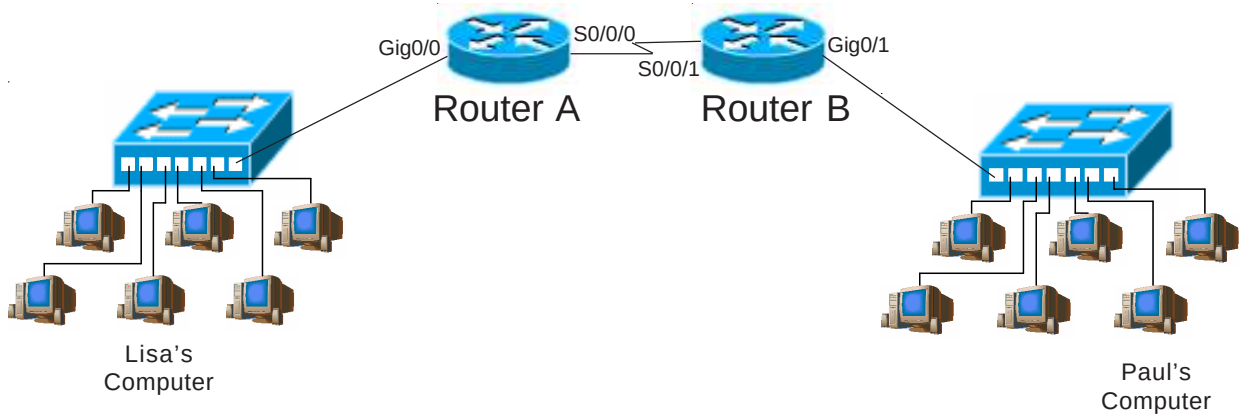


If you place the ACL on Router D to block traffic from Router A, it will work. However, Routers B and C will have to route the packet before it is finally blocked at Router D. This increases the volume of useless network traffic.

Extended Access List Placement Sample Problems



In order to permit packets from Juan's computer to arrive at Jan's computer you would place the extended access list at router interface Gig0/0.



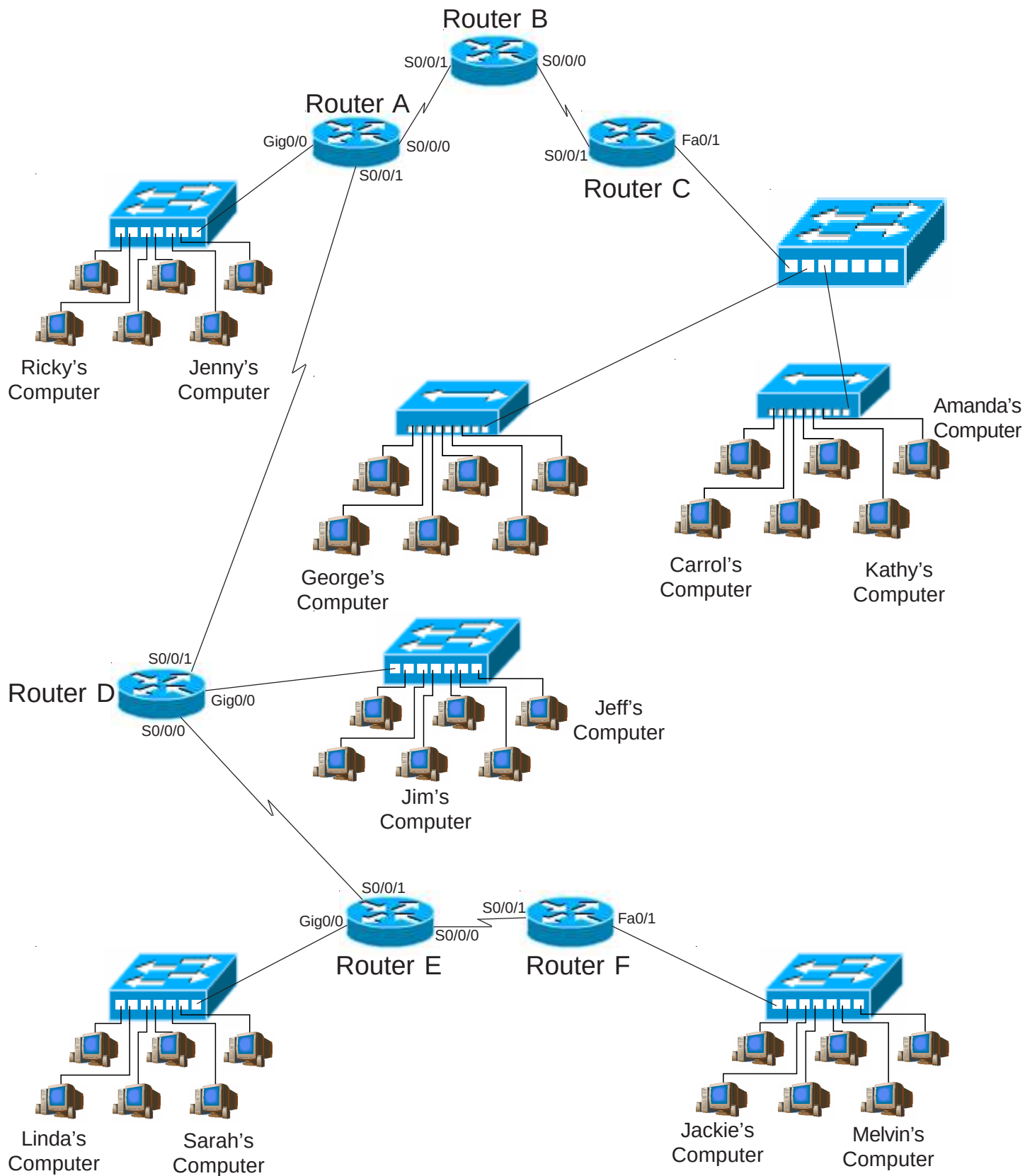
Lisa has been sending unnecessary information to Paul. Where would you place the extended ACL to deny all traffic from Lisa to Paul?

Router Name Router A Interface Gig0/0

Where would you place the extended ACL to deny traffic from Paul to Lisa?

Router Name Router B Interface Gig0/1

Extended Access List Placement



Extended Access List Placement

1. Where would you place an ACL to deny traffic from Jeff's computer from reaching George's computer?

Router Name Router D
Interface Gig0/0

2. Where would you place an extended access list to permit traffic from Jackie's computer to reach Linda's computer?

Router Name Router F
Interface FA0/1

3. Where would you place an extended access list to deny traffic to Carrol's computer from Ricky's computer?

Router Name _____
Interface _____

4. Where would you place an extended access list to deny traffic to Sarah's computer from Jackie's computer?

Router Name _____
Interface _____

5. Where would you place an extended access list to permit traffic from Carrol's computer to reach Jeff's computer?

Router Name _____
Interface _____

6. Where would you place an extended access list to deny traffic from Melvin's computer from reaching Jeff and Jim's computer?

Router Name _____
Interface _____

7. Where would you place an extended access list to permit traffic from George's computer to reach Jeff's computer?

Router Name _____
Interface _____

8. Where would you place an extended access list to permit traffic from Jim's computer to reach Carrol and Amanda's computer?

Router Name _____
Interface _____

9. Where would you place an ACL to deny traffic from Linda's computer from reaching Kathy's computer?

Router Name _____
Interface _____

10. Where would you place an extended access list to deny traffic to Jenny's computer from Sarah's computer?

Router Name _____
Interface _____

11. Where would you place an extended access list to permit traffic from George's computer to reach Linda and Sarah's computer?

Router Name _____
Interface _____

12. Where would you place an extended access list to deny traffic from Linda's computer from reaching Jenny's computer?

Router Name _____
Interface _____

Choosing to Filter Incoming or Outgoing Packets

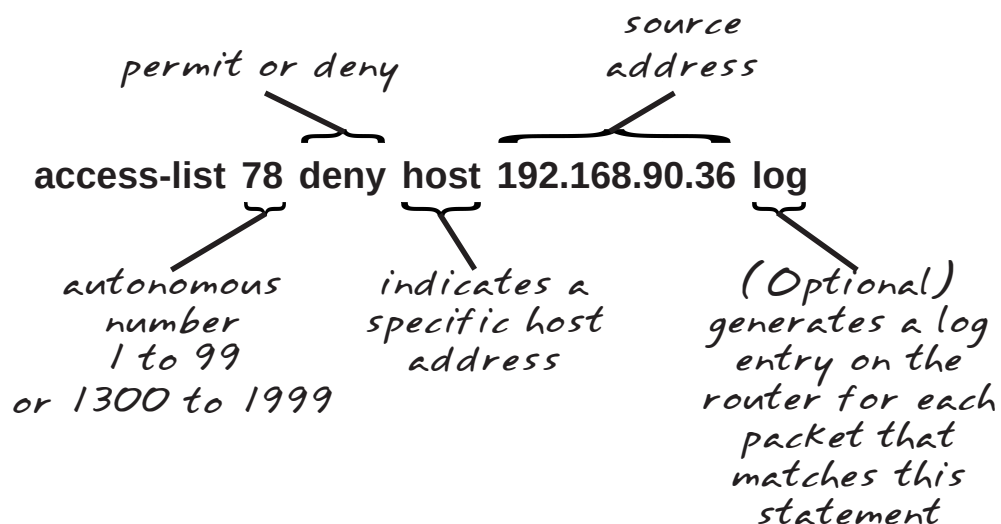
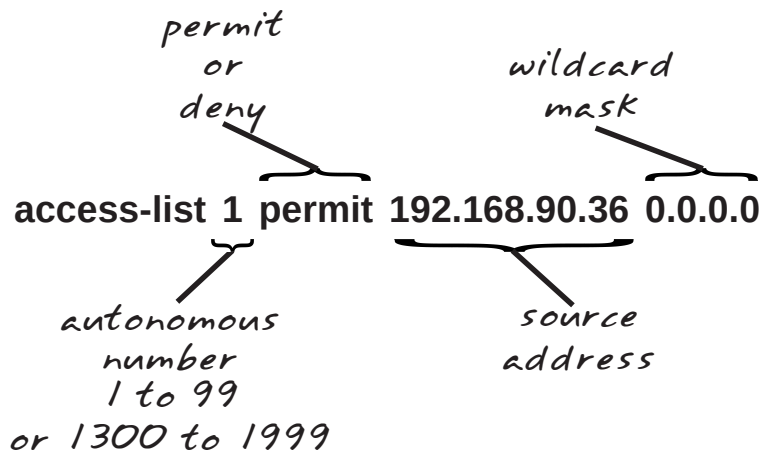
Access Lists on your incoming port...

- ...requires less CPU processing.
- ...filters and denies packets before the router has to make a routing decision.

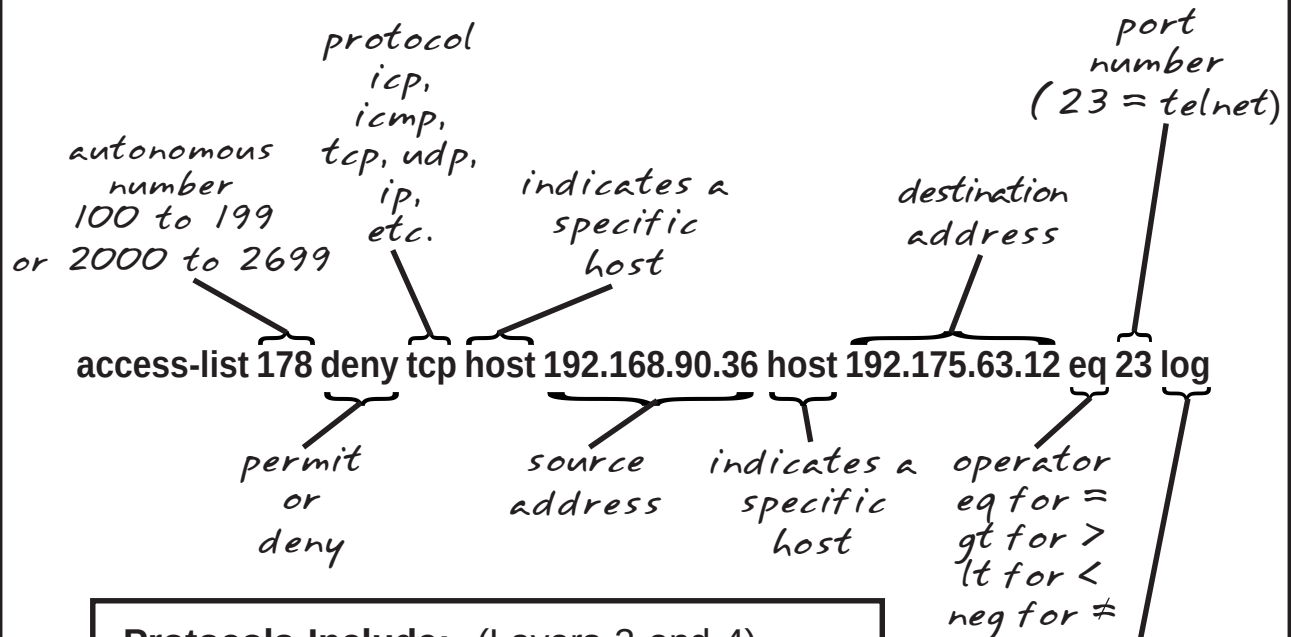
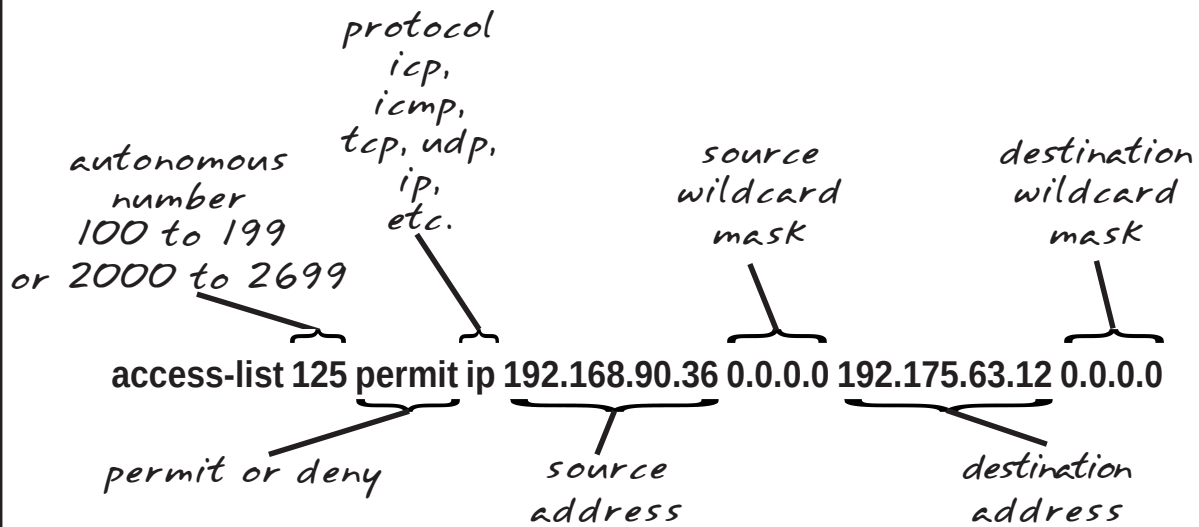
Access Lists on your outgoing port...

- ...are outbound by default unless otherwise specified.
- ...increases the CPU processing time because the routing decision is made and the packet switched to the correct outgoing port before it is tested against the ACL.

Breakdown of a Standard ACL Statement



Breakdown of an Extended ACL Statement



Protocols Include: (Layers 3 and 4)

IP	IGMP	IPINIP
TCP	GRE	OSPF
UDP	IGRP	NOS
ICMP	EIGRP	Integer 0-255

To match any internet protocol use IP.

What are Named Access Control Lists?

Named ACLs...

...are standard or extended ACLs which have an alphanumeric name instead of a number. (ie. 1-99 or 100-199)

Named Access Lists Information

Named Access Lists...

- ...identify ACLs with an intuitive name instead of a number.
- ...eliminate the limits imposed by using numbered ACLs. (798 for standard and 799 for extended)
- ...names should be typed in all CAPS to make it easier to see.
- ...provide the ability to modify your ACLs without deleting and reloading the revised access list. It will only allow you to add statements to the end of the existing statements.
- ...are not compatible with any IOS prior to Release 11.2.
- ...can not repeat the same name on multiple ACLs.

Applying a Standard Named Access List called "GEORGE"

Write a named standard access list called "GEORGE" on Router A, interface E1 to block Melvin's computer from sending information to Kathy's computer; but will allow all other traffic.

Place the access list at:

Router Name: Router A

Interface: E1

Access-list Name: GEORGE

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# ip access-list standard GEORGE
Router(config-std-nacl)# deny host 72.16.70.35
Router(config-std-nacl)# permit any
Router(config-std-nacl)# interface gig0/1
Router(config-if)# ip access-group george out
Router(config-if)# exit
Router(config)# exit
```

Applying an extended Named Access List called "GRACIE"

Write a named extended access list called "GRACIE" on Router A, Interface E0 called "Gracie" to deny HTTP traffic intended for web server 192.168.207.27, but will permit all other HTTP traffic to reach the only the 192.168.207.0 network. Deny all other IP traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: E0
Access-list Name: GRACIE

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# ip access-list extended GRACIE
Router(config-ext-nacl)# deny tcp any host 192.168.207.27 eq www
Router(config-ext-nacl)# permit tcp any 192.168.207.0 0.0.0.255 eq www
Router(config-ext-nacl)# interface gig0/1
Router(config-if)# ip access-group gracie in
Router(config-if)# exit
Router(config)# exit
```

Choices for Using Wildcard Masks

Wildcard masks are usually set up to do one of four things:

1. Match a specific host.
2. Match an entire subnet.
3. Match a specific range.
4. Match all addresses.

1. Matching a specific host.

For standard access lists:

Access-List 10 permit 192.168.150.50 0.0.0.0

or

Access-List 10 permit 192.168.150.50 (standard ACL's
assume a 0.0.0.0 mask)

or

Access-List 10 permit host 192.168.150.50

For extended access lists:

Access-list 110 deny ip 192.168.150.50 0.0.0.0 any

or

Access-list 110 deny ip host 192.168.150.50 any

2. Matching an entire subnet

Example 1

Address: 192.168.50.0 Subnet Mask: 255.255.255.0

Access-list 25 deny 192.168.50.0 0.0.0.255

Example 2

Address: 172.16.0.0 Subnet Mask: 255.255.0.0

Access-list 12 permit 172.16.0.0 0.0.255.255

Example 3

Address: 10.0.0.0 Subnet Mask: 255.0.0.0

Access-list 125 deny udp 10.0.0.0 0.255.255.255 any

3. Match a specific range

Example 1

Address: 10.250.50.112 Subnet Mask: 255.255.255.224

255.255.255.255
Custom Subnet mask: -255.255.255.224
Wildcard: 0. 0. 0. 31

Access-list 125 permit udp 10.250.50.112 0.0.0.31 any

Example 2

Address Range: 192.168.16.0 to 192.168.16.127

192.168.16.127
-192.168.16. 0
Wildcard: 0. 0. 0.127

Access-list 125 deny ip 192.168.16.0 0.0.0.127 any
(This ACL would block the lower half of the subnet.)

Example 3

Address: 172.250.16.32 to 172.250.31.63

172.250.31. 63
-172.250.16. 32
Wildcard: 0. 0.15. 31

Access-list 125 permit ip 172.250.16.32 0.0.15.31 any

4. Match everyone.

For standard access lists:

Access-List 15 permit any

or

Access-List 15 deny 0.0.0.0 255.255.255.255

For extended access lists:

Access-List 175 permit ip any any

or

Access-List 175 deny tcp 0.0.0.0 255.255.255.255 any

Creating Wildcard Masks

- ❑ Just like a subnet mask the wildcard mask tells the router what part of the address to check or ignore. Zero (0) must match exactly, one (1) will be ignored.
- ❑ The source address can be a single address, a range of addresses, or an entire subnet.
- ❑ As a rule of thumb the wildcard mask is the inverse of the subnet mask.

Example #1:

IP Address and subnet mask: 204.100.100.0 255.255.255.0

IP Address and wildcard mask: 204.100.100.0 0.0.0.255

- ❑ All zero's (or 0.0.0.0) means the address must match exactly.

Example #2:

10.10.150.95 0.0.0.0 (This address must match exactly.)

- ❑ One's will be ignored.

Example #3:

10.10.150.95 0.0.0.255 (Any 10.10.150.0 subnet address will match.
10.10.150.0 to 10.10.150.255)

- ❑ This also works with subnets.

Example #4:

IP Address and subnet mask: 192.170.25.30 255.255.255.224

IP Address and wildcard mask: 192.170.25.30 0.0.0.31
(Subtract the subnet mask from
255.255.255.255 to create the wildcard)

Do the math... $255 - 255 = 0$ (This is the inverse of the subnet mask.)
 $255 - 224 = 31$

Example #5:

IP Address and subnet mask: 172.24.128.0 255.255.128.0

IP Address and wildcard mask: 172.24.128.0 0.0.127.255

Do the math... $255 - 255 = 0$ (This is the inverse of the subnet mask.)
 $255 - 128 = 127$
 $255 - 0 = 255$

Wildcard Mask Problems

1. Create a wildcard mask to match this exact address.
IP Address: 192.168.25.70
Subnet Mask: 255.255.255.0 0 . 0 . 0 . 0
2. Create a wildcard mask to match this range.
IP Address: 210.150.10.0
Subnet Mask: 255.255.255.0 0 . 0 . 0 . 255
3. Create a wildcard mask to match this host.
IP Address: 195.190.10.35
Subnet Mask: 255.255.255.0 _____
4. Create a wildcard mask to match this range.
IP Address: 172.16.0.0
Subnet Mask: 255.255.0.0 _____
5. Create a wildcard mask to match this range.
IP Address: 10.0.0.0
Subnet Mask: 255.0.0.0 _____
6. Create a wildcard mask to match this exact address.
IP Address: 165.100.0.130
Subnet Mask: 255.255.255.192 _____
7. Create a wildcard mask to match this range.
IP Address: 192.10.10.16
Subnet Mask: 255.255.255.224 _____
8. Create a wildcard mask to match this range.
IP Address: 171.50.75.128
Subnet Mask: 255.255.255.192 _____
9. Create a wildcard mask to match this host.
IP Address: 10.250.30.2
Subnet Mask: 255.0.0.0 _____
10. Create a wildcard mask to match this range.
IP Address: 210.150.28.16
Subnet Mask: 255.255.255.240 _____
11. Create a wildcard mask to match this range.
IP Address: 172.18.0.0
Subnet Mask: 255.255.224.0 _____
12. Create a wildcard mask to match this range.
IP Address: 135.35.230.32
Subnet Mask: 255.255.255.248 _____

Basic Wildcard Mask Problems

Based on the given information list the range of source addresses for each ACE statement.

1. access-list 10 permit 192.168.150.50 0.0.0.0

Answer: 192.168.150.50

2. access-list 5 permit any

Answer: Any address

3. access-list 125 deny tcp 195.223.50.0 0.0.0.63 host 172.168.10.1 fragments

Answer: 195.223.50.0 to 195.223.50.63

4. access-list 11 deny 210.10.10.0 0.0.0.255

Answer: _____

5. access-list 108 deny ip 192.220.10.0 0.0.0.15 172.32.4.0 0.0.0.255

Answer: _____

6. access-list 171 deny any host 175.18.24.10 fragments

Answer: _____

7. access-list 105 permit 192.168.15.0 0.0.0.255 any

Answer: _____

8. access-list 109 permit tcp 172.16.10.0 0.0.0.255 host 192.168.10.1 eq 80

Answer: _____

9. access-list 111 permit ip any any

Answer: _____

10. access-list 195 permit udp 172.30.12.0 0.0.0.127 172.50.10.0 0.0.0.255

Answer: _____

11. **access-list 110 permit ip 192.168.15.0 0.0.0.3 192.168.30.10 0.0.0.0**

Answer: _____

12. **access-list 120 permit ip 192.168.15.0 0.0.0.7 192.168.30.10 0.0.0.0**

Answer: _____

13. **access-list 130 permit ip 192.168.15.0 0.0.0.15 192.168.30.10 0.0.0.0**

Answer: _____

14. **access-list 140 permit ip 192.168.15.0 0.0.0.31 192.168.30.10 0.0.0.0**

Answer: _____

15. **access-list 150 permit ip 192.168.15.0 0.0.0.63 192.168.30.10 0.0.0.0**

Answer: _____

16. **access-list 101 Permit ip 192.168.15.0 0.0.0.127 192.168.30.10 0.0.0.0**

Answer: _____

17. **access-list 185 permit ip 192.168.15.0 0.0.0.255 192.168.30.0 0.0.0.255**

Answer: _____

18. **access-list 160 deny udp 172.16.0.0 0.0.1.255 172.18.10.18 0.0.0.0 gt 22**

Answer: _____

19. **access-list 195 permit icmp 172.85.0.0 0.0.15.255 172.50.10.0 0.0.0.255**

Answer: _____

20. **access-list 10 permit 175.15.120.0 0.0.0.255**

Answer: _____

21. **access-list 190 permit tcp 192.15.10.0 0.0.0.31 any**

Answer: _____

22. **access-list 100 permit ip 10.0.0.0 0.255.255.255 172.50.10.0 0.0.0.255**

Answer: _____

(Slightly More) **Advanced Wildcard Mask Problems**

Not every address range ends with a zero making it simple to determine the address range. Using basic subnetting you can adjust the host portion of the IP address with the wildcard mask to indicate specific ranges.

IP Address: 192 . 100 . 10 . 0
Custom Subnet Mask: 255.255.255.240

Address Ranges: 192.100.10.0 to 192.100.10.15
192.100.10.16 to 192.100.10.31
192.100.10.32 to 192.100.10.47 (Range in the sample below)
192.100.10.48 to 192.100.10.63
192.100.10.64 to 192.100.10.79
192.100.10.80 to 192.100.10.95
192.100.10.96 to 192.100.10.111
192.100.10.112 to 192.100.10.127
192.100.10.128 to 192.100.10.143
192.100.10.144 to 192.100.10.159
192.100.10.160 to 192.100.10.175
192.100.10.176 to 192.100.10.191
192.100.10.192 to 192.100.10.207
192.100.10.208 to 192.100.10.223
192.100.10.224 to 192.100.10.239
192.100.10.240 to 192.100.10.255

Based on this sample to deny or permit the third subnet we would create the correct wildcard mask by inverting the subnet mask from 255.255.255.240 to 0.0.0.15. The third address range would be 192.100.10.32 to 192.100.10.47 .

A standard ACE statement would be written as:

access-list 10 permit 192.100.10.32 0.0.0.15

The wildcard mask indicates that address range 0 to 15 will be permitted. If the source address was 192.100.10.0 it would indicate the first 16 addresses (0 to 15). Since the source address is 192.100.10.32, it indicates that the address range will be from 192.100.10.32 to 192.100.10.47 (32 to 47). The IP address indicates the starting point for the wildcard mask to begin counting up from.

This technique gives you greater control with the addresses you want to permit or deny. It is a **Best Practice** to stay within standard address ranges.

Based on the given information list the range of source addresses for each ACE statement.

1. access-list 10 permit 192.100.10.64 0.0.0.15

Answer: 192.100.10.64 to 192.100.10.79

2. access-list 5 permit 172.16.128.0 0.0.63.255

Answer: 172.16.128.0 to 172.16.191.255

3. access-list 125 deny tcp 192.100.10.208 0.0.0.15 host 192.168.10.1 log

Answer: _____

4. access-list 11 deny 210.48.72.192 0.0.0.31

Answer: _____

5. access-list 108 deny ip 192.168.5.184 0.0.0.7 192.64.4.0 0.0.0.255

Answer: _____

6. access-list 171 deny 172.32.128.0 0.0.15.255 any

Answer: _____

7. access-list 105 permit 165.50.196.0 0.0.3.255 any

Answer: _____

8. access-list 109 permit tcp 172.16.128.0 0.0.127.255 host 172.16.20.1

Answer: _____

9. access-list 111 permit ip 192.168.1.236 0.0.0.3 any

Answer: _____

10. access-list 195 permit udp 172.32.160.0.0 0.0.31.255 172.100.10.0 0.0.0.255

Answer: _____

11. access-list 110 permit ip 10.64.0.0 0.63.255.255 172.168.40.10 0.0.0.0

Answer: _____

Wildcard Mask Problems

Based on the given information list the range of destination addresses for each ACE statement.

1. **access-list 125 deny tcp 195.223.50.0 0.0.0.63 host 172.168.10.1 fragments**

Answer: 172.168.10.1

2. **access-list 115 permit any any**

Answer: Any address

3. **access-list 150 permit ip 192.168.30.10 0.0.0.0 192.168.15.0 0.0.0.63**

Answer: 192.168.15.0 to 192.168.15.63

4. **access-list 120 deny tcp 172.32.4.0 0.0.0.255 192.220.10.0 0.0.0.15**

Answer: _____

5. **access-list 108 deny ip 192.220.10.0 0.0.0.15 172.32.4.0 0.0.0.255**

Answer: _____

6. **access-list 101 deny ip 140.130.110.100 0.0.0.0 0.0.0.0 255.255.255.255**

Answer: _____

7. **access-list 105 permit any 192.168.15.0 0.0.0.255**

Answer: _____

8. **access-list 120 permit ip 192.168.15.10 0.0.0.0 172.16.40.0 0.0.3.255**

Answer: _____

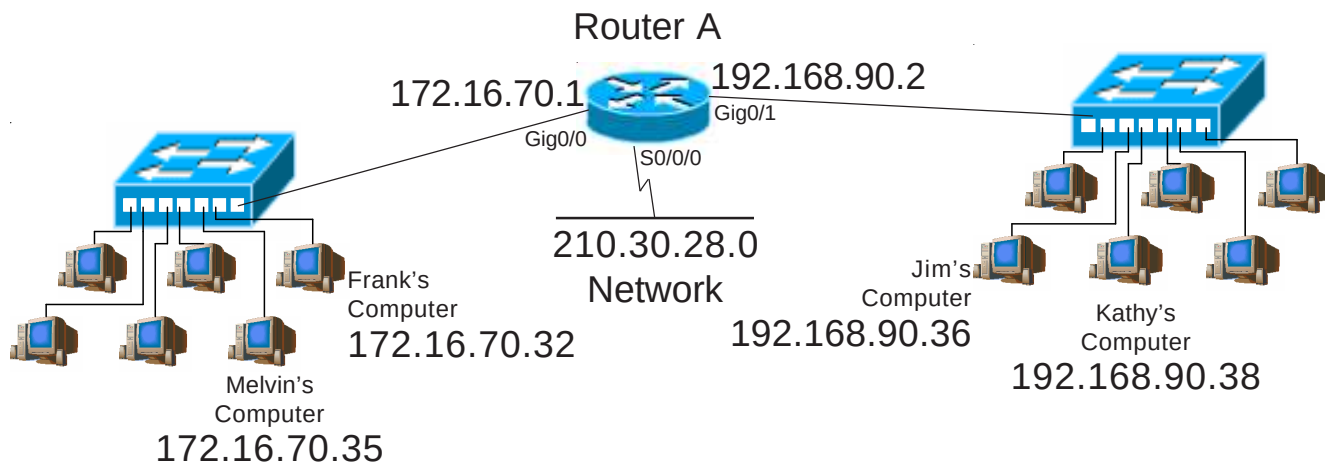
9. **access-list 160 deny udp 172.16.0.0 0.0.1.255 172.18.104.0 0.0.7.255 eq 21**

Answer: _____

10. **access-list 150 permit ip 192.168.30.0 0.0.0.63 192.168.15.96 0.0.0.31**

Answer: _____

Writing Standard Access Lists...



Standard Access List Sample #1

Write a standard access list to block Melvin's computer from sending information to Kathy's computer; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: Gig0/1
 Access-list #: 10

Note:

A standard access list blocks only the source address. Melvin will also be blocked from sending information to Jim or anyone else on the 192.168.90.0 network.

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 10 deny 172.16.70.35
                        or
                        access-list 10 deny 172.16.70.35 0.0.0.0
                        or
                        access-list 10 deny host 172.16.70.35
Router(config)# access-list 10 permit 0.0.0.0 255.255.255.255
                        or
                        access-list 10 permit any
Router(config)# interface gig0/1
Router(config-if)# ip access-group 10 out
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

Router# *show configuration*

(This will show which access groups are associated with particular interfaces)

Router# *show access list 10*

(This will show detailed information about this ACL)

Standard Access List Sample #2

Include a remark with each statement of your ACL. Write a standard access list to block Jim's computer from sending information to Frank's computer; but will allow all other traffic from the 192.168.90.0 network. Permit all traffic from the 210.30.28.0 network to reach the 172.16.70.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 28

[Writing and installing an ACL]

```
Router# configure terminal
Router(config)# access-list 28 remark Block Jim from reaching Frank
Router(config)# access-list 28 deny 192.168.90.36
                        or
                        access-list 28 deny 192.168.90.36 0.0.0.0
                        or
                        access-list 28 deny host 192.168.90.36
Router(config)# access-list 28 remark Allow all other traffic
Router(config)# access-list 28 permit 192.168.90.0 0.0.0.255
Router(config)# access-list 28 remark Allow all traffic
Router(config)# access-list 28 permit 210.30.28.0 0.0.0.255
Router(config)# interface Gig0/0
Router(config-if)# ip access-group 28 out
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Remark Command]

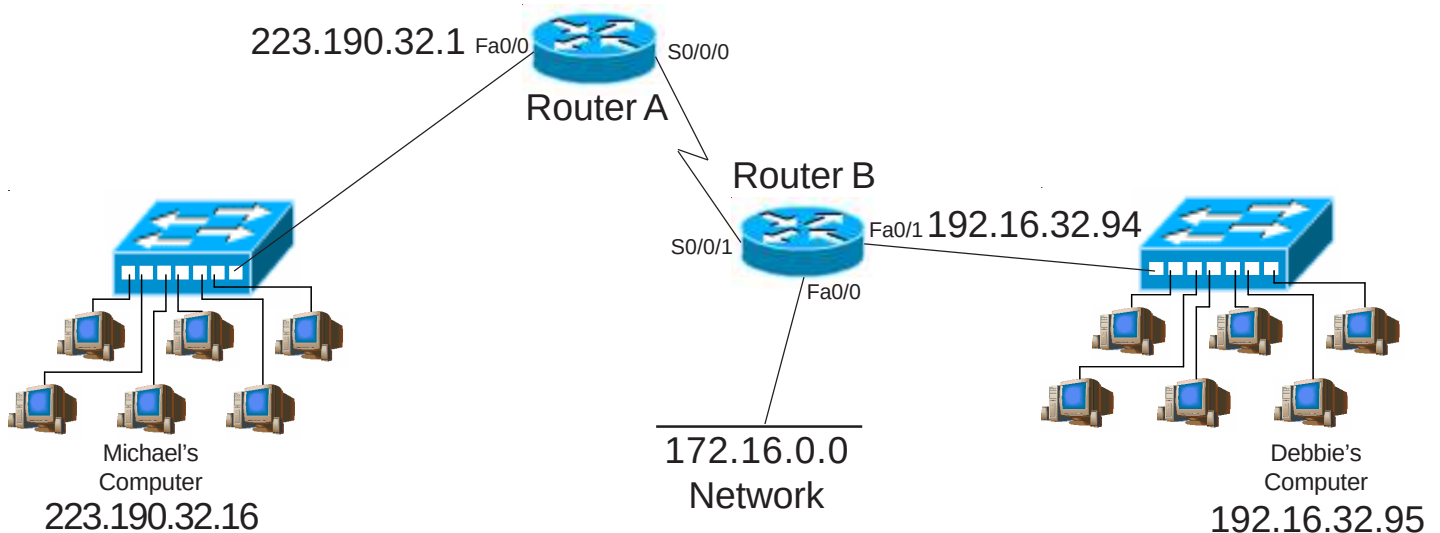
The remark command allows you to place text within the ACL so it can be viewed after it is inserted on the router. It can be viewed using the show run or any command that lists the ACL.

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 28 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 28 out
Router(config-if)# exit
Router(config)# no access-list 28
Router(config)# exit
```

Standard Access List Problem #1

Write a standard access list to block Debbie's computer from receiving information from Michael's computer; but will allow all other traffic. List all the command line options for this problem. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____
or

or

Router(config)# _____
or

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #2

Write a standard access list to permit Debbie's computer to receive information from Michael's computer; but will deny all other traffic from the 223.190.32.0 network. Block all traffic from the 172.16.0.0 network. Permit all other traffic. List all the command line options for this problem. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

```
Router(config)# _____
```

or

or

or

```
Router(config)#
```

```
Router(config)#
```

```
Router(config)#
```

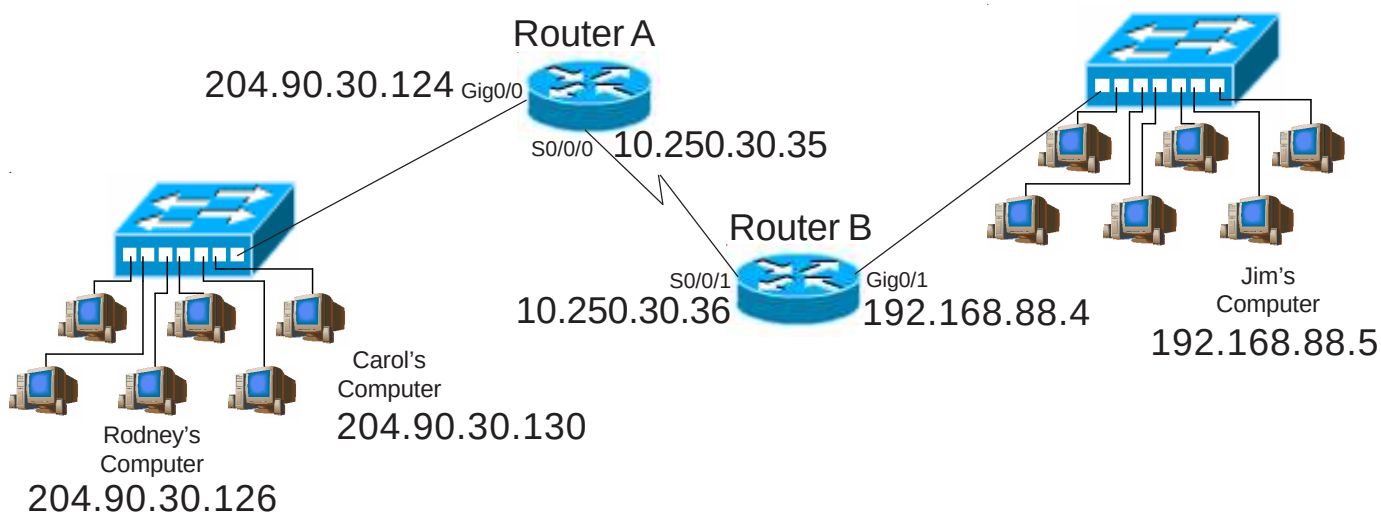
or

```
Router(config)# interface _____
```

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

```
Router(config-if)# exit
```

```
Router(config)# exit
```



Standard Access List Problem #3

Write a standard access list to block Rodney and Carol's computer from sending information to Jim's computer; but will allow all other traffic from the 204.90.30.0 network. Block all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #4

Include a remark with each statement of your ACL. Using a minimum number of commands write a standard access list named "Ralph" to block Carol's computer from sending information to Jim's computer; but will permit Jim to receive data from Rodney. Block the upper half of the 204.90.30.0 range from reaching Jim's computer while permitting the lower half of the range. Block all other traffic. For help with blocking the upper half of the range review page 13 or the wildcard mask problems on pages 16 and 17. For help with named ACLs review pages 12 and 13. For help with the remark command review page 23.

Place the access list at:

Router Name: _____

Interface: _____

Access-list Name: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

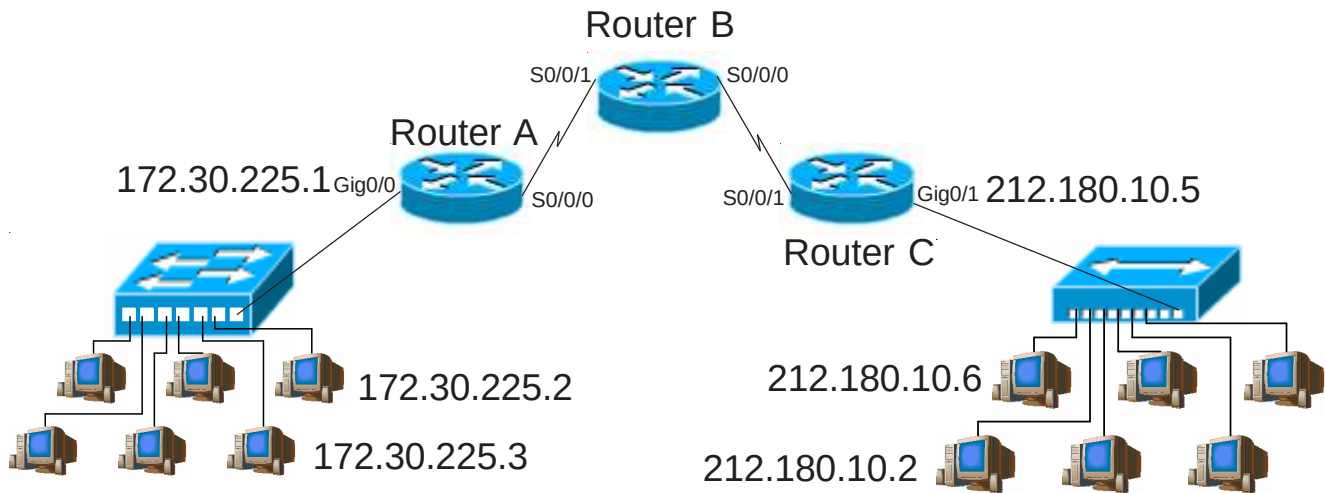
Router(config-std-nacl)# _____

Router(config-std-nacl)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*



Standard Access List Problem #5

Write a standard access list to block 172.30.225.2 and 172.30.225.3 from sending information to the 212.180.10.0 network; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #6

Add a remark to each statement explaining its purpose. Write a standard access list to block and log 212.180.10.2 from sending information to the 172.30.225.0 network. Permit and log 212.180.10.6 to send data to the 172.30.225.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written. Check the example on page 10 for help with the logging option. For help with the remark command review page 23.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

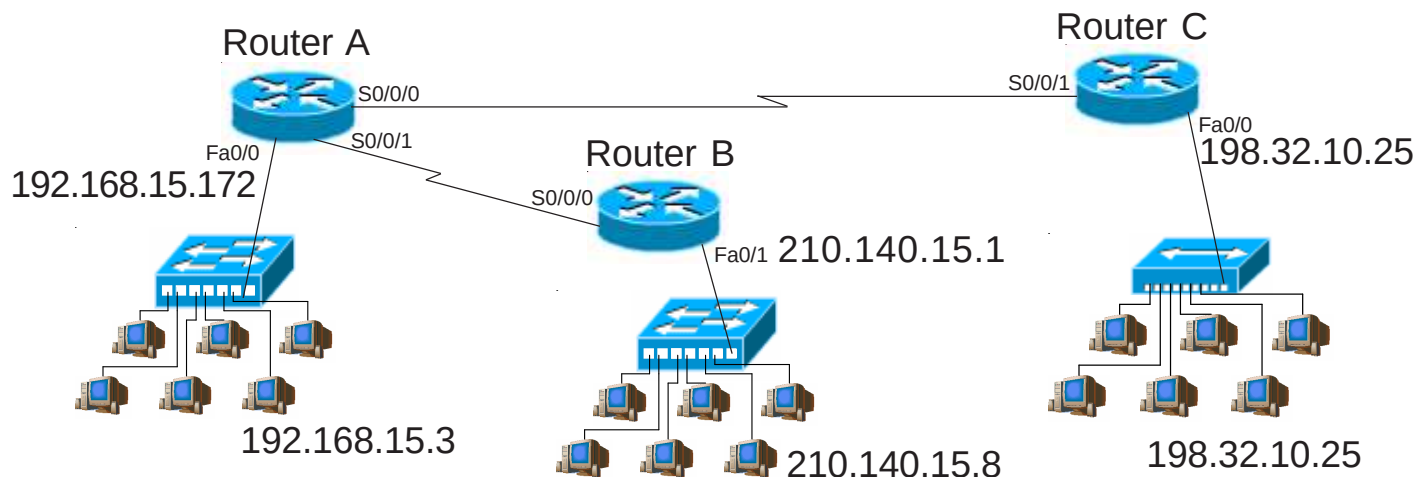
Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*



Standard Access List Problem #7

Write a standard access list to block the addresses 192.168.15.1 to 192.168.15.31 from sending information to the 210.140.15.0 network. Do not permit any traffic from 198.32.10.25 to reach the 210.140.15.0 network. Permit all other traffic. For help with this problem review page 13 or the wildcard mask problems on pages 16 and 17.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #8

Write a standard named access list called "CISCO_LAB_A" to permit traffic from the lower half of the 198.32.10.0 network to reach 192.168.15.0 network; block the upper half of the addresses. Allow host 198.32.10.192 to reach network 192.168.15.0. Permit all other traffic. For help with this problem review page 13 or the wildcard masks problems on pages 16 and 17. For assistance with named ACLs review pages 12 and 13.

Place the access list at:

Router Name: _____

Interface: _____

Access-list Name: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config-std-nacl)# _____

Router(config-std-nacl)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #9

Write a standard access list to block network 192.168.255.0 from receiving information from the following addresses: 10.250.1.1, 10.250.2.1, 10.250.4.1, and the entire 10.250.3.0 255.255.255.0 network. Allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A

Interface: Fa0/0

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

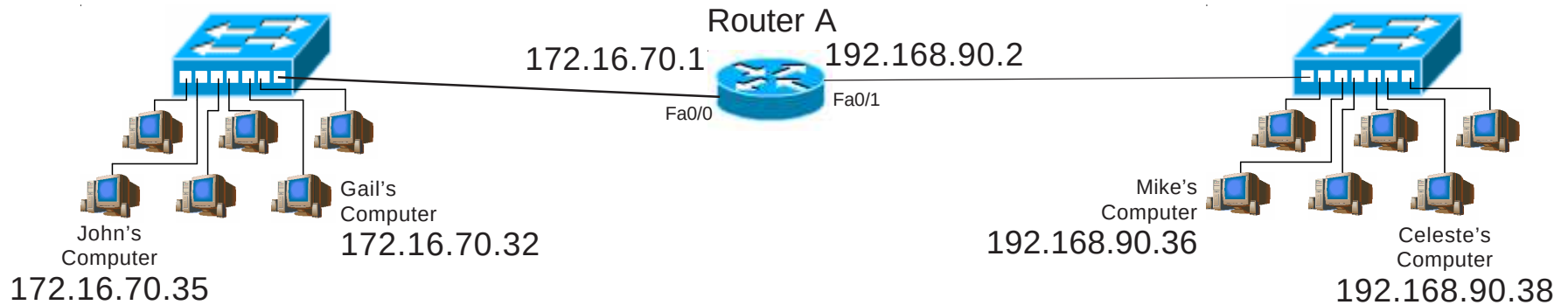
Router(config)# *interface fa0/0*

Router(config-if)# *ip access-group _____ in or out (circle one)*

Router(config-if)# *exit*

Router(config)# *exit*

Writing Extended Access Lists...



Extended Access List Sample #1

Deny/Permit Specific Addresses

Write an extended access list to prevent John's computer from sending information to Mike's computer; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: FA0/0
 Access-list #: 110

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 110 deny ip 172.16.70.35 0.0.0.0 192.168.90.36 0.0.0.0
or
access-list 110 deny ip host 172.16.70.35 host 192.168.90.36
Router(config)# access-list 110 permit ip any any
or
access-list 110 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/0
Router(config-if)# ip access-group 110 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

```
Router# show configuration (This will show which access groups
                             are associated with particular interfaces)

Router# show access list 110 (This will show detailed information
                              about this ACL)
```

Extended Access List Sample #2

Deny/Permit Specific Addresses

Write an extended access list to block the 172.16.70.0 network from receiving information from Mike's computer at 192.168.90.36. Block the lower half of the ip addresses from 192.168.90.0 network from reaching Gail's computer at 172.16.70.32. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: FA0/1
Access-list #: 135

[Writing and installing an ACL]

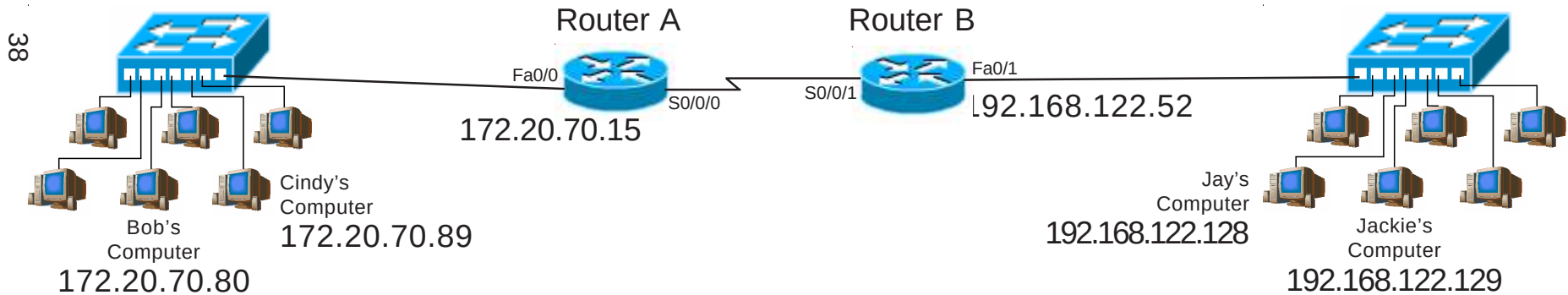
```
Router# configure terminal
Router(config)# access-list 135 deny ip 192.168.90.36 0.0.0.0 172.16.70.0 0.0.0.255
                        or
                        access-list 135 deny ip host 192.168.90.36 172.16.70.0 0.0.0.255
Router(config)# access-list 135 deny ip 192.168.90.0 0.0.0.127 172.16.70.32 0.0.0.0
                        or
                        access-list 135 deny ip 192.168.90.0 0.0.0.127 host 172.16.70.32
Router(config)# access-list 135 permit ip any any
                        or
                        access-list 135 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/1
Router(config-if)# ip access-group 135 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface fa0/1
Router(config-if)# no ip access-group 135 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface fa0/1
Router(config-if)# no ip access-group 135 out
Router(config-if)# exit
Router(config)# no access-list 135
Router(config)# exit
```



Extended Access List Problem #1 Deny/Permit Specific Addresses

Write an extended access list to prevent Jay's computer from receiving information from Cindy's computer. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*

Extended Access List Problem #2

Deny/Permit Specific Addresses

Write an extended access list to block the 172.20.70.0 255.255.255.0 network from receiving information from Jackie's computer at 192.168.122.129. Block the lower half of the ip addresses from 192.168.122.0 network from reaching Cindy's computer at 172.20.70.89. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# _____

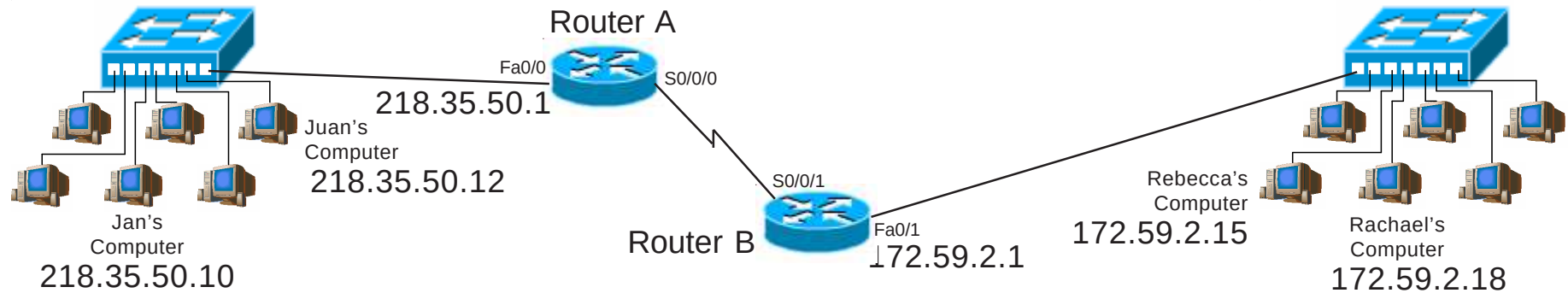
Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #3

Deny/Permit Specific Addresses

Write a named extended access list called "LAB_166" to permit Jan's computer at 218.35.50.10 to receive packets from Rachael's computer at 172.59.2.18; but not Rebecca's computer at 172.59.2.15. Deny all other packets. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list Name: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config-ext-nacl)# _____

Router(config-ext-nacl)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #4

Deny/Permit Specific Addresses

Write an extended access list to allow Juan's computer at 218.35.50.12 to send information to Rebecca's computer at 172.59.2.15; but not Rachael's computer at 172.59.2.18. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)_____

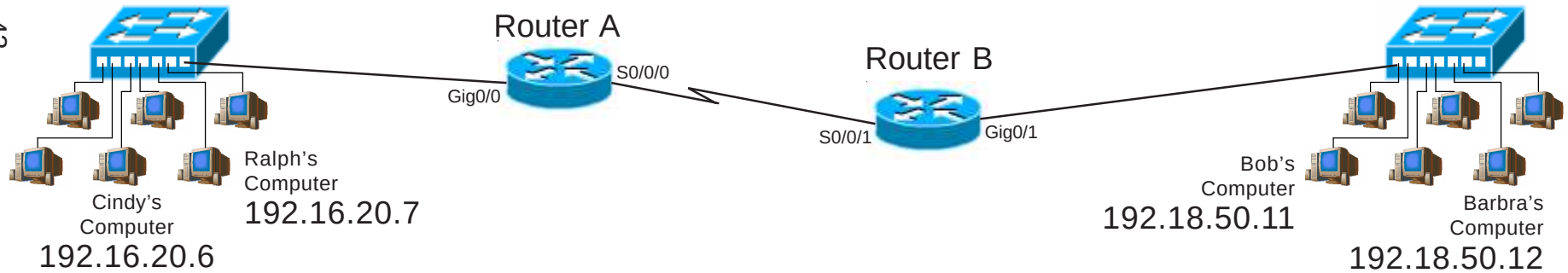
Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router((config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Sample #3

Deny/Permit Entire Ranges

Write an extended access list to permit the 192.16.20.0 network to receive packets from the 192.18.50.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Gig0/1
 Access-list #: 111

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 111 permit ip 192.18.50.0 0.0.0.255 192.16.20.0 0.0.0.255
Router(config)# access-list 111 deny ip any any
or
Router(config)# access-list 111 deny ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface gig0/1
Router(config-if)# ip access-group 111 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

Router# *show configuration* (This will show which access groups are associated with particular interfaces)

Router# *show access list 111* (This will show detailed information about this ACL)

Extended Access List Sample #4

Deny/Permit Entire Ranges

Add a remark to each statement. Write an extended access list to block the 192.18.50.0 network from receiving information from the 192.16.20.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 188

[Remark Command]

The remark command allows you to place text within the ACL so it can be viewed after it is inserted on the router. It can be viewed using the show run or any command that lists the ACEs.

[Writing and installing an ACL]

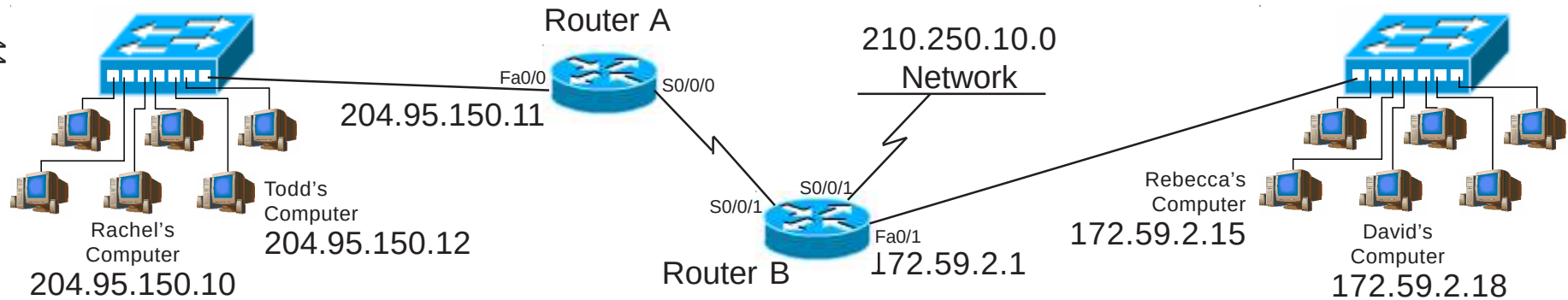
```
Router# configure terminal
Router(config)# access-list 188 remark block all traffic from the Science lab
Router(config)# access-list 188 deny ip 192.16.20.0 0.0.0.255 192.18.50.0 0.0.0.255
Router(config)# access-list 188 remark allow everyone else unrestricted access
Router(config)# access-list 188 permit ip any any
                        or
                        access-list 188 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface gig0/0
Router(config-if)# ip access-group 188 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 188 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 188 out
Router(config-if)# exit
Router(config)# no access-list 188
Router(config)# exit
```



Extended Access List Problem #5

Deny/Permit Entire Ranges

Include a remark with each statement of your ACL. Write an extended access list to permit network 204.95.150.0 to send packets to network 172.59.0.0, but not to the 210.250.10.0 network. Permit all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #6

Deny/Permit Entire Ranges

Write an extended access list to allow Rachel's computer at 204.95.150.10 to receive information from the 172.59.2.0 255.255.255.0 network. Deny all other hosts on the 204.95.150.0 network access from the 172.59.2.0 255.255.255.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# _____

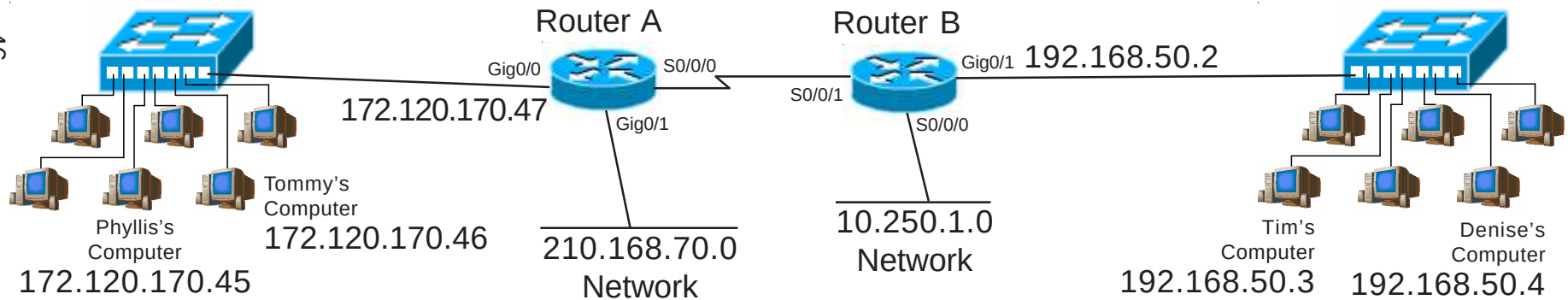
Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #7 Deny/Permit Entire Ranges

Write a named extended access list called "Godzilla" to prevent the 172.120.0.0 network from sending information to the 210.168.70.0, and 10.250.1.0 255.255.255.0 networks; but will permit traffic to the 192.168.50.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list Name: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config-ext-nacl)# _____

Router(config-ext-nacl)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #8

Deny/Permit Entire Ranges

Assuming default subnet masks write an extended access list to permit Tim at 192.168.50.3 to receive data from the 172.120.0.0 network. Allow the 192.168.50.0 network to receive information from Phyllis's computer at 172.120.170.45. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# _____

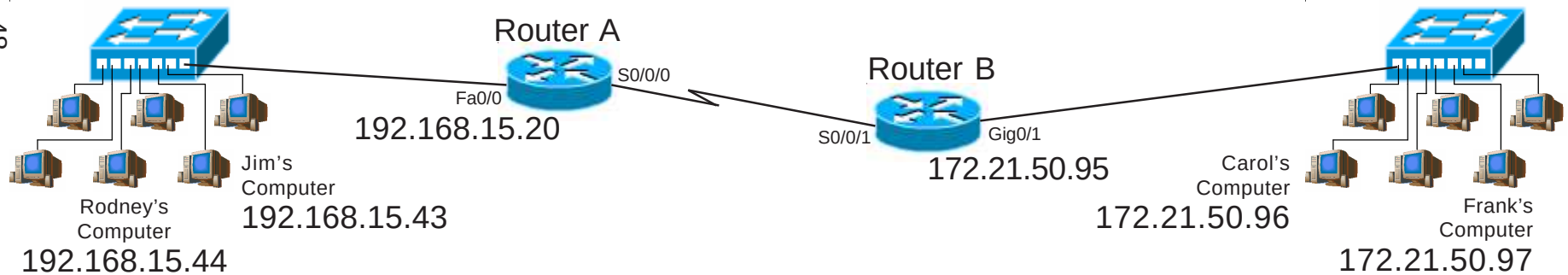
Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Sample #5

Deny/Permit a Range of Addresses

Write an extended access list to deny the first 15 usable addresses of the 192.168.15.0 network from reaching the 172.21.0.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: Fa0/0
 Access-list #: 185

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 185 deny ip 192.168.15.0 0.0.0.15 172.21.50.0 0.0.255.255
Router(config)# access-list 185 permit ip any any
or
Router(config)# access-list 185 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/0
Router(config-if)# ip access-group 185 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

Router# *show configuration* (This will show which access groups are associated with particular interfaces)

Router# *show access list 185* (This will show detailed information about this ACL)

Extended Access List Sample #6

Deny/Permit a Range of Addresses

Write an extended access list which will allow the lower half of 192.168.15.0 network access to the 172.21.50.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Fa0/0
Access-list #: 121

[Writing and installing an ACL]

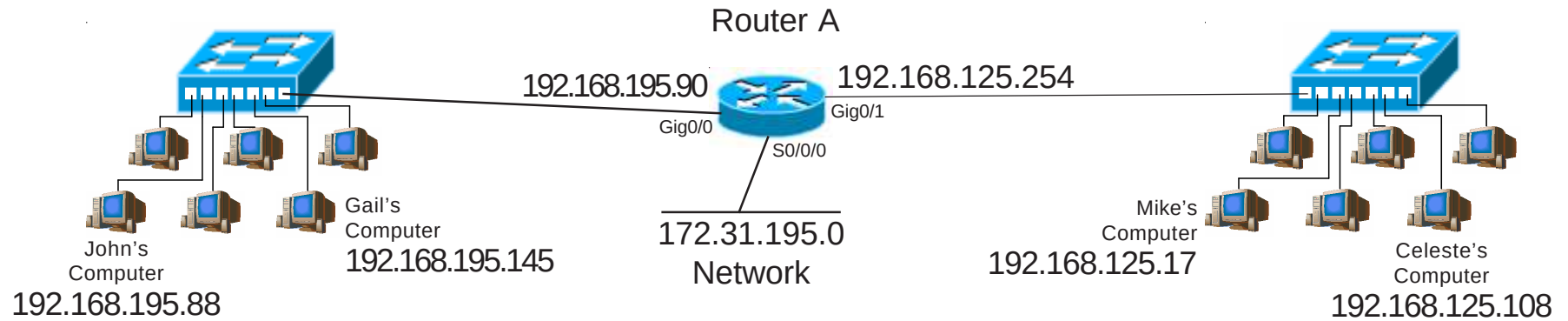
```
Router# configure terminal
Router(config)# access-list 121 permit ip 192.168.15.0 0.0.0.127 172.21.50.0 0.0.0.255
Router(config)# access-list 121 deny ip any any
                        or
                        access-list 121 deny ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/0
Router(config-if)# ip access-group 121 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface fa0/0
Router(config-if)# no ip access-group 121 in
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface fa0/0
Router(config-if)# no ip access-group 121 in
Router(config-if)# exit
Router(config)# no access-list 121
Router(config)# exit
```

Extended Access List Problem #9 Deny/Permit a Range of Addresses

Write an extended access list to prevent the first 31 usable addresses in the 192.168.125.0 network from reaching the 192.168.195.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Extended Access List Problem #10 Deny/Permit a Range of Addresses

Include a remark with each statement of your ACL. Write a named extended access list called "Media_Center" to permit the range of addresses from 172.31.195.1 through 172.31.195.7 to send data to the 192.168.125.0 network. Deny all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list Name: _____

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# _____

Router(config-ext-nacl)# _____

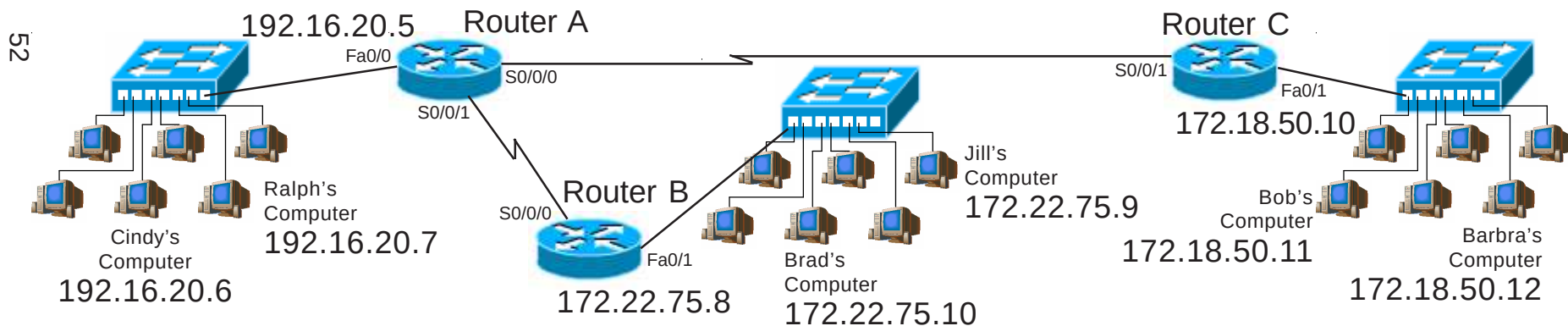
Router(config-ext-nacl)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #11. Deny/Permit a Range of Addresses

Write an extended access list to permit the first 3 usable addresses in the 192.16.20.0 network to reach the 172.22.75.0 network. Deny the addresses from 192.16.20.4 through 192.16.20.31 from reaching the 172.22.75.0 network. Permit all other traffic. Keep in mind that there are multiple ways this ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Extended Access List Problem #12 Deny/Permit a Range of Addresses

Write an extended access list to deny the addresses from 172.22.75.8 through 172.22.75.127 from sending data to the 172.18.50.0 network. Deny the first half of the addresses from the 172.22.75.0 network from reaching the 192.16.20.0 network. Permit all other traffic. Keep in mind that there are multiple ways this ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# _____

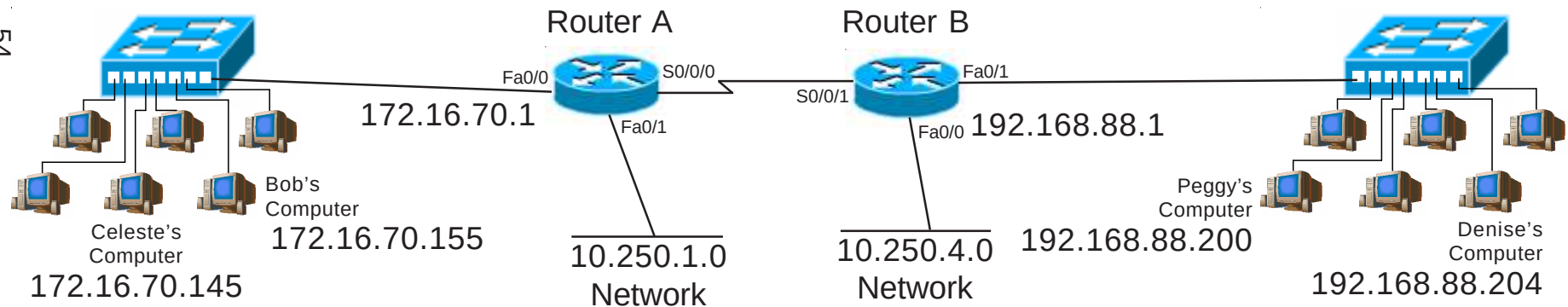
Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #13 Deny/Permit a Range of Addresses

Include a remark with each statement of your ACL. Write an extended access list to permit the first 63 usable addresses in the 192.168.88.0 network to reach the lower half of the addresses in the 172.16.70.0 network; but not the upper half. Deny all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Extended Access List Problem #14 Deny/Permit a Range of Addresses

Write an extended access list to deny the addresses from 10.250.1.0 through 10.250.1.63 from sending data to Denise's computer. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# _____

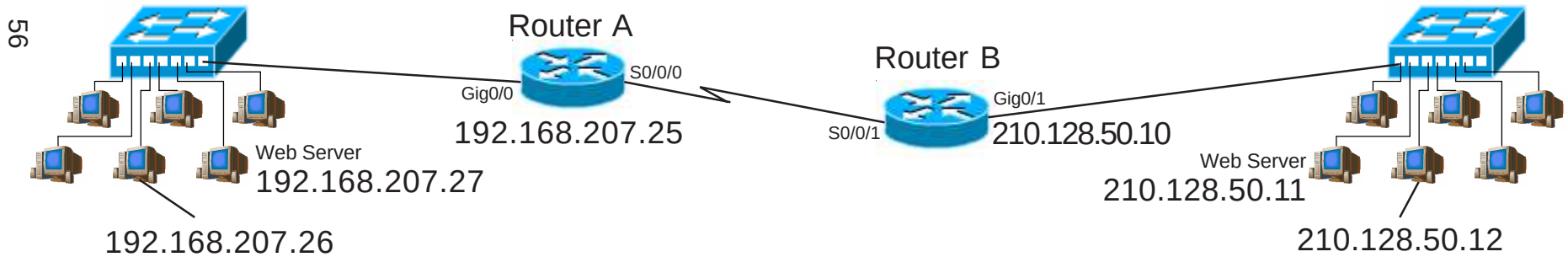
Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Sample #7

Deny/Permit Port Numbers

Write an extended access list to deny HTTP traffic intended for web server 192.168.207.27 from all other networks, but will permit all other HTTP traffic to reach the 192.168.207.0 network. Deny all other IP traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
Interface: Gig0/1
Access-list #: 198

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 198 deny tcp any 192.168.207.27 0.0.0.0 eq www
or
access-list 198 deny tcp any host 192.168.207.27 eq www
Router(config)# access-list 198 permit tcp any 192.168.207.0 0.0.0.255 eq www
Router(config)# interface gig0/1
Router(config-if)# ip access-group 198 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

```
Router# show configuration (This will show which access groups are associated
with particular interfaces)

Router# show access list 198 (This will show detailed information about this ACL)
```

Extended Access List Sample #8

Deny/Permit Port Numbers

Write an extended access list on Router B to deny pings between hosts on the 210.128.50.0 and the 192.168.207.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
Interface: Gig0/1
Access-list #: 134

[Writing and installing an ACL]

```
Router# configure terminal
Router(config)# access-list 134 deny icmp 210.128.50.0 0.0.0.255 192.168.207.0 0.0.0.255
Router(config)# access-list 134 permit ip any any
Router(config)# interface gig0/1
Router(config-if)# ip access-group 134 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

Hint:

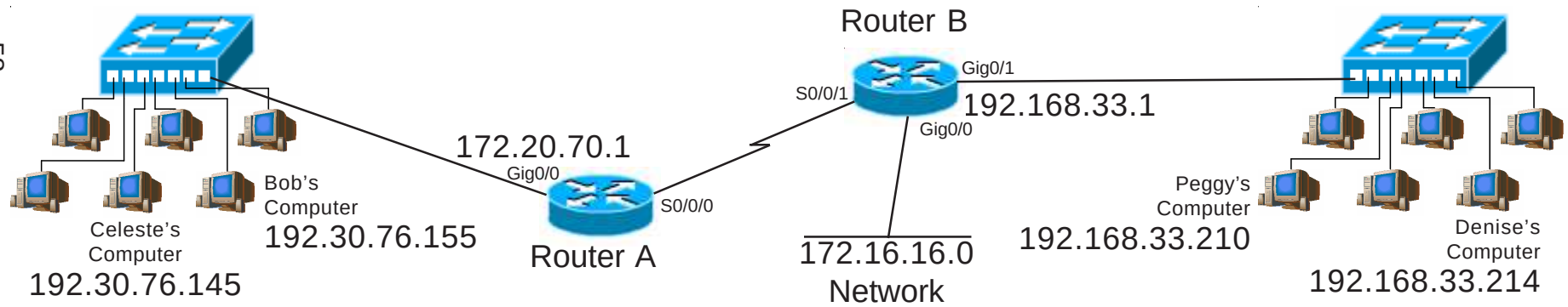
It's OK to use multiple protocols in the same ACL.

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface Gig0/1
Router(config-if)# no ip access-group 134 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface Gig0/1
Router(config-if)# no ip access-group 134 out
Router(config-if)# exit
Router(config)# no access-list 134
Router(config)# exit
```

Extended Access List Sample #9

Deny/Permit Port Numbers

Write an Extended access list to permit Denise's computer to use TFTP with Bob's computer. Deny all other traffic from the 192.168.33.0 network to the 192.30.76.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Gig0/1
 Access-list #: 145

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 145 permit udp 192.168.33.214 0.0.0.0 192.30.76.155 0.0.0.0 eq tftp
or
access-list 145 permit udp host 192.168.33.214 host 192.30.76.155 eq tftp
Router(config)# interface Gig0/1
Router(config-if)# ip access-group 145 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

Router# *show configuration* (This will show which access groups are associated with particular interfaces)

Router# *show access list 45* (This will show detailed information about this ACL)

Extended Access List Sample #10

Deny/Permit Port Numbers

Write an extended access list to deny FTP traffic from ip addresses 192.30.76.0 through 192.30.76.13 to any destination. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 155

[Writing and installing an ACL]

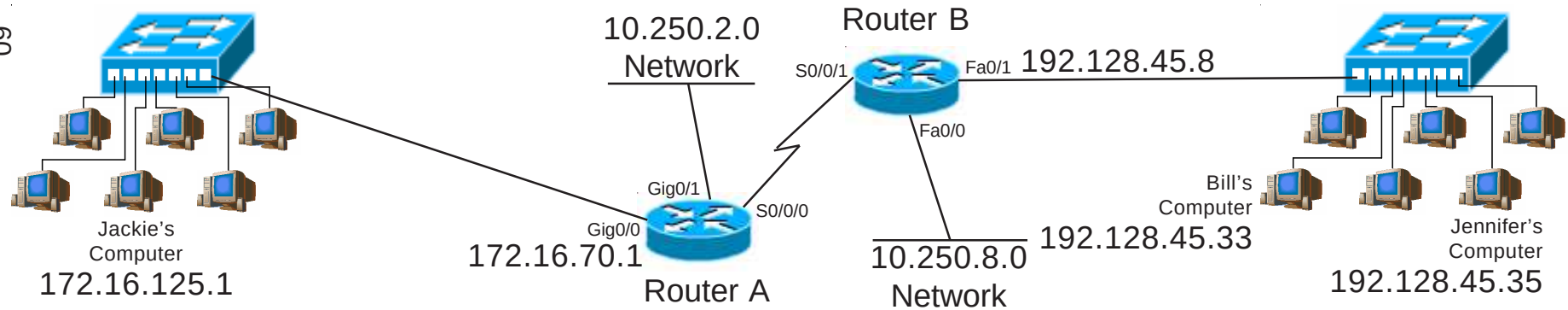
```
Router# configure terminal
Router(config)# access-list 155 deny tcp 192.30.76.0 0.0.0.8 any eq ftp (Blocks 0 to 7)
Router(config)# access-list 155 deny tcp 192.30.76.8 0.0.0.4 any eq ftp (Blocks 8 to 11)
Router(config)# access-list 155 deny tcp 192.30.76.12 0.0.0.1 any eq ftp (Blocks 12 to 13)
Router(config)# access-list 155 permit ip any any
                        or
                        access-list 155 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface gig0/0
Router(config-if)# ip access-group 155 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

Hint:

It's OK to use multiple protocols in the same ACL.

The first three TCP statements could be shortened to:

```
access-list 155 permit tcp 192.30.76.14 0.0.0.1 any eq ftp (Permits 14 and 15)
access-list 155 deny tcp 192.30.76.8 0.0.0.15 any eq ftp (Blocks 0 to 15)
```



Extended Access List Problem #15 Deny/Permit a Port Numbers

Write an extended access list to permit ICMP traffic from the 192.128.45.0 network to reach the 172.16.125.0 255.255.255.0 and 10.250.2.0 255.255.255.0 networks. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Extended Access List Problem #16 Deny/Permit a Port Numbers

Write a named extended access list called "PEGGYS_LAB" to deny telnet from 10.250.8.0 through 10.250.8.127 from reaching the 192.128.45.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list Name: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config-std-nacl)# _____

Router(config-std-nacl)# _____

Router(config-ext-nacl)# *interface* _____

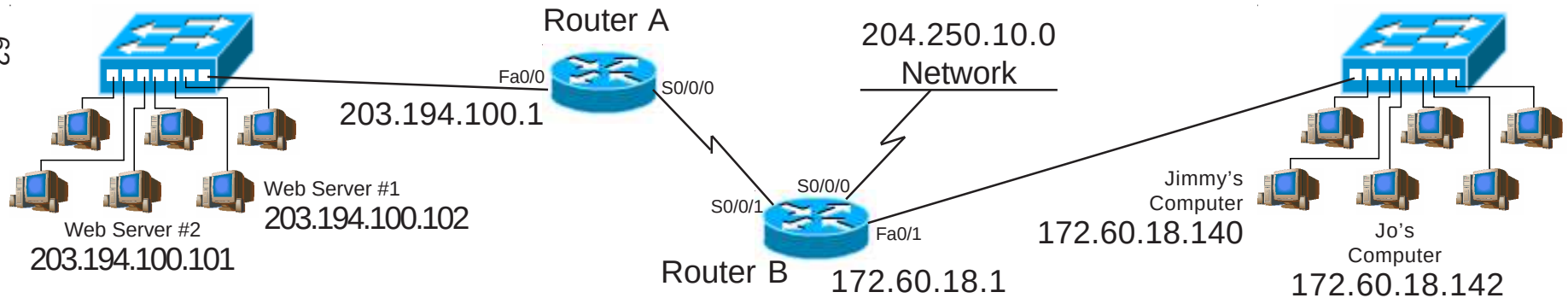
Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*

Router# *copy run start*



Extended Access List Problem #17 Deny/Permit Port Numbers

Write an access list to deny Jimmy's computer from sending ftp packets to Web Server 1, but permit ftp to Web Server #2. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #18

Deny/Permit Port Numbers

Write an extended access list to deny all HTTP traffic intended for the web server at 203.194.100.102 from the 172.66.0.0 network. Permit all other HTTP traffic from the 204.250.10.0 and 172.60.0.0 networks to any other web servers. Deny all other IP traffic to the 203.194.100.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

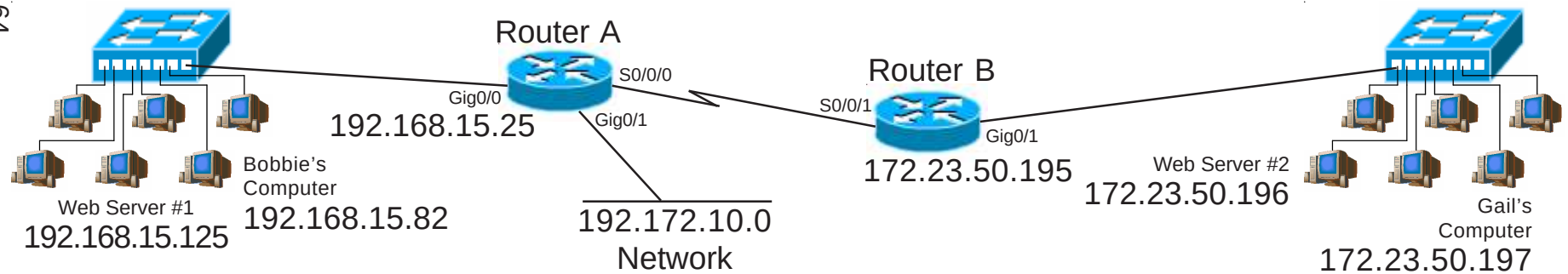
Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #19

Deny/Permit Port Numbers

Include a remark with each statement of your ACL. Write an extended access list to permit TFTP traffic from all hosts on the 192.168.15.0 network. Deny all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #20

Deny/Permit Port Numbers

Write an extended access list that permits web traffic from web server #2 at 172.23.50.196 to reach everyone on the 192.168.15.0 network. Deny all other IP traffic going to the 192.172.10.0, and 192.168.15.0 networks from the 172.25.50.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# _____

Router(config)# *interface* _____

Router(config-if)# *ip access-group* _____ *in or out* (circle one)

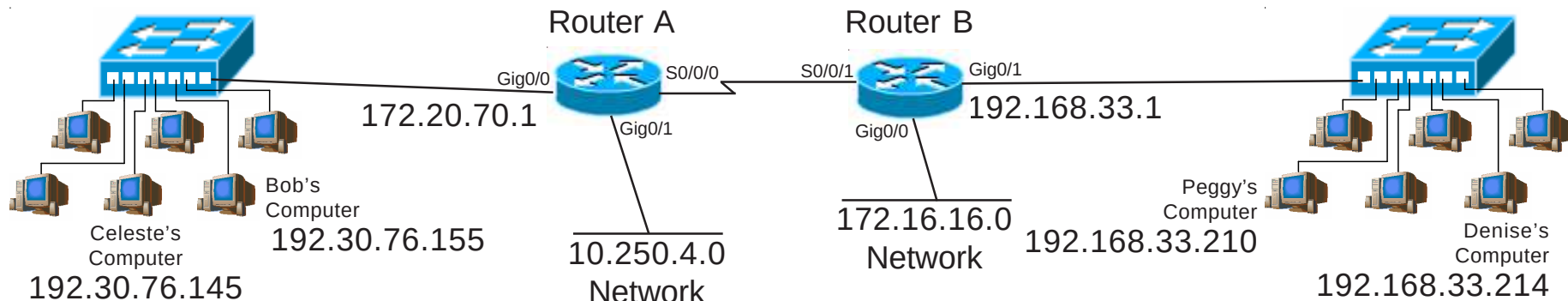
Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*

Writing Access Lists to Restrict Telnet Access...

Restricting access to telnet can be a very usefull option. Telnet is considered a very insecure protocol because it sends passwords through the network in clear-text. By switching from the *access-group* command to the *access-class* command you can increase your security by allowing only those users through that you want to use telnet. The *access-class* command also allows you to apply this access list to the vty connections.



Standard Access List Sample #11 Deny/Permit Telnet

Write a standard access list to permit Denise's and Bob's computers to telnet into Router B. Deny all other telnet traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B

Interface: line VTY 0 4

Access-list #: 45

(using line VTY 0 4 instead of an interface like E1 allows you to apply this access list to all VTY lines with one statement)

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 45 permit 192.168.33.214 0.0.0.0
```

or

```
access-list 45 permit host 192.168.33.214
Router(config)# access-list 45 permit 192.30.76.155 0.0.0.0
```

or

```
access-list 45 permit host 192.30.76.155
```

```
Router(config)# line vty 0 4
Router(config-line)# access-class 45 in
```

```
Router(config-line)# exit
```

```
Router(config)# exit
```

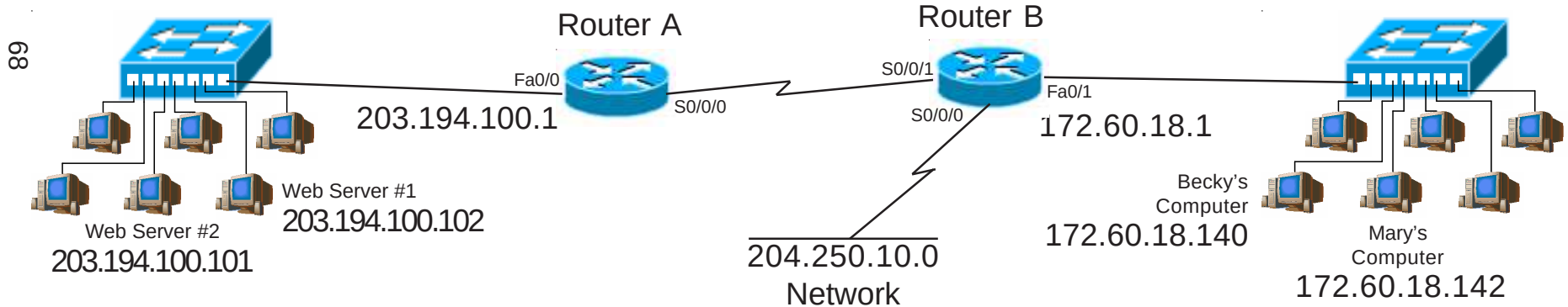
[Viewing information about existing ACL's]

```
Router# show configuration
```

(This will show which access groups are associated with particular interfaces)

```
Router# show access list 45
```

(This will show detailed information about this ACL)



Access List Problem #21 Deny/Permit Telnet

Write a standard access list to permit Becky and Mary's computer to telnet into Router B. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# _____

Router(config-line)# *access-class* _____ *in or out* (circle one)

Router(config-line)# *exit*

Router(config)# *exit*

Access List Problem #22

Deny/Permit Telnet

Write a standard access list to permit which will permit Web Server #1 to telnet into Router A. Log the telnet attempts. Deny all other telnet access. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

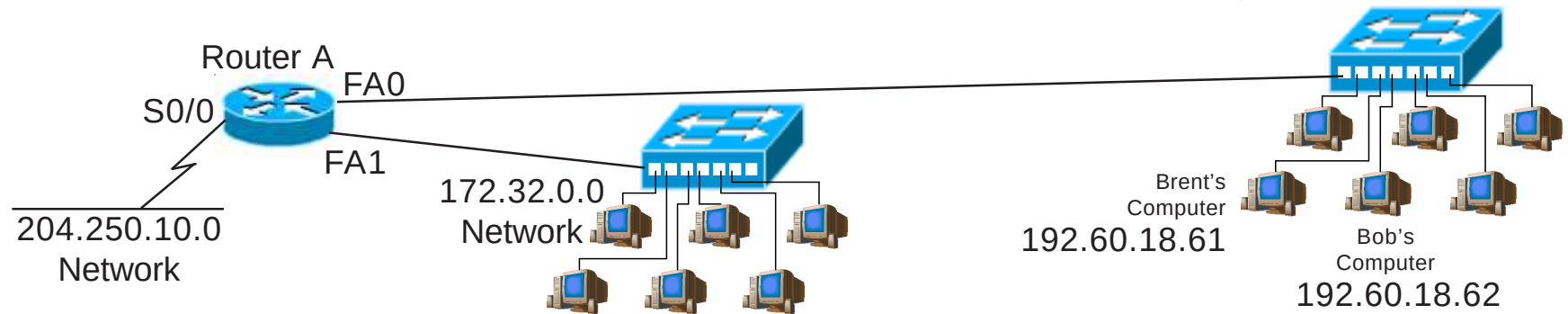
Router(config)# _____

Router(config)# _____

Router(config-line)# *access-class* _____ *in or out* (circle one)

Router(config-line)# *exit*

Router(config)# *exit*



Access List Problem #23

Deny/Permit Telnet

Write a standard access list to deny Brent and Bob's computer telnet access to into Router A. Permit all other telnet traffic from the 192.60.18.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: _____

Interface: _____

Access-list #: _____

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# _____

Router(config)# _____

Router(config-line)# *access-class* _____ *in or out* (circle one)

Router(config-line)# *exit*

Router(config)# *exit*

Reference

Port Numbers

Port numbers are now assigned by the ICANN (Internet Corporation for Assigned Names and Numbers). Commonly used TCP and UDP applications are assigned a port number; such as: HTTP - 80, POP3 - 110, FTP - 20. When an application communicates with another application on another node on the internet, it specifies that application in each data transmission by using its port number. You can also type the name (ie. Telnet) instead of the port number (ie. 23). Port numbers range from 0 to 65536 and are divided into three ranges:

Well Known Ports	0 to 1,023
Registered Ports	1,024 to 49,151
Dynamic and/or Private Ports	49,152 to 65,535

Below is a short list of some commonly used ports. For a complete list of port numbers go to <http://www.iana.org/assignments/port-numbers>.

Some commonly used port numbers:

0	Reserved	
1	TCPMUX	(TCP Port Service Multiplexer)
5	RJE	(Remote Job Entry)
7	ECHO	
9	DISCARD	
11	SYSTAT	(Active users)
13	DAYTIME	
17	QUOTE	(Quote of the day)
18	MSP	(Message Send Protocol)
19	CHARGEN	(Character generator)
20	FTP-DATA	(File Transfer Protocol - Data)
21	FTP	(File Transfer Protocol - Control)
22	SSH	(Remote Login Protocol)
23	Telnet	(Terminal Connection)
25	SMTP	(Simple Mail Transfer Protocol)
29	MSG ICP	
37	TIME	
39	RLP	(Resource Location Protocol)
42	NAMESERV	(Host Name Server)

43	NICNAME	(Who Is)
49	LOGIN	(Login Host Protocol)
53	DNS	(Domain Name Server)
67	BOOTP	(Bootstrap Protocol Server)
68	BOOTPS	(Bootstrap Protocol Client)
69	TFTP	(Trivial File Transfer Protocol)
70	GOPHER	(Gopher Services)
75		(Any Private Dial-out Service)
79	FINGER	
80	HTTP	(Hypertext Transfer Protocol)
95	SUPDUP	(SUPDUP Protocol)
101	HOSTNAME	(NIC Host Name Server)
108	SNAGAS	(SNA Gateway Access Server)
109	POP2	(Post Office Protocol - Version 2)
110	POP3	(Post Office Protocol - Version 3)
113	AUTH	(Authentication Service)
115	SFTP	(Simple File Transfer Protocol)
117	UUCP-PATH	(UUCP Path Service)
118	SQLSERV	(SQL Services)
119	NNTP	(Newsgroup)
123	NTP	(Network Time Protocol)
137	NetBIOS-NS	(NetBIOS Name Service)
139	NetBIOS-SSN	(NetBIOS Session Service)
143	IMAP	(Interim Mail Access Protocol)
150	SQL-NET	(NetBIOS Session Service)
156	SQLSRV	(SQL Service)
161	SNMP	(Simple Network Management Protocol)
179	BGP	(Border Gateway Protocol)
190	GACP	(Gateway Access Control Protocol)
194	IRC	(Internet Relay Chat)
197	DLS	(Directory Location Service)
389	LDAP	(Lightweight Directory Access Protocol)
396	NETWARE-IP	(Novell Netware over IP)
443	HTTPS	(HTTP MCom)
444	SNPP	(Simple Network Paging Protocol)
445	Microsoft-DS	
458	Apple QuickTime	
546	DHCP Client	
547	DHCP Server	
563	SNEWS	
569	MSN	

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31	16-23	8-11	4-5
					24-31	12-15	6-7
						16-19	8-9
						20-23	10-11
		64-127	32-63	32-47	32-39	24-27	12-13
					40-47	28-31	14-15
				48-63	48-55	32-35	16-17
					56-63	36-39	18-19
						40-43	20-21
						44-47	22-23
			64-95	64-79	64-71	48-51	24-25
					72-79	52-55	26-27
				80-95	80-87	56-59	28-29
					88-95	60-63	30-31
						64-67	32-33
			96-127	96-111	96-103	68-71	34-35
					104-111	72-75	36-37
				112-127	112-119	76-79	38-39
					120-127	80-83	40-41
						84-87	42-43
						88-91	44-45
	128-255	128-191	128-159	128-143	128-135	92-95	46-47
					136-143	96-99	48-49
				144-159	144-151	100-103	50-51
					152-159	104-107	52-53
						108-111	54-55
						112-115	56-57
			160-191	160-175	160-167	116-119	58-59
					168-175	120-123	60-61
				176-191	176-183	124-127	62-63
					184-191	128-131	64-65
						132-135	66-67
			192-255	192-207	192-199	136-139	68-69
					200-207	140-143	70-71
					208-215	144-147	72-73
				208-223	216-223	148-151	74-75
						152-155	76-77
						156-159	78-79
						160-163	80-81
			224-255	224-239	224-231	164-167	82-83
					232-239	168-171	84-85
				240-255	240-247	172-175	86-87
					248-255	176-179	88-89
						180-183	90-91
						184-187	92-93
						188-191	94-95
						192-195	96-97
						196-199	98-99
						200-203	100-101
						204-207	102-103
						208-211	104-105
						212-215	106-107
						216-219	108-109
						220-223	110-111
						224-227	112-113
						228-231	114-115
						232-235	116-117
						236-239	118-119
						240-243	120-121
						244-247	122-123
						248-251	124-125
						252-255	126-127
							128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

Class B Addresses

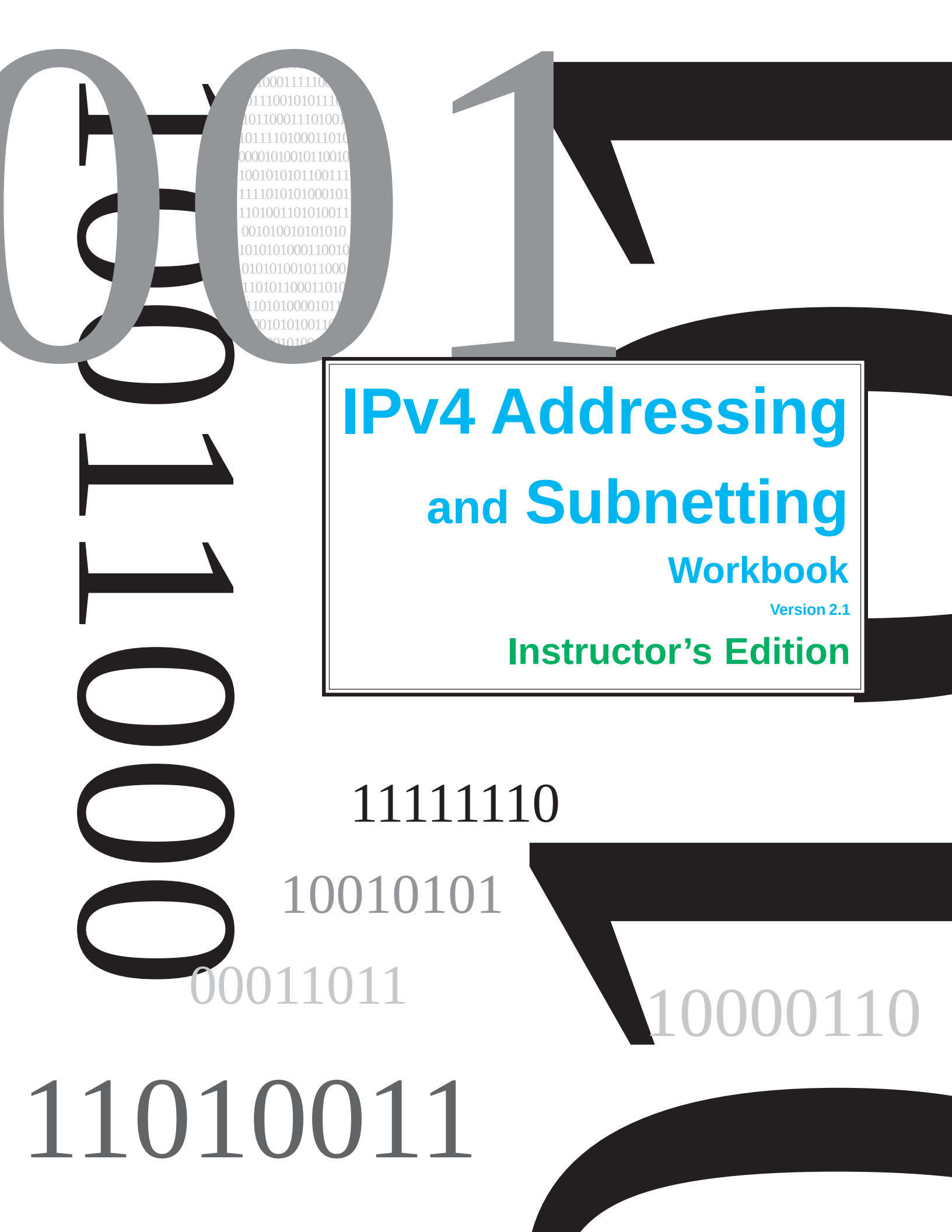
VLSM Chart 16-23 Bits (3rd octet)

/16	/17	/18	/19	/20	/21	/22	/23		
255.255.0.0 65,536 Hosts	255.255.128.0 32,768 Hosts	255.255.192.0 16,384 Hosts	255.255.224.0 8,192 Hosts	255.255.240.0 4,096 Hosts	255.255.248.0 2,048 Hosts	255.255.252.0 1,024 Hosts	255.255.254.0 512 Hosts		
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1		
					8-15	4-7	2-3		
				16-31	16-23	8-11	4-5		
					24-31	12-15	6-7		
					32-63	32-47	16-19	18-19	10-11
							20-23	14-15	12-13
							24-27	16-17	14-15
							28-31	18-19	16-17
			48-63	32-35		20-21	18-19		
				36-39		22-23	20-21		
				40-43		24-25	22-23		
				44-47		26-27	24-25		
			64-95	64-79	48-51	28-29	26-27		
					52-55	30-31	28-29		
					56-59	32-33	30-31		
					60-63	34-35	32-33		
				80-95	64-67	36-37	34-35		
					68-71	38-39	36-37		
					72-75	40-41	38-39		
					76-79	42-43	40-41		
			64-127	96-127	96-111	80-83	44-45	42-43	
						84-87	46-47	44-45	
						88-91	48-49	46-47	
						92-95	50-51	48-49	
		112-127			96-99	52-53	50-51		
					100-103	54-55	52-53		
					104-107	56-57	54-55		
					108-111	58-59	56-57		
		128-159		128-143	112-115	60-61	60-61		
					116-119	62-63	62-63		
					120-123	64-65	64-65		
					124-127	66-67	66-67		
				144-159	128-131	68-69	68-69		
					132-135	70-71	70-71		
					136-139	72-73	72-73		
					140-143	74-75	74-75		
		128-191	160-191	160-175	144-147	76-77	76-77		
					148-151	78-79	78-79		
					152-155	80-81	80-81		
					156-159	82-83	82-83		
				176-191	160-163	84-85	84-85		
					164-167	86-87	86-87		
					168-171	88-89	88-89		
					172-175	90-91	90-91		
			192-223	192-207	176-179	92-93	92-93		
					180-183	94-95	94-95		
					184-187	96-97	96-97		
					188-191	98-99	98-99		
				208-223	192-195	100-101	100-101		
					196-199	102-103	102-103		
					200-203	104-105	104-105		
					204-207	106-107	106-107		
			192-255	224-255	224-239	208-211	108-109	108-109	
						212-215	110-111	110-111	
						216-219	112-113	112-113	
						220-223	114-115	114-115	
					240-255	224-227	116-117	116-117	
						228-231	118-119	118-119	
						232-235	120-121	120-121	
						236-239	122-123	122-123	
		248-255		240-243	124-125	124-125			
				244-247	126-127	126-127			
				248-249	128-129	128-129			
				252-253	130-131	130-131			
	250-251			132-133	132-133				
	252-253			134-135	134-135				
	254-255			136-137	136-137				
	256-257			138-139	138-139				

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255



IPv4 Addressing and Subnetting Workbook

Version 2.1

Instructor's Edition

11111110

10010101

00011011

10000110

11010011

IPv4 Address Classes

Class A	1 – 127	Leading bit pattern	0	00000000.00000000.00000000.00000000 Network . Host . Host . Host
Class B	128 – 191	Leading bit pattern	10	10000000.00000000.00000000.00000000 Network . Network . Host . Host
Class C	192 – 223	Leading bit pattern	110	11000000.00000000.00000000.00000000 Network . Network . Network . Host
Class D	224 – 239	(Reserved for multicast)		
Class E	240 – 255	(Reserved for experimental, used for research)		

Speciality Address Ranges

Loopback -	Only the single 127.0.0.1 address is used, addresses 127.0.0.0 to 127.255.255.255 are reserved. Any address within this block will loop back to the local host.
Link-Local Addresses -	IPv4 addresses in the address block 169.254.0.0 to 169.254.255.255 (169.254.0.0/16) are designated as link-local addresses.
TEST-NET Addresses -	The address block 192.0.2.0 to 192.0.2.255 (192.0.2.0/24) is set aside for teaching and learning purposes.
Experimental Addresses -	The addresses in the block 240.0.0.0 to 255.255.255.254 are listed as reserved for future use (RFC 3330).

Private Address Space

Class A	10.0.0.0 to 10.255.255.255
Class B	172.16.0.0 to 172.31.255.255
Class C	192.168.0.0 to 192.168.255.255

Default Subnet Masks

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

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Special Thanks to Melvin Baker and Jim Dorsch for taking the time to check this workbook for errors,
and to everyone who has sent in suggestions to improve the series.
Inside Cover

Binary To Decimal Conversion

128	64	32	16	8	4	2	1	Answers	Scratch Area
1	0	0	1	0	0	1	0	<u>146</u>	<u>128</u> 64
0	1	1	1	0	1	1	1	<u>119</u>	16 32
1	1	1	1	1	1	1	1	<u>255</u>	<u>2</u> 16
1	1	0	0	0	1	0	1	<u>197</u>	<u>146</u> 4
1	1	1	1	0	1	1	0	<u>246</u>	2
0	0	0	1	0	0	1	1	<u>19</u>	<u>1</u> 119
1	0	0	0	0	0	0	1	<u>129</u>	
0	0	1	1	0	0	0	1	<u>49</u>	
0	1	1	1	1	0	0	0	<u>120</u>	
1	1	1	1	0	0	0	0	<u>240</u>	
0	0	1	1	1	0	1	1	<u>59</u>	
0	0	0	0	0	1	1	1	<u>7</u>	
							00011011	<u>27</u>	
							10101010	<u>170</u>	
							01101111	<u>111</u>	
							11111000	<u>248</u>	
							00100000	<u>32</u>	
							01010101	<u>85</u>	
							00111110	<u>62</u>	
							00000011	<u>3</u>	
							11101101	<u>237</u>	
							11000000	<u>192</u>	

Decimal To Binary Conversion

Use all 8 bits for each problem

128	64	32	16	8	4	2	1 =	255	Scratch Area	
1	1	1	0	1	1	1	0	238	238	34
0	0	1	0	0	0	1	0	34	-128	-32
0	1	1	1	1	0	1	1	123	110	2
0	0	1	1	0	0	1	0	50	-64	-2
1	1	1	1	1	1	1	1	255	46	0
1	1	0	0	1	0	0	0	200	-32	
0	0	0	0	1	0	1	0	10	14	
1	0	0	0	1	0	1	0	138	-8	
0	0	0	0	0	0	0	1	1	6	
0	0	0	0	1	1	0	1	13	-4	
1	1	1	1	1	0	1	0	250	2	
0	1	1	0	1	0	1	1	107	-2	
1	1	1	0	0	0	0	0	224	0	
0	1	1	1	0	0	1	0	114		
1	1	0	0	0	0	0	0	192		
1	0	1	0	1	1	0	0	172		
0	1	1	0	0	1	0	0	100		
0	1	1	1	0	1	1	1	119		
0	0	1	1	1	0	0	1	57		
0	1	1	0	0	0	1	0	98		
1	0	1	1	0	0	1	1	179		
0	0	0	0	0	0	1	0	2		

Address Class Identification

Address	Class
10.250.1.1	<u>A</u>
150.10.15.0	<u>B</u>
192.14.2.0	<u>C</u>
148.17.9.1	<u>B</u>
193.42.1.1	<u>C</u>
126.8.156.0	<u>A</u>
220.200.23.1	<u>C</u>
230.230.45.58	<u>D</u>
177.100.18.4	<u>B</u>
119.18.45.0	<u>A</u>
249.240.80.78	<u>E</u>
199.155.77.56	<u>C</u>
117.89.56.45	<u>A</u>
215.45.45.0	<u>C</u>
199.200.15.0	<u>C</u>
95.0.21.90	<u>A</u>
33.0.0.0	<u>A</u>
158.98.80.0	<u>B</u>
219.21.56.0	<u>C</u>

Network & Host Identification

Circle the network portion
of these addresses:

177.100.18.4

119.18.45.0

209.240.80.78

199.155.77.56

117.89.56.45

215.45.45.0

192.200.15.0

95.0.21.90

33.0.0.0

158.98.80.0

217.21.56.0

10.250.1.1

150.10.15.0

192.14.2.0

148.17.9.1

193.42.1.1

126.8.156.0

220.200.23.1

Circle the host portion of
these addresses:

10.15.123.50

171.2.199.31

198.125.87.177

223.250.200.222

17.45.222.45

126.201.54.231

191.41.35.112

155.25.169.227

192.15.155.2

123.102.45.254

148.17.9.155

100.25.1.1

195.0.21.98

25.250.135.46

171.102.77.77

55.250.5.5

218.155.230.14

10.250.1.1

Network Addresses

Using the IP address and subnet mask shown write out the network address:

188.10.18.2 255.255.0.0	<u>188 . 10 . 0 . 0</u>
10.10.48.80 255.255.255.0	<u>10 . 10 . 48 . 0</u>
192.149.24.191 255.255.255.0	<u>192 . 149 . 24 . 0</u>
150.203.23.19 255.255.0.0	<u>150 . 203 . 0 . 0</u>
10.10.10.10 255.0.0.0	<u>10 . 0 . 0 . 0</u>
186.13.23.110 255.255.255.0	<u>186 . 13 . 23 . 0</u>
223.69.230.250 255.255.0.0	<u>223 . 69 . 0 . 0</u>
200.120.135.15 255.255.255.0	<u>200 . 120 . 135 . 0</u>
27.125.200.151 255.0.0.0	<u>27 . 0 . 0 . 0</u>
199.20.150.35 255.255.255.0	<u>199 . 20 . 150 . 0</u>
191.55.165.135 255.255.255.0	<u>191 . 55 . 165 . 0</u>
28.212.250.254 255.255.0.0	<u>28 . 212 . 0 . 0</u>

Host Addresses

Using the IP address and subnet mask shown write out the host address:

188.10.18.2 255.255.0.0	<u>0 . 0 . 18 . 2</u>
10.10.48.80 255.255.255.0	<u>0 . 0 . 0 . 80</u>
222.49.49.11 255.255.255.0	<u>0 . 0 . 0 . 11</u>
128.23.230.19 255.255.0.0	<u>0 . 0 . 230 . 19</u>
10.10.10.10 255.0.0.0	<u>0 . 10 . 10 . 10</u>
200.113.123.11 255.255.255.0	<u>0 . 0 . 0 . 11</u>
223.169.23.20 255.255.0.0	<u>0 . 0 . 23 . 20</u>
203.20.35.215 255.255.255.0	<u>0 . 0 . 0 . 215</u>
117.15.2.51 255.0.0.0	<u>0 . 15 . 2 . 51</u>
199.120.15.135 255.255.255.0	<u>0 . 0 . 0 . 135</u>
191.55.165.135 255.255.255.0	<u>0 . 0 . 0 . 135</u>
48.21.25.54 255.255.0.0	<u>0 . 0 . 25 . 54</u>

Default Subnet Masks

Write the correct default subnet mask for each of the following addresses:

177.100.18.4	<u>255 . 255 . 0 . 0</u>
119.18.45.0	<u>255 . 0 . 0 . 0</u>
191.249.234.191	<u>255 . 255 . 0 . 0</u>
223.23.223.109	<u>255 . 255 . 255 . 0</u>
10.10.250.1	<u>255 . 0 . 0 . 0</u>
126.123.23.1	<u>255 . 0 . 0 . 0</u>
223.69.230.250	<u>255 . 255 . 255 . 0</u>
192.12.35.105	<u>255 . 255 . 255 . 0</u>
77.251.200.51	<u>255 . 0 . 0 . 0</u>
189.210.50.1	<u>255 . 255 . 0 . 0</u>
88.45.65.35	<u>255 . 0 . 0 . 0</u>
128.212.250.254	<u>255 . 255 . 0 . 0</u>
193.100.77.83	<u>255 . 255 . 255 . 0</u>
125.125.250.1	<u>255 . 0 . 0 . 0</u>
1.1.10.50	<u>255 . 0 . 0 . 0</u>
220.90.130.45	<u>255 . 255 . 255 . 0</u>
134.125.34.9	<u>255 . 255 . 0 . 0</u>
95.250.91.99	<u>255 . 0 . 0 . 0</u>

ANDING with Default subnet masks

Every IP address must be accompanied by a subnet mask. By now you should be able to look at an IP address and tell what class it is. Unfortunately your computer doesn't think that way. For your computer to determine the network and subnet portion of an IP address it must "AND" the IP address with the subnet mask.

Default Subnet Masks:

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

ANDING Equations:

1 AND 1	= 1
1 AND 0	= 0
0 AND 1	= 0
0 AND 0	= 0

Sample:

What you see...

IP Address: 192 . 100 . 10 . 33

What you can figure out in your head...

Address Class:	C
Network Portion:	<u>192 . 100 . 10</u> . 33
Host Portion:	192 . 100 . 10 . <u>33</u>

In order for your computer to get the same information it must AND the IP address with the subnet mask in binary.

	Network	Host	
IP Address:	1 1 0 0 0 0 0 0 . 0 1 1 0 0 1 0 0 . 0 0 0 0 1 0 1 0	0 0 1 0 0 0 0 1	(192 . 100 . 10 . 33)
Default Subnet Mask:	1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1	0 0 0 0 0 0 0 0	(255 . 255 . 255 . 0)
AND:	1 1 0 0 0 0 0 0 . 0 1 1 0 0 1 0 0 . 0 0 0 0 1 0 1 0	0 0 0 0 0 0 0 0	(192 . 100 . 10 . 0)

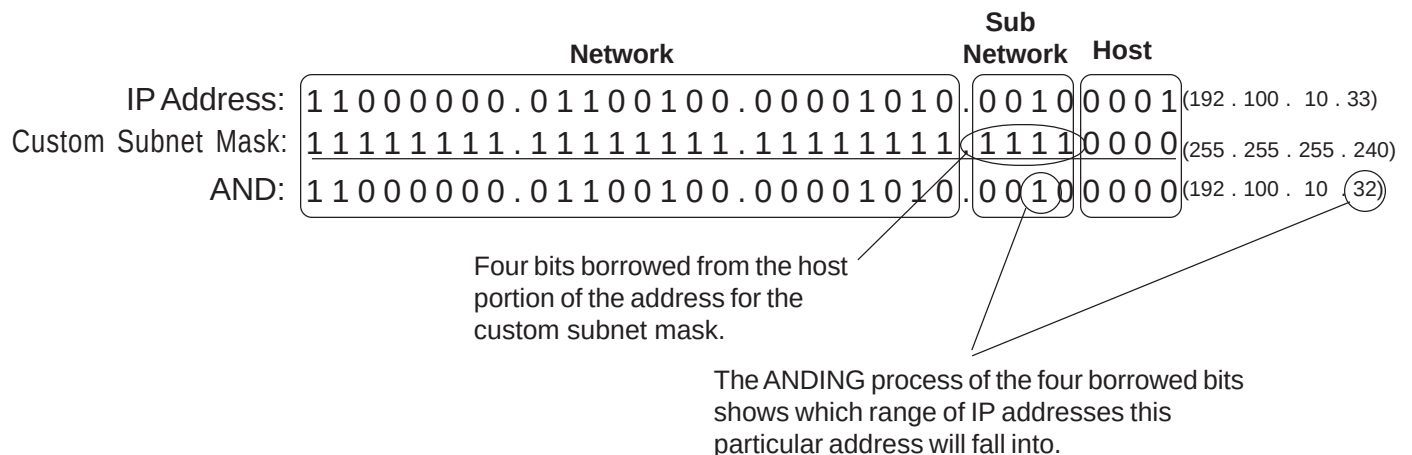
ANDING with the default subnet mask allows your computer to figure out the network portion of the address.

ANDING with Custom subnet masks

When you take a single network such as 192.100.10.0 and divide it into five smaller networks (192.100.10.16, 192.100.10.32, 192.100.10.48, 192.100.10.64, 192.100.10.80) the outside world still sees the network as 192.100.10.0, but the internal computers and routers see five smaller subnetworks. Each independent of the other. This can only be accomplished by using a custom subnet mask. A custom subnet mask borrows bits from the host portion of the address to create a subnetwork address between the network and host portions of an IP address. In this example each range has 14 usable addresses in it. The computer must still AND the IP address against the custom subnet mask to see what the network portion is and which subnetwork it belongs to.

IP Address: 192 . 100 . 10 . 0
Custom Subnet Mask: 255.255.255.240

Address Ranges: 192.10.10.0 to 192.100.10.15
 192.100.10.16 to 192.100.10.31
 192.100.10.32 to 192.100.10.47 (Range in the sample below)
 192.100.10.48 to 192.100.10.63
 192.100.10.64 to 192.100.10.79
 192.100.10.80 to 192.100.10.95
 192.100.10.96 to 192.100.10.111
 192.100.10.112 to 192.100.10.127
 192.100.10.128 to 192.100.10.143
 192.100.10.144 to 192.100.10.159
 192.100.10.160 to 192.100.10.175
 192.100.10.176 to 192.100.10.191
 192.100.10.192 to 192.100.10.207
 192.100.10.208 to 192.100.10.223
 192.100.10.224 to 192.100.10.239
 192.100.10.240 to 192.100.10.255



In the next set of problems you will determine the necessary information to determine the correct subnet mask for a variety of IP addresses.

How to determine the number of subnets and the number of hosts per subnet

The formula that can provide this basic information:

$$\text{Number of subnets} = 2^s$$

$$\text{Number of usable hosts per subnet} = 2^h - 2$$

This formula calculates the number of hosts or subnets based on the number of binary bits used. For example if you borrow three bits from the host portion of the address use the *number of subnets* formula to determine the total number of subnets gained by borrowing the three bits. This would be 2^3 or $2 \times 2 \times 2 = 8$ subnets

To determine the number of hosts per subnet you would take the number of binary bits used in the host portion and apply this to the *number of hosts per subnet* formula. If five bits are in the host portion of the address this would be 2^5 or $2 \times 2 \times 2 \times 2 \times 2 = 32$ hosts.

When dealing with the *number of hosts per subnet* you have to subtract two addresses from the range. The first address in every range is the subnet number. The last address in every range is the broadcast address. These two addresses cannot be assigned to any device in the network which is why you have to subtract two addresses to find the number of usable addresses in each range.

For example, if two bits are borrowed for the network portion of the address you can easily determine the number of subnets and hosts per subnets using the formula.

195. 223 . 50 . 0 0 | 0 0 0 0 0 0

The number of subnets created by borrowing 2 bits is 2^2 or $2 \times 2 = 4$ subnets.

The number of hosts created by leaving 6 bits is $2^6 - 2$ or $2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64 - 2 = 62$ usable hosts per subnet.

Custom Subnet Problems

Custom Subnet Masks

Problem 1

Number of needed subnets **14**
 Number of needed usable hosts **14**
 Network Address **192.10.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 240

Total number of subnets 16

Total number of host addresses 16

Number of usable addresses 14

Number of bits borrowed 4

Show your work for Problem 1 in the space below.

	256	128	64	32	16	8	4	2	1	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
192 . 10 . 10 . 0	0	0	0	0	0	0	0	0	0	

Add the binary value numbers to the left of the line to create the custom subnet mask.

128
64
32
+16
240

16	Observe the total number of hosts.
-2	
14	Subtract 2 for the number of usable hosts.

Custom Subnet Masks

Problem 2

Number of needed subnets **1000**

Number of needed usable hosts **60**

Network Address **165.100.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

Show your work for Problem 2 in the space below.

Number of Hosts -	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets -	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values -	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
	165	100	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Add the binary value numbers to the left of the line to create the custom subnet mask.

128	128
64	+64
32	192
16	
8	
4	
2	
+1	
255	

Observe the total number of hosts.

Subtract 2 for the number of usable hosts.

64
-2
62

Custom Subnet Masks

Problem 3

Network Address **148.75.0.0 /26**

/26 indicates the total number of bits used for the network and subnetwork portion of the address. All bits remaining belong to the host portion of the address.

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

Show your work for **Problem 3** in the space below.

	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Hosts	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Number of Subnets	-	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768
Binary values	-	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2
		148	75	0	0	0	0	0	0	0	0	0	0	0	0	0

Add the binary value numbers to the left of the line to create the custom subnet mask.

128	128
64	+64
32	192
16	
8	
4	
2	
+1	
255	

64	Observe the total number of hosts.
-2	
62	Subtract 2 for the number of usable hosts.

1024	
-2	
1,022	Subtract 2 for the total number of subnets to get the usable number of subnets.

Custom Subnet Masks

Problem 4

Number of needed subnets **6**
 Number of needed usable hosts **30**
 Network Address **195.85.8.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 224

Total number of subnets 8

Total number of host addresses 32

Number of usable addresses 30

Number of bits borrowed 3

Show your work for Problem 5 in the space below.

Number of Subnets	256 128 64			32 16 8 4 2					-	Number of Hosts
	2	4	8	16	32	64	128	256		
	128	64	32	16	8	4	2	1	-	Binary values
195 . 85 . 8 .	0	0	0	0	0	0	0	0		

$$\begin{array}{r}
 128 \\
 64 \\
 +32 \\
 \hline
 224
 \end{array}
 \qquad
 \begin{array}{r}
 32 \\
 -2 \\
 \hline
 30
 \end{array}
 \qquad
 \begin{array}{r}
 8 \\
 -2 \\
 \hline
 6
 \end{array}$$

Custom Subnet Masks

Problem 5

Number of needed subnets **6**
 Number of needed usable hosts **30**
 Network Address **210.100.56.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 224

Total number of subnets 8

Total number of host addresses 32

Number of usable addresses 30

Number of bits borrowed 3

Show your work for Problem 4 in the space below.

								Number of						
			256	128	64	32	16	8	4	2	-	Hosts		
Number of Subnets	-	2	4	8	16	32	64	128	256					
		128	64	32	16	8	4	2	1	-	Binary values			
210 . 100 . 56 .			0	0	0	0	0	0	0	0				
128														
64			8											
+32			-2	32										
224			6	-2	30									

Custom Subnet Masks

Problem 6

Number of needed subnets **126**
 Number of needed usable hosts **131,070**
 Network Address **118.0.0.0**

Address class A

Default subnet mask 255 . 0 . 0 . 0

Custom subnet mask 255 . 254 . 0 . 0

Total number of subnets 128

Total number of host addresses 131,072

Number of usable addresses 131,070

Number of bits borrowed 7

Show your work for **Problem 6** in the space below.

Number of Hosts	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576	2097152	4194304	8388608	16777216	33554432	67108864	134217728	268435456	536870912	1073741824	2147483648	4294967296	8589934592	17179869184	34359738368	68719476736	137438953472	274877906944	549755813888	1099511627776	2199023255552	4398046511104	8796093022208	17592186044416	35184372088832	70368744177664	140737488355328	281474976710656	562949953421312	1125899906842624	2251799813685248	4503599627370496	9007199254740992	18014398509481984	36028797018963968	72057594037927936	144115188075855872	288230376151711744	576460752303423488	1152921504606846976	2305843009213693952	4611686018427387904	9223372036854775808	18446744073709551616	36893488147419103232	73786976294838206464	147573952589676412928	295147905179352825856	590295810358705651712	1180591620717411303424	2361183241434822606848	4722366482869645213696	9444732965739290427392	18889465931478580854784	37778931862957161709568	75557863725914323419136	151115727451828646838272	302231454903657293676544	604462909807314587353088	1208925819614629174706176	2417851639229258349412352	4835703278458516698824704	9671406556917033397649408	19342813113834066795298816	38685626227668133590597632	77371252455336267181195264	154742504910672534362390528	309485009821345068724781056	618970019642690137449562112	1237940039285380274899124224	2475880078570760549798248448	4951760157141521099596496896	9903520314283042199192993792	19807040628566084398385987584	39614081257132168796771975168	79228162514264337593543950336	158456325028528675187087900672	316912650057057350374175801344	633825300114114700748351602688	1267650600228229401496703205376	2535301200456458802993406410752	5070602400912917605986812821504	10141204801825835211973625643008	20282409603651670423947251286016	40564819207303340847894502572032	81129638414606681695789005144064	162259276829213363391578010288128	324518553658426726783156020576256	649037107316853453566312041152512	1298074214633706907132624082305024	2596148429267413814265248164610048	5192296858534827628530496329220096	10384593717069655257060992658440192	20769187434139310514121985316880384	41538374868278621028243970633760768	83076749736557242056487941267521536	166153499473114484112975882535043072	332306998946228968225951765070086144	664613997892457936451903530140172288	1329227995784915872903807060280344576	2658455991569831745807614120560689152	5316911983139663491615228241121378304	10633823966279326983230456482242756608	21267647932558653966460912964485513216	42535295865117307932921825928971026432	85070591730234615865843651857942052864	170141183460469231731687303715884105728	340282366920938463463374607431768211456	680564733841876926926749214863536422912	1361129467683753853853498429727072845824	272225893536750770770699685945414569152	544451787073501541541399371890829138304	1088903574147003083082798743781658276608	2177807148294006166165597487563316553216	4355614296588012332331194975126633106432	8711228593176024664662389950253266212864	17422457186352049329324779900506532425728	34844914372704098658649559801013064851456	69689828745408197317299119602026129702912	139379657490816394634598239204052259405824	278759314981632789269196478408104518811648	557518629963265578538392956816209037623296	1115037259926531157076785913632418075246592	2230074519853062314153571827264836150493184	4460149039706124628307143654529672300986368	8920298079412249256614287309059344601972736	17840596158824498513228574618118689203945472	35681192317648997026457149236237378407890848	71362384635297994052914298472474756815781696	142724769270595988105828596944949513631563392	285449538541191976211657193889899027263126784	570899077082383952423314387779798054526253568	1141798154164767904846628775559596109052507136	2283596308329535809693257551119192218105014272	4567192616659071619386515102238384436210028544	9134385233318143238773030204476768872420057088	18268770466636286477546060408953537744840114176	36537540933272572955092120817907075489680228352	73075081866545145910184241635814150979360456704	146150163733090291820368483271628301958720913408	292300327466180583640736966543256603917441826816	584600654932361167281473933086513207834883653632	1169201309864722334562947866173026415669767307264	2338402619729444669125895732346052831339534614528	4676805239458889338251791464692105662679069229056	9353610478917778676503582929384211325358138458112	18707220957835557353007165858768422650716276916224	37414441915671114706014331717536845301432553832448	74828883831342229412028663435073690602865107664896	149657767662684458824057326870147381205730215329792	299315535325368917648114653740294762411460430659584	598631070650737835296229307480589524822920861319168	1197262141301475670592458614961179049645841722638336	2394524282602951341184917229922358099291683445276672	4789048565205902682369834459844716198583366890553344	9578097130411805364739668919689432397166733781106688	19156194260823610729479337839378864794333467562213376	38312388521647221458958675678757729588666935124426752	76624777043294442917917351357515459177333870248853504	153249554086588885835834702715030918354667740497707008	306499108173177771671669405430061836709335480995414016	612998216346355543343338810860123673418670961990828224	122599643269271108668667762172024734683734192398164544	245199286538542217337335524344049469367468384796329088	490398573077084434674671048688098938734936769592658176	980797146154168869349342097376197877469873539185316352	1961594292308337738698684194752395754939747078370632704	3923188584616675477397368389504791509879494156741265408	7846377169233350954794736779009583019758988313482530816	15692754338466701909589473558019166039517976266965061312	31385508676933403819178947116038332079035952533930122624	62771017353866807638357894232076664158071905067860245248	125542034707733615276715788464153328316143810135720490496	251084069415467230553431576928306656632287620271440980992	502168138830934461106863153856613313264575240542881961984	1004336277661868922213726307713226626529150481085763923968	2008672555323737844427452615426453253058300962171527847936	4017345110647475688854905230852906506116601924343055695872	8034690221294951377709810461705813012233203848686111391744	1606938044258990275541962092341162602446640767337222278368	3213876088517980551083924184682325204893281534674444556736	6427752177035961102167848369364650409786563069348889113472	12855504354071922204335696738729300819573126138697778268944	25711008708143844408671393477458601639146252277395556537888	51422017416287688817342786954917203278292504554791113075776	102844034832575377634685573909834406556585009109582226151552	205688069665150755269371147819668813113170018219164452303104	411376139330301510538742295639337626226340036438328904606208	822752278660603021077484591278675252452680072876657809212416	1645504557321206042154969182557350504905360145753315618424832	3291009114642412084309938365114701009810720291506631236849664	6582018229284824168619876730229402019621440583013262473699328	1316403645856964833723975346045880403924288116602652495398656	2632807291713929667447950692091760807848576233205304990797312	5265614583427859334895901384183521615697152466410609981594624	10531229166855718669791802768367043231394304932821219963189248	21062458333711437339583605536734086462788609865642439926378496	42124916667422874679167211073468172925577219731284879852756992	84249833334845749358334422146936345851154439462569759705513984	168499666669691498716668844293872691702308878925139519411027968	336999333339382997433337688587745383404617757850279038822055936	673998666678765994866675377175490766809235515700558007644111872	1347997333357531989733350754350981533618471031401160153288223648	269599466671506397946670150870196306723694206280232030657644736	539198933343012795893340301740392613447388412560464061315289472	1078397866686025591786680603480785226894776825120928122630578944	2156795733372051183573361206961570453789553650241856245261157888	4313591466744102367146722413923140907579107300483712490522315776	8627182933488204734293444827846281815158214600967424981044631552	1725436586697640946858688965569256363031642920193484996208926304	3450873173395281893717377931138512726063285840386969992417852608	6901746346790563787434755862277025452126571680773939984835705216	13803492693581127574869511724554050904253143361547879969671410432	27606985387162255149739023449108101808506286723095759939342820864	55213970774324510299478046898216203617012573446191519878685641728	110427941548649020598956093796432407234025146892383039757371283456	220855883097298041197912187592864814468050293784766079514742566912	441711766194596082395824375185729628936100587569532159029485133824	883423532389192164791648750371459257872201175139064318058970267648	1766847064778384329583297500742918515744402350278128636117940535296	3533694129556768659166595001485837031488804700556257272235881070592	7067388259113537318333190002971674062977609401112514544471762141184	14134776518227074636666380005943348125955218802225029088943524282368	28269553036454149273332760011886696251910437604450058177887048564736	56539106072908298546665520023773392503820875208900116355774097129472	113078212145816597093331040047546785007641750417800232711548194258944	226156424291633194186662080095093570015283500835600465423096388517888	452312848583266388373324160190187140030567001671200930846192777035776	904625697166532776746648320380374280061134003342401861692385554071552	1809251394333065553493296640760748560122268006684803723384771108143104	3618502788666131106986593281521497120244536013369607446769422216286208	7237005577332262213973186563042994240489072026739214893538844432572416	14474011154664524427946373126085988480978144053478429787077688865144832	28948022309329048855892746252171976961956288106956859574155377730289664	57896044618658097711785492504343953923912576213913719148310755460579328	115792089237316195423570985008687907847825152427827438296621510921158656	231584178474632390847141970017375815695650304855654876593243021842317312	463168356949264781694283940034751631391300609711309753186486043684634624	926336713898529563388567880069503262782601219422619506372972087369269248	1852673427797059126777135760139006525565202438845239012745944174738538496	3705346855594118253554271520278013051130404877690478025491888349477076992	7410693711188236507108543040556026102260809755380956050983776698954153984	14821387422376473014217086081112052204521619510761912101967553397908307968	29642774844752946028434172162224104409043239021523824203935106795816615936	59285549689505892056868344324448208818086478043047648407870213591633231872	118571099379011784113736688648896417636172956086095296815740427183266463744	23714
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Custom Subnet Masks

Problem 7

Number of needed subnets **2000**
 Number of needed usable hosts **15**
 Network Address **178.100.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 224

Total number of subnets 2,048

Total number of host addresses 32

Number of usable addresses 30

Number of bits borrowed 11

Show your work for Problem 7 in the space below.

Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
178 . 100 . 0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

$$\begin{array}{r}
 128 \\
 64 \\
 32 \\
 16 \\
 8 \\
 4 \\
 2 \\
 +1 \\
 \hline
 255
 \end{array}
 \qquad
 \begin{array}{r}
 2,048 \\
 -2 \\
 \hline
 2,046
 \end{array}
 \qquad
 \begin{array}{r}
 32 \\
 -2 \\
 \hline
 30
 \end{array}$$

Custom Subnet Masks

Problem 8

Number of needed subnets **3**

Number of needed usable hosts **45**

Network Address **200.175.14.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 4

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 2

Show your work for Problem 8 in the space below.

	256	128	64	32	16	8	4	2	1	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
200 . 175 . 14 .	0	0	0	0	0	0	0	0		

128	4	64
+64	-2	-2
240	2	62

Custom Subnet Masks

Problem 9

Number of needed subnets **60**
 Number of needed usable hosts **1,000**
 Network Address **128.77.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 252 . 0

Total number of subnets 64

Total number of host addresses 1,024

Number of usable addresses 1,022

Number of bits borrowed 6

Show your work for Problem 9 in the space below.

Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
	128	77	0	0	0	0	0	0	0	0	0	0	0	0	0	0

128		
64		
32		
16		
8	64	1,024
+4	-2	-2
252	62	1,022

Custom Subnet Masks

Problem 10

Number of needed usable hosts **60**

Network Address **198.100.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 4

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 2

Show your work for Problem 10 in the space below.

	256	128	64	32	16	8	4	2	1	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
198 . 100 . 10 .	0	0	0	0	0	0	0	0	0	

128	64	4
+64	-2	-2
192	62	2

Custom Subnet Masks

Problem 11

Number of needed subnets **250**

Network Address **101.0.0.0**

Address class A

Default subnet mask *255 . 0 . 0 . 0*

Custom subnet mask *255 . 255 . 0 . 0*

Total number of subnets 256

Total number of host addresses **65,536**

Number of usable addresses 65,534

Number of bits borrowed 8

Show your work for **Problem 11** in the space below.

Number of Hosts	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576	2097152	4194304	8388608	16777216	33554432	67108864	134217728	268435456	536870912	1073741824	2147483648	4294967296	8589934592	17179869184	34359738368	68719476736	137438953472	274877906944	549755813888	1099511627776	2199023255552	4398046511104	8796093022208	17592186044416	35184372088832	70368744177664	140737488355328	281474976710656	562949953421312	1125899906842624	2251799813685248	4503599627370496	9007199254740992	18014398509481984	36028797018963968	72057594037927936	144115188075855872	288230376151711744	576460752303423488	1152921504606846976	2305843009213693952	4611686018427387904	9223372036854775808	18446744073709551616	36893488147419103232	73786976294838206464	147573952589676412928	295147905179352825856	590295810358705651712	1180591620717411303424	2361183241434822606848	4722366482869645213696	9444732965739290427392	18889465931478580854784	37778931862957161709568	75557863725914323419136	151115727451828646838272	302231454903657293676544	604462909807314587353088	1208925819614629174706176	2417851639229258349412352	4835703278458516698824704	9671406556917033397649408	19342813113834066795298816	38685626227668133590597632	77371252455336267181195264	154742504910672534362390528	309485009821345068724781056	618970019642690137449562112	1237940039285380274899124224	2475880078570760549798248448	4951760157141521099596496896	9903520314283042199192993792	19807040628566084398385987584	39614081257132168796771975168	79228162514264337593543950336	158456325028528675187087900672	316912650057057350374175801344	633825300114114700748351602688	1267650600228229401496703205376	2535301200456458802993406410752	5070602400912917605986812821504	10141204801825835211973625643008	20282409603651670423947251286016	40564819207303340847894502572032	81129638414606681695789005144064	162259276829213363391578010288128	324518553658426726783156020576256	649037107316853453566312041152512	1298074214633706907132624082305024	2596148429267413814265248164610048	5192296858534827628530496329220096	10384593717069655257060992658440192	20769187434139310514121985316880384	41538374868278621028243970633760768	83076749736557242056487941267521536	166153499473114484112975882535043072	332306998946228968225951765070086144	664613997892457936451903530140172288	1329227995784915872903807060280344576	2658455991569831745807614120560689152	5316911983139663491615228241121378304	10633823966279326983230456482242756608	21267647932558653966460912964485513216	42535295865117307932921825928971026432	85070591730234615865843651857942052864	17014118346046923173168730371588410528	34028236692093846346337460743176821056	68056473384187692692674921486353642112	136112946768375385385349842972707284224	272225893536750770770699685945414568448	544451787073501541541399371890829136896	1088903574147003083082798743781658273792	2177807148294006166165597487563316547584	4355614296588012332331194975126633095168	8711228593176024664662389950253266190336	17422457186352049329324779900506532380672	34844914372704098658649559801013064761344	6968982874540819731729911960202612952288	13937965749081639463459823920405225904576	27875931498163278926919647840810451809152	55751862996326557853839295681620903618304	111503725992653115707678591363241807236608	223007451985306231415357182726483614473216	4460149039
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Custom Subnet Masks

Problem 12

Number of needed subnets **5**

Network Address **218.35.50.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 224

Total number of subnets 8

Total number of host addresses 32

Number of usable addresses 30

Number of bits borrowed 3

Show your work for Problem 12 in the space below.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
218 . 35 . 50 .	0	0	0	0	0	0	0	0		

128		
64	64	4
+32	-2	-2
<u>224</u>	<u>62</u>	<u>2</u>

Custom Subnet Masks

Problem 13

Number of needed usable hosts **25**

Network Address **218.35.50.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 224

Total number of subnets 8

Total number of host addresses 32

Number of usable addresses 30

Number of bits borrowed 3

Show your work for Problem 13 in the space below.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
218 . 35 . 50 .	0	0	0	0	0	0	0	0		

128		
64	8	32
+32	-2	-2
224	6	30

Custom Subnet Masks

Problem 14

Number of needed subnets **10**

Network Address **172.59.0.0**

Address class **B**

Default subnet mask **255 . 255 . 0 . 0**

Custom subnet mask **255 . 255 . 240 . 0**

Total number of subnets **16**

Total number of host addresses **4,096**

Number of usable addresses **4,094**

Number of bits borrowed **4**

Show your work for Problem 14 in the space below.

Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	2	4	8	16	32	64	128	256	512	1024	2048	4096	8,192	16,384	32,768	65,536
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
	172	59	0	0	0	0	0	0	0	0	0	0	0	0	0	0

$$\begin{array}{r}
 128 \\
 64 \\
 32 \\
 +16 \\
 \hline
 240
 \end{array}
 \qquad
 \begin{array}{r}
 16 \\
 -2 \\
 \hline
 14
 \end{array}
 \qquad
 \begin{array}{r}
 4,096 \\
 -2 \\
 \hline
 4,094
 \end{array}$$

Custom Subnet Masks

Problem 15

Number of needed usable hosts **50**

Network Address **172.59.0.0**

Address class **B**

Default subnet mask **255 . 255 . 0 . 0**

Custom subnet mask **255 . 255 . 255 . 192**

Total number of subnets **1,024**

Total number of host addresses **64**

Number of usable addresses **62**

Number of bits borrowed **10**

Show your work for Problem 15 in the space below.

Number of Hosts	-	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	-	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	-	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		172	59	0	0	0	0	0	0	0	0	0	0	0	0	0	0

128			
64			
32			
16			
8			
4			
2	128	64	1,024
+1	+64	-2	-2
255	192	62	1,022

Custom Subnet Masks

Problem 16

Number of needed usable hosts **29**

Network Address **23.0.0.0**

Address class **A**

Default subnet mask **255 . 0 . 0 . 0**

Custom subnet mask **255 . 255 . 255 . 224**

Total number of subnets **524,288**

Total number of host addresses **32**

Number of usable addresses **30**

Number of bits borrowed **19**

Show your work for Problem 16 in the space below.

Number of Hosts	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576	2097152	4194304	8388608	16777216	33554432	67108864	134217728	268435456	536870912	1073741824	2147483648	4294967296	8589934592	17179869184	34359738368	68719476736	137438953472	274877906944	549755813888	1099511627776	2199023255552	4398046511104	8796093022208	17592186044416	35184372088832	70368744177664	140737488355328	281474976710656	562949953421312	1125899906842624	2251799813685248	4503599627370496	9007199254740992	18014398509481984	36028797018963968	72057594037927936	144115188075855872	288230376151711744	576460752303423488	1152921504606846976	2305843009213693952	4611686018427387904	9223372036854775808	18446744073709551616	36893488147419103232	73786976294838206464	147573952589676412928	295147905179352825856	590295810358705651712	1180591620717411303424	2361183241434822606848	4722366482869645213696	9444732965739290427392	18889465931478580854784	37778931862957161709568	75557863725914323419136	151115727451828646838272	302231454903657293676544	604462909807314587353088	1208925819614629174706176	2417851639229258349412352	4835703278458516698824704	9671406556917033397649408	19342813113834066795298816	38685626227668133590597632	77371252455336267181195264	154742504910672534362390528	309485009821345068724781056	618970019642690137449562112	1237940039285380274899124224	2475880078570760549798248448	4951760157141521099596496896	9903520314283042199192993792	19807040628566084398385987584	39614081257132168796771975168	79228162514264337593543950336	158456325028528675187087900672	316912650057057350374175801344	633825300114114700748351602688	1267650600228229401496703205376	2535301200456458802993406410752	5070602400912917605986812821504	10141204801825835211973625643008	20282409603651670423947251286016	40564819207303340847894502572032	81129638414606681695789005144064	162259276829213363391578010288128	324518553658426726783156020576256	649037107316853453566312041152512	1298074214633706907132624082305024	2596148429267413814265248164610048	5192296858534827628530496329220096	10384593717069655257060992658440192	20769187434139310514121985316880384	41538374868278621028243970633760768	83076749736557242056487941267521536	166153499473114484112975882535043072	332306998946228968225951765070086144	664613997892457936451903530140172288	1329227995784915872903807060280344576	2658455991569831745807614120560689152	5316911983139663491615228241121378304	10633823966279326983230456482242756608	21267647932558653966460912964485513216	42535295865117307932921825928971026432	85070591730234615865843651857942052864	170141183460469231731687303715884105728	340282366920938463463374607431768211456	680564733841876926926749214863536422912	1361129467683753853853498429727072845824	272225893536750770770699685945414569152	544451787073501541541399371890829138304	1088903574147003083082798743781658276608	2177807148294006166165597487563316553216	4355614296588012332331194975126633106432	8711228593176024664662389950253266212864	17422457186352049329324779900506532425728	34844914372704098658649559801013064851456	69689828745408197317299119602026129702912	139379657490816394634598239204052259405824	278759314981632789269196478408104518811648	557518629963265578538392956816209037623296	1115037259926531157076785913632418075246592	2230074519853062314153571827264836150493184	4460149039706124628307143654529672300986368	8920298079412249256614287309059344601972736	17840596158824498513228574618118689203945472	35681192317648997026457149236237378407890848	71362384635297994052914298472474756815781696	142724769270595988105828596944949513631563392	285449538541191976211657193889899027263126784	570899077082383952423314387779798054526253568	1141798154164767904846628775559596109052507136	2283596308329535809693257551119192218105014272	4567192616659071619386515102238384436210028544	9134385233318143238773030204476768872420057088	18268770466636286477546060408953537744840114176	36537540933272572955092120817907075489680228352	73075081866545145910184241635814150979360456704	146150163733090291820368483271628301958720913408	292300327466180583640736966543256603917441826816	584600654932361167281473933086513207834883653632	1169201309864722334562947866173026415669767307264	2338402619729444669125895732346052831339534614528	4676805239458889338251791464692105662679069229056	9353610478917778676503582929384211325358138458112	18707220957835557353007165858768422650716276916224	37414441915671114706014331717536845301432553832448	74828883831342229412028663435073690602865107664896	149657767662684458824057326870147381205730215329792	299315535325368917648114653740294762411460430659584	598631070650737835296229307480589524822920861319168	1197262141301475670592458614961179049645841722638336	2394524282602951341184917229922358099291683445276672	4789048565205902682369834459844716198583366890553344	9578097130411805364739668919689432397166733781106688	19156194260823610729479337839378864794333467562213376	38312388521647221458958675678757729588666935124426752	76624777043294442917917351357515459177333870248853504	153249554086588885835834702715030918354667740497707008	306499108173177771671669405430061836709335480995414016	612998216346355543343338810860123673418670961990828032	1225996432692711086686677621720247346837341923981665664	24519928653854221733733552434404946936746838479633312	49039857307708443467467104868809893873493676959266624	98079714615416886934934209737619787746987353918533248	196159429230833773869868419475239575493974707837066496	392318858461667547739736838950479150987949415674132992	784637716923335095479473677900958301975898831348265984	1569275433846670190958947355801916603951797662696531968	3138550867693340381917894711603833207903595325393063936	6277101735386680763835789423207666415807190650786127872	12554203470773361527671578846415332831614381301572255744	25108406941546723055343157692830665663228762603144511488	50216813883093446110686315385661331326457525206289022976	100433627766186892221372630771322662652915050412578045952	200867255532373784442745261542645325305830100825156101104	401734511064747568885490523085290650611660201650312202208	803469022129495137770981046170581301223320403300624404416	1606938044258990275541962092341162602446640806601248808832	3213876088517980551083924184682325204893281613202497617664	6427752177035961102167848369364650409786563226404995235328	12855504354071922204335696738729300819573126452809904706656	25711008708143844408671393477458601639146252905619809413312	51422017416287688817342786954917203278292505811239618826624	102844034832575377634685573909834406556585011622479237653248	205688069665150755269371147819668813113170023244958475306496	411376139330301510538742295639337626226340046489916950612992	822752278660603021077484591278675252452680092979833901225984	1645504557321206042154969182557350504905360185959667802451968	3291009114642412084309938365114701009810720371919335604903936	6582018229284824168619876730229402019621440743838671209807872	13164036458569648337239753460458804039242881487677342419615744	26328072917139296674479506920917608078485762975354684839231488	52656145834278593348959013841835216156971525950709369678462976	105312291668557186697918027683670432313943051901418739356925952	210624583337114373395836055367340864627886103802837478713851904	421249166674228746791672110734681729255772207605674957427703808	842498333348457493583344221469363458511544415211349914855407616	1684996666696914987166688442938726917023088830422699829710815232	3369993333393829974333376885877453834046177660845399659421630464	6739986666787659948666753771754907668092355321690799318843260928	13479973333575319897333507543509815336184710643381598637686521856	26959946667150639794667015087019630672369421286763197275373043712	53919893334301279589334030174039261344738842573526394550746087424	107839786668602559178668060348078522689477685147052789101492174848	215679573337205118357336120696157045378955370294105578202984349696	431359146674410236714672241392314090757910740588211156405968699392	862718293348820473429344482784628181515821481176422312811937398784	1725436586697640946858688965569256363031642962352844625623874797568	3450873173395281893717377931138512726063285924705689251247749595136	6901746346790563787434755862277025452126571849411378502495499190272	13803492693581127574869511724554050904253143698822757004990998380544	27606985387162255149739023449108101808506287397645514009981996761088	55213970774324510299478046898216203617012574795291028019963993522176	110427941548649020598956093796432407234025149590582056039927987044352	220855883097298041197912187592864814468050299181164112079855974088704	441711766194596082395824375185729628936100598362328224159711948177408	883423532389192164791648750371459257872201196724656448319423896354816	1766847064778384329583297500742918515744402393449312896638847792709632	3533694129556768659166595001485837031488804786898625793277695585419264	7067388259113537318333190002971674062977609573797251586555391170838528	14134776518227074636666380005943348125955219147594503173110782341677056	28269553036454149273332760011886696251910438295189006346221564683354112	56539106072908298546665520023773392503820876590378012692443129366708224	113078212145816597093331040047546785007641753180756025384886258733416448	226156424291633194186662080095093570015283506361512050769772517466832896	452312848583266388373324160190187140030567012723024101539545034933665792	904625697166532776746648320380374280061134025446048203079090069867331584	1809251394333065553493296640760748560122268050892096406158180139734663168	3618502788666131106986593281521497120244536101784192812316360279469326336	7237005577332262213973186563042994240489072203568385624632720558938652672	14474011154664524427946373126085988480978144407136771249265441117877305344	28948022309329048855892746252171976961956288814273542498530882235754610688	57896044618658097711785492504343953923912577628547084997061764471509221376	115792089237316195423570985008687907847825155257094169994123528943018442752	231584178474632390847141970017375815695650310514188339988247057886036885504	463168356949264781694283940034751631391300621028376679976494115772073771008	926336713898529563388567880069503262782601242056753359952988231544147542016	1852673427797059126777135760139006525565202484113506719905976463088284084032	3705346855594118253554271520278013051130404968227013439811952926176568168064	7410693711188236507108543040556026102260809936454026879623905852353136336128	14821387422376473014217086081112052204521619872908053759247811704706272672256	29642774844752946028434172162224104409043239745816107518495623409412545344512	5928554968950589
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Subnetting

Problem 1

Number of needed subnets **14**

Number of needed usable hosts **14**

Network Address **192.10.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 240

Total number of subnets 16

Total number of host addresses 16

Number of usable addresses 14

Number of bits borrowed 4

What is the 4th subnet range? 192.10.10.48 to 192.10.10.63

What is the subnet number for the 8th subnet? 192 . 10 . 10 . 112

What is the subnet broadcast address for the 13th subnet? 192 . 10 . 10 . 207

What are the assignable addresses for the 9th subnet? 192.10.10.129 to 192.10.10.142

Show your work for Problem 1 in the space below.

Number of Subnets	256 128 64 32				16 8 4 2				Number of Hosts	
	2	4	8	16	32	64	128	256		
	128	64	32	16	8	4	2	1	Binary values	
192.10.10.0	0	0	0	0	0	0	0	0		
(1)	0	0	0	0	192.10.10.0				to	192.10.10.15
(2)	0	0	0	1	192.10.10.16				to	192.10.10.31
(3)	0	0	1	0	192.10.10.32				to	192.10.10.47
(4)	0	0	1	1	192.10.10.48				to	192.10.10.63
(5)	0	1	0	0	192.10.10.64				to	192.10.10.79
(6)	0	1	0	1	192.10.10.80				to	192.10.10.95
(7)	0	1	1	0	192.10.10.96				to	192.10.10.111
(8)	0	1	1	1	192.10.10.112				to	192.10.10.127
(9)	1	0	0	0	192.10.10.128				to	192.10.10.143
(10)	1	0	0	1	192.10.10.144				to	192.10.10.159
(11)	1	0	1	0	192.10.10.160				to	192.10.10.175
(12)	1	0	1	1	192.10.10.176				to	192.10.10.191
(13)	1	1	0	0	192.10.10.192				to	192.10.10.207
(14)	1	1	0	1	192.10.10.208				to	192.10.10.223
(15)	1	1	1	0	192.10.10.224				to	192.10.10.239
(16)	1	1	1	1	192.10.10.240				to	192.10.10.255

$$\begin{array}{r}
 128 \\
 64 \\
 32 \\
 +16 \\
 \hline
 \text{Custom subnet mask } 240
 \end{array}$$

$$\begin{array}{r}
 16 \\
 -2 \\
 \hline
 \text{Usable subnets } 14
 \end{array}$$

$$\begin{array}{r}
 16 \\
 -2 \\
 \hline
 \text{Usable hosts } 14
 \end{array}$$

The binary value of the last bit borrowed is the range. In this problem the range is 16.

The first address in each subnet range is the subnet number.

The last address in each subnet range is the subnet broadcast address.

Subnetting

Problem 2

Number of needed subnets **1000**

Number of needed usable hosts **60**

Network Address **165.100.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

What is the 15th subnet range? 165.100.3.128 to 165.100.3.191

What is the subnet number for the 6th subnet? 165 . 100 . 1 . 64

What is the subnet broadcast address for the 6th subnet? 165 . 100 . 1 . 127

What are the assignable addresses for the 9th subnet? 165.100.2.1 to 165.100.0.62

Show your work for Problem 2 in the space below.

Number of Hosts -	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets -	2	4	8	16	32	64	128	256	512	1,024	2,048	4,096	8,192	16,384	32,768	65,536
Binary values -	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
165 . 100 . 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Usable hosts	64	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2
Custom subnet mask	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
	192	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
	255	255	255	255	255	255	255	255	255	255	255	255	255	255	255	255
The binary value of the last bit borrowed is the range. In this problem the range is 64. The first address in each subnet range is the subnet number. The last address in each subnet range is the subnet broadcast address.																
(1023)	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
(1024)	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Down to																
(1023)	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
(1024)	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
165.100.255.128 to 165.100.255.191																
165.100.255.192 to 165.100.255.255																

Subnetting

Problem 3

Hint: It is possible to borrow one bit to create two subnets.

Number of needed subnets **2**

Network Address **195.223.50.0**

Address class **C**

Default subnet mask **255 . 255 . 255 . 0**

Custom subnet mask **255 . 255 . 255 . 128**

Total number of subnets **2**

Total number of host addresses **128**

Number of usable addresses **126**

Number of bits borrowed **1**

What is the 2nd subnet range? **195.223.50.128 - 195.223.50.255**

What is the subnet number for the 2nd subnet? **195.223.50.128**

What is the subnet broadcast address for the 1st subnet? **195.223.50.127**

What are the assignable addresses for the 1st subnet? **195.223.50.1 - 195.223.50.126**

Show your work for Problem 3 in the space below.

		256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256		
		128	64	32	16	8	4	2	1	-	Binary values
195. 223 . 50 . 0			0	0	0	0	0	0	0	0	
(1)		0	195.223.50.0 to 195.223.50.127								
(2)		1	195.223.50.128 to 195.223.50.255								

Subnetting

Problem 4

Number of needed subnets **750**

Network Address **190.35.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

What is the 15th subnet range? 190.35.3.128 to 190.35.3.191

What is the subnet number for the 13th subnet? 190.35.3.0

What is the subnet broadcast address for the 10th subnet? 190.35.2.127

What are the assignable addresses for the 6th subnet? 190.35.1.65 to 190.35.1.126

Show your work for Problem 4 in the space below.

Number of Hosts -	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets -	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values -	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
190.35.0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(1)	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
(2)	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	1
(3)	32	16	8	4	2	1	128	64	32	16	8	4	2	1	1	1
(4)	16	8	4	2	1	128	64	32	16	8	4	2	1	1	1	1
(5)	8	4	2	1	128	64	32	16	8	4	2	1	1	1	1	1
(6)	4	2	1	128	64	32	16	8	4	2	1	1	1	1	1	1
(7)	2	1	128	64	32	16	8	4	2	1	1	1	1	1	1	1
(8)	1	128	64	32	16	8	4	2	1	1	1	1	1	1	1	1
(9)	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
(10)	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	1
(11)	32	16	8	4	2	1	128	64	32	16	8	4	2	1	1	1
(12)	16	8	4	2	1	128	64	32	16	8	4	2	1	1	1	1
(13)	8	4	2	1	128	64	32	16	8	4	2	1	1	1	1	1
(14)	4	2	1	128	64	32	16	8	4	2	1	1	1	1	1	1
(15)	2	1	128	64	32	16	8	4	2	1	1	1	1	1	1	1
(16)	1	128	64	32	16	8	4	2	1	1	1	1	1	1	1	1
190.35.0.63	190	35	0	63	to											
190.35.0.127	190	35	0	127	to											
190.35.0.191	190	35	0	191	to											
190.35.0.255	190	35	0	255	to											
190.35.1.63	190	35	1	63	to											
190.35.1.127	190	35	1	127	to											
190.35.1.191	190	35	1	191	to											
190.35.1.255	190	35	1	255	to											
190.35.2.63	190	35	2	63	to											
190.35.2.127	190	35	2	127	to											
190.35.2.191	190	35	2	191	to											
190.35.2.255	190	35	2	255	to											
190.35.3.63	190	35	3	63	to											
190.35.3.127	190	35	3	127	to											
190.35.3.191	190	35	3	191	to											
190.35.3.255	190	35	3	255	to											

Subnetting

Problem 5

Number of needed usable hosts **6**

Network Address **126.0.0.0**

Address class A

Default subnet mask 255 . 0 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 248

Total number of subnets 2,097,152

Total number of host addresses 8

Number of usable addresses 6

Number of bits borrowed 21

What is the 2nd subnet range? 126.0.0.8 to 126.0.0.15

What is the subnet number for the 5th subnet? 126.0.0.32

What is the subnet broadcast address for the 7th subnet? 126.0.0.55

What are the assignable addresses for the 10th subnet? 126.0.0.73 to 126.0.0.78

Show your work for Problem 5 in the space below.

Number of Hosts		Number of Subnets															
		Binary values															
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
4,194,304		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
2097,152		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1048,576		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
524,288		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
262,144		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
131,072		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
65,536		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
32,768		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
16,384		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
8,192		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
4,096		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
2,048		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1,024		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
512		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
		128	64	32	16	8	4	2	1	128	64	32	16	8			

Subnetting

Problem 6

Number of needed subnets **10**

Network Address **192.70.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 240

Total number of subnets 16

Total number of host addresses 16

Number of usable addresses 14

Number of bits borrowed 4

What is the 9th subnet range? 192.70.10.128 to 192.70.10.143

What is the subnet number for the 4th subnet? 192.70.10.48

What is the subnet broadcast address for the 12th subnet? 192.70.10.191

What are the assignable addresses for the 10th subnet? 192.70.10.145 to 192.70.10.158

Show your work for Problem 6 in the space below.

												Number of
								256	128	64	32	Hosts
Number of Subnets								2	4	8	16	
								32	64	128	256	
								128	64	32	16	
								8	4	2	1	Binary values
192	.	70	.	10	.	0	0	0	0	0	0	
(1)							0	192.70.10.0	to	192.70.10.15		
(2)							1	192.70.10.16	to	192.70.10.31		
(3)					1	0		192.70.10.32	to	192.70.10.47		
(4)					1	1		192.70.10.48	to	192.70.10.63		
(5)				1	0	0		192.70.10.64	to	192.70.10.79		
(6)				1	0	1		192.70.10.80	to	192.70.10.95		
(7)				1	1	0		192.70.10.96	to	192.70.10.111		
(8)				1	1	1		192.70.10.112	to	192.70.10.127		
(9)	1	0	0	0				192.70.10.128	to	192.70.10.143		
(10)	1	0	0	1				192.70.10.144	to	192.70.10.159		
(11)	1	0	1	0				192.70.10.160	to	192.70.10.175		
(12)	1	0	1	1				192.70.10.176	to	192.70.10.191		
(13)	1	1	0	0				192.70.10.192	to	192.70.10.207		
(14)	1	1	0	1				192.70.10.208	to	192.70.10.223		
(15)	1	1	1	0				192.70.10.224	to	192.70.10.239		
(16)	1	1	1	1				192.70.10.240	to	192.70.10.255		

$$\begin{array}{r}
 128 \\
 +64 \\
 \hline
 240
 \end{array}
 \qquad
 \begin{array}{r}
 16 \\
 -2 \\
 \hline
 14
 \end{array}$$

Subnetting

Problem 7

Network Address **10.0.0.0 /16**

Address class A

Default subnet mask 255 . 0 . 0 . 0

Custom subnet mask 255 . 255 . 0 . 0

Total number of subnets 256

Total number of host addresses 65,536

Number of usable addresses 65,534

Number of bits borrowed 8

What is the 11th subnet range? 10.10.0.0 to 10.10.255.255

What is the subnet number for the 6th subnet? 10.5.0.0

What is the subnet broadcast address for the 2nd subnet? 10.1.255.255

What are the assignable addresses for the 9th subnet? 10.8.0.1 to 10.8.255.254

Show your work for Problem 7 in the space below.

Number of Hosts	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576	2097152	4194304	8388608	16777216	33554432	67108864	134217728	268435456	536870912	1073741824	2147483648	4294967296	8589934592	17179869184	34359738368	68719476736	137438953472	274877906944	549755813888	1099511627776	2199023255552	4398046511104	8796093022208	17592186044416	35184372088832	70368744177664	140737488355328	281474976710656	562949953421312	1125899906842624	2251799813685248	4503599627370496	9007199254740992	18014398509481984	36028797018963968	72057594037927936	144115188075855872	288230376151711744	576460752303423488	1152921504606846976	2305843009213693952	4611686018427387904	9223372036854775808	18446744073709551616	36893488147419103232	73786976294838206464	147573952589676412928	295147905179352825856	590295810358705651712	1180591620717411303424	2361183241434822606848	4722366482869645213696	9444732965739290427392	18889465931478580854784	37778931862957161709568	75557863725914323419136	151115727451828646838272	302231454903657293676544	604462909807314587353088	1208925819614629174706176	2417851639229258349412352	4835703278458516698824704	9671406556917033397649408	19342813113834066795298816	38685626227668133590597632	77371252455336267181195264	154742504910672534362390528	309485009821345068724781056	618970019642690137449562112	1237940039285380274899124224	2475880078570760549798248448	4951760157141521099596496896	9903520314283042199192993792	19807040628566084398385987584	39614081257132168796771975168	79228162514264337593543950336	158456325028528675187087900672	316912650057057350374175801344	633825300114114700748351602688	1267650600228229401496703205376	2535301200456458802993406410752	5070602400912917605986812821504	10141204801825835211973625643008	20282409603651670423947251286016	40564819207303340847894502572032	81129638414606681695789005144064	162259276829213363391578010288128	324518553658426726783156020576256	649037107316853453566312041152512	1298074214633706907132624082305024	2596148429267413814265248164610048	5192296858534827628530496329220096	10384593717069655257060992658440192	20769187434139310514121985316880384	41538374868278621028243970633760768	83076749736557242056487941267521536	166153499473114484112975882535043072	332306998946228968225951765070086144	664613997892457936451903530140172288	1329227995784915872903807060280344576	2658455991569831745807614120560689152	5316911983139663491615228241121378304	10633823966279326983230456482242756608	21267647932558653966460912964485513216	42535295865117307932921825928971026432	85070591730234615865843651857942052864	170141183460469231731687303715884105728	340282366920938463463374607431768211456	680564733841876926926749214863536422912	1361129467683753853853498429727072845824	272225893536750770770699685945414569152	544451787073501541541399371890829138304	1088903574147003083082798743781658276608	2177807148294006166165597487563316553216	4355614296588012332331194975126633106432	8711228593176024664662389950253266212864	17422457186352049329324779900506532425728	34844914372704098658649559801013064851456	69689828745408197317299119602026129702912	139379657490816394634598239204052259405824	278759314981632789269196478408104518811648	557518629963265578538392956816209037623296	1115037259926531157076785913632418075246592	2230074519853062314153571827264836150493184	4460149039706124628307143654529672300986368	8920298079412249256614287309059344601972736	17840596158824498513228574618118689203945472	35681192317648997026457149236237378407890848	71362384635297994052914298472474756815781696	142724769270595988105828596944949513631563392	285449538541191976211657193889899027263126784	570899077082383952423314387779798054526253568	1141798154164767904846628775559596109052507136	2283596308329535809693257551119192218105014272	4567192616659071619386515102238384436210028544	9134385233318143238773030204476768872420057088	18268770466636286477546060408953537744840114176	36537540933272572955092120817907075489680228352	73075081866545145910184241635814150979360456704	146150163733090291820368483271628301958720913408	292300327466180583640736966543256603917441826816	584600654932361167281473933086513207834883653632	1169201309864722334562947866173026415669767307264	2338402619729444669125895732346052831339534614528	4676805239458889338251791464692105662679069229056	9353610478917778676503582929384211325358138458112	18707220957835557353007165858768422650716276916224	37414441915671114706014331717536845301432553832448	74828883831342229412028663435073690602865107664896	149657767662684458824057326870147381205730215329792	299315535325368917648114653740294762411460430659584	598631070650737835296229307480589524822920861319168	1197262141301475670592458614961179049645841722638336	2394524282602951341184917229922358099291683445276672	4789048565205902682369834459844716198583366890553344	9578097130411805364739668919689432397166733781106688	19156194260823610729479337839378864794333467562213376	38312388521647221458958675678757729588666935124426752	76624777043294442917917351357515459177333870248853504	153249554086588885835834702715030918354667740497707008	306499108173177771671669405430061836709335480995414016	612998216346355543343338810860123673418670961990828032	1225996432692711086686677621720247346837341923981665664	2451992865385422173373355243440494693674683847963331328	4903985730770844346746710486880989387349367695926662656	9807971461541688693493420973761978774698735391853325312	19615942923083377386986841947523957549397470783706650624	39231885846166754773973683895047915098794941567413301248	78463771692333509547947367790095830197589883134826602496	156927543384667019095894735580191660395179766269653204992	31385508676933403819178947116038332079035953253930640992	62771017353866807638357894232076664158071906507861281984	125542034707733615276715788464153328316143813015722563968	251084069415467230553431576928306656632287626031445127936	50216813883093446110686315385661331326457525206289025584	100433627766186892221372630771322662652915050412578051168	200867255532373784442745261542645325305830100825156102336	401734511064747568885490523085290650611660201650312204672	803469022129495137770981046170581301223320403300624409344	1606938044258990275541962092341162602446640806601248818688	32138760885179805510839241846823252048932816132024977376	64277521770359611021678483693646504097865632264049954752	128555043540719222043356967387293008195731264528099909504	257110087081438444086713934774586016391462529056199819008	514220174162876888173427869549172032782925058112399638016	1028440348325753776346855739098344065565850116224799276032	2056880696651507552693711478196688131131700232449598552064	4113761393303015105387422956393376262263400464899197104128	8227522786606030210774845912786752524526800929798394208256	16455045573212060421549691825573505049053601859596788416512	32910091146424120843099383651147010098107203719193576833024	65820182292848241686198767302294020196214407438387153666048	131640364585696483372397534604588040392428814876774307292096	263280729171392966744795069209176080784857629753548614584192	526561458342785933489590138418352161569715259507097229168384	1053122916685771866979180276836704323139430519014194458336768	2106245833371543733958360553673408646278861038028388916673536	4212491666743087467916721107346817292557722076056777833347072	8424983333486174935833442214693634585115444152113555666694144	16849966666972349871666884429387269170230888304227111133388288	33699933333944699743333768858774538340461776608454222266776576	67399866667889399486667537717549076680923553216908444533553152	134799733335778798973335075435098153361847106437968889067106304	269599466671557597946670150870196306723694212875937778134212608	539198933343115195893340301740392613447388425751875556268425216	1078397866686230391786680603480785226894776851503751112536850432	2156795733372460783573361206961570453789553703007502225073700672	4313591466744921567146722413923140907579107406015004450147401344	8627182933489843134293444827846281815158214812030008900294802688	17254365866979686268586889655692563630316429624060017805889605376	34508731733959372537173779311385127260632859248120035611779210752	69017463467918745074347558622770254521265718496240071223558421504	138034926935837490148695117245540509042531436992480142447116843008	276069853871674980297390234491081018085062873984960284894233686016	552139707743349960594780468982162036170125747969920569788467372032	1104279415486699921189560937964324072340251495939841139576934744064	2208558830973399842379121875928648144680502991879682279153869488128	4417117661946799684758243751857296289361005983759364558307738976256	8834235323893599369516487503714592578722011967518729116615477952512	17668470647787198739032975007429185157444023935037458233230955905024	35336941295574397478065950014858370314888047870074916466461911810048	70673882591148794956131900029716740629776095740149832932923823620096	141347765182297589912263800059433481259552191480299658565847647240192	282695530364595179824527600118866962519104382960599317131755294480384	565391060729190359649055200237733925038208765921198634263510588960768	1130782121458380719298110400475467850076417531842397268527021177921536	2261564242916761438596220800950935700152835063684794537054042355843072	4523128485833522877192441601901871400305670127369589074108084711686144	9046256971667045754384883203803742800611340254739178148216169423372288	18092513943334091508769766407607485601222680509478356296432338846744576	36185027886668183017539532815214971202445361018956712592864677693489152	72370055773336366035079065630429942404890722037913425185729355386978304	144740111546672732070158131260859884809781444075826850371458710773956608	289480223093345464140316262521719769619562888151653700742917421547913216	578960446186690928280632525043439539239125776303307401485834843095826432	1157920892373381856561265050086879078478251552606614802971669686191652864	2315841784746763713122530100173758156956503105213229605943339372383305728	4631683569493527426245060200347516313913006210426459211886678744766611456	9263367138987054852490120400695032627826012420852918423773357489533222912	18526734277974109704980240801390065255652024841705836847546714979066445824	37053468555948219409960481602780130511304049683411673695093429958132891648	74106937111896438819920963205560261022608099366823347390186859916265783296	148213874223792877639841926411120522045216198733646694780373719832531566592	296427748447585755279683852822241044090432397467293389560747439665063133184	592855496895171510559367705644482088180864794934586779121494879330126266368	1185710993790343021118735411288964176361729589869173558242989758660252532736	2371421987580686042237470822577928352723459179738347116485979517320505065472	4742843975161372084474941645155856705446918359476694232971959034641010130944	9485687950322744168949883290311713410893836718953388465943918069282020261888	18971375900645488337899766580623426821787673437906776931887836388564040523776	37942751801290976675799533161246853643575346875813553863775672777128081047552	75885503602581953351599066322493707287150693751627107727551345554256162095104	151771007205163906703198132644987414574301387503254215455102691108524184190208	30354201441032781340639626528997482914860277500650843
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Subnetting

Problem 8

Number of needed subnets **5**

Network Address **172.50.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 224 . 0

Total number of subnets 8

Total number of host addresses 8,192

Number of usable addresses 8,190

Number of bits borrowed 3

What is the 4th subnet range? 172.50.96.0 to 172.50.127.255

What is the subnet number for the 5th subnet? 172.50.128.0

What is the subnet broadcast address for the 6th subnet? 172.50.191.255

What are the assignable addresses for the 3rd subnet? 172.50.64.1 to 172.50.95.254

Show your work for Problem 8 in the space below.

Number of Hosts -	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets -	2	4	8	16	32	64	128	256	512	1,024	2,048	4,096	8,192	16,384	32,768	65,536
Binary values -	128	64	32	16	8	4	2	1	1	1	1	1	1	1	1	1
	172	50	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(1)	0	172	50	0	0	0	0	0	0	0	0	0	0	0	0	0
(2)	1	172	50	0	32	0	0	0	0	0	0	0	0	0	0	0
(3)	1	172	50	0	64	0	0	0	0	0	0	0	0	0	0	0
(4)	1	172	50	0	96	0	0	0	0	0	0	0	0	0	0	0
(5)	1	172	50	0	128	0	0	0	0	0	0	0	0	0	0	0
(6)	1	172	50	0	160	0	0	0	0	0	0	0	0	0	0	0
(7)	1	172	50	0	192	0	0	0	0	0	0	0	0	0	0	0
(8)	1	172	50	0	224	0	0	0	0	0	0	0	0	0	0	0

128

64

+32

224

8,192

-2

8,190

Subnetting

Problem 9

Number of needed usable hosts **28**

Network Address **172.50.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 224

Total number of subnets 2,048

Total number of host addresses 32

Number of usable addresses 30

Number of bits borrowed 11

What is the 2nd subnet range? 172.50.0.32 to 172.50.0.63

What is the subnet number for the 10th subnet? 172.50.1.32

What is the subnet broadcast address for the 4th subnet? 172.50.0.127

What are the assignable addresses for the 6th subnet? 172.50.0.161 to 172.50.0.190

Show your work for Problem 9 in the space below.

Number of Hosts -	512	1,024	2,048	4,096	8,192	16,384	32,768	65,536	32	16	8	4	2
Number of Subnets -	-----	-----	-----	-----	-----	-----	-----	-----	4,096	8,192	16,384	32,768	65,536
Binary values -	128	64	32	16	8	4	2	1	16	8	4	2	1
172.50.0.0	0	0	0	0	0	0	0	0	0	0	0	0	0
128	(1)								172.50.0.0	172.50.0.0	172.50.0.0	172.50.0.0	172.50.0.0
64	(2)								172.50.0.32	172.50.0.32	172.50.0.32	172.50.0.32	172.50.0.32
32	(3)								172.50.0.64	172.50.0.64	172.50.0.64	172.50.0.64	172.50.0.64
16	(4)								172.50.0.96	172.50.0.96	172.50.0.96	172.50.0.96	172.50.0.96
8	(5)								172.50.0.128	172.50.0.128	172.50.0.128	172.50.0.128	172.50.0.128
4	(6)								172.50.0.160	172.50.0.160	172.50.0.160	172.50.0.160	172.50.0.160
2	(7)								172.50.0.192	172.50.0.192	172.50.0.192	172.50.0.192	172.50.0.192
+1	(8)								172.50.0.224	172.50.0.224	172.50.0.224	172.50.0.224	172.50.0.224
252	(9)								172.50.1.0	172.50.1.0	172.50.1.0	172.50.1.0	172.50.1.0
	(10)								172.50.1.32	172.50.1.32	172.50.1.32	172.50.1.32	172.50.1.32
	(11)								172.50.1.64	172.50.1.64	172.50.1.64	172.50.1.64	172.50.1.64
	(12)								172.50.1.96	172.50.1.96	172.50.1.96	172.50.1.96	172.50.1.96
	(13)								172.50.1.128	172.50.1.128	172.50.1.128	172.50.1.128	172.50.1.128
	(14)								172.50.1.160	172.50.1.160	172.50.1.160	172.50.1.160	172.50.1.160
	(15)								172.50.1.192	172.50.1.192	172.50.1.192	172.50.1.192	172.50.1.192
	(16)								172.50.1.224	172.50.1.224	172.50.1.224	172.50.1.224	172.50.1.224

Subnetting

Problem 10

Number of needed subnets **45**

Network Address **220.100.100.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 252

Total number of subnets 64

Total number of host addresses 4

Number of usable addresses 2

Number of bits borrowed 6

What is the 5th subnet range? 220.100.100.16 to 220.100.100.19

What is the subnet number for the 4th subnet? 220.100.100.12

What is the subnet broadcast address for the 13th subnet? 220.100.100.51

What are the assignable addresses for the 12th subnet? 220.100.100.45 to 220.100.100.46

Show your work for Problem 10 in the space below.

Number of Subnets				Number of Hosts			
256	128	64	32	16	8	4	2
-	2	4	8	16	32	64	128
128	64	32	16	8	4	2	1
220 . 100 . 100 .	0	0	0	0	0	0	0
Binary values							
(1)	0					220.100.100.0	to 220.100.100.3
(2)					1	220.100.100.4	to 220.100.100.7
(3)			1		0	220.100.100.8	to 220.100.100.11
(4)			1		1	220.100.100.12	to 220.100.100.15
(5)			1	0	0	220.100.100.16	to 220.100.100.19
(6)			1	0	1	220.100.100.20	to 220.100.100.23
(7)			1	1	0	220.100.100.24	to 220.100.100.27
(8)			1	1	1	220.100.100.28	to 220.100.100.31
(9)	1		0	0	0	220.100.100.32	to 220.100.100.35
(10)	1		0	0	1	220.100.100.36	to 220.100.100.39
(11)	1		0	1	0	220.100.100.40	to 220.100.100.43
(12)	1		0	1	1	220.100.100.44	to 220.100.100.47
(13)	1		1	0	0	220.100.100.48	to 220.100.100.51
(14)	1		1	0	1	220.100.100.52	to 220.100.100.55
(15)	1		1	1	0	220.100.100.56	to 220.100.100.59
(16)	1		1	1	1	220.100.100.60	to 220.100.100.63

128
64
32
16
8
+4
252

4
-2
2

Subnetting

Problem 11

Number of needed usable hosts **8,000**

Network Address **135.70.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 224 . 0

Total number of subnets 8

Total number of host addresses 8,192

Number of usable addresses 8,190

Number of bits borrowed 3

What is the 6th subnet range? 135.70.160.0 to 135.70.191.255

What is the subnet number for the 7th subnet? 135.70.192.0

What is the subnet broadcast address for the 3rd subnet? 135.70.95.255

What are the assignable addresses for the 5th subnet? 135.70.128.1 to 135.70.159.254

Show your work for **Problem 11** in the space below.

[illegible]

Subnetting

Problem 12

Number of needed usable hosts **45**

Network Address **198.125.50.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 4

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 2

What is the 2nd subnet range? 198.125.50.64 to 198.125.50.127

What is the subnet number for the 2nd subnet? 198.125.50.64

What is the subnet broadcast address for the 4th subnet? 198.125.50.255

What are the assignable addresses for the 3rd subnet? 198.125.50.129 to 198.125.50.190

Show your work for Problem 12 in the space below.

Number of Subnets	256 128			64 32 16 8 4 2							Number of Hosts
	-	2	4	8	16	32	64	128	256		
		128	64	32	16	8	4	2	1	-	Binary values
198 . 125 . 50 .	0	0	0	0	0	0	0	0	0		
(1)		0								to	198.125.50.63
(2)		1								to	198.125.50.127
(3)	1	0								to	198.125.50.191
(4)	1	1								to	198.125.50.255

$$\begin{array}{r} 128 \\ +64 \\ \hline 192 \end{array}$$

$$\begin{array}{r} 64 \\ -2 \\ \hline 62 \end{array}$$

Subnetting

Problem 13

Network Address **165.200.0.0 /26**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

What is the 10th subnet range? 165.200.2.64 to 165.200.2.127

What is the subnet number for the 11th subnet? 165.200.2.128

What is the subnet broadcast address for the 1023rd subnet? 165.200.255.191

What are the assignable addresses for the 1022nd subnet? 165.200.255.65 to 165.200.255.126

Show your work for **Problem 13** in the space below.

Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
165.200.0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
128	(1)															to 165.200.0.63
64	(2)															to 165.200.0.127
32	(3)															to 165.200.0.191
16	(4)															to 165.200.0.255
8	(5)															to 165.200.1.63
4	(6)															to 165.200.1.127
2	(7)															to 165.200.1.191
+1	(8)															to 165.200.1.255
252	(9)															to 165.200.2.63
	(10)															to 165.200.2.127
	(11)															to 165.200.2.191
	(12)															to 165.200.2.255
64	(13)															to 165.200.3.63
-2	(14)															to 165.200.3.127
62	(15)															to 165.200.3.191
	(16)															to 165.200.3.255
(1021)	1	1	1	1	1	1	1	1	1	0	1	165.200.255.64	to 165.200.255.127			
(1022)	1	1	1	1	1	1	1	1	1	1	0	165.200.155.128	to 165.200.255.191			
(1023)	1	1	1	1	1	1	1	1	1	1	1	165.200.255.192	to 165.200.255.255			

Subnetting

Problem 14

Number of needed usable hosts **16**

Network Address **200.10.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 224

Total number of subnets 8

Total number of host addresses 32

Number of usable addresses 30

Number of bits borrowed 3

What is the 7th subnet range? 200.10.10.192 to 200.10.10.223

What is the subnet number for the 5th subnet? 200.10.10.128

What is the subnet broadcast address for the 4th subnet? 200.10.10.127

What are the assignable addresses for the 6th subnet? 200.10.10.161 to 200.10.10.190

Show your work for Problem 14 in the space below.

												Number of					
							256	128	64	32	16	8	4	2	-	Hosts	
Number of Subnets							-	2	4	8	16	32	64	128	256		
							128	64	32	16	8	4	2	1	-	Binary values	
200 . 10 . 10 . 0 0 0										0	0	0	0	0	0		
(1)									0	200.10.10.0					to	200.10.10.31	
(2)									1	200.10.10.32					to	200.10.10.63	
(3)								1	0	200.10.10.64					to	200.10.10.95	
(4)								1	1	200.10.10.96					to	200.10.10.127	
(5)							1	0	0	200.10.10.128					to	200.10.10.159	
(6)							1	0	1	200.10.10.160					to	200.10.10.191	
(7)							1	1	0	200.10.10.192					to	200.10.10.223	
(8)							1	1	1	200.10.10.224					to	200.10.10.255	

$$\begin{array}{r}
 128 \\
 64 \\
 +32 \\
 \hline
 224
 \end{array}
 \qquad
 \begin{array}{r}
 32 \\
 -2 \\
 \hline
 30
 \end{array}$$

Subnetting

Problem 15

Network Address **93.0.0.0 /19**

Address class A

Default subnet mask 255 . 0 . 0 . 0

Custom subnet mask 255 . 255 . 224 . 0

Total number of subnets 2,048

Total number of host addresses 8,192

Number of usable addresses 8,190

Number of bits borrowed 11

What is the 15th subnet range? 93.1.192.0 to 93.1.223.255

What is the subnet number for the 9th subnet? 93.1.0.0

What is the subnet broadcast address for the 7th subnet? 93.0.223.255

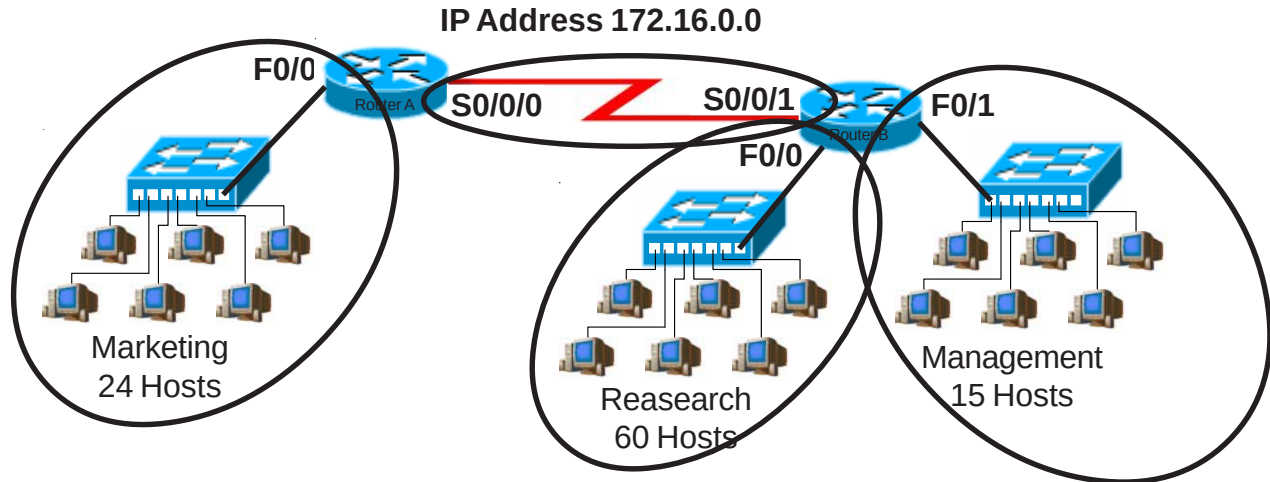
What are the assignable addresses for the 12th subnet? 93.1.96.1 to 93.1.127.254

Show your work for **Problem 15** in the space below.

Number of Hosts	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576	2097152	4194304	8388608	16777216	33554432	67108864	134217728	268435456	536870912	1073741824	2147483648	4294967296	8589934592	17179869184	34359738368	68719476736	137438953472	274877906944	549755813888	1099511627776	2199023255552	4398046511104	8796093022208	17592186044416	35184372088832	70368744177664	140737488355328	281474976710656	562949953421312	1125899906842624	2251799813685248	4503599627370496	9007199254740992	18014398509481984	36028797018963968	72057594037927936	144115188075855872	288230376151711744	576460752303423488	1152921504606846976	2305843009213693952	4611686018427387904	9223372036854775808	18446744073709551616	36893488147419103232	73786976294838206464	147573952589676412928	295147905179352825856	590295810358705651712	1180591620717411303424	2361183241434822606848	4722366482869645213696	9444732965739290427392	18889465931478580854784	37778931862957161709568	75557863725914323419136	151115727451828646838272	302231454903657293676544	604462909807314587353088	1208925819614629174706176	2417851639229258349412352	4835703278458516698824704	9671406556917033397649408	19342813113834066795298816	38685626227668133590597632	77371252455336267181195264	154742504910672534362390528	309485009821345068724781056	618970019642690137449562112	1237940039285380274899124224	2475880078570760549798248448	4951760157141521099596496896	9903520314283042199192993792	19807040628566084398385987584	39614081257132168796771975168	79228162514264337593543950336	158456325028528675187087900672	316912650057057350374175801344	633825300114114700748351602688	1267650600228229401496703205376	2535301200456458802993406410752	5070602400912917605986812821504	10141204801825835211973625643008	20282409603651670423947251286016	40564819207303340847894502572032	81129638414606681695789005144064	162259276829213363391578010288128	324518553658426726783156020576256	649037107316853453566312041152512	1298074214633706907132624082305024	2596148429267413814265248164610048	5192296858534827628530496329220096	10384593717069655257060992658440192	20769187434139310514121985316880384	41538374868278621028243970633760768	83076749736557242056487941267521536	166153499473114484112975882535043072	332306998946228968225951765070086144	664613997892457936451903530140172288	1329227995784915872903807060280344704	2658455991569831745807614120560689408	5316911983139663491615228241121378816	10633823966279326983230456482242757632	21267647932558653966460912964485515264	42535295865117307932921825928971030528	85070591730234615865843651857942061056	170141183460469231731687303715884122112	340282366920938463463374607431768244224	680564733841876926926749214863536488448	1361129467683753853853498429727072976896	2722258935367507707706996859454145953792	5444517870735015415413993718908291907584	10889035741470030830827987437816583815168	21778071482940061661655974875633167630336	43556142965880123323311949751266335260672	87112285931760246646623899502532670521344	174224571863520493293247799005065341042688	348449143727040986586495598010130682085376	696898287454081973172991196020261364170752	1393796574908163946345982392040522728341504	2787593149816327892691964784081045456683008	5575186299632655785383929568162090913366016	1115037259926531157076785913632418182673216	2230074519853062314153571827264836365346432	4460149039706124628307143654529672730692864	8920298079412249256614287309059345461385728	1784059615882449
-----------------	---	---	---	---	----	----	----	-----	-----	-----	------	------	------	------	-------	-------	-------	--------	--------	--------	---------	---------	---------	---------	----------	----------	----------	-----------	-----------	-----------	------------	------------	------------	------------	-------------	-------------	-------------	--------------	--------------	--------------	---------------	---------------	---------------	---------------	----------------	----------------	----------------	-----------------	-----------------	-----------------	------------------	------------------	------------------	------------------	-------------------	-------------------	-------------------	--------------------	--------------------	--------------------	---------------------	---------------------	---------------------	---------------------	----------------------	----------------------	----------------------	-----------------------	-----------------------	-----------------------	------------------------	------------------------	------------------------	------------------------	-------------------------	-------------------------	-------------------------	--------------------------	--------------------------	--------------------------	---------------------------	---------------------------	---------------------------	---------------------------	----------------------------	----------------------------	----------------------------	-----------------------------	-----------------------------	-----------------------------	------------------------------	------------------------------	------------------------------	------------------------------	-------------------------------	-------------------------------	-------------------------------	--------------------------------	--------------------------------	--------------------------------	---------------------------------	---------------------------------	---------------------------------	----------------------------------	----------------------------------	----------------------------------	----------------------------------	-----------------------------------	-----------------------------------	-----------------------------------	------------------------------------	------------------------------------	------------------------------------	-------------------------------------	-------------------------------------	-------------------------------------	-------------------------------------	--------------------------------------	--------------------------------------	--------------------------------------	---------------------------------------	---------------------------------------	---------------------------------------	--	--	--	--	---	---	---	--	--	--	---	---	---	---	--	--	--	---	---	---	---	---	---	---	------------------

Practical Subnetting 1

Based on the information in the graphic shown, design a network addressing scheme that will supply the minimum number of subnets, and allow enough extra subnets and hosts for 100% growth in both areas. Circle each subnet on the graphic and answer the questions below.



Address class	<i>B</i>
Custom subnet mask	<i>255.255.224.0</i>
Minimum number of subnets needed	<i>4</i>
Extra subnets required for 100% growth (Round up to the next whole number)	<i>+</i> <i>4</i>
Total number of subnets needed	<i>=</i> <i>8</i>
Number of host addresses in the largest subnet group	<i>60</i>
Number of addresses needed for 100% growth in the largest subnet (Round up to the next whole number)	<i>+</i> <i>60</i>
Total number of address needed for the largest subnet	<i>=</i> <i>120</i>

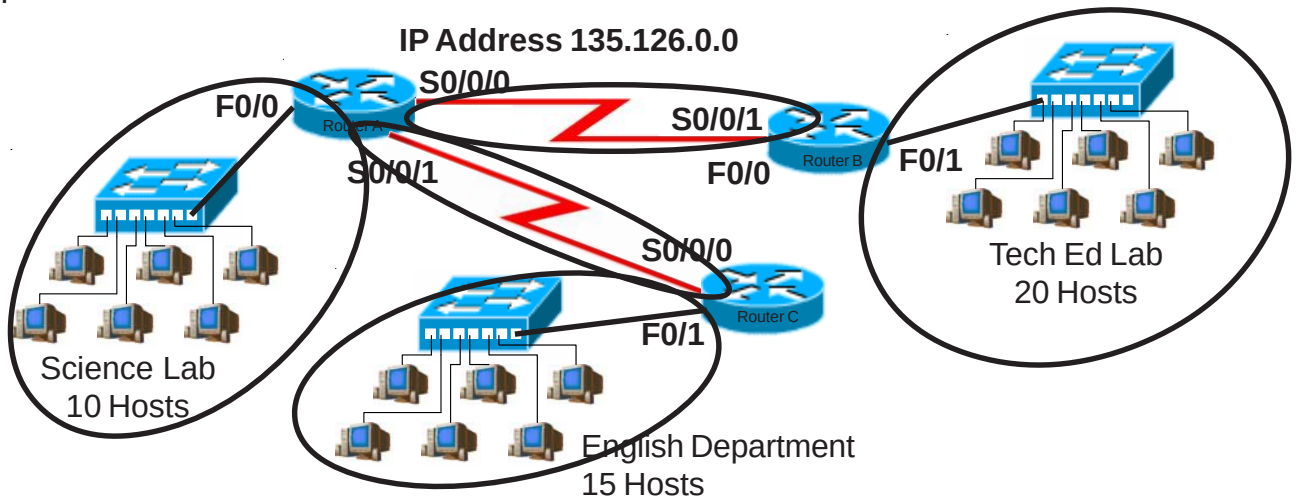
Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Research	<i>172.16.0.0 to 172.31.255</i>
IP address range for Marketing	<i>172.16.32.0 to 172.63.255</i>
IP address range for Management	<i>172.16.64.0 to 172.95.255</i>
IP address range for Router A to Router B serial connection	<i>172.16.96.0 to 172.127.255</i>

[illegible]

Practical Subnetting 2

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 30% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class	<u>B</u>
Custom subnet mask	<u>255.255.255.224</u>
Minimum number of subnets needed	<u>5</u>
Extra subnets required for 30% growth (Round up to the next whole number)	<u>+ 2</u>
Total number of subnets needed	<u>= 7</u>
Number of host addresses in the largest subnet group	<u>20</u>
Number of addresses needed for 30% growth in the largest subnet (Round up to the next whole number)	<u>+ 6</u>
Total number of address needed for the largest subnet	<u>= 26</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

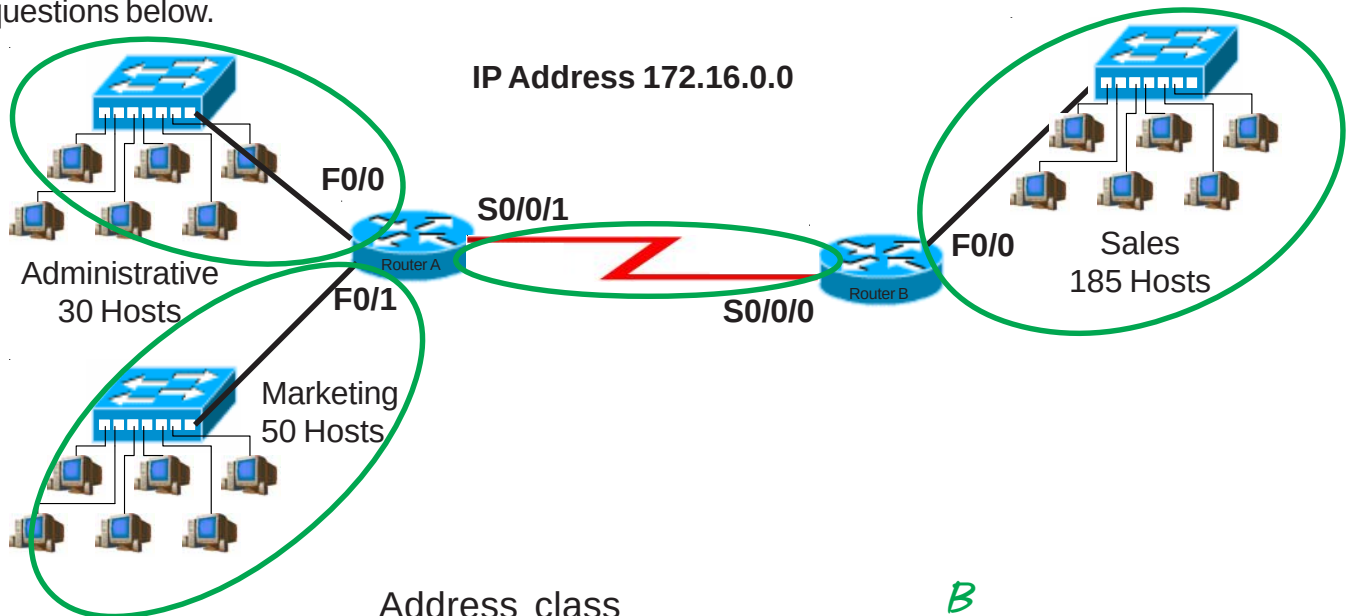
IP address range for Tech Ed	<u>135.126.0.0 to 135.126.0.31</u>
IP address range for English	<u>135.126.0.32 to 135.126.0.63</u>
IP address range for Science	<u>135.126.0.64 to 135.126.0.95</u>
IP address range for Router A to Router B serial connection	<u>135.126.0.96 to 135.126.0.127</u>
IP address range for Router A to Router C serial connection	<u>135.126.0.128 to 135.126.0.159</u>

Show your work for Problem 2 in the space below.

Number of Hosts -	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets -	2	4	8	16	32	64	128	256	512	1,024	2,048	4,096	8,192	16,384	32,768	65,536
Binary values -	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
135.126.0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(1)	.															
(2)																
(3)																
(4)																
(5)																
(6)																
(7)																
(8)																
(9)																
(10)																
(11)																
(12)																
(13)																
(14)																
(15)																
(16)																
5																
x.3																
1.5																
(Round up to 2)																
20																
x.3																
6																
135.126.0.31	to															
135.126.0.63	to															
135.126.0.95	to															
135.126.0.127	to															
135.126.0.159	to															
135.126.0.191	to															
135.126.0.223	to															
135.126.0.255	to															
135.126.1.31	to															
135.126.1.63	to															
135.126.1.95	to															
135.126.1.127	to															
135.126.1.159	to															
135.126.1.191	to															
135.126.1.223	to															
135.126.1.255	to															

Practical Subnetting 3

Based on the information in the graphic shown, design a classfull network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 25% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class	<u>B</u>
Custom subnet mask	<u>255.255.255.0</u>
Minimum number of subnets needed	<u>4</u>
Extra subnets required for 25% growth (Round up to the next whole number)	<u>+ 1</u>
Total number of subnets needed	<u>= 5</u>
Number of host addresses in the largest subnet group	<u>185</u>
Number of addresses needed for 25% growth in the largest subnet (Round up to the next whole number)	<u>+ 47</u>
Total number of address needed for the largest subnet	<u>= 232</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Sales	<u>172.16.0.0 to 172.16.0.255</u>
IP address range for Marketing	<u>172.16.1.0 to 172.16.1.255</u>
IP address range for Administrative	<u>172.16.2.0 to 172.16.2.255</u>
IP address range for Router A to Router B serial connection	<u>172.16.3.0 to 172.16.3.255</u>

Show your work for Problem 3 in the space below.

Number of
Hosts

65,536

32,768

16,384

8,192

4,096

2,048

1,024

512

Number of
Subnets

2

4

8

16

32

64

128

256

Binary values

128

64

32

16

8

4

2

1

172.16.0.0

0

0

0

0

0

0

0

0

4

x.25

1

225

x.25

56.25

(Round up to 57)

(1)

(2)

(3)

(4)

(5)

(6)

(7)

(8)

(9)

(10)

(11)

(12)

(13)

(14)

(15)

(16)

0

1

1

1

0

0

1

1

0

0

0

1

0

1

0

1

172.16.0.0

to

172.16.0.255

172.16.1.0

to

172.16.1.255

172.16.2.0

to

172.16.2.255

172.16.3.0

to

172.16.3.255

172.16.4.0

to

172.16.4.255

172.16.5.0

to

172.16.5.255

172.16.6.0

to

172.16.6.255

172.16.7.0

to

172.16.7.255

172.16.8.0

to

172.16.8.255

172.16.9.0

to

172.16.9.255

172.16.10.0

to

172.16.10.255

172.16.11.0

to

172.16.11.255

172.16.12.0

to

172.16.12.255

172.16.13.0

to

172.16.13.255

172.16.14.0

to

172.16.14.255

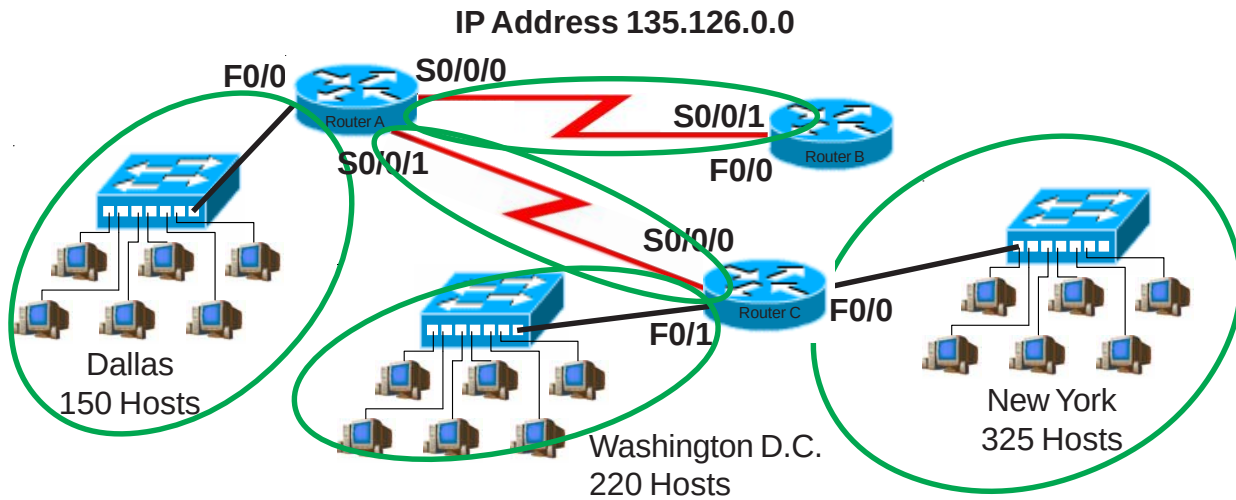
172.16.15.0

to

172.16.15.255

Practical Subnetting 4

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of subnets**, and allow enough extra subnets and hosts for 70% growth in all areas. Circle each subnet on the graphic and answer the questions below. Circle each subnet on the graphic and answer the questions below.



Address class	<u>B</u>
Custom subnet mask	<u>255.255.240.0</u>
Minimum number of subnets needed	<u>5</u>
Extra subnets required for 70% growth (Round up to the next whole number)	<u>+ 4</u>
Total number of subnets needed	<u>= 9</u>
Number of host addresses in the largest subnet group	<u>325</u>
Number of addresses needed for 70% growth in the largest subnet (Round up to the next whole number)	<u>+ 228</u>
Total number of address needed for the largest subnet	<u>= 553</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for New York	<u>135.126.0.0 to 135.126.15.255</u>
IP address range for Washington D. C.	<u>135.126.16.0 to 135.126.31.255</u>
IP address range for Dallas	<u>135.126.32.0 to 135.126.47.255</u>
IP address range for Router A to Router B serial connection	<u>135.126.48.0 to 135.126.63.255</u>
IP address range for Router A to Router C serial connection	<u>135.126.64.0 to 135.126.79.255</u>

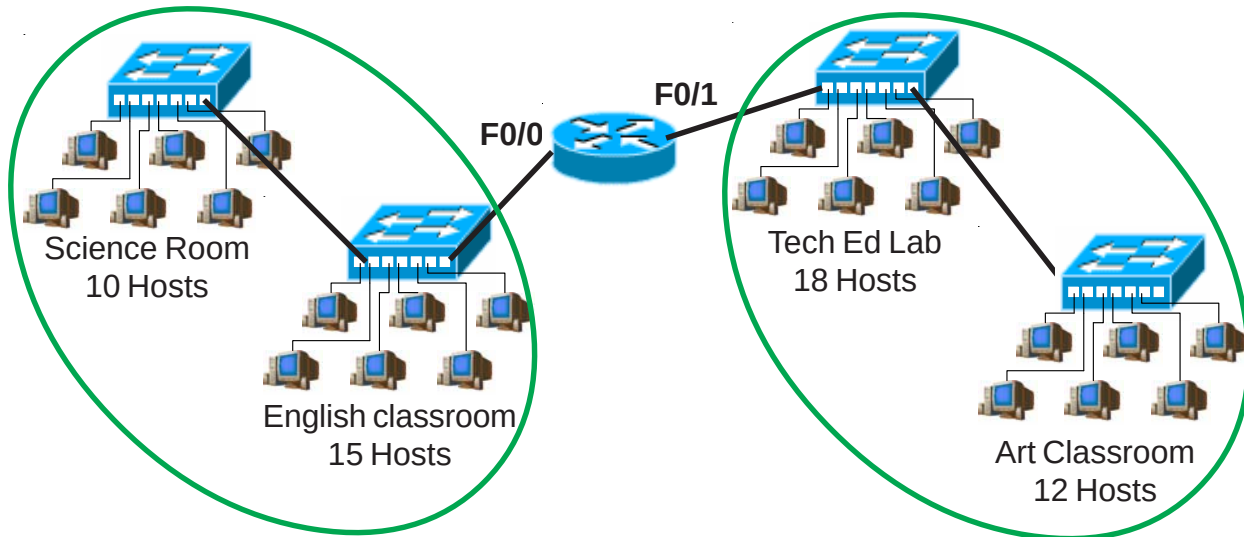
Show your work for Problem 4 in the space below.

Number of Hosts -	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets -	2	4	8	16	32	64	128	256	512	1,024	2,048	4,096	8,192	16,384	32,768	65,536
Binary values -	128	64	32	16	8	4	2	1	.	.	.	.	.	.	.	.
135.126.0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(1)	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
(2)	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(3)	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
(4)	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(5)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(6)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(7)	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(8)	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
(9)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(10)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(11)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(12)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(13)	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(14)	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(15)	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
(16)	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
135.126.0.0	135.126.0.0	135.126.16.0	135.126.32.0	135.126.48.0	135.126.64.0	135.126.80.0	135.126.96.0	135.126.112.0	135.126.128.0	135.126.144.0	135.126.160.0	135.126.176.0	135.126.192.0	135.126.208.0	135.126.224.0	135.126.240.0
to	to	to	to	to	to	to	to	to	to	to	to	to	to	to	to	to
135.126.15.255	135.126.31.255	135.126.47.255	135.126.63.255	135.126.79.255	135.126.95.255	135.126.111.255	135.126.127.255	135.126.143.255	135.126.159.255	135.126.175.255	135.126.191.255	135.126.207.255	135.126.223.255	135.126.239.255	135.126.255.255	135.126.255.255

Practical Subnetting 5

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 100% growth in all areas. Circle each subnet on the graphic and answer the questions below.

IP Address 210.15.10.0



Address class	<u>C</u>
Custom subnet mask	<u>255.255.255.192</u>
Minimum number of subnets needed	<u>2</u>
Extra subnets required for 100% growth (Round up to the next whole number)	<u>+ 2</u>
Total number of subnets needed	<u>= 4</u>
Number of host addresses in the largest subnet group	<u>30</u>
Number of addresses needed for 100% growth in the largest subnet (Round up to the next whole number)	<u>+ 30</u>
Total number of address needed for the largest subnet	<u>= 60</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Router F0/0 Port 210.15.10.0 to 210.15.10.63

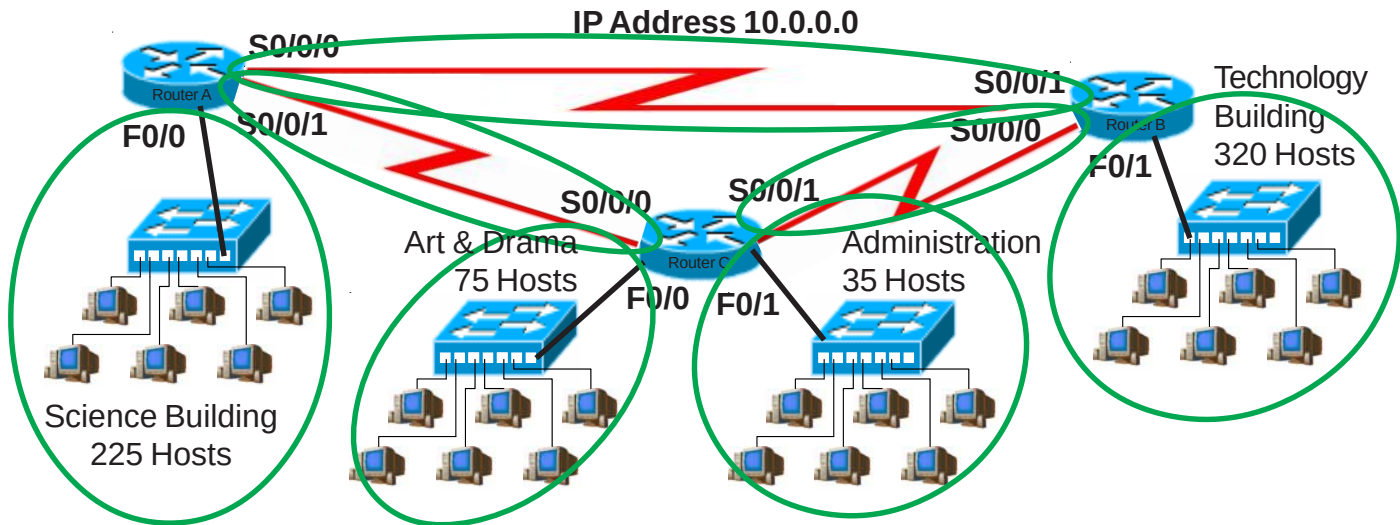
IP address range for Router F0/1 Port 210.15.10.64 to 210.15.10.127

Show your work for Problem 5 in the space below.

Number of Subnets		256	128	64	32	16	8	4	2	-	Number of Hosts	
		-	2	4	8	16	32	64	128	256		
		128	64	32	16	8	4	2	1	-	Binary values	
210.15.10.0 0		0	0	0	0	0	0	0	0	0		
(1)	0	210.15.10.0									to	210.15.10.63
(2)	1	210.15.10.64									to	210.15.10.127
(3)	1 0	210.15.10.128									to	210.15.10.191
(4)	1 1	210.15.10.192									to	210.15.10.255

Practical Subnetting 6

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of subnets**, and allow enough extra subnets and hosts for 20% growth in all areas. Circle each subnet on the graphic and answer the questions below. Circle each subnet on the graphic and answer the questions below.



Address class	<u>A</u>
Custom subnet mask	<u>255.240.0.0</u>
Minimum number of subnets needed	<u>7</u>
Extra subnets required for 20% growth (Round up to the next whole number)	<u>+ 2</u>
Total number of subnets needed	<u>= 9</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

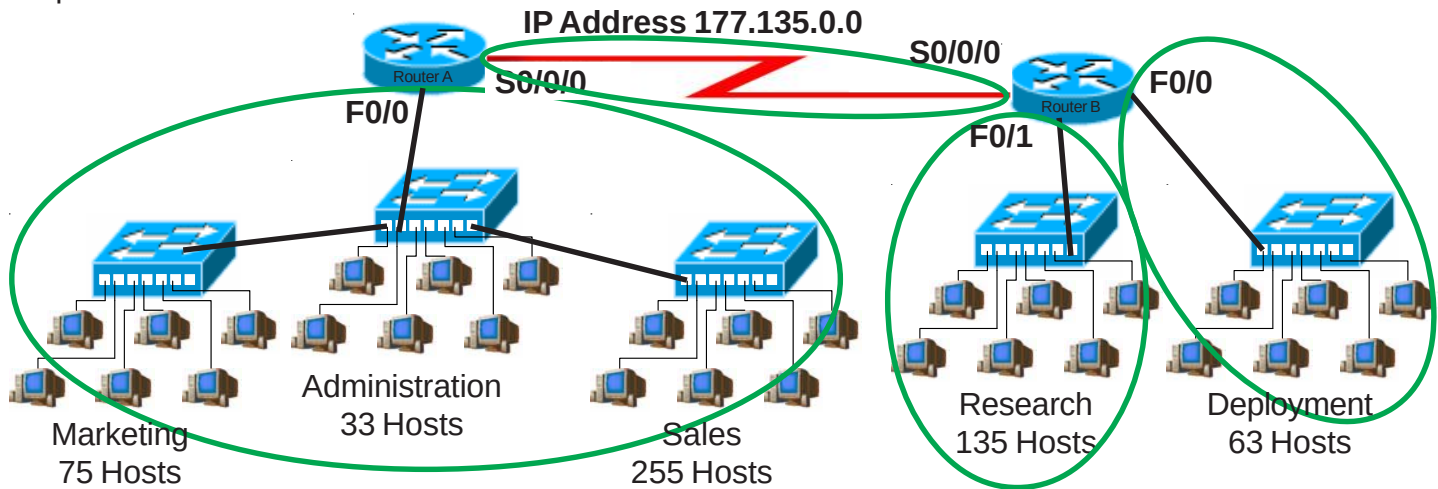
IP address range for Technology	<u>10.0.0.0 to 10.15.255.255</u>
IP address range for Science	<u>10.16.0.0 to 10.31.255.255</u>
IP address range for Arts & Drama	<u>10.32.0.0 to 10.47.255.255</u>
IP Address range Administration	<u>10.48.0.0 to 10.63.255.255</u>
IP address range for Router A to Router B serial connection	<u>10.64.0.0 to 10.79.255.255</u>
IP address range for Router A to Router C serial connection	<u>10.80.0.0 to 10.95.255.255</u>
IP address range for Router B to Router C serial connection	<u>10.96.0.0 to 10.111.255.255</u>

Show your work for Problem 6 in the space below.

Number of Hosts		1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576	2097152	4194304	8388608	16777216	33554432	67108864	134217728	268435456	536870912	1073741824	2147483648	4294967296	8589934592	17179869184	34359738368	68719476736	137438953472	274877906944	549755813888	1099511627776	2199023255552	4398046511104	8796093022208	17592186044416	35184372088832	70368744177664	140737488355328	281474976710656	562949953421312	1125899906842624	2251799813685248	4503599627370496	9007199254740992	18014398509481984	36028797018963968	72057594037927936	144115188075855872	288230376151711744	576460752303423488	1152921504606846976	2305843009213693952	4611686018427387904	9223372036854775808	18446744073709551616	36893488147419103232	73786976294838206464	147573952589676412928	295147905179352825856	590295810358705651712	1180591620717411303424	2361183241434822606848	4722366482869645213696	9444732965739290427392	18889465931478580854784	37778931862957161709568	75557863725914323419136	151115727451828646838272	302231454903657293676544	604462909807314587353088	1208925819614629174706176	2417851639229258349412352	4835703278458516698824704	9671406556917033397649408	19342813113834066795298816	38685626227668133590597632	77371252455336267181195264	154742504910672534362390528	309485009821345068724781056	618970019642690137449562112	1237940039285380274899124224	2475880078570760549798248448	4951760157141521099596496896	9903520314283042199192993792	19807040628566084398385987584	39614081257132168796771975168	79228162514264337593543950336	158456325028528675187087900672	316912650057057350374175801344	633825300114114700748351602688	1267650600228229401496703205376	2535301200456458802993406410752	5070602400912917605986812821504	10141204801825835211973625643008	20282409603651670423947251286016	40564819207303340847894502572032	81129638414606681695789005144064	162259276829213363391578010288128	324518553658426726783156020576256	649037107316853453566312041152512	1298074214633706907132624082305024	2596148429267413814265248164610048	5192296858534827628530496329220096	10384593717069655257060992658440192	20769187434139310514121985316880384	41538374868278621028243970633760768	83076749736557242056487941267521536	166153499473114484112975882535043072	332306998946228968225951765070086144	664613997892457936451903530140172288	1329227995784915872903807060280344576	2658455991569831745807614120560689152	5316911983139663491615228241121378304	10633823966279326983230456482242756608	21267647932558653966460912964485513216	42535295865117307932921825928971026432	85070591730234615865843651857942052864	170141183460469231731687303715884105728	340282366920938463463374607431768211456	680564733841876926926749214863536422912	1361129467683753853853498429727072845824	272225893536750770770699685945414569152	544451787073501541541399371890829138304	1088903574147003083082798743781658276608	2177807148294006166165597487563316553216	4355614296588012332331194975126633106432	8711228593176024664662389950253266212864	17422457186352049329324779900506532425728	34844914372704098658649559801013064851456	69689828745408197317299119602026129702912	139379657490816394634598239204052259405824	278759314981632789269196478408104518811648	557518629963265578538392956816209037623296	1115037259926531157076785913632418075246592	2230074519853062314153571827264836150493184	4460149039706124628307143654529672300986368	8920298079412249256614287309059344601972736	17840596158824498513228574618118689203945472	35681192317648997026457149236237378407890848	71362384635297994052914298472474756815781696	142724769270595988105828596944949513631563392	285449538541191976211657193889899027263126784	570899077082383952423314387779798054526253568	1141798154164767904846628775559596109052507136	2283596308329535809693257551119192218105014272	4567192616659071619386515102238384436210028544	9134385233318143238773030204476768872420057088	18268770466636286477546060408953537744840114176	36537540933272572955092120817907075489680228352	73075081866545145910184241635814150979360456704	146150163733090291820368483271628301958720913408	292300327466180583640736966543256603917441826816	584600654932361167281473933086513207834883653632	1169201309864722334562947866173026415669767307264	2338402619729444669125895732346052831339534614528	4676805239458889338251791464692105662679069229056	9353610478917778676503582929384211325358138458112	18707220957835557353007165858768422650716276916224	37414441915671114706014331717536845301432553832448	74828883831342229412028663435073690602865107664896	149657767662684458824057326870147381205730215329792	299315535325368917648114653740294762411460430659584	598631070650737835296229307480589524822920861319168	1197262141301475670592458614961179049645841722638336	2394524282602951341184917229922358099291683445276672	4789048565205902682369834459844716198583366890553344	9578097130411805364739668919689432397166733781106688	19156194260823610729479337839378864794333467562213376	38312388521647221458958675678757729588666935124426752	76624777043294442917917351357515459177333870248853504	153249554086588885835834702715030918354667740497707008	306499108173177771671669405430061836709335480995414016	612998216346355543343338810860123673418670961990828032	1225996432692711086686677621720247346837341923981665664	2451992865385422173373355243440494693674683847963331296	4903985730770844346746710486880989387349367695926662592	9807971461541688693493420973761978774698735391853325184	19615942923083377386986841947523957549397470783706650368	39231885846166754773973683895047915098794941567413300736	78463771692333509547947367790095830197589883134826601472	156927543384667019095894735580191660395179766269653202944	313855086769334038191789471160383320790359532539306405888	627710173538668076383578942320766641580719065078612811776	1255420347077336152767157884641533283161438130157225623552	251084069415467230553431576928306656632287626031445124704	502168138830934461106863153856613313264575252062890249408	1004336277661868922213726307713226626529150504125780498016	2008672555323737844427452615426453253058301008251560996032	4017345110647475688854905230852906506116602016503121992064	8034690221294951377709810461705813012233204033006243984128	16069380442589902755419620923411626024466408066012487968256	32138760885179805510839241846823252048932816132024975936512	64277521770359611021678483693646504097865632264049951873024	128555043540719222043356967387293008195731264528099903746048	257110087081438444086713934774586016391462529056199807492096	514220174162876888173427869549172032782925058112399614984192	1028440348325753776346855739098344065565850116224799229968384	2056880696651507552693711478196688131131700232449598459936768	4113761393303015105387422956393376262263400464899196919873536	8227522786606030210774845912786752524526800929798393839747072	16455045573212060421549691825573505049053601859596787679494144	32910091146424120843099383651147010098107203719193575358988288	65820182292848241686198767302294020196214407438387150717976576	131640364585696483372397534604588040392428814876774301435953152	263280729171392966744795069209176080784857629753548602871906304	526561458342785933489590138418352161569715259507097205743812608	1053122916685771866979180276836704323139430519014194411487625216	2106245833371543733958360553673408646278861038028388822975250432	4212491666743087467916721107346817292557722076056777645950500864	8424983333486174935833442214693634585115444152113555291901001728	16849966666972349871666884429387269170230888304227110583802003456	33699933333944699743333768858774538340461776608454221167604006912	67399866667889399486667537717549076680923553216908442335208013824	13479973333577879897333507543509815336184710643381688467041602768	26959946667155759794667015087019630672369421286763376934083205536	53919893334311519589334030174039261344738842573526753868166411072	107839786668623039178668060348078522689477685147053507736332822144	215679573337246078357336120696157045378955370294107015472665644288	431359146674492156714672241392314090757910740588214030945331288576	862718293348984313429344482784628181515821481176428061890662577152	1725436586697968626858688965569256363031642962352856123781325154304	3450873173395937253717377931138512726063285924705712247562650308608	6901746346791874507434755862277025452126571849411424495125300617216	1380349269358374901486951172455405090425314369882284899025060123424	2760698538716749802973902344910810180850628739764569798050120246848	5521397077433499605947804689821620361701257479529139596100240493696	11042794154866999211895609379643240723402514959058279192200480987392	22085588309733998423791218759286481446805029918116558384400961974784	44171176619467996847582437518572962893610059836233116768801923949568	88342353238935993695164875037145925787220119672466233537603847899136	176684706477871987390329750074291851574440239344932467075207695798272	353369412955743974780659500148583703148880478689864934150415391596544	706738825911487949561319000297167406297760957379729868300830783193088	1413477651822975899122638000594334812595521914759459736601661566386176	2826955303645951798245276001188669625191043829518919473203323132772352	5653910607291903596490552002377339250382087659037838946406646265544704	11307821214583807192981104004754678500764175318075677892813292531089408	22615642429167614385962208009509357001528350636151355785626585062178816	45231284858335228771924416019018714003056701272302711571253170124357632	90462569716670457543848832038037428006113402544605423142506340248715264	180925139433340915087697664076074856012226805089210846285012680497430528	361850278866681830175395328152149712024453610178421692570025360994861056	723700557733363660350790656304299424048907220356843385140050721989722112	1447401115466727320701581312608598848097814440713686770280101443979444224	2894802230933454641403162625217197696195628881427373540560202887958888448	5789604461866909282806325250434395392391257762854747081120405775917776896	11579208923733818565612650500868790784782515525709494162240811551835553792	23158417847467637131225301001737581569565031051418988324481623103671107584	46316835694935274262450602003475163139130062022837976648963246207342215168	92633671389870548524901204006950326278260124045675953297926492414684430336	185267342779741097049802408013900652556520248091351906595852984829368860672	370534685559482194099604816027801305113040496182703813191705969658737721344	741069371118964388199209632055602610226080992365407626383411939317475442688	1482138742237928776398419264111205220452161984730815252766823878634950885376	2964277484475857552796838528222410440904323969461630505533647757269901770752	5928554968951715105593677056444820881808647938923261011067295514539803541504	11857109937903430211187354112889641763617295877846522022134591029079607083008	23714219875806860422374708225779283527234591755693044044269182058159214166016	47428439751613720844749416451558567054469183511386088088538364116318428332032	94856879503227441689498832903117134108938367022772176177076728232636856664064	189713759006454883378997665806234268217876734045544352354153456465273713328128	379427518012909766757995331612468536435753468091088704708306912930547
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Practical Subnetting 7

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 125% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class	<u>B</u>
Custom subnet mask	<u>255.255.252.0</u>
Minimum number of subnets needed	<u>4</u>
Extra subnets required for 125% growth (Round up to the next whole number)	<u>+ 5</u>
Total number of subnets needed	<u>= 9</u>
Number of host addresses in the largest subnet group	<u>363</u>
Number of addresses needed for 125% growth in the largest subnet (Round up to the next whole number)	<u>+ 454</u>
Total number of address needed for the largest subnet	<u>= 817</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

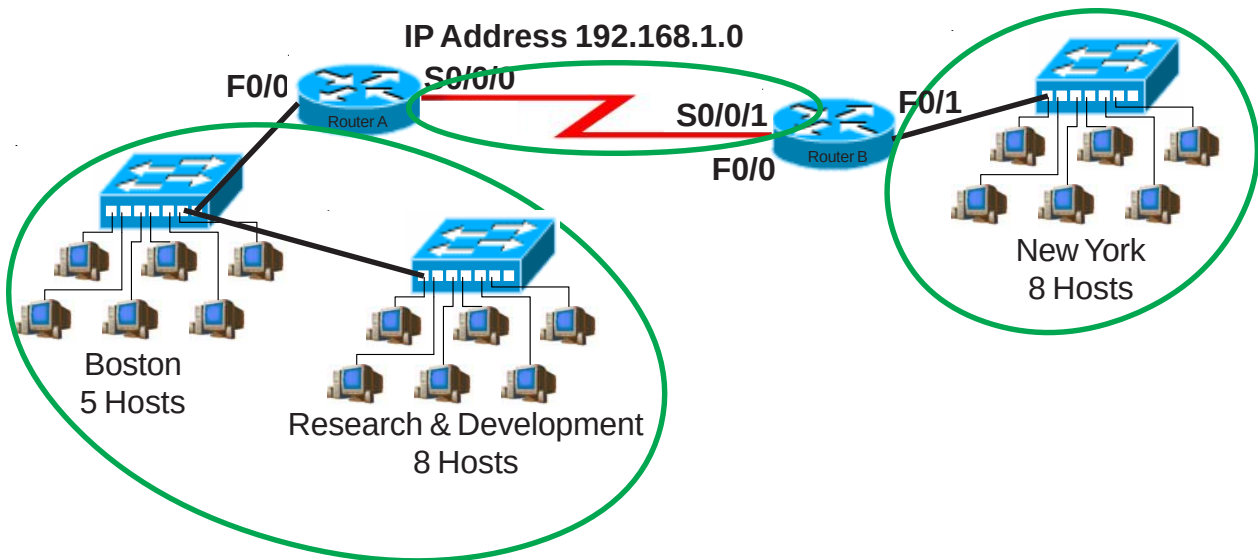
IP address range for Router A Port F0/0	<u>177.135.0.0 to 177.135.3.255</u>
IP address range for Research	<u>177.135.4.0 to 177.135.7.255</u>
IP address range for Deployment	<u>177.135.8.0 to 177.135.11.255</u>
IP address range for Router A to Router B serial connection	<u>177.135.12.0 to 177.135.15.255</u>

Show your work for Problem 7 in the space below.

Number of Hosts -	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets -	2	4	8	16	32	64	128	256	512	1,024	2,048	4,096	8,192	16,384	32,768	65,536
Binary values -	128	64	32	16	8	4	2	1	0	0	0	0	0	0	0	0
177.135 .	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(1)	.															
(2)																
(3)																
(4)																
(5)																
(6)																
(7)																
(8)																
(9)																
(10)																
(11)																
(12)																
(13)																
(14)																
(15)																
(16)																
177.135.0.0	to	177.135.3.255														
177.135.4.0	to	177.135.7.255														
177.135.8.0	to	177.135.11.255														
177.135.12.0	to	177.135.15.255														
177.135.16.0	to	177.135.19.255														
177.135.20.0	to	177.135.23.255														
177.135.24.0	to	177.135.27.255														
177.135.28.0	to	177.135.31.255														
177.135.32.0	to	177.135.35.255														
177.135.36.0	to	177.135.39.255														
177.135.40.0	to	177.135.43.255														
177.135.44.0	to	177.135.47.255														
177.135.48.0	to	177.135.51.255														
177.135.52.0	to	177.135.55.255														
177.135.56.0	to	177.135.59.255														
177.135.60.0	to	177.135.63.255														

Practical Subnetting 8

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number subnets**, and allow enough extra subnets and hosts for 85% growth in all areas. Circle each subnet on the graphic and answer the questions below. Circle each subnet on the graphic and answer the questions below.



Address class

C

Custom subnet mask

255.255.255.224

Minimum number of subnets needed

3

Extra subnets required for 85% growth
(Round up to the next whole number)

+ 3

Total number of subnets needed

= 6

Number of host addresses
in the largest subnet group

13

Number of addresses needed for
85% growth in the largest subnet
(Round up to the next whole number)

+ 12

Total number of address
needed for the largest subnet

= 25

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Router A F0/0

192.168.1.0 to 192.168.1.31

IP address range for New York

192.168.1.32 to 192.168.1.63

IP address range for Router A
to Router B serial connection

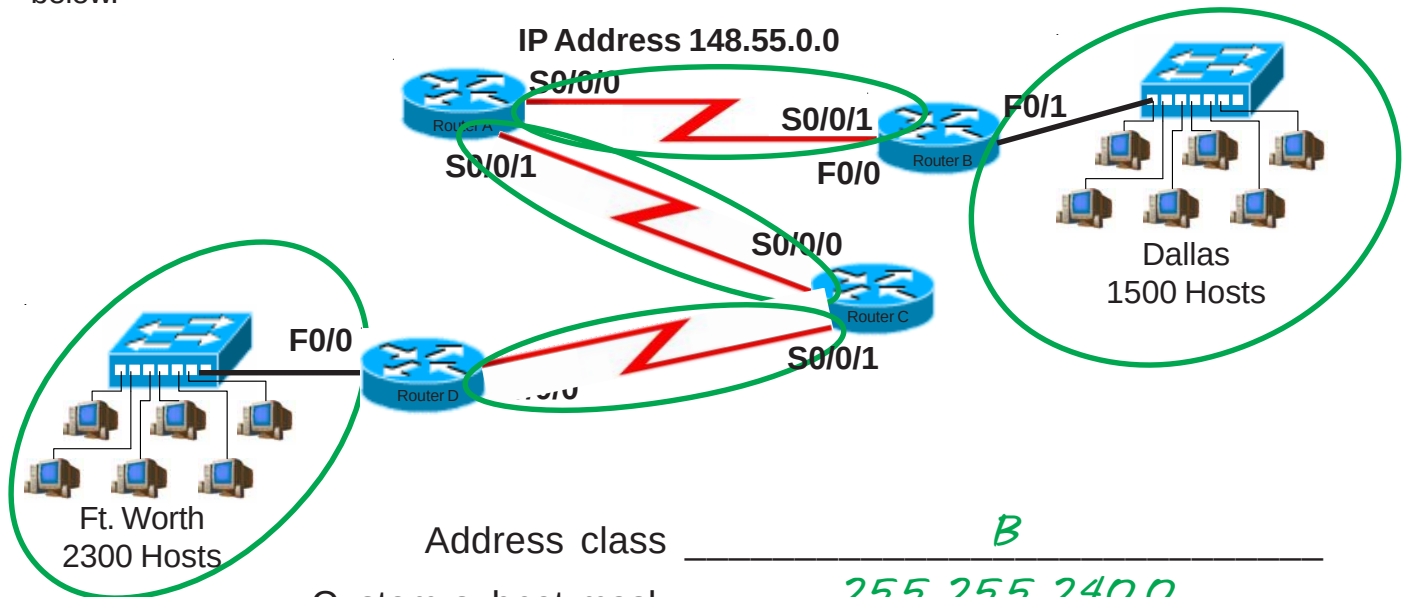
192.168.1.64 to 192.168.1.95

Show your work for Problem 8 in the space below.

Number of Subnets	256	128	64	32	16	8	4	2	-	Number of Hosts
	2	4	8	16	32	64	128	256		
	128	64	32	16	8	4	2	1	-	Binary values
192.168.1.0 0 0 0 0 0										
(1)			0	192.168.1.0			to 192.168.1.31			
(2)			1	192.168.1.32			to 192.168.1.63			
(3)		1	0	192.168.1.64			to 192.168.1.95			
(4)		1	1	192.168.1.96			to 192.168.1.127			
(5)	1	0	0	192.168.1.128			to 192.168.1.159			
(6)	1	0	1	192.168.1.160			to 192.168.1.191			
(7)	1	1	0	192.168.1.192			to 192.168.1.223			
(8)	1	1	1	192.168.1.224			to 192.168.1.255			

Practical Subnetting 9

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 15% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class	<u>B</u>
Custom subnet mask	<u>255.255.240.0</u>
Minimum number of subnets needed	<u>5</u>
Extra subnets required for 15% growth (Round up to the next whole number)	<u>+ 1</u>
Total number of subnets needed	<u>= 6</u>
Number of host addresses in the largest subnet group	<u>2300</u>
Number of addresses needed for 15% growth in the largest subnet (Round up to the next whole number)	<u>+ 345</u>
Total number of address needed for the largest subnet	<u>= 2645</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

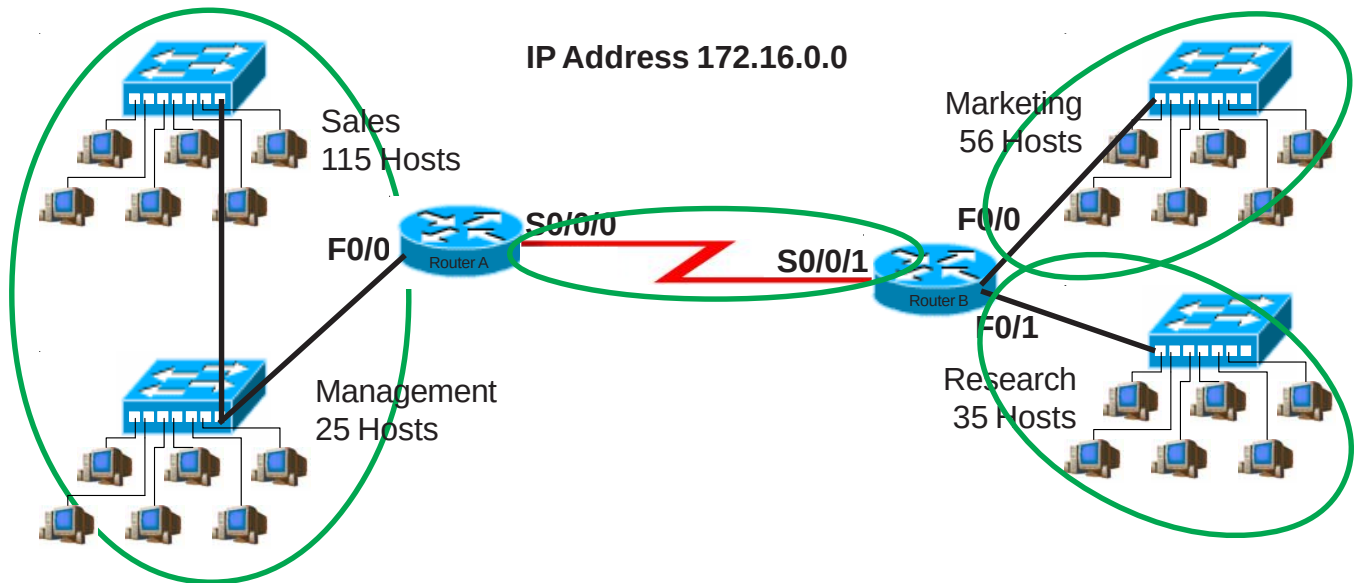
IP address range for Ft. Worth	<u>148.55.0.0 to 148.55.15.255</u>
IP address range for Dallas	<u>148.55.16.0 to 148.55.31.255</u>
IP address range for Router A to Router B serial connection	<u>148.55.32.0 to 148.55.47.255</u>
IP address range for Router A to Router C serial connection	<u>148.55.48.0 to 148.55.63.255</u>
IP address range for Router C to Router D serial connection	<u>148.55.64.0 to 148.55.79.255</u>

Show your work for Problem 9 in the space below.

Number of Hosts -	2	4	8	16	32	64	128	256	512	1,024	2,048	4,096	8,192	16,384	32,768	65,536
Number of Subnets -	1	2	4	8	16	32	64	128	256	512	1,024	2,048	4,096	8,192	16,384	32,768
Binary values -	128	64	32	16	8	4	2	1								
148.55.0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
(1)	.															
(2)																
(3)																
(4)																
(5)																
(6)																
(7)																
(8)																
(9)																
(10)																
(11)																
(12)																
(13)																
(14)																
(15)																
(16)																

Practical Subnetting 10

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of subnets**, and allow enough extra subnets and hosts for 110% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class	<u>B</u>
Custom subnet mask	<u>255.255.254.0</u>
Minimum number of subnets needed	<u>4</u>
Extra subnets required for 110% growth (Round up to the next whole number)	<u>+ 5</u>
Total number of subnets needed	<u>= 9</u>
Number of host addresses in the largest subnet group	<u>140</u>
Number of addresses needed for 110% growth in the largest subnet (Round up to the next whole number)	<u>+ 154</u>
Total number of address needed for the largest subnet	<u>= 294</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Sales/Management	<u>172.16.0.0 to 172.16.1.255</u>
IP address range for Marketing	<u>172.16.2.0 to 172.16.3.255</u>
IP address range for Research	<u>172.16.4.0 to 172.16.5.255</u>
IP address range for Router A to Router B serial connection	<u>172.16.6.0 to 172.16.7.255</u>

Show your work for Problem 10 in the space below.

[illegible]

Valid and Non-Valid IP Addresses

Using the material in this workbook identify which of the addresses below are correct and usable. If they are not usable addresses explain why.

IP Address: 0.230.190.192

Subnet Mask: 255.0.0.0

Reference Page Inside Front Cover

The network ID cannot be 0.

IP Address: 192.10.10.1

Subnet Mask: 255.255.255.0

Reference Pages 28-29

OK

IP Address: 245.150.190.10

Subnet Mask: 255.255.255.0

Reference Page Inside Front Cover

245 is reserved for experimental use.

IP Address: 135.70.191.255

Subnet Mask: 255.255.254.0

Reference Pages 48-49

This is the broadcast address for this range.

IP Address: 127.100.100.10

Subnet Mask: 255.0.0.0

Reference Pages Inside Front Cover

127 is reserved for loopback testing.

IP Address: 93.0.128.1

Subnet Mask: 255.255.224.0

Reference Pages 56-57

OK

IP Address: 200.10.10.128

Subnet Mask: 255.255.255.224

Reference Pages 54-55

This is the subnet address for the 3rd usable range of 200.10.10.0

IP Address: 165.100.255.189

Subnet Mask: 255.255.255.192

Reference Pages 30-31

OK

IP Address: 190.35.0.10

Subnet Mask: 255.255.255.192

Reference Pages 34-35

This address is taken from the first range for this subnet which is invalid.

IP Address: 218.35.50.195

Subnet Mask: 255.255.0.0

Reference Page Inside Front Cover

This has a class B subnet mask.

IP Address: 200.10.10.175 /22

Reference Pages 54-55 and/or Inside Front Cover

A class C address must use a minimum of 24 bits.

IP Address: 135.70.255.255

Subnet Mask: 255.255.224.0

Reference Pages 48-49

This is a broadcast address.

Reference Charts and Support Materials

Class A Addresses
VLSM Chart 8-15 Bits (2nd octet)

[illegible]

Class B Addresses
VLSM Chart 16-23 Bits (3rd octet)

16	17	18	19	20	21	22	23
16.01 seats	17.01 seats	18.01 seats	19.01 seats	20.01 seats	21.01 seats	22.01 seats	23.01 seats
			9-81	3-15	6-15	2-12	
				15-21	5-15	2-12	
		9-81		15-21	5-15	2-12	
			9-81	15-21	5-15	2-12	
				30-47	30-47	30-47	
				41-47	41-47	41-47	
				48-55	48-55	48-55	
				56-65	56-65	56-65	
				66-71	66-71	66-71	
				72-75	72-75	72-75	
				76-79	76-79	76-79	
				80-87	80-87	80-87	
				88-95	88-95	88-95	
				96-111	96-111	96-111	
				112-127	112-127	112-127	
				128-143	128-143	128-143	
				144-161	144-161	144-161	
				162-179	162-179	162-179	
				180-195	180-195	180-195	
				196-211	196-211	196-211	
				212-229	212-229	212-229	
				230-245	230-245	230-245	
				246-261	246-261	246-261	
				262-279	262-279	262-279	
				280-295	280-295	280-295	
				296-311	296-311	296-311	
				312-327	312-327	312-327	
				328-343	328-343	328-343	
				344-359	344-359	344-359	
				360-375	360-375	360-375	
				376-391	376-391	376-391	
				392-407	392-407	392-407	
				408-423	408-423	408-423	
				424-439	424-439	424-439	
				440-455	440-455	440-455	
				456-471	456-471	456-471	
				472-487	472-487	472-487	
				488-503	488-503	488-503	
				504-519	504-519	504-519	
				520-535	520-535	520-535	
				536-551	536-551	536-551	
				552-567	552-567	552-567	
				568-583	568-583	568-583	
				584-599	584-599	584-599	
				600-615	600-615	600-615	
				616-631	616-631	616-631	
				632-647	632-647	632-647	
				648-663	648-663	648-663	
				664-679	664-679	664-679	
				680-695	680-695	680-695	
				696-711	696-711	696-711	
				712-727	712-727	712-727	
				728-743	728-743	728-743	
				744-759	744-759	744-759	
				760-775	760-775	760-775	
				776-791	776-791	776-791	
				792-807	792-807	792-807	
				808-823	808-823	808-823	
				824-839	824-839	824-839	
				840-855	840-855	840-855	
				856-871	856-871	856-871	
				872-887	872-887	872-887	
				888-903	888-903	888-903	
				904-919	904-919	904-919	

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

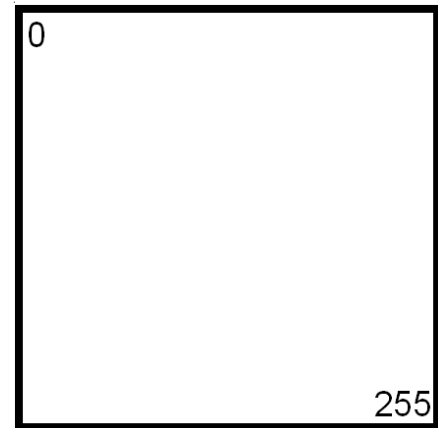
[illegible]

Visualizing Subnets Using The Box Method

The box method is the simplest way to visualize the breakdown of subnets and addresses into smaller sizes.

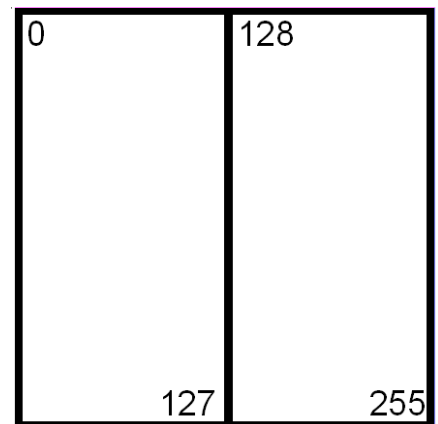
Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



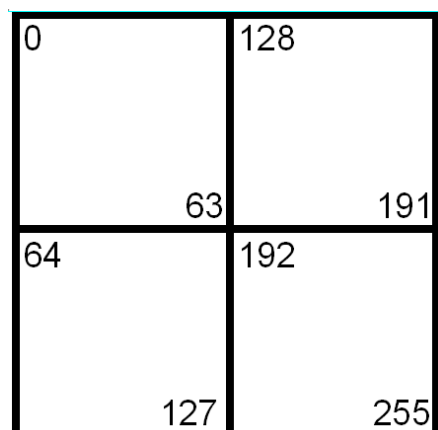
Split the box in half and you get two subnets with 128 addresses,

/25
255.255.255.128
128 Hosts
2 Subnets



Divide the box into quarters and you get four subnets with 64 addresses,

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you get eight subnets with 32 addresses,

/27
255.255.255.224
32 Hosts
8 Subnets

0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

Split the boxes in half again and you get sixteen subnets with sixteen addresses,

/28
255.255.255.240
16 Hosts
16 Subnets

0	32	128	160
15	47	143	175
16	48	144	176
31	63	159	191
64	96	192	224
79	111	207	239
80	112	208	240
95	127	223	255

The next split gives you thirty two subnets with eight addresses,

/29
255.255.255.248
8 Hosts
32 Subnets

0	8	32	40	128	136	160	168
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
71	79	103	111	199	207	321	239
80	88	112	120	208	216	240	248
87	95	119	127	215	223	247	255

The last split gives sixty four subnets with four addresses each,

/30
255.255.255.252
4 Hosts
64 Subnets

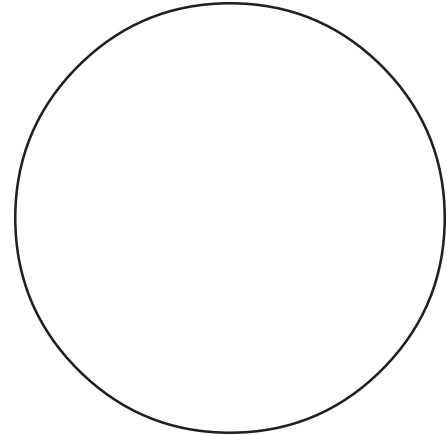
0	8	32	40	128	136	160	168
3	11	35	43	131	139	163	171
4	12	36	44	132	140	164	172
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
19	27	51	59	147	155	179	187
20	28	52	60	148	156	180	188
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
67	75	99	107	195	203	227	235
68	76	100	108	196	204	228	236
71	79	103	111	199	207	321	239
80	88	112	120	208	216	240	248
83	91	115	123	211	219	243	251
84	92	116	124	212	220	244	252
87	95	119	127	215	223	247	255

Visualizing Subnets Using The Circle Method

The circle method is another method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the different sections of the circle you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

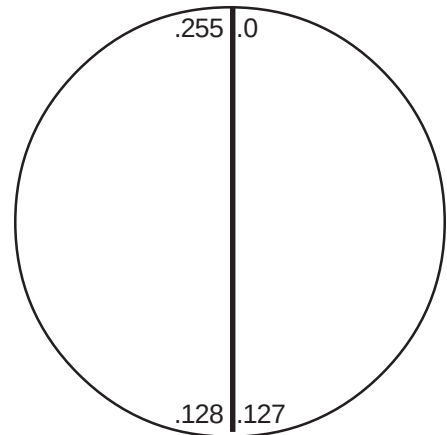
Start with a circle. The whole circle is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



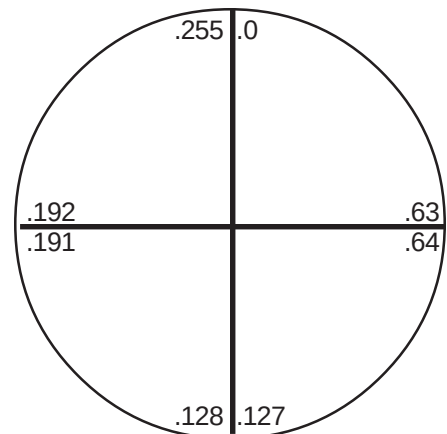
Split the circle in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



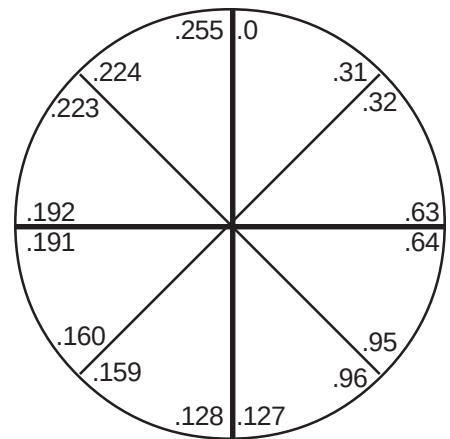
Divide the circle into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



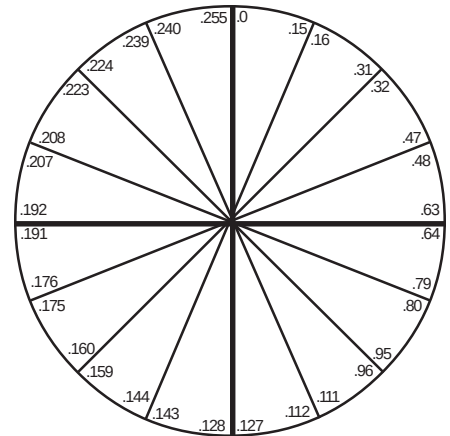
Split each quarter and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets



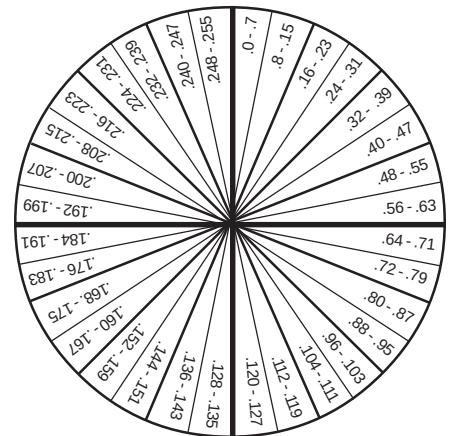
Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets



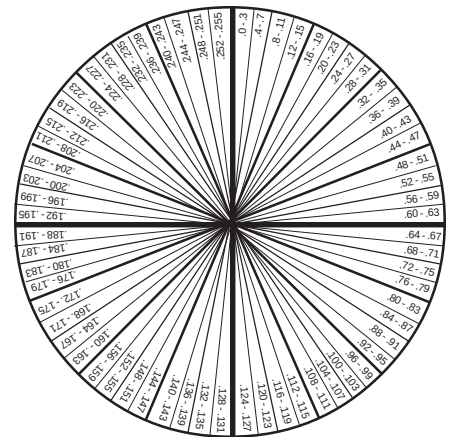
The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets



The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets



Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31	16-23	8-11	4-5
						12-15	6-7
					24-31	16-19	8-9
						20-23	10-11
		32-63	32-47	32-39	24-27	22-23	12-13
					28-31	24-25	14-15
				40-47	32-35	26-27	16-17
					36-39	28-29	18-19
			48-63	48-55	40-43	30-31	20-21
					44-47	32-33	22-23
				56-63	48-51	34-35	24-25
					52-55	36-37	26-27
		64-127	64-95	64-79	56-59	38-39	28-29
					60-63	40-41	30-31
				80-95	64-67	42-43	32-33
					68-71	44-45	34-35
			96-111	88-95	48-49	46-47	36-37
					50-51	48-49	38-39
				96-103	52-53	50-51	40-41
					100-103	52-53	42-43
			96-127	104-111	54-55	54-55	44-45
					106-107	56-57	46-47
				112-119	108-111	58-59	48-49
					112-113	60-61	50-51
		128-255	128-159	128-143	114-115	62-63	52-53
					116-117	64-65	54-55
				144-151	118-119	66-67	56-57
					120-123	68-69	58-59
			160-191	160-175	122-123	70-71	60-61
					124-127	72-73	62-63
				176-191	126-127	74-75	64-65
					128-131	76-77	66-67
		192-255	192-223	192-207	132-135	78-79	68-69
					136-139	80-81	70-71
				208-223	140-143	82-83	72-73
					144-147	84-85	74-75
			224-255	224-231	146-147	86-87	76-77
					148-151	88-89	78-79
				240-247	152-155	90-91	80-81
					156-159	92-93	82-83
			240-255	248-251	160-163	94-95	84-85
					164-167	96-97	86-87
				248-255	168-171	98-99	88-89
					172-175	100-101	90-91

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					4-7	4-5	2-3
					8-15	8-11	6-7
					16-23	12-15	8-9
					24-31	16-19	10-11
					32-39	20-23	12-13
		32-63	16-31	16-31	24-27	24-25	14-15
					28-31	30-31	16-17
					32-35	32-33	18-19
					36-39	40-43	20-21
					40-47	44-47	22-23
					48-55	48-51	24-25
			32-47	32-47	52-55	56-57	26-27
					56-59	58-59	28-29
					60-63	60-63	30-31
					64-71	64-67	32-33
					72-79	68-71	34-35
					80-87	72-75	36-37
		64-127	64-95	64-79	84-87	76-79	38-39
					88-95	80-83	40-41
					96-103	84-87	42-43
					104-111	88-91	44-45
					112-119	92-95	46-47
					120-127	96-99	48-49
			96-127	96-111	100-103	100-103	50-51
					104-107	104-107	52-53
					108-111	108-111	54-55
					112-115	112-115	56-57
					116-119	116-119	58-59
					120-123	120-123	60-61
			128-159	128-143	124-127	124-127	62-63
					128-131	128-129	64-65
					132-135	132-133	66-67
					136-139	136-137	68-69
					140-143	140-141	70-71
					144-147	144-145	72-73
					148-151	148-149	74-75
					152-155	152-153	76-77
					156-159	156-157	78-79
					160-163	160-161	80-81
					164-167	164-165	82-83
					168-171	168-169	84-85
		160-191	160-175	160-175	172-175	172-173	86-87
					176-179	176-177	88-89
					180-183	180-181	90-91
					184-187	184-185	92-93
					188-191	188-189	94-95
					192-195	192-193	96-97
		192-255	192-207	192-207	196-199	196-197	98-99
					200-203	200-201	100-101
					204-207	204-205	102-103
					208-211	208-209	104-105
					212-215	212-213	106-107
					216-219	216-217	108-109
			208-223	208-223	220-223	220-221	110-111
					224-227	224-225	112-113
					228-231	228-229	114-115
					232-235	232-233	116-117
					236-239	236-237	118-119
					240-243	240-241	120-121
		224-255	224-239	224-239	244-247	244-245	122-123
					248-251	248-249	124-125
				240-255	252-255	252-253	126-127
						254-255	128-129
			240-255	240-255			130-131
							132-133
							134-135
							136-137

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
					4-7	4-7
				16-31	8-15	8-11
					12-15	12-15
		64-127	32-63	32-47	16-23	16-19
					20-23	20-23
				48-63	24-31	24-27
					28-31	28-31
			64-95	64-79	32-39	32-35
					36-39	36-39
				80-95	40-47	40-43
					44-47	44-47
		96-127	96-111	96-103	48-55	48-51
					52-55	52-55
				104-111	56-59	56-59
					60-63	60-63
			112-127	112-119	64-67	64-67
					68-71	68-71
				120-127	72-75	72-75
					76-79	76-79
	128-255	128-191	128-159	128-143	80-83	80-83
					84-87	84-87
				144-159	88-91	88-91
					92-95	92-95
		160-191	160-175	160-167	96-99	96-99
					100-103	100-103
				176-191	104-107	104-107
					108-111	108-111
		192-255	192-223	192-207	112-115	112-115
					116-119	116-119
				208-223	120-123	120-123
					124-127	124-127
		224-255	224-239	224-231	128-131	128-131
					132-135	132-135
				232-239	136-139	136-139
					140-143	140-143
			240-255	240-247	144-147	144-147
					148-151	148-151
				248-255	152-155	152-155
					156-159	156-159
			240-255	240-247	160-163	160-163
					164-167	164-167
				248-255	168-171	168-171
					172-175	172-175
			240-255	240-247	176-179	176-179
					180-183	180-183
				248-255	184-187	184-187
					188-191	188-191
			240-255	240-247	192-195	192-195
					196-199	196-199
				248-255	200-203	200-203
					204-207	204-207
			240-255	240-247	208-211	208-211
					212-215	212-215
				248-255	216-219	216-219
					220-223	220-223
			240-255	240-247	224-227	224-227
					228-231	228-231
				248-255	232-235	232-235
					236-239	236-239
			240-255	240-247	240-243	240-243
					244-247	244-247
				248-255	248-251	248-251
					252-255	252-255

Class A Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/8	0	255.0.0.0	1	16,777,216	16,777,214
/9	1	255.128.0.0	2	8,388,608	8,388,606
/10	2	255.192.0.0	4	4,194,304	4,194,302
/11	3	255.224.0.0	8	2,097,152	2,097,150
/12	4	255.240.0.0	16	1,048,576	1,048,574
/13	5	255.248.0.0	32	524,288	524,286
/14	6	255.252.0.0	64	262,144	262,142
/15	7	255.254.0.0	128	131,072	131,070
/16	8	255.255.0.0	256	65,536	65,534
/17	9	255.255.128.0	512	32,768	32,766
/18	10	255.255.192.0	1,024	16,384	16,382
/19	11	255.255.224.0	2,048	8,192	8,190
/20	12	255.255.240.0	4,096	4,096	4,094
/21	13	255.255.248.0	8,192	2,048	2,046
/22	14	255.255.252.0	16,384	1,024	1,022
/23	15	255.255.254.0	32,768	512	510
/24	16	255.255.255.0	65,536	256	254
/25	17	255.255.255.128	131,072	128	126
/26	18	255.255.255.192	262,144	64	62
/27	19	255.255.255.224	524,288	32	30
/28	20	255.255.255.240	1,048,576	16	14
/29	21	255.255.255.248	2,097,152	8	6
/30	22	255.255.255.252	4,194,304	4	2

Class B Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/16	0	255.255.0.0	1	65,536	65,534
/17	1	255.255.128.0	2	32,768	32,766
/18	2	255.255.192.0	4	16,384	16,382
/19	3	255.255.224.0	8	8,192	8,190
/20	4	255.255.240.0	16	4,096	4,094
/21	5	255.255.248.0	32	2,048	2,046
/22	6	255.255.252.0	64	1,024	1,022
/23	7	255.255.254.0	128	512	510
/24	8	255.255.255.0	256	256	254
/25	9	255.255.255.128	512	128	126
/26	10	255.255.255.192	1,024	64	62
/27	11	255.255.255.224	2,048	32	30
/28	12	255.255.255.240	4,096	16	14
/29	13	255.255.255.248	8,192	8	6
/30	14	255.255.255.252	16,384	4	2

Class C Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/24	0	255.255.255.0	1	256	254
/25	1	255.255.255.128	2	128	126
/26	2	255.255.255.192	4	64	62
/27	3	255.255.255.224	8	32	30
/28	4	255.255.255.240	16	16	14
/29	5	255.255.255.248	32	8	6
/30	6	255.255.255.252	64	4	2

IPv6 Addressing and Subnetting Workbook

Version 1

Instructor's Edition

Global Routing Prefix

Subnet ID

Interface ID

Hexadecimal

2001:0DB9:F000

Types of IPv6 Addresses

Unspecified, Loopback, Embedded IPv4

Unspecified address is an all 0 address and cannot be assigned to an interface. It would be typed as ::. This is only used as a source address to indicate the absence of an actual address.

Loopback Address is all 0's except for the last bit, which is 1. It would be typed as ::1. It operates the same as the IPv4 127.0.0.1 loopback address.

IPv4 Embedded addresses are IPv6 addresses with an IPv4 address embedded in the low-order 32 bits. They are used to transition networks from IPv4 to IPv6.

Address Range:

0000:0000:0000:0000:0000:0000:0000:0000/8 to 00FF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/8

Global Unicast

Global Unicast addresses are used to uniquely identify a specific interface on a host and can be used as a public address on the internet.

Address Range:

2000:0000:0000:0000:0000:0000:0000:0000/3 to 3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/3

Unique local Unicast

Unique local Unicast addresses are roughly the same as IPv4 private addresses.

Address Range:

FC00:0000:0000:0000:0000:0000:0000:0000/7 to FDFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/7

Link-local Unicast

Link-local addresses are unicast addresses that are limited to a point to point connection within a local network. Routers will not forward packets with a link-local address.

Address Range:

FE80:0000:0000:0000:0000:0000:0000:0000/10 to FEBF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/10

Multicast

Multicast addresses are used to send a single packet to multiple destinations simultaneously.

Address Range:

FF00:0000:0000:0000:0000:0000:0000:0000/8 to FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/8

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Special Thanks to Melvin Baker and Jim Dorsch for taking the time to check this workbook for errors,
and to everyone who has sent in suggestions to improve the series.

Inside Cover

A Brief History of TCP/IP Versions

TCP version 1 through TCP version 3 were developed as test versions and not widely used. Contrary to popular belief there was never an IPv1, IPv2, or IPv3. The version numbers were kept intact to avoid confusion when the TCP protocol was split into TCP and IP.

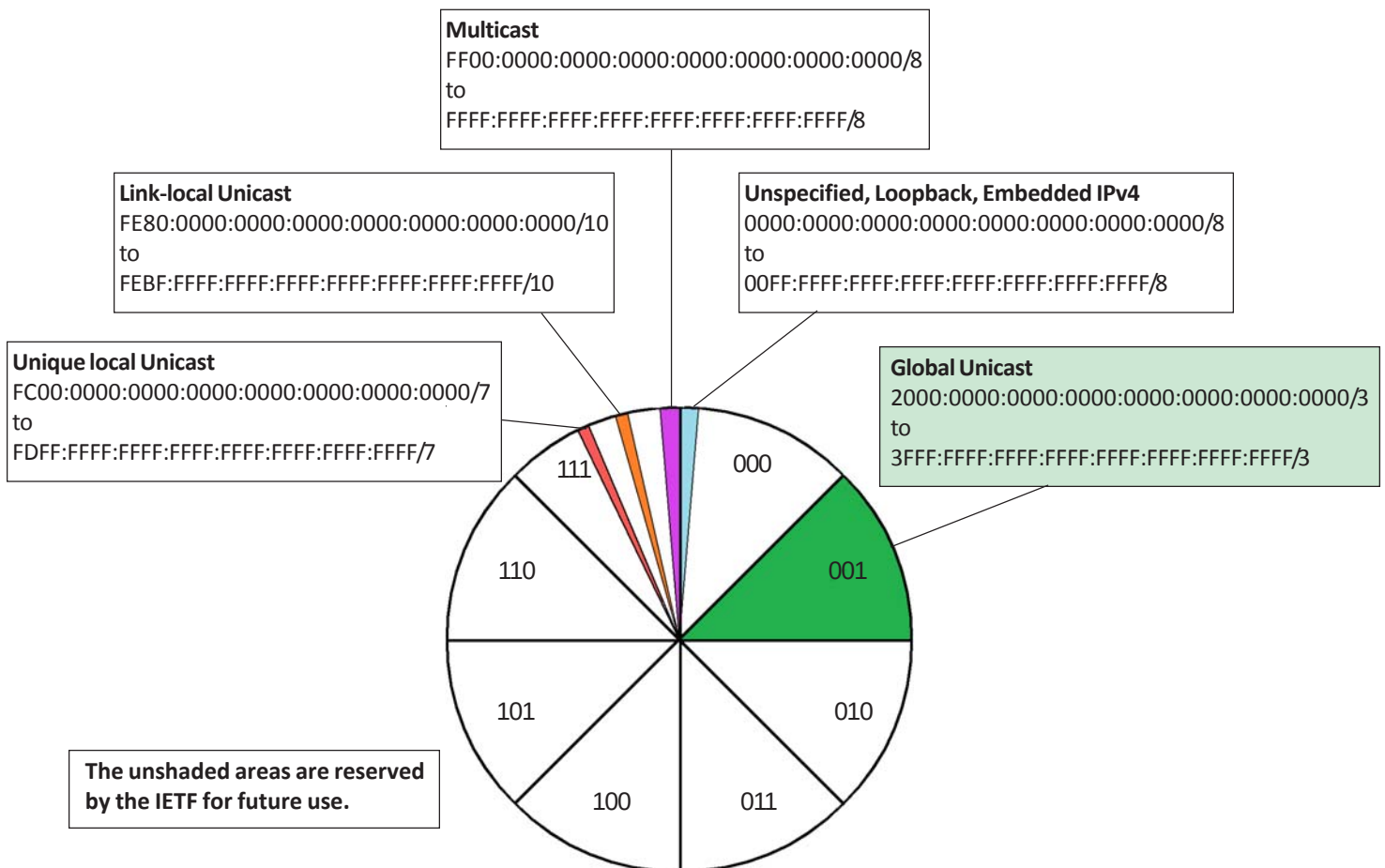
- 1973 - TCP version 1 was developed and documented in RFC 675. At this time IP was part of TCP.
- 1977 - TCP version 2 was developed and documented in March. In August of 1977 it was decided that the TCP protocol was going in the wrong direction.
- 1978 - TCP and IP were split into two separate protocols. Both TCP and IP were part of version 3.
- 1980 - Early development of IPv4 defined in RFC 760.
- 1981 - The current version of IPv4 is defined in RFC 791, 792 and 793. It was the first widely used version of the Internet Protocol.
- 1983 - On January 1, 1983, TCP/IP protocols became the only approved protocol on the ARPANET, replacing the earlier NCP protocol. This was known as flag day.
- 1984 - The number of hosts on the internet breaks 1000.
- 1987 - Hosts on the internet exceeds 10,000.
- 1989 - Host accessing the internet surpasses 100,000.
- 1990 - IPv5 relates to an experimental TCP/IP protocol called the Internet Stream Protocol, Version 2, originally defined in RFC 1190. This protocol was a peer of IPv4 and was designed to work with voice conversations and conferences with delay and bandwidth guarantees. These packets were assigned IP version 5 to differentiate them from "normal" IPv4 packets. This protocol was never introduced to the public, and was always considered experimental. To be sure there would be no confusion, version 5 was skipped over in favor of version 6.
- 1992 - The number of hosts on the internet breaks 1,000,000.
- 1995 - IPv6, introduced as IP Next Generation, was presented in RFC 1883.
- 1997 - The number of hosts using the internet exceeds 19,000,000.
- 1998 - The more fully developed IPv6 obsoletes RFC 1883 with the updated RFC 2460.

IPv4 has been well established for years. IPv6 is still in flux as it undergoes growing pains with changes and adjustments to the rules as it is being implemented.

IPv6

There are 340,282,366,920,938,463,463,374,607,431,768,211,456 possible IPv6 addresses.

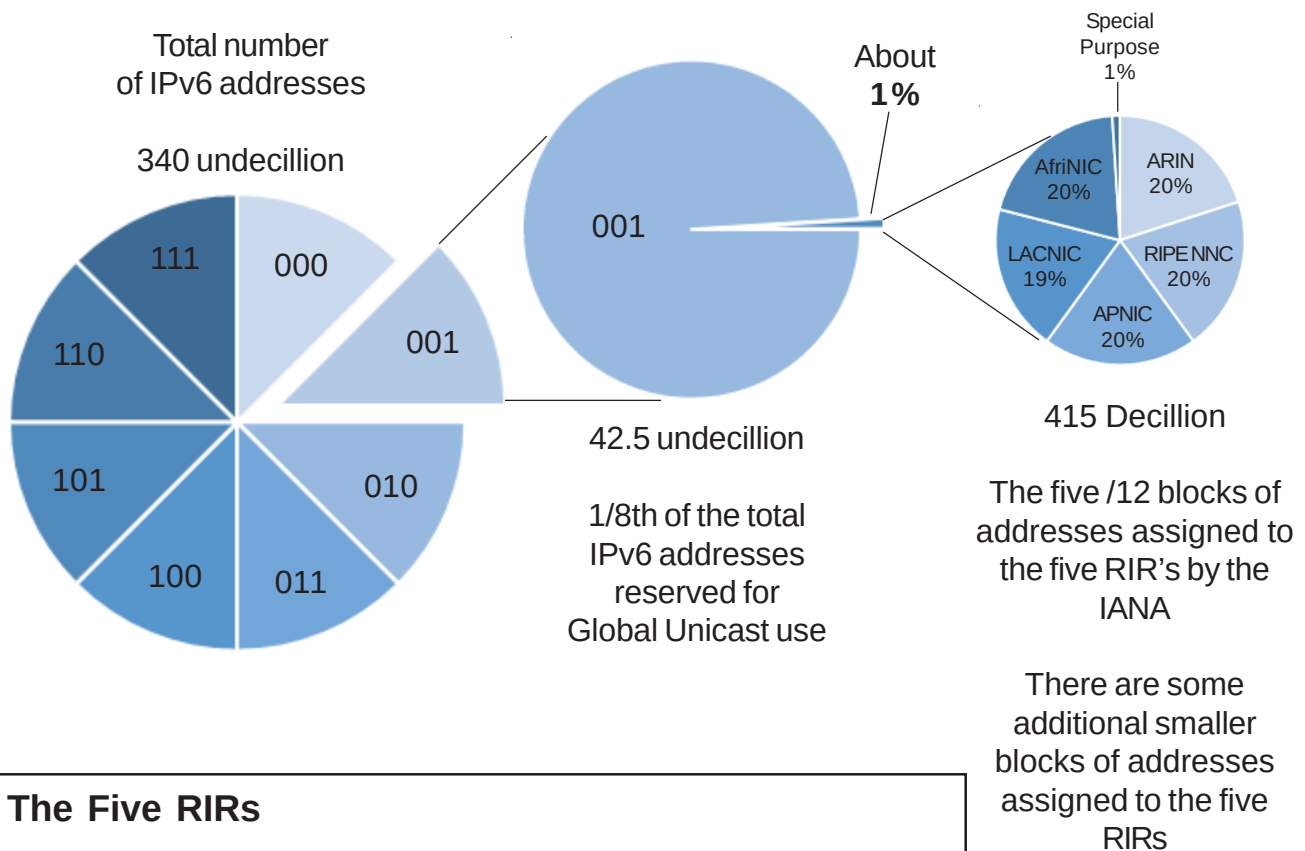
If you want to actually say the number it is three hundred and forty undecillion, two hundred and eighty-two decillion, three hundred and sixty-six nonillion, nine hundred and twenty octillion, nine hundred and thirty-eight septillion, four hundred and sixty-three sextillion, four hundred and sixty-three quintillion, three hundred and seventy-four quadrillion, six hundred and seven trillion, four hundred and thirty-one billion, seven hundred and sixty-eight million, two hundred and eleven thousand, four hundred and fifty-six. (or you can have Windows Narrator say it for you.)



The Internet Assigned Numbers Authority (IANA) divided the available IPv6 addresses into eight equal segments based on the three leading bits of the addresses (000, 001, 010, 011, 100, 101, 110, and 111). Only one eighth of the total available addresses have been reserved for use as global unicast addresses. Four smaller subgroups have been made available for unique local unicast, link-local unicast, multicast, and (unspecified, loopback, embedded IPv4).

IPv6 by the Numbers

340,282,366,920,938,463,463,374,607,431,768,211,456 Total number of IPv6 Addresses
 42,535,295,865,117,307,932,921,825,928,971,026,432 1/8 or the reserved Global Unicast addresses
 415,383,748,682,786,210,282,439,706,337,607,680 The five /12 ranges assigned to the RIRs
 7,119,157,000 Estimated world population
 58,347,322,398,253,923,924,200,534 Estimated number of IPv6 addresses per person
 (That's over 58 septillion addresses per person and doesn't include the additional smaller blocks of addresses assigned to the five RIRs by the IANA)



The Five RIRs

The **Regional Internet Registry** is an organization that manages the allocation and registration of internet number resources world wide. It has evolved over time to divide the world into five areas, or RIRs.

AfriNIC - African Network Information Centre

ARIN - American Registry for Internet Numbers

APNIC - Asia-Pacific Network Information Centre

LACNIC - Latin America and Caribbean Network Information Centre

RIPE NCC - Réseaux IP Européens Network Coordination Centre

There is a chart in the Reference Section that has all of these listed.

To make IPv6 addresses a little less imposing, two rules were developed to make them easier to work with. Rule 1: Omission of the Leading 0s, and Rule 2: Omission of the all-0 Hextets.

Rule 1: Omission of the Leading 0s

Rule 1 allows you to remove all the leading 0s in each individual hextet.

Sample 1 Unspecified address

Preferred Format: **0000:0000:0000:0000:0000:0000:0000:0000**
 Leading 0's removed: 0: 0: 0: 0: 0: 0: 0: 0
 or
 0:0:0:0:0:0:0:0

Sample 2 - Loopback Address

Preferred Format: **0000:0000:0000:0000:0000:0000:0000:0001**
 Leading 0's removed: 0: 0: 0: 0: 0: 0: 0: 1
 or
 0:0:0:0:0:0:0:1

Sample 3 – Global Unicast Address

Preferred Format: **2000:0000:0000:0000:0000:0000:0000:0001**
 Leading 0's removed: 2000: 0: 0: 0: 0: 0: 0: 1
 or
 2000:0:0:0:0:0:0:1

Sample 4 – Global Unicast Address

Preferred Format: **2001:00FE:ACAD:2013:0000:0000:00AA:0271**
 Leading 0's removed: 2001: FE:ACAD:2013: 0: 0: AA: 271
 or
 2001: FE:ACAD:2013:0:0:AA:271

Sample 5 – Unique local Unicast Address

Preferred Format: **FC80:0000:0000:ACAD:0000:0000:0000:0001**
 Leading 0's removed: FC80: 0: 0:ACAD: 0: 0: 0: 1
 or
 FC80:0:0:ACAD:0:0:0:1

Sample 6 – Link-local Address

Preferred Format: **FE80:ACAD:0000:0197:0000:0000:0000:FF01**
 Leading 0's removed: FE80:ACAD: 0: 197: 0: 0: 0:FF01
 or
 FE80:ACAD:0:197:0:0:0:FF01

Sample 7 – Multicast Address

Preferred Format: **FF00:0000:0000:ACAD:0000:0000:FE00:0721**
 Leading 0's removed: FF00: 0: 0:ACAD: 0: 0:FE00: 721
 or
 FF00:0:0:ACAD:0:0:FE00:721

Rule 1: Omission of the Leading 0s Problems

Using Rule 1 reduce the IPv6 addresses to their shortened form.

1. 0000:0000:0000:0000:0000:0000:0000:0000

0:0:0:0:0:0:0:0

2. 0000:0000:0000:0000:0000:0000:0000:0001

0:0:0:0:0:0:0:1

3. 2000:0000:0000:0000:0000:ABCD:0000:0025

2000:0:0:0:0:ABCD:0:25

4. 3F00:0090:0000:0000:0000:0098:0000:0001

3F00:90:0:0:0:98:0:1

5. 2001:3756:0005:0000:ACAD:0000:0000:0025

2001:3756:5:0:ACAD:0:0:25

6. 3FFF:FF00:0000:0000:ACAD:0000:0000:0127

3FFF.FF00:0:0:ACAD:0:0:127

7. 2001:0000:0000:ABCD:FFFF:0000:0000:0001

2001:0:0:ABCD:FFFF:0:0:1

8. 3ABC:0001:ACAD:0000:0000:1234:0000:0005

3ABC:1:ACAD:0:0:1234:0:5

9. FC00:0000:0000:0000:3E00:1275:0000:0034

FC00:0:0:0:3E00:1275:0:34

10. FE95:FC6C:C540:0000:0000:0000:0000:9800

FE95:FC6C:C540:0:0:0:0:9800

11. FF00:ACAD:0000:0000:1234:0000:0000:0001

FF00:ACAD:0:0:1234:0:0:1

Rule 2: Omission of the All-0 Hextets

Rule 2 uses a double colon :: to represent a single contiguous set of all zero hextets. It can only be used once in any IPv6 address.

Sample 1 Unspecified address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0000

Contiguous 0's removed: ::

Sample 2 - Loopback Address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0001

Contiguous 0's removed: ::0001

Sample 3 – Global Unicast Address

Preferred Format: 2000:0000:0000:0000:0000:0000:0000:0001

Contiguous 0's removed: 2000: :0001

or

2000::0001

Sample 4 – Global Unicast Address

Preferred Format: 2001:00FE:ACAD:2013:0000:0000:00AA:0271

Contiguous 0's removed: 2001:00FE:ACAD:2013: :00AA:0271

or

2001:00FE:ACAD:2013::00AA:0271

Sample 5 – Unique local Unicast Address

Preferred Format: FC80:0000:0000:ACAD:0000:0000:0000:0001

Contiguous 0's removed: FC80:0000:0000:ACAD: :0001

or

FC80:0000:0000:ACAD::0001

Sample 6 – Link-local Address

Preferred Format: FE80:ACAD:0000:0197:0000:0000:0000:FF01

Contiguous 0's removed: FE80:ACAD:0000:0197: :FF01

or

FE80:ACAD:0000:0197::FF01

Sample 7 – Multicast Address

Preferred Format: FF00:0000:0000:ACAD:0000:0000:FE00:0721

Contiguous 0's removed: FF00: :ACAD:0000:0000:FE00:0721 (Option #1)

FF00:0000:0000:ACAD: :FE00:0721 (Option #2)

or

FF00::ACAD:0000:0000:FE00:0721 (Option #1)

FF00:0000:0000:ACAD::FE00:0721 (Option #2)

Rule 2: Omission of the All-0 Hextets Problems

Using Rule 2 reduce the IPv6 addresses to their shortened form.

1. 0000:0000:0000:0000:0000:0000:0000:0000

::

2. 0000:0000:0000:0000:0000:0000:0000:0001

:::0001

3. 2000:0000:0000:0000:0000:ABCD:0000:0025

2000::ABCD:0000:0025

4. 3F00:0090:0000:0000:0000:0098:0000:0001

3F00:0090::0098:0000:0001

5. 2001:3756:0005:0000:ACAD:0000:0000:0025

2001:3756:0005:0000:ACAD::0025

6. 3FFF:FF00:0000:0000:ACAD:0025:0000:0127

3FFF.FF00::ACAD:0025:0000:0127

7. 2001:ACAD:0000:ABCD:FFFF:0000:0000:0001

2001:ACAD:0000:ABCD:FFFF::0001

8. 3ABC:0001:ACAD:0000:0000:1234:0000:0005

3ABC:0001:ACAD::1234:0000:0005

9. FC00:0000:0000:0000:3E00:1275:0000:0034

FC00::3E00:1275:0000:0034

10. FE95:FC6C:C540:0000:0000:0000:0000:9800

FE95:FC6C:C540::9800

11. FF00:ACAD:0000:0000:1234:0000:0000:0001

FF00:ACAD::1234:0000:0000:0001 or FF00:ACAD:0000:0000:1234::0001

Combining Rule 1 and Rule 2

To reduce the size of IPv6 address even more you can combine Rule 1 with Rule 2.

Sample 1 Unspecified address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0000
Combined reduction: ::

Sample 2 - Loopback Address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0001
Combined reduction: ::1

Sample 3 – Global Unicast Address

Preferred Format: 2000:0000:0000:0000:0000:0000:0000:0001
Combined reduction: 2000: :1
or
2000::1

Sample 4 – Global Unicast Address

Preferred Format: 2001:00FE:ACAD:2013:0000:0000:00AA:0271
Combined reduction: 2001: FE:ACAD:2013: : AA: 271
or
2001:FE:ACAD:2013::AA:271

Sample 5 – Unique local Unicast Address

Preferred Format: FC80:0000:0000:ACAD:0000:0000:0000:0001
Combined reduction: FC80: 0: 0:ACAD: : 1
or
FC80:0:0:ACAD::1

Sample 6 – Link-local Address

Preferred Format: FE80:ACAD:0000:0197:0000:0000:0000:FF01
Combined reduction: FE80:ACAD:0 : 197: :FF01
or
FE80:ACAD:0:197::FF01

Sample 7 – Multicast Address

Preferred Format: FF00:0000:0000:ACAD:0000:0000:FE00:0721
Combined reduction: FF00: :ACAD: 0: 0:FE00: 721 (Option #1)
FF00: 0: 0:ACAD: :FE00: 721 (Option #2)
or
FF00::ACAD:0:0:FE00:721 (Option #1)
FF00:0:0:ACAD::FE00:721 (Option #2)

Combining Rule 1 and Rule 2 Problems

Using Rule 1 and 2 reduce the IPv6 addresses to their shortest form.

1. 0000:0000:0000:0000:0000:0000:0000:0000

::

2. 0000:0000:0000:0000:0000:0000:0000:0001

::1

3. 2000:0000:0000:0000:0000:ABCD:0000:0025

2000::ABCD:0:25

4. 3F00:0090:0000:0000:0000:0098:0000:0001

3F00:0090::98:0:1

5. 2001:3756:0005:0000:ACAD:0000:0000:0025

2001:3756:5:0:ACAD::25

6. 3FFF:FF00:0000:0000:ACAD:0025:0000:0127

3FFF:FF00::ACAD:25:0:127

7. 2001:ACAD:0000:ABCD:FFFF:0000:0000:0001

2001:ACAD:0:ABCD:FFFF::1

8. 3ABC:0001:ACAD:0000:0000:1234:0000:0005

3ABC:1:ACAD::1234:0:5

9. FC00:0000:0000:0000:3E00:1275:0000:0034

FC00::3E00:1275:0:34

10. FE95:FC6C:C540:0000:0000:0000:0000:9800

FE95:FC6C:C540::9800

11. FF00:ACAD:0000:0000:1234:0000:0000:0001

FF00:ACAD::1234:0:0:1 or FF00:ACAD:0:0:1234::1

Reverting Reduced Address Problems

The following addresses have been shorted using Rule1 and/or Rule 2. Expand them back to their preferred format.

Sample: FF00:ACAD:ABCD:0:1234::1

FF00:ACAD:ABCD:0000:1234:0000:0000:0001

1. 2000::1

2000:0000:0000:0000:0000:0000:0000:0001

2. ::1

0000:0000:0000:0000:0000:0000:0000:0001

3. 2001:0:0:0:ABCD:0:127

2001:0000:0000:0000:0000:ABCD:0000:0127

4. 3E80:0070::0098:0000:0001

3E80:0070:0000:0000:0000:0098:0000:0001

5. 2FFF:38:5:0:ACAD::5

2FFF:0038:0005:0000:ACAD:0000:0000:0005

6. 3FFF::ACAD:25:0:100

3FFF:0000:0000:0000:ACAD:0025:0000:0100

7. 2002:ACAD:0:1BCD:FFFF::4

2002:ACAD:0000:1BCD:FFFF:0000:0000:0004

8. 3FAA:0025:ACAD::ABCD:0000:0005

3FAA:0025:ACAD:0000:0000:ABCD:0000:0005

9. FFFF::4E00:1235:0:34

FFFF:0000:0000:0000:4E00:1235:0000:0034

10. 3E01:6C:40::9800

3E01:006C:0040:0000:0000:0000:0000:9800

IPv6 Address Categories

All IPv6 addresses fall into one of three categories

Unicast -

Unicast addresses identify a unique interface on an IPv6 device.
It is a one to one connection between a source and destination.

Examples of IPv6 Unicast addresses include:

Global Unicast -

Similar to a public IPv4 address
Can be used as a public address on the internet
Globally unique
Can be static or dynamic

Link-Local -

Only used on a local network link to uniquely identify a host
Not routable on the public internet
No router will forward a link-local address
Every IPv6 enabled networked device is required to have a link-local address
Multiple interfaces on the same device can have the same link-local address

Loopback - (::1/128)

Used by a host to ping itself to test the TCP/IP stack
It cannot be assigned to a physical interface

Unspecified Address - (::/128)

Is only used as a source address to indicate the absence of an actual address

Unique Local -

Unique local addresses are roughly the same as IPv4 private addresses

Embedded IPv4 -

Used to transition IPv4 networks into IPv6

Multicast -

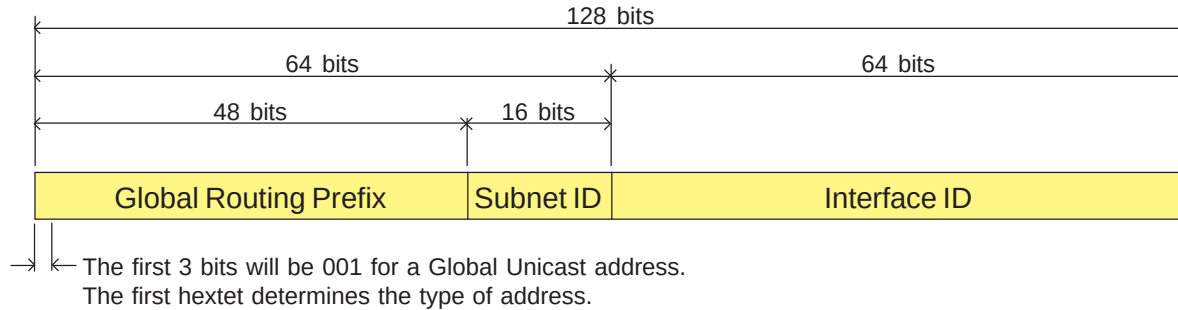
Multicast addresses are used to send a single packet to multiple destinations simultaneously.

Anycast -

Anycast addresses are described as a one-to-nearest or one-to-one-of-many packet delivery. For example, all routers in the same network will be assigned the same Anycast address. A packet sent to that address will only be delivered to the closest router with that address based on routing protocol metrics. Anycast addresses can be pulled from Global Unicast, Site-Local, or Link-Local address ranges. The first address and the last 128 addresses in a /64 Global Unicast range are reserved as the Subnet-Router Anycast Address.

There are no broadcast addresses in IPv6.

Global Unicast IPv6 Address



Global Routing Prefix -

This is assigned by the ISP to a customer or site.

The Global Routing Prefix is determined by the *prefix-length* notation. (example /48 or /64).

This is similar to the network portion of an IPv4 address.

Subnet ID -

This is similar to the subnet portion of an IPv4 address.

The difference is in IPv4 the subnet is borrowed from the host portion of the address.

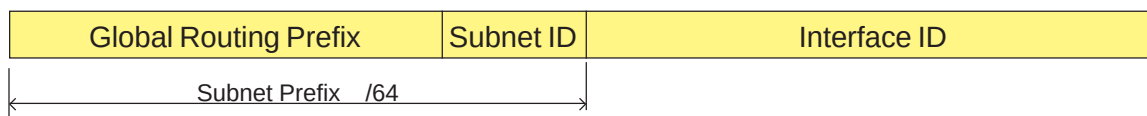
In IPv6 the Subnet ID is a separate field (/48 to /64) and not necessarily part of the Interface ID.

Interface ID -

The Interface ID uniquely identifies a interface on the local subnet.

Subnet Prefix

The Subnet Prefix is the address space used by the Global Routing Prefix and the Subnet ID, and can range from 0 to 128. The preferred Subnet Prefix length is /48 to /64 for customers or sites.



RFC 4291 recommends that the Interface-ID or host portion of your IPv6 address be 64 bits.

A minimum /64 prefix length is required to support Stateless Address Auto-configuration.

IPv6 Prefix Length vs IPv4 CIDR

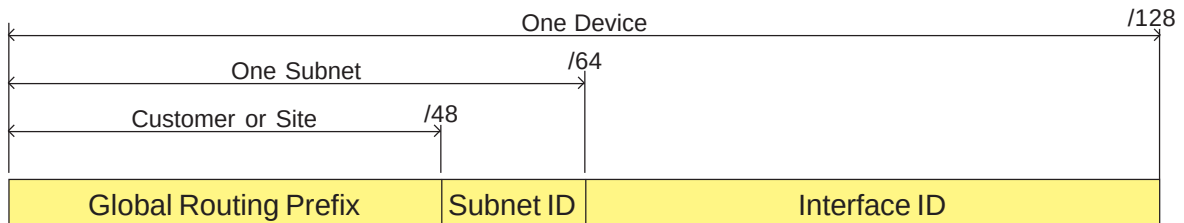
In IPv4 the network portion of the IP address is indicated with a dotted-decimal subnet mask; such as 255.255.255.0. It can also be identified with a CIDR (classless interdomain routing) notation; such as /24.

IPv6 does not use either of these terms to indicate the network portion of an IPv6 address. The network portion of the address is indicated with a **Prefix Length** at the end of the address. While a /48 or /56 looks like a CIDR notation it is not a classless interdomain routing notation. The prefix length indicates the number of nibbles or bits used in the network or subnetwork portion of an IPv6 address.

Global Unicast Prefix Allocations

In 2001, RFC 3177 was written to provide recommendations for how IPv6 Global Unicast addresses should be allocated to customers or Sites.

“In particular, it recommends the assignment of /48 in the general case, /64 when it is known that one and only one subnet is needed and /128 when it is absolutely known that one and only one device is connecting.”

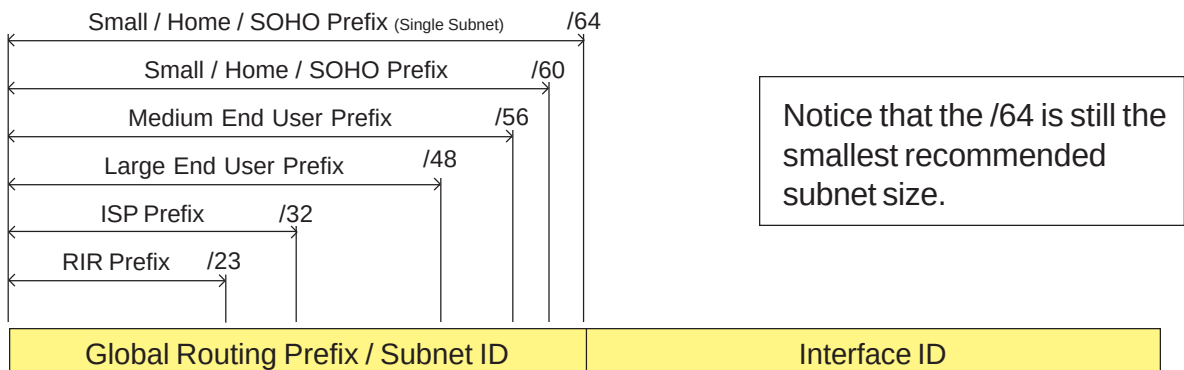


A /48 prefix allows each customer or Site to have 1,208,925,819,614,629,174,706,176 addresses. These can be used as a single subnet, or up to 65,536 subnets with a /64 prefix-length.

The Regional Internet Registries (RIRs) adopted RFC 3177 recommendation in 2002, but began reconsidering the policy in 2005. In March of 2011 RFC 6177 obsoleted RFC 3177 with a new recommendation.

“The exact choice of how much address space to assign end sites is an issue for the operational community.”

This gives local ISPs more options when assigning IPv6 addresses to their customers or Sites.



Typical IPv6 Prefix Assignments:

<u>Subscribers</u>	<u>Prefix-Length</u>	<u># of Subnets at /64</u>	<u>Total number of Possible Addresses</u>
RIR	/23	2,199,023,255,552	40,564,819,207,303,340,847,894,502,572,032
Service Provider (*LIR) -	/32	4,294,967,296	79,228,162,514,264,337,593,543,950,336
Large End User -	/48	65,536	1,208,925,819,614,629,174,706,176
Medium End User -	/56	256	4,722,366,482,869,645,213,696
Small / Home / SOHO -	/60	16	295,147,905,179,352,825,856
Small / Home / SOHO -	/64	1	18,446,744,073,709,551,616

*Local Internet registry

Subnetting on the Nibble Boundary

One of the design themes driving IPv6 was to keep subnetting as simple as possible. The fourth hexet is reserved for subnetting, giving network administrators multiple options for developing a network plan. In order to keep IPv6 addressing as simple as possible it is a **Best Practice** to subnet on the nibble boundary.

Every IPv6 address is comprised of 128 bits, which is represented with 32 hexadecimal numbers.

2001:ACAD:1234:0000:0000:0000:0000:0000/48

Showing this address with 128 binary characters makes it difficult to read and almost impossible for most people to work with.

0010000000000001:1010110010101101:0001001000110100:0000000000000000:
0000000000000000:0000000000000000:0000000000000000:0000000000000000

Each hexadecimal number in an IPv6 address represents 4 bits or a Nibble. An IPv6 address is composed of 32 hexadecimal numbers or 32 Nibbles.

2001:ACAD:1234:0000:0000:0000:0000:0000/48

IPv6 can be subnetted just like IPv4 using individual binary bits. To keep subnetting as simple as possible it is a **Best Practice** to borrow 4 bits, or one Nibble at a time.

Nibble Boundary Subnets

(Subnetting on the Nibble Boundary)

Prefix	# of /64 subnets
/48	65,536
/52	4096
/56	256
/60	16
/64	1

It is a Best Practice to subnet on the Nibble Boundary.

Subnets Based on Individual Binary Bits

(Subnetting within a Nibble)

Prefix subnets	# of /64
/48	65,536
/49	32,768
/50	16,384
/51	8,192
/52	4,096
/53	2,048
/54	1,024
/55	512
/56	256
/57	128
/58	64
/59	32
/60	16
/61	8
/62	4
/63	2
/64	1

Site ID and Sub-Site IDs

Subnetting on the nibble boundary allows you to easily set up proper route aggregation and summarization to use for each subnet or location. It also allows for easier deployment of firewalls based on location or network users. In order to do this you need to assign Site ID's and Sub-Site ID's as necessary. The Site ID is the first address in the subnet you have assigned to a specific location or user group. Sub-Site ID's come into play if you subnet a location or user group into smaller subnets. The first address in each range becomes the Sub-Site ID.

Subnetting on the Nibble boundary gives you these subnetting options.

/48 No Nibbles	/52 1 Nibble	/56 2 Nibbles	/60 3 Nibbles	/64 4 Nibbles
/48 - 1 Subnet				
/52 - 16 Subnets	/52 - 1 Subnet			
/56 - 256 Subnets	/56 - 16 Subnets	/56 - 1 Subnets		
/60 - 4096 Subnets	/60 - 256 Subnets	/60 - 16 Subnets	/60 - 1 Subnets	
/64 - 65,536 Subnets	/64 - 4096 Subnets	/64 - 256 Subnets	/64 - 16 Subnets	/64 - 1 Subnet

As an example, your company has two offices, and within each office there are several groups you want on separate subnets. The groups include: Infrastructure, Management, Marketing, Finance, Research, Warehouse, and Sales.

Your ISP has assigned your company the IPv6 address 2001:ACAD:1234::/48. You will need one Site ID for each office. A /52 Subnet Prefix will give you 16 subnets, or you could use a /56 Subnet Prefix and have 256 subnets. For our purposes we'll use the /56 Subnet Prefix.

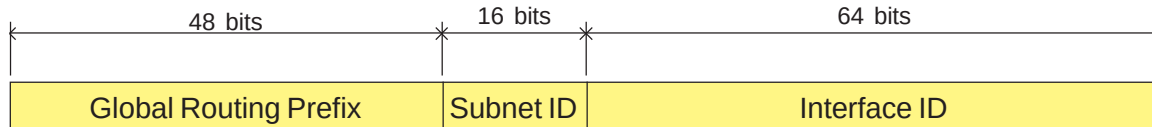
Our original IPv6 Range was: 2001:ACAD:1234::/48. Subnetting this address with a /56 Subnet Prefix will take two nibbles from the Subnet ID and give you the following address ranges:

2001:ACAD:1234:**0000**::/56 Save this range for over all infrastructure needs.
 2001:ACAD:1234:**0100**::/56 This becomes the Site ID for Office 1.
 2001:ACAD:1234:**0200**::/56 This becomes the Site ID for Office 2.
 2001:ACAD:1234:**0300**::/56
 (Lots of subnets omitted)
 2001:ACAD:1234:**FF00**::/56

Office 1 needs subnets for Infrastructure, Management, and Sales. Subnet the Site ID for Office 1 with a /60 Subnet Prefix.

2001:ACAD:1234:01**00**::/60 This becomes the Sub-Site ID for Infrastructure needs.
 2001:ACAD:1234:01**10**::/60 This becomes the Sub-Site ID for Managment.
 2001:ACAD:1234:01**20**::/60 This becomes the Sub-Site ID for Sales.
 (Lots of subnets omitted)
 2001:ACAD:1234:01**F0**::/60

Subnetting IPv6



Unlike IPv4 which requires you to borrow bits from the host portion, IPv6 has 16 bits or four hexadecimal numbers built into the address specifically allocated for creating subnets. A /48 address will allow you to have a single subnet or up to 65,536 subnets.

Lets say that your ISP has assigned you the 2001:ACAD:1234::/48 IPv6 address.

Global Routing Prefix	Subnet ID	Interface ID
2001:ACAD:1234:	0000	0000:0000:0000:0000

Basic subnetting in IPv6 was designed to make subnetting a very simple process. Start at the /64 bit and start counting up until you've used all the available bits in the subnet ID.

It really is that simple!

2001:ACAD:1234:0000:0000:0000:0000:0000/64	1st subnet
2001:ACAD:1234:0001:0000:0000:0000:0000/64	2nd subnet
2001:ACAD:1234:0002:0000:0000:0000:0000/64	3rd subnet
2001:ACAD:1234:0003:0000:0000:0000:0000/64	4th subnet
2001:ACAD:1234:0004:0000:0000:0000:0000/64	5th subnet
2001:ACAD:1234:0005:0000:0000:0000:0000/64	6th subnet
2001:ACAD:1234:0006:0000:0000:0000:0000/64	7th subnet
2001:ACAD:1234:0007:0000:0000:0000:0000/64	8th subnet
2001:ACAD:1234:0008:0000:0000:0000:0000/64	9th subnet
2001:ACAD:1234:0009:0000:0000:0000:0000/64	10th subnet
2001:ACAD:1234:000A:0000:0000:0000:0000/64	11th subnet
2001:ACAD:1234:000B:0000:0000:0000:0000/64	12th subnet
2001:ACAD:1234:000C:0000:0000:0000:0000/64	13th subnet
2001:ACAD:1234:000D:0000:0000:0000:0000/64	14th subnet
2001:ACAD:1234:000E:0000:0000:0000:0000/64	15th subnet
2001:ACAD:1234:000F:0000:0000:0000:0000/64	16th subnet
2001:ACAD:1234:0010:0000:0000:0000:0000/64	17th subnet
2001:ACAD:1234:0011:0000:0000:0000:0000/64	18th subnet
2001:ACAD:1234:0012:0000:0000:0000:0000/64	19th subnet
2001:ACAD:1234:0013:0000:0000:0000:0000/64	20th subnet

Each subnet contains over 18 quintillion addresses.

(Lots of subnets omitted for space.)

2001:ACAD:1234:FFFC:0000:0000:0000:0000/64	65,533rd subnet
2001:ACAD:1234:FFFD:0000:0000:0000:0000/64	65,534th subnet
2001:ACAD:1234:FFFE:0000:0000:0000:0000/64	65,535th subnet
2001:ACAD:1234:FFFF:0000:0000:0000:0000/64	65,536th subnet

A Medium End user might receive a /56 IPv6 address from their ISP.

Global Routing Prefix	Subnet ID	Interface ID	
2001:ACAD:1234:12	00	0000:0000:0000:0000	
2001:ACAD:1234:12	00	0000:0000:0000:0000/64	1st subnet
2001:ACAD:1234:12	01	0000:0000:0000:0000/64	2nd subnet
2001:ACAD:1234:12	02	0000:0000:0000:0000/64	3rd subnet
2001:ACAD:1234:12	03	0000:0000:0000:0000/64	4th subnet
(Lots of subnets omitted for space.)			
2001:ACAD:1234:12	FD	0000:0000:0000:0000/64	254th subnet
2001:ACAD:1234:12	FE	0000:0000:0000:0000/64	255th subnet
2001:ACAD:1234:12	FF	0000:0000:0000:0000/64	256th subnet

Each subnet contains over 18 quintillion addresses.

A Small End user might receive a /60 IPv6 address from their ISP.

Global Routing Prefix	Subnet ID	Interface ID
2001:ACAD:1234:123 0 :0000:0000:0000:0000		
2001:ACAD:1234:123 0 :0000:0000:0000:0000/64	1st subnet	
2001:ACAD:1234:123 1 :0000:0000:0000:0000/64	2nd subnet	
2001:ACAD:1234:123 2 :0000:0000:0000:0000/64	3rd subnet	
2001:ACAD:1234:123 3 :0000:0000:0000:0000/64	4th subnet	
2001:ACAD:1234:123 4 :0000:0000:0000:0000/64	5th subnet	
2001:ACAD:1234:123 5 :0000:0000:0000:0000/64	6th subnet	
2001:ACAD:1234:123 6 :0000:0000:0000:0000/64	7th subnet	
2001:ACAD:1234:123 7 :0000:0000:0000:0000/64	8th subnet	
2001:ACAD:1234:123 8 :0000:0000:0000:0000/64	9th subnet	
2001:ACAD:1234:123 9 :0000:0000:0000:0000/64	10th subnet	
2001:ACAD:1234:123 A :0000:0000:0000:0000/64	11th subnet	
2001:ACAD:1234:123 B :0000:0000:0000:0000/64	12th subnet	
2001:ACAD:1234:123 C :0000:0000:0000:0000/64	13th subnet	
2001:ACAD:1234:123 D :0000:0000:0000:0000/64	14th subnet	
2001:ACAD:1234:123 E :0000:0000:0000:0000/64	15th subnet	
2001:ACAD:1234:123 F :0000:0000:0000:0000/64	16th subnet	

Each subnet contains over 18 quintillion addresses.

A Home or single Site might receive a /64 IPv6 address from their ISP.

Global Routing Prefix	Interface ID
2001:ACAD:1234:1234	0000:0000:0000:0000
2001:ACAD:1234:1234:0000:0000:0000:0000/64	1 subnet

Over 18 quintillion addresses

Common Prefix's and Number of Subnets

Using the Subnet ID, the common Subnet Prefix's available from your ISP are /48, /52, /56, /60, and /64. The ISP could assign a lower Prefix Length, but most business will not need more than 65,536 subnets per Site.

The commonly available /64 subnets are:

/48	65,536 subnets
/52	4096 subnets
/56	256 subnets
/60	16 subnets
/64	1 subnet

Notice that the Subnet ID always changes by four even though we're only using a single hexadecimal character.

Each hexadecimal character equals 4 binary numbers.

(/48, /52, /56, /60, /64)

Each /64 subnet contains over 18 quintillion addresses.

With IPv4 the main concern was using the fewest possible number of addresses through creative subnetting.

The primary concern with IPv6 is making sure you have a prefix length that will cover all the subnets your Site will require and allow for future growth.

Basic Subnetting Problems

Sample Problem

Your ISP has given you the IPv6 address 2001:FE12:A231::/48.

How many /64 subnets are available with this address? 65,536

What are the first six /64 subnets?

2001:FE12:A231::/64

2001:FE12:A231:1::/64

2001:FE12:A231:2::/64

2001:FE12:A231:3::/64

2001:FE12:A231:4::/64

2001:FE12:A231:5::/64

Problem 1

Your ISP has given you the IPv6 address 2000:ACAD:1234:6600::/56.

How many /64 subnets are available with this address? 256

What are the first four /64 subnets?

2000:ACAD:1234:6600::/64

2000:ACAD:1234:6601::/64

2000:ACAD:1234:6602::/64

2000:ACAD:1234:6603::/64

What are the last two /64 subnets in this range?

2000:ACAD:1234:66FE::/64

2000:ACAD:1234:66FF::/64

Problem 2

Your ISP has given you the IPv6 address 3FFF:5801:DEAF::/48.

How many /64 subnets are available with this address? 65,536

What are the first four /64 subnets?

3FFF:5801:DEAF::/64

3FFF:5801:DEAF:1::/64

3FFF:5801:DEAF:2::/64

3FFF:5801:DEAF:3::/64

What are the last two /64 subnets in this range?

3FFF:5801:DEAF:FFFE::/64

3FFF:5801:DEAF:FFFF::/64

Problem 3

Your ISP has given you the IPv6 address 2001:ACAD:5678:1840::/60.

How many /64 subnets are available with this address? 16

Complete the /64 subnets in this range. (The ISP's Global Routing Prefix is already printed for you.)

2001:ACAD:5678:1840::/64

2001:ACAD:5678:1841::/64

2001:ACAD:5678:1842::/64

2001:ACAD:5678:1843::/64

2001:ACAD:5678:1844::/64

2001:ACAD:5678:1845::/64

2001:ACAD:5678:1846::/64

2001:ACAD:5678:1847::/64

2001:ACAD:5678:1848::/64

2001:ACAD:5678:1849::/64

2001:ACAD:5678:184A::/64

2001:ACAD:5678:184B::/64

2001:ACAD:5678:184C::/64

2001:ACAD:5678:184D::/64

2001:ACAD:5678:184E::/64

2001:ACAD:5678:184F::/64

Problem 4

Your ISP has given you the IPv6 address 3100:6523:AD14:8000::/52.

How many /64 subnets are available with this address? 4096

What are the first four /64 subnets? (The ISP's Global Routing Prefix is already printed for you.)

3100:6523:AD14:8000::/64

3100:6523:AD14:8001::/64

3100:6523:AD14:8002::/64

3100:6523:AD14:8003::/64

What are the last two /64 subnets in this range?

3100:6523:AD14:8FFE::/64

3100:6523:AD14:8FFF::/64

Problem 5

Your ISP has given you the IPv6 address 2100:89:4500::/48.

How many /64 subnets are available with this address? 65,536

What are the first four /64 subnets? (The ISP's Global Routing Prefix is already printed for you.)

2100:89:4500::/64

2100:89:4500:1::/64

2100:89:4500:2::/64

2100:89:4500:3::/64

What are the last two /64 subnets in this range?

2100:89:4500:FFFE::/64

2100:89:4500:FFFF::/64

IPv6 Subnetting Best Practices

IPv6 was designed to make subnetting as simple as possible by using ridiculously large blocks of addresses. Yes, it's wasteful! Most experts agree that IPv6 will last for 100 years and IP will be replaced before we run out of IPv6 address space.

So until the rules and/or the consensus of the experts change, these are the common subnets and best practices you will be working with. Information for subnets can be found in RFC & RTC documents, plus some additional resources listed in the Reference section of this workbook.

The common /64 subnets are:

You can negotiate with your ISP for a larger block

/48	65,536 subnets
/52	4096 subnets
/56	256 subnets
/60	16 subnets
/64	1 subnet

Specific Address Rules and Best Practices:

Point-To-Point Connections -

Best Practice: Use a /64 address range for these two addresses.

In rare cases this approach might present an issue with certain router setups where addresses are bounced back and forth between routers. Reducing the number of addresses in the range can also help avoid neighbor cache exhaustion attacks. (RFC 6164 - section 5)

Acceptable options include:

- /127 -is recommended in RFC 6164. You must disable the router's Subnet-Router Anycast option to avoid issues with the all routers anycast address.
- /126 -is discussed in RFC 2526. The /126 allows the all-zero reserved Anycast address to be avoided.
- /120 - allows all the reserved anycast address to be avoided.
- /112 - allows you to avoid the reserved anycast address, and gives you the entire four-digit hex value after the last colon.

Of all these options the /120 is probably the best choice since it avoids any issues with the Anycast addresses.

It is recommended that you reserve the entire /64 network with each of the above options.

/128 Single Address Subnets -

Best Practice: reserve the first subnet to use for infrastructure needs, such as loopback addresses, etc..

Acceptable option:

Allocate a full /64 range of addresses for each loopback address, but assign it a /128 subnet Prefix.

Anycast addresses -

Best Practice: Don't use the very first address or the last 128 addresses in any /64 network. These can only be assigned to an interface as an Anycast address.

Anycast addresses can be pulled from Global Unicast, Site-Local or Link-Local address ranges. Any address assigned to more than one interface on a subnet becomes an Anycast address. Anycast addresses can only be used by network devices, not a host. No anycast address can be used as the source address of an IPv6 packet.

The first address in every /64 subnet range is reserved for special use. The Interface ID is all zero's. (Example: 2001:ACAD:1234:5678:0000:0000:0000:0000/64) This address is the Subnet-Anycast Address. These addresses are typically used by different protocols such as Mobile IPv6. (RFC 4291)

The last eight bits in every /64 subnet range are reserved for Anycast addresses. These bits are 10000000 to 11111111. This means you can not assign any addresses if the last hextet falls between FF80 and FFFF unless it is a Anycast address. (RFC 2526)

If you try to apply the last 128 addresses to a router without setting them up as an Anycast address you will get the following error message.

```
Router(config-if)#ipv6 address 2001:ACAD:1234:5678:FFFF:FFFF:FFFF:FFFF/64
% 2001:DB8:1:1:FFFF:FFFF:FFFF:FFFF/64 should not be configured on FastEthernet0/0, a reserved anycast
```

Developing an Address Plan

(or IPv6 Subnetting in the Real World)

There is no one right way for developing an IPv6 addressing plan, but the recommended general guidelines include the following:

Step 1: Decide how you are going to divide your network:

- a. by location
- b. by user groups



Administration

Users Include:
Administration
Staff



Academic

Users Include:
Staff
Students



Dormitory

Users Include:
Staff
Students

Subnetting by Location:

To divide by location you would need four subnets. One for each building and one for the overall network infrastructure needs. You also need to hold several extra subnets in reserve for later growth.

Advantages:

This allows you to optimize your routing tables. All the networks within each location will aggregate to a single route.

Subnetting by User Groups:

To subnet the network by user groups you would need four subnets. One for Administration, Staff, and Students, plus one for overall network infrastructure needs. You also need to hold several extra subnets in reserve for later growth.

Advantages:

Subnetting by user groups makes it much easier to implement a security policy. Grouping by usage also helps track addresses for allocation and management.

Best Practice:

Subnetting by either location or user is acceptable. However, with the emphasis on network security, most networks are better served by subnetting user groups. It makes it much easier to maintain a higher level of security.

Step 2: Determine how many primary and secondary subnets your Site will need.

- Create the primary subnets first.
- Then create secondary subnets.

Subnetting by Location:

Primary Subnets: Quantity 4

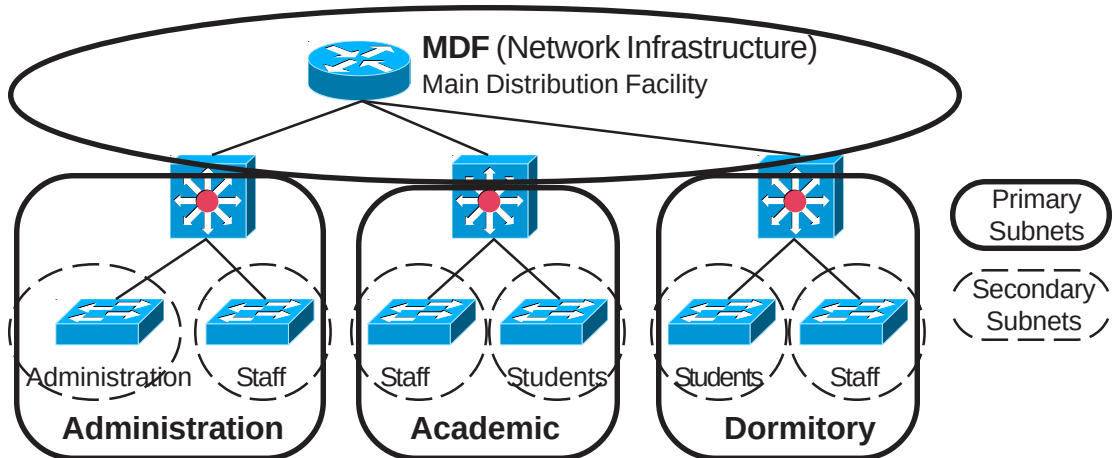
With three buildings you will need four primary subnets. One for each building and one for the overall infrastructure needs.

Secondary Subnets: Quantity 6

Administration will need two secondary subnets: Administration and Staff.

Academic will need two secondary subnets: Staff and Students.

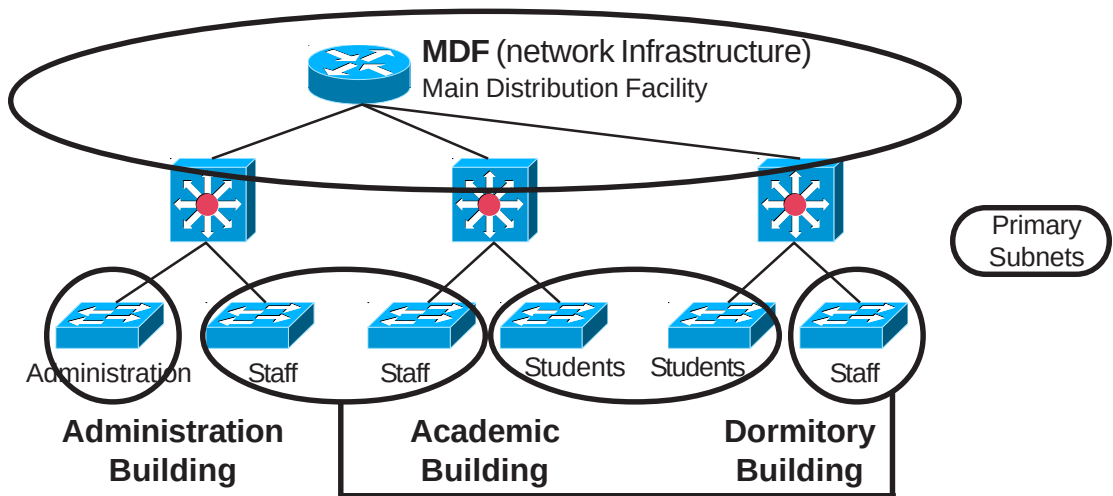
Dormitory will need two secondary subnets: Staff and Students.



Subnetting by User Groups:

Primary Subnets: Quantity 4

With three user groups you will need four subnets. One for each group and one for the overall infrastructure needs.



Step 3: Based on the number of primary and secondary subnets needed assign the address ranges. The ISP has assigned you 2001:ACAD:1234::/48.

Subnetting Options:

/48 No Nibbles	/52 1 Nibble	/56 2 Nibbles	/60 3 Nibbles	/64 4 Nibbles
/48 - 1 Subnet				
/52 - 16 Subnets	/52 - 1 Subnet			
/56 - 256 Subnets	/56 - 16 Subnets	/56 - 1 Subnets		
/60 - 4096 Subnets	/60 - 256 Subnets	/60 - 16 Subnets	/60 - 1 Subnets	
/64 - 65,536 Subnets	/64 - 4096 Subnets	/64 - 256 Subnets	/64 - 16 Subnets	/64 - 1 Subnet

Subnetting by Location:

Primary Subnets: Quantity 4

With three buildings you will need four primary subnets. One for each building and one for the overall infrastructure needs.

Secondary Subnets: Quantity 6

Administration will need two secondary subnets: Administration and Staff.

Academic will need two secondary subnets: Staff and Students.

Dormitory will need two secondary subnets: Staff and Students.

Take the addresses assigned to you by the ISP use one nibble and subnet it into 16 subnets using a /52 Subnet Prefix. This will give you the 4 primary subnets required with several to spare for future growth.

2001:ACAD:1234::/48 becomes:

- ▣ 2001:ACAD:1234::/52 Site ID for over all infrastructure needs.
- ▣ 2001:ACAD:1234:1000::/52 Site ID for the Administration Building.
- ▣ 2001:ACAD:1234:2000::/52 Site ID for the Academic Building.
- ▣ 2001:ACAD:1234:3000::/52 Site ID for the Dormitory.
- ▣ 2001:ACAD:1234:4000::/52
- (Subnets omitted for space.)
- ▣ 2001:ACAD:1234:F000::/52

Site IDs and Sub-Site IDs will be the addresses found in the routing tables.

Take the second, third, and forth ranges and subnet them again by using the next Nibble with a /56 Subnet Prefix. This will create 16 subnets for each location.

Administration Building Site ID 2001:ACAD:1234:1000::/52 becomes:

- ▣ 2001:ACAD:1234:1000::/52 Administration Building Site ID
 - ▣ 2001:ACAD:1234:1000::/56 Sub-Site ID for infrastructure needs.
 - ▣ 2001:ACAD:1234:1100::/56 Sub-Site ID for Administration
 - ▣ 2001:ACAD:1234:1200::/56 Sub-Site ID for Staff

Academic Building Site ID 2001:ACAD:1234:2000::/52 becomes:

- ▣ 2001:ACAD:1234:2000::/52 Academic Building Site ID
 - ▣ 2001:ACAD:1234:2000::/56 Sub-Site ID for infrastructure needs.
 - ▣ 2001:ACAD:1234:2100::/56 Sub-Site ID for Students
 - ▣ 2001:ACAD:1234:2200::/56 Sub-Site ID for Staff

Dormitory Building Site ID 2001:ACAD:1234:3000::/52 becomes:

- ▣ 2001:ACAD:1234:3000::/52 Dormitory Building Site ID
 - ▣ 2001:ACAD:1234:3000::/56 Sub-Site ID for infrastructure needs.
 - ▣ 2001:ACAD:1234:3100::/56 Sub-Site ID for Students
 - ▣ 2001:ACAD:1234:3200::/56 Sub-Site ID for Staff

Subnetting by User Group:

Primary Subnets: Quantity 4

With three user groups you will need four primary subnets. One for each group and one for the overall infrastructure needs. In this example no secondary subnets are required.

Take the address assigned to you by the ISP use one nibble and subnet it into 16 subnets using a /52 Subnet Prefix. This will give you the 4 primary subnets required with several to spare for future growth.

2001:ACAD:1234::/48 becomes:

- ▣ 2001:ACAD:1234::/52 Site ID for over all infrastructure needs.
- ▣ 2001:ACAD:1234:1000::/52 Site ID for the Administration employees.
- ▣ 2001:ACAD:1234:2000::/52 Site ID for the Staff.
- ▣ 2001:ACAD:1234:3000::/52 Site ID for the Students.
- ▣ 2001:ACAD:1234:4000::/52
- (Subnets omitted for space.)
- ▣ 2001:ACAD:1234:F000::/52

IPv6 Subnetting Problems

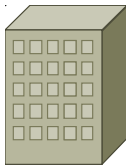
Subnetting on the Nibble Boundary

Sample Problem 1

Using the **minimum number of subnets required** for the primary and secondary sites, design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

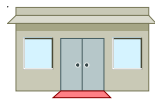
A coffee shop is opening three new stores and a central office/warehouse in your community and needs an IPv6 network plan developed. Each store will need secure network access for three groups: managers, the registers (which will monitor inventory and revenue), and wireless access for guests. The central office/warehouse will need secure network access for several departments: Managers, Finance, Human Relations.

The ISP has given the company 2000:FE23:0054::/48.



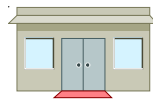
Central Office

Users Include:
Mangers
Finance
Human Relations



Store 1

Users Include:
Mangers
Registers
Wireless



Store 2

Users Include:
Mangers
Registers
Wireless



Store 3

Users Include:
Mangers
Registers
Wireless

Subnets Based on User Groups

ISP Address: 2000:FE23:0054::/48

- ❑ Infrastructure Site ID: 2000:FE23:0054::/52
- ❑ Managers Site ID: 2000:FE23:0054:1000::/52
- ❑ Finance Site ID: 2000:FE23:0054:2000::/52
- ❑ Human Relations Site ID: 2000:FE23:0054:3000::/52
- ❑ Registers Site ID: 2000:FE23:0054:4000::/52
- ❑ Wireless Site ID: 2000:FE23:0054:5000::/52

Subnets Based on Location

ISP Address: 2000:FE23:0054::/48

- ▣ Infrastructure Site ID: 2000:FE23:0054::/52
- ▣ Central Office Site ID: 2000:FE23:0054:1000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:1000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:1100::/56
- ▣ Finance Sub-Site ID: 2000:FE23:0054:1200::/56
- ▣ Human Relations Sub-Site ID: 2000:FE23:0054:1300::/56
- ▣ Store 1 Site ID: 2000:FE23:0054:2000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:2000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:2100::/56
- ▣ Registers Sub-Site ID: 2000:FE23:0054:2200::/56
- ▣ Wireless Sub-Site ID: 2000:FE23:0054:2300::/56
- ▣ Store 2 Site ID: 2000:FE23:0054:3000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:3000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:3100::/56
- ▣ Registers Sub-Site ID: 2000:FE23:0054:3200::/56
- ▣ Wireless Sub-Site ID: 2000:FE23:0054:3300::/56
- ▣ Store 3 Site ID: 2000:FE23:0054:4000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:4000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:4100::/56
- ▣ Registers Sub-Site ID: 2000:FE23:0054:4200::/56
- ▣ Wireless Sub-Site ID: 2000:FE23:0054:4300::/56

IPv6 Subnetting Problems

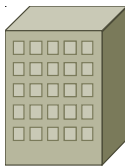
Subnetting on the Nibble Boundary

Sample Problem 2

Using the **minimum number of subnets required** for the primary and secondary sites, design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

A Medical Building is opening in your community and needs an IPv6 network plan developed.

The ISP has given the company 2001:5378:8801::/48.



Medical Building

First Floor Rooms

Patient Check-in
Emergency Room

Users Include:
Administrators
Staff
Guests

Second Floor Rooms

Nurses Station
Ward A

Users Include:
Staff
Guests

Subnets Based on User Groups

ISP Address: 2001:5378:8801::/48

- ▣ Infrastructure Site ID: 2001:5378:8801::/52
- ▣ Administrators Site ID: 2001:5378:8801:1000::/52
- ▣ Staff Site ID: 2001:5378:8801:2000::/52
- ▣ Guests Site ID: 2001:5378:8801:3000::/52

Subnets Based on Location

ISP Address: 2001:5378:8801::/48

- ▣ Infrastructure Site ID: 2001:5378:8801::/52
- ▣ First Floor Site ID: 2001:5378:8801:1000::/52
- ▣ Infrastructure Sub-Site ID: 2001:5378:8801:1000::/56
- ▣ Patient Check-in Sub-Site ID: 2001:5378:8801:1100::/56
- ▣ Administrators Sub-Site ID: 2001:5378:8801:1100::/60
- ▣ Staff Sub-Site ID: 2001:5378:8801:1110::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:1120::/60
- ▣ Emergency Room Sub-Site ID: 2001:5378:8801:1200::/56
- ▣ Administrators Sub-Site ID: 2001:5378:8801:1200::/60
- ▣ Staff Sub-Site ID: 2001:5378:8801:1210::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:1220::/60
- ▣ Second Floor Site ID: 2001:5378:8801:2000::/52
- ▣ Infrastructure Sub-Site ID: 2001:5378:8801:2000::/56
- ▣ Nurses Station Sub-Site ID: 2001:5378:8801:2100::/56
- ▣ Staff Sub-Site ID: 2001:5378:8801:2100::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:2110::/60
- ▣ Ward A Sub-Site ID: 2001:5378:8801:2200::/56
- ▣ Staff Sub-Site ID: 2001:5378:8801:2200::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:2210::/60

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 1

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

The XYZ Film company is setting up a new office and needs an IPv6 network plan developed. The company will have the following departments:

The ISP has given the company 2001:EE00:2575::/48.

Administration Building

Users Include:

Sales
Accounting
Distribution

Production Building

Users Include:

Casting
Editing

Subnets Based on User Groups

ISP Address: 2001:EE00:2575::/48

- ☐ Infrastructure Site ID: 2001:EE00:2575::/52
- ☐ Sales Site ID: 2001:EE00:2575:1000::/52
- ☐ Accounting Site ID: 2001:EE00:2575:2000::/52
- ☐ Distribution Site ID: 2001:EE00:2575:3000::/52
- ☐ Casting Site ID: 2001:EE00:2575:4000::/52
- ☐ Editing Site ID: 2001:EE00:2575:5000::/52

Subnets Based on Location

ISP Address: 2001:EE00:2575::/48

▣ Infrastructure Site ID: 2001:EE00:2575::/52

▣ Administration Site ID: 2001:EE00:2575:1000::/52

▣ Infrastructure Sub-Site ID: 2001:EE00:2575:1000::/56

▣ Sales Sub-Site ID: 2001:EE00:2575:1100::/56

▣ Accounting Sub-Site ID: 2001:EE00:2575:1200::/56

▣ Distribution Sub-Site ID: 2001:EE00:2575:1300::/56

▣ Production Site ID: 2001:EE00:2575:2000::/52

▣ Infrastructure Sub-Site ID: 2001:EE00:2575:2000::/56

▣ Casting Sub-Site ID: 2001:EE00:2575:2100::/56

▣ Editing Sub-Site ID: 2001:EE00:2575:2200::/56

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 2

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

A new medical supply company is opening and needs an IPv6 network plan developed. The company has three buildings and will include the following user groups:

The ISP has given the company 3F01:ABCD:8875::/48.

Building 1

Users Include:
Management
Sales

Building 2

Users Include:
Management
Human Resources

Building 3

Users Include:
Management
Warehouse

Subnets Based on User Groups

ISP Address: 3F01:ABCD:8875::/48

- ▣ Infrastructure Site ID: 3F01:ABCD:8875::/52
- ▣ Management Site ID: 3F01:ABCD:8875:1000::/52
- ▣ Sales Site ID: 3F01:ABCD:8875:2000::/52
- ▣ Human Resources Site ID: 3F01:ABCD:8875:3000::/52
- ▣ Warehouse Site ID: 3F01:ABCD:8875:4000::/52

Subnets Based on Location

ISP Address: 3F01:ABCD:8875::/48

▣ Infrastructure Site ID: 3F01:ABCD:8875::/52

▣ Building 1 Site ID: 3F01:ABCD:8875:1000::/52

▣ Infrastructure Sub-Site ID: 3F01:ABCD:8875:1000::/56

▣ Management Sub-Site ID: 3F01:ABCD:8875:1100::/56

▣ Sales Sub-Site ID: 3F01:ABCD:8875:1200::/56

▣ Building 2 Site ID: 3F01:ABCD:8875:2000::/52

▣ Infrastructure Sub-Site ID: 3F01:ABCD:8875:2000::/56

▣ Management Sub-Site ID: 3F01:ABCD:8875:2100::/56

▣ Human Resources Sub-Site ID: 3F01:ABCD:8875:2200::/56

▣ Building 2 Site ID: 3F01:ABCD:8875:3000::/52

▣ Infrastructure Sub-Site ID: 3F01:ABCD:8875:3000::/56

▣ Management Sub-Site ID: 3F01:ABCD:8875:3100::/56

▣ Warehouse Sub-Site ID: 3F01:ABCD:8875:3200::/56

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 3

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

A paper supply company needs an IPv6 network plan developed. The company has two buildings and will include the following user groups and sub-user groups:

The ISP has given the company 2001:CA21:9000::/48.

Building A

Management Groups

Human Resources

Sales:

Wholesale

Retail

Building B

Production Groups

Warehouse

Shipping:

Domestic

Worldwide

Subnets Based on User Groups

ISP Address: 2001:CA21:9000::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2001:CA21:9000::/52

☐ Management Groups Site ID: 2001:CA21:9000:1000::/52

☐ HR Sub-Site ID: 2001:CA21:9000:1000::/56

☐ Sales Sub-Site ID: 2001:CA21:9000:1100::/56

☐ Wholesale Sub- Site ID: 2001:CA21:9000:1100::/60

☐ Retail Sub- Site ID: 2001:CA21:9000:1110::/60

☐ Production Groups Site ID: 2001:CA21:9000:2000::/52

☐ Warehouse Sub-Site ID: 2001:CA21:9000:2000::/56

☐ Shipping Sub-Site ID: 2001:CA21:9000:2100::/56

☐ Domestic Sub-Site ID: 2001:CA21:9000:2100::/60

☐ Worldwide Sub-Site ID: 2001:CA21:9000:2110::/60

Subnets Based on Location

ISP Address: 2001:CA21:9000::/48 (The ISP's Global Routing Prefix is already printed for you.)

- ☐ Infrastructure Site ID: 2001:CA21:9000::/52
- ☐ Building A Site ID: 2001:CA21:9000:1000::/52
- ☐ Infrastructure Sub-Site ID: 2001:CA21:9000:1000::/56
- ☐ HR Sub-Site ID: 2001:CA21:9000:1100::/56
- ☐ Sales Sub-Site ID: 2001:CA21:9000:1200::/56
- ☐ Wholesale Sub-Site ID: 2001:CA21:9000:1200::/60
- ☐ Retail Sub-Site ID: 2001:CA21:9000:1210::/60
- ☐ Building B Site ID: 2001:CA21:9000:2000::/52
- ☐ Infrastructure Sub-Site ID: 2001:CA21:9000:2000::/56
- ☐ Warehouse Sub-Site ID: 2001:CA21:9000:2100::/56
- ☐ Shipping Sub-Site ID: 2001:CA21:9000:2200::/56
- ☐ Domestic Sub-Site ID: 2001:CA21:9000:2200::/60
- ☐ Worldwide Sub-Site ID: 2001:CA21:9000:2210::/60

IPv6 Subnetting Problems

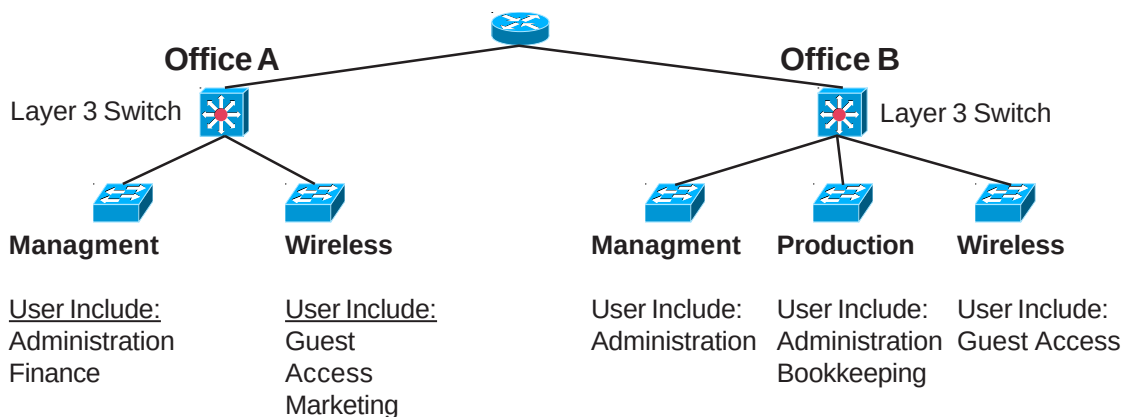
Subnetting on the Nibble Boundary

Problem 4

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

A company is setting up a new server farm and needs an IPv6 network plan developed. The Core Router and the direct connections with layer 3 switches will pull their IPv6 addresses from the Infrastructure Site ID range.

The ISP has given the company 2000:ACAD:1145::/48.



Subnets Based on User Groups

☐ ISP Address: 2000:ACAD:1145::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2000:ACAD:1145::/52

☐ Administration Site ID: 2000:ACAD:1145:1000::/52

☐ Finance Site ID: 2000:ACAD:1145:2000::/52

☐ Guest Access Site ID: 2000:ACAD:1145:3000::/52

☐ Marketing Site ID: 2000:ACAD:1145:4000::/52

☐ Bookkeeping Site ID: 2000:ACAD:1145:5000::/52

Subnets Based on Location

ISP Address: 2000:ACAD:1145::/48 (The ISP's Global Routing Prefix is already printed for you.)

- ☐ Infrastructure Site ID: 2000:ACAD:1145::/52
- ☐ Office A Site ID: 2000:ACAD:1145:1000::/52
- ☐ Infrastructure Sub-Site ID: 2000:ACAD:1145:1000::/56
- ☐ Management Sub-Site ID: 2000:ACAD:1145:1100::/56
- ☐ Administration Sub-Site ID: 2000:ACAD:1145:1100::/60
- ☐ Finance Sub-Site ID: 2000:ACAD:1145:1110::/60
- ☐ Wireless Access Sub-Site ID: 2000:ACAD:1145:1200::/56
- ☐ Guest Access Sub-Site ID: 2000:ACAD:1145:1200::/60
- ☐ Marketing Sub-Site ID: 2000:ACAD:1145:1210::/60
- ☐ Office B Site ID: 2000:ACAD:1145:2000::/52
- ☐ Infrastructure Sub-Site ID: 2000:ACAD:1145:2000::/56
- ☐ Management Sub-Site ID: 2000:ACAD:1145:2100::/56
- ☐ Administration Sub-Site ID: 2000:ACAD:1145:2100::/60
- ☐ Production Sub-Site ID: 2000:ACAD:1145:2200::/56
- ☐ Administration Sub-Site ID: 2000:ACAD:1145:2200::/60
- ☐ Bookkeeping Sub-Site ID: 2000:ACAD:1145:2210::/60
- ☐ Wireless Access Sub-Site ID: 2000:ACAD:1145:2300::/56
- ☐ Guest Access Sub-Site ID: 2000:ACAD:1145:2300::/60

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 5

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

The company has multiple floors in a high rise building and will include the following user groups and sub-user groups:

The ISP has given the company 3F01:AA07:3907::/48.

33rd Floor

Manufacturing Groups

Marketing
Inventory
Shipping

34th Floor

Admin Groups

Human Resources (HR):
Hiring
Benefits
Financial:
Purchasing
Sales

Subnets Based on User Groups

☐ ISP Address: 3F01:AA07:3907::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 3F01:AA07:3907::/52

☐ Manufacturing Groups Site ID: 3F01:AA07:3907:1000::/52

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:1000::/56

☐ Marketing Sub-Site ID: 3F01:AA07:3907:1100::/56

☐ Inventory Sub-Site ID: 3F01:AA07:3907:1200::/56

☐ Shipping Sub-Site ID: 3F01:AA07:3907:1300::/56

☐ Admin Groups Site ID: 3F01:AA07:3907:2000::/52

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:2000::/56

☐ HR Sub-Site ID: 3F01:AA07:3907:2100::/56

☐ Hiring Sub-Site ID: 3F01:AA07:3907:2100::/60

☐ Benefits Sub-Site ID: 3F01:AA07:3907:2110::/60

☐ Financial Sub-Site ID: 3F01:AA07:3907:2200::/56

☐ Purchasing Sub-Site ID: 3F01:AA07:3907:2200::/60

☐ Sales Sub-Site ID: 3F01:AA07:3907:2210::/60

Subnets Based on Location

ISP Address: 3F01:AA07:3907::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 3F01:AA07:3907::/52

☐ 33rd Floor Site ID: 3F01:AA07:3907:1000::/52

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:1000::/56

☐ Manufacturing Groups Sub-Site ID: 3F01:AA07:3907:1100::/56

☐ Marketing Sub-Site ID: 3F01:AA07:3907:1100::/60

☐ Inventory Sub-Site ID: 3F01:AA07:3907:1110::/60

☐ Shipping Sub-Site ID: 3F01:AA07:3907:1120::/60

☐ 34th Floor Site ID: 3F01:AA07:3907:2000::/52

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:2000::/56

☐ Admin Groups Sub-Site ID: 3F01:AA07:3907:2100::/56

☐ HR Sub-Site ID: 3F01:AA07:3907:2100::/60

☐ Hiring Sub-Site ID: 3F01:AA07:3907:2100::/64

☐ Benefits Sub-Site ID: 3F01:AA07:3907:2101::/64

☐ Financial Sub-Site ID: 3F01:AA07:3907:2200::/60

☐ Purchasing Sub-Site ID: 3F01:AA07:3907:2200::/64

☐ Sales Sub-Site ID: 3F01:AA07:3907:2201::/64

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 6

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

This medium sized company will include the following user groups and sub-user groups:

The ISP has given the company 2001:0:17::/52

Management

Users Include:
Laboratory
Administrators
Staff

Finance

Users Include:
Staff

Marketing Department

Users Include:
Advertising:
Radio
TV
Web
Sales:
Retail
Wholesale

Subnets Based on User Groups

☐ ISP Address: 2001:0:17::/52 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2001:0:17::/56

☐ Administrators Site ID: 2001:0:17:100::/56

☐ Staff Site ID: 2001:0:17:200::/56

☐ Advertising Site ID: 2001:0:17:300::/56

☐ Infrastructure Sub-Site ID: 2001:0:17:300::/60

☐ Radio Sub-Site ID: 2001:0:17:310::/60

☐ TV Sub- Site ID: 2001:0:17:320::/60

☐ Web Sub- Site ID: 2001:0:17:330::/60

☐ Sales Site ID: 2001:0:17:400::/56

☐ Infrastructure Sub-Site ID: 2001:0:17:400::/60

☐ Retail Sub-Site ID: 2001:0:17:410::/60

☐ Wholesale Sub-Site ID: 2001:0:17:420::/60

Subnets Based on Location

▣ ISP Address: 2001:0:17::/52 (The ISP's Global Routing Prefix is already printed for you.)

▣ Infrastructure Site ID: 2001:0:17::/56

▣ Management Site ID: 2001:0:17:100::/56

▣ Infrastructure Sub-Site ID: 2001:0:17:100::/60

▣ Administrators Sub-Site ID: 2001:0:17:110::/60

▣ Staff Sub-Site ID: 2001:0:17:120::/60

▣ Finance Site ID: 2001:0:17:200::/56

▣ Infrastructure Sub-Site ID: 2001:0:17:200::/60

▣ Staff Sub-Site ID: 2001:0:17:210::/60

▣ Marketing Dept Site ID: 2001:0:17:300::/56

▣ Infrastructure Sub-Site ID: 2001:0:17:300::/60

▣ Advertising Sub-Site ID: 2001:0:17:310::/60

▣ Radio Sub-Site ID: 2001:0:17:310::/64

▣ TV Sub-Site ID: 2001:0:17:311::/64

▣ Web Sub-Site ID: 2001:0:17:312::/64

▣ Sales Sub-Site ID: 2001:0:17:320::/60

▣ Retail Sub-Site ID: 2001:0:17:320::/64

▣ Wholesale Sub-Site ID: 2001:0:17:321::/64

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 7

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

This medium sized company is setting up a new IPv6 addressing plan which will include the following user groups and sub-user groups:

The ISP has given the company 3F00:3589:0:5000::/52

Office A

Users Include:
Management
Human Relations (HR):
 Record Keeping
 Insurance

Office B

Users Include:
Management
Finance:
 Sales

Office C

Users Include:
Management
Purchasing:
 Inventory
 Distribution

Subnets Based on User Groups

ISP Address: 3F00:3589:0:5000::/52 (The ISP's Global Routing Prefix is already printed for you.)

- ▣ Infrastructure Site ID: 3F00:3589:0:5000::/56
- ▣ Management Site ID: 3F00:3589:0:5100::/56
- ▣ HR Site ID: 3F00:3589:0:5200::/56
 - ▣ Infrastructure Sub-Site ID: 3F00:3589:0:5200::/60
 - ▣ Record Keeping Sub-Site ID: 3F00:3589:0:5210::/60
 - ▣ Insurance Site ID: 3F00:3589:0:5220::/60
- ▣ Finance Site ID: 3F00:3589:0:5300::/56
 - ▣ Infrastructure Sub-Site ID: 3F00:3589:0:5300::/60
 - ▣ Sales Sub-Site ID: 3F00:3589:0:5310::/60
- ▣ Purchasing Sub-Site ID: 3F00:3589:0:5400::/56
 - ▣ Inventory Sub-Site ID: 3F00:3589:0:5400::/60
 - ▣ Distribution Sub-Site ID: 3F00:3589:0:5410::/60

Subnets Based on Location

ISP Address: 3F00:3589:0:5000::/52 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 3F00:3589:0:5000::/56

☐ Office A Site ID: 3F00:3589:0:5100::/56

☐ Infrastructure Sub-Site ID: 3F00:3589:0:5100::/60

☐ Management Sub-Site ID: 3F00:3589:0:5110::/60

☐ HR Sub- Site ID: 3F00:3589:0:5120::/60

☐ Record Keeping Sub-Site ID: 3F00:3589:0:5120::/64

☐ Insurance Sub-Site ID: 3F00:3589:0:5121::/64

☐ Office B Site ID: 3F00:3589:0:5200::/56

☐ Infrastructure Sub-Site ID: 3F00:3589:0:5200::/60

☐ Management Sub-Site ID: 3F00:3589:0:5210::/60

☐ Finance Sub-Site ID: 3F00:3589:0:5220::/60

☐ Sales Sub-Site ID: 3F00:3589:0:5220::/64

☐ Office C Site ID: 3F00:3589:0:5300::/56

☐ Infrastructure Sub-Site ID: 3F00:3589:0:5300::/60

☐ Management Sub-Site ID: 3F00:3589:0:5310::/60

☐ Purchasing Sub-Site ID: 3F00:3589:0:5320::/60

☐ Inventory Sub-Site ID: 3F00:3589:0:5320::/64

☐ Distribution Sub-Site ID: 3F00:3589:0:5321::/64

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 8

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

A Health Care facility is upgrading their network to IPv6 and will include the following user groups and sub-user groups:

The ISP has given the company 2000:2531:FE00::/48.

Emergency

Users Include:
Nurses/Staff
Laboratory
Obstetrics
Pediactric

Admissions

Users Include:
Nurses/Staff
Records

Patient Wards

Users Include:
Ward A:
Nurses/Staff
Guest WIFI
Ward B:
Nurses/Staff
Guest WIFI

Subnets Based on User Groups

ISP Address: 2000:2531:FE00::/48 (The ISP's Global Routing Prefix is already printed for you.)

- ▣ Infrastructure Site ID: 2000:2531:FE00::/52
- ▣ Nurses/Staff Site ID: 2000:2531:FE00:1000::/52
- ▣ Laboratory Site ID: 2000:2531:FE00:2000::/52
- ▣ Obstetrics Site ID: 2000:2531:FE00:3000::/52
- ▣ Pediatric Site ID: 2000:2531:FE00:4000::/52
- ▣ Records Site ID: 2000:2531:FE00:5000::/52
- ▣ Guest WIFI Site ID: 2000:2531:FE00:6000::/52

Subnets Based on Location

ISP Address: 2000:2531:FE00::/48 (The ISP's Global Routing Prefix is already printed for you.)

- ☐ Infrastructure Site ID: 2000:2531:FE00::/52
- ☐ Emergency Site ID: 2000:2531:FE00:1000::/52
- ☐ Infrastructure Sub-Site ID: 2000:2531:FE00:1000::/56
- ☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:1100::/56
- ☐ Laboratory Sub-Site ID: 2000:2531:FE00:1200::/56
- ☐ Obstetrics Sub-Site ID: 2000:2531:FE00:1300::/56
- ☐ Pediatric Sub-Site ID: 2000:2531:FE00:1400::/56
- ☐ Admissions Site ID: 2000:2531:FE00:2000::/52
- ☐ Infrastructure Sub-Site ID: 2000:2531:FE00:2000::/56
- ☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:2100::/56
- ☐ Records Sub-Site ID: 2000:2531:FE00:2200::/56
- ☐ Patient Wards Site ID: 2000:2531:FE00:3000::/52
- ☐ Infrastructure Sub-Site ID: 2000:2531:FE00:3000::/56
- ☐ Ward A Sub-Site ID: 2000:2531:FE00:3100::/56
- ☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:3100::/60
- ☐ Guest WIFI Sub-Site ID: 2000:2531:FE00:3110::/60
- ☐ Ward B Sub-Site ID: 2000:2531:FE00:3200::/56
- ☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:3200::/60
- ☐ Guest WIFI Sub-Site ID: 2000:2531:FE00:3210::/60

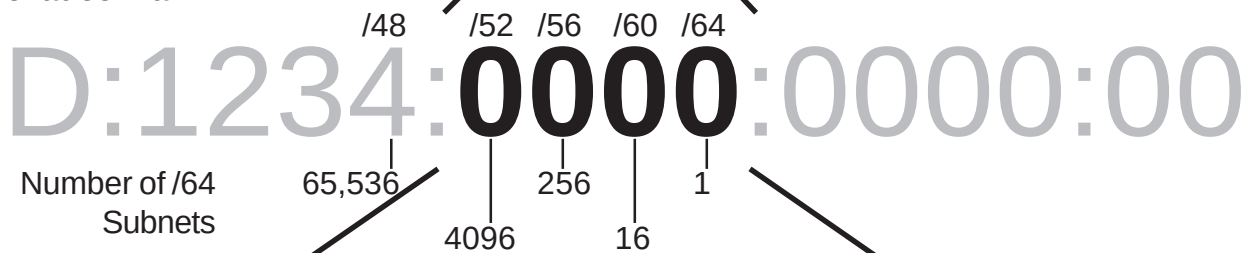
Subnetting Within a Nibble

It is a Best Practice to subnet on the Nibble Boundary. However, subnetting within a Nibble is an acceptable practice. It tends to make subnetting, implementation, and troubleshooting more difficult.

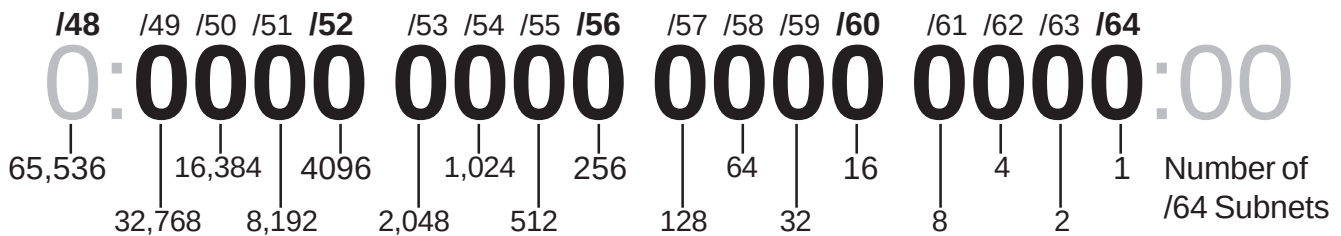
Subnet ID Break Down

2001:ACAD:1234:0000:0000:0000:0000:0000/48

Hexadecimal



Binary

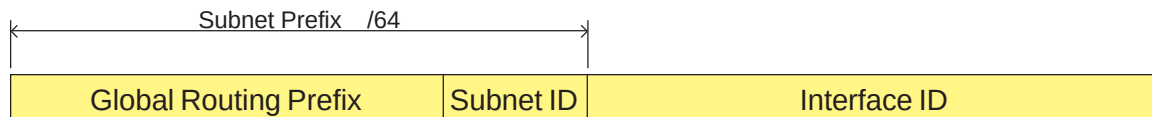


Subnetting Beyond the /64 Boundary

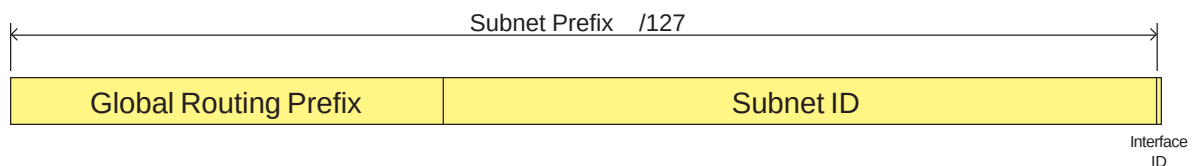
What happens if you need more subnets than the 16 bit Subnet Prefix will allow? Or your IPv4 address conservatism kicks in and you decide that using a /64 Global Routing Prefix with over 18 quintillion addresses for a single point-to-point serial connection is more than you can handle; there are some options.

IPv6 was designed to be very flexible. The Subnet Prefix is the address space used by the Global Routing Prefix and the Subnet ID, and can range from 0 to 128.

Standard /48 Subnet Prefix



Point-To-Point /127 Subnet Prefix



Just as you could borrow host bits in IPv4, you can borrow Interface ID bits in IPv6. This allows you to create more subnets with fewer addresses. Before you get too excited, there are a few rules and best practices you need to take into account.

Borrowing bits from the Interface ID should only be done on network infrastructure links. Loopback addresses, point-to-point links, addresses that are usually statically assigned.

Any subnet that includes end devices needs to stay on a /64 or lower prefix. This would be computers, tablets, smart phones, servers, printers, anything that might be on a subnet that connects to the internet.

According to RFC 5375, a /64 prefix is required to support a number of benefits offered by IPv6; such as:

- Stateless Address Autoconfiguration
- Neighbor Discovery (ND)
- Secure Neighborhood Discovery (SEND)
- privacy extensions
- parts of Mobile IPv6
- PIM-SM with Embedded-RP
- SHIM6 [SHIM6]

Plus a number of other features currently in development, or being proposed, which will rely on a /64 prefix.

The bottom line is that IPv6 was designed to waste an unbelievable amount of addresses and it's OK. So while it is possible to subnet beyond the /64 subnet prefix it is not recommended.

Define the Following IPv6 Terms:

(Use the definitions/explanations from this workbook)

1. Global Routing Prefix -

This is assigned by the ISP to a customer or site. The Global Routing Prefix is determined by the prefix-length notation. (example /48 or /64). This is similar to the network portion of an IPv4 address.

2. Subnet ID -

This is similar to the subnet portion of an IPv4 address. The difference is in IPv4 the subnet is borrowed from the host portion of the address. In IPv6 the Subnet ID is a separate field (/48 to /64) and not necessarily part of the Interface ID.

3. Interface ID -

The Interface ID uniquely identifies a interface on the local subnet.

4. Subnet Prefix -

The Subnet Prefix is the address space used by the Global Routing Prefix and the Subnet ID, and can range from 0 to 128. The preferred Subnet Prefix length is /48 to /64 for customers or sites.

5. Nibble -

Each hexadecimal number in an IPv6 address represents 4 bits or a Nibble. An IPv6 address is composed of 32 hexadecimal numbers or 32 Nibbles.

6. Unspecified address -

Unspecified address is an all 0 address and cannot be assigned to an interface. It would be typed as ::. This is only used as a source address to indicate the absence of an actual address.

7. Loopback address -

Loopback Address is all 0's except for the last bit, which is 1. It would be typed as ::1. It operates the same as the IPv4 127.0.0.1 loopback address.

8. Ipv4 Embedded address -

IPv4 Embedded addresses are IPv6 addresses with an IPv4 address embedded in the low-order 32 bits. They are used to transition networks from IPv4 to IPv6.

9. Global Unicast address -

Global Unicast addresses are used to uniquely identify a specific interface on a host and can be used as a public address on the internet.

10. Unique Local address -

Unique local Unicast addresses are roughly the same as IPv4 private addresses.

11. Multicast address -

Multicast addresses are used to send a single packet to multiple destinations simultaneously.

12. Unicast address -

Unicast addresses identify a unique interface on an IPv6 device. It is a one to one connection between a source and destination.

13. Anycast address -

Anycast addresses are described as a one-to-nearest or one-to-one-of-many packet delivery. For example, all routers in the same network will be assigned the same Anycast address. A packet sent to that address will only be delivered to the closest router with that address based on routing protocol metrics. Anycast addresses can be pulled from Global Unicast, Site-Local or Link-Local address ranges. The first address and the last 128 addresses in a /64 Global Unicast range are reserved as the Subnet-Router Anycast Address.

14. Site ID -

The Site ID is the first address in the subnet you have assigned to a specific location or user group.

15. Sub-Site ID -

Sub-Site ID's come into play if you subnet a location or user group into smaller subnets. The first address in each range becomes the Sub-Site ID.

16. Prefix Length -

The prefix length indicates the number of nibbles or bits used in the network or subnetwork portion of an IPv6 address.

What Type of IPv6 Address is this?

Base on the information supplied on the inside front cover of this workbook identify the following IPv6 addresses as either: Unspecified, Loopback, Embedded IPv4, Global Unicast, Unique local Unicast, Link-local Unicast, or Multicast.

1. 2000:ACAD:1234::/48 Global Unicast
2. 0000:0000:0000:0000:0000:0000:0000:0000 Unspecified
3. FE80:ACAD:1234::/48 Link-local Unicast
4. FDFF:8771:3321::/48 Unique local Unicast
5. FFCD:984:1::/48 Multicast
6. 3F98::/48 Global Unicast
7. ::1 Loopback
9. 3000::0001/64 Global Unicast
10. FEA1:8934:3021:8945:1234:ACAD:FE23:0001/64 Link-Local Unicast
11. 00AB:2307:4829::/56 IPv4 Embedded address
13. FF45:6543:ACAD::/60 Multicast
14. 2ABC:ACAD:AAAA:0000:0000:0000:0000:00001/64 Global Unicast
15. FC12:0000:ACAD:1234:5678:9101:1121:3141/48 Unique local Unicast
16. 2345:FE66:FECD:9999:2365::1/52 Global Unicast
17. :: Unspecified
18. 0000:0000:0000:0000:0000:0000:0000:0001 Loopback
19. FFF8:0000:00001::0023/64 Multicast
20. 0023:5935:F441::/48 IPv4 Embedded address
22. 2001:ABCD:1234:FFFF:ACAD::45/60 Global Unicast
23. 3211:FCAB:EEEE::/48 Global Unicast
24. FCCC:25:1::/48 Unique local Unicast

Reference Section

IANA IPv6 Global Unicast Address Allocations to the RIRs

Prefix	Designation	Date	Whois	Status
2001:0000::/23	IANA	7/1/1999	whois.iana.org	ALLOCATED
2001:0200::/23	APNIC	7/1/1999	whois.apnic.net	ALLOCATED
2001:0400::/23	ARIN	7/1/1999	whois.arin.net	ALLOCATED
2001:0600::/23	RIPE NCC	7/1/1999	whois.ripe.net	ALLOCATED
2001:0800::/23	RIPE NCC	5/2/2002	whois.ripe.net	ALLOCATED
2001:0a00::/23	RIPE NCC	11/2/2002	whois.ripe.net	ALLOCATED
2001:0c00::/23	APNIC	5/2/2002	whois.apnic.net	ALLOCATED
2001:0e00::/23	APNIC	1/1/2003	whois.apnic.net	ALLOCATED
2001:1200::/23	LACNIC	11/1/2002	whois.lacnic.net	ALLOCATED
2001:1400::/23	RIPE NCC	2/1/2003	whois.ripe.net	ALLOCATED
2001:1600::/23	RIPE NCC	7/1/2003	whois.ripe.net	ALLOCATED
2001:1800::/23	ARIN	4/1/2003	whois.arin.net	ALLOCATED
2001:1a00::/23	RIPE NCC	1/1/2004	whois.ripe.net	ALLOCATED
2001:1c00::/22	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:2000::/20	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:3000::/21	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:3800::/22	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:3c00::/22	IANA			RESERVED
2001:4000::/23	RIPE NCC	6/11/2004	whois.ripe.net	ALLOCATED
2001:4200::/23	AFRINIC	6/1/2004	whois.afrinic.net	ALLOCATED
2001:4400::/23	APNIC	6/11/2004	whois.apnic.net	ALLOCATED
2001:4600::/23	RIPE NCC	8/17/2004	whois.ripe.net	ALLOCATED
2001:4800::/23	ARIN	8/24/2004	whois.arin.net	ALLOCATED
2001:4a00::/23	RIPE NCC	10/15/2004	whois.ripe.net	ALLOCATED
2001:4c00::/23	RIPE NCC	12/17/2004	whois.ripe.net	ALLOCATED
2001:5000::/20	RIPE NCC	9/10/2004	whois.ripe.net	ALLOCATED
2001:8000::/19	APNIC	11/30/2004	whois.apnic.net	ALLOCATED
2001:a000::/20	APNIC	11/30/2004	whois.apnic.net	ALLOCATED
2001:b000::/20	APNIC	3/8/2006	whois.apnic.net	ALLOCATED
2002:0000::/16	6to4	2/1/2001		ALLOCATED
2003:0000::/18	RIPE NCC	1/12/2005	whois.ripe.net	ALLOCATED
2400:0000::/12	APNIC	10/3/2006	whois.apnic.net	ALLOCATED
2600:0000::/12	ARIN	10/3/2006	whois.arin.net	ALLOCATED
2610:0000::/23	ARIN	11/17/2005	whois.arin.net	ALLOCATED
2620:0000::/23	ARIN	9/12/2006	whois.arin.net	ALLOCATED
2800:0000::/12	LACNIC	10/3/2006	whois.lacnic.net	ALLOCATED
2a00:0000::/12	RIPE NCC	10/3/2006	whois.ripe.net	ALLOCATED
2c00:0000::/12	AFRINIC	10/3/2006	whois.afrinic.net	ALLOCATED
2d00:0000::/8	IANA	7/1/1999		RESERVED
2e00:0000::/7	IANA	7/1/1999		RESERVED
3000:0000::/4	IANA	7/1/1999		RESERVED
3ffe::/16	IANA	2008-04		RESERVED
5f00::/8	IANA	2008-04		RESERVED

Prefix-Length	Number of IPs		
/128	1	/68	1,152,921,504,606,840,000
/127	2	/67	2,305,843,009,213,690,000
/126	4	/66	4,611,686,018,427,380,000
/125	8	/65	9,223,372,036,854,770,000
/124	16	/64 - Residential	18,446,744,073,709,500,000
/123	32	/63	36,893,488,147,419,100,000
/122	64	/62	73,786,976,294,838,200,000
/121	128	/61	147,573,952,589,676,000,000
/120	256	/60 - Residential	295,147,905,179,352,000,000
/119	512	/59	590,295,810,358,705,000,000
/118	1,024	/58	1,180,591,620,717,410,000,000
/117	2,048	/57	2,361,183,241,434,820,000,000
/116	4,096	/56 - Medium	4,722,366,482,869,640,000,000
/115	8,192	/55	9,444,732,965,739,290,000,000
/114	16,384	/54	18,889,465,931,478,500,000,000
/113	32,768	/53	37,778,931,862,957,100,000,000
/112	65,536	/52	75,557,863,725,914,300,000,000
/111	131,072	/51	151,115,727,451,828,000,000,000
/110	262,144	/50	302,231,454,903,657,000,000,000
/109	524,288	/49	604,462,909,807,314,000,000,000
/108	1,048,576	/48 - Large	1,208,925,819,614,620,000,000,000
/107	2,097,152	/47	2,417,851,639,229,250,000,000,000
/106	4,194,304	/46	4,835,703,278,458,510,000,000,000
/105	8,388,608	/45	9,671,406,556,917,030,000,000,000
/104	16,777,216	/44	19,342,813,113,834,000,000,000,000
/103	33,554,432	/43	38,685,626,227,668,100,000,000,000
/102	67,108,864	/42	77,371,252,455,336,200,000,000,000
/101	134,217,728	/41	154,742,504,910,672,000,000,000,000
/100	268,435,456	/40	309,485,009,821,345,000,000,000,000
/99	536,870,912	/39	618,970,019,642,690,000,000,000,000
/98	1,073,741,824	/38	1,237,940,039,285,380,000,000,000,000
/97	2,147,483,648	/37	2,475,880,078,570,760,000,000,000,000
/96	4,294,967,296	/36	4,951,760,157,141,520,000,000,000,000
/95	8,589,934,592	/35	9,903,520,314,283,040,000,000,000,000
/94	17,179,869,184	/34	19,807,040,628,566,000,000,000,000,000
/93	34,359,738,368	/33	39,614,081,257,132,100,000,000,000,000
/92	68,719,476,736	/32 - Service LIR	79,228,162,514,264,300,000,000,000,000
/91	137,438,953,472	/31	158,456,325,028,528,000,000,000,000,000
/90	274,877,906,944	/30	316,912,650,057,057,000,000,000,000,000
/89	549,755,813,888	/29	633,825,300,114,114,000,000,000,000,000
/88	1,099,511,627,776	/28	1,267,650,600,228,220,000,000,000,000,000
/87	2,199,023,255,552	/27	2,535,301,200,456,450,000,000,000,000,000
/86	4,398,046,511,104	/26	5,070,602,400,912,910,000,000,000,000,000
/85	8,796,093,022,208	/25	10,141,204,801,825,800,000,000,000,000,000
/84	17,592,186,044,416	/24	20,282,409,603,651,600,000,000,000,000,000
/83	35,184,372,088,832	/23 - ISP	40,564,819,207,303,300,000,000,000,000,000
/82	70,368,744,177,664	/22	81,129,638,414,606,600,000,000,000,000,000
/81	140,737,488,355,328	/21	162,259,276,829,213,000,000,000,000,000,000
/80	281,474,976,710,656	/20	324,518,553,658,426,000,000,000,000,000,000
/79	562,949,953,421,312	/19	649,037,107,316,853,000,000,000,000,000,000
/78	1,125,899,906,842,620	/18	1,298,074,214,633,700,000,000,000,000,000,000
/77	2,251,799,813,685,240	/17	2,596,148,429,267,410,000,000,000,000,000,000
/76	4,503,599,627,370,490	/16	5,192,296,858,534,820,000,000,000,000,000,000
/75	9,007,199,254,740,990	/15	10,384,593,717,069,600,000,000,000,000,000,000
/74	18,014,398,509,481,900	/14	20,769,187,434,139,300,000,000,000,000,000,000
/73	36,028,797,018,963,900	/13	41,538,374,868,278,600,000,000,000,000,000,000
/72	72,057,594,037,927,900	/12	83,076,749,736,557,200,000,000,000,000,000,000
/71	144,115,188,075,855,000	/11	166,153,499,473,114,000,000,000,000,000,000,000
/70	288,230,376,151,711,000	/10	332,306,998,946,228,000,000,000,000,000,000,000
/69	576,460,752,303,423,000	/9	664,613,997,892,457,000,000,000,000,000,000,000
		/8	1,329,227,995,784,910,000,000,000,000,000,000,000

IPv6 Resources

Web Sites:

ARIN IPv6 Wiki

<http://www.getipv6.info/display/IPv6/IPv6+Info+Home>

Cisco Support Community IPv6 Subnetting - Overview and Case Study

<https://supportforums.cisco.com/docs/DOC-17232>

Videos:

IPv6 for CCNAs with Anthony Sequeira

Video Series by the Cisco Learning Network - Parts 1, 2 & 3

<https://learningnetwork.cisco.com/docs/DOC-20357>

PDF Resources:

Preparing An IPv6 Address Plan, Version 2, 18 September 2013

https://www.ripe.net/lir-services/training/material/IPv6-for-LIRs-Training-Course/IPv6_addr_plan4.pdf

Best Current Operational Practices - IPv6 Subnetting (v1)

http://www.ipbcop.org/wp-content/uploads/2012/02/BCOP-IPv6_Subnetting.pdf

6net An IPv6 Deployment Guide by The European 6NET Consortium

<http://www.6net.org/book/deployment-guide.pdf>

IPv6 Addressing At-A-Glance By Cisco

http://www.cisco.com/en/US/technologies/tk648/tk872/technologies_white_paper0900aecd8026003d.pdf

IPv6 Implementation Guide, Cisco IOS Release 15.2M&T

<http://www.cisco.com/en/US/docs/ios-xml/ios/ipv6/configuration/15-2mt/ipv6-15-2mt-book.pdf>

Printed Books:

IPv6 Fundamentals A Straightforward Approach to Understanding IPv6

By Rick Graziani

ISBN-13: 978-1-58714-313-7

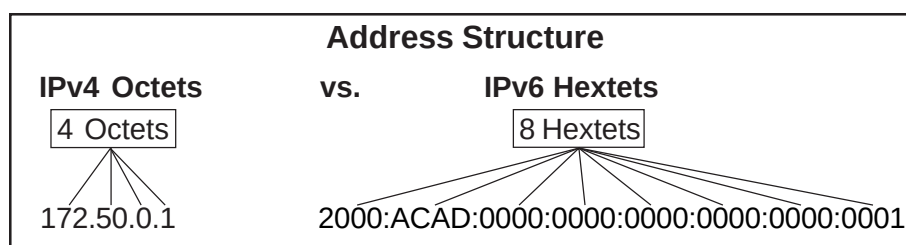
Understanding IPv6 Third Edition

By Joseph Davies

ISBN: 978-0-7356-5914-8

Available subnets within the Subnet ID using the Nibble boundary.

/48 No Nibbles	/52 1 Nibble	/56 2 Nibbles	/60 3 Nibbles	/64 4 Nibbles
/48 - 1 Subnet				
/52 - 16 Subnets	/52 - 1 Subnet			
/56 - 256 Subnets	/56 - 16 Subnets	/56 - 1 Subnets		
/60 - 4096 Subnets	/60 - 256 Subnets	/60 - 16 Subnets	/60 - 1 Subnets	
/64 - 65,536 Subnets	/64 - 4096 Subnets	/64 - 256 Subnets	/64 - 16 Subnets	/64 - 1 Subnet



What is a Site?

A Site = one building
A home, apartment, or house = a Site
A campus with 10 buildings = 10 Sites
A single building with 15 businesses = 15 Sites

Conversion Chart

Decimal (Base 10)	Hexadecimal (Base 16)	Binary (Base 2)
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

VLSM

Variable-Length Subnet Mask

IPv4

Workbook

Version 2.3

Instructor's Edition

192.168.

192.168.10.96

192.168.10.126

172.31.15.0

10.250.1.0

IPv4 Address Classes

Class A	1 – 127	Leading bit pattern	0	00000000.00000000.00000000.00000000 Network . Host . Host . Host
Class B	128 – 191	Leading bit pattern	10	10000000.00000000.00000000.00000000 Network . Network . Host . Host
Class C	192 – 223	Leading bit pattern	110	11000000.00000000.00000000.00000000 Network . Network . Network . Host
Class D	224 – 239	(Reserved for multicast)		
Class E	240 – 255	(Reserved for experimental, used for research)		

Speciality Address Ranges

Loopback -	Only the single 127.0.0.1 address is used, addresses 127.0.0.0 to 127.255.255.255 are reserved. Any address within this block will loop back to the local host.
Link-Local Addresses -	IPv4 addresses in the address block 169.254.0.0 to 169.254.255.255 (169.254.0.0/16) are designated as link-local addresses.
TEST-NET Addresses -	The address block 192.0.2.0 to 192.0.2.255 (192.0.2.0/24) is set aside for teaching and learning purposes.
Experimental Addresses -	The addresses in the block 240.0.0.0 to 255.255.255.254 are listed as reserved for future use (RFC 3330).

Private Address Space

Class A	10.0.0.0 to 10.255.255.255
Class B	172.16.0.0 to 172.31.255.255
Class C	192.168.0.0 to 192.168.255.255

Default Subnet Masks

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

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Special Thanks to Melvin Baker and Jim Dorsch for taking the time to check this workbook for errors,
and to everyone who has sent in suggestions to improve the series.

Classful vs. Classless Subnetting

When you're subnetting an IP address for a network you have two options: classful and classless. Classful subnetting is the simplest method. It tends to be the most wasteful because it uses more addresses than are necessary. In classful subnetting you use the same subnet mask for each subnet, and all the subnets have the same number of addresses in them.

Classless addressing allows you to use different subnet masks and create subnets tailored to the number of users in each group. This technique is referred to as VLSM, Variable Length Subnet Masks.

What is VLSM

Variable Length Subnet Masks allow you a much tighter control over your addressing scheme. If you use a class C address with a default subnet mask you end up with one subnet containing 256 addresses. By using VLSM you can adjust the number of subnets and number of addresses depending on the specific needs of your network. The same rules apply to a class A or B addresses.

VLSM is supported by the following protocols: RIP version 2, OSPF, EIGRP, Dual IS-IS, and BGP. You need to configure your router for Variable Length Subnet Masks by setting up one of these protocols. Then configure the subnet masks of the various interfaces in the IP address interface sub-command.

Benefits of VLSM

- Allows efficient use of address space
- Allows the use of multiple subnet mask lengths
- Breaks up an address block into smaller custom blocks
- Allows for route summarization
- Provides more flexibility in network design
- Supports hierarchical enterprise networks

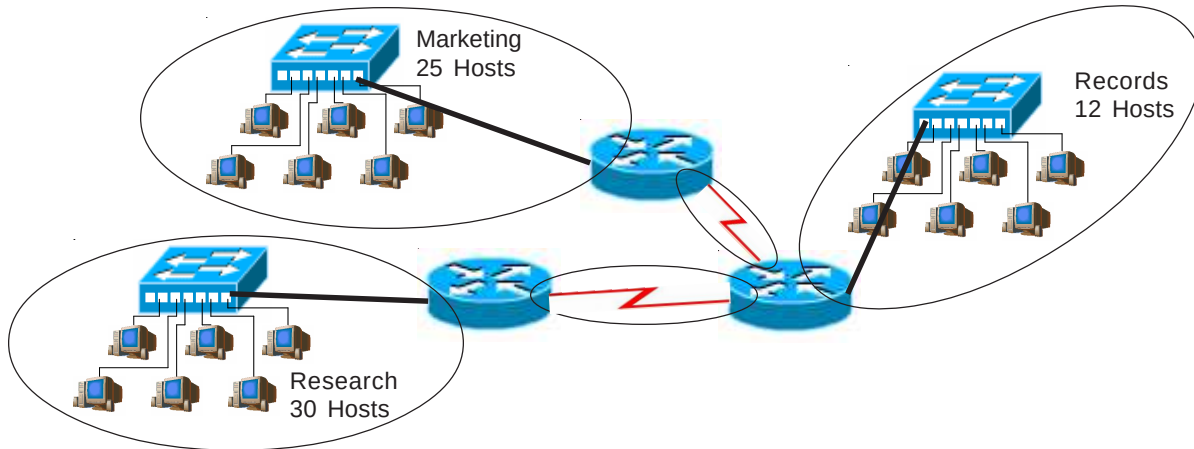
This workbook explores three different methods to figure out sub-subnets: the box method, the circle method, and a VLSM chart.

Classful Subnetting Example

When you're subnetting an IP address for a network you have two options: classful and classless. Classful subnetting is the simplest method. It also tends to be the most wasteful because it uses more addresses than are necessary. In classful subnetting you use the same subnet mask for each subnet, and all the subnets have the same number of addresses in them.

In this example you need five subnets, each one containing 30 hosts. The serial connections only require two address each so you are wasting 28 usable addresses in each of the serial subnet ranges.

IP Address: 192.168.1.0



The Box Method for visualizing subnets

Classful Subnet Ranges

```

192.168.1.0      to 192.168.1.31  /27
192.168.1.32    to 192.168.1.63  /27
192.168.1.64    to 192.168.1.95  /27
192.168.1.96    to 192.168.1.127 /27
192.168.1.128   to 192.168.1.159 /27
192.168.1.160   to 192.168.1.191 /27
192.168.1.192   to 192.168.1.223 /27
192.168.1.224   to 192.168.1.255 /27

                                     /27
                                     255.255.255.224
                                     32 Hosts
                                     8 Subnets

```

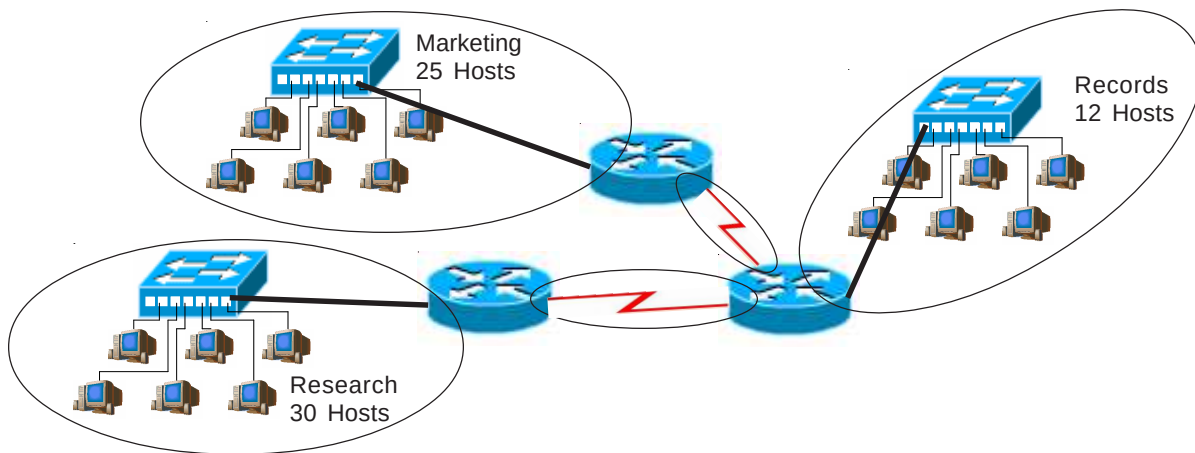
0 31	32 63	128 159	160 191
64 95	96 127	192 223	224 255

Classless Subnetting Example

Classless addressing allows you to use different subnet masks and create subnets tailored to the number of users in each subnetwork. There are fewer wasted IP addresses using smaller subnets.

In this example you need at total of five subnets, two containing 30 hosts, one containing 12 hosts, and two serial connections that only require two usable addresses each.

IP Address: 192.168.1.0

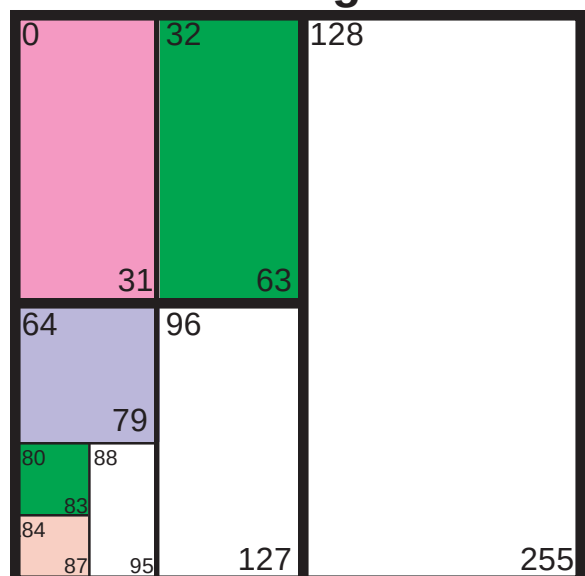


By adjusting the subnet masks you can cut your address usage by almost half in this example. This type of subnetting requires a network protocol which will support it such as: RIP version 2, EIGRP, OSPF, or BGP.

The Box Method for visualizing subnets

Classless Subnet Ranges

192.168.1.0	to	192.168.1.31	/27
192.168.1.32	to	192.168.1.63	/27
192.168.1.64	to	192.168.1.79	/28
192.168.1.80	to	192.168.1.87	/30
192.168.1.88	to	192.168.1.95	/29
192.168.1.96	to	192.168.1.127	/27
192.168.1.128	to	192.168.1.255	/25

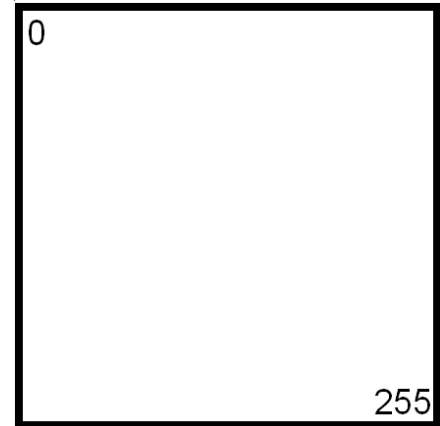


Visualizing Subnets Using The Box Method

The box method is a simple way to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

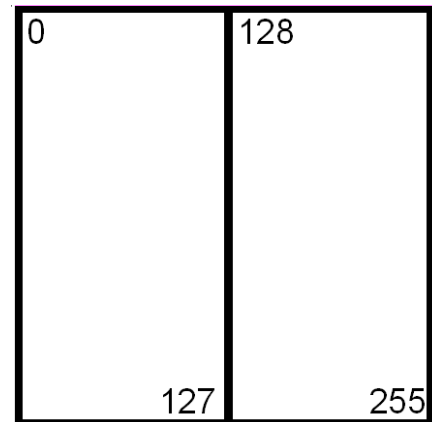
Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



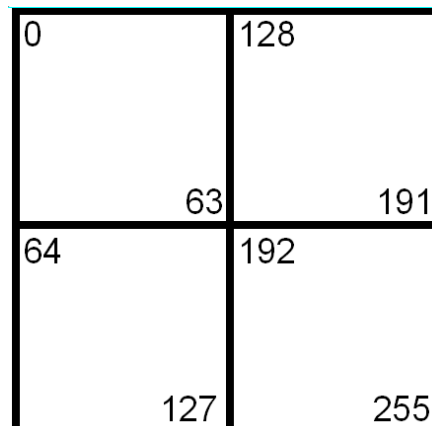
Split the box in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



Divide the box into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets

0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets

0	32	128	160
15	47	143	175
16	48	144	176
31	63	159	191
64	96	192	224
79	111	207	239
80	112	208	240
95	127	223	255

The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets

0	8	32	40	128	136	160	168
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
87	95	119	127	215	223	247	255

The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets

0	8	32	40	128	136	160	168
3	11	35	43	131	139	163	171
4	12	36	44	132	140	164	172
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
19	27	51	59	147	155	179	187
20	28	52	60	148	156	180	188
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
67	75	99	107	195	203	227	235
68	76	100	108	196	204	228	236
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
83	91	115	123	211	219	243	251
84	92	116	124	212	220	244	252
87	95	119	127	215	223	247	255

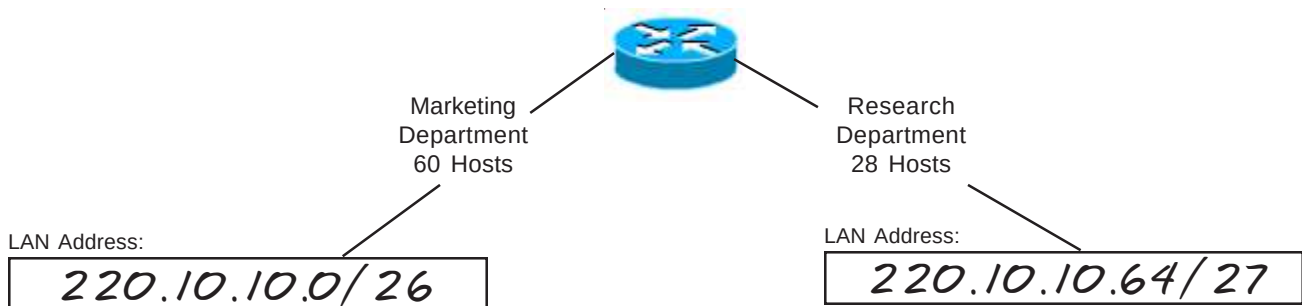
VLSM Addressing

Box Method

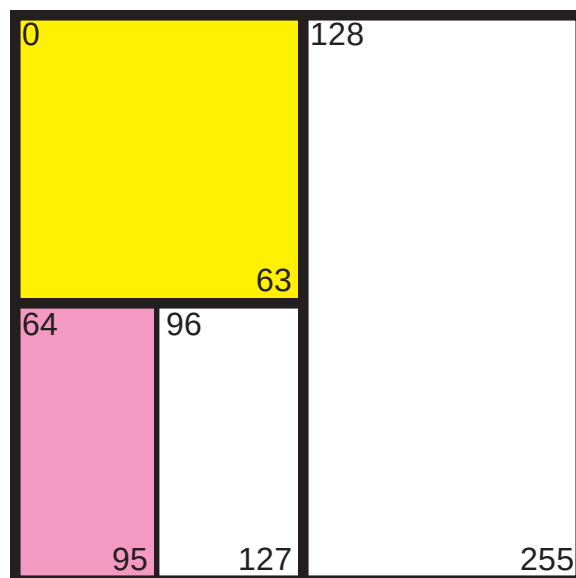
(Sample)

Problem 1

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This business will be using the class C address 220.10.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.



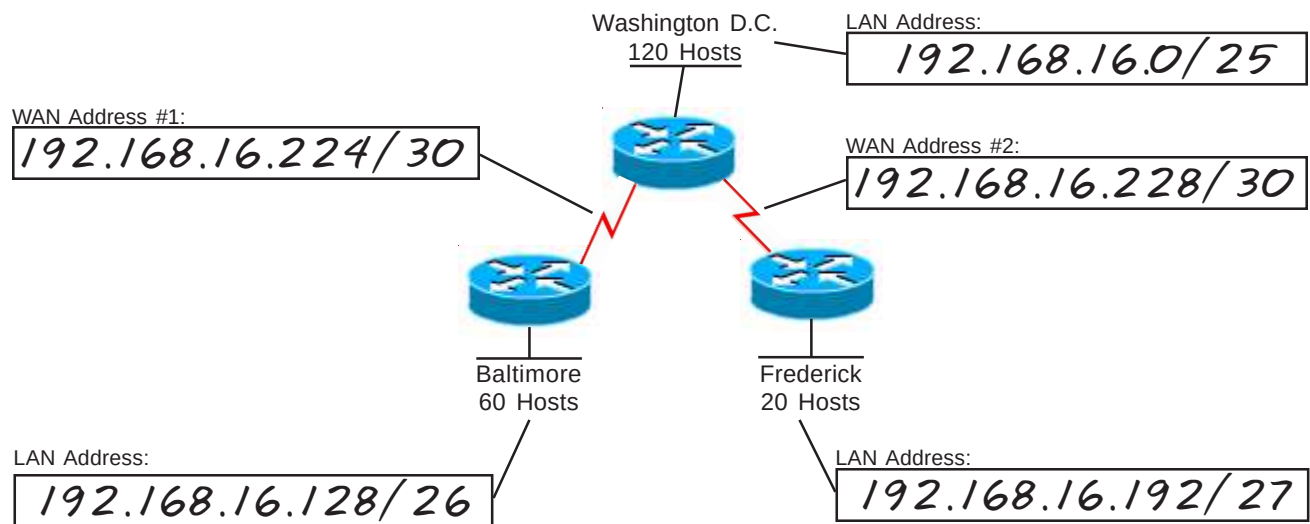
VLSM Addressing

Box Method

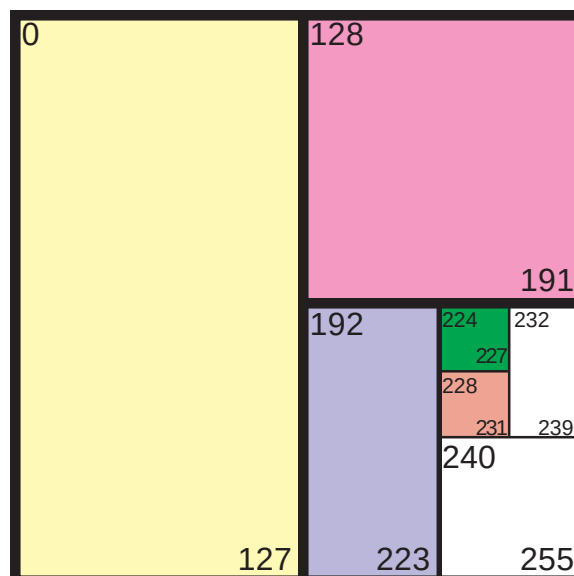
(Sample)

Problem 2

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.16.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

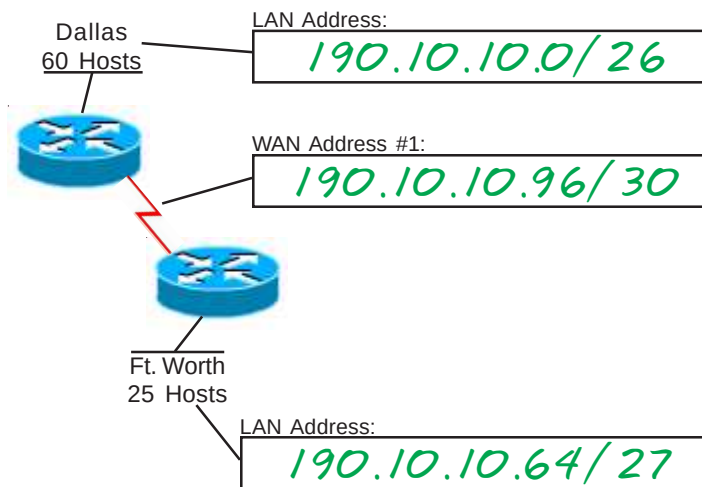


VLSM Addressing

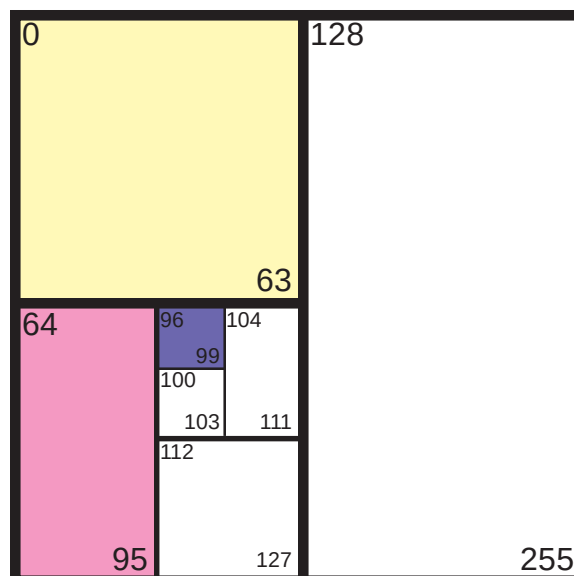
Box Method

Problem 3

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 190.10.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

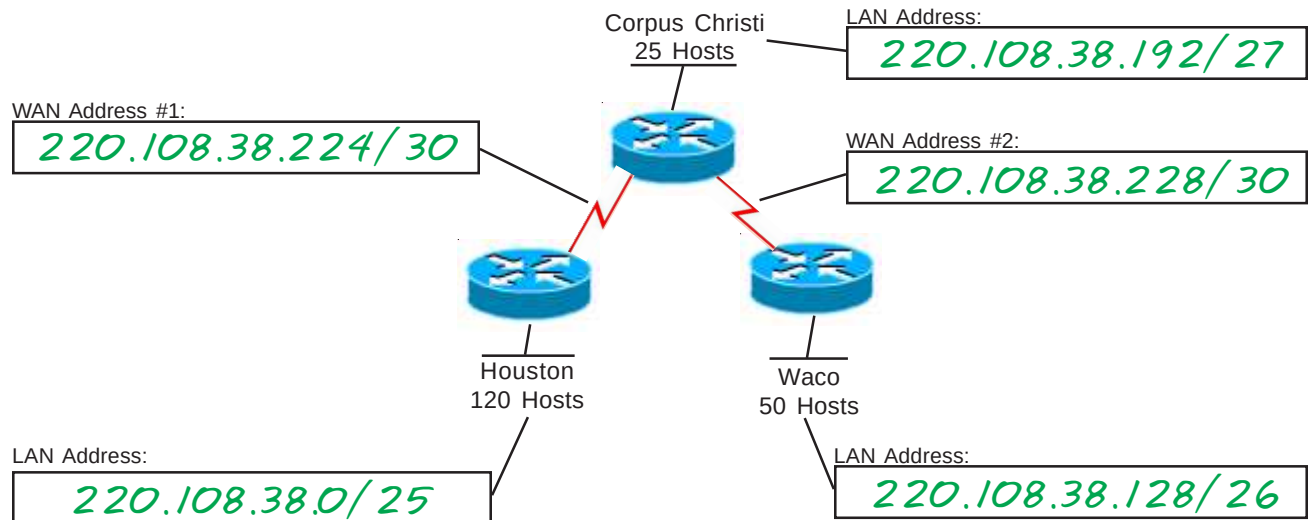


VLSM Addressing

Box Method

Problem 4

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 220.108.38.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

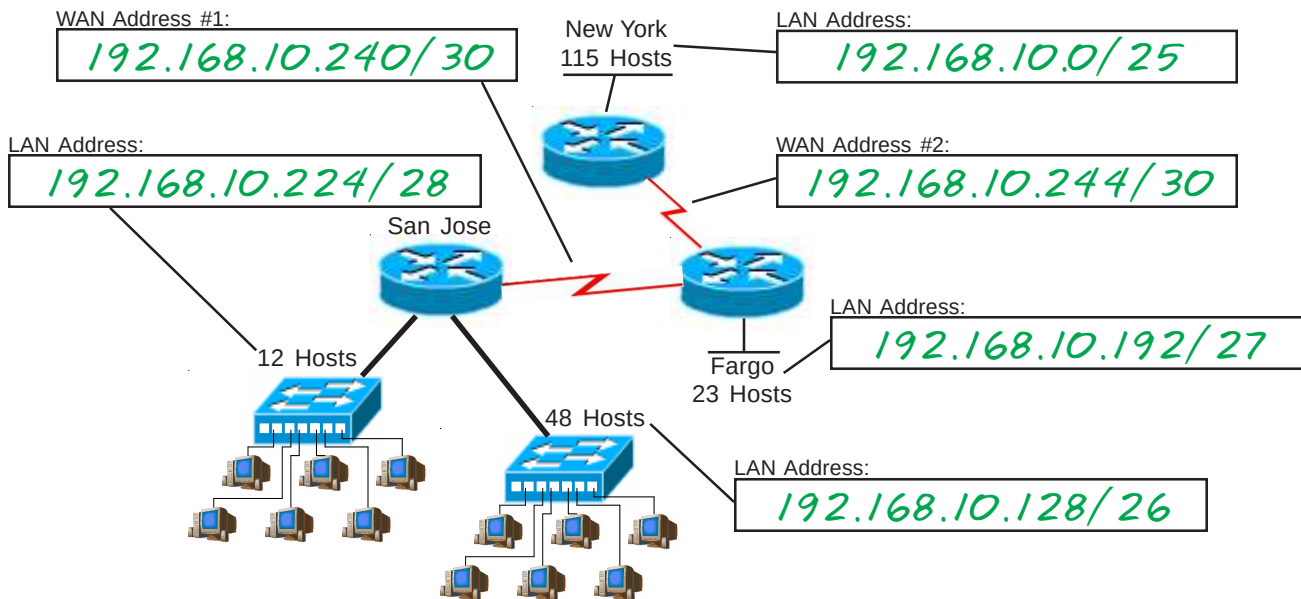
0	8	32	40	128	136	160	168
	3	11	35	43	131	139	163
4	12	36	44	132	140	164	172
	7	15	39	47	135	143	167
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247

VLSM Addressing

Box Method

Problem 5

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

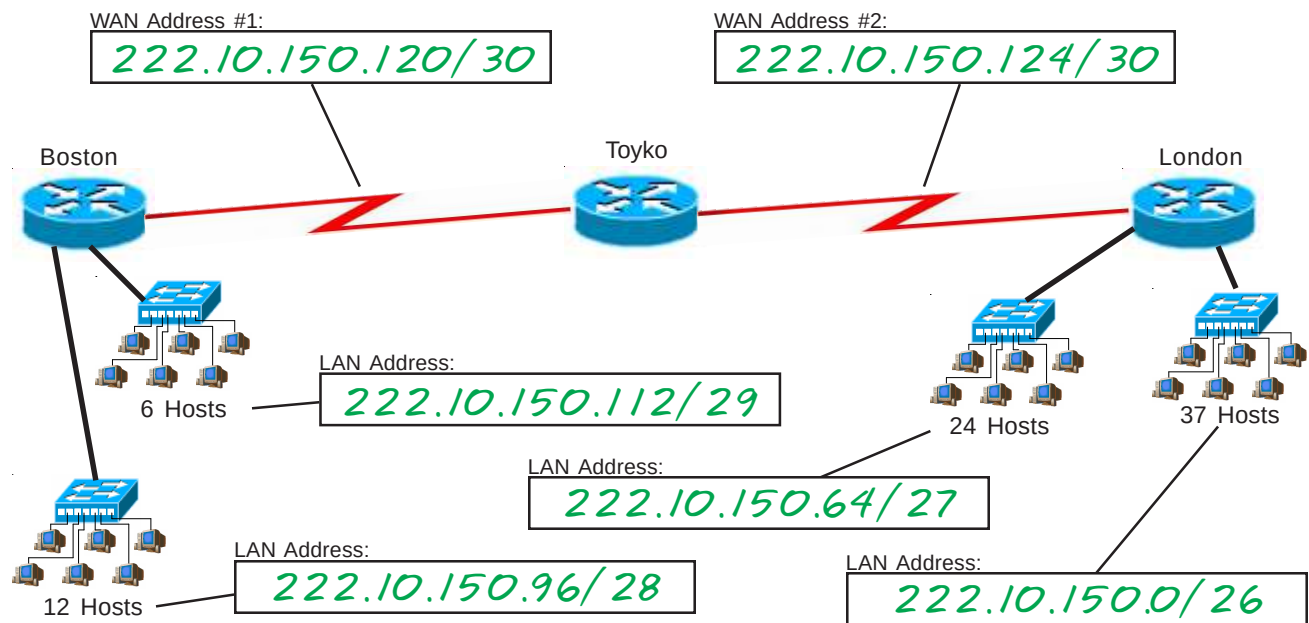
0	8	32	40	128	136	160	168
	3	11	35	131	139	163	171
4	12	36	44	132	140	164	172
	7	15	39	135	143	167	175
16	24	48	56	144	152	176	184
	19	27	51	147	155	179	187
20	28	52	60	148	156	180	188
	23	31	55	151	159	183	191
64	72	96	104	192	200	224	232
	67	75	99	195	203	227	235
68	76	100	108	196	204	228	236
	71	79	103	199	207	231	239
80	88	112	120	208	216	240	248
	83	91	115	211	219	243	251
84	92	116	124	212	220	244	252
	87	95	119	215	223	247	255

VLSM Addressing

Box Method

Problem 6

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 222.10.150.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

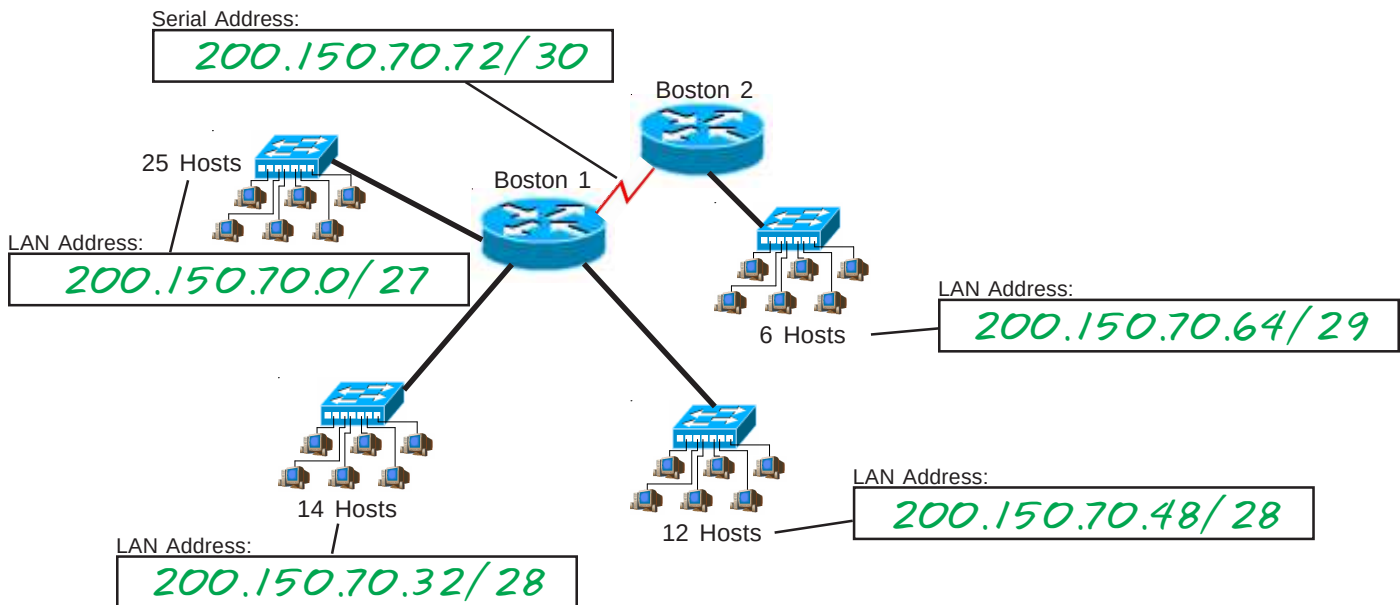
0	32	128	160
31	63	159	191
64	96	192	224
95	111	223	255
	112		
	120		
	123		
	124		
	119		
	127		

VLSM Addressing

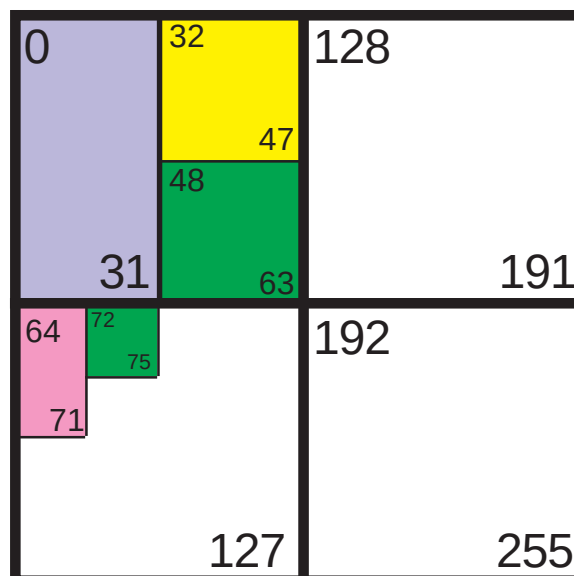
Box Method

Problem 7

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and subnet mask in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 200.150.70.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

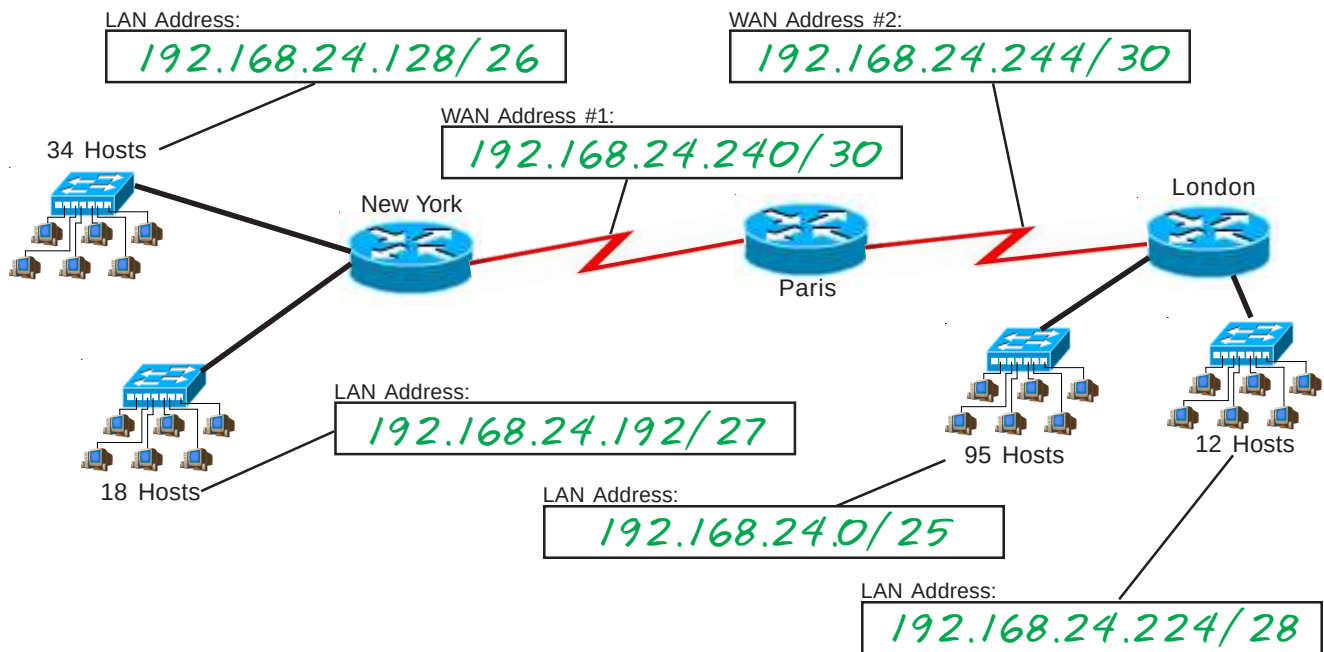


VLSM Addressing

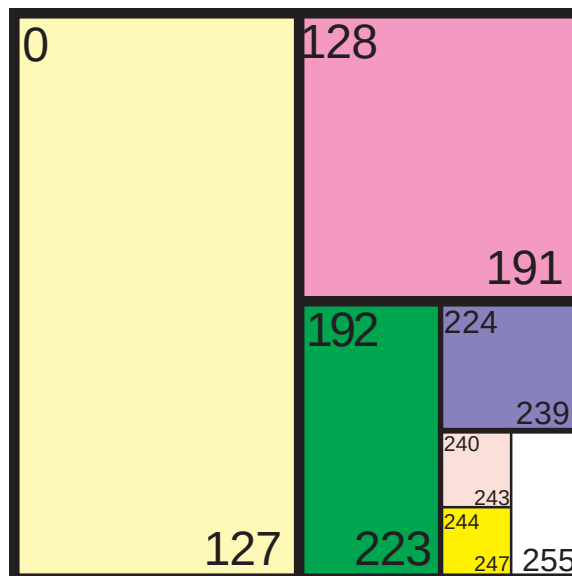
Box Method

Problem 8

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and subnet mask in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.24.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.



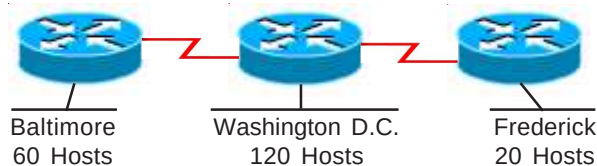
Configuring Subnets Using The Numeric Method

The Numeric method extends the technique from the IP Addressing and Subnetting workbook to allow for classless subnetting. Start by listing the needed hosts per subnets from largest to smallest.

Step 1

In this sample the needed host groups are:

- 120 Washington D. C.
- 60 Baltimore
- 20 Frederick
- 2 Serial #1
- 2 Serial #2



Step 2

Lay out the IP address the same way you would in the IP Addressing and Subnetting workbook showing the binary values of the bits, the number of subnets, and the number of hosts.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
192.10.10.0	0	0	0	0	0	0	0	0	0	

Step 3

Draw a line allowing for 120 hosts. Using the value of the last bit borrowed or magic number, write in your first range. Then add the subnet address to start the second range.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
192.10.10.0	0	0	0	0	0	0	0	0	0	

	Address Ranges		
Hosts	192.168.16.0	to	192.168.16.127
120	192.168.16.128	to	
			CIDR /25

Draw a second line for the 60 hosts. Use the value of the last bit borrowed or magic number to lay out the second largest subnet. Add the subnet address which will begin the third range.

Step 5

Continue the same process until all the subnets are accounted for. The CIDR is the total number of bits borrowed for the network and subnetwork portion of the address.

15

Numeric Method

Problem 11

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 200.20.20.0. Start with your largest groups first.

LAN Address:

200.20.20.0/25

Marketing
Department
100 Hosts\



Marketing
Department
50 Hosts

Department
25 Hosts

LAN Address:

200.20.20.128/26

LAN Address:

200.20.20.192/27

Show your work for **Problem 11** in the space below.

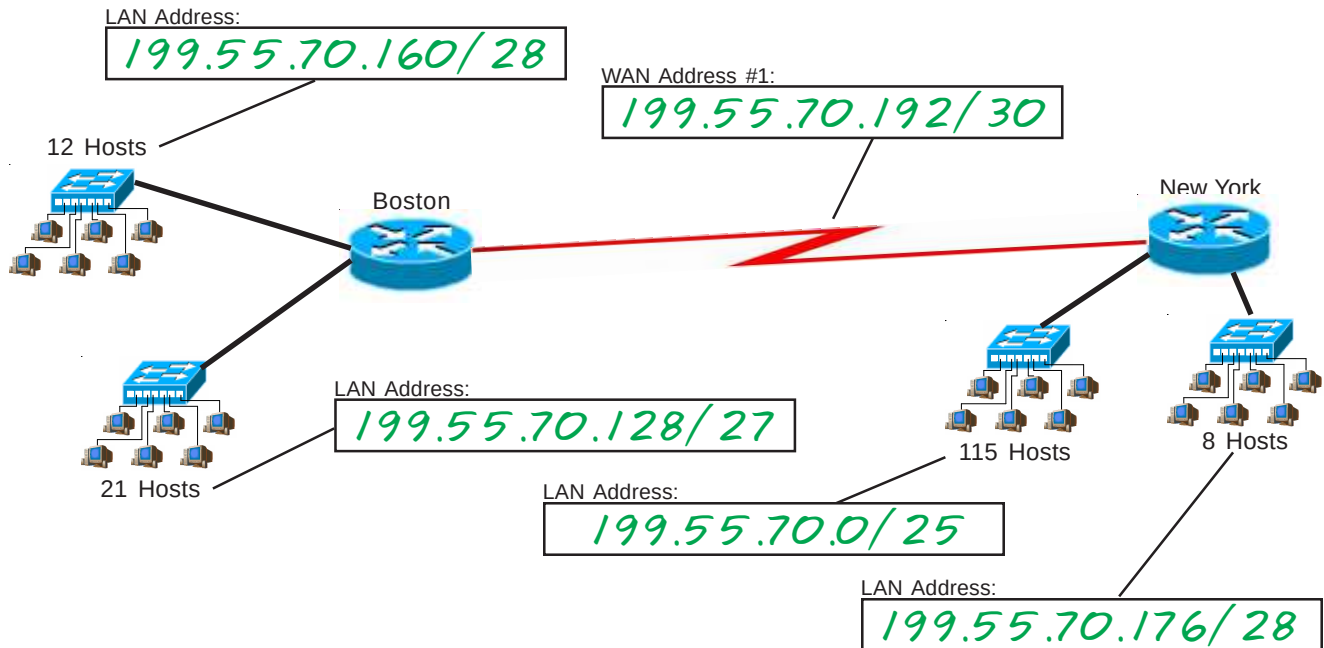
[illegible]

<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>
100	200.20.20.0 to 200.20.20.127	/25
50	200.20.20.128 to 200.20.20.191	/26
25	200.20.20.192 to 200.20.20.223	/27

Numeric Method

Problem 12

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 199.55.70.0. Start with your largest groups first.



Show your work for Problem 12 in the space below.

	115 Hosts		21 Hosts		12 Hosts		2 Hosts			
Number of Subnets	256	128	64	32	16	8	4	2	-	Number of Hosts
	2	4	8	16	32	64	128	256		
	128	64	32	16	8	4	2	1	-	Binary values
199.55.70.0	0	0	0	0	0	0	0	0		

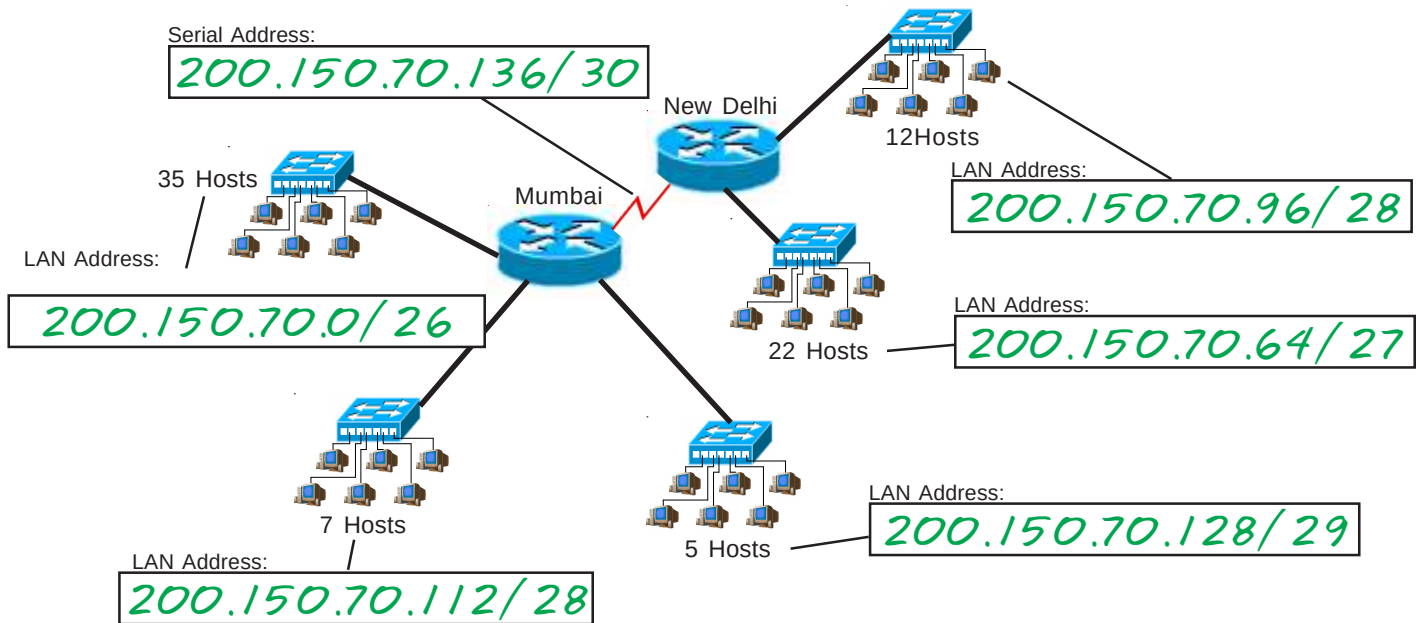
Hosts	Address Ranges		CIDR
115	199.55.70.0	to 199.55.70.127	/25
21	199.55.70.128	to 199.55.70.159	/27
12	199.55.70.160	to 199.55.70.175	/28
8	199.55.70.176	to 199.55.70.191	/28
2	199.55.70.192	to 199.55.70.195	/30

VLSM Addressing

Numeric Method

Problem 13

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 200.150.70.0. Start with your largest groups first.



Show your work for Problem 13 in the space below.

			22	7	5					
			35	12	12	2				
			Hosts	Hosts	Hosts	Hosts				
Number of	256	128	64	32	16	8	4	2	-	Number of
Subnets	-	2	4	8	16	32	64	128	256	Hosts
	128	64	32	16	8	4	2	1	-	Binary values
200. 150 . 70 . 0	0	0	0	0	0	0	0	0	0	

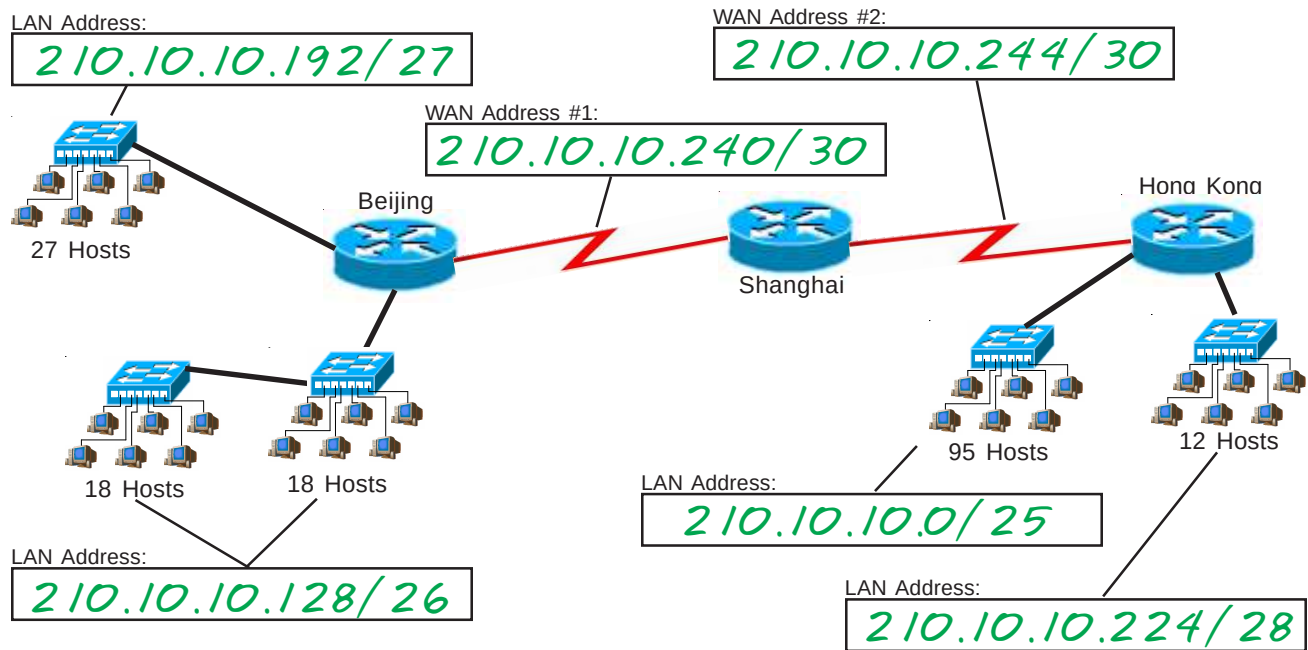
Hosts	Address Ranges		CIDR
35	200.150.70.0	to 200.150.70.63	/26
22	200.150.70.64	to 200.150.70.95	/27
12	200.150.70.96	to 200.150.70.111	/28
7	200.150.70.112	to 200.150.70.127	/28
5	200.150.70.128	to 200.150.70.135	/29
2	200.150.70.136	to 200.150.70.139	/30

VLSM Addressing

Numeric Method

Problem 14

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 210.10.10.0. Start with your largest groups first.



Show your work for **Problem 14** in the space below.

	95	36	27	12	2	2			
	Hosts	Hosts	Hosts	Hosts	Hosts	Hosts			
Number of Subnets	256	128	64	32	16	8	4	2	Number of Hosts
	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	Binary values
	210	10	10	0	0	0	0	0	0

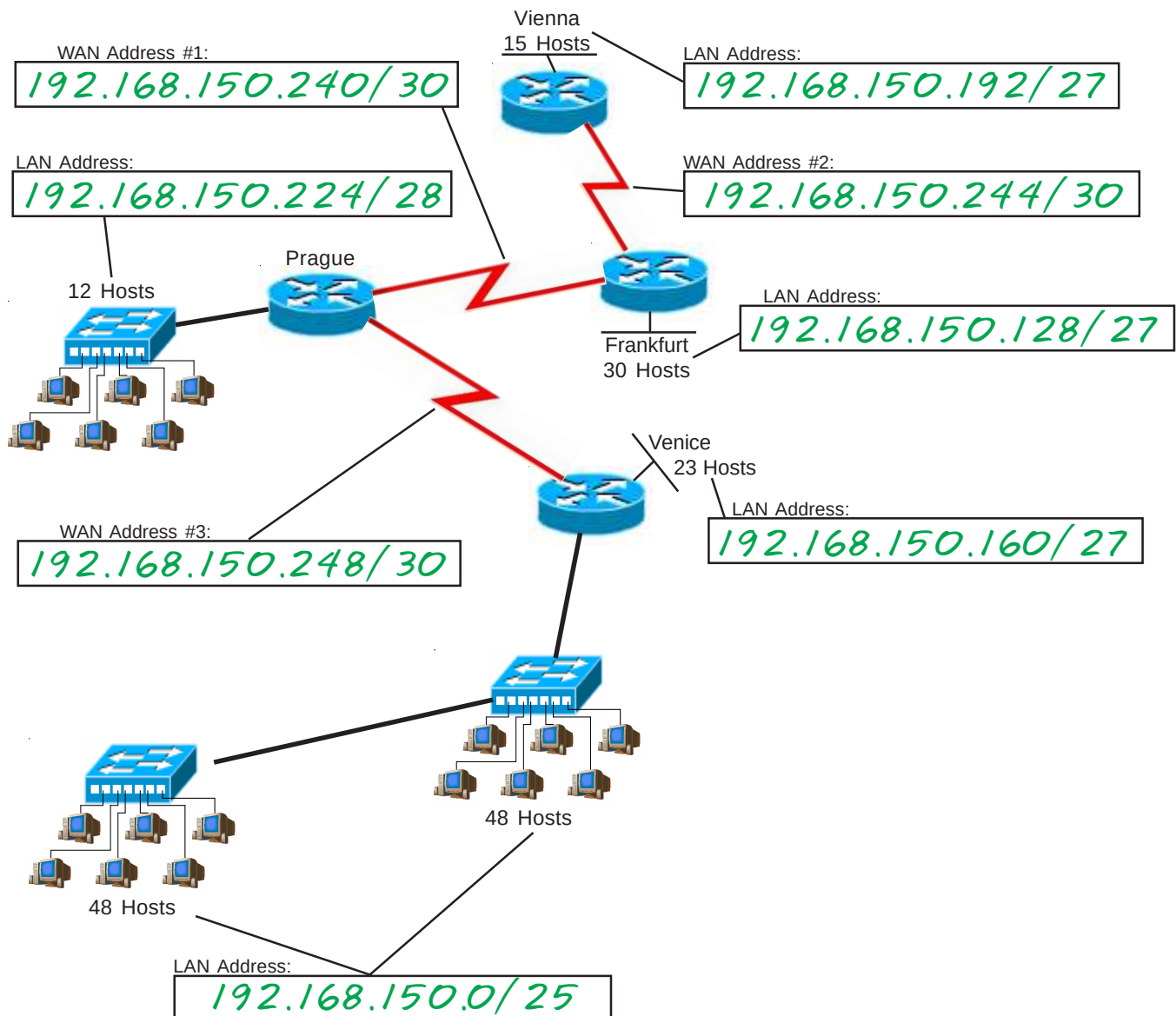
Hosts	Address Ranges	CIDR
95	210.10.10.0 to 210.10.10.127	/25
36	210.10.10.128 to 210.10.10.191	/26
27	210.10.10.192 to 210.10.10.223	/27
12	210.10.10.224 to 210.10.10.239	/28
2	210.10.10.240 to 210.10.10.243	/30
2	210.10.10.244 to 210.10.10.247	/30

VLSM Addressing

Numeric Method

Problem 15

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 192.168.150.0. Start with your largest groups first.



Show your work for Problem 15 in the space below.

[illegible]

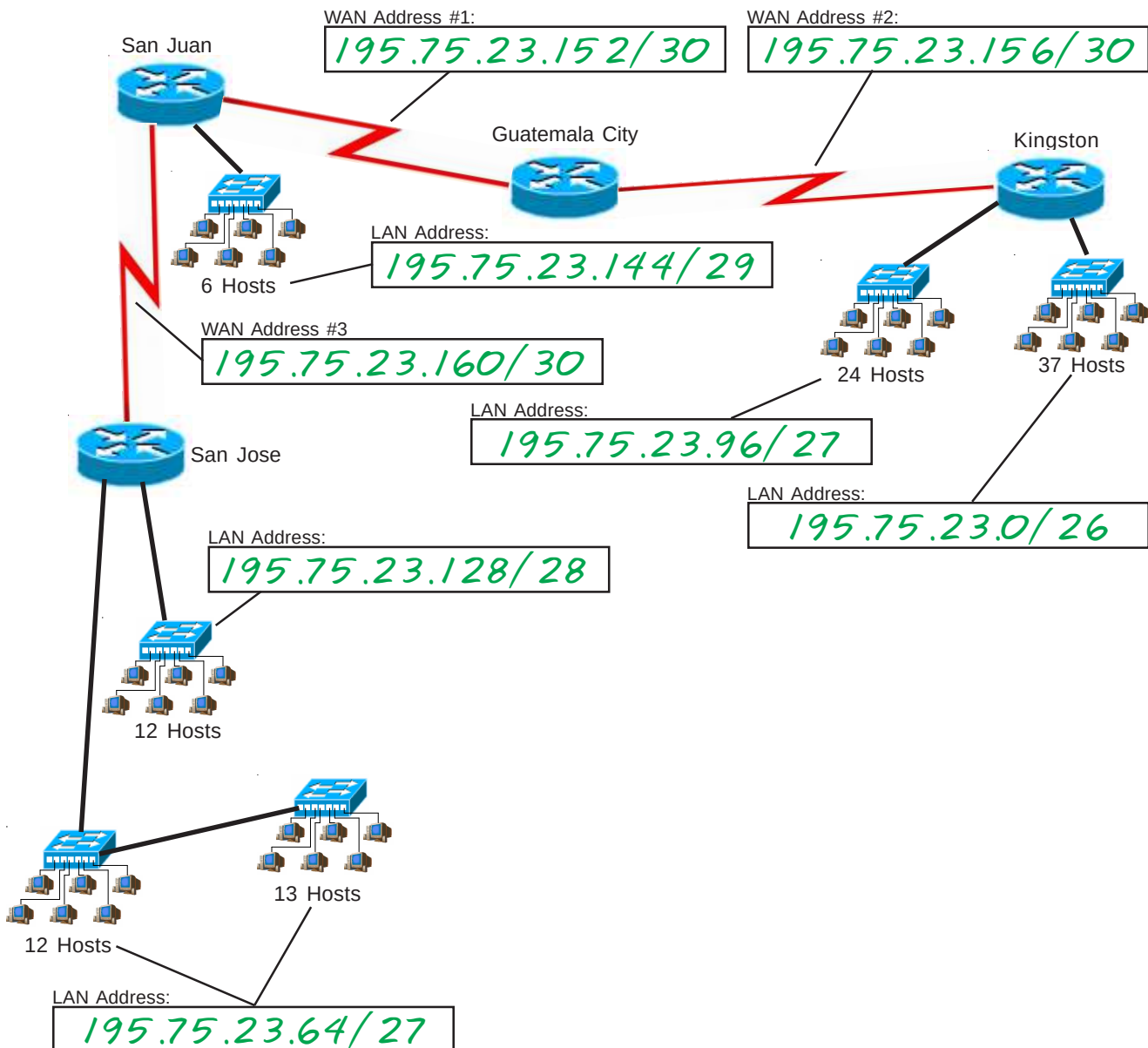
<u>Hosts</u>	<u>Address Ranges</u>		<u>CIDR</u>
96	192.168.150.0	to 192.168.150.127	/25
30	192.168.150.128	to 192.168.150.159	/27
23	192.168.150.160	to 192.168.150.191	/27
15	192.168.150.192	to 192.168.150.223	/27
12	192.168.150.224	to 192.168.150.239	/28
2	192.168.150.240	to 192.168.150.243	/30
2	192.168.150.244	to 192.168.150.247	/30
2	192.168.150.248	to 192.168.150.251	/30

VLSM Addressing

Numeric Method

Problem 16

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 195.75.23.0. Start with your largest groups first.



Show your work for **Problem 16** in the space below.

Number of Subnets										Number of Hosts									
256	128	64	32	16	8	4	2	1	0	256	128	64	32	16	8	4	2	1	0
2	4	8	16	32	64	128	256			2	4	8	16	32	64	128	256		
128	64	32	16	8	4	2	1	0		128	64	32	16	8	4	2	1	0	
195	75	23	0	0	0	0	0	0		195	75	23	0	0	0	0	0	0	

Hosts	Address Ranges		CIDR
37	195.75.23.0	to 195.75.23.63	/26
25	195.75.23.64	to 195.75.23.95	/27
24	195.75.23.96	to 195.75.23.127	/27
12	195.75.23.128	to 195.75.23.143	/28
6	195.75.23.144	to 195.75.23.151	/29
2	195.75.23.152	to 195.75.23.155	/30
2	195.75.23.156	to 195.75.23.159	/30
2	195.75.23.160	to 195.75.23.163	/30

Visualizing Subnets Using a VLSM Chart

The VLSM chart is the third method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You can adjust each sub-subnet to the correct size needed.

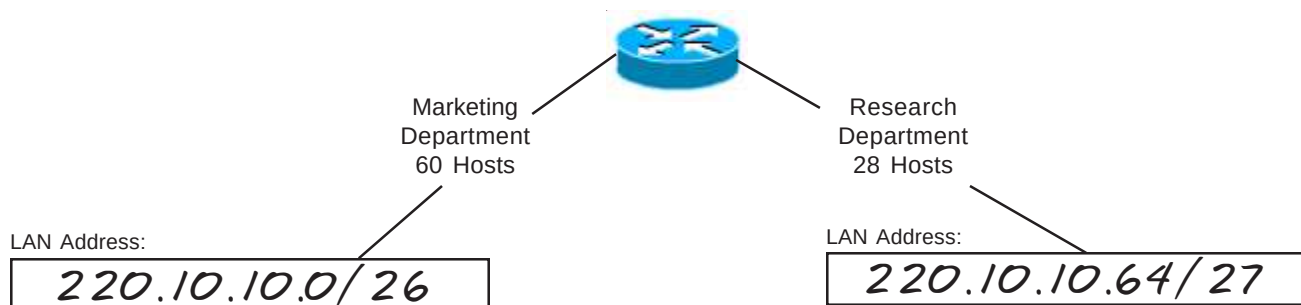
VLSM Addressing

VLSM Chart Method

(Sample)

Problem 17

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This business will be using the class C address 220.10.10.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

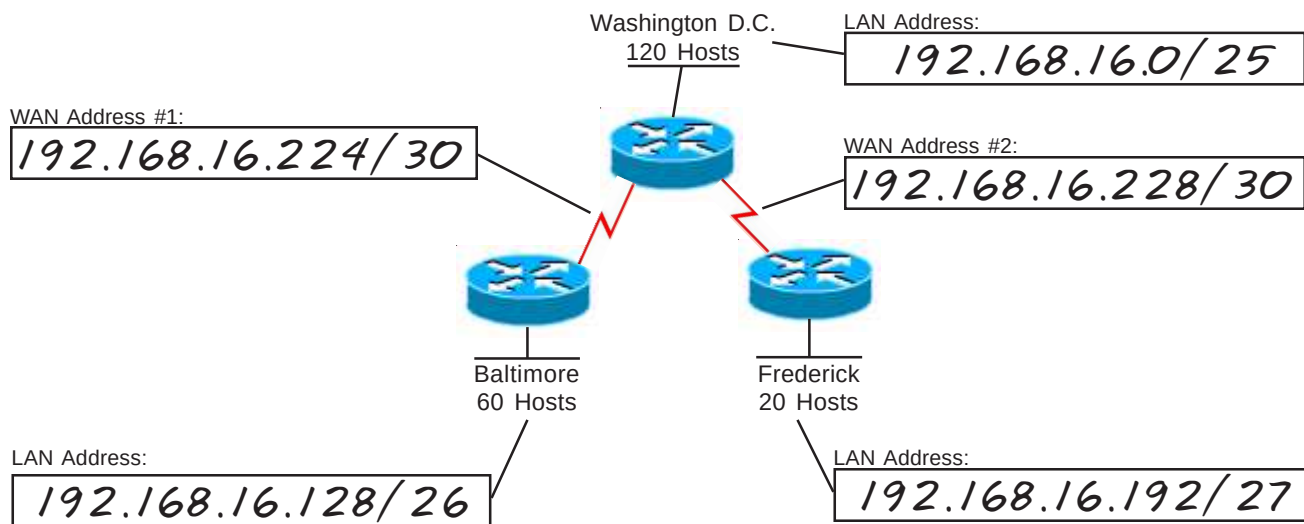
VLSM Addressing

VLSM Chart Method

(Sample)

Problem 18

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 192.168.16.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

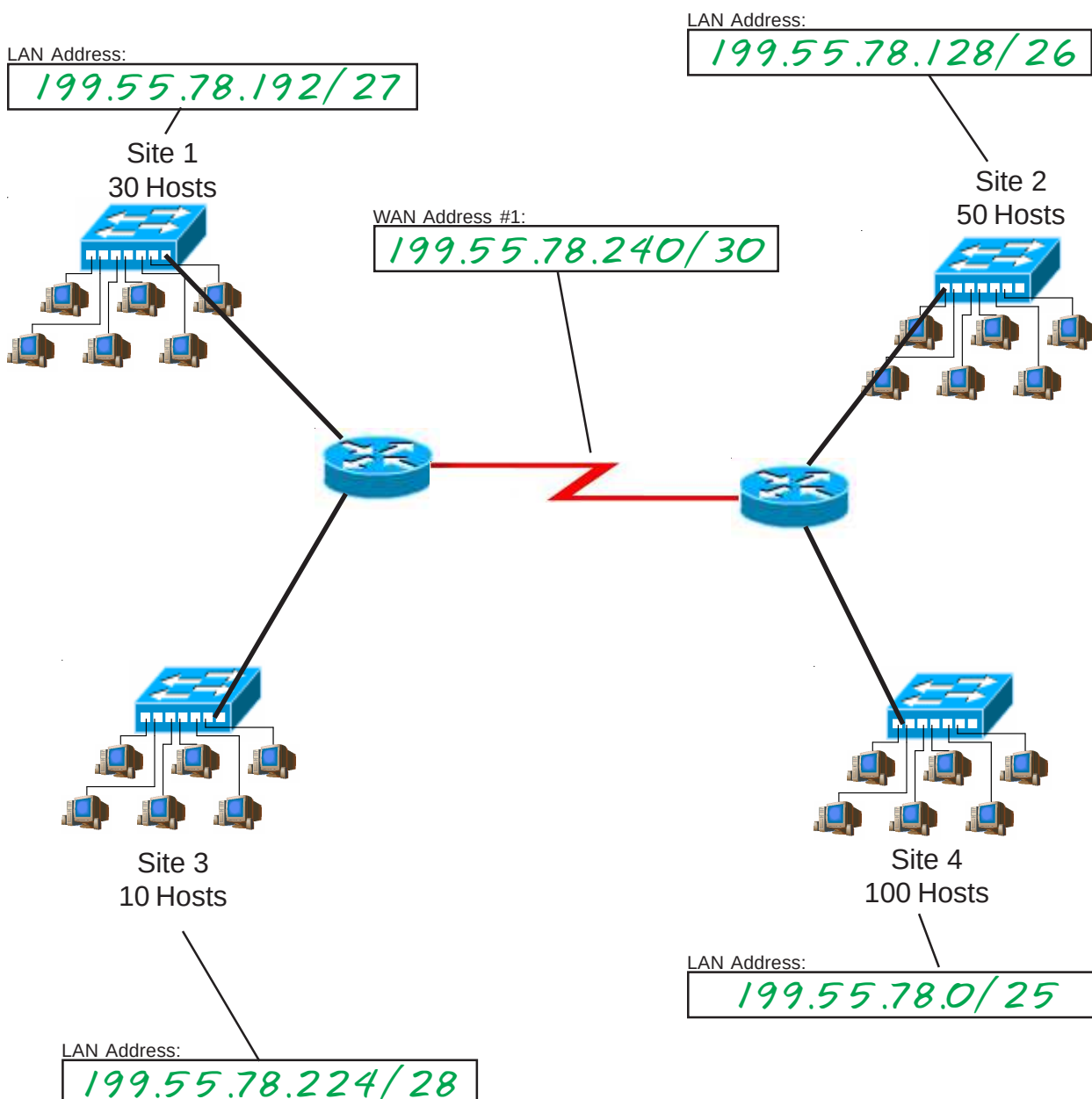
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
		32-63	32-47	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
				96-111	96-103	96-99 100-103
					104-111	104-107 108-111
			96-127	112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 19

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 199.55.78.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

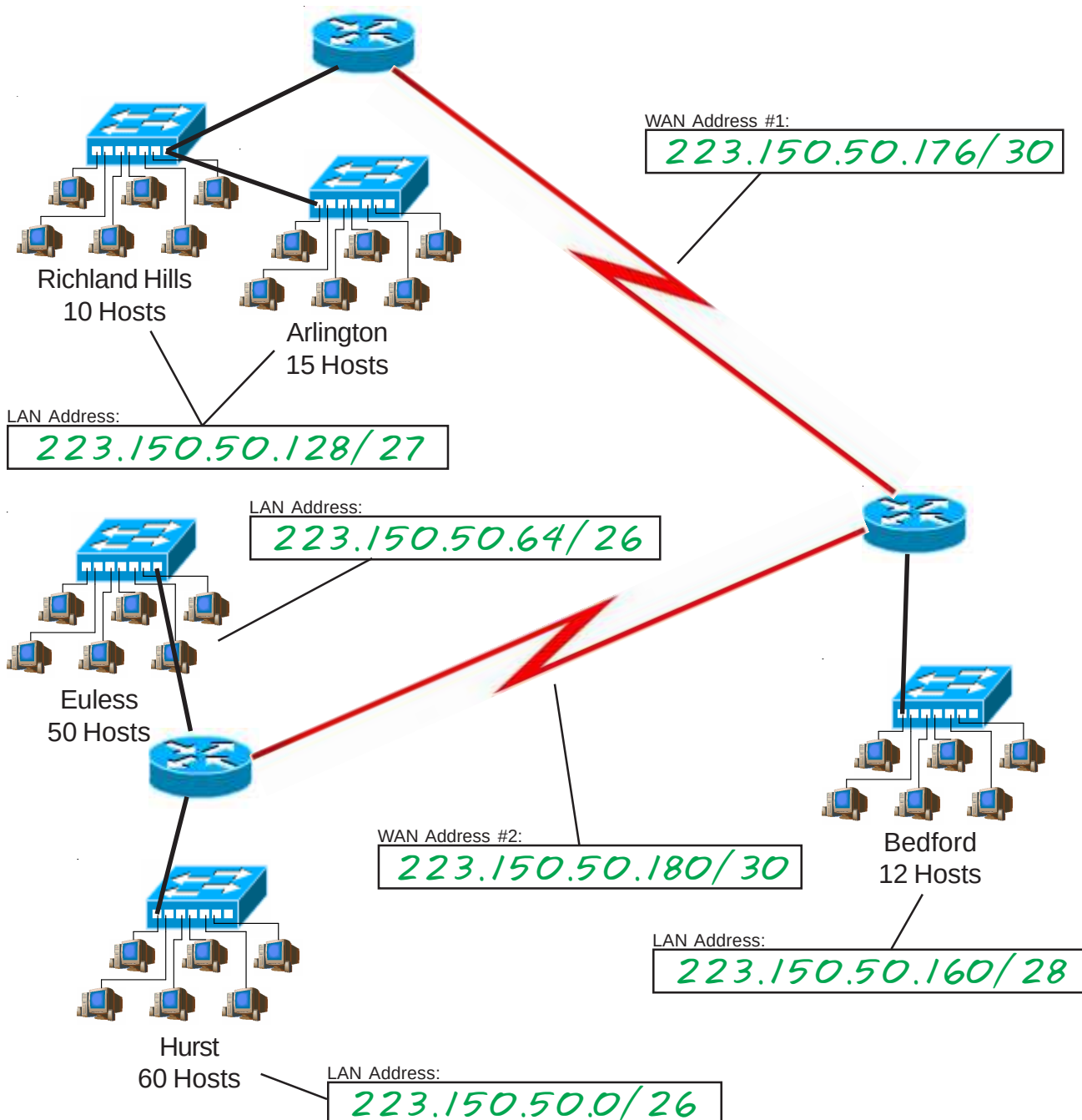
/24	/25	/26	/27	/28	/29	/30			
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts			
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7			
					16-31	8-15	8-11 12-15		
						32-47	16-23	16-19 20-23	
							24-31	24-27 28-31	
				64-127			32-63	48-63	32-39
					40-47				40-43 44-47
					48-55	48-51 52-55			
					56-63	56-59 60-63			
			64-95		64-79	80-95	64-71	64-67 68-71	
							72-79	72-75 76-79	
							80-87	80-83 84-87	
							88-95	88-91 92-95	
				96-127	96-111	96-103	96-99 100-103		
						104-111	104-107 108-111		
						112-119	112-115 116-119		
						120-127	120-123 124-127		
		128-255	128-191	128-159	128-143	128-135	128-131 132-135		
						136-143	136-139 140-143		
						144-159	144-151	144-147 148-151	
							152-159	152-155 156-159	
					160-191		160-175	16-167	160-163 164-167
								168-175	168-171 172-175
						176-191		176-183	176-179 180-183
								184-191	184-187 188-191
				192-255	192-223		192-207	192-199	192-195 196-199
								200-207	200-203 204-207
						208-223		208-215	208-211 212-215
								216-223	216-219 220-223
					224-255		224-239	224-231	224-227 228-231
								232-239	232-235 236-239
						240-255		240-247	240-243 244-247
								248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 20

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 223.150.50.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

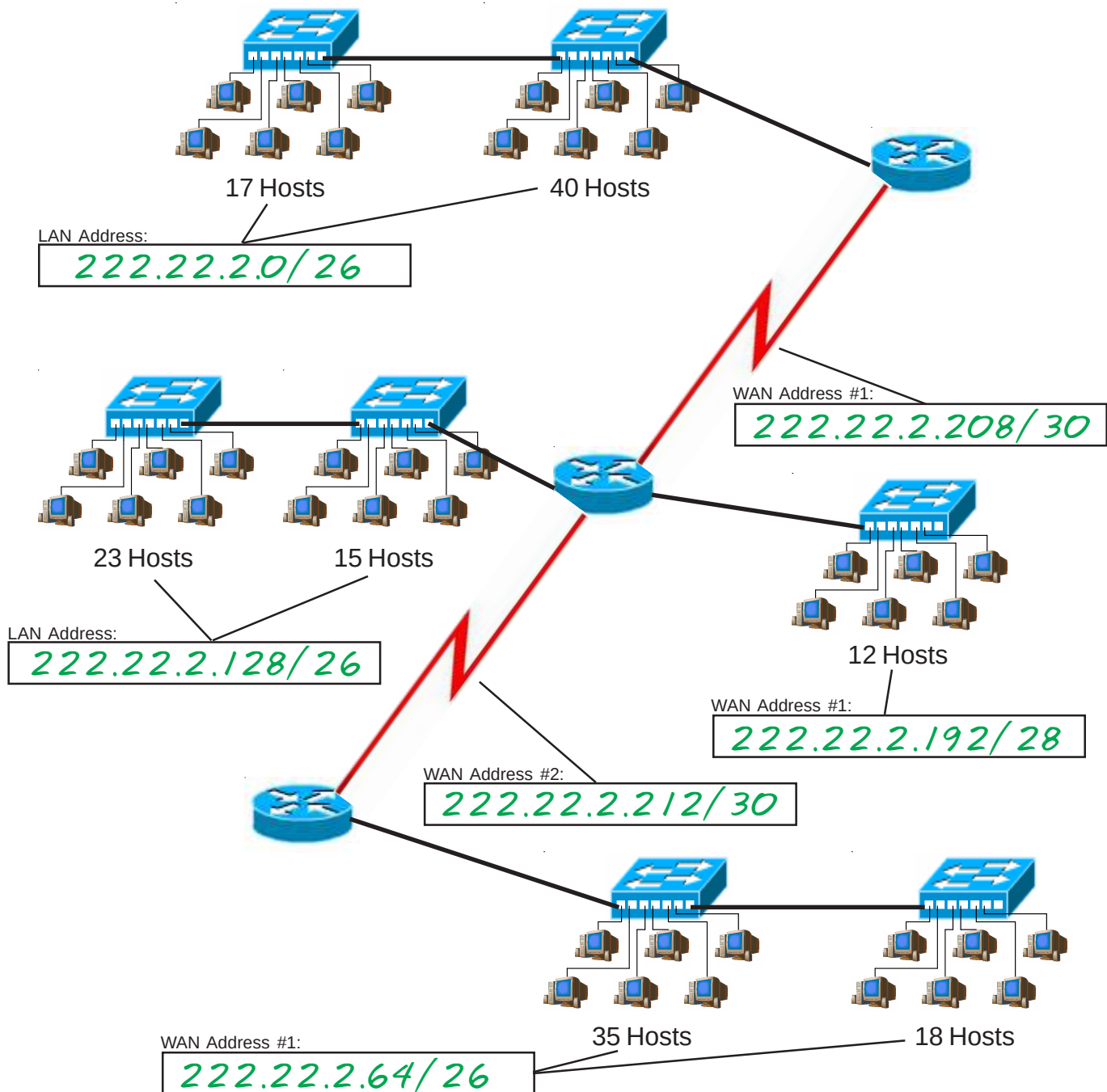
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 21

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 222.22.2.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

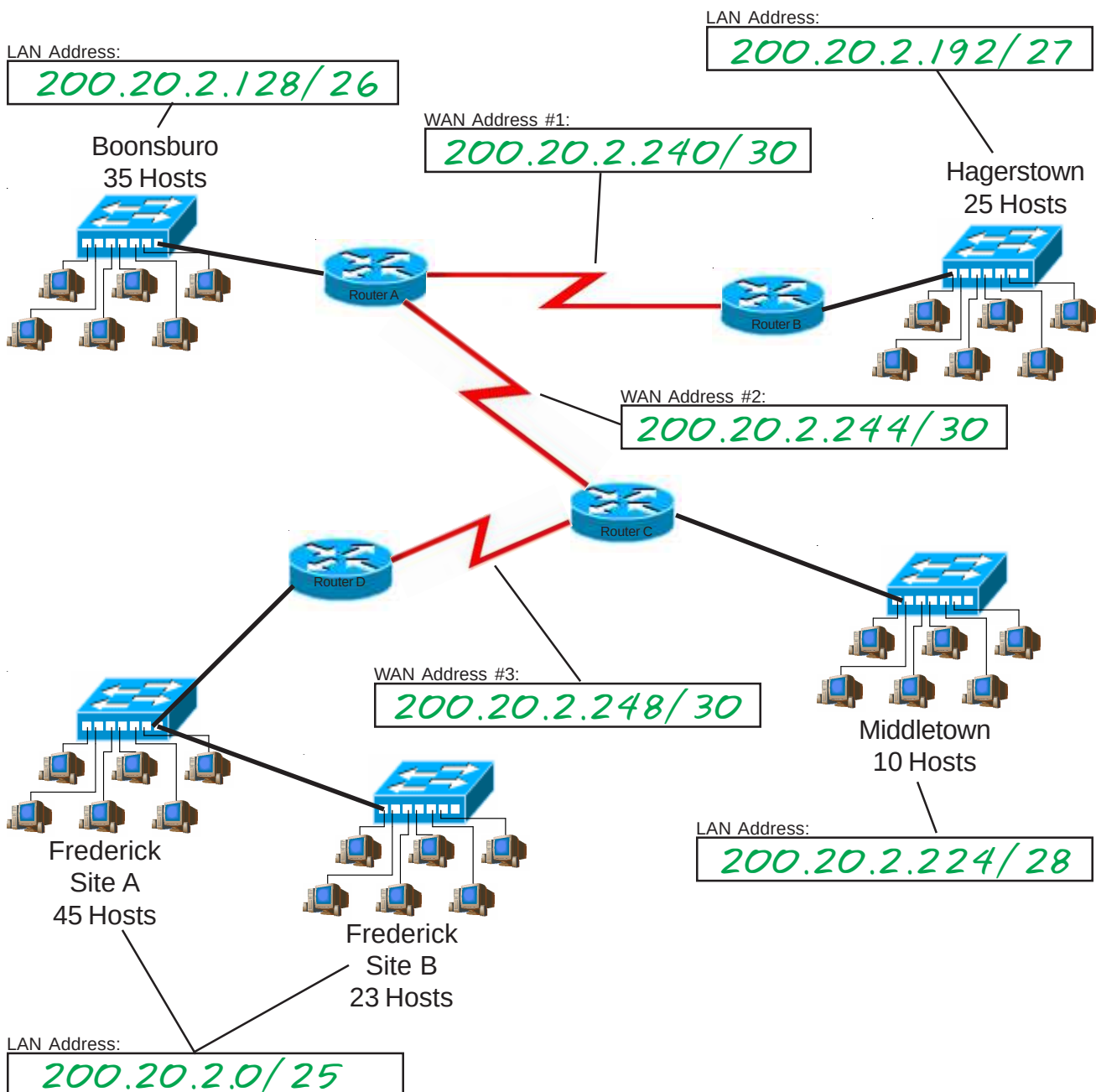
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 22

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 200.20.2.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

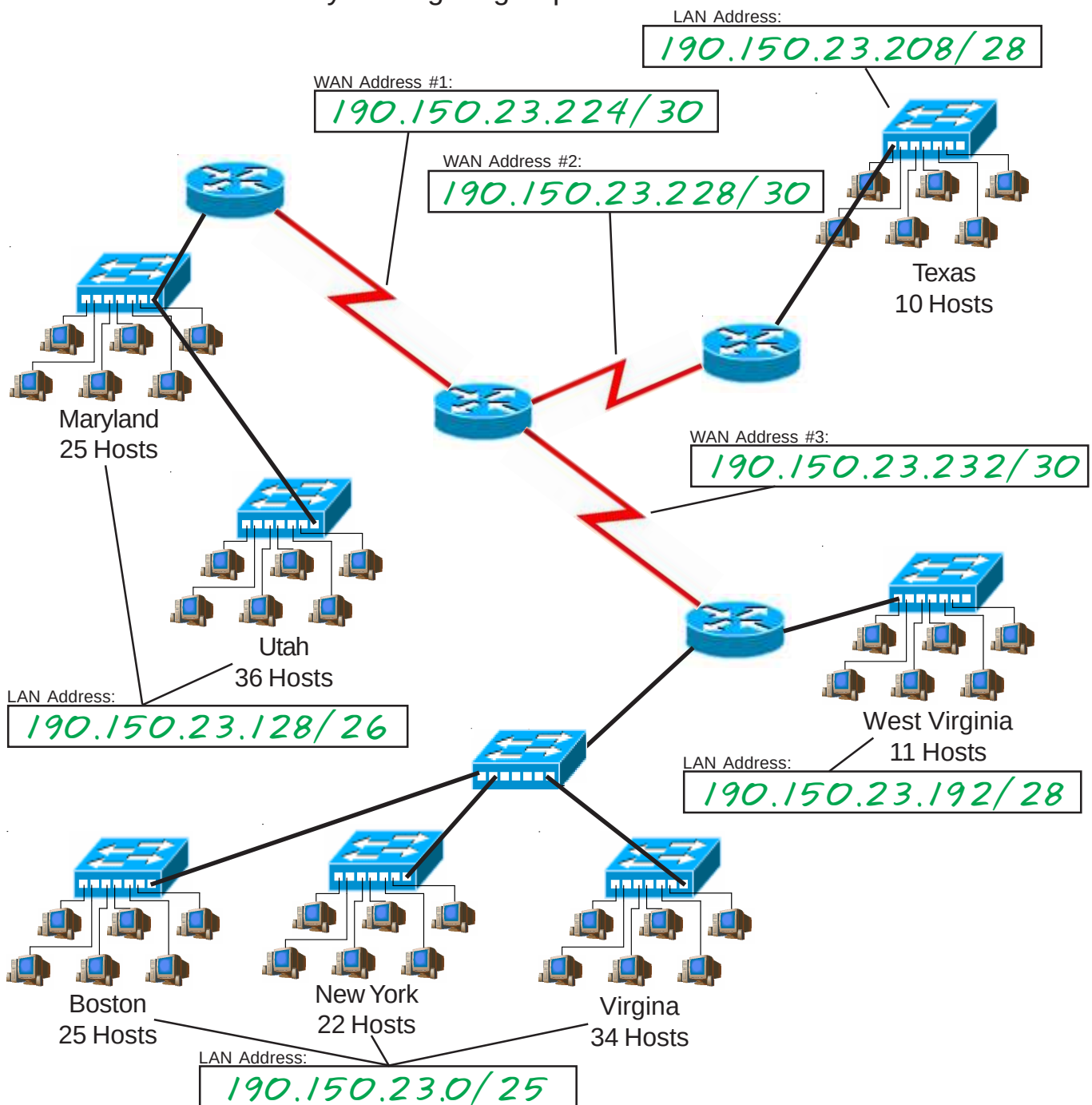
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 23

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 190.150.23.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

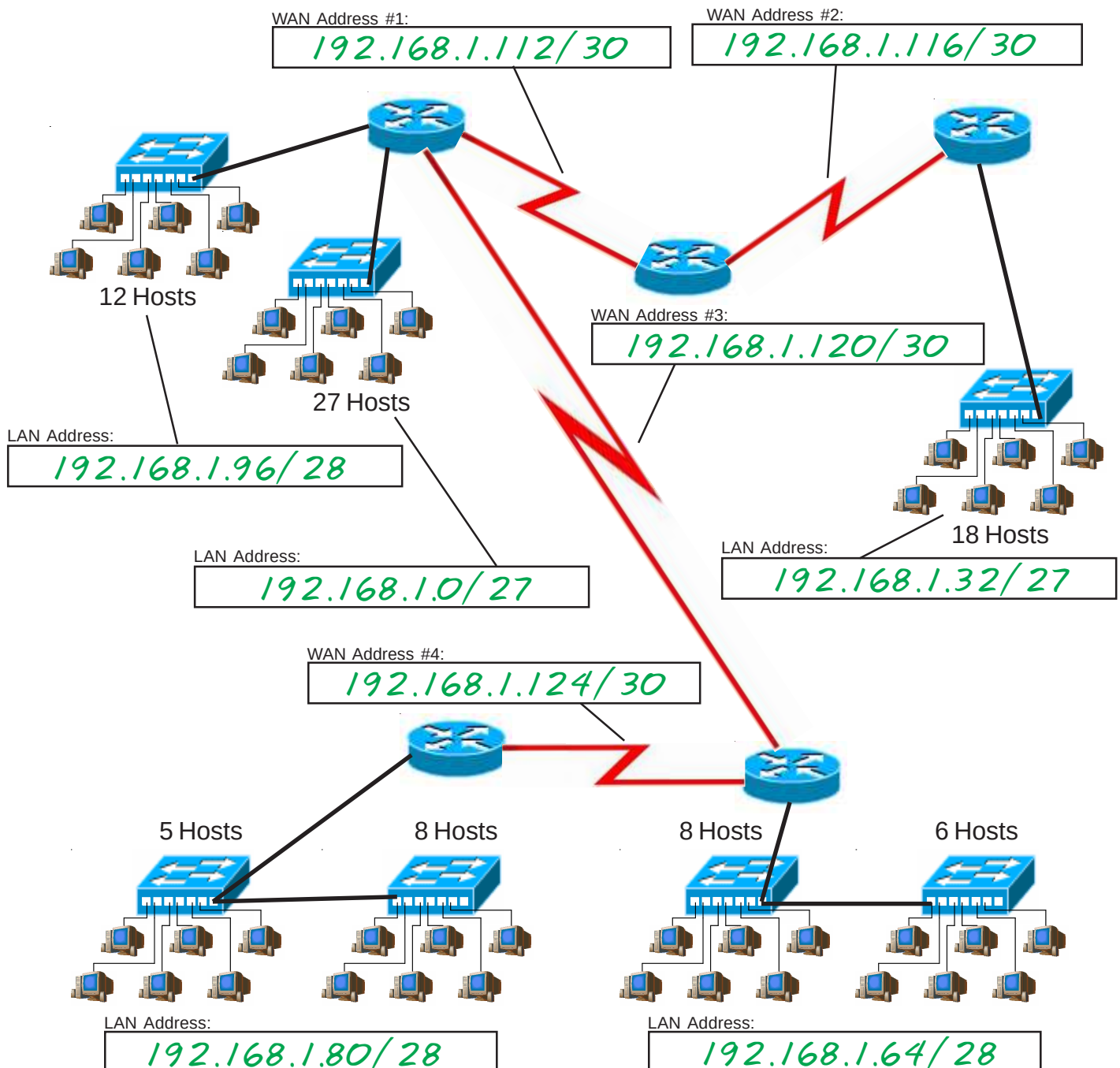
/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-103	96-99	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	16-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

VLSM Addressing

VLSM Chart Method

Problem 24

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 192.168.1.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

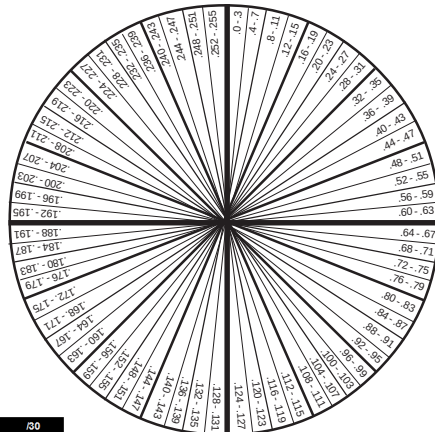
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

Practical VLSM Problems

Use the VLSM method of your choice to complete the following problems.

(Reference pages are in the back of the workbook.)

0	8	32	40	128	136	160	168
	3	11	35	43	131	139	163
4	12	36	44	132	140	164	172
	7	15	39	47	135	143	167
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247
							255



VLSM Chart 24-30 Bits (4th octet)

/24 256 Hosts	/25 128 Hosts	/26 64 Hosts	/27 32 Hosts	/28 16 Hosts	/29 8 Hosts	/30 4 Hosts
0 - 255	0 - 127	0 - 63	0 - 31	0 - 15	0 - 7	0 - 3
				8 - 15	8 - 15	8 - 15
				16 - 23	16 - 23	16 - 23
				24 - 31	24 - 31	24 - 31
				32 - 39	32 - 39	32 - 39
				40 - 47	40 - 47	40 - 47
				48 - 55	48 - 55	48 - 55
				56 - 63	56 - 63	56 - 63
				64 - 71	64 - 71	64 - 71
				72 - 79	72 - 79	72 - 79
				80 - 87	80 - 87	80 - 87
				88 - 95	88 - 95	88 - 95
				96 - 103	96 - 103	96 - 103
				104 - 111	104 - 111	104 - 111
				112 - 119	112 - 119	112 - 119
				120 - 127	120 - 127	120 - 127
				128 - 135	128 - 135	128 - 135
				136 - 143	136 - 143	136 - 143
				144 - 151	144 - 151	144 - 151
				152 - 159	152 - 159	152 - 159
				160 - 167	160 - 167	160 - 167
				168 - 175	168 - 175	168 - 175
				176 - 183	176 - 183	176 - 183
				184 - 191	184 - 191	184 - 191
				192 - 199	192 - 199	192 - 199
				200 - 207	200 - 207	200 - 207
				208 - 215	208 - 215	208 - 215
				216 - 223	216 - 223	216 - 223
				224 - 231	224 - 231	224 - 231
				232 - 239	232 - 239	232 - 239
				240 - 247	240 - 247	240 - 247
				248 - 255	248 - 255	248 - 255

VLSM Addressing

(Sample)

Problem 25

You are developing a school network with the class C address 192.168.2.0/24. There will be three computer labs with 30 computers each that need to be on different sub-subnets. Forty eight classrooms with one computer each that will comprise a single sub-subnet. The administrative office and guidance office contain a total of seven computers which will need to be grouped together. Plan for four more mini labs with six computers to each sub-subnetwork. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.2.0	/26	192.168.2.1	192.168.2.62	192.168.2.63
2	192.168.2.64	/27	192.168.2.65	192.168.2.94	192.168.2.95
3	192.168.2.96	/27	192.168.2.97	192.168.2.126	192.168.2.127
4	192.168.2.128	/27	192.168.2.129	192.168.2.158	192.168.2.159
5	192.168.2.160	/28	192.168.2.161	192.168.2.174	192.168.2.175
6	192.168.2.176	/29	192.168.2.177	192.168.2.182	192.168.2.183
7	192.168.2.184	/29	192.168.2.185	192.168.2.190	192.168.2.191
8	192.168.2.192	/29	192.168.2.193	192.168.2.198	192.168.2.199
9	192.168.2.200	/29	192.168.2.201	192.168.2.206	192.168.2.207
10					
11					
12					
13					
14					

VLSM Addressing

(Sample)

Problem 26

You are setting up a small business network with the class C address 220.55.80.0/24. The marketing division will need 12 computers. Research and development needs 27 computers. The reception area will need two computers. Management requires 19 computers. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	220.55.80.0	/27	220.55.80.1	220.55.80.30	220.55.80.31
2	220.55.80.32	/27	220.55.80.	220.55.80.62	220.55.80.63
3	220.55.80.64	/28	220.55.80.65	220.55.80.78	220.55.80.79
4	220.55.80.80	/30	220.55.80.81	220.55.80.82	220.55.80.83
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 27

You are setting up a medium sized network with the class C address 222.37.34.0/24. Marketing needs 29 computers. Research and development needs 110 computers. Bookkeeping will use 12 computers. The reception area will need three computers. Management requires 60 computers. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	222.37.34.0	/25	222.37.34.1	222.37.34.126	222.37.34.127
2	222.37.34.128	/26	222.37.34.129	222.37.34.190	222.37.34.191
3	222.37.34.192	/27	222.37.34.193	222.37.34.222	222.37.34.223
4	222.37.34.224	/28	222.37.34.225	222.37.34.238	222.37.34.239
5	222.37.34.240	/29	222.37.34.241	222.37.34.246	222.37.34.247
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VLSM Addressing

Problem 28

A shipping company needs to set up its network across several locations. The Denver office needs six computers. The Waco office needs 22 computers. The Fargo office will need five computers. The WAN links between all three locations need to be included in the solution. Using the IP address 192.168.10.0/24 divide the network using VLSM. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.10.0	/27	192.168.10.1	192.168.10.30	192.168.10.31
2	192.168.10.32	/29	192.168.10.33	192.168.10.38	192.168.10.39
3	192.168.10.40	/29	192.168.10.41	192.168.10.46	192.168.10.47
4	192.168.10.48	/30	192.168.10.49	192.168.10.50	192.168.10.51
5	192.168.10.52	/30	192.168.10.53	192.168.10.54	192.168.10.55
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VLSM Addressing

Problem 29

A new school is being built in the local school district. It will have three computer labs with 28 computers each. There will be 58 classrooms with 2 computers each that need to be on one sub-subnet. The office staff and administrators will need 7 computers. The guidance and attendance office will have 5 computers. The school has been given the address 223.145.75.0/24. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	223.145.75.0	/25	223.145.75.1	223.145.75.126	223.145.75.127
2	223.145.75.128	/27	223.145.75.129	223.145.75.158	223.145.75.159
3	223.145.75.160	/27	223.145.75.161	223.145.75.190	223.145.75.191
4	223.145.75.192	/27	223.145.75.193	223.145.75.222	223.145.75.223
5	223.145.75.224	/28	223.145.75.225	223.145.75.238	223.145.75.239
6	223.145.75.240	/29	223.145.75.241	223.145.75.246	223.145.75.247
7				2	
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VLSM Addressing

Problem 30

A local college is setting up a campus wide network. The technology wing will be on its own network address of 192.168.250.0/24. The office wing will include 15 computers. There are 2 labs of 20 computers each, 2 labs of 30 computers each and one lab of 35 computers. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.250.0	/26	92.168.250.1	92.168.250.62	92.168.250.63
2	92.168.250.64	/27	92.168.250.65	92.168.250.94	92.168.250.95
3	92.168.250.96	/27	92.168.250.97	92.168.250.126	92.168.250.127
4	92.168.250.128	/27	92.168.250.129	92.168.250.158	92.168.250.159
5	92.168.250.160	/27	92.168.250.161	92.168.250.190	92.168.250.191
6	92.168.250.192	/27	92.168.250.193	92.168.250.222	92.168.250.223
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VLSM Addressing

Problem 31

You are setting up a network for a company in four locations. Location A has 8 computers. Location B has 122 computers. Location C has 4 computers. Location D has 55 computers. There is a WAN connection between all four locations. Complete the information required below using the class C address 192.168.10.0. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.10.0	/25	192.168.10.1	192.168.10.126	192.168.10.127
2	192.168.10.128	/26	192.168.10.129	192.168.10.190	192.168.10.191
3	192.168.10.192	/28	192.168.10.193	192.168.10.206	192.168.10.207
4	192.168.10.208	/29	192.168.10.209	192.168.10.214	192.168.10.215
5	192.168.10.216	/30	192.168.10.217	192.168.10.218	192.168.10.219
6	192.168.10.220	/30	192.168.10.221	192.168.10.222	192.168.10.223
7	192.168.10.224	/30	192.168.10.225	192.168.10.226	192.168.10.227
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10					
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12					
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14					

VLSM Addressing

Problem 32

A college dormitory is being remodeled. A new network is being installed. There are 50 dorm rooms with two drops each that will be on one sub-subnet. The offices will have 5 drops. The reception desk will have three drops. A small study hall will include 30 drops. Using the IP address 192.168.12.0/24 complete the information required below using VLSM. Work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.12.0	/25	192.168.12.1	192.168.12.126	192.168.12.127
2	192.168.12.128	/27	192.168.12.129	192.168.12.158	192.168.12.159
3	192.168.12.160	/29	192.168.12.161	192.168.12.166	192.168.12.167
4	192.168.12.168	/29	192.168.12.169	192.168.12.174	192.168.12.175
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8					
9					
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11					
12					
13					
14					

VLSM Addressing

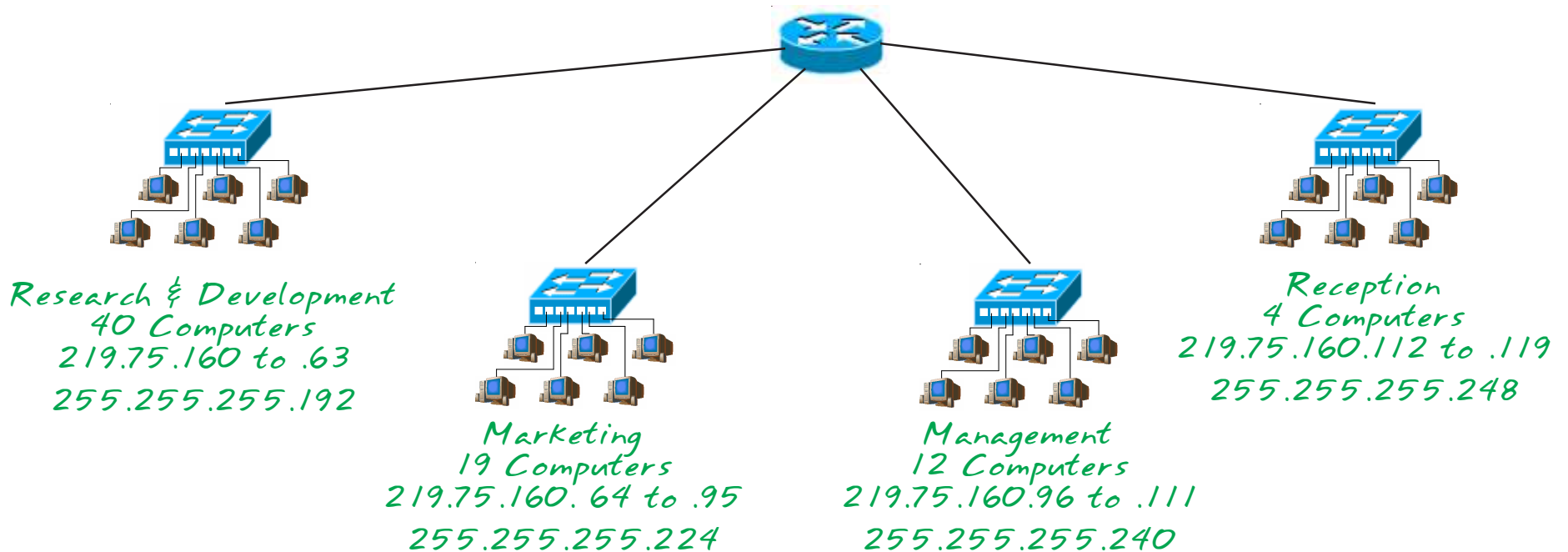
Problem 33

You are setting up a business network with the class C address 219.75.160.0/24. The marketing division will need 19 computers. Research and development needs 40 computers. The reception area will need four computers. Management requires 12 computers. Divide the network using variable length subnet information. On the opposite page draw a detailed map of this network. Include the name and sub-subnet IP addresses for each branch of the network with the subnet mask. One router with four ethernet ports will be used for this network.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	219.75.160.0	/26	219.75.160.1	219.75.160.62	219.75.160.63
2	219.75.160.64	/27	219.75.160.65	219.75.160.94	219.75.160.95
3	219.75.160.96	/28	219.75.160.97	219.75.160.110	219.75.160.111
4	219.75.160.112	/29	219.75.160.113	219.75.160.118	219.75.160.119
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Problem 33 - Detailed Map

Draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network, and the subnet mask.



VLSM Addressing

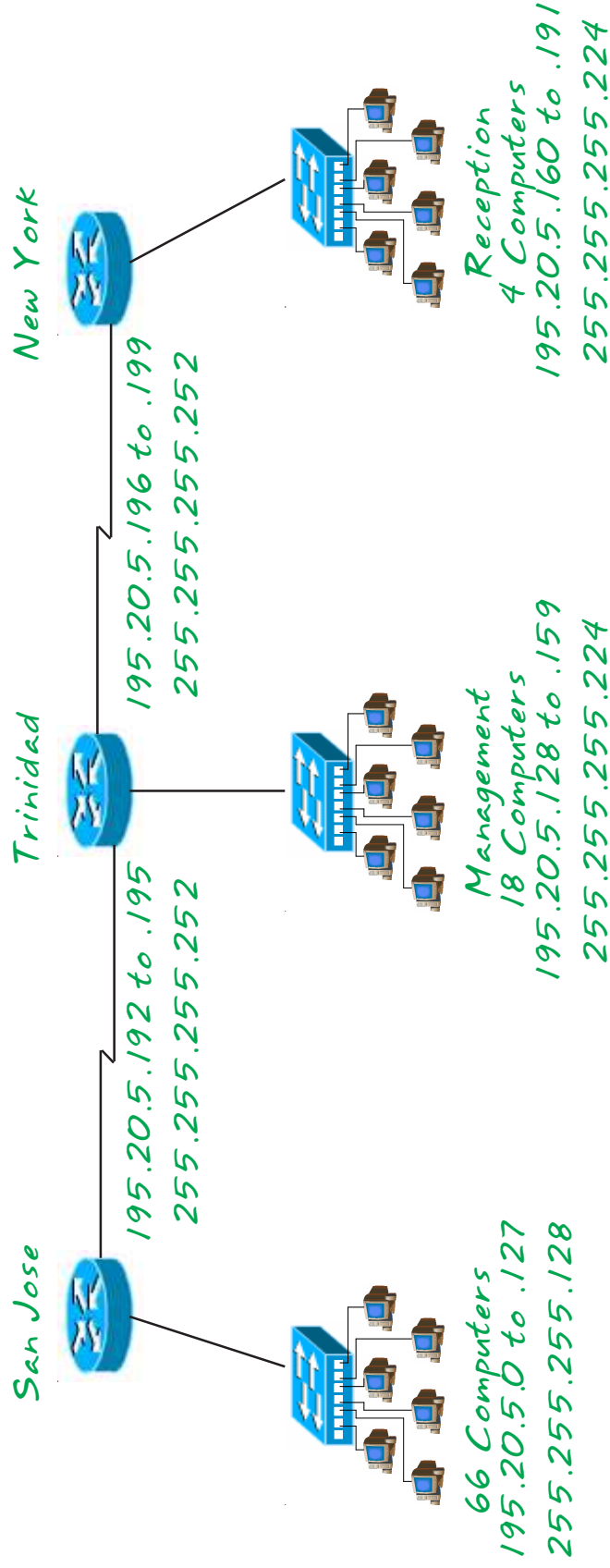
Problem 34

A small company needs to set up its network across several locations. The New York branch office needs 15 computers. The San Jose office needs 66 computers. The Trinidad office will need 18 computers. You will need two WAN links between the routers. Using the IP address 195.20.5.0/24 divide the network using VLSM. On the opposite page draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network. Label the WAN links with the same information. Complete the information required below. Work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	195.20.5.0	/25	195.20.5.1	195.20.5.126	195.20.5.127
2	195.20.5.128	/27	195.20.5.129	195.20.5.158	195.20.5.159
3	195.20.5.160	/27	195.20.5.161	195.20.5.190	195.20.5.191
4	195.20.5.192	/30	195.20.5.193	195.20.5.194	195.20.5.195
5	195.20.5.196	/30	195.20.5.197	195.20.5.198	195.20.5.199
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Problem 34 - Detailed Map

Draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network.



Class A and B VLSM Problems

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Class A Addresses
VLSM Chart 8-15 Bits (2nd octet)

10	11	12	13	14	15
255.0.0.0 8,192,384 hosts	255.128.0.0 4,096,384 hosts	255.192.0.0 2,048,384 hosts	255.240.0.0 1,024,384 hosts	255.288.0.0 512,384 hosts	255.336.0.0 256,384 hosts
0-255	0-127	0-63	0-5	6-11	12-15
			16-23	24-31	32-39
			40-47	48-55	56-63
		64-127	64-71	72-79	80-87
			88-95	96-103	104-111
			112-119	120-127	
	128-255	128-191	128-135	136-143	144-151
			152-159	160-167	168-175
			176-183	184-191	192-199
			200-207	208-215	216-223
		192-255	224-231	232-239	240-247
			248-255		

Class B Addresses
VLSM Chart 16-23 Bits (3rd octet)

16	17	18	19	20	21	22	23
255.255.0.0 65,536 hosts	255.255.128.0 32,768 hosts	255.255.192.0 16,384 hosts	255.255.224.0 8,192 hosts	255.255.240.0 4,096 hosts	255.255.248.0 2,048 hosts	255.255.252.0 1,024 hosts	255.255.254.0 512 hosts
0-255	0-127	0-63	0-5	6-11	12-15	16-23	24-31
			32-39	40-47	48-55	56-63	64-71
			72-79	80-87	88-95	96-103	104-111
		64-127	112-119	120-127			
	128-255	128-191	128-135	136-143	144-151	152-159	160-167
			168-175	176-183	184-191	192-199	200-207
			208-215	216-223	224-231	232-239	240-247
			248-255				
		192-255					

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

24	25	26	27	28	29	30
255.255.255.0 256 hosts	255.255.255.128 128 hosts	255.255.255.192 64 hosts	255.255.255.224 32 hosts	255.255.255.240 16 hosts	255.255.255.248 8 hosts	255.255.255.252 4 hosts
0-255	0-127	0-63	0-5	6-11	12-15	16-23
			24-31	32-39	40-47	48-55
			56-63	64-71	72-79	80-87
		64-127	88-95	96-103	104-111	112-119
			120-127			
	128-255	128-191	128-135	136-143	144-151	152-159
			160-167	168-175	176-183	184-191
			192-199	200-207	208-215	216-223
			224-231	232-239	240-247	248-255
		192-255				

VLSM

with

Class A and B Addresses

We've gone over the practical applications of using VLSM on class C addresses. The same approach works with class A and B addresses. For example an ISP may have a class A address which it needs to subnet between its customers. Each customer may need to take their addresses and subnet them again in order to use them more effectively. The real trick to this is to remember which octet of the IP address you are working with.

Sample Problem 35

Part 1 of 3

Use the **Class A** address chart to break down the address for different ISP customers. At this stage of the problem you are creating subnets using the second octet of the IP address.

ISP Addresses 15.0.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Customer #1	8 million	<i>15.0.0.0 to 15.127.255.255</i>	<i>/9</i>
Customer #2	2 million	<i>15.128.0.0 to 15.159.255.255</i>	<i>/11</i>
Customer #3	2,000,000	<i>15.160.0.0 to 15.191.255.255</i>	<i>/11</i>
Customer #4	1,000,000	<i>15.192.0.0 to 15.207.255.255</i>	<i>/12</i>
Customer #5	500,000	<i>15.208.0.0 to 15.215.255.255</i>	<i>/13</i>
Customer #6	450,000	<i>15.216.0.0 to 15.223.255.255</i>	<i>/13</i>
Customer #7	200,000	<i>15.224.0.0 to 15.227.255.255</i>	<i>/14</i>
Customer #8	130,000	<i>15.228.0.0 to 15.229.255.255</i>	<i>/15</i>
Customer #9	100,000	<i>15.230.0.0 to 15.231.255.255</i>	<i>/15</i>

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
				16-31		16-19	8-9
					16-23	20-23	10-11
						24-27	12-13
					24-31	28-31	14-15
			32-63	32-47		32-35	16-17
					32-39	36-39	18-19
						40-43	20-21
					40-47	44-47	22-23
				48-63		48-51	24-25
					48-55	52-55	26-27
						56-59	28-29
					56-63	60-63	30-31
		64-127	64-95	64-79	64-71	64-67	32-33
						68-71	34-35
					72-79	72-75	36-37
						76-79	38-39
			80-95	80-87	80-83	84-87	40-41
						88-91	42-43
					88-95	92-95	44-45
						96-99	46-47
			96-127	96-103	100-103	104-107	48-49
						108-111	50-51
				104-111	104-107	112-115	52-53
						116-119	54-55
				112-127	112-119	120-123	56-57
						124-127	58-59
					120-127	128-131	60-61
						132-135	62-63
	128-255	128-191	128-159	128-143	128-135	128-131	64-65
						132-135	66-67
					136-143	136-139	68-69
						140-143	70-71
			144-159	144-151	144-147	148-151	72-73
						152-155	74-75
					152-159	156-159	76-77
						160-163	78-79
			160-191	160-175	160-167	164-167	80-81
						168-171	82-83
					168-175	172-175	84-85
						176-179	86-87
				176-191	176-183	180-183	88-89
						184-187	90-91
					184-191	188-191	92-93
						192-195	94-95
		192-255	192-223	192-207	192-199	196-199	96-97
						200-203	98-99
					200-207	204-207	100-101
				208-223	208-215	212-215	102-103
						216-219	104-105
					216-223	220-223	106-107
			224-255	224-239		224-227	108-109
					224-231	228-231	110-111
						232-235	112-113
					232-239	236-239	114-115
				240-255	240-247	244-247	116-117
						248-251	118-119
					248-255	252-255	120-121
							122-123
							124-125
							126-127
							128-129
							130-131
							132-133
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							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Sample Problem 35

Part 2 of 3

Customer #5 has a total of 524,288 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for their different clients. At this stage of the problem you are creating sub-subnets with the third octet of the IP address.

ISP Addresses 15.208.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Client #1	7,500	<i>15.208.0.0 to 15.208.31.255</i>	<i>/19</i>
Client #2	5,000	<i>15.208.32.0 to 15.208.63.255</i>	<i>/19</i>
Client #3	4,500	<i>15.208.64.0 to 15.208.95.255</i>	<i>/19</i>
Client #4	2,000	<i>15.208.96.0 to 15.208.103.255</i>	<i>/21</i>
Client #5	1,450	<i>15.208.104.0 to 15.208.111.255</i>	<i>/21</i>
Client #6	1,150	<i>15.208.112.0 to 15.208.119.255</i>	<i>/21</i>
Client #7	900	<i>15.208.120.0 to 15.208.123.255</i>	<i>/22</i>
Client #8	750	<i>15.208.124.0 to 15.208.127.255</i>	<i>/22</i>
Client #9	450	<i>15.208.128.0 to 15.208.129.255</i>	<i>/23</i>

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
						8-11	4-5
						12-15	6-7
						16-19	8-9
				16-31	16-23	20-23	10-11
					24-31	28-31	12-13
						32-35	14-15
						36-39	16-17
						40-43	18-19
		64-127	32-63	32-47	40-47	44-47	20-21
					48-55	52-55	22-23
					56-63	60-63	24-25
						64-67	26-27
						68-71	28-29
			64-95	64-79	72-79	76-79	30-31
					80-87	84-87	32-33
					88-95	92-95	34-35
						96-99	36-37
						100-103	38-39
			96-127	96-111	104-111	108-111	40-41
					112-119	116-119	42-43
				112-127	120-127	124-127	44-45
						128-129	46-47
						130-131	48-49
	128-255	128-191	128-159	128-143	128-135	132-135	50-51
					136-143	140-143	52-53
						144-147	54-55
						148-151	56-57
						152-155	58-59
			160-191	160-175	160-167	164-167	60-61
					168-175	172-175	62-63
						176-179	64-65
						180-183	66-67
						184-187	68-69
		192-255	192-223	192-207	192-199	196-199	70-71
					200-207	204-207	72-73
						208-211	74-75
						212-215	76-77
						216-219	78-79
			224-255	224-239	224-231	228-231	80-81
					232-239	236-239	82-83
						240-243	84-85
						244-247	86-87
						248-251	88-89
				240-255	248-255	252-255	90-91
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							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Sample Problem 35

Part 3 of 3

Client #8 has a total of 1,024 addresses. Use the **Class C** address chart to break down the sub-subnetwork addresses for their different branch offices. At this stage of the problem you are creating sub-subnets with the fourth octet of the IP address.

ISP Addresses 15.208.124.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Branch #1	100	<i>15.208.124.0 to 15.208.124.127</i>	<i>/ 25</i>
Branch #2	55	<i>15.208.124.128 to 15.208.124.191</i>	<i>/ 26</i>
Branch #3	25	<i>15.208.124.192 to 15.208.124.223</i>	<i>/ 27</i>
Branch #4	6	<i>15.208.124.224 to 15.208.124.231</i>	<i>/ 29</i>
Branch #5	4	<i>15.208.124.232 to 15.208.124.239</i>	<i>/ 29</i>
Branch #6	2	<i>15.208.124.240 to 15.208.124.243</i>	<i>/ 30</i>
Branch #7	2	<i>15.208.124.244 to 15.208.124.247</i>	<i>/ 30</i>
Branch #8	2	<i>15.208.124.248 to 15.208.124.251</i>	<i>/ 30</i>
Branch #9	2	<i>15.208.124.252 to 15.208.124.255</i>	<i>/ 30</i>

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

VLSM

with

Class A and B Addresses

Problem 36

Part 1 of 3

The company you are working for is using the IP address 110.0.0.0 sub-subneted for multiple offices around the world. Use the **Class A** address chart to break down the sub-subnetwork addresses for the different offices.

At this stage of the problem you are creating sub-subnets with the third octet of the IP address. Remember which octet of the IP address you are working in.

Company Address 110.0.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Moskva	3,050,000	<i>110.0.0.0 to 110.63.255.255</i>	<i>/10</i>
New York	1,540,000	<i>110.64.0.0 to 110.95.255.255</i>	<i>/11</i>
St. Petersburg	1,075,000	<i>110.96.0.0 to 110.127.255.255</i>	<i>/11</i>
London	975,000	<i>110.128.0.0 to 110.143.255.255</i>	<i>/12</i>
Ekaterinoburg	525,000	<i>110.144.0.0 to 110.159.255.255</i>	<i>/12</i>
Munchen	450,000	<i>110.160.0.0 to 110.167.255.255</i>	<i>/13</i>
Napoli	150,000	<i>110.168.0.0 to 110.171.255.255</i>	<i>/14</i>
Birmingham	130,000	<i>110.172.0.0 to 110.173.255.255</i>	<i>/15</i>
Rotterdam	95,000	<i>110.174.0.0 to 110.175.255.255</i>	<i>/15</i>

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8	/9	/10	/11	/12	/13	/14	/15	
255.0.0.0 16,777,216 Hosts	255.128.0.0 8,388,608 Hosts	255.192.0.0 4,194,304 Hosts	255.224.0.0 2,097,152 Hosts	255.240.0.0 1,048,576 Hosts	255.248.0.0 524,288 Hosts	255.252.0.0 262,144 Hosts	255.254.0.0 131,072 Hosts	
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1	
						4-7	4-5	
				16-31	8-15	8-11	6-7	
						12-15	8-9	
					32-47	16-23	16-19	10-11
							20-23	12-13
			24-31			24-27	20-23	14-15
							28-31	22-23
				32-63		32-39	32-35	24-25
							36-39	26-27
					40-47	40-43	36-39	28-29
							44-47	30-31
			48-63			48-55	48-51	32-33
							52-55	34-35
				56-63		56-59	56-59	36-37
							60-63	38-39
					64-71	64-67	64-67	40-41
							68-71	42-43
			72-79			72-75	72-75	44-45
							76-79	46-47
				80-87		80-83	80-83	48-49
							84-87	50-51
					88-95	88-91	88-91	52-53
							92-95	54-55
		96-103	96-99			96-99	56-57	
						100-103	58-59	
			104-111	104-107		104-107	60-61	
						108-111	62-63	
				112-119	112-115	112-115	64-65	
						116-119	66-67	
		120-127			120-123	120-123	68-69	
						124-127	70-71	
			128-135		128-131	128-131	72-73	
						132-135	74-75	
				136-143	136-139	136-139	76-77	
						140-143	78-79	
		144-151			144-147	144-147	80-81	
						148-151	82-83	
			152-159		152-155	152-155	84-85	
						156-159	86-87	
				160-167	160-163	160-163	88-89	
						164-167	90-91	
		168-175			168-171	168-171	92-93	
						172-175	94-95	
			176-179		176-179	176-179	96-97	
						180-183	98-99	
				184-191	184-187	184-187	100-101	
						188-191	102-103	
	192-199	192-199			192-199	104-105		
					196-199	106-107		
		200-207	200-203		200-203	108-109		
					204-207	110-111		
			208-215	208-211	208-211	112-113		
					212-215	114-115		
	216-223			216-219	216-219	116-117		
					220-223	118-119		
		224-231		224-227	224-227	120-121		
					228-231	122-123		
			232-239	232-235	232-235	124-125		
					236-239	126-127		
	240-247			240-243	240-243	128-129		
					244-247	130-131		
		248-255		248-251	248-251	132-133		
					252-255	134-135		
			252-255	252-255	252-255	136-137		
						138-139		
					140-141			
					142-143			
				144-145				
				146-147				
			148-149					
			150-151					
			152-153					
			154-155					
			156-157					
			158-159					
			160-161					
			162-163					
			164-165					
			166-167					
			168-169					
			170-171					
			172-173					
			174-175					
			176-177					
			178-179					
			180-181					
			182-183					
			184-185					
			186-187					
			188-189					
			190-191					
			192-193					
			194-195					
			196-197					
			198-199					
			200-201					
			202-203					
			204-205					
			206-207					
			208-209					
			210-211					
			212-213					
			214-215					
			216-217					
			218-219					
			220-221					
			222-223					
			224-225					
			226-227					
			228-229					
			230-231					
			232-233					
			234-235					
			236-237					
			238-239					
			240-241					
			242-243					
			244-245					
			246-247					
			248-249					
			250-251					
			252-253					
			254-255					

VLSM

with

Class A and B Addresses

Problem 36

Part 2 of 3

London in part 1 of this problem has been given 1,048,576 hosts, with the address range of 110.128.0.0 to 110.143.255.255 /12 (255.240.0.0).

Based on the information below supply the required address ranges and subnet masks for each office. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different areas of the network.

London
Address Range 110.128.0.0 to 110.143.255.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Office #1	6,450	<i>110.128.0.0 to 110.128.31.255</i>	<i>/19</i>
Office #2	3,780	<i>110.128.32.0 to 110.128.47.255</i>	<i>/20</i>
Office #3	2,750	<i>110.128.48.0 to 110.128.63.255</i>	<i>/20</i>
Office #4	2,000	<i>110.128.64.0 to 110.128.71.255</i>	<i>/21</i>
Office #5	1,000	<i>110.128.72.0 to 110.128.75.255</i>	<i>/22</i>
Office #6	845	<i>110.128.76.0 to 110.128.79.255</i>	<i>/22</i>
Office #7	500	<i>110.128.80.0 to 110.128.81.255</i>	<i>/23</i>
Office #8	450	<i>110.128.82.0 to 110.128.83.255</i>	<i>/23</i>
Office #9	300	<i>110.128.84.0 to 110.128.85.255</i>	<i>/23</i>

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31	16-23	8-11	4-5
						12-15	6-7
					24-31	16-19	8-9
						20-23	10-11
				32-63	32-39	24-27	12-13
						28-31	14-15
					40-47	32-35	16-17
						36-39	18-19
				48-63	48-55	40-43	20-21
						44-47	22-23
					56-63	52-55	24-25
						60-63	26-27
				64-95	64-71	64-67	28-29
						68-71	30-31
					80-87	72-75	32-33
						76-79	34-35
				80-95	80-83	80-83	36-37
						84-87	38-39
					88-95	88-91	40-41
						92-95	42-43
				96-111	96-103	96-99	44-45
						100-103	46-47
					104-111	104-107	48-49
						108-111	50-51
				112-127	112-119	112-115	52-53
						116-119	54-55
					120-127	120-123	56-57
						124-127	58-59
	128-255	128-191	128-159	128-143	128-135	128-131	60-61
					136-143	132-135	62-63
				144-159	144-151	136-139	64-65
						140-143	66-67
					152-159	144-147	68-69
						148-151	70-71
			160-191	160-175	160-167	152-155	72-73
						156-159	74-75
				168-175	168-171	160-163	76-77
						164-167	78-79
			176-191	176-183	176-179	168-171	80-81
						180-183	82-83
				184-191	184-187	172-175	84-85
						176-179	86-87
			192-255	192-207	192-199	188-191	88-89
						192-195	90-91
					200-207	196-199	92-93
						200-203	94-95
				208-223	208-215	204-207	96-97
						208-211	98-99
					216-223	212-215	100-101
						216-219	102-103
				224-239	224-231	220-223	104-105
						224-227	106-107
					232-239	228-231	108-109
						232-235	110-111
				240-255	240-247	236-239	112-113
						240-243	114-115
					248-255	244-247	116-117
						248-251	118-119
						252-255	120-121
							122-123
							124-125
							126-127
							128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Problem 36

Part 3 of 3

Office #7 in part 2 of this problem has been given 512 hosts, with the address range of 110.128.80.0 / 23 (255.255.254.0).

Based on the information below supply the required address ranges and subnet masks for each school area. Use the **Class C** address chart to break down the sub-subnetwork addresses for the different areas of the network. Hint: Another way to look at this problem is to see that with the third octet range of 80 to 81 you have access to 2 groups of 255 addresses or two Class C VLSM charts.

Office #7
Address Range 110.128.80.0 to 110.128.81.255

Customer Name	Number of Addresses	Address Range	CIDR
1st Floor	125	<i>110.128.80.0 to 110.128.80.127</i>	<i>/25</i>
2nd Floor	75	<i>110.128.80.128 to 110.128.80.255</i>	<i>/25</i>
5th Floor	50	<i>110.128.81.0 to 110.128.81.63</i>	<i>/26</i>
8th Floor	45	<i>110.128.81.64 to 110.128.81.127</i>	<i>/26</i>
4th Floor	30	<i>110.128.81.128 to 110.128.81.159</i>	<i>/27</i>
Basement	14	<i>110.128.81.160 to 110.128.81.175</i>	<i>/28</i>
7th Floor	12	<i>110.128.81.176 to 110.128.81.191</i>	<i>/28</i>
3rd Floor	6	<i>110.128.81.192 to 110.128.81.199</i>	<i>/29</i>
6th Floor	4	<i>110.128.81.200 to 110.128.81.207</i>	<i>/29</i>

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	160-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM

with

Class A and B Addresses

Problem 37

Part 1 of 3

The school system you are working for is using the private address of 172.32.0.0 to subnet the entire district. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different schools and offices.

At this stage of the problem you are creating sub-subnets with the third octet of the IP address. Remember which octet of the IP address you are working in.

School System Address 172.32.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
North High	2,400	<i>172.32.0.0 to 172.32.15.255</i>	<i>/20</i>
South High	2,000	<i>172.32.16.0 to 172.32.23.255</i>	<i>/21</i>
North Middle	1,200	<i>172.32.24.0 to 172.32.31.255</i>	<i>/21</i>
South Middle	1,000	<i>172.32.32.0 to 172.32.35.255</i>	<i>/22</i>
Central Elem.	550	<i>172.32.36.0 to 172.32.39.255</i>	<i>/22</i>
Southern Elem.	475	<i>172.32.40.0 to 172.32.41.255</i>	<i>/23</i>
Eastern Elem.	450	<i>172.32.42.0 to 172.32.43.255</i>	<i>/23</i>
Central Office	400	<i>172.32.44.0 to 172.32.45.255</i>	<i>/23</i>
Western Elem.	300	<i>172.32.46.0 to 172.32.47.255</i>	<i>/23</i>

Show your work for **Problem 36** in the space below.

	128	64	32	16	8	4	2	1
Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512
Number of Subnets	2	4	8	16	32	64	128	256
Binary values	128	64	32	16	8	4	2	1

<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>
2,400	172.32.0.0 to 172.32.15.255	/20
2,000	172.32.16.0 to 172.32.23.255	/21
1,200	172.32.24.0 to 172.32.31.255	/21
1,000	172.32.32.0 to 172.32.35.255	/22
550	172.32.36.0 to 172.32.39.255	/22
475	172.32.40.0 to 172.32.41.255	/23
450	172.32.42.0 to 172.32.43.255	/23
400	172.32.44.0 to 172.32.45.255	/23
300	172.32.46.0 to 172.32.47.255	/23

VLSM

with

Class A and B Addresses

Problem 37

Part 2 of 3

Eastern Elementary has been given 512 hosts, with the address range of 172.32.42.0 / 21 (255.255.248.0).

Based on the information below supply the required address ranges and subnet masks for each school area. Use the **Class C** address chart to break down the sub-subnetworks.

Hint:

Another way to look at this problem is to see that with the third octet range of 42 to 43 you have access to 2 groups of 255 addresses (172.32.42.0 and 172.32.43.0). Think in terms of having two Class C VLSM charts.

Eastern Elementary School
Address Range 172.32.42.0 to 172.32.43.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Students	250	<i>172.32.42.0 to 172.32.42.255</i>	<i>/24</i>
Printers	45	<i>172.32.43.0 to 172.32.43.63</i>	<i>/26</i>
Staff	40	<i>172.32.43.64 to 172.32.43.127</i>	<i>/26</i>
Network Devices	25	<i>172.32.43.128 to 172.32.43.159</i>	<i>/27</i>
Administrative	12	<i>172.32.43.160 to 172.32.43.175</i>	<i>/28</i>

Show your work for Problem 36 in the space below.

[illegible][illegible]

VLSM

with

Class A and B Addresses

Problem 37

Part 3 of 3

South High in part 1 of this problem has been given 2,048 hosts, with the address range of 172.32.16.0 / 21 (255.255.248.0).

Based on the information below supply the required address ranges and subnet masks for each school area. Use both the **Class B** and **Class C** address charts to break down the sub-subnetwork addresses for the different areas of the network.

Hint:

With this problem you are creating sub-subnets with both the third and fourth octets of the IP address. You may need to use the Class B VLSM chart for the *Students* addressing information. All the other addresses will be using the Class C VLSM chart. Another way to look at this problem is to see that with the third octet range of 16 to 23 you have access to 8 groups of 255 addresses or eight Class C VLSM charts.

South High School

Address Range 172.32.16.0 to 172.32.23.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Students	1,000	172.32.16.0 to 172.32.19.255	/22
Network Devices	250	172.32.20.0 to 172.32.20.255	/24
Printers	200	172.32.21.0 to 172.32.21.255	/24
Staff	150	172.32.22.0 to 172.32.22.255	/24
Administrative	50	172.32.23.0 to 172.32.23.63	/26

Show your work for Problem 36 in the space below.

Reminder - Keep within the South High School address range.

		1000						150 200 250			50						
Number of Hosts	1	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1

<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>
1,000	172.32.16.0 to 172.32.19.255	/22
250	172.32.20.0 to 172.32.20.255	/24
200	172.32.21.0 to 172.32.21.255	/24
150	172.32.22.0 to 172.32.22.255	/24
50	172.32.23.0 to 172.32.23.63	/26

VLSM

with

Class A and B Addresses

Problem 38

Part 1 of 4

Use the **Class A** address chart to break down the address for different business customers by country. At this stage of this problem you are creating subnets in the second octet of the IP address.

Addresses 75.0.0.0

Customer Name	Number of Addresses	Address Range	CIDR
United States	6.5 million	<i>75.0.0.0 to 75.127.255.255</i>	<i>/9</i>
China	4 million	<i>75.128.0.0 to 75.191.255.255</i>	<i>/10</i>
Japan	1 million	<i>75.192.0.0 to 75.207.255.255</i>	<i>/12</i>
Germany	500,000	<i>75.208.0.0 to 75.215.255.255</i>	<i>/13</i>
Russia	455,000	<i>75.216.0.0 to 75.223.255.255</i>	<i>/13</i>
Australia	450,000	<i>75.224.0.0 to 75.231.255.255</i>	<i>/13</i>
Brazil	125,000	<i>75.232.0.0 to 75.233.255.255</i>	<i>/15</i>
Canda	90,000	<i>75.234.0.0 to 75.235.255.255</i>	<i>/15</i>
Denmark	88,000	<i>75.236.0.0 to 75.237.255.255</i>	<i>/15</i>

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31	16-23	8-11	4-5
					24-31	12-15	6-7
			32-63	32-47	32-39	16-19	8-9
					40-47	20-23	10-11
				48-63	48-55	24-27	12-13
					56-63	28-31	14-15
			64-127	64-79	64-71	32-35	16-17
					72-79	36-39	18-19
				80-95	80-87	40-43	20-21
					88-95	44-47	22-23
				96-111	96-103	48-49	24-25
					104-111	50-51	26-27
				112-127	112-119	52-55	28-29
					120-127	56-59	30-31
			128-191	128-143	128-135	60-63	32-33
					136-143	64-67	34-35
				144-159	144-151	68-71	36-37
					152-159	72-75	38-39
				160-191	160-167	76-79	40-41
					168-175	80-83	42-43
					176-183	84-87	44-45
				176-191	184-191	88-91	46-47
					192-199	92-95	48-49
			192-255	192-207	192-199	96-99	50-51
					200-207	100-103	52-53
				208-223	208-215	104-107	54-55
					216-223	108-111	56-57
				224-239	224-231	112-115	58-59
					232-239	116-119	60-61
				240-255	240-247	120-123	62-63
					248-255	124-127	64-65
						128-131	66-67
						132-135	68-69
						136-139	70-71
						140-143	72-73
						144-147	74-75
						148-151	76-77
						152-155	78-79
						156-159	80-81
						160-163	82-83
						164-167	84-85
						168-171	86-87
						172-175	88-89
						176-179	90-91
						180-183	92-93
						184-187	94-95
						188-191	96-97
						192-195	98-99
						196-199	100-101
						200-203	102-103
						204-207	104-105
						208-211	106-107
						212-215	108-109
						216-219	110-111
						220-223	112-113
						224-227	114-115
						228-231	116-117
						232-235	118-119
						236-239	120-121
						240-243	122-123
						244-247	124-125
						248-251	126-127
						252-255	128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
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							180-181
							182-183
							184-185
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							192-193
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							196-197
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							200-201
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							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 2 of 4

The United States customers have a total of 8,388,608 addresses. Use the **Class A** address chart to break down the sub-subnetwork addresses for their different areas. At this stage of this problem you are creating sub-subnets in the second octet of the IP address.

Addresses Range: 75.0.0.0 to 75.127.255.255

Customer Name	Number of Addresses	Address Range	CIDR
Client #1	1,950,000	75.0.0.0 to 75.31.255.255	/11
Client #2	1,000,000	75.32.0.0 to 75.47.255.255	/12
Client #3	950,000	75.48.0.0 to 75.63.255.255	/12
Client #4	700,000	75.64.0.0 to 75.79.255.255	/12
Client #5	550,000	75.80.0.0 to 75.95.255.255	/12
Client #6	500,000	75.96.0.0 to 75.103.255.255	/13
Client #7	450,000	75.104.0.0 to 75.111.255.255	/13

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts			
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1			
					8-15	4-7	2-3			
				16-31	16-23	8-11	4-5			
					24-31	12-15	6-7			
					32-63	32-39	16-19	8-9		
						40-47	20-23	10-11		
			48-63			48-55	24-27	12-13		
				56-63		28-31	14-15			
			64-127	64-95	64-79	64-71	32-35	16-17		
						72-79	36-39	18-19		
					80-95	80-87	40-43	20-21		
						88-95	44-47	22-23		
					96-127	96-111	96-103	48-51	24-25	
		104-111					52-55	26-27		
		112-127		112-119		56-59	28-29			
				120-127		60-63	30-31			
				128-255		128-191	128-159	128-135	64-67	32-33
								136-143	68-71	34-35
		144-159			144-151			72-75	36-37	
					152-159			76-79	38-39	
		160-191	160-175		160-167			80-83	40-41	
					168-175			84-87	42-43	
			176-191		176-183		88-91	44-45		
					184-191		92-95	46-47		
					192-255		192-223	192-207	192-199	96-99
	200-207								100-103	50-51
	208-223							208-215	104-107	52-53
		216-223	108-111					54-55		
		224-255	224-239					224-231	112-115	56-57
						232-239		116-119	58-59	
	240-255		240-247			120-123	60-61			
			248-255			124-127	62-63			
			192-255			192-223	192-207	192-199	64-65	64-65
								200-207	66-67	66-67
	208-223	208-215					68-69	68-69		
		216-223					70-71	70-71		
		224-255			224-239		224-231	72-73	72-73	
							232-239	74-75	74-75	
	240-255				240-247	76-77	76-77			
					248-255	78-79	78-79			
					192-255	192-223	192-207	192-199	80-81	80-81
				200-207				82-83	82-83	
	208-223	208-215		84-85			84-85			
		216-223		86-87			86-87			
		224-255	224-239	224-231			88-89	88-89		
				232-239			90-91	90-91		
	240-255		240-247	92-93		92-93				
			248-255	94-95		94-95				
			192-255	192-223		192-207	192-199	96-97	96-97	
							200-207	98-99	98-99	
	208-223	208-215				100-101	100-101			
		216-223				102-103	102-103			
		224-255			224-239	224-231	104-105	104-105		
						232-239	106-107	106-107		
	240-255			240-247	108-109	108-109				
				248-255	110-111	110-111				
				192-255	192-223	192-207	192-199	112-113	112-113	
							200-207	114-115	114-115	
	208-223	208-215				116-117	116-117			
		216-223				118-119	118-119			
		224-255	224-239			224-231	120-121	120-121		
						232-239	122-123	122-123		
	240-255		240-247		124-125	124-125				
			248-255		126-127	126-127				
			192-255		192-223	192-207	192-199	128-129	128-129	
							200-207	130-131	130-131	
	208-223	208-215				132-133	132-133			
		216-223				134-135	134-135			
		224-255		224-239		224-231	136-137	136-137		
						232-239	138-139	138-139		
	240-255			240-247	140-141	140-141				
				248-255	142-143	142-143				
				192-255	192-223	192-207	192-199	144-145	144-145	
							200-207	146-147	146-147	
	208-223	208-215				148-149	148-149			
		216-223				150-151	150-151			
		224-255	224-239			224-231	152-153	152-153		
						232-239	154-155	154-155		
	240-255		240-247		156-157	156-157				
			248-255		158-159	158-159				
			192-255		192-223	192-207	192-199	160-161	160-161	
							200-207	162-163	162-163	
	208-223	208-215				164-165	164-165			
		216-223				166-167	166-167			
		224-255		224-239		224-231	168-169	168-169		
						232-239	170-171	170-171		
	240-255			240-247	172-173	172-173				
				248-255	174-175	174-175				
				192-255	192-223	192-207	192-199	176-177	176-177	
							200-207	178-179	178-179	
	208-223	208-215				180-181	180-181			
		216-223				182-183	182-183			
		224-255	224-239			224-231	184-185	184-185		
						232-239	186-187	186-187		
	240-255		240-247		188-189	188-189				
			248-255		190-191	190-191				
			192-255		192-223	192-207	192-199	192-193	192-193	
							200-207	194-195	194-195	
	208-223	208-215				196-197	196-197			
		216-223				198-199	198-199			
		224-255		224-239		224-231	200-201	200-201		
						232-239	202-203	202-203		
	240-255			240-247	204-205	204-205				
				248-255	206-207	206-207				
				192-255	192-223	192-207	192-199	208-209	208-209	
							200-207	210-211	210-211	
	208-223	208-215				212-213	212-213			
		216-223				214-215	214-215			
		224-255	224-239			224-231	216-217	216-217		
						232-239	218-219	218-219		
	240-255		240-247		220-221	220-221				
			248-255		222-223	222-223				
			192-255		192-223	192-207	192-199	224-225	224-225	
							200-207	226-227	226-227	
	208-223	208-215				228-229	228-229			
		216-223				230-231	230-231			
		224-255		224-239		224-231	232-233	232-233		
						232-239	234-235	234-235		
	240-255			240-247	236-237	236-237				
				248-255	238-239	238-239				
				192-255	192-223	192-207	192-199	240-241	240-241	
							200-207	242-243	242-243	
	208-223	208-215				244-245	244-245			
		216-223				246-247	246-247			
		224-255	224-239			224-231	248-249	248-249		
						232-239	250-251	250-251		
	240-255		240-247		252-253	252-253				
248-255			254-255		254-255					

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 3 of 4

Client #7 has a total of 524,288 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for their different clients. At this stage of this problem you are creating sub-subnets in the third or forth octet of the IP address.

Hint: Another way to look at this problem is to see that with the second octet range of 104 to 111 you have access to 8 groups of 65,536 addresses or 8 Class B VLSM charts.

ISP Addresses 75.104.0.0 to 75.111.255.255

Customer Name	Number of Addresses	Address Range	CIDR
Office #1	60,000	<i>75.104.0.0 to 75.104.255.255</i>	<i>/16</i>
Office #2	45,000	<i>75.105.0.0 to 75.105.255.255</i>	<i>/16</i>
Office #3	30,000	<i>75.106.0.0 to 75.106.127.255</i>	<i>/17</i>
Office #4	24,000	<i>75.106.128.0 to 75.106.255.255</i>	<i>/17</i>
Office #5	15,000	<i>75.107.0.0 to 75.107.63.255</i>	<i>/18</i>
Office #6	10,000	<i>75.107.64.0 to 75.107.127.255</i>	<i>/18</i>
Office #7	8,000	<i>75.107.128.0 to 75.107.159.255</i>	<i>/19</i>
Office #8	2,000	<i>75.107.160.0 to 75.107.167.255</i>	<i>/21</i>
Office #9	1,000	<i>75.107.168.0 to 75.107.171.255</i>	<i>/22</i>

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
							6-7
							8-9
							10-11
				16-31		12-15	12-13
						16-19	14-15
					16-23	20-23	16-17
							18-19
					24-31	24-27	20-21
						28-31	22-23
			32-63	32-47		32-35	24-25
						36-39	26-27
					40-47	40-43	28-29
						44-47	30-31
						48-51	32-33
						52-55	34-35
				48-63		56-59	36-37
						60-63	38-39
		64-127	64-95	64-79	64-71	64-67	40-41
						68-71	42-43
					72-79	72-75	44-45
						76-79	46-47
			80-95	80-87		80-83	48-49
						84-87	50-51
					88-95	88-91	52-53
						92-95	54-55
			96-127	96-103		96-99	56-57
						100-103	58-59
					104-111	104-107	60-61
						108-111	62-63
				112-127		112-115	64-65
						116-119	66-67
					120-127	120-123	68-69
						124-127	70-71
	128-255	128-191	128-159	128-143		128-131	72-73
						132-135	74-75
					136-143	136-139	76-77
						140-143	78-79
			144-159	144-151		144-147	80-81
						148-151	82-83
					152-159	152-155	84-85
						156-159	86-87
			160-191	160-175	160-167	160-163	88-89
						164-167	90-91
					168-175	168-171	92-93
						172-175	94-95
				176-191	176-183	176-179	96-97
						180-183	98-99
					184-191	184-187	100-101
						188-191	102-103
		192-255	192-223	192-207	192-199	192-195	104-105
						196-199	106-107
					200-207	200-203	108-109
						204-207	110-111
			208-223	208-215		208-211	112-113
						212-215	114-115
					216-223	216-219	116-117
						220-223	118-119
			224-255	224-239	224-231	224-227	120-121
						228-231	122-123
					232-239	232-235	124-125
						236-239	126-127
				240-255	240-247	240-243	128-129
						244-247	130-131
					248-255	248-251	132-133
						252-255	134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
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							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 4 of 4

Office #7 from part 3 of 4 has a total of 8,192 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different branch offices. At this stage of this problem you are creating sub-subnets in the third octet of the IP address.

Hint: Remember that the range of this problem is between 128 and 159 in the third octet. Your subnetting will start in the middle of the chart not at the top for this range.

ISP Addresses 75.107.128.0 to 75.107.159.255

Customer Name	Number of Addresses	Address Range	CIDR
Branch #1	4,000	<i>75.107.128.0 to 75.107.143.255</i>	<i>/20</i>
Branch #2	2,000	<i>75.107.144.0 to 75.107.151.255</i>	<i>/21</i>
Branch #3	1,000	<i>75.107.152.0 to 75.107.155.255</i>	<i>/22</i>
Branch #4	500	<i>75.107.156.0 to 75.107.157.255</i>	<i>/23</i>
Branch #5	450	<i>75.107.158.0 to 75.107.159.255</i>	<i>/23</i>

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
			16-31	16-31	16-23	16-19	8-9
						20-23	10-11
					24-31	24-27	12-13
						28-31	14-15
			32-63	32-47	32-39	32-35	16-17
						36-39	18-19
					40-47	40-43	20-21
						44-47	22-23
				48-63	48-55	48-51	24-25
						52-55	26-27
					56-63	56-59	28-29
						60-63	30-31
		64-127	64-95	64-79	64-71	64-67	32-33
						68-71	34-35
					72-79	72-75	36-37
						76-79	38-39
			80-95	80-87	80-83	80-81	40-41
						84-87	42-43
					88-95	88-91	44-45
						92-95	46-47
			96-127	96-103	96-99	96-97	48-49
						100-103	50-51
					104-111	104-107	52-53
						108-111	54-55
				112-127	112-119	112-115	56-57
						116-119	58-59
					120-127	120-123	60-61
						124-127	62-63
	128-255	128-191	See Hint 128-159	128-143	128-135	128-131	64-65
						132-135	66-67
					136-143	136-139	68-69
						140-143	70-71
			144-159	144-151	144-147	144-145	72-73
						148-151	74-75
					152-159	152-155	76-77
						156-159	78-79
			160-191	160-175	160-167	160-163	80-81
						164-167	82-83
					168-175	168-171	84-85
						172-175	86-87
				176-191	176-183	176-179	88-89
						180-183	90-91
					184-191	184-187	92-93
						188-191	94-95
		192-255	192-223	192-207	192-199	192-195	96-97
						196-199	98-99
					200-207	200-203	100-101
						204-207	102-103
				208-223	208-215	208-211	104-105
						212-215	106-107
					216-223	216-219	108-109
						220-223	110-111
			224-255	224-231	224-227	224-223	112-113
						228-231	114-115
					232-239	232-235	116-117
						236-239	118-119
				240-255	240-247	240-243	120-121
						244-247	122-123
					248-255	248-251	124-125
						252-255	126-127

Reference Charts and Support Materials

Class A Addresses
VLSM Chart 8-15 Bits (2nd octet)

[illegible]

Class B Addresses
VLSM Chart 16-23 Bits (3rd octet)

16 161-200 16,100 hours	17 161-200 17,700 hours	18 161-200 18,300 hours	19 161-200 19,000 hours	20 161-200 20,000 hours	21 161-200 21,000 hours	22 161-200 22,000 hours	23 161-200 23,000 hours
				0-15	0-15	0-15	0-15
					16-30	16-30	16-30
					31-45	31-45	31-45
					46-60	46-60	46-60
					61-75	61-75	61-75
					76-90	76-90	76-90
					91-105	91-105	91-105
					106-120	106-120	106-120
					121-135	121-135	121-135
					136-150	136-150	136-150
					151-165	151-165	151-165
					166-180	166-180	166-180
					181-195	181-195	181-195
					196-210	196-210	196-210
					211-225	211-225	211-225
					226-240	226-240	226-240
					241-255	241-255	241-255
					256-270	256-270	256-270
					271-285	271-285	271-285
					286-300	286-300	286-300
					301-315	301-315	301-315
					316-330	316-330	316-330
					331-345	331-345	331-345
					346-360	346-360	346-360
					361-375	361-375	361-375
					376-390	376-390	376-390
					391-405	391-405	391-405
					406-420	406-420	406-420
					421-435	421-435	421-435
					436-450	436-450	436-450
					451-465	451-465	451-465
					466-480	466-480	466-480
					481-495	481-495	481-495
					496-510	496-510	496-510
					511-525	511-525	511-525
					526-540	526-540	526-540
					541-555	541-555	541-555
					556-570	556-570	556-570
					571-585	571-585	571-585
					586-600	586-600	586-600
					601-615	601-615	601-615
					616-630	616-630	616-630
					631-645	631-645	631-645
					646-660	646-660	646-660
					661-675	661-675	661-675
					676-690	676-690	676-690
					691-705	691-705	691-705
					706-720	706-720	706-720
					721-735	721-735	721-735
					736-750	736-750	736-750
					751-765	751-765	751-765
					766-780	766-780	766-780
					781-795	781-795	781-795
					796-810	796-810	796-810
					811-825	811-825	811-825
					826-840	826-840	826-840
					841-855	841-855	841-855
					856-870	856-870	856-870
					871-885	871-885	871-885
					886-900	886-900	886-900
					901-915	901-915	901-915
					916-930	916-930	916-930
					931-945	931-945	931-945
					946-960	946-960	946-960
					961-975	961-975	961-975
					976-990	976-990	976-990
					991-1005	991-1005	991-1005
					1006-1020	1006-1020	1006-1020
					1021-1035	1021-1035	1021-1035
					1036-1050	1036-1050	1036-1050
					1051-1065	1051-1065	1051-1065
					1066-1080	1066-1080	1066-1080
					1081-1095	1081-1095	1081-1095
					1096-1110	1096-1110	1096-1110
					1111-1125	1111-1125	1111-1125
					1126-1140	1126-1140	1126-1140
					1141-1155	1141-1155	1141-1155
					1156-1170	1156-1170	1156-1170
					1171-1185	1171-1185	1171-1185
					1186-1200	1186-1200	1186-1200
					1201-1215	1201-1215	1201-1215
					1216-1230	1216-1230	1216-1230
					1231-1245	1231-1245	1231-1245
					1246-1260	1246-1260	1246-1260
					1261-1275	1261-1275	1261-1275
					1276-1290	1276-1290	1276-1290
					1291-1305	1291-1305	1291-1305
					1306-1320	1306-1320	1306-1320
					1321-1335	1321-1335	1321-1335
					1336-1350	1336-1350	1336-1350
					1351-1365	1351-1365	1351-1365
					1366-1380	1366-1380	1366-1380
					1381-1395	1381-1395	1381-1395
					1396-1410	1396-1410	1396-1410
					1411-1425	1411-1425	1411-1425
					1426-1440	1426-1440	1426-1440
					1441-1455	1441-1455	1441-1455
					1456-1470	1456-1470	1456-1470
					1471-1485	1471-1485	1471-1485
					1486-1500	1486-1500	1486-1500
					1501-1515	1501-1515	1501-1515
					1516-1530	1516-1530	1516-1530
					1531-1545	1531-1545	1531-1545
					1546-1560	1546-1560	1546-1560
					1561-1575	1561-1575	1561-1575
					1576-1590	1576-1590	1576-1590
					1591-1605	1591-1605	1591-1605
					1606-1620	1606-1620	1606-1620
					1621-1635	1621-1635	1621-1635
					1636-1650	1636-1650	1636-1650
					1651-1665	1651-1665	1651-1665
					1666-1680	1666-1680	1666-1680
					1681-1695	1681-1695	1681-1695
					1696-1710	1696-1710	1696-1710
					1711-1725	1711-1725	1711-1725
					1726-1740	1726-1740	1726-1740
					1741-1755	1741-1755	1741-1755
					1756-1770	1756-1770	1756-1770
					1771-1785	1771-1785	1771-1785
					1786-1800	1786-1800	1786-1800
					1801-1815	1801-1815	1801-1815
					1816-1830	1816-1830	1816-1830
					1831-1845	1831-1845	1831-1845
					1846-1860	1846-1860	1846-1860
					1861-1875	1861-1875	1861-1875
					1876-1890	1876-1890	1876-1890
					1891-1905	1891-1905	1891-1905
					1906-1920	1906-1920	1906-1920
					1921-1935	1921-1935	1921-1935
					1936-1950	1936-1950	1936-1950
					1951-1965	1951-1965	1951-1965
					1966-1980	1966-1980	1966-1980
					1981-1995	1981-1995	1981-1995
					1996-2010	1996-2010	1996-2010
					2011-2025	2011-2025	2011-2025
					2026-2040	2026-2040	2026-2040
					2041-2055	2041-2055	2041-2055
					2056-2070	2056-2070	2056-2070
					2071-2085	2071-2085	2071-2085
					2086-2100	2086-2100	2086-2100
					2101-2115	2101-2115	2101-2115
					2116-2130	2116-2130	2116-2130
					2131-2145	2131-2145	2131-2145
					2146-2160	2146-2160	2146-2160
					2161-2175	2161-2175	2161-2175
					2176-2190	2176-2190	2176-2190
					2191-2205	2191-2205	2191-2205
					2206-2220	2206-2220	2206-2220
					2221-2235	2221-2235	2221-2235
					2236-2250	2236-2250	2236-2250
					2251-2265	2251-2265	2251-2265
					2266-2280	2266-2280	2266-2280
					2281-2295	2281-2295	2281-2295
					2296-2310	2296-2310	2296-2310
					2311-2325	2311-2325	2311-2325
					2326-2340	2326-2340	2326-2340
					2341-2355	2341-2355	2341-2355
					2356-2370	2356-2370	2356-2370
					2371-2385	2371-2385	2371-2385
					2386-2400	2386-2400	2386-2400
					2401-2415	2401-2415	2401-2415
					2416-2430	2416-2430	2416-2430
					2431-2445	2431-2445	2431-2445
					2446-2460	2446-2460	2446-2460
					2461-2475	2461-2475	2461-2475
					2476-2490	2476-2490	2476-2490
					2491-2505	2491-2505	2491-2505
					2506-2520	2506-2520	2506-2520
					2521-2535	2521-2535	2521-2535
					2536-2550	2536-2550	2536-2550
					2551-2565	2551-2565	2551-2565
					2566-2580	2566-2580	2566-2580
					2581-2595	2581-2595	2581-2595
					2596-2610	2596-2610	2596-2610
					2611-2625	2611-2625	2611-2625
					2626-2640	2626-2640	2626-2640
					2641-2655	2641-2655	2641-2655
					2656-2670	2656-2670	2656-2670
					2671-2685	2671-2685	2671-2685
					2686-2700	2686-2700	2686-2700
					2701-2715	2701-2715	2701-2715
					2716-2730	2716-2730	2716-2730
					2731-2745	2731-2745	2731-2745
					2746-2760	2746-2760	2746-2760
					2761-2775	2761-2775	2761-2775
					2776-2790	2776-2790	2776-2790
					2791-2805	2791-2805	2791-2805
					2806-2820	2806-2820	2806-2820
					2821-2835	2821-2835	2821-2835
					2836-2850	2836-2850	2836-2850
					2851-2865</		

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

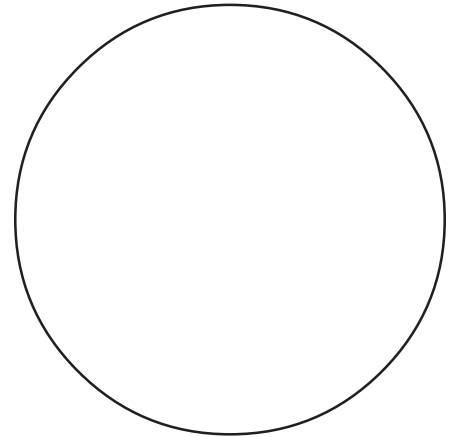
D4	D5	D6	D7	D8	D9	D10
205,206,207,208	209,210,211,212	213,214,215,216	217,218,219,220	221,222,223,224	225,226,227,228	229,230,231,232
			0-15	0-7	0-7	0-7
				8-16	8-16	8-16
		0-31	16-23	16-23	16-23	16-23
			24-31	24-31	24-31	24-31
			32-39	32-39	32-39	32-39
		32-63	40-47	40-47	40-47	40-47
			48-55	48-55	48-55	48-55
			56-63	56-63	56-63	56-63
			64-71	64-71	64-71	64-71
			72-79	72-79	72-79	72-79
		64-95	80-87	80-87	80-87	80-87
			88-95	88-95	88-95	88-95
			96-103	96-103	96-103	96-103
			104-111	104-111	104-111	104-111
			112-119	112-119	112-119	112-119
			120-127	120-127	120-127	120-127
			128-135	128-135	128-135	128-135
			136-143	136-143	136-143	136-143
			144-151	144-151	144-151	144-151
			152-159	152-159	152-159	152-159
			160-167	160-167	160-167	160-167
			168-175	168-175	168-175	168-175
			176-183	176-183	176-183	176-183
			184-191	184-191	184-191	184-191
			192-199	192-199	192-199	192-199
			200-207	200-207	200-207	200-207
			208-215	208-215	208-215	208-215
			216-223	216-223	216-223	216-223
			224-231	224-231	224-231	224-231
			232-239	232-239	232-239	232-239
			240-247	240-247	240-247	240-247
			248-255	248-255	248-255	248-255

Visualizing Subnets Using The Circle Method

The circle method is another method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the different sections of the circle you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

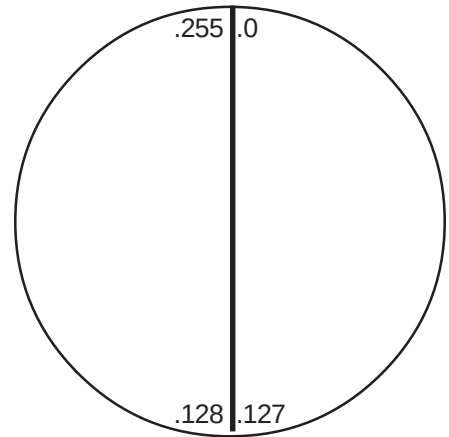
Start with a circle. The whole circle is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



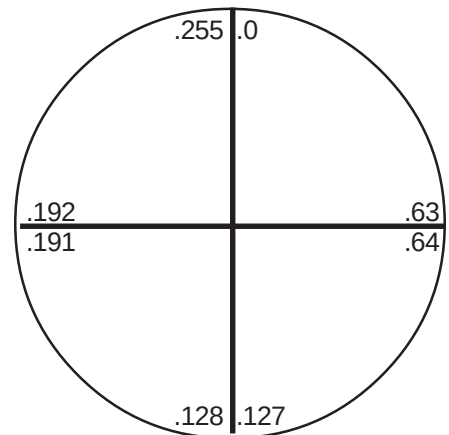
Split the circle in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



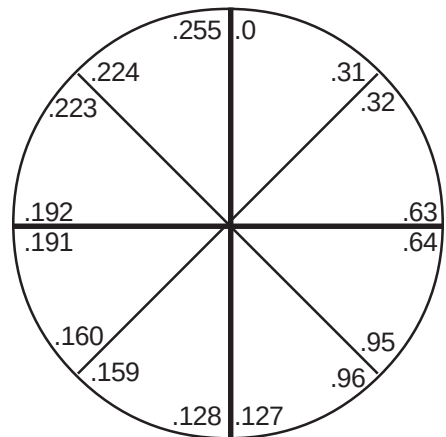
Divide the circle into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



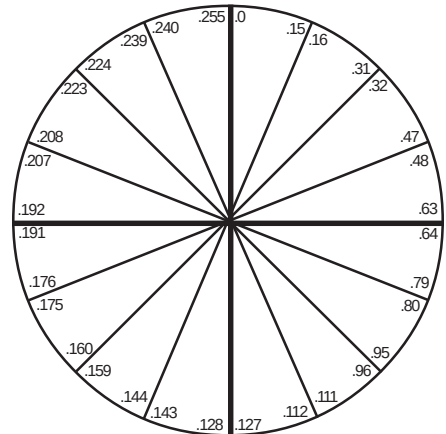
Split each quarter and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets



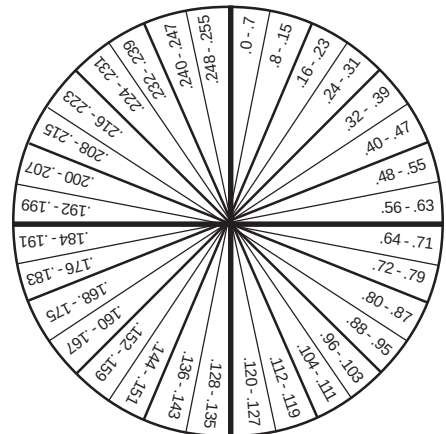
Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets



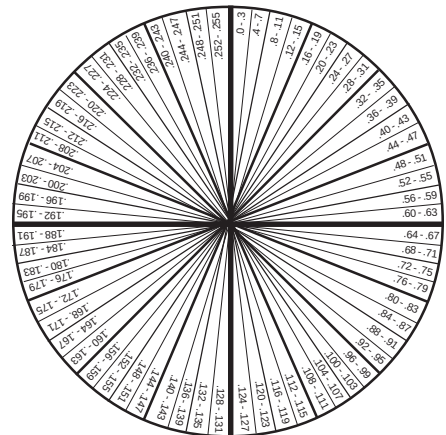
The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets



The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets

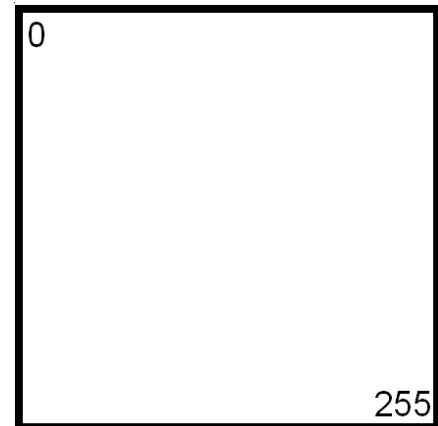


Visualizing Subnets Using The Box Method

The box method is a simple way to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

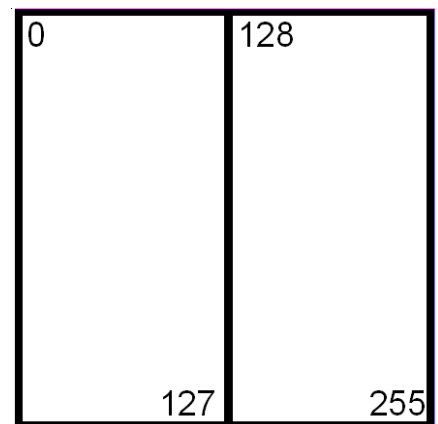
Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



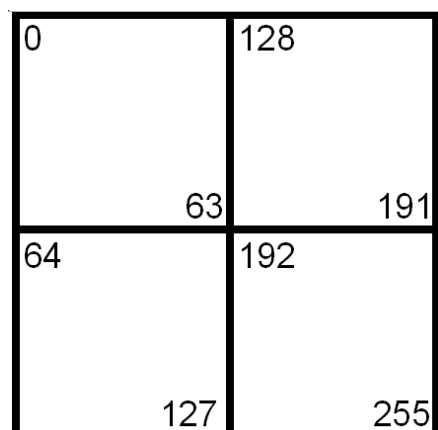
Split the box in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



Divide the box into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets

0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets

0	32	128	160
15	47	143	175
16	48	144	176
31	63	159	191
64	96	192	224
79	111	207	239
80	112	208	240
95	127	223	255

The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets

0	8	32	40	128	136	160	168
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
71	79	103	111	199	207	321	239
80	88	112	120	208	216	240	248
87	95	119	127	215	223	247	255

The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets

0	8	32	40	128	136	160	168
3	11	35	43	131	139	163	171
4	12	36	44	132	140	164	172
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
19	27	51	59	147	155	179	187
20	28	52	60	148	156	180	188
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
67	75	99	107	195	203	227	235
68	76	100	108	196	204	228	236
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
83	91	115	123	211	219	243	251
84	92	116	124	212	220	244	252
87	95	119	127	215	223	247	255

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
						16-19	8-9
						20-23	10-11
						24-27	12-13
						28-31	14-15
						32-35	16-17
						36-39	18-19
						40-43	20-21
						44-47	22-23
						48-51	24-25
						52-55	26-27
						56-59	28-29
						60-63	30-31
		64-127	32-63	32-47	32-39	32-35	32-33
						40-43	34-35
					48-55	48-51	36-37
						52-55	38-39
				48-63	56-63	56-59	40-41
						60-63	42-43
			64-95	64-79	64-71	64-67	44-45
						68-71	46-47
					72-79	72-75	48-49
						76-79	50-51
				80-95	80-87	80-83	52-53
						84-87	54-55
				96-111	88-95	88-91	56-57
						92-95	58-59
					96-103	96-99	60-61
						100-103	62-63
			96-127	104-111	104-107	104-107	64-65
						108-111	66-67
				112-127	112-119	112-115	68-69
						116-119	70-71
				120-127	120-123	120-123	72-73
						124-127	74-75
		128-255	128-159	128-143	128-135	128-131	76-77
						132-135	78-79
					136-143	136-139	80-81
						140-143	82-83
				144-159	144-151	144-147	84-85
						148-151	86-87
				160-175	160-167	160-167	88-89
					168-175	168-171	90-91
			160-191	176-191	176-183	176-179	92-93
						180-183	94-95
				184-191	184-187	184-187	96-97
						188-191	98-99
			192-255	192-207	192-199	192-195	100-101
						196-199	102-103
				200-207	200-203	200-203	104-105
						204-207	106-107
				208-215	208-211	208-211	108-109
						212-215	110-111
				216-223	216-219	216-219	112-113
						220-223	114-115
				224-231	224-227	224-227	116-117
						228-231	118-119
				232-239	232-235	232-235	120-121
						236-239	122-123
			224-255	240-247	240-243	240-243	124-125
						244-247	126-127
				248-255	248-251	248-251	128-129
						252-255	130-131

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31		8-11	4-5
						12-15	6-7
					16-23	16-19	8-9
					24-31	20-23	10-11
		32-63	32-47			24-27	12-13
					32-39	28-31	14-15
					40-47	32-35	16-17
				48-63		36-39	18-19
					48-55	40-43	20-21
					56-63	44-47	22-23
		64-127	64-95			48-51	24-25
					64-71	52-55	26-27
				80-95	72-79	56-59	28-29
					80-87	60-63	30-31
					88-95	64-67	32-33
				96-111	96-103	68-71	34-35
					104-111	72-75	36-37
			112-127			76-79	38-39
					112-119	80-83	40-41
				120-127	120-123	84-87	42-43
					124-127	88-91	44-45
	128-255	128-191	128-159	128-143		92-95	46-47
					128-135	96-99	48-49
				144-159	136-143	100-103	50-51
					144-151	104-107	52-53
					152-159	108-111	54-55
				160-191	160-167	112-115	56-57
					168-175	116-119	58-59
				176-191	176-183	120-123	60-61
					184-191	124-127	62-63
				192-207	192-199	128-131	64-65
					200-207	132-135	66-67
		192-255	192-223			136-139	68-69
					208-215	140-143	70-71
				208-223	216-223	144-147	72-73
					224-231	148-151	74-75
					232-239	152-155	76-77
				240-255	240-247	156-159	78-79
					248-255	160-163	80-81
						164-167	82-83
						168-171	84-85
						172-175	86-87
						176-179	88-89
						180-183	90-91
						184-187	92-93
						188-191	94-95
						192-195	96-97
						196-199	98-99
						200-203	100-101
						204-207	102-103
						208-211	104-105
						212-215	106-107
						216-219	108-109
						220-223	110-111
						224-227	112-113
						228-231	114-115
						232-235	116-117
						236-239	118-119
						240-243	120-121
						244-247	122-123
						248-251	124-125
						252-255	126-127
							128-129
							130-131
							132-133
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							226-227
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							230-231
							232-233
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							250-251
							252-253
							254-255

Class C Addresses

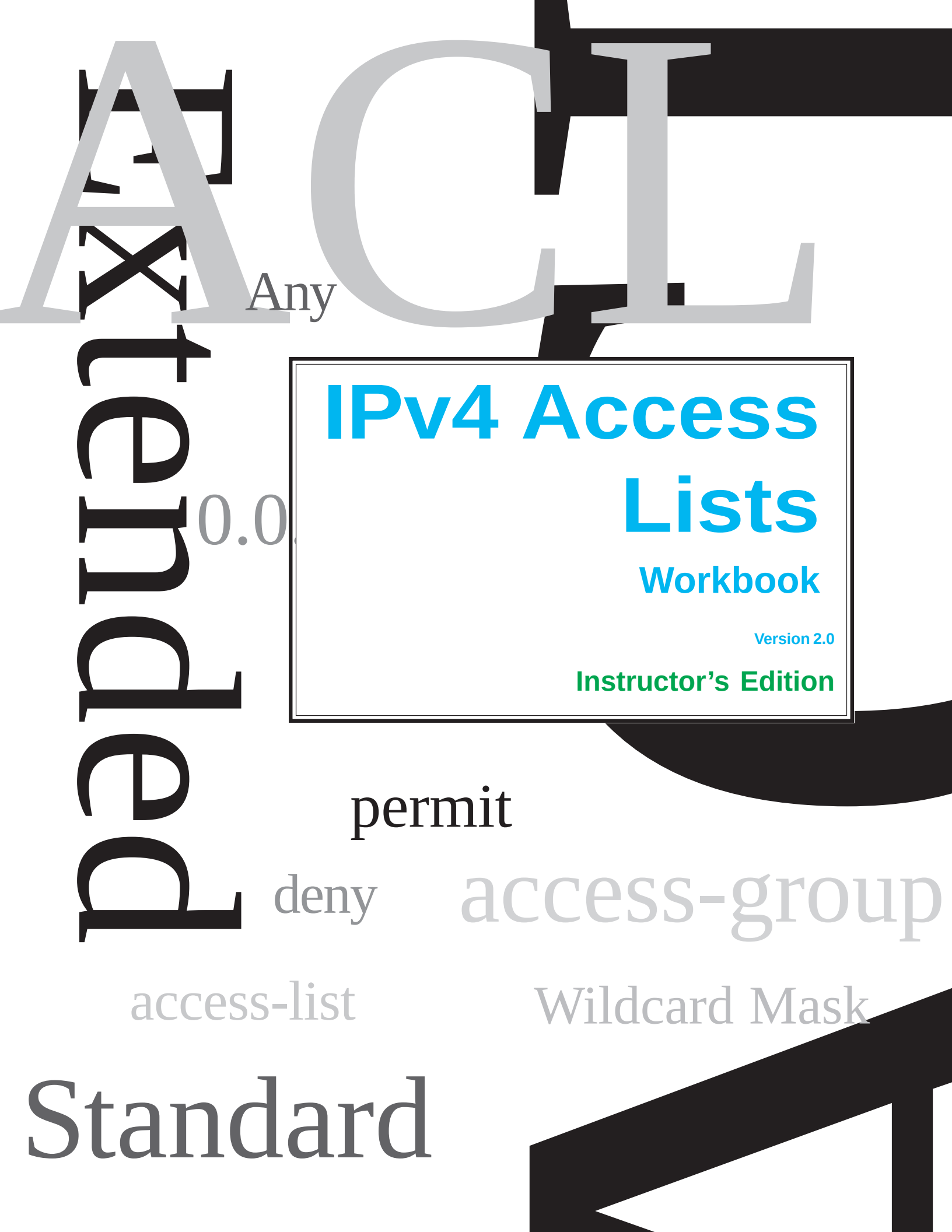
VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

Class A Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/8	0	255.0.0.0	1	16,777,216	16,777,214
/9	1	255.128.0.0	2	8,388,608	8,388,606
/10	2	255.192.0.0	4	4,194,304	4,194,302
/11	3	255.224.0.0	8	2,097,152	2,097,150
/12	4	255.240.0.0	16	1,048,576	1,048,574
/13	5	255.248.0.0	32	524,288	524,286
/14	6	255.252.0.0	64	262,144	262,142
/15	7	255.254.0.0	128	131,072	131,070
/16	8	255.255.0.0	256	65,536	65,534
/17	9	255.255.128.0	512	32,768	32,766
/18	10	255.255.192.0	1,024	16,384	16,382
/19	11	255.255.224.0	2,048	8,192	8,190
/20	12	255.255.240.0	4,096	4,096	4,094
/21	13	255.255.248.0	8,192	2,048	2,046
/22	14	255.255.252.0	16,384	1,024	1,022
/23	15	255.255.254.0	32,768	512	510
/24	16	255.255.255.0	65,536	256	254
/25	17	255.255.255.128	131,072	128	126
/26	18	255.255.255.192	262,144	64	62
/27	19	255.255.255.224	524,288	32	30
/28	20	255.255.255.240	1,048,576	16	14
/29	21	255.255.255.248	2,097,152	8	6
/30	22	255.255.255.252	4,194,304	4	2

Class B Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/16	0	255.255.0.0	1	65,536	65,534
/17	1	255.255.128.0	2	32,768	32,766
/18	2	255.255.192.0	4	16,384	16,382
/19	3	255.255.224.0	8	8,192	8,190
/20	4	255.255.240.0	16	4,096	4,094
/21	5	255.255.248.0	32	2,048	2,046
/22	6	255.255.252.0	64	1,024	1,022
/23	7	255.255.254.0	128	512	510
/24	8	255.255.255.0	256	256	254
/25	9	255.255.255.128	512	128	126
/26	10	255.255.255.192	1,024	64	62
/27	11	255.255.255.224	2,048	32	30
/28	12	255.255.255.240	4,096	16	14
/29	13	255.255.255.248	8,192	8	6
/30	14	255.255.255.252	16,384	4	2

Class C Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/24	0	255.255.255.0	1	256	254
/25	1	255.255.255.128	2	128	126
/26	2	255.255.255.192	4	64	62
/27	3	255.255.255.224	8	32	30
/28	4	255.255.255.240	16	16	14
/29	5	255.255.255.248	32	8	6
/30	6	255.255.255.252	64	4	2



IPv4 Access Lists

Workbook

Version 2.0

Instructor's Edition

permit

deny

access-group

access-list

Wildcard Mask

Standard

Access-List Numbers

IP Standard	1	to	99
IP Extended	100	to	199
Ethernet Type Code	200	to	299
Ethernet Address	700	to	799
DECnet and Extended DECnet	300	to	399
XNS	400	to	499
Extended XNS	500	to	599
Appletalk	600	to	699
48-bit MAC Addresses	700	to	799
IPX Standard	800	to	899
IPX Extended	900	to	999
IPX SAP (service advertisement protocol)	1000	to	1099
IPX SAP SPX	1000	to	1099
Extended 48-bit MAC Addresses	1100	to	1199
IPX NLSP	1200	to	1299
IP Standard, expanded range	1300	to	1999
IP Extended, expanded range	2000	to	2699
SS7 (voice)	2700	to	2999
Standard Vines	1	to	100
Extended Vines	101	to	200
Simple Vines	201	to	300
Transparent bridging (protocol type)	200	to	299
Transparent bridging (vendor type)	700	to	799
Extended Transparent bridging	1100	to	1199
Source-route bridging (protocol type)	200	to	299
Source-route bridging (vendor type)	700	to	799

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Special Thanks to Melvin Baker, Jim Dorsch, and Brent Sieling
for taking the time to check this workbook for errors, and making suggestions for improvements.

Instructors (and anyone else for that matter) please do not post the Instructors version on public websites.
When you do this you are giving everyone else worldwide the answers. Yes, students look for answers this way.
It also discourages others; myself included, from posting high quality materials.

What are Access Control Lists?

ACLs...

...are a sequential list of instructions that tell a router which packets to permit or deny.

General Access Lists Information

Access Lists...

...are read sequentially.

...are set up so that as soon as the packet matches a statement it stops comparing and permits or denies the packet.

...need to be written to take care of the most abundant traffic first.

...must be configured on your router before you can deny packets.

...can be written for all supported routed protocols; but each routed protocol must have a different ACL for each interface.

...must be applied to an interface to work.

How routers use Access Lists

(Outbound Port - Default)

- ❑ The router checks to see if the packet is routable. If it is it looks up the route in its routing table.
- ❑ The router then checks for an ACL on that outbound interface.
- ❑ If there is no ACL the router switches the packet out that interface to its destination.
- ❑ If there is an ACL the router checks the packet against the access list statements sequentially. Then permits or denies each packet as it is matched.
- ❑ If the packet does not match any statement written in the ACL it is denied because there is an implicit “deny any” statement at the end of every ACL.

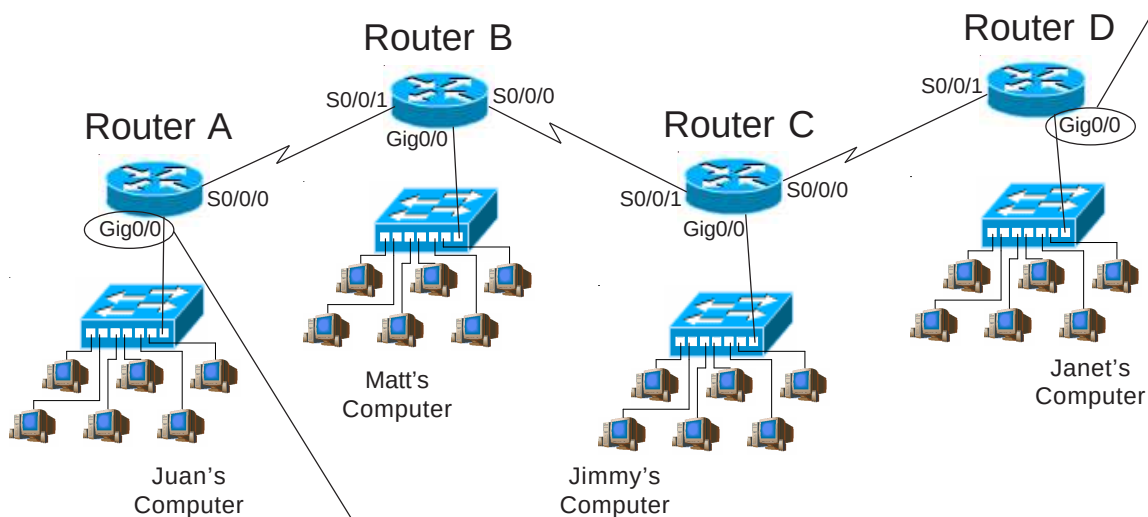
Standard Access Lists

Standard Access Lists...

- ...are numbered from 1 to 99 or 1300 to 1999.
- ...filter (permit or deny) only source addresses.
- ...do not have any destination information so it must be placed as close to the destination as possible.
- ...work at layer 3 of the OSI model.

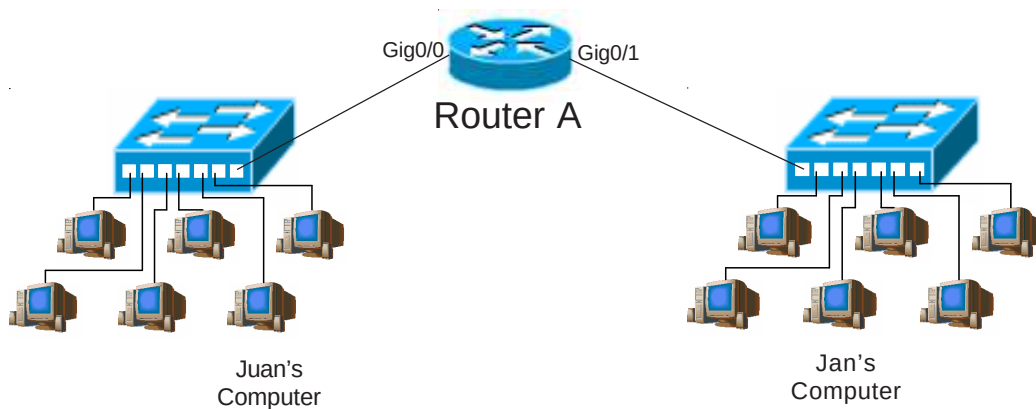
Why standard ACLs are placed close to the destination.

If you want to block traffic from Juan's computer from reaching Janet's computer with a standard access list you would place the ACL close to the destination on Router D, interface Gig0/0. Since it's using only the source address to permit or deny packets the ACL here will not effect packets reaching Routers B, or C.

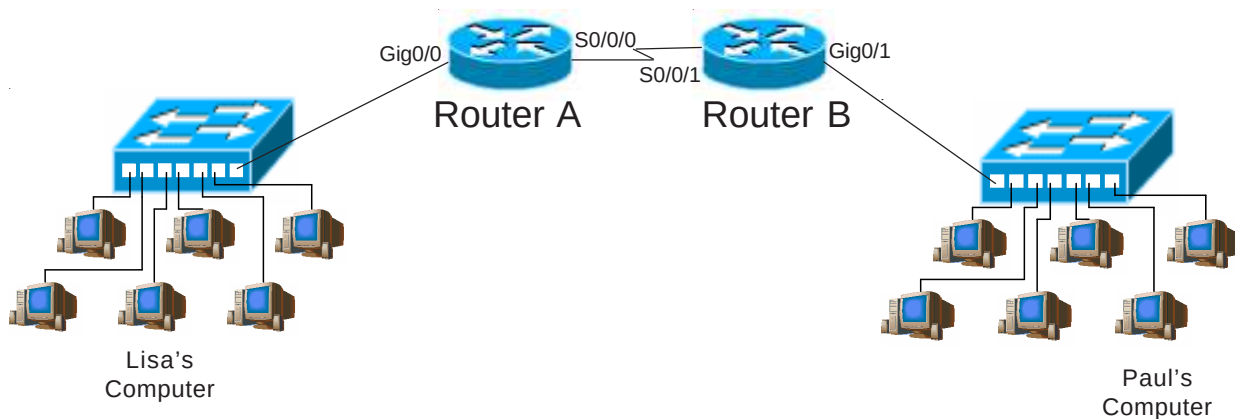


If you place the ACL on router A to block traffic to Router D it will also block all packets going to Routers B, and C; because all the packets will have the same source address.

Standard Access List Placement Sample Problems



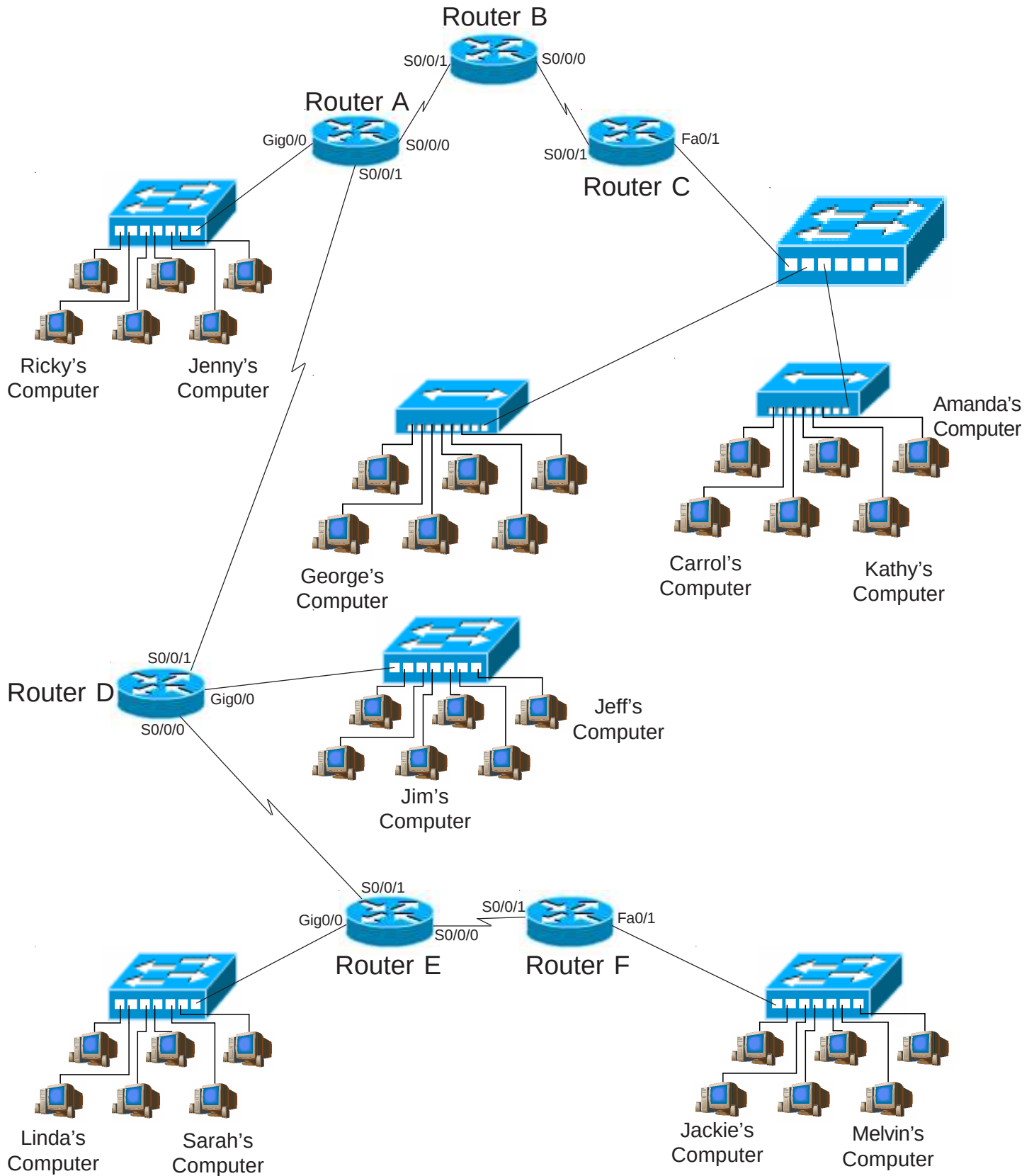
In order to permit packets from Juan's computer to arrive at Jan's computer you would place the standard access list at router interface Gig0/1.



Lisa has been sending unnecessary information to Paul. Where would you place the standard ACL to deny all traffic from Lisa to Paul?
Router Name Router B Interface Gig0/1

Where would you place the standard ACL to deny traffic from Paul to Lisa?
Router Name Router A Interface Gig0/0

Standard Access List Placement



Standard Access List Placement

1. Where would you place a standard access list to permit traffic from Ricky's computer to reach Jeff's computer?

Router Name Router D
Interface Gig0/0

2. Where would you place a standard access list to deny traffic from Melvin's computer from reaching Jenny's computer?

Router Name Router A
Interface Gig0/0

3. Where would you place a standard access list to deny traffic to Carrol's computer from Sarah's computer?

Router Name Router C
Interface Fa0/1

4. Where would you place a standard access list to permit traffic to Ricky's computer from Jeff's computer?

Router Name Router A
Interface Gig0/0

5. Where would you place a standard access list to deny traffic from Amanda's computer from reaching Jeff and Jim's computer?

Router Name Router D
Interface Gig0/0

6. Where would you place a standard access list to permit traffic from Jackie's computer to reach Linda's computer?

Router Name Router E
Interface Gig0/0

7. Where would you place a standard access list to permit traffic from Ricky's computer to reach Carrol and Amanda's computer?

Router Name Router C
Interface FA0/1

8. Where would you place a standard access list to deny traffic to Jenny's computer from Jackie's computer?

Router Name Router A
Interface Gig0/0

9. Where would you place a standard access list to permit traffic from George's computer to reach Linda and Sarah's computer?

Router Name Router E
Interface Gig0/0

10. Where would you place an ACL to deny traffic from Jeff's computer from reaching George's computer?

Router Name Router C
Interface FA0/1

11. Where would you place a standard access list to deny traffic to Sarah's computer from Ricky's computer?

Router Name Router E
Interface Gig0/0

12. Where would you place an ACL to deny traffic from Linda's computer from reaching Jackie's computer?

Router Name Router F
Interface FA0/1

Extended Access Lists

Extended Access Lists...

...are numbered from 100 to 199 or 2000 to 2699.

...filter (permit or deny) based on the:

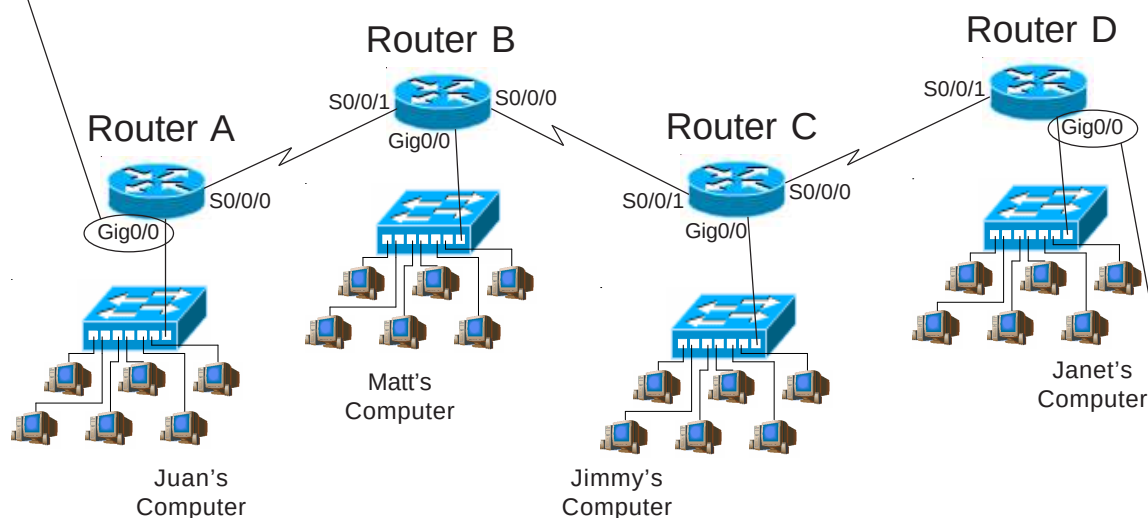
- source address
- destination address
- protocol
- application / port number

... are placed close to the source.

...work at both layer 3 and 4 of the OSI model.

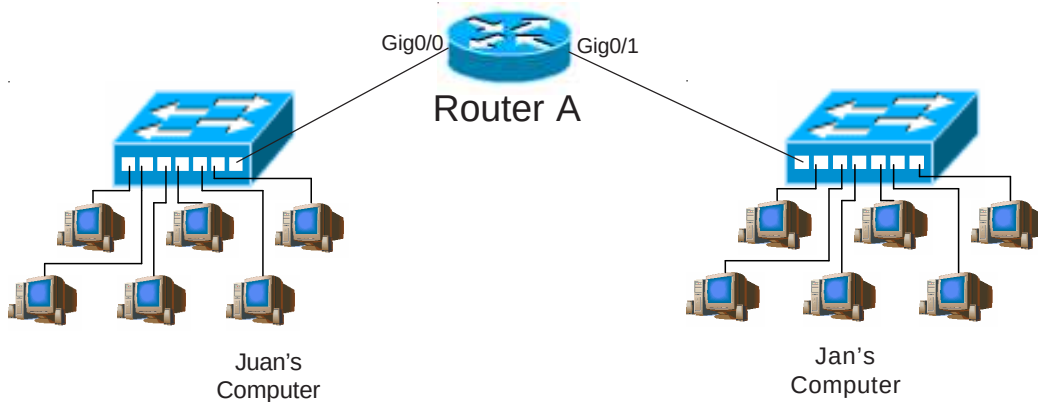
Why extended ACLs are placed close to the source.

If you want to deny traffic from Juan's computer from reaching Janet's computer with an extended access list you would place the ACL **close to the source** on Router A, interface Gig0/0. Since it can permit or deny based on the destination address it can reduce backbone overhead and not affect traffic to Routers B or C.

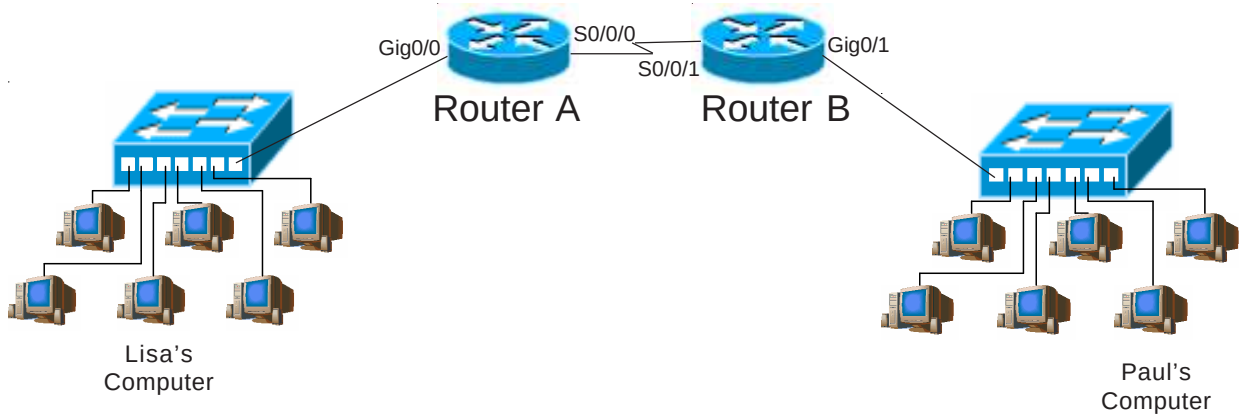


If you place the ACL on Router D to block traffic from Router A, it will work. However, Routers B and C will have to route the packet before it is finally blocked at Router D. This increases the volume of useless network traffic.

Extended Access List Placement Sample Problems



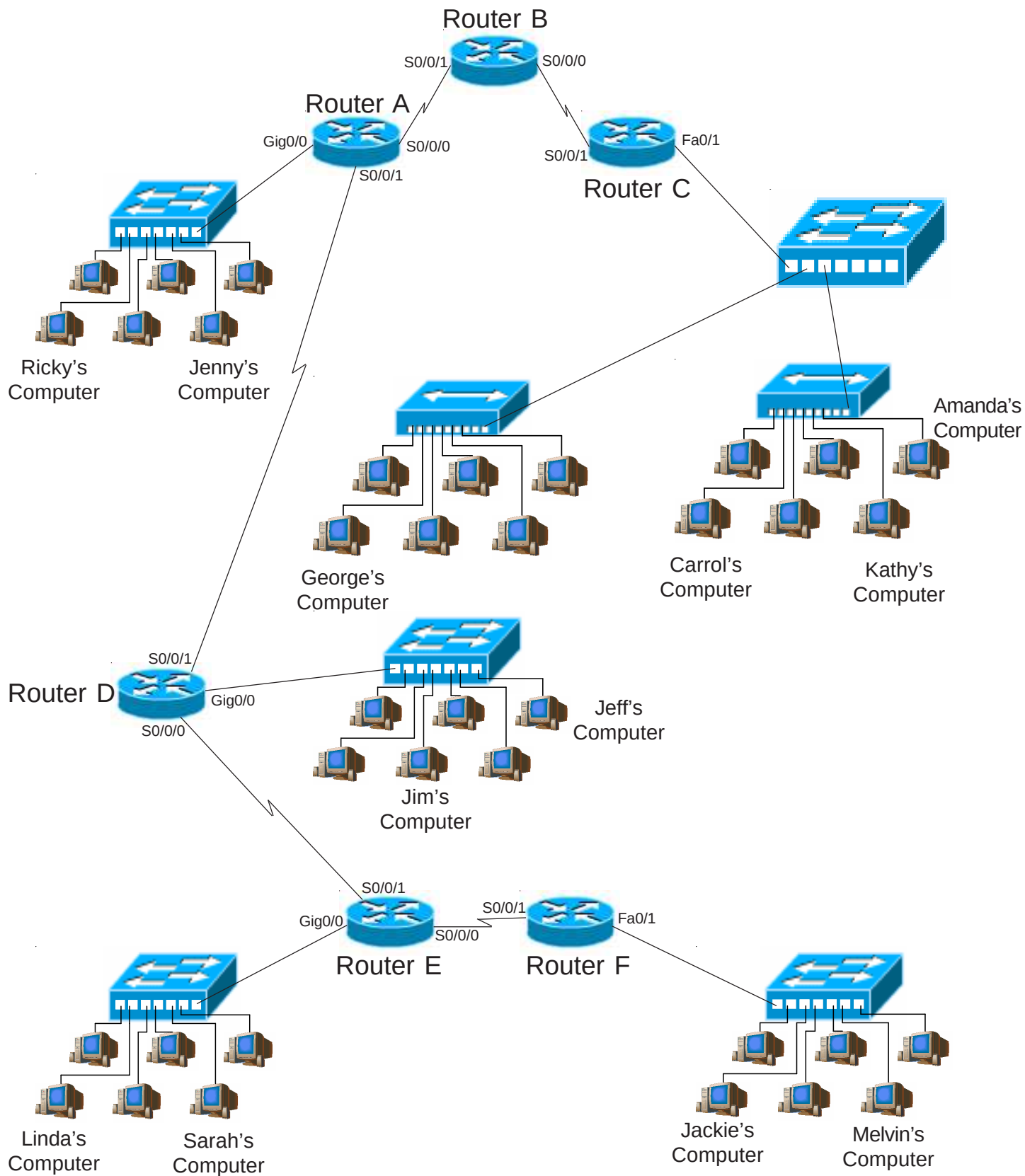
In order to permit packets from Juan's computer to arrive at Jan's computer you would place the extended access list at router interface Gig0/0.



Lisa has been sending unnecessary information to Paul. Where would you place the extended ACL to deny all traffic from Lisa to Paul?
Router Name Router A Interface Gig0/0

Where would you place the extended ACL to deny traffic from Paul to Lisa?
Router Name Router B Interface Gig0/1

Extended Access List Placement



Extended Access List Placement

1. Where would you place an ACL to deny traffic from Jeff's computer from reaching George's computer?

Router Name Router D
Interface Gig0/0

2. Where would you place an extended access list to permit traffic from Jackie's computer to reach Linda's computer?

Router Name Router F
Interface FA0/1

3. Where would you place an extended access list to deny traffic to Carrol's computer from Ricky's computer?

Router Name Router A
Interface Gig0/0

4. Where would you place an extended access list to deny traffic to Sarah's computer from Jackie's computer?

Router Name Router F
Interface Gig0/0

5. Where would you place an extended access list to permit traffic from Carrol's computer to reach Jeff's computer?

Router Name Router C
Interface FA0/1

6. Where would you place an extended access list to deny traffic from Melvin's computer from reaching Jeff and Jim's computer?

Router Name Router F
Interface FA0/1

7. Where would you place an extended access list to permit traffic from George's computer to reach Jeff's computer?

Router Name Router C
Interface FA0/1

8. Where would you place an extended access list to permit traffic from Jim's computer to reach Carrol and Amanda's computer?

Router Name Router D
Interface Gig0/0

9. Where would you place an ACL to deny traffic from Linda's computer from reaching Kathy's computer?

Router Name Router E
Interface Gig0/0

10. Where would you place an extended access list to deny traffic to Jenny's computer from Sarah's computer?

Router Name Router E
Interface Gig0/0

11. Where would you place an extended access list to permit traffic from George's computer to reach Linda and Sarah's computer?

Router Name Router C
Interface FA0/1

12. Where would you place an extended access list to deny traffic from Linda's computer from reaching Jenny's computer?

Router Name Router E
Interface Gig0/0

Choosing to Filter Incoming or Outgoing Packets

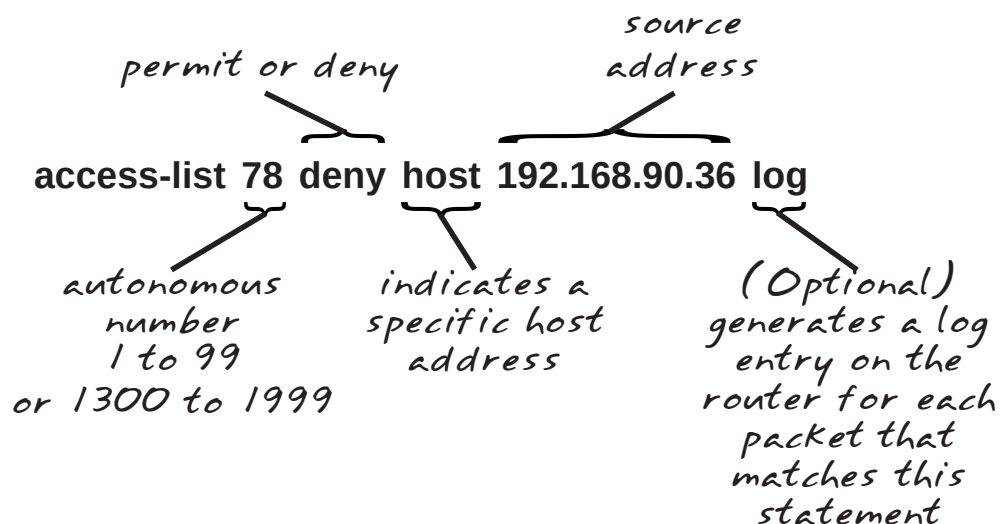
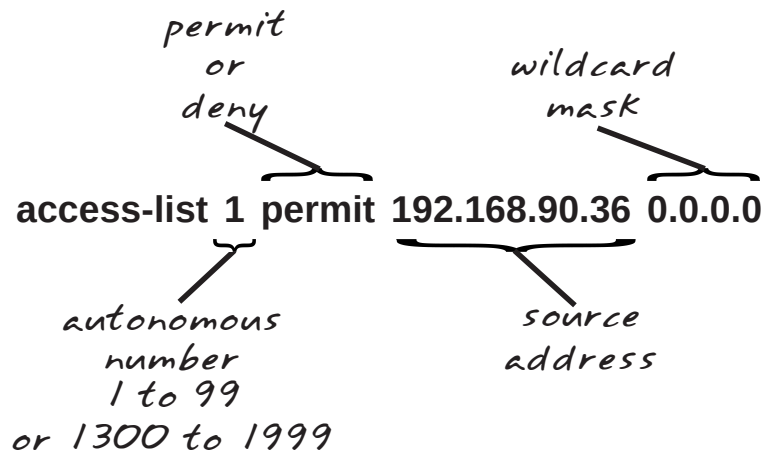
Access Lists on your incoming port...

- ...requires less CPU processing.
- ...filters and denies packets before the router has to make a routing decision.

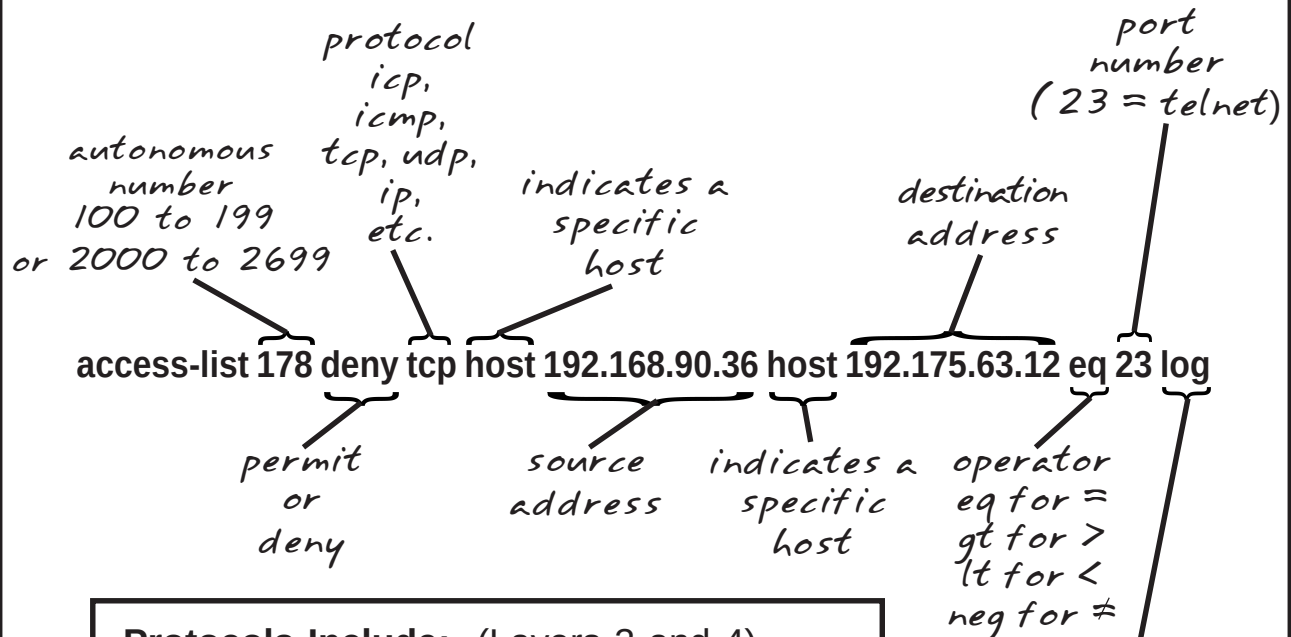
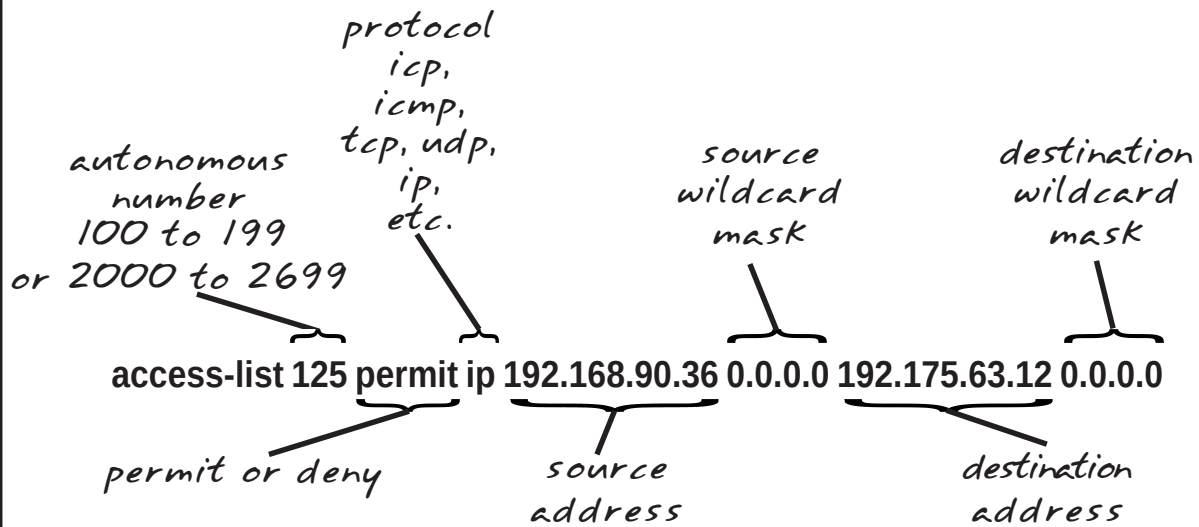
Access Lists on your outgoing port...

- ...are outbound by default unless otherwise specified.
- ...increases the CPU processing time because the routing decision is made and the packet switched to the correct outgoing port before it is tested against the ACL.

Breakdown of a Standard ACL Statement



Breakdown of an Extended ACL Statement



Protocols Include: (Layers 3 and 4)

IP	IGMP	IPINIP
TCP	GRE	OSPF
UDP	IGRP	NOS
ICMP	EIGRP	Integer 0-255

To match any internet protocol use IP.

What are Named Access Control Lists?

Named ACLs...

...are standard or extended ACLs which have an alphanumeric name instead of a number. (ie. 1-99 or 100-199)

Named Access Lists Information

Named Access Lists...

- ...identify ACLs with an intuitive name instead of a number.
- ...eliminate the limits imposed by using numbered ACLs.
(798 for standard and 799 for extended)
- ...names should be typed in all CAPS to make it easier to see.
- ...provide the ability to modify your ACLs without deleting and reloading the revised access list. It will only allow you to add statements to the end of the existing statements.
- ...are not compatible with any IOS prior to Release 11.2.
- ...can not repeat the same name on multiple ACLs.

Applying a Standard Named Access List called "GEORGE"

Write a named standard access list called "GEORGE" on Router A, interface E1 to block Melvin's computer from sending information to Kathy's computer; but will allow all other traffic.

Place the access list at:

Router Name: Router A

Interface: E1

Access-list Name: GEORGE

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# ip access-list standard GEORGE
Router(config-std-nacl)# deny host 72.16.70.35
Router(config-std-nacl)# permit any
Router(config-std-nacl)# interface gig0/1
Router(config-if)# ip access-group george out
Router(config-if)# exit
Router(config)# exit
```

Applying an extended Named Access List called "GRACIE"

Write a named extended access list called "GRACIE" on Router A, Interface E0 called "Gracie" to deny HTTP traffic intended for web server 192.168.207.27, but will permit all other HTTP traffic to reach the only the 192.168.207.0 network. Deny all other IP traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: E0
Access-list Name: GRACIE

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# ip access-list extended GRACIE
Router(config-ext-nacl)# deny tcp any host 192.168.207.27 eq www
Router(config-ext-nacl)# permit tcp any 192.168.207.0 0.0.0.255 eq www
Router(config-ext-nacl)# interface gig0/1
Router(config-if)# ip access-group gracie in
Router(config-if)# exit
Router(config)# exit
```


Choices for Using Wildcard Masks

Wildcard masks are usually set up to do one of four things:

1. Match a specific host.
2. Match an entire subnet.
3. Match a specific range.
4. Match all addresses.

1. Matching a specific host.

For standard access lists:

Access-List 10 permit 192.168.150.50 0.0.0.0

or

Access-List 10 permit 192.168.150.50 (standard ACL's
assume a 0.0.0.0 mask)

or

Access-List 10 permit host 192.168.150.50

For extended access lists:

Access-list 110 deny ip 192.168.150.50 0.0.0.0 any

or

Access-list 110 deny ip host 192.168.150.50 any

2. Matching an entire subnet

Example 1

Address: 192.168.50.0 Subnet Mask: 255.255.255.0

Access-list 25 deny 192.168.50.0 0.0.0.255

Example 2

Address: 172.16.0.0 Subnet Mask: 255.255.0.0

Access-list 12 permit 172.16.0.0 0.0.255.255

Example 3

Address: 10.0.0.0 Subnet Mask: 255.0.0.0

Access-list 125 deny udp 10.0.0.0 0.255.255.255 any

3. Match a specific range

Example 1

Address: 10.250.50.112 Subnet Mask: 255.255.255.224

255.255.255.255
Custom Subnet mask: -255.255.255.224
Wildcard: 0. 0. 0. 31

Access-list 125 permit udp 10.250.50.112 0.0.0.31 any

Example 2

Address Range: 192.168.16.0 to 192.168.16.127

192.168.16.127
-192.168.16. 0
Wildcard: 0. 0. 0.127

Access-list 125 deny ip 192.168.16.0 0.0.0.127 any
(This ACL would block the lower half of the subnet.)

Example 3

Address: 172.250.16.32 to 172.250.31.63

172.250.31. 63
-172.250.16. 32
Wildcard: 0. 0.15. 31

Access-list 125 permit ip 172.250.16.32 0.0.15.31 any

4. Match everyone.

For standard access lists:

Access-List 15 permit any

or

Access-List 15 deny 0.0.0.0 255.255.255.255

For extended access lists:

Access-List 175 permit ip any any

or

Access-List 175 deny tcp 0.0.0.0 255.255.255.255 any

Creating Wildcard Masks

- ❑ Just like a subnet mask the wildcard mask tells the router what part of the address to check or ignore. Zero (0) must match exactly, one (1) will be ignored.
- ❑ The source address can be a single address, a range of addresses, or an entire subnet.
- ❑ As a rule of thumb the wildcard mask is the inverse of the subnet mask.

Example #1:

IP Address and subnet mask: 204.100.100.0 255.255.255.0

IP Address and wildcard mask: 204.100.100.0 0.0.0.255

- ❑ All zero's (or 0.0.0.0) means the address must match exactly.

Example #2:

10.10.150.95 0.0.0.0 (This address must match exactly.)

- ❑ One's will be ignored.

Example #3:

10.10.150.95 0.0.0.255 (Any 10.10.150.0 subnet address will match.
10.10.150.0 to 10.10.150.255)

- ❑ This also works with subnets.

Example #4:

IP Address and subnet mask: 192.170.25.30 255.255.255.224

IP Address and wildcard mask: 192.170.25.30 0.0.0.31
(Subtract the subnet mask from
255.255.255.255 to create the wildcard)

Do the math... $255 - 255 = 0$ (This is the inverse of the subnet mask.)
 $255 - 224 = 31$

Example #5:

IP Address and subnet mask: 172.24.128.0 255.255.128.0

IP Address and wildcard mask: 172.24.128.0 0.0.127.255

Do the math... $255 - 255 = 0$ (This is the inverse of the subnet mask.)
 $255 - 128 = 127$
 $255 - 0 = 255$

Wildcard Mask Problems

1. Create a wildcard mask to match this exact address.
IP Address: 192.168.25.70
Subnet Mask: 255.255.255.0 0 . 0 . 0 . 0
2. Create a wildcard mask to match this range.
IP Address: 210.150.10.0
Subnet Mask: 255.255.255.0 0 . 0 . 0 . 255
3. Create a wildcard mask to match this host.
IP Address: 195.190.10.35
Subnet Mask: 255.255.255.0 0 . 0 . 0 . 0
4. Create a wildcard mask to match this range.
IP Address: 172.16.0.0
Subnet Mask: 255.255.0.0 0 . 0 . 255 . 255
5. Create a wildcard mask to match this range.
IP Address: 10.0.0.0
Subnet Mask: 255.0.0.0 0 . 255 . 255 . 255
6. Create a wildcard mask to match this exact address.
IP Address: 165.100.0.130
Subnet Mask: 255.255.255.192 0 . 0 . 0 . 0
7. Create a wildcard mask to match this range.
IP Address: 192.10.10.16
Subnet Mask: 255.255.255.224 0 . 0 . 0 . 31
8. Create a wildcard mask to match this range.
IP Address: 171.50.75.128
Subnet Mask: 255.255.255.192 0 . 0 . 0 . 63
9. Create a wildcard mask to match this host.
IP Address: 10.250.30.2
Subnet Mask: 255.0.0.0 0 . 0 . 0 . 0
10. Create a wildcard mask to match this range.
IP Address: 210.150.28.16
Subnet Mask: 255.255.255.240 0 . 0 . 0 . 15
11. Create a wildcard mask to match this range.
IP Address: 172.18.0.0
Subnet Mask: 255.255.224.0 0 . 0 . 31 . 255
12. Create a wildcard mask to match this range.
IP Address: 135.35.230.32
Subnet Mask: 255.255.255.248 0 . 0 . 0 . 7

Basic Wildcard Mask Problems

Based on the given information list the range of source addresses for each ACE statement.

1. access-list 10 permit **192.168.150.50 0.0.0.0**

Answer: 192.168.150.50

2. access-list 5 permit **any**

Answer: Any address

3. access-list 125 deny tcp **195.223.50.0 0.0.0.63** host 172.168.10.1 fragments

Answer: 195.223.50.0 to 195.223.50.63

4. access-list 11 deny 210.10.10.0 0.0.0.255

Answer: 210.10.10.0 to 210.10.10.255

5. access-list 108 deny ip 192.220.10.0 0.0.0.15 172.32.4.0 0.0.0.255

Answer: 192.220.10.0 to 192.220.10.15

6. access-list 171 deny any host 175.18.24.10 fragments

Answer: Any Address

7. access-list 105 permit 192.168.15.0 0.0.0.255 any

Answer: 192.168.15.0 to 192.168.15.255

8. access-list 109 permit tcp 172.16.10.0 0.0.0.255 host 192.168.10.1 eq 80

Answer: 172.16.10.0 to 172.16.10.255

9. access-list 111 permit ip any any

Answer: Any Address

10. access-list 195 permit udp 172.30.12.0 0.0.0.127 172.50.10.0 0.0.0.255

Answer: 172.30.12.0 to 172.30.12.127

11. access-list 110 permit ip 192.168.15.0 0.0.0.3 192.168.30.10 0.0.0.0

Answer: 192.168.15.0 to 192.168.15.3

12. access-list 120 permit ip 192.168.15.0 0.0.0.7 192.168.30.10 0.0.0.0

Answer: 192.168.15.0 to 192.168.15.7

13. access-list 130 permit ip 192.168.15.0 0.0.0.15 192.168.30.10 0.0.0.0

Answer: 192.168.15.0 to 192.168.15.15

14. access-list 140 permit ip 192.168.15.0 0.0.0.31 192.168.30.10 0.0.0.0

Answer: 192.168.15.0 to 192.168.15.31

15. access-list 150 permit ip 192.168.15.0 0.0.0.63 192.168.30.10 0.0.0.0

Answer: 192.168.15.0 to 192.168.15.63

16. access-list 101 Permit ip 192.168.15.0 0.0.0.127 192.168.30.10 0.0.0.0

Answer: 192.168.15.0 to 192.168.15.127

17. access-list 185 permit ip 192.168.15.0 0.0.0.255 192.168.30.0 0.0.0.255

Answer: 192.168.15.0 to 192.168.15.255

18. access-list 160 deny udp 172.16.0.0 0.0.1.255 172.18.10.18 0.0.0.0 gt 22

Answer: 172.16.0.0 to 172.16.1.255

19. access-list 195 permit icmp 172.85.0.0 0.0.15.255 172.50.10.0 0.0.0.255

Answer: 172.85.0.0 to 172.85.15.255

20. access-list 10 permit 175.15.120.0 0.0.0.255

Answer: 175.15.120.0 to 175.15.120.255

21. access-list 190 permit tcp 192.15.10.0 0.0.0.31 any

Answer: 192.15.10.0 to 192.15.10.31

22. access-list 100 permit ip 10.0.0.0 0.255.255.255 172.50.10.0 0.0.0.255

Answer: 10.0.0.0 to 10.255.255.255

(Slightly More) **Advanced Wildcard Mask Problems**

Not every address range ends with a zero making it simple to determine the address range. Using basic subnetting you can adjust the host portion of the IP address with the wildcard mask to indicate specific ranges.

IP Address: 192 . 100 . 10 . 0
Custom Subnet Mask: 255.255.255.240

Address Ranges: 192.100.10.0 to 192.100.10.15
192.100.10.16 to 192.100.10.31
192.100.10.32 to 192.100.10.47 (Range in the sample below)
192.100.10.48 to 192.100.10.63
192.100.10.64 to 192.100.10.79
192.100.10.80 to 192.100.10.95
192.100.10.96 to 192.100.10.111
192.100.10.112 to 192.100.10.127
192.100.10.128 to 192.100.10.143
192.100.10.144 to 192.100.10.159
192.100.10.160 to 192.100.10.175
192.100.10.176 to 192.100.10.191
192.100.10.192 to 192.100.10.207
192.100.10.208 to 192.100.10.223
192.100.10.224 to 192.100.10.239
192.100.10.240 to 192.100.10.255

Based on this sample to deny or permit the third subnet we would create the correct wildcard mask by inverting the subnet mask from 255.255.255.240 to 0.0.0.15. The third address range would be 192.100.10.32 to 192.100.10.47 .

A standard ACE statement would be written as:

access-list 10 permit 192.100.10.32 0.0.0.15

The wildcard mask indicates that address range 0 to 15 will be permitted. If the source address was 192.100.10.0 it would indicate the first 16 addresses (0 to 15). Since the source address is 192.100.10.32, it indicates that the address range will be from 192.100.10.32 to 192.100.10.47 (32 to 47). The IP address indicates the starting point for the wildcard mask to begin counting up from.

This technique gives you greater control with the addresses you want to permit or deny. It is a **Best Practice** to stay within standard address ranges.

Based on the given information list the range of source addresses for each ACE statement.

1. access-list 10 permit 192.100.10.64 0.0.0.15

Answer: 192.100.10.64 to 192.100.10.79

2. access-list 5 permit 172.16.128.0 0.0.63.255

Answer: 172.16.128.0 to 172.16.191.255

3. access-list 125 deny tcp 192.100.10.208 0.0.0.15 host 192.168.10.1 log

Answer: 192.100.10.208 to 192.100.10.223

4. access-list 11 deny 210.48.72.192 0.0.0.31

Answer: 210.48.72.192 to 210.48.72.223

5. access-list 108 deny ip 192.168.5.184 0.0.0.7 192.64.4.0 0.0.0.255

Answer: 192.168.5.184 to 192.168.5.191

6. access-list 171 deny 172.32.128.0 0.0.15.255 any

Answer: 172.32.128.0 to 172.32.143.255

7. access-list 105 permit 165.50.196.0 0.0.3.255 any

Answer: 165.50.196.0 to 165.50.199.255

8. access-list 109 permit tcp 172.16.128.0 0.0.127.255 host 172.16.20.1

Answer: 172.16.128.0 to 172.16.255.255

9. access-list 111 permit ip 192.168.1.236 0.0.0.3 any

Answer: 192.168.1.236 to 192.168.1.239

10. access-list 195 permit udp 172.32.160.0.0 0.0.31.255 172.100.10.0 0.0.0.255

Answer: 172.32.160.0 to 172.32.191.255

11. access-list 110 permit ip 10.64.0.0 0.63.255.255 172.168.40.10 0.0.0.0

Answer: 10.64.0.0 to 10.127.255.255

Wildcard Mask Problems

Based on the given information list the range of destination addresses for each ACE statement.

1. access-list 125 deny tcp 195.223.50.0 0.0.0.63 host 172.168.10.1 fragments

Answer: 172.168.10.1

2. access-list 115 permit any any

Answer: Any address

3. access-list 150 permit ip 192.168.30.10 0.0.0.0 192.168.15.0 0.0.0.63

Answer: 192.168.15.0 to 192.168.15.63

4. access-list 120 deny tcp 172.32.4.0 0.0.0.255 192.220.10.0 0.0.0.15

Answer: 192.220.10.0 to 192.220.10.15

5. access-list 108 deny ip 192.220.10.0 0.0.0.15 172.32.4.0 0.0.0.255

Answer: 172.32.4.0 to 172.32.4.255

6. access-list 101 deny ip 140.130.110.100 0.0.0.0 0.0.0.0 255.255.255.255

Answer: Any Address

7. access-list 105 permit any 192.168.15.0 0.0.0.255

Answer: 192.168.15.0 to 192.168.15.255

8. access-list 120 permit ip 192.168.15.10 0.0.0.0 172.16.40.0 0.0.3.255

Answer: 172.16.40.0 to 172.16.43.255

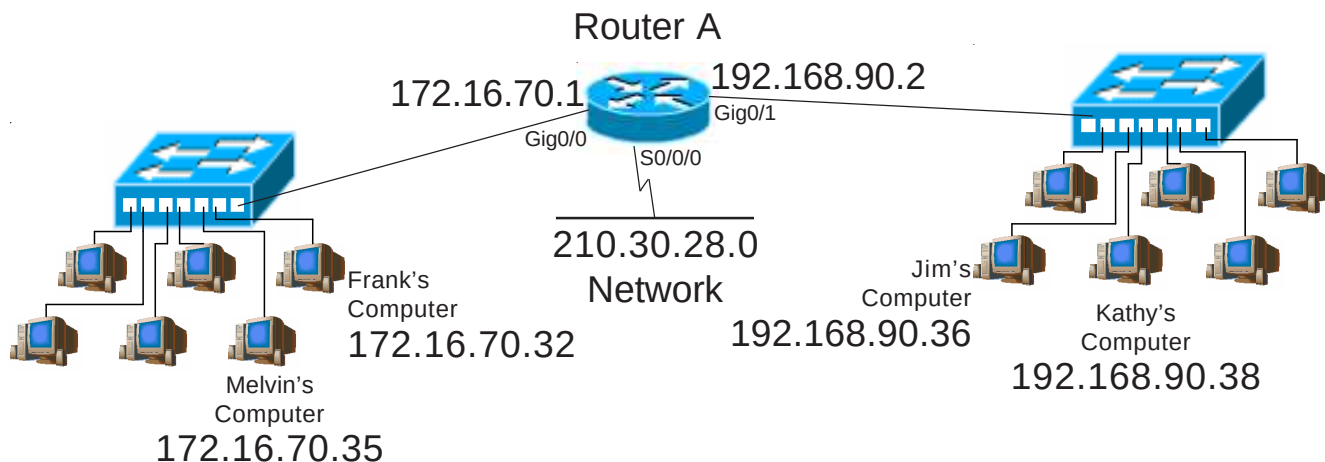
9. access-list 160 deny udp 172.16.0.0 0.0.1.255 172.18.104.0 0.0.7.255 eq 21

Answer: 172.18.104.0 to 172.18.111.255

10. access-list 150 permit ip 192.168.30.0 0.0.0.63 192.168.15.96 0.0.0.31

Answer: 192.168.15.96 to 192.168.15.127

Writing Standard Access Lists...



Standard Access List Sample #1

Write a standard access list to block Melvin's computer from sending information to Kathy's computer; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: Gig0/1
 Access-list #: 10

Note:

A standard access list blocks only the source address. Melvin will also be blocked from sending information to Jim or anyone else on the 192.168.90.0 network.

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 10 deny 172.16.70.35
                        or
                        access-list 10 deny 172.16.70.35 0.0.0.0
                        or
                        access-list 10 deny host 172.16.70.35
Router(config)# access-list 10 permit 0.0.0.0 255.255.255.255
                        or
                        access-list 10 permit any
Router(config)# interface gig0/1
Router(config-if)# ip access-group 10 out
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

Router# *show configuration* (This will show which access groups are associated with particular interfaces)

Router# *show access list 10* (This will show detailed information about this ACL)

Standard Access List Sample #2

Include a remark with each statement of your ACL. Write a standard access list to block Jim's computer from sending information to Frank's computer; but will allow all other traffic from the 192.168.90.0 network. Permit all traffic from the 210.30.28.0 network to reach the 172.16.70.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 28

[Writing and installing an ACL]

```
Router# configure terminal
Router(config)# access-list 28 remark Block Jim from reaching Frank
Router(config)# access-list 28 deny 192.168.90.36
                        or
                    access-list 28 deny 192.168.90.36 0.0.0.0
                        or
                    access-list 28 deny host 192.168.90.36
Router(config)# access-list 28 remark Allow all other traffic
Router(config)# access-list 28 permit 192.168.90.0 0.0.0.255
Router(config)# access-list 28 remark Allow all traffic
Router(config)# access-list 28 permit 210.30.28.0 0.0.0.255
Router(config)# interface Gig0/0
Router(config-if)# ip access-group 28 out
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Remark Command]

The remark command allows you to place text within the ACL so it can be viewed after it is inserted on the router. It can be viewed using the show run or any command that lists the ACL.

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 28 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 28 out
Router(config-if)# exit
Router(config)# no access-list 28
Router(config)# exit
```


Standard Access List Problem #2

Write a standard access list to permit Debbie's computer to receive information from Michael's computer; but will deny all other traffic from the 223.190.32.0 network. Block all traffic from the 172.16.0.0 network. Permit all other traffic. List all the command line options for this problem. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B

Interface: Fa0/1

Access-list #: 40 (1-99)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *access-list 40 permit 223.190.32.16*
or
access-list 40 permit host 223.190.32.16
or
access-list 40 permit 223.190.32.16 0.0.0.0

Router(config)# *access-list 40 deny 223.190.32.0 0.0.0.255*

Router(config)# *access-list 40 deny 172.16.0.0 0.0.255.255*

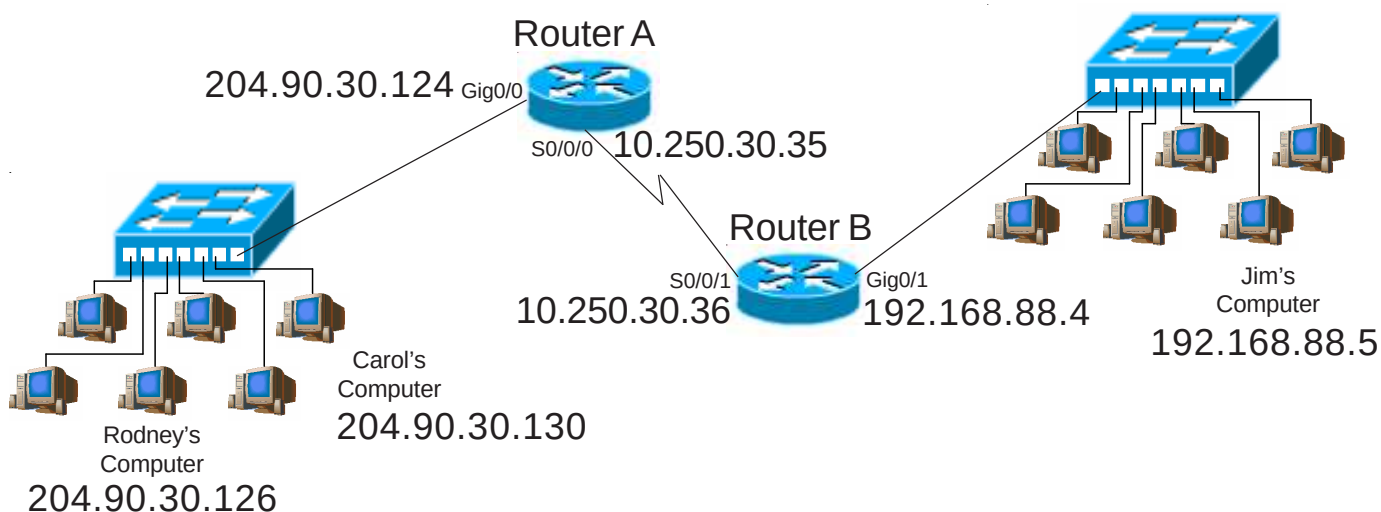
Router(config)# *access-list 40 permit any*
or
access-list 40 permit 0.0.0.0 255.255.255.255

Router(config)# *interface* *fa0/1*

Router(config-if)# *ip access-group* *40* in or *out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*



Standard Access List Problem #3

Write a standard access list to block Rodney and Carol's computer from sending information to Jim's computer; but will allow all other traffic from the 204.90.30.0 network. Block all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Gig0/1
 Access-list #: 45 (1-99)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *access-list 45 deny 204.90.30.130*
or
access-list 45 deny host 204.90.30.130
or
access-list 45 deny 204.90.30.130 0.0.0.0

access-list 45 deny 204.90.30.126
or
access-list 45 deny host 204.90.30.126
or
access-list 45 deny 204.90.30.126 0.0.0.0

access-list 45 permit 204.90.30.0 0.0.0.255

Router(config)# *interface gig0/1*

Router(config-if)# *ip access-group 45 in or out (circle one)*
 Router(config-if)# *exit*
 Router(config)# *exit*

Standard Access List Problem #4

Include a remark with each statement of your ACL. Using a minimum number of commands write a standard access list named "Ralph" to block Carol's computer from sending information to Jim's computer; but will permit Jim to receive data from Rodney. Block the upper half of the 204.90.30.0 range from reaching Jim's computer while permitting the lower half of the range. Block all other traffic. For help with blocking the upper half of the range review page 13 or the wildcard mask problems on pages 16 and 17. For help with named ACLs review pages 12 and 13. For help with the remark command review page 23.

Place the access list at:

Router Name: Router B

Interface: Gig0/1

Access-list Name: Ralph

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# ip access-list standard Ralph

Router(config-std-nacl)# remark Permits the lower half of the range

remark while blocking all other traffic

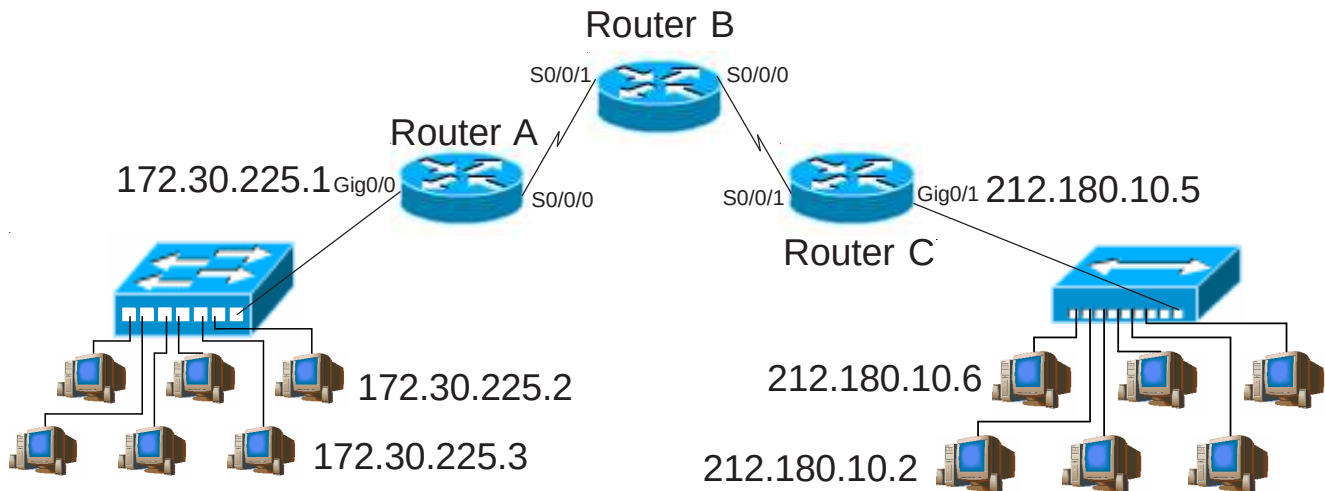
permit 204.90.30.0 0.0.0.127

Router(config-std-nacl)# interface gig0/1

Router(config-if)# ip access-group Ralph in or out (circle one)

Router(config-if)# exit

Router(config)# exit



Standard Access List Problem #5

Write a standard access list to block 172.30.225.2 and 172.30.225.3 from sending information to the 212.180.10.0 network; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router C
 Interface: Gig0/1
 Access-list #: 55 (1-99)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *access-list 55 deny 172.30.225.2*
or
access-list 55 deny host 172.30.225.2
or
access-list 55 deny 172.30.225.2 0.0.0.0

access-list 55 deny 172.30.225.3
or
access-list 55 deny host 172.30.225.3
or
access-list 55 deny 172.30.225.3 0.0.0.0

access-list 55 permit any

This ACL could be shortened to these two statements.

access-list 55 deny 172.30.225.2 0.0.0.1
access-list 55 permit any

Router(config)# *interface gig0/1*

Router(config-if)# *ip access-group 55* in or out (circle one)
 Router(config-if)# *exit*
 Router(config)# *exit*

Standard Access List Problem #6

Add a remark to each statement explaining its purpose. Write a standard access list to block and log 212.180.10.2 from sending information to the 172.30.225.0 network. Permit and log 212.180.10.6 to send data to the 172.30.225.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written. Check the example on page 10 for help with the logging option. For help with the remark command review page 23.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 60 (1-99)

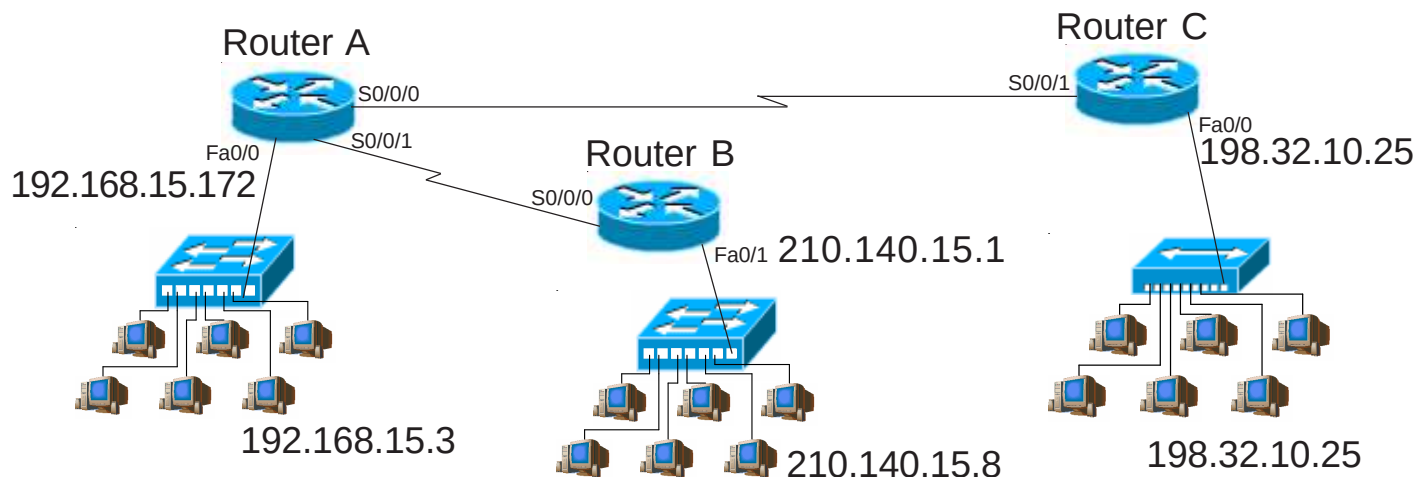
[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *access-list 60 remark block and log this user*
or access-list 60 deny 212.180.10.2 log
or access-list 60 deny host 212.180.10.2 log
or access-list 60 deny 212.180.10.2 0.0.0.0 log
access-list 60 remark allow and log this user
access-list 60 permit 212.180.10.6 log
or access-list 60 permit host 212.180.10.6 log
or access-list 60 permit 212.180.10.6 0.0.0.0 log

Router(config)# *interface gig0/0*

Router(config-if)# *ip access-group* *60* *in or* *out* (circle one)
Router(config-if)# *exit*
Router(config)# *exit*



Standard Access List Problem #7

Write a standard access list to block the addresses 192.168.15.1 to 192.168.15.31 from sending information to the 210.140.15.0 network. Do not permit any traffic from 198.32.10.25 to reach the 210.140.15.0 network. Permit all other traffic. For help with this problem review page 13 or the wildcard mask problems on pages 16 and 17.

Place the access list at:

Router Name: Router B
 Interface: Fa0/1
 Access-list #: 65 (1-99)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# access-list 65 deny 192.168.15.0 0.0.0.31
access-list 65 deny 198.32.10.25
 or
access-list 65 deny host 198.32.10.25
 or
access-list 65 deny 198.32.10.25 0.0.0.0
access-list 65 permit any

Router(config)# *interface fa0/1*

Router(config-if)# *ip access-group 65 in or out (circle one)*

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #8

Write a standard named access list called "CISCO_LAB_A" to permit traffic from the lower half of the 198.32.10.0 network to reach 192.168.15.0 network; block the upper half of the addresses. Allow host 198.32.10.192 to reach network 192.168.15.0. Permit all other traffic. For help with this problem review page 13 or the wildcard masks problems on pages 16 and 17. For assistance with named ACLs review pages 12 and 13.

Place the access list at:

Router Name: Router A

Interface: Fa0/0

Access-list Name: CISCO_LAB_A

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *ip access-list standard CISCO_LAB_A*

Router(config-std-nacl)# *permit 198.32.10.0 0.0.0.127*

permit host 198.32.10.192
or
permit 198.32.10.192 0.0.0.0
or
permit 198.32.10.192

deny 198.32.10.0 0.0.0.255

permit any

An alternate (and shorter) version of this:

Permit 198.32.10.192 0.0.0.0

Deny 198.32.10.128 0.0.0.127

Permit Any

Router(config-std-nacl)# *interface* *fa0/0*

Router(config-if)# *ip access-group* *CISCO_LAB_A* in or *out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #9

Write a standard access list to block network 192.168.255.0 from receiving information from the following addresses: 10.250.1.1, 10.250.2.1, 10.250.4.1, and the entire 10.250.3.0 255.255.255.0 network. Allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Fa0/0
Access-list #: 75 (1-99)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

```
Router(config)# access-list 75 deny 10.250.1.1  
or  
access-list 75 deny host 10.250.1.1  
or  
access-list 75 deny 10.250.1.1 0.0.0.0  
_____  
access-list 75 deny 10.250.2.1  
or  
access-list 75 deny host 10.250.2.1  
or  
access-list 75 deny 10.250.2.1 0.0.0.0  
_____  
access-list 75 deny 10.250.4.1  
or  
access-list 75 deny host 10.250.4.1  
or  
access-list 75 deny 10.250.4.1 0.0.0.0  
_____  
access-list 75 deny 10.250.3.0 0.0.0.255  
_____  
access-list 75 permit any  
_____
```

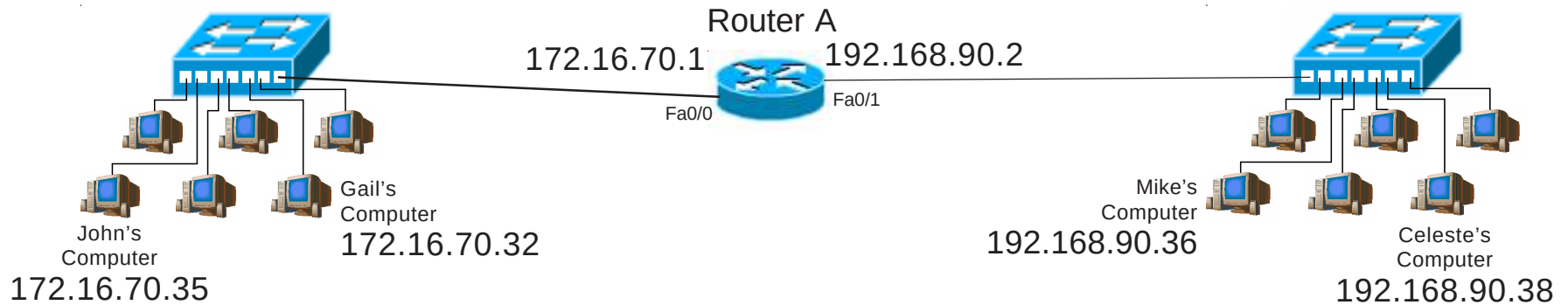
Router(config)# *interface fa0/0*

Router(config-if)# *ip access-group 75* in or out (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Writing Extended Access Lists...



Extended Access List Sample #1

Deny/Permit Specific Addresses

Write an extended access list to prevent John's computer from sending information to Mike's computer; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: FA0/0
 Access-list #: 110

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 110 deny ip 172.16.70.35 0.0.0.0 192.168.90.36 0.0.0.0
or
access-list 110 deny ip host 172.16.70.35 host 192.168.90.36
Router(config)# access-list 110 permit ip any any
or
access-list 110 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/0
Router(config-if)# ip access-group 110 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

```
Router# show configuration (This will show which access groups
                             are associated with particular interfaces)

Router# show access list 110 (This will show detailed information
                              about this ACL)
```

Extended Access List Sample #2

Deny/Permit Specific Addresses

Write an extended access list to block the 172.16.70.0 network from receiving information from Mike's computer at 192.168.90.36. Block the lower half of the ip addresses from 192.168.90.0 network from reaching Gail's computer at 172.16.70.32. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: FA0/1
Access-list #: 135

[Writing and installing an ACL]

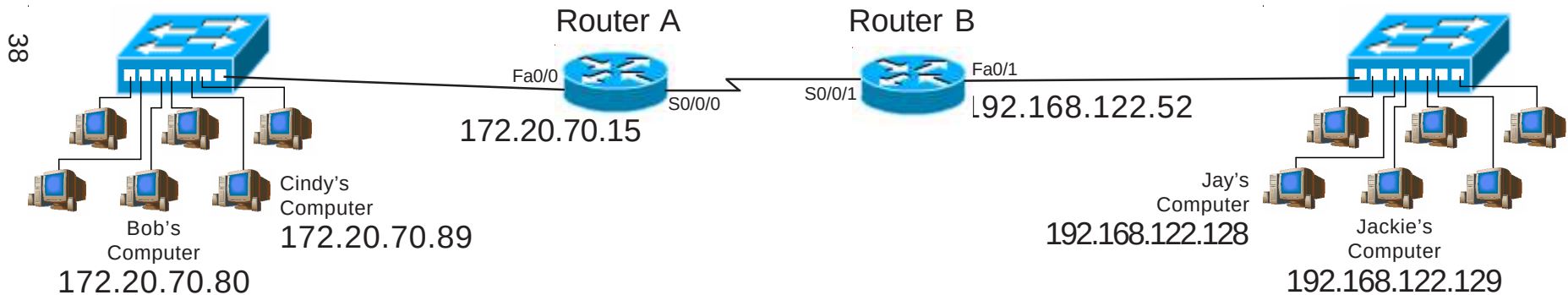
```
Router# configure terminal
Router(config)# access-list 135 deny ip 192.168.90.36 0.0.0.0 172.16.70.0 0.0.0.255
                        or
                        access-list 135 deny ip host 192.168.90.36 172.16.70.0 0.0.0.255
Router(config)# access-list 135 deny ip 192.168.90.0 0.0.0.127 172.16.70.32 0.0.0.0
                        or
                        access-list 135 deny ip 192.168.90.0 0.0.0.127 host 172.16.70.32
Router(config)# access-list 135 permit ip any any
                        or
                        access-list 135 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/1
Router(config-if)# ip access-group 135 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface fa0/1
Router(config-if)# no ip access-group 135 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface fa0/1
Router(config-if)# no ip access-group 135 out
Router(config-if)# exit
Router(config)# no access-list 135
Router(config)# exit
```

Extended Access List Problem #1 Deny/Permit Specific Addresses

Write an extended access list to prevent Jay's computer from receiving information from Cindy's computer. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Fa0/0
Access-list #: 105 (100-199)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *access-list 105 deny ip host 172.20.70.89 host 192.168.122.128*
or
access-list 105 deny ip 172.30.225.2 0.0.0.0 192.168.122.128 0.0.0.0
access-list 105 permit ip any any

Router(config)# *interface fa0/0*
Router(config-if)# *ip access-group 110 in or out (circle one)*
Router(config-if)# *exit*
Router(config)# *exit*
Router# *copy run start*

Extended Access List Problem #2

Deny/Permit Specific Addresses

Write an extended access list to block the 172.20.70.0 255.255.255.0 network from receiving information from Jackie's computer at 192.168.122.129. Block the lower half of the ip addresses from 192.168.122.0 network from reaching Cindy's computer at 172.20.70.89. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
Interface: Fa0/1
Access-list #: 110 (100-199)

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# *access-list 110 deny ip host 192.168.122.129 172.20.70.0 0.0.0.255*

or

access-list 110 deny ip 192.168.122.129 0.0.0.0 172.20.70.0 0.0.0.255

or

access-list 110 deny ip 192.168.122.0 0.0.0.127 host 172.20.70.89

access-list 110 deny ip 192.168.122.0 0.0.0.127 172.20.70.89 0.0.0.0

access-list 110 permit ip any any

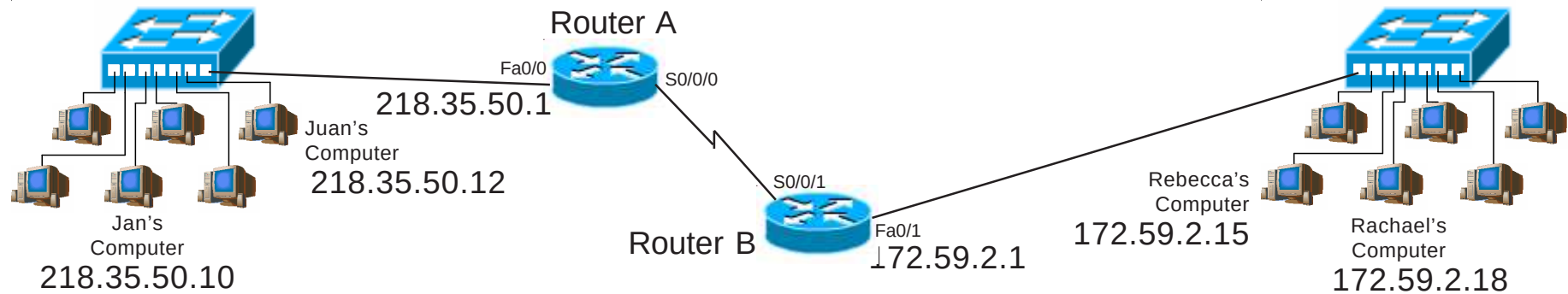
Router(config)# *interface fa0/1*

Router(config-if)# *ip access-group 110* in or out (circle one)

Router(config-if)# *exit*

39 Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #3

Deny/Permit Specific Addresses

Write a named extended access list called "LAB_166" to permit Jan's computer at 218.35.50.10 to receive packets from Rachael's computer at 172.59.2.18; but not Rebecca's computer at 172.59.2.15. Deny all other packets. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B

Interface: Gig0/1

Access-list Name: LAB_166

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *ip access-list extended LAB_166*

Router(config-ext-nacl)# *permit ip host 172.59.2.18 host 218.35.50.10*

or

permit ip 172.59.2.18 0.0.0.0 218.35.50.10 0.0.0.0

Router(config-ext-nacl)# *interface gig0/1*

Router(config-if)# *ip access-group LAB_166* in or out (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #4

Deny/Permit Specific Addresses

Write an extended access list to allow Juan's computer at 218.35.50.12 to send information to Rebecca's computer at 172.59.2.15; but not Rachael's computer at 172.59.2.18. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 120 (100-199)

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# *access-list 120 deny ip host 218.35.50.12 host 172.59.2.18*

or

access-list 120 deny ip 218.35.50.12 0.0.0.0 172.59.2.18 0.0.0.0

access-list 120 permit ip any any

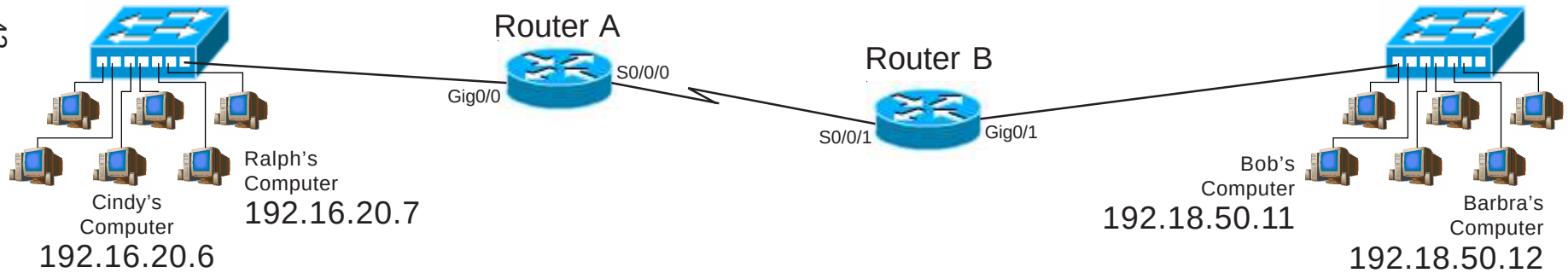
Router(config)# *interface gig0/0*

Router(config-if)# *ip access-group 115* in or out (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Sample #3

Deny/Permit Entire Ranges

Write an extended access list to permit the 192.16.20.0 network to receive packets from the 192.18.50.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Gig0/1
 Access-list #: 111

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 111 permit ip 192.18.50.0 0.0.0.255 192.16.20.0 0.0.0.255
Router(config)# access-list 111 deny ip any any
or
Router(config)# access-list 111 deny ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface gig0/1
Router(config-if)# ip access-group 111 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

Router# *show configuration* (This will show which access groups are associated with particular interfaces)

Router# *show access list 111* (This will show detailed information about this ACL)

Extended Access List Sample #4

Deny/Permit Entire Ranges

Add a remark to each statement. Write an extended access list to block the 192.18.50.0 network from receiving information from the 192.16.20.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 188

[Remark Command]

The remark command allows you to place text within the ACL so it can be viewed after it is inserted on the router. It can be viewed using the show run or any command that lists the ACEs.

[Writing and installing an ACL]

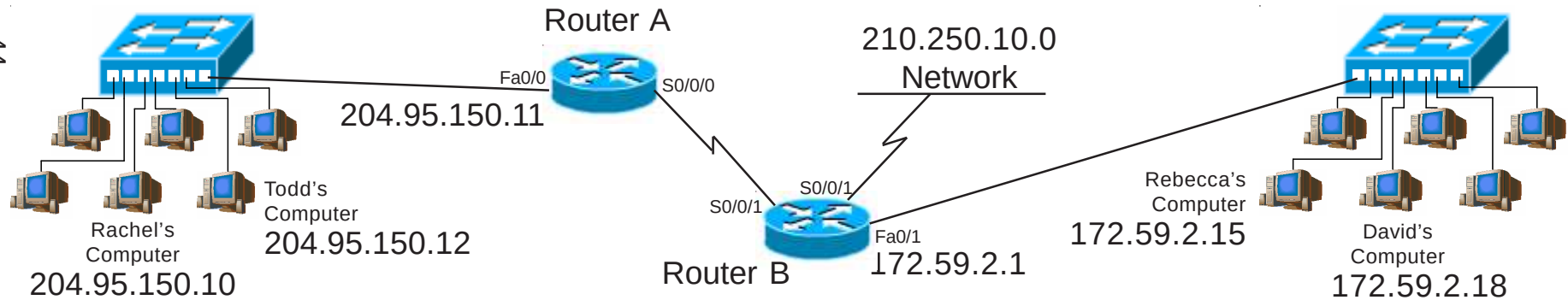
```
Router# configure terminal
Router(config)# access-list 188 remark block all traffic from the Science lab
Router(config)# access-list 188 deny ip 192.16.20.0 0.0.0.255 192.18.50.0 0.0.0.255
Router(config)# access-list 188 remark allow everyone else unrestricted access
Router(config)# access-list 188 permit ip any any
                        or
                        access-list 188 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface gig0/0
Router(config-if)# ip access-group 188 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 188 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 188 out
Router(config-if)# exit
Router(config)# no access-list 188
Router(config)# exit
```



Extended Access List Problem #5

Deny/Permit Entire Ranges

Include a remark with each statement of your ACL. Write an extended access list to permit network 204.95.150.0 to send packets to network 172.59.0.0, but not to the 210.250.10.0 network. Permit all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: Fa0/0
 Access-list #: 125 (100-199)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# access-list 125 remark Keep Marketing from bothering Research

access-list 125 deny ip 204.95.150.0 0.0.0.255 210.250.10.0 0.0.0.255

access-list 125 remark allow all other traffic

access-list 125 permit ip any any

Router(config)# *interface fa0/0*

Router(config-if)# *ip access-group 125* in or out (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #6

Deny/Permit Entire Ranges

Write an extended access list to allow Rachel's computer at 204.95.150.10 to receive information from the 172.59.2.0 255.255.255.0 network. Deny all other hosts on the 204.95.150.0 network access from the 172.59.2.0 255.255.255.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
Interface: Fa0/1
Access-list #: 130 (100-199)

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# *access-list 130 permit ip 172.59.2.0 0.0.0.255 host 204.95.150.10*

or

access-list 130 permit ip 172.59.2.0 0.0.0.255 204.95.150.10 0.0.0.0

access-list 130 deny ip 172.59.2.0 0.0.0.255 204.95.150.0 0.0.0.255

access-list 130 permit ip any any

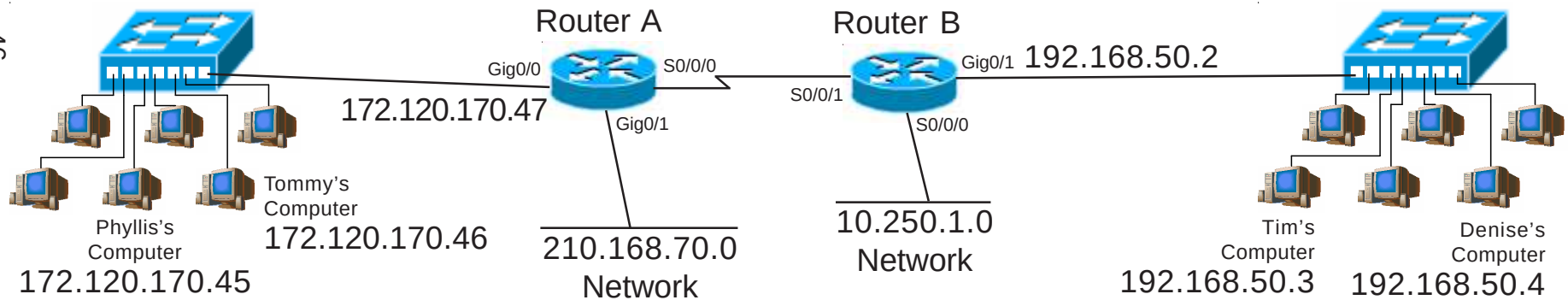
Router(config)# *interface Fa0/1*

Router(config-if)# *ip access-group 130* in or out (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #7 Deny/Permit Entire Ranges

Write a named extended access list called "Godzilla" to prevent the 172.120.0.0 network from sending information to the 210.168.70.0, and 10.250.1.0 255.255.255.0 networks; but will permit traffic to the 192.168.50.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A

Interface: Gig0/0

Access-list Name: GODZILLA

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# ip access-list extended GODZILLA

Router(config-ext-nacl)# deny ip 172.120.0.0 0.0.255.255 210.168.70.0 0.0.0.255

deny ip 172.120.0.0 0.0.255.255 10.250.1.0 0.0.0.255

permit ip any any

Router(config-ext-nacl)# interface gig0/0

Router(config-if)# ip access-group GODZILLA in or out (circle one)

Router(config-if)# exit

Router(config)# exit

Extended Access List Problem #8

Deny/Permit Entire Ranges

Assuming default subnet masks write an extended access list to permit Tim at 192.168.50.3 to receive data from the 172.120.0.0 network. Allow the 192.168.50.0 network to receive information from Phyllis's computer at 172.120.170.45. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 140 (100-199)

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# *access-list 140 permit ip 172.120.0.0 0.0.255.255 host 192.168.50.3*

or
access-list 140 permit ip 172.120.0.0 0.0.255.255 192.168.50.3 0.0.0.0

access-list 140 permit ip host 172.120.170.45 192.168.50.0 0.0.0.255

or
access-list 140 permit ip 172.120.170.45 0.0.0.0 192.168.50.0 0.0.0.255

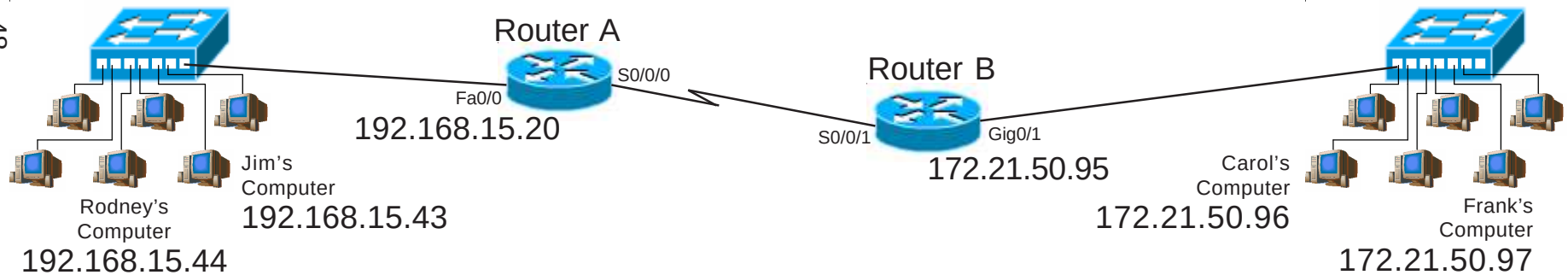
Router(config)# *interface gig0/0*

Router(config-if)# *ip access-group 140* in *or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Sample #5

Deny/Permit a Range of Addresses

Write an extended access list to deny the first 15 usable addresses of the 192.168.15.0 network from reaching the 172.21.0.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: Fa0/0
 Access-list #: 185

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 185 deny ip 192.168.15.0 0.0.0.15 172.21.50.0 0.0.255.255
Router(config)# access-list 185 permit ip any any
or
Router(config)# access-list 185 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/0
Router(config-if)# ip access-group 185 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

Router# *show configuration* (This will show which access groups are associated with particular interfaces)

Router# *show access list 185* (This will show detailed information about this ACL)

Extended Access List Sample #6

Deny/Permit a Range of Addresses

Write an extended access list which will allow the lower half of 192.168.15.0 network access to the 172.21.50.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Fa0/0
Access-list #: 121

[Writing and installing an ACL]

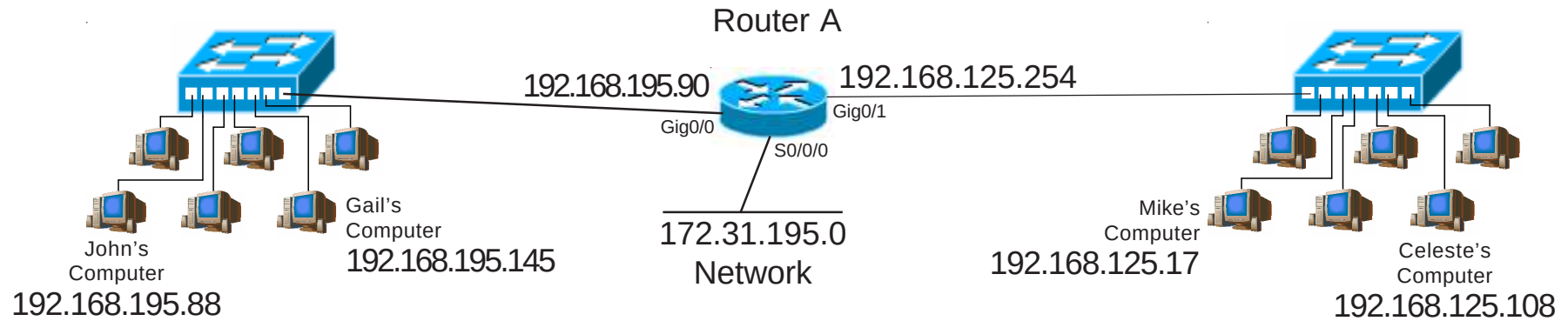
```
Router# configure terminal
Router(config)# access-list 121 permit ip 192.168.15.0 0.0.0.127 172.21.50.0 0.0.0.255
Router(config)# access-list 121 deny ip any any
                        or
                        access-list 121 deny ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/0
Router(config-if)# ip access-group 121 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface fa0/0
Router(config-if)# no ip access-group 121 in
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface fa0/0
Router(config-if)# no ip access-group 121 in
Router(config-if)# exit
Router(config)# no access-list 121
Router(config)# exit
```



Extended Access List Problem #9 Deny/Permit a Range of Addresses

Write an extended access list to prevent the first 31 usable addresses in the 192.168.125.0 network from reaching the 192.168.195.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: Gig0/1
 Access-list #: 145 (100-199)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# access-list 145 deny ip 192.168.125.0 0.0.0.31 192.168.195.0 0.0.0.255

access-list 145 permit ip any any

Router(config)# interface gig0/1

Router(config-if)# ip access-group 145 in or out (circle one)

Router(config-if)# exit

Extended Access List Problem #10 Deny/Permit a Range of Addresses

Include a remark with each statement of your ACL. Write a named extended access list called "Media_Center" to permit the range of addresses from 172.31.195.1 through 172.31.195.7 to send data to the 192.168.125.0 network. Deny all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: SO/0/0
Access-list Name: MEDIA_CENTER

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# ip access-list extended MEDIA_CENTER

Router(config-ext-nacl)# remark Allow the first seven addresses from the Media Center

permit ip 172.31.195.0 0.0.0.7 192.168.125.0 0.0.0.255

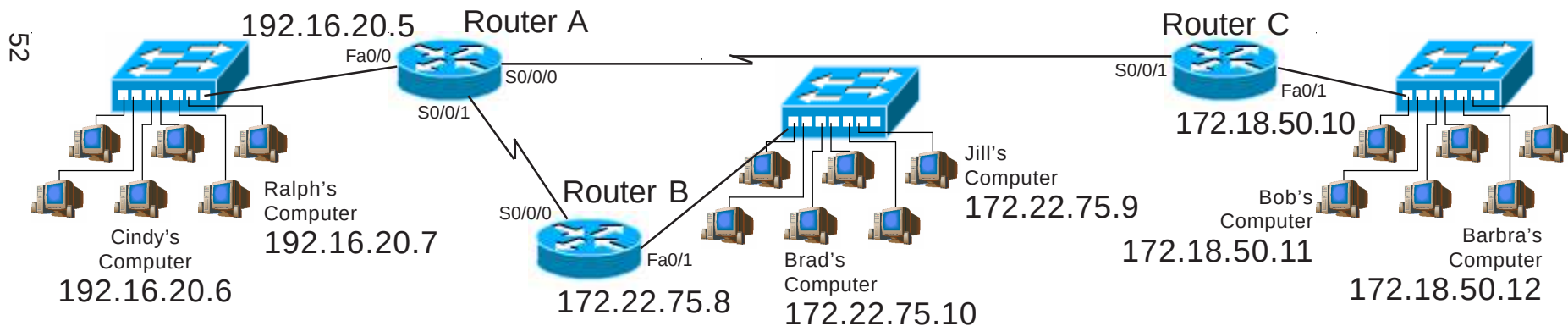
Router(config-ext-nacl)# *interface* SO/0/0

Router(config-if)# *ip access-group* MEDIA_CENTER in or out (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #11. Deny/Permit a Range of Addresses

Write an extended access list to permit the first 3 usable addresses in the 192.16.20.0 network to reach the 172.22.75.0 network. Deny the addresses from 192.16.20.4 through 192.16.20.31 from reaching the 172.22.75.0 network. Permit all other traffic. Keep in mind that there are multiple ways this ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: Fa0/0
 Access-list #: 155 (100-199)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# access-list 155 permit ip 192.16.20.0 0.0.0.3 172.22.75.0 0.0.0.255

access-list 155 deny ip 192.16.20.0 0.0.0.31 172.22.75.0 0.0.0.255

access-list 155 permit ip any any

Router(config)# interface fa0/0

Router(config-if)# ip access-group 155 in or out (circle one)

Router(config-if)# exit

Extended Access List Problem #12 Deny/Permit a Range of Addresses

Write an extended access list to deny the addresses from 172.22.75.8 through 172.22.75.127 from sending data to the 172.18.50.0 network. Deny the first half of the addresses from the 172.22.75.0 network from reaching the 192.16.20.0 network. Permit all other traffic. Keep in mind that there are multiple ways this ACL can be written.

Place the access list at:

Router Name: Router B
Interface: Fa0/1
Access-list #: 160 (100-199)

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# *access-list 160 permit ip 172.22.75.0 0.0.0.7 172.18.50.0 0.0.0.255*

access-list 160 deny ip 172.22.75.0 0.0.0.127 192.16.20.0 0.0.0.255

access-list 160 permit ip any any

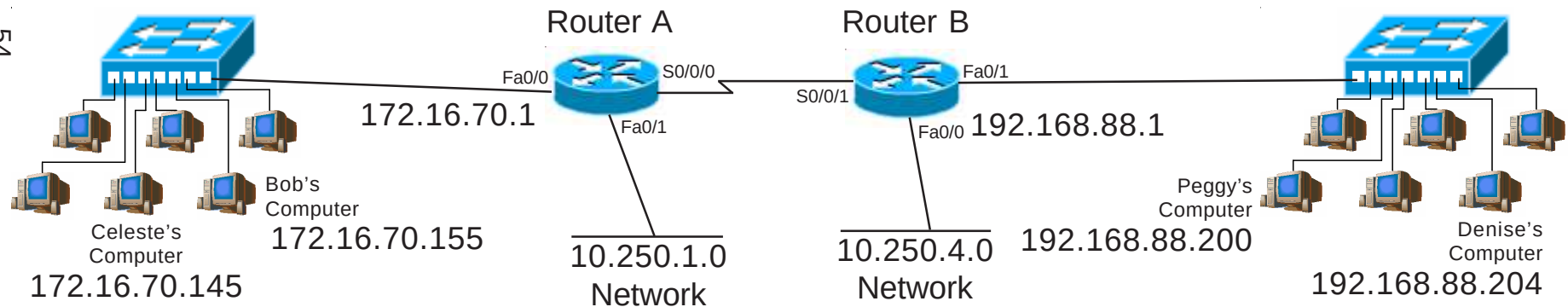
Router(config)# *interface fa0/1*

Router(config-if)# *ip access-group 160* *in* or out (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #13 Deny/Permit a Range of Addresses

Include a remark with each statement of your ACL. Write an extended access list to permit the first 63 usable addresses in the 192.168.88.0 network to reach the lower half of the addresses in the 172.16.70.0 network; but not the upper half. Deny all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Fa0/1
 Access-list #: 165 (100-199)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# access-list 165 remark block the upper half of the addresses from passing
access-list 165 permit ip 192.168.88.0 0.0.0.63 172.16.70.0 0.0.0.127

Router(config)# interface fa0/1
 Router(config-if)# ip access-group 165 in or out (circle one)
 Router(config-if)# exit

Extended Access List Problem #14 Deny/Permit a Range of Addresses

Write an extended access list to deny the addresses from 10.250.1.0 through 10.250.1.63 from sending data to Denise's computer. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Fa0/1
Access-list #: 170 (100-199)

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# *access-list 170 deny ip 10.250.1.0 0.0.0.63 host 192.168.88.204*

or

access-list 170 deny ip 10.250.1.0 0.0.0.63 192.168.88.204 0.0.0.0

access-list 170 permit ip any any

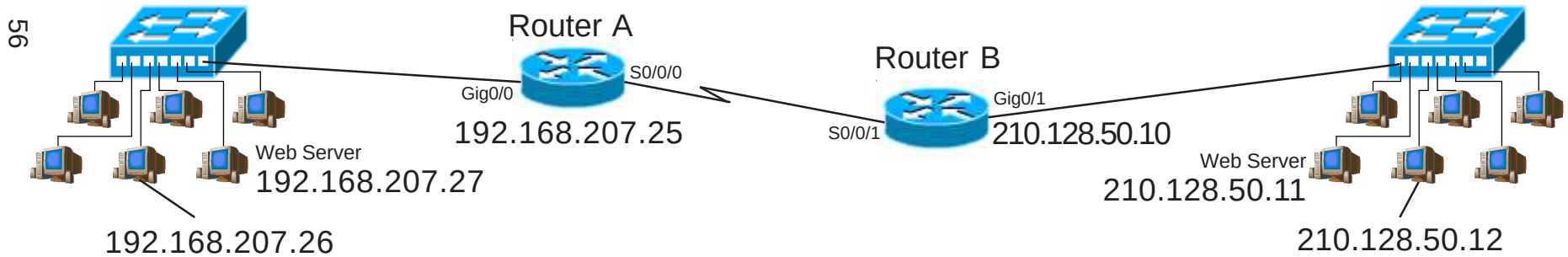
Router(config)# *interface fa0/1*

Router(config-if)# *ip access-group 170* in or out (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Sample #7

Deny/Permit Port Numbers

Write an extended access list to deny HTTP traffic intended for web server 192.168.207.27 from all other networks, but will permit all other HTTP traffic to reach the 192.168.207.0 network. Deny all other IP traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Gig0/1
 Access-list #: 198

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 198 deny tcp any 192.168.207.27 0.0.0.0 eq www
or
access-list 198 deny tcp any host 192.168.207.27 eq www
Router(config)# access-list 198 permit tcp any 192.168.207.0 0.0.0.255 eq www
Router(config)# interface gig0/1
Router(config-if)# ip access-group 198 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

```
Router# show configuration (This will show which access groups are associated
with particular interfaces)

Router# show access list 198 (This will show detailed information about this ACL)
```

Extended Access List Sample #8

Deny/Permit Port Numbers

Write an extended access list on Router B to deny pings between hosts on the 210.128.50.0 and the 192.168.207.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
Interface: Gig0/1
Access-list #: 134

[Writing and installing an ACL]

```
Router# configure terminal
Router(config)# access-list 134 deny icmp 210.128.50.0 0.0.0.255 192.168.207.0 0.0.0.255
Router(config)# access-list 134 permit ip any any
Router(config)# interface gig0/1
Router(config-if)# ip access-group 134 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

Hint:

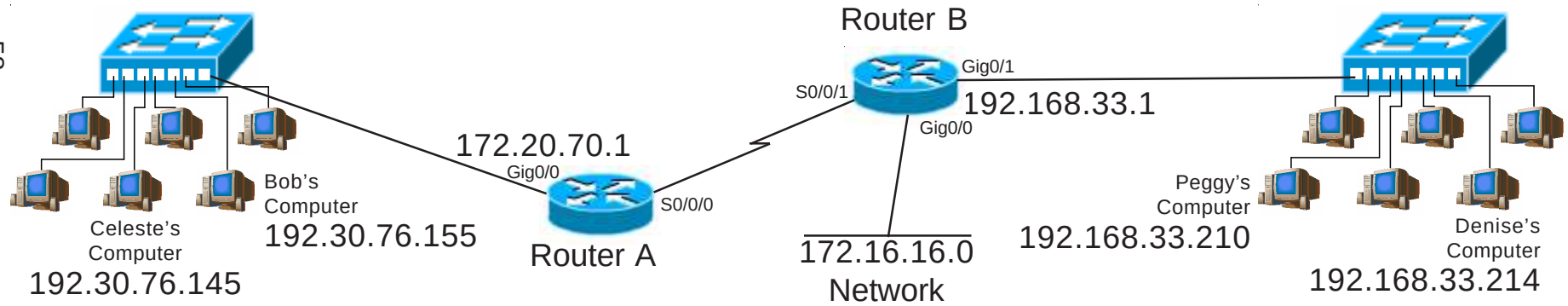
It's OK to use multiple protocols in the same ACL.

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface Gig0/1
Router(config-if)# no ip access-group 134 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface Gig0/1
Router(config-if)# no ip access-group 134 out
Router(config-if)# exit
Router(config)# no access-list 134
Router(config)# exit
```



Extended Access List Sample #9

Deny/Permit Port Numbers

Write an Extended access list to permit Denise's computer to use TFTP with Bob's computer. Deny all other traffic from the 192.168.33.0 network to the 192.30.76.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Gig0/1
 Access-list #: 145

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 145 permit udp 192.168.33.214 0.0.0.0 192.30.76.155 0.0.0.0 eq tftp
or
access-list 145 permit udp host 192.168.33.214 host 192.30.76.155 eq tftp
Router(config)# interface Gig0/1
Router(config-if)# ip access-group 145 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

Router# *show configuration* (This will show which access groups are associated with particular interfaces)

Router# *show access list 45* (This will show detailed information about this ACL)

Extended Access List Sample #10

Deny/Permit Port Numbers

Write an extended access list to deny FTP traffic from ip addresses 192.30.76.0 through 192.30.76.13 to any destination. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 155

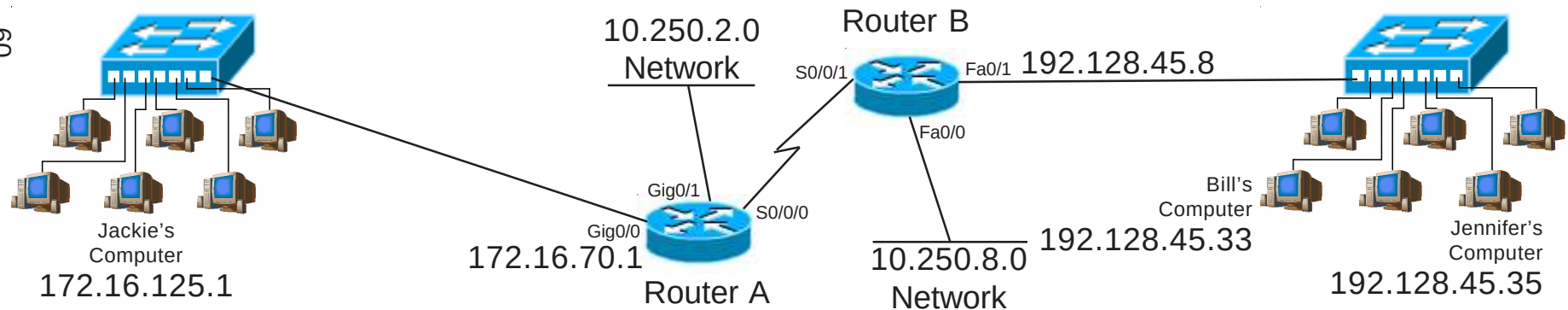
[Writing and installing an ACL]

```
Router# configure terminal
Router(config)# access-list 155 deny tcp 192.30.76.0 0.0.0.8 any eq ftp (Blocks 0 to 7)
Router(config)# access-list 155 deny tcp 192.30.76.8 0.0.0.4 any eq ftp (Blocks 8 to 11)
Router(config)# access-list 155 deny tcp 192.30.76.12 0.0.0.1 any eq ftp (Blocks 12 to 13)
Router(config)# access-list 155 permit ip any any
                        or
                        access-list 155 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface gig0/0
Router(config-if)# ip access-group 155 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

Hint:
It's OK to use multiple protocols
in the same ACL.

The first three TCP statements could be shortened to:

```
access-list 155 permit tcp 192.30.76.14 0.0.0.1 any eq ftp (Permits 14 and 15)
access-list 155 deny tcp 192.30.76.8 0.0.0.15 any eq ftp (Blocks 0 to 15)
```



Extended Access List Problem #15 Deny/Permit a Port Numbers

Write an extended access list to permit ICMP traffic from the 192.128.45.0 network to reach the 172.16.125.0 255.255.255.0 and 10.250.2.0 255.255.255.0 networks. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Fa0/1
 Access-list #: 175 (100-199)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# access-list 175 permit icmp 192.128.45.0 0.0.0.255 172.16.125.0 0.0.0.255
access-list 175 permit icmp 192.128.45.0 0.0.0.255 10.250.2.0 0.0.0.255

Router(config)# *interface fa0/1*
 Router(config-if)# *ip access-group 175 in* or out (circle one)
 Router(config-if)# *exit*

Extended Access List Problem #16

Deny/Permit a Port Numbers

Write a named extended access list called "PEGGYS_LAB" to deny telnet from 10.250.8.0 through 10.250.8.127 from reaching the 192.128.45.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B

Interface: Fa0/0

Access-list Name: PEGGYS_LAB

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# ip access-list extended PEGGYS_LAB

Router(config-std-nacl)# deny tcp 10.250.8.0 0.0.0.127 192.128.45.0 0.0.0.255 eq 23

Router(config-std-nacl)# permit ip any any

Router(config-ext-nacl)# interface fa0/0

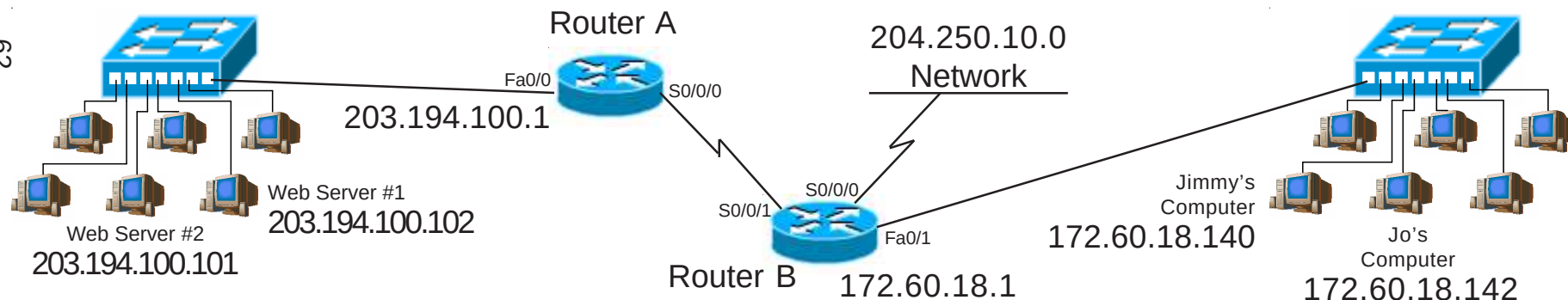
Router(config-if)# ip access-group PEGGYS_LAB in or out (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*

Router# *copy run start*



Extended Access List Problem #17

Deny/Permit Port Numbers

Write an access list to deny Jimmy's computer from sending ftp packets to Web Server 1, but permit ftp to Web Server #2. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Fa0/1
 Access-list #: 150 (1-99)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *access-list 150 deny tcp host 172.60.18.140 host 203.194.100.102 eq ftp*
or
access-list 150 deny tcp 172.60.18.140 0.0.0.0 203.194.100.102 0.0.0.0 eq ftp

Router(config)# *access-list 150 permit ip any any*
or
access-list 150 permit 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255

Router(config)# *interface Fa0/1*
 Router(config-if)# *ip access-group 150 in or out (circle one)*
 Router(config-if)# *exit*
 Router(config)# *exit*

Extended Access List Problem #18

Deny/Permit Port Numbers

Write an extended access list to deny all HTTP traffic intended for the web server at 203.194.100.102 from the 172.66.0.0 network. Permit all other HTTP traffic from the 204.250.10.0 and 172.60.0.0 networks to any other web servers. Deny all other IP traffic to the 203.194.100.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
Interface: S0/0/1
Access-list #: 185 (100-199)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *access-list 185 deny tcp any host 203.194.100.102 eq 80*

or

access-list 185 deny tcp any 203.194.100.102 0.0.0.0 eq 80

access-list 185 permit tcp any any eq 80

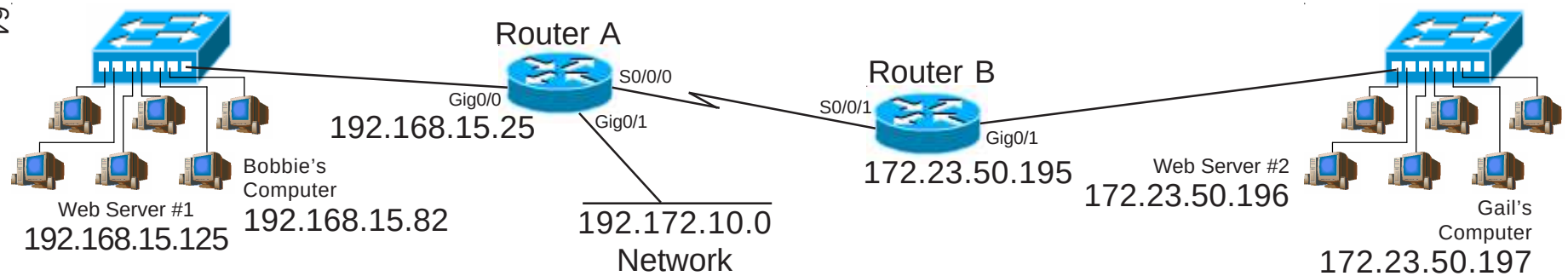
Router(config)# *interface S0/0/1*

Router(config-if)# *ip access-group 185 in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #19

Deny/Permit Port Numbers

Include a remark with each statement of your ACL. Write an extended access list to permit TFTP traffic from all hosts on the 192.168.15.0 network. Deny all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: Gig0/0
 Access-list #: 190 (100-199)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# access-list 175 remark allow ftp from the 192.168.15.0 network

access-list 175 permit udp 192.168.15.0 0.0.0.255 any eq tftp

Router(config)# interface gig0/0

Router(config-if)# ip access-group 190 in or out (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #20

Deny/Permit Port Numbers

Write an extended access list that permits web traffic from web server #2 at 172.23.50.196 to reach everyone on the 192.168.15.0 network. Deny all other IP traffic going to the 192.172.10.0, and 192.168.15.0 networks from the 172.25.50.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B

Interface: Gig0/1

Access-list #: 195 (100-199)

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)# *access-list 195 permit tcp host 172.23.50.196 192.168.15.0 0.0.0.255 eq 80*

or

access-list 195 permit tcp 172.23.50.196 0.0.0.0 192.168.15.0 0.0.0.255 eq 80

Router(config)# *interface gig0/1*

Router(config-if)# *ip access-group 195* in or out (circle one)

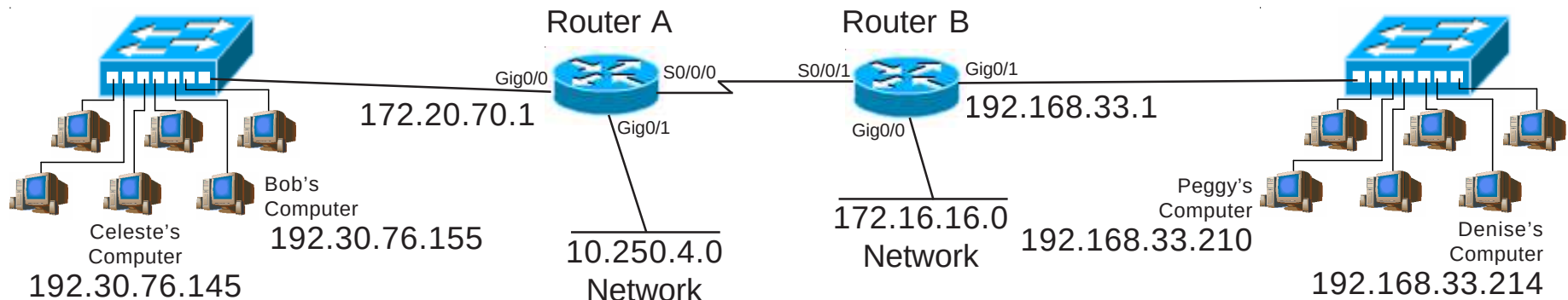
Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*

Writing Access Lists to Restrict Telnet Access...

Restricting access to telnet can be a very usefull option. Telnet is considered a very insecure protocol because it sends passwords through the network in clear-text. By switching from the *access-group* command to the *access-class* command you can increase your security by allowing only those users through that you want to use telnet. The *access-class* command also allows you to apply this access list to the vty connections.



Standard Access List Sample #11 Deny/Permit Telnet

Write a standard access list to permit Denise's and Bob's computers to telnet into Router B. Deny all other telnet traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B

Interface: line VTY 0 4

Access-list #: 45

(using line VTY 0 4 instead of an interface like E1 allows you to apply this access list to all VTY lines with one statement)

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 45 permit 192.168.33.214 0.0.0.0
```

or

```
Router(config)# access-list 45 permit host 192.168.33.214
Router(config)# access-list 45 permit 192.30.76.155 0.0.0.0
```

or

```
Router(config)# access-list 45 permit host 192.30.76.155
```

```
Router(config)# line vty 0 4
```

```
Router(config-line)# access-class 45 in
```

```
Router(config-line)# exit
```

```
Router(config)# exit
```

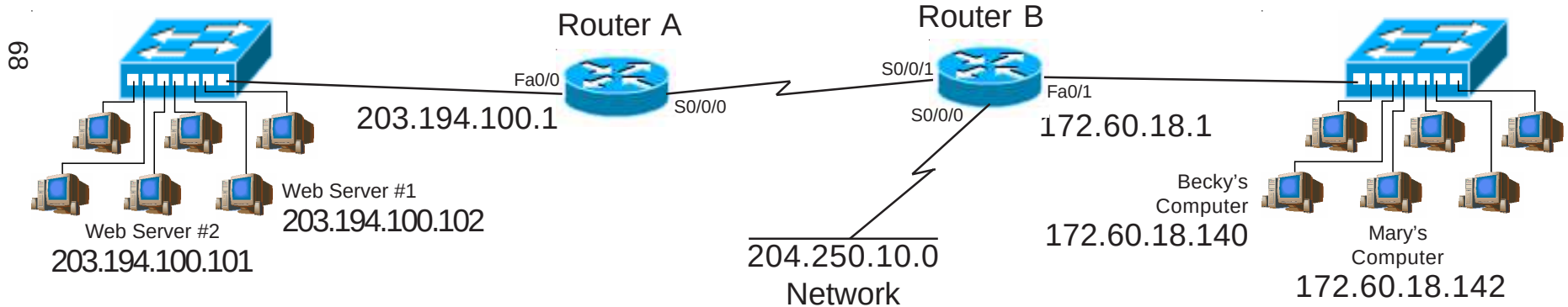
[Viewing information about existing ACL's]

```
Router# show configuration
```

(This will show which access groups are associated with particular interfaces)

```
Router# show access list 45
```

(This will show detailed information about this ACL)



Access List Problem #21

Deny/Permit Telnet

Write a standard access list to permit Becky and Mary's computer to telnet into Router B. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: line vty 0 4
 Access-list #: 50 (1-99)

These ACE statements could be shortened to:

access-list 50 permit 172.60.18.140 0.0.0.2

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *access-list 50 permit 172.60.18.140*
or
access-list 50 permit host 172.60.18.140
or
access-list 50 permit 172.60.18.140 0.0.0.0

access-list 50 permit 172.60.18.142
or
access-list 50 permit host 172.60.18.142
or
access-list 50 permit 172.60.18.142 0.0.0.0

Router(config)# *line vty 0 4*
 Router(config-line)# *access-class* *50* *in* *or out* (circle one)
 Router(config-line)# *exit*
 Router(config)# *exit*

Access List Problem #22

Deny/Permit Telnet

Write a standard access list to permit which will permit Web Server #1 to telnet into Router A. Log the telnet attempts. Deny all other telnet access. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: line vty 0 4
Access-list #: 60 (1-99)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)# *access-list 60 permit 203.194.100.102 0.0.0.0 log*
or

access-list 60 permit 203.194.100.102 log
or

access-list 60 permit host 203.194.100.102 log

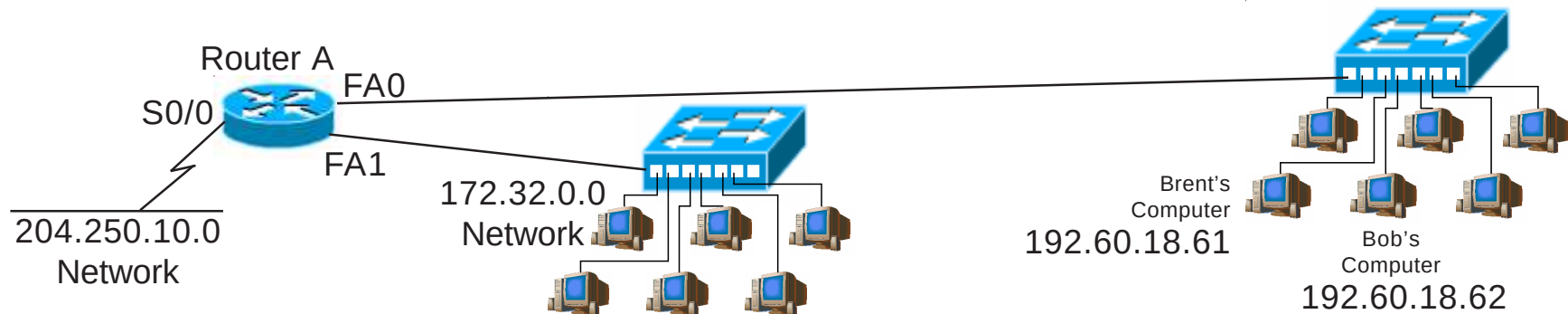
Router(config) *access-list 60 deny any log* (You have to add the implicit deny any log for this to log)

Router(config)# *line vty 0 4*

Router(config-line)# *access-class* *60* *in* *or out* (circle one)

Router(config-line)# *exit*

Router(config)# *exit*



Access List Problem #23

Deny/Permit Telnet

Write a standard access list to deny Brent and Bob's computer telnet access into Router A. Permit all other telnet traffic from the 192.60.18.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A

Interface: line vty 0 4

Access-list #: 70 (1-99)

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

```
Router(config)# access-list 70 deny 192.60.18.61
or
access-list 70 deny host 192.60.18.61
or
access-list 70 deny 192.60.18.61 0.0.0.0


---


access-list 70 deny 192.60.18.62
or
access-list 70 deny host 192.60.18.62
or
access-list 70 deny 192.60.18.62 0.0.0.0


---


access-list 70 permit 192.60.18.0 0.0.0.255 any
```

Router(config)# line vty 0 4

Router(config-line)# *access-class* 70 in or out (circle one)

Router(config-line)# *exit*

Router(config)# *exit*

Reference

Port Numbers

Port numbers are now assigned by the ICANN (Internet Corporation for Assigned Names and Numbers). Commonly used TCP and UDP applications are assigned a port number; such as: HTTP - 80, POP3 - 110, FTP - 20. When an application communicates with another application on another node on the internet, it specifies that application in each data transmission by using its port number. You can also type the name (ie. Telnet) instead of the port number (ie. 23). Port numbers range from 0 to 65536 and are divided into three ranges:

Well Known Ports	0 to 1,023
Registered Ports	1,024 to 49,151
Dynamic and/or Private Ports	49,152 to 65,535

Below is a short list of some commonly used ports. For a complete list of port numbers go to <http://www.iana.org/assignments/port-numbers>.

Some commonly used port numbers:

0	Reserved	
1	TCPMUX	(TCP Port Service Multiplexer)
5	RJE	(Remote Job Entry)
7	ECHO	
9	DISCARD	
11	SYSTAT	(Active users)
13	DAYTIME	
17	QUOTE	(Quote of the day)
18	MSP	(Message Send Protocol)
19	CHARGEN	(Character generator)
20	FTP-DATA	(File Transfer Protocol - Data)
21	FTP	(File Transfer Protocol - Control)
22	SSH	(Remote Login Protocol)
23	Telnet	(Terminal Connection)
25	SMTP	(Simple Mail Transfer Protocol)
29	MSG ICP	
37	TIME	
39	RLP	(Resource Location Protocol)
42	NAMESERV	(Host Name Server)

43	NICNAME	(Who Is)
49	LOGIN	(Login Host Protocol)
53	DNS	(Domain Name Server)
67	BOOTP	(Bootstrap Protocol Server)
68	BOOTPS	(Bootstrap Protocol Client)
69	TFTP	(Trivial File Transfer Protocol)
70	GOPHER	(Gopher Services)
75		(Any Private Dial-out Service)
79	FINGER	
80	HTTP	(Hypertext Transfer Protocol)
95	SUPDUP	(SUPDUP Protocol)
101	HOSTNAME	(NIC Host Name Server)
108	SNAGAS	(SNA Gateway Access Server)
109	POP2	(Post Office Protocol - Version 2)
110	POP3	(Post Office Protocol - Version 3)
113	AUTH	(Authentication Service)
115	SFTP	(Simple File Transfer Protocol)
117	UUCP-PATH	(UUCP Path Service)
118	SQLSERV	(SQL Services)
119	NNTP	(Newsgroup)
123	NTP	(Network Time Protocol)
137	NetBIOS-NS	(NetBIOS Name Service)
139	NetBIOS-SSN	(NetBIOS Session Service)
143	IMAP	(Interim Mail Access Protocol)
150	SQL-NET	(NetBIOS Session Service)
156	SQLSRV	(SQL Service)
161	SNMP	(Simple Network Management Protocol)
179	BGP	(Border Gateway Protocol)
190	GACP	(Gateway Access Control Protocol)
194	IRC	(Internet Relay Chat)
197	DLS	(Directory Location Service)
389	LDAP	(Lightweight Directory Access Protocol)
396	NETWARE-IP	(Novell Netware over IP)
443	HTTPS	(HTTP MCom)
444	SNPP	(Simple Network Paging Protocol)
445	Microsoft-DS	
458	Apple QuickTime	
546	DHCP Client	
547	DHCP Server	
563	SNEWS	
569	MSN	

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31	16-23	8-11	4-5
					24-31	12-15	6-7
						16-19	8-9
						20-23	10-11
		64-127	32-63	32-47	32-39	24-27	12-13
					40-47	28-31	14-15
				48-63	48-55	32-35	16-17
					56-63	36-39	18-19
						40-43	20-21
						44-47	22-23
			64-95	64-79	64-71	48-51	24-25
					72-79	52-55	26-27
				80-95	80-87	56-59	28-29
					88-95	60-63	30-31
						64-67	32-33
			96-127	96-111	96-103	68-71	34-35
					104-111	72-75	36-37
				112-127	112-119	76-79	38-39
					120-127	80-83	40-41
						84-87	42-43
						88-91	44-45
	128-255	128-191	128-159	128-143	128-135	92-95	46-47
					136-143	96-99	48-49
				144-159	144-151	100-103	50-51
					152-159	104-107	52-53
						108-111	54-55
						112-115	56-57
			160-191	160-175	160-167	116-119	58-59
					168-175	120-123	60-61
				176-191	176-183	124-127	62-63
					184-191	128-131	64-65
						132-135	66-67
		192-255	192-223	192-207	192-199	136-139	68-69
					200-207	140-143	70-71
				208-223	208-215	144-147	72-73
					216-223	148-151	74-75
						152-155	76-77
						156-159	78-79
			224-255	224-239	224-231	160-163	80-81
					232-239	164-167	82-83
				240-255	240-247	168-171	84-85
					248-255	172-175	86-87
						176-179	88-89
						180-183	90-91
						184-187	92-93
						188-191	94-95
						192-195	96-97
						196-199	98-99
						200-203	100-101
						204-207	102-103
						208-211	104-105
						212-215	106-107
						216-219	108-109
						220-223	110-111
						224-227	112-113
						228-231	114-115
						232-235	116-117
						236-239	118-119
						240-243	120-121
						244-247	122-123
						248-251	124-125
						252-255	126-127
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							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
						8-11	4-5
						12-15	6-7
			16-31	16-31	16-23	16-19	8-9
					20-23	20-21	10-11
					24-31	24-27	12-13
						28-31	14-15
			32-63	32-47	32-39	32-35	16-17
					36-39	36-37	18-19
					40-47	40-43	20-21
						44-47	22-23
				48-63	48-55	48-51	24-25
					52-55	52-53	26-27
					56-63	56-59	28-29
						60-63	30-31
			64-127	64-79	64-71	64-67	32-33
					72-79	68-71	34-35
						72-75	36-37
						76-79	38-39
				80-95	80-87	80-83	40-41
					88-95	84-87	42-43
						88-91	44-45
						92-95	46-47
				96-111	96-103	96-99	48-49
					104-111	100-103	50-51
						104-107	52-53
						108-111	54-55
			112-127	112-119	112-115	112-115	56-57
					116-119	116-119	58-59
					120-127	120-123	60-61
						124-127	62-63
			128-159	128-143	128-135	128-131	64-65
					136-143	132-135	66-67
						136-139	68-69
						140-143	70-71
				144-159	144-151	144-147	72-73
					148-151	148-149	74-75
					152-155	152-155	76-77
					156-159	156-159	78-79
			160-191	160-175	160-167	160-163	80-81
					164-167	164-167	82-83
					168-175	168-171	84-85
					172-175	172-175	86-87
				176-191	176-183	176-179	88-89
					180-183	180-183	90-91
					184-187	184-187	92-93
					188-191	188-191	94-95
			192-255	192-207	192-199	192-195	96-97
					200-207	196-199	98-99
						200-203	100-101
						204-207	102-103
				208-223	208-215	208-211	104-105
					212-215	212-215	106-107
					216-219	216-219	108-109
					220-223	220-223	110-111
				224-239	224-231	224-227	112-113
					228-231	228-231	114-115
					232-235	232-235	116-117
					236-239	236-239	118-119
			240-255	240-247	240-243	240-243	120-121
					244-247	244-247	122-123
					248-251	248-249	124-125
					252-255	252-255	126-127
				248-255	248-251	248-251	128-129
					252-255	252-255	130-131
							132-133
							134-135

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

IPv4 Addressing and Subnetting Workbook

Version 2.1

11111110

10010101

00011011

10000110

11010011

Student Name:

IPv4 Address Classes

Class A	1 – 127	Leading bit pattern	0	00000000.00000000.00000000.00000000 Network . Host . Host . Host
Class B	128 – 191	Leading bit pattern	10	10000000.00000000.00000000.00000000 Network . Network . Host . Host
Class C	192 – 223	Leading bit pattern	110	11000000.00000000.00000000.00000000 Network . Network . Network . Host
Class D	224 – 239	(Reserved for multicast)		
Class E	240 – 255	(Reserved for experimental, used for research)		

Speciality Address Ranges

Loopback -	Only the single 127.0.0.1 address is used, addresses 127.0.0.0 to 127.255.255.255 are reserved. Any address within this block will loop back to the local host.
Link-Local Addresses -	IPv4 addresses in the address block 169.254.0.0 to 169.254.255.255 (169.254.0.0/16) are designated as link-local addresses.
TEST-NET Addresses -	The address block 192.0.2.0 to 192.0.2.255 (192.0.2.0/24) is set aside for teaching and learning purposes.
Experimental Addresses -	The addresses in the block 240.0.0.0 to 255.255.255.254 are listed as reserved for future use (RFC 3330).

Private Address Space

Class A	10.0.0.0 to 10.255.255.255
Class B	172.16.0.0 to 172.31.255.255
Class C	192.168.0.0 to 192.168.255.255

Default Subnet Masks

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

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Special Thanks to Melvin Baker and Jim Dorsch for taking the time to check this workbook for errors, and to everyone who has sent in suggestions to improve the series.

Binary To Decimal Conversion

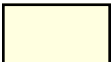
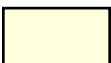
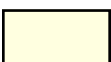
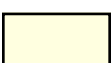
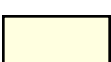
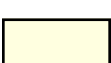
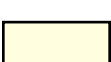
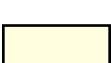
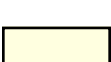
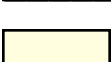
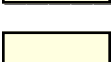
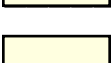
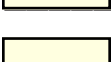
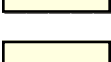
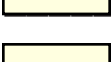
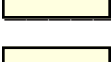
128	64	32	16	8	4	2	1	Answers	Scratch Area	
1	0	0	1	0	0	1	0	<u>146</u>	128	64
0	1	1	1	0	1	1	1	<u>119</u>	16	32
1	1	1	1	1	1	1	1		2	16
1	1	0	0	0	1	0	1		146	4
1	1	1	1	0	1	1	0			2
0	0	0	1	0	0	1	1			1
1	0	0	0	0	0	0	1			<u>119</u>
0	0	1	1	0	0	0	1			
0	1	1	1	1	0	0	0			
1	1	1	1	0	0	0	0			
0	0	1	1	1	0	1	1			
0	0	0	0	0	1	1	1			
						00011011				
						10101010				
						01101111				
						11111000				
						00100000				
						01010101				
						00111110				
						00000011				
						11101101				
						11000000				

Decimal To Binary Conversion

Use all 8 bits for each problem

128	64	32	16	8	4	2	1 =	255		Scratch Area
1	1	1	0	1	1	1	0	238	238	34
0	0	1	0	0	0	1	0	34	-128	-32
								123	110	2
								50	-64	-2
								255	46	0
								200	-32	
								10	14	
								138	-8	
								1	6	
								13	-4	
								250	2	
								107	-2	
								224	0	
								114		
								192		
								172		
								100		
								119		
								57		
								98		
								179		
								2		

Address Class Identification

Address	Class
10.250.1.1	<u>A</u>
150.10.15.0	<u>B</u>
192.14.2.0	
148.17.9.1	
193.42.1.1	
126.8.156.0	
220.200.23.1	
230.230.45.58	
177.100.18.4	
119.18.45.0	
249.240.80.78	
199.155.77.56	
117.89.56.45	
215.45.45.0	
199.200.15.0	
95.0.21.90	
33.0.0.0	
158.98.80.0	
219.21.56.0	

Network Addresses

Using the IP address and subnet mask shown write out the network address:

188.10.18.2	<i>188 . 10 . 0 . 0</i>
255.255.0.0	

10.10.48.80	<i>10 . 10 . 48 . 0</i>
255.255.255.0	

192.149.24.191	
255.255.255.0	

150.203.23.19	
255.255.0.0	

10.10.10.10	
255.0.0.0	

186.13.23.110	
255.255.255.0	

223.69.230.250	
255.255.0.0	

200.120.135.15	
255.255.255.0	

27.125.200.151	
255.0.0.0	

199.20.150.35	
255.255.255.0	

191.55.165.135	
255.255.255.0	

28.212.250.254	
255.255.0.0	

Default Subnet Masks

Write the correct default subnet mask for each of the following addresses:

177.100.18.4	<u>255 . 255 . 0 . 0</u>
119.18.45.0	<u>255 . 0 . 0 . 0</u>
191.249.234.191	
223.23.223.109	
10.10.250.1	
126.123.23.1	
223.69.230.250	
192.12.35.105	
77.251.200.51	
189.210.50.1	
88.45.65.35	
128.212.250.254	
193.100.77.83	
125.125.250.1	
1.1.10.50	
220.90.130.45	
134.125.34.9	
95.250.91.99	

ANDING with Default subnet masks

Every IP address must be accompanied by a subnet mask. By now you should be able to look at an IP address and tell what class it is. Unfortunately your computer doesn't think that way. For your computer to determine the network and subnet portion of an IP address it must "AND" the IP address with the subnet mask.

Default Subnet Masks:

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

ANDING Equations:

1 AND 1	= 1
1 AND 0	= 0
0 AND 1	= 0
0 AND 0	= 0

Sample:

What you see...

IP Address: 192 . 100 . 10 . 33

What you can figure out in your head...

Address Class:	C
Network Portion:	<u>192 . 100 . 10</u> . 33
Host Portion:	192 . 100 . 10 . <u>33</u>

In order for your computer to get the same information it must AND the IP address with the subnet mask in binary.

	Network	Host	
IP Address:	1 1 0 0 0 0 0 0 . 0 1 1 0 0 1 0 0 . 0 0 0 0 1 0 1 0	0 0 1 0 0 0 0 1	(192 . 100 . 10 . 33)
Default Subnet Mask:	1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1	0 0 0 0 0 0 0 0	(255 . 255 . 255 . 0)
AND:	1 1 0 0 0 0 0 0 . 0 1 1 0 0 1 0 0 . 0 0 0 0 1 0 1 0	0 0 0 0 0 0 0 0	(192 . 100 . 10 . 0)

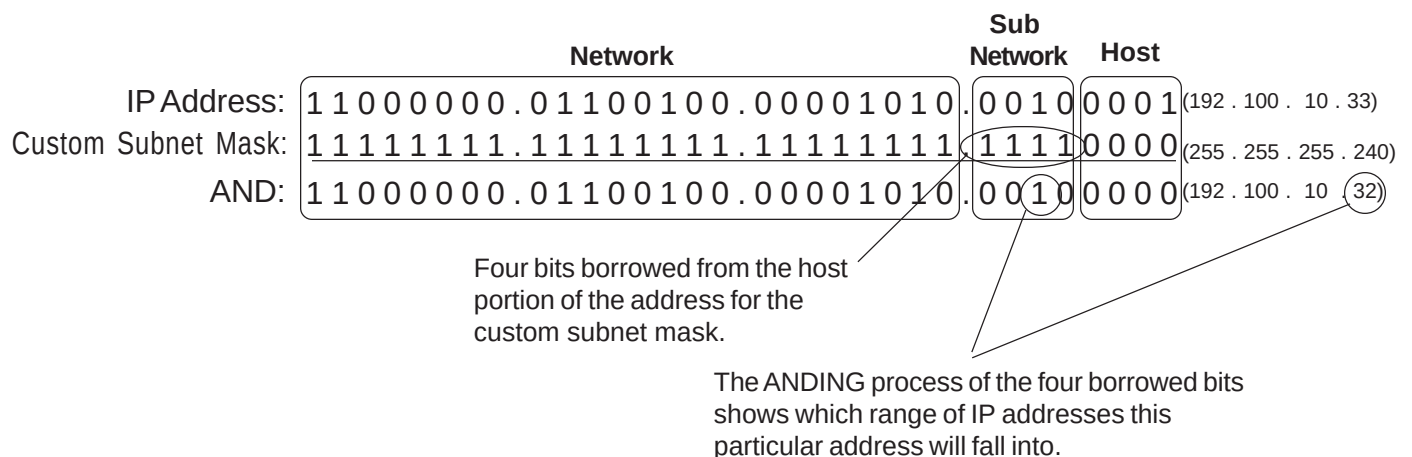
ANDING with the default subnet mask allows your computer to figure out the network portion of the address.

ANDING with Custom subnet masks

When you take a single network such as 192.100.10.0 and divide it into five smaller networks (192.100.10.16, 192.100.10.32, 192.100.10.48, 192.100.10.64, 192.100.10.80) the outside world still sees the network as 192.100.10.0, but the internal computers and routers see five smaller subnetworks. Each independent of the other. This can only be accomplished by using a custom subnet mask. A custom subnet mask borrows bits from the host portion of the address to create a subnetwork address between the network and host portions of an IP address. In this example each range has 14 usable addresses in it. The computer must still AND the IP address against the custom subnet mask to see what the network portion is and which subnetwork it belongs to.

IP Address: 192 . 100 . 10 . 0
Custom Subnet Mask: 255.255.255.240

Address Ranges: 192.10.10.0 to 192.100.10.15
 192.100.10.16 to 192.100.10.31
 192.100.10.32 to 192.100.10.47 (Range in the sample below)
 192.100.10.48 to 192.100.10.63
 192.100.10.64 to 192.100.10.79
 192.100.10.80 to 192.100.10.95
 192.100.10.96 to 192.100.10.111
 192.100.10.112 to 192.100.10.127
 192.100.10.128 to 192.100.10.143
 192.100.10.144 to 192.100.10.159
 192.100.10.160 to 192.100.10.175
 192.100.10.176 to 192.100.10.191
 192.100.10.192 to 192.100.10.207
 192.100.10.208 to 192.100.10.223
 192.100.10.224 to 192.100.10.239
 192.100.10.240 to 192.100.10.255



In the next set of problems you will determine the necessary information to determine the correct subnet mask for a variety of IP addresses.

How to determine the number of subnets and the number of hosts per subnet

The formula that can provide this basic information:

$$\text{Number of subnets} = 2^s$$

$$\text{Number of usable hosts per subnet} = 2^h - 2$$

This formula calculates the number of hosts or subnets based on the number of binary bits used. For example if you borrow three bits from the host portion of the address use the *number of subnets* formula to determine the total number of subnets gained by borrowing the three bits. This would be 2^3 or $2 \times 2 \times 2 = 8$ subnets

To determine the number of hosts per subnet you would take the number of binary bits used in the host portion and apply this to the *number of hosts per subnet* formula. If five bits are in the host portion of the address this would be 2^5 or $2 \times 2 \times 2 \times 2 \times 2 = 32$ hosts.

When dealing with the *number of hosts per subnet* you have to subtract two addresses from the range. The first address in every range is the subnet number. The last address in every range is the broadcast address. These two addresses cannot be assigned to any device in the network which is why you have to subtract two addresses to find the number of usable addresses in each range.

For example, if two bits are borrowed for the network portion of the address you can easily determine the number of subnets and hosts per subnets using the formula.

195. 223 . 50 . 0 0 | 0 0 0 0 0 0

The number of subnets created by borrowing 2 bits is 2^2 or $2 \times 2 = 4$ subnets.

The number of hosts created by leaving 6 bits is $2^6 - 2$ or $2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64 - 2 = 62$ usable hosts per subnet.

Custom Subnet Problems

Custom Subnet Masks

Problem 1

Number of needed subnets **14**
 Number of needed usable hosts **14**
 Network Address **192.10.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 240

Total number of subnets 16

Total number of host addresses 16

Number of usable addresses 14

Number of bits borrowed 4

Show your work for Problem 1 in the space below.

		256	128	64	32	16	8	4	2	1	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256		
		128	64	32	16	8	4	2	1	-	Binary values
192 . 10 . 10 . 0		0	0	0	0	0	0	0	0		

Add the binary value numbers to the left of the line to create the custom subnet mask.

	128
	64
	32
	+16
	240

16	Observe the total number of hosts.
-2	
14	Subtract 2 for the number of usable hosts.

Custom Subnet Masks

Problem 2

Number of needed subnets **1000**

Number of needed usable hosts **60**

Network Address **165.100.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

Show your work for Problem 2 in the space below.

Number of Hosts -	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets -	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values -	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
	165	100	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Add the binary value numbers to the left of the line to create the custom subnet mask.

128	128
64	+64
32	192
16	
8	
4	
2	
+1	
255	

Observe the total number of hosts.

Subtract 2 for the number of usable hosts.

64
-2
62

Custom Subnet Masks

Problem 3

Network Address **148.75.0.0 /26**

/26 indicates the total number of bits used for the network and subnetwork portion of the address. All bits remaining belong to the host portion of the address.

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

Show your work for Problem 3 in the space below.

	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Hosts	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Number of Subnets	-	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768
Binary values	-	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2
		148	75	0	0	0	0	0	0	0	0	0	0	0	0	0

Add the binary value numbers to the left of the line to create the custom subnet mask.

128	128
64	+64
32	192
16	
8	
4	
2	
+1	
255	

64	Observe the total number of hosts.
-2	
62	Subtract 2 for the number of usable hosts.

1024	
-2	
1,022	Subtract 2 for the total number of subnets to get the usable number of subnets.

Custom Subnet Masks

Problem 4

Number of needed subnets **6**
 Number of needed usable hosts **30**
 Network Address **195.85.8.0**

Address class	
Default subnet mask	
Custom subnet mask	
Total number of subnets	
Total number of host addresses	
Number of usable addresses	
Number of bits borrowed	

Show your work for Problem 5 in the space below.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets -	2	4	8	16	32	64	128	256		
	128	64	32	16	8	4	2	1	-	Binary values
195 . 85 . 8 .	0	0	0	0	0	0	0	0		

Custom Subnet Masks

Problem 5

Number of needed subnets **6**
 Number of needed usable hosts **30**
 Network Address **210.100.56.0**

Address class	
Default subnet mask	
Custom subnet mask	
Total number of subnets	
Total number of host addresses	
Number of usable addresses	
Number of bits borrowed	

Show your work for Problem 4 in the space below.

<i>Number of</i>	256	128	64	32	16	8	4	2	-	<i>Number of</i>
<i>Subnets</i>	2	4	8	16	32	64	128	256		<i>Hosts</i>
	128	64	32	16	8	4	2	1	-	<i>Binary values</i>
210 . 100 . 56 .	0	0	0	0	0	0	0	0		

Custom Subnet Masks

Problem 6

Number of needed subnets **126**
 Number of needed usable hosts **131,070**
 Network Address **118.0.0.0**

Address class	
Default subnet mask	
Custom subnet mask	
Total number of subnets	
Total number of host addresses	
Number of usable addresses	
Number of bits borrowed	

Show your work for Problem 6 in the space below.

Number of Hosts	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576	2097152	4194304	8388608	16777216	33554432	67108864	134217728	268435456	536870912	1073741824	2147483648	4294967296	8589934592	17179869184	34359738368	68719476736	137438953472	274877906944	549755813888	1099511627776	2199023255552	4398046511104	8796093022208	17592186044416	35184372088832	70368744177664	140737488355328	281474976710656	562949953421312	1125899906842624	2251799813685248	4503599627370496	9007199254740992	18014398509481984	36028797018963968	72057594037927936	144115188075855872	288230376151711744	576460752303423488	1152921504606846976	2305843009213693952	4611686018427387904	9223372036854775808	18446744073709551616	36893488147419103232	73786976294838206464	147573952589676412928	295147905179352825856	590295810358705651712	1180591620717411303424	2361183241434822606848	4722366482869645213696	9444732965739290427392	18889465931478580854784	37778931862957161709568	75557863725914323419136	151115727451828646838272	302231454903657293676544	604462909807314587353088	1208925819614629174706176	2417851639229258349412352	4835703278458516698824704	9671406556917033397649408	19342813113834066795298816	38685626227668133590597632	77371252455336267181195264	154742504910672534362390528	309485009821345068724781056	618970019642690137449562112	1237940039285380274899124224	2475880078570760549798248448	4951760157141521099596496896	9903520314283042199192993792	19807040628566084398385987584	39614081257132168796771975168	79228162514264337593543950336	158456325028528675187087900672	316912650057057350374175801344	633825300114114700748351602688	1267650600228229401496703205376	2535301200456458802993406410752	5070602400912917605986812821504	10141204801825835211973625643008	20282409603651670423947251286016	40564819207303340847894502572032	81129638414606681695789005144064	162259276829213363391578010288128	324518553658426726783156020576256	649037107316853453566312041152512	1298074214633706907132624082305024	2596148429267413814265248164610048	5192296858534827628530496329220096	10384593717069655257060992658440192	20769187434139310514121985316880384	41538374868278621028243970633760768	83076749736557242056487941267521536	166153499473114484112975882535043072	332306998946228968225951765070086144	664613997892457936451903530140172288	1329227995784915872903807060280344576	2658455991569831745807614120560689152	5316911983139663491615228241121378304	10633823966279326983230456482242756608	21267647932558653966460912964485513216	42535295865117307932921825928971026432	85070591730234615865843651857942052864	170141183460469231731687303715884105728	340282366920938463463374607431768211456	680564733841876926926749214863536422912	1361129467683753853853498429727072845824	272225893536750770770699685945414569152	544451787073501541541399371890829138304	1088903574147003083082798743781658276608	2177807148294006166165597487563316553216	4355614296588012332331194975126633106432	8711228593176024664662389950253266212864	17422457186352049329324779900506532425728	34844914372704098658649559801013064851456	69689828745408197317299119602026129702912	139379657490816394634598239204052259405824	278759314981632789269196478408104518811648	557518629963265578538392956816209037623296	1115037259926531157076785913632418075246592	2230074519853062314153571827264836150493184	4460149039706124628307143654529672300986368	8920298079412249256614287309059344601972736	17840596158824498513228574618118689203945472	35681192317648997026457149236237378407890848	71362384635297994052914298472474756815781696	142724769270595988105828596944949513631563392	285449538541191976211657193889899027263126784	570899077082383952423314387779798054526253568	1141798154164767904846628775559596109052507136	2283596308329535809693257551119192218105014272	4567192616659071619386515102238384436210028544	9134385233318143238773030204476768872420057088	18268770466636286477546060408953537744840114176	36537540933272572955092120817907075489680228352	73075081866545145910184241635814150979360456704	146150163733090291820368483271628301958720913408	292300327466180583640736966543256603917441826816	584600654932361167281473933086513207834883653632	1169201309864722334562947866173026415669767307264	2338402619729444669125895732346052831339534614528	4676805239458889338251791464692105662679069229056	9353610478917778676503582929384211325358138458112	18707220957835557353007165858768422650716276916224	37414441915671114706014331717536845301432553832448	74828883831342229412028663435073690602865107664896	149657767662684458824057326870147381205730215329792	299315535325368917648114653740294762411460430659584	598631070650737835296229307480589524822920861319168	1197262141301475670592458614961179049645841722638336	2394524282602951341184917229922358099291683445276672	4789048565205902682369834459844716198583366890553344	9578097130411805364739668919689432397166733781106688	19156194260823610729479337839378864794333467562213376	38312388521647221458958675678757729588666935124426752	76624777043294442917917351357515459177333870248853504	153249554086588885835834702715030918354667740497707008	306499108173177771671669405430061836709335480995414016	612998216346355543343338810860123673418670961990828032	1225996432692711086686677621720247346837341923981646064	2451992865385422173373355243440494693674683847963292128	4903985730770844346746710486880989387349367695926584256	9807971461541688693493420973761978774698735391853168512	19615942923083377386986841947523957549397470783706336032	39231885846166754773973683895047915098794941567412672064	78463771692333509547947367790095830197589883134825344128	156927543384667019095894735580191660395179766269650688256	313855086769334038191789471160383320790359532539301376512	627710173538668076383578942320766641580719065078602753024	1255420347077336152767157884641533283161438130157205506048	2510840694154672305534315769283066566322876260314411012096	5021681388309344611068631538566133132645752520628822024192	10043362776618689222137263077132266265291505041257644048384	20086725553237378444274526154264532530583010082515288096768	40173451106474756888549052308529065061166020165030576193536	80346902212949513777098104617058130122332040330061152707072	160693804425899027554196209234116260244664080660122304344144	321387608851798055108392418468232520489328161320244608688288	642775217703596110216784836936465040978656322640489217376576	1285550435407192220433569673872930081957312645280978434753152	2571100870814384440867139347745860163914625290561956869506304	5142201741628768881734278695491720327829250581123913739012608	10284403483257537763468557390983440655658501162247827478025216	20568806966515075526937114781966881311317002324495654956050432	41137613933030151053874229563933762622634004648991309912100864	82275227866060302107748459127867525245268009297982619824201728	164550455732120604215496918255735050490536018595965239648403456	329100911464241208430993836511470100981072037191930479296806912	658201822928482416861987673022940201962144074383860958593613824	1316403645856964833723975346045880403924288148767721917187227648	2632807291713929667447950692091760807848576297535443834374455296	5265614583427859334895901384183521615697152595070887668748910592	10531229166855718669791802768367043231394305190141775337497821184	21062458333711437339583605536734086462788610380283550674995642368	42124916667422874679167211073468172925577220760567101349991284736	84249833334845749358334422146936345851154441521134202699982574472	168499666669691498716668844293872691702308883042268405399965148944	336999333339382997433337688587745383404617766084536810799930297888	673998666678765994866675377175490766809235532169073621599860595776	1347997333357531989733350754350981533618471064338147243199721191552	2695994666715063979466701508701963067236942128676294486399442383104	5391989333430127958933403017403926134473884257352588972798884766208	10783978666860255917866806034807852268947768514705177945597769532416	21567957333720511835733612069615704537895537029410355891195539064832	43135914667441023671467224139231409075791074058820711782391078129664	86271829334882047342934448278462818151582148117641423564782156259328	172543658669764094685868896556925636303164296235282847129564312518656	345087317339528189371737793113851272606328592470565694259128625037312	690174634679056378743475586227702545212657184941131388518257250074624	1380349269358112757486951172455405090425314369882262777036514500149248	2760698538716225514973902344910810180850628739764525554073029000298496	5521397077432451029947804689821620361701257479529051108146058000596992	11042794154864902059895609379643240723402514959058102216292116001193984	22085588309729804119791218759286481446805029918116204432584232002387968	44171176619459608239582437518572962893610059836232408865168464004775936	88342353238919216479164875037145925787220119672464817730336928009551872	176684706477838432958329750074291851574440239344929635460673856019103744	353369412955676865916659500148583703148880478689859270921347712038207488	706738825911353731833319000297167406297760957379718541842695424076414976	1413477651822707463666638000594334812595521914759437083685390848152899952	2826955303645414927333276001188669625191043829518874167370781696305799904	5653910607290829854666552002377339250382087659037748334741563392611599808	11307821214581659709333104004754678500764175318075496669483126785223199616	22615642429163319418666208009509357001528350636150993338966253570446399232	45231284858326638837332416019018714003056701272301986677932507140892798464	90462569716653277674664832038037428006113402544603973355865014281785596928	180925139433306555349329664076074856012226805089207946711730028563571193856	361850278866613110698659328152149712024453610178415893423460057127142387712	723700557733226221397318656304299424048907220356831786846920114254284775424	1447401115466452442794637312608598848097814440713663573693840228508569550848	2894802230932904885589274625217197696195628881427327147387680457017139101696	5789604461865809771178549250434395392391257762854654294775360914034278203392	11579208923731619542357098500868790784782515525709308589550721828068556406784	23158417847463239084714197001737581569565031051418617179101443656137112813568	46316835694926478169428394003475163139130062102837234358202887312274225627136	92633671389852956338856788006950326278260124205674468716405774624548451254272	185267342779705912677713576013900652556520248411348937432811549249096902508544	370534685559411825355427152027801305113040496822697874865623098498193805017088	741069371118823650710854304055602610226080993645395749731246196996387610034176	1482138742237647301421708608111205220452161987290791499462492393992775220068352	2964277484475294602843417216222410440904323974581582998924984787985550440136704	592855496895058920568683443244482088180864794916316599784996957
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Custom Subnet Masks

Problem 7

Number of needed subnets **2000**

Number of needed usable hosts **15**

Network Address **178.100.0.0**

Address class	
Default subnet mask	
Custom subnet mask	
Total number of subnets	
Total number of host addresses	
Number of usable addresses	
Number of bits borrowed	

Show your work for Problem 7 in the space below.

Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
178 . 100 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0																

Custom Subnet Masks

Problem 8

Number of needed subnets **3**

Number of needed usable hosts **45**

Network Address **200.175.14.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

Show your work for Problem 8 in the space below.

Custom Subnet Masks

Problem 9

Number of needed subnets **60**

Number of needed usable hosts **1,000**

Network Address **128.77.0.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

Show your work for Problem 9 in the space below.

Custom Subnet Masks

Problem 10

Number of needed usable hosts **60**

Network Address **198.100.10.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

Show your work for Problem 10 in the space below.

Custom Subnet Masks

Problem 11

Number of needed subnets **250**

Network Address **101.0.0.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

Show your work for Problem 11 in the space below.

Custom Subnet Masks

Problem 12

Number of needed subnets **5**

Network Address **218.35.50.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

Show your work for Problem 12 in the space below.

Custom Subnet Masks

Problem 13

Number of needed usable hosts **25**

Network Address **218.35.50.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

Show your work for Problem 13 in the space below.

Custom Subnet Masks

Problem 14

Number of needed subnets **10**

Network Address **172.59.0.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

Show your work for Problem 14 in the space below.

Custom Subnet Masks

Problem 15

Number of needed usable hosts **50**

Network Address **172.59.0.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

Show your work for Problem 15 in the space below.

Custom Subnet Masks

Problem 16

Number of needed usable hosts **29**

Network Address **23.0.0.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

Show your work for Problem 16 in the space below.

Subnetting

Problem 1

Number of needed subnets **14**

Number of needed usable hosts **14**

Network Address **192.10.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 240

Total number of subnets 16

Total number of host addresses 16

Number of usable addresses 14

Number of bits borrowed 4

What is the 4th subnet range? 192.10.10.48 to 192.10.10.63

What is the subnet number for the 8th subnet? 192 . 10 . 10 . 112

What is the subnet broadcast address for the 13th subnet? 192 . 10 . 10 . 207

What are the assignable addresses for the 9th subnet? 192.10.10.129 to 192.10.10.142

Show your work for Problem 1 in the space below.

Number of Subnets	256 128 64 32				16 8 4 2				Number of Hosts	
	2	4	8	16	32	64	128	256		
	128	64	32	16	8	4	2	1	Binary values	
192.10.10.0	0	0	0	0	0	0	0	0		
(1)	0	0	0	0	192.10.10.0				to	192.10.10.15
(2)	0	0	0	1	192.10.10.16				to	192.10.10.31
(3)	0	0	1	0	192.10.10.32				to	192.10.10.47
(4)	0	0	1	1	192.10.10.48				to	192.10.10.63
(5)	0	1	0	0	192.10.10.64				to	192.10.10.79
(6)	0	1	0	1	192.10.10.80				to	192.10.10.95
(7)	0	1	1	0	192.10.10.96				to	192.10.10.111
(8)	0	1	1	1	192.10.10.112				to	192.10.10.127
(9)	1	0	0	0	192.10.10.128				to	192.10.10.143
(10)	1	0	0	1	192.10.10.144				to	192.10.10.159
(11)	1	0	1	0	192.10.10.160				to	192.10.10.175
(12)	1	0	1	1	192.10.10.176				to	192.10.10.191
(13)	1	1	0	0	192.10.10.192				to	192.10.10.207
(14)	1	1	0	1	192.10.10.208				to	192.10.10.223
(15)	1	1	1	0	192.10.10.224				to	192.10.10.239
(16)	1	1	1	1	192.10.10.240				to	192.10.10.255

$$\begin{array}{r}
 128 \\
 64 \\
 32 \\
 +16 \\
 \hline
 \text{Custom subnet mask } 240
 \end{array}$$

$$\begin{array}{r}
 16 \\
 -2 \\
 \hline
 \text{Usable subnets } 14
 \end{array}$$

$$\begin{array}{r}
 16 \\
 -2 \\
 \hline
 \text{Usable hosts } 14
 \end{array}$$

The binary value of the last bit borrowed is the range. In this problem the range is 16.

The first address in each subnet range is the subnet number.

The last address in each subnet range is the subnet broadcast address.

Subnetting

Problem 2

Number of needed subnets **1000**

Number of needed usable hosts **60**

Network Address **165.100.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

What is the 15th
subnet range? 165.100.3.128 to 165.100.3.191

What is the subnet number
for the 6th subnet? 165 . 100 . 1 . 64

What is the subnet
broadcast address for
the 6th subnet? 165 . 100 . 1 . 127

What are the assignable
addresses for the 9th
subnet? 165.100.2.1 to 165.100.0.62

31

[illegible]

Subnetting

Problem 3

Hint: It is possible to borrow one bit to create two subnets.

Number of needed subnets **2**

Network Address **195.223.50.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 2nd
subnet range?

What is the subnet number
for the 2nd subnet?

What is the subnet
broadcast address for
the 1st subnet?

What are the assignable
addresses for the 1st
subnet?

Show your work for Problem 3 in the space below.

		256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256		
		128	64	32	16	8	4	2	1	-	Binary values
195. 223 . 50 . 0		0	0	0	0	0	0	0	0		

Subnetting

Problem 4

Number of needed subnets **750**

Network Address **190.35.0.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 15th
subnet range?

What is the subnet number
for the 13th subnet?

What is the subnet
broadcast address for
the 10th subnet?

What are the assignable
addresses for the 6th
subnet?

Subnetting

Problem 5

Number of needed usable hosts **6**

Network Address **126.0.0.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 2nd
subnet range?

What is the subnet number
for the 5th subnet?

What is the subnet
broadcast address for
the 7th subnet?

What are the assignable
addresses for the 10th
subnet?

Subnetting

Problem 6

Number of needed subnets **10**

Network Address **192.70.10.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 9th
subnet range?

What is the subnet number
for the 4th subnet?

What is the subnet
broadcast address for
the 12th subnet?

What are the assignable
addresses for the 10th
subnet?

Subnetting

Problem 7

Network Address **10.0.0.0 /16**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 11th
subnet range?

What is the subnet number
for the 6th subnet?

What is the subnet
broadcast address for
the 2nd subnet?

What are the assignable
addresses for the 9th
subnet?

Subnetting

Problem 8

Number of needed subnets **5**

Network Address **172.50.0.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 4th
subnet range?

What is the subnet number
for the 5th subnet?

What is the subnet
broadcast address for
the 6th subnet?

What are the assignable
addresses for the 3rd
subnet?

Subnetting

Problem 9

Number of needed usable hosts **28**

Network Address **172.50.0.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 2nd
subnet range?

What is the subnet number
for the 10th subnet?

What is the subnet broadcast
address for
the 4th subnet?

What are the assignable
addresses for the 6th
subnet?

Subnetting

Problem 10

Number of needed subnets **45**

Network Address **220.100.100.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 5th
subnet range?

What is the subnet number
for the 4th subnet?

What is the subnet
broadcast address for
the 13th subnet?

What are the assignable
addresses for the 12th
subnet?

Subnetting

Problem 11

Number of needed usable hosts **8,000**

Network Address **135.70.0.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 6th
subnet range?

What is the subnet number
for the 7th subnet?

What is the subnet
broadcast address for
the 3rd subnet?

What are the assignable
addresses for the 5th
subnet?

Subnetting

Problem 12

Number of needed usable hosts **45**

Network Address **198.125.50.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 2nd
subnet range?

What is the subnet number
for the 2nd subnet?

What is the subnet
broadcast address for
the 4th subnet?

What are the assignable
addresses for the 3rd
subnet?

Subnetting

Problem 13

Network Address **165.200.0.0 /26**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 10th
subnet range?

What is the subnet number
for the 11th subnet?

What is the subnet
broadcast address for
the 1023rd subnet?

What are the assignable
addresses for the 1022nd
subnet?

Subnetting

Problem 14

Number of needed usable hosts **16**

Network Address **200.10.10.0**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 7th
subnet range?

What is the subnet number
for the 5th subnet?

What is the subnet
broadcast address for
the 4th subnet?

What are the assignable
addresses for the 6th
subnet?

Subnetting

Problem 15

Network Address **93.0.0.0 /19**

Address class

Default subnet mask

Custom subnet mask

Total number of subnets

Total number of host addresses

Number of usable addresses

Number of bits borrowed

What is the 15th
subnet range?

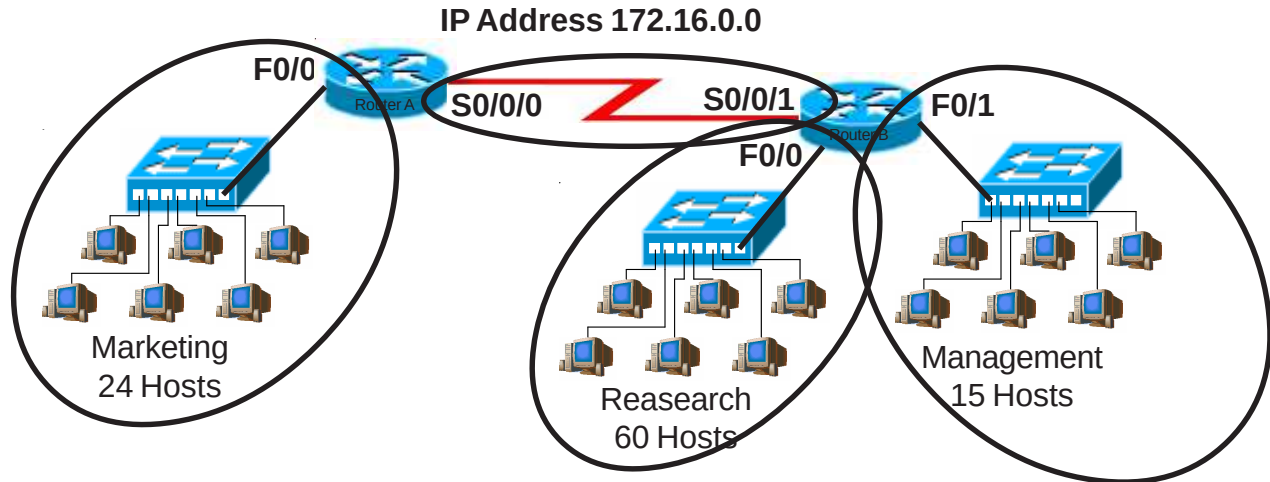
What is the subnet number
for the 9th subnet?

What is the subnet
broadcast address for
the 7th subnet?

What are the assignable
addresses for the 12th
subnet?

Practical Subnetting 1

Based on the information in the graphic shown, design a network addressing scheme that will supply the minimum number of subnets, and allow enough extra subnets and hosts for 100% growth in both areas. Circle each subnet on the graphic and answer the questions below.



Address class	<u>B</u>
Custom subnet mask	<u>255.255.224.0</u>
Minimum number of subnets needed	<u>4</u>
Extra subnets required for 100% growth (Round up to the next whole number)	<u>+ 4</u>
Total number of subnets needed	<u>= 8</u>
Number of host addresses in the largest subnet group	<u>60</u>
Number of addresses needed for 100% growth in the largest subnet (Round up to the next whole number)	<u>+ 60</u>
Total number of address needed for the largest subnet	<u>= 120</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Research	<u>172.16.0.0 to 172.31.255</u>
IP address range for Marketing	<u>172.16.32.0 to 172.63.255</u>
IP address range for Management	<u>172.16.64.0 to 172.95.255</u>
IP address range for Router A to Router B serial connection	<u>172.16.96.0 to 172.127.255</u>

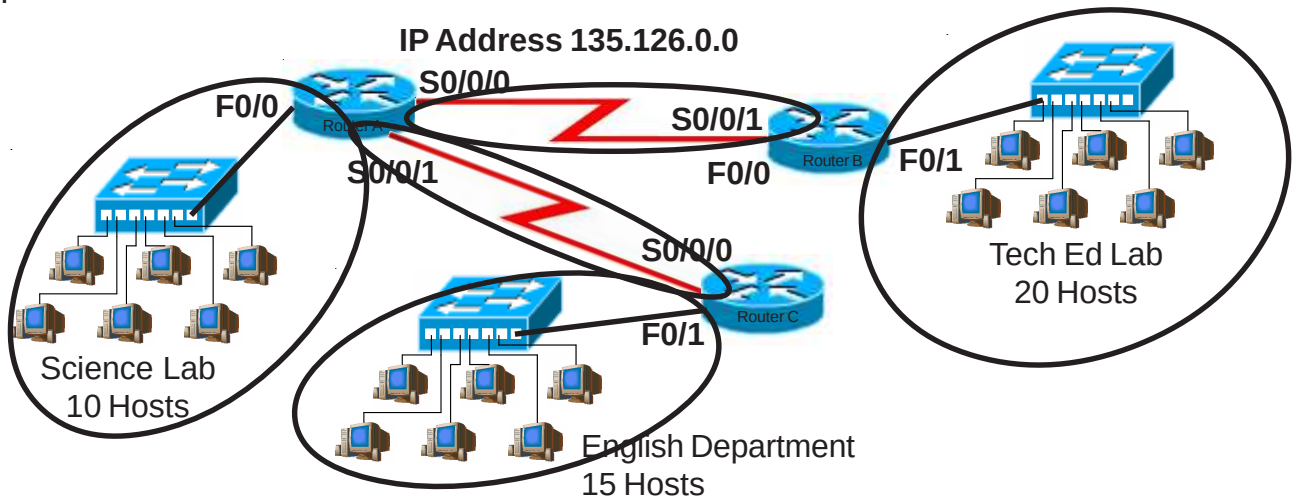
47

$$\begin{array}{r} 4 \\ \times 1.0 \\ \hline 4 \end{array}$$

$$\begin{array}{r} 60 \\ \times 1.0 \\ \hline 60 \end{array}$$

Practical Subnetting 2

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 30% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class	<u>B</u>
Custom subnet mask	<u>255.255.255.224</u>
Minimum number of subnets needed	<u>5</u>
Extra subnets required for 30% growth (Round up to the next whole number)	<u>+ 2</u>
Total number of subnets needed	<u>= 7</u>
Number of host addresses in the largest subnet group	<u>20</u>
Number of addresses needed for 30% growth in the largest subnet (Round up to the next whole number)	<u>+ 6</u>
Total number of address needed for the largest subnet	<u>= 26</u>

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Tech Ed	<u>135.126.0.0 to 135.126.0.31</u>
IP address range for English	<u>135.126.0.32 to 135.126.0.63</u>
IP address range for Science	<u>135.126.0.64 to 135.126.0.95</u>
IP address range for Router A to Router B serial connection	<u>135.126.0.96 to 135.126.0.127</u>
IP address range for Router A to Router C serial connection	<u>135.126.0.128 to 135.126.0.159</u>

Show your work for Problem 2 in the space below.

Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
	135	126	0	0	0	0	0	0	0	0	0	0	0	0	0	0

$$\begin{array}{r} 5 \\ \times 3 \\ \hline 15 \end{array}$$

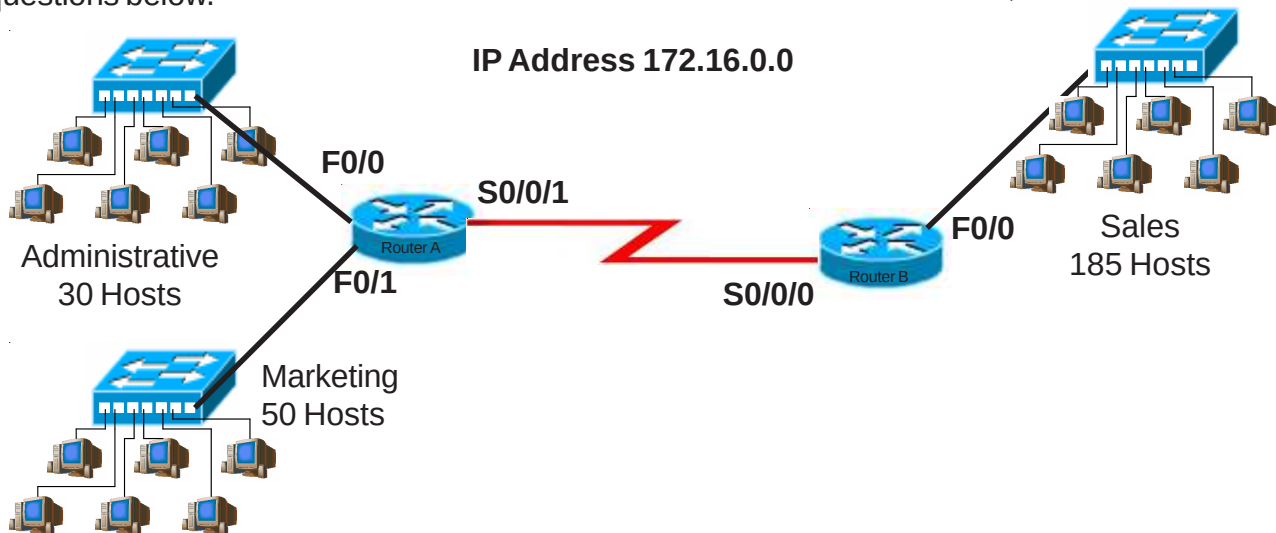
(Round up to 2)

$$\begin{array}{r} 20 \\ \times 3 \\ \hline 6 \end{array}$$

(1)				0	135.126.0.0	to	135.126.0.31
(2)				1	135.126.0.32	to	135.126.0.63
(3)			1	0	135.126.0.64	to	135.126.0.95
(4)			1	1	135.126.0.96	to	135.126.0.127
(5)		1	0	0	135.126.0.128	to	135.126.0.159
(6)		1	0	1	135.126.0.160	to	135.126.0.191
(7)		1	1	0	135.126.0.192	to	135.126.0.223
(8)		1	1	1	135.126.0.224	to	135.126.0.255
(9)	1	0	0	0	135.126.1.0	to	135.126.1.31
(10)	1	0	0	1	135.126.1.32	to	135.126.1.63
(11)	1	0	1	0	135.126.1.64	to	135.126.1.95
(12)	1	0	1	1	135.126.1.96	to	135.126.1.127
(13)	1	1	0	0	135.126.1.128	to	135.126.1.159
(14)	1	1	0	1	135.126.1.160	to	135.126.1.191
(15)	1	1	1	0	135.126.1.192	to	135.126.1.223
(16)	1	1	1	1	135.126.1.224	to	135.126.1.255

Practical Subnetting 3

Based on the information in the graphic shown, design a classfull network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 25% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class

Custom subnet mask

Minimum number of subnets needed

Extra subnets required for 25% growth
(Round up to the next whole number)

Total number of subnets needed

Number of host addresses
in the largest subnet group

Number of addresses needed for
25% growth in the largest subnet
(Round up to the next whole number)

Total number of address
needed for the largest subnet

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Sales

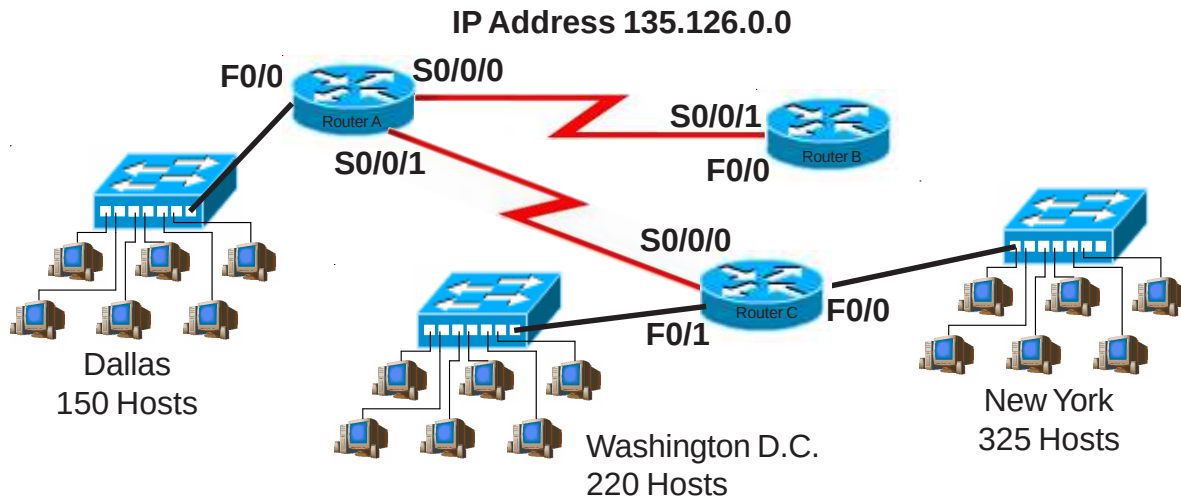
IP address range for Marketing

IP address range for Administrative

IP address range for Router A
to Router B serial connection

Practical Subnetting 4

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of subnets**, and allow enough extra subnets and hosts for 70% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class

Custom subnet mask

Minimum number of subnets needed

Extra subnets required for 70% growth
(Round up to the next whole number)

Total number of subnets needed

Number of host addresses
in the largest subnet group

Number of addresses needed for
70% growth in the largest subnet
(Round up to the next whole number)

Total number of address
needed for the largest subnet

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for New York

IP address range for Washington D. C.

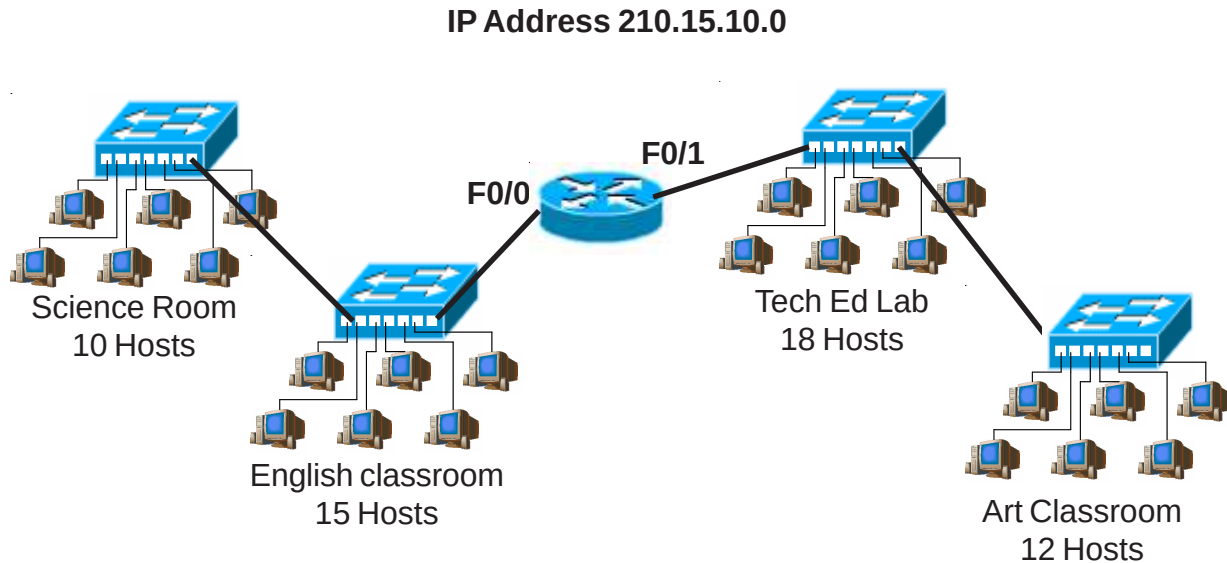
IP address range for Dallas

IP address range for Router A
to Router B serial connection

IP address range for Router A
to Router C serial connection

Practical Subnetting 5

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 100% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class

Custom subnet mask

Minimum number of subnets needed

Extra subnets required for 100% growth

(Round up to the next whole number)

Total number of subnets needed

Number of host addresses
in the largest subnet group

Number of addresses needed for
100% growth in the largest subnet

(Round up to the next whole number)

Total number of address
needed for the largest subnet

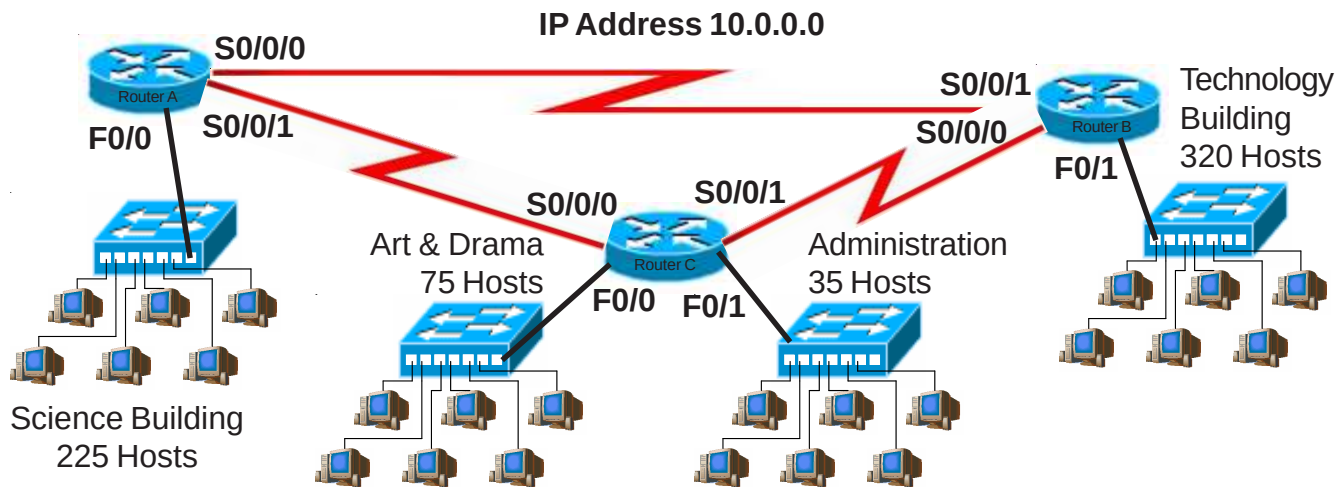
Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Router F0/0 Port

IP address range for Router F0/1 Port

Practical Subnetting 6

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of subnets**, and allow enough extra subnets and hosts for 20% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class

Custom subnet mask

Minimum number of subnets needed

Extra subnets required for 20% growth
(Round up to the next whole number)

Total number of subnets needed

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Technology

IP address range for Science

IP address range for Arts & Drama

IP Address range Administration

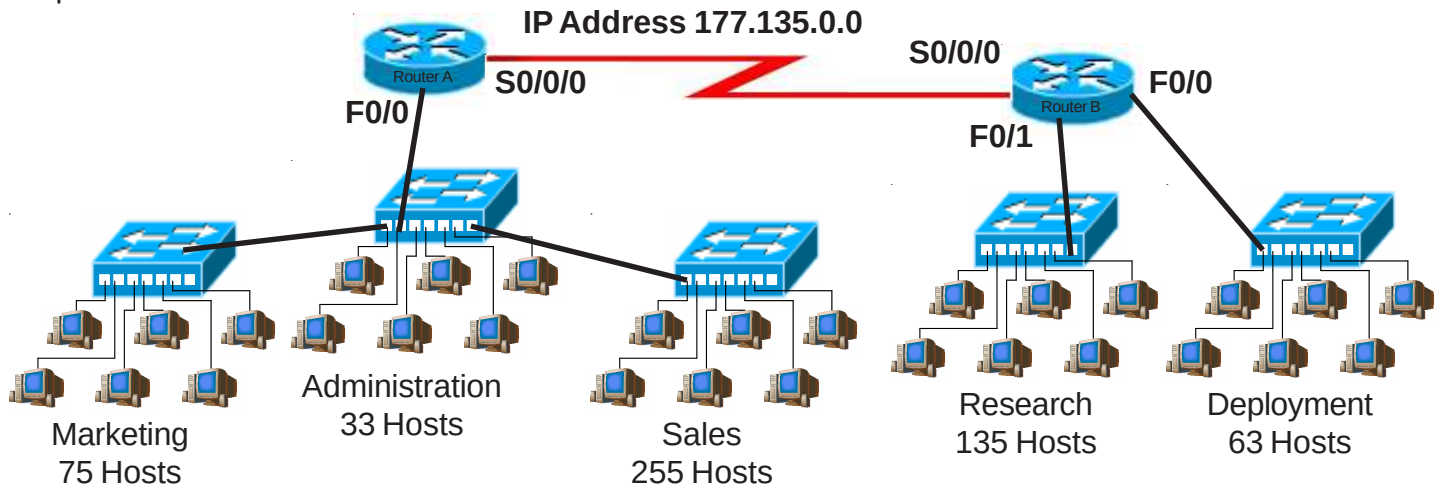
IP address range for Router A
to Router B serial connection

IP address range for Router A
to Router C serial connection

IP address range for Router B
to Router C serial connection

Practical Subnetting 7

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 125% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class

Custom subnet mask

Minimum number of subnets needed

Extra subnets required for 125% growth
(Round up to the next whole number)

Total number of subnets needed

Number of host addresses
in the largest subnet group

Number of addresses needed for
125% growth in the largest subnet
(Round up to the next whole number)

Total number of address
needed for the largest subnet

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Router A Port F0/0

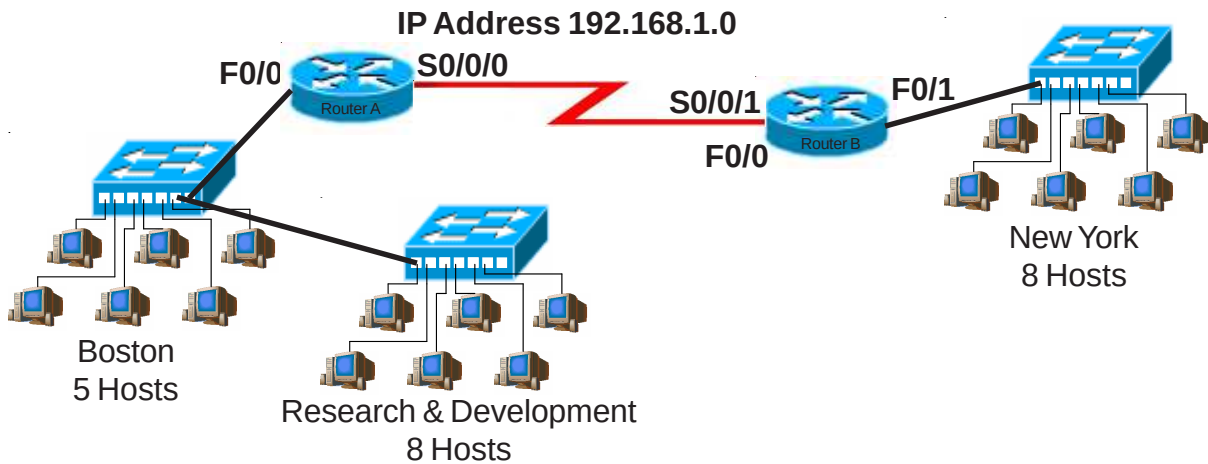
IP address range for Research

IP address range for Deployment

IP address range for Router A
to Router B serial connection

Practical Subnetting 8

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number subnets**, and allow enough extra subnets and hosts for 85% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class

Custom subnet mask

Minimum number of subnets needed

Extra subnets required for 85% growth
(Round up to the next whole number)

Total number of subnets needed

Number of host addresses
in the largest subnet group

Number of addresses needed for
85% growth in the largest subnet
(Round up to the next whole number)

Total number of address
needed for the largest subnet

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

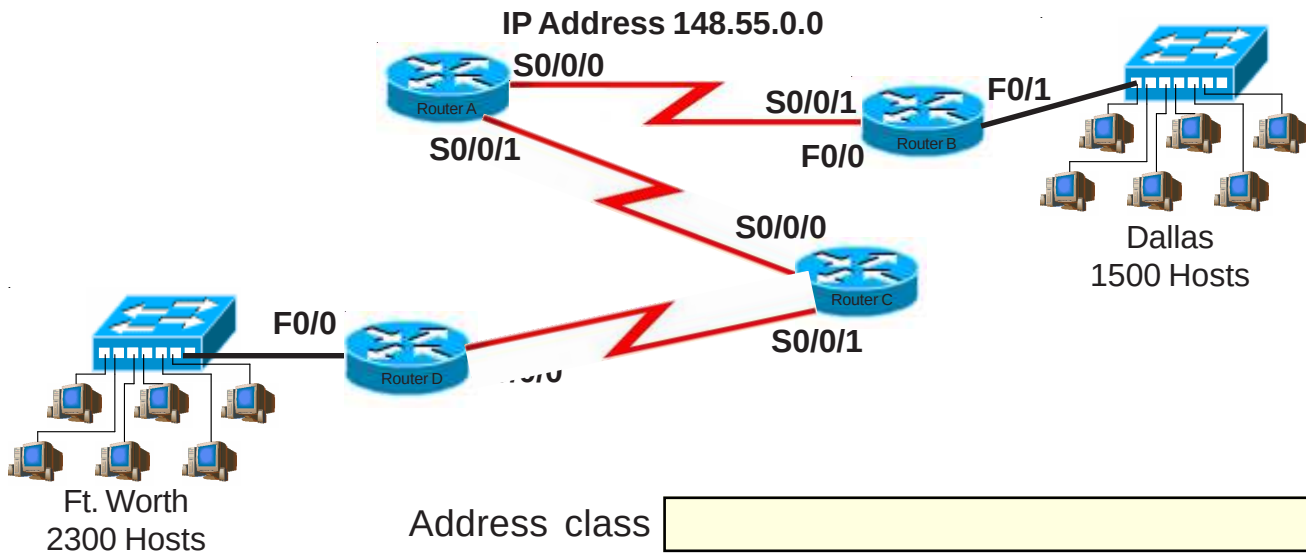
IP address range for Router A F0/0

IP address range for New York

IP address range for Router A
to Router B serial connection

Practical Subnetting 9

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of hosts per subnet**, and allow enough extra subnets and hosts for 15% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class

Custom subnet mask

Minimum number of subnets needed

Extra subnets required for 15% growth
(Round up to the next whole number)

Total number of subnets needed

Number of host addresses
in the largest subnet group

Number of addresses needed for
15% growth in the largest subnet
(Round up to the next whole number)

Total number of address
needed for the largest subnet

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Ft. Worth

IP address range for Dallas

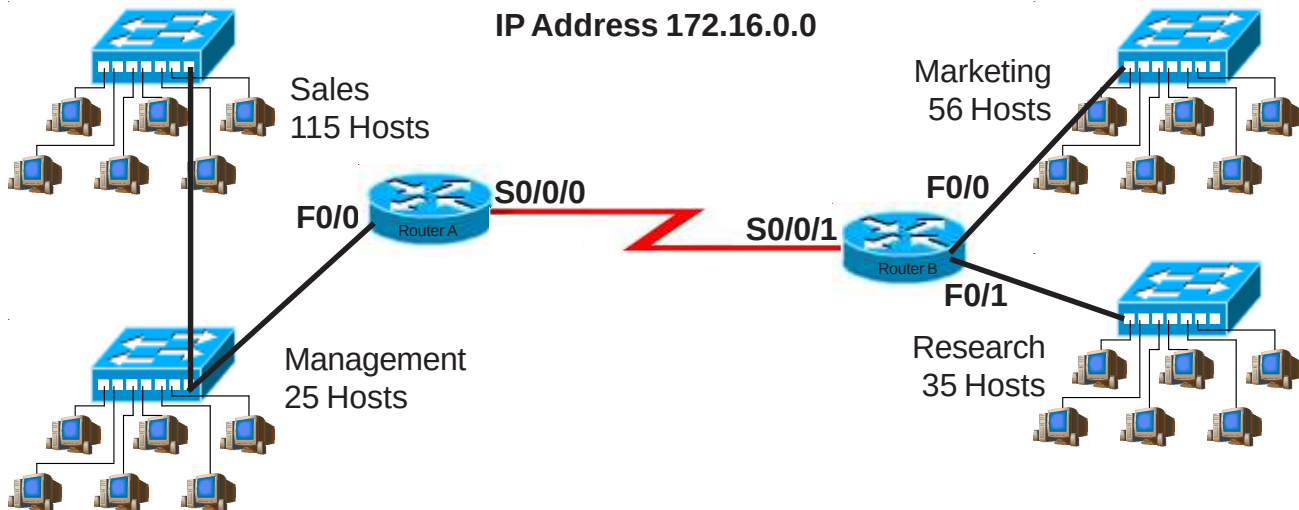
IP address range for Router A
to Router B serial connection

IP address range for Router A
to Router C serial connection

IP address range for Router C
to Router D serial connection

Practical Subnetting 10

Based on the information in the graphic shown, design a network addressing scheme that will supply the **minimum number of subnets**, and allow enough extra subnets and hosts for 110% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class

Custom subnet mask

Minimum number of subnets needed

Extra subnets required for 110% growth

(Round up to the next whole number)

Total number of subnets needed

Number of host addresses
in the largest subnet group

Number of addresses needed for
110% growth in the largest subnet

(Round up to the next whole number)

Total number of address
needed for the largest subnet

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Sales/Managemnt

IP address range for Marketing

IP address range for Research

IP address range for Router A
to Router B serial connection

Valid and Non-Valid IP Addresses

Using the material in this workbook identify which of the addresses below are correct and usable. If they are not usable addresses explain why.

IP Address: 0.230.190.192

Subnet Mask: 255.0.0.0

Reference Page Inside Front Cover

The network ID cannot be 0.

IP Address: 192.10.10.1

Subnet Mask: 255.255.255.0

Reference Pages 28-29

OK

IP Address: 245.150.190.10

Subnet Mask: 255.255.255.0

Reference Page Inside Front Cover

IP Address: 135.70.191.255

Subnet Mask: 255.255.254.0

Reference Pages 48-49

IP Address: 127.100.100.10

Subnet Mask: 255.0.0.0

Reference Pages Inside Front Cover

IP Address: 93.0.128.1

Subnet Mask: 255.255.224.0

Reference Pages 56-57

IP Address: 200.10.10.128

Subnet Mask: 255.255.255.224

Reference Pages 54-55

IP Address: 165.100.255.189

Subnet Mask: 255.255.255.192

Reference Pages 30-31

IP Address: 190.35.0.10

Subnet Mask: 255.255.255.192

Reference Pages 34-35

IP Address: 218.35.50.195

Subnet Mask: 255.255.0.0

Reference Page Inside Front Cover

IP Address: 200.10.10.175 /22

Reference Pages 54-55 and/or Inside Front Cover

IP Address: 135.70.255.255

Subnet Mask: 255.255.224.0

Reference Pages 48-49

Reference Charts and Support Materials

Class A Addresses
VLSM Chart 8-15 Bits (2nd octet)

[illegible]

Class B Addresses
VLSM Chart 16-23 Bits (3rd octet)

16 161-200 16,100 hours	17 161-200 17,700 hours	18 161-200 18,300 hours	19 161-200 19,000 hours	20 161-200 20,000 hours	21 161-200 21,000 hours	22 161-200 22,000 hours	23 161-200 23,000 hours
				0-15	0-15	0-15	0-15
					16-30	16-30	16-30
					31-45	31-45	31-45
					46-60	46-60	46-60
					61-75	61-75	61-75
					76-90	76-90	76-90
					91-105	91-105	91-105
					106-120	106-120	106-120
					121-135	121-135	121-135
					136-150	136-150	136-150
					151-165	151-165	151-165
					166-180	166-180	166-180
					181-195	181-195	181-195
					196-210	196-210	196-210
					211-225	211-225	211-225
					226-240	226-240	226-240
					241-255	241-255	241-255
					256-270	256-270	256-270
					271-285	271-285	271-285
					286-300	286-300	286-300
					301-315	301-315	301-315
					316-330	316-330	316-330
					331-345	331-345	331-345
					346-360	346-360	346-360
					361-375	361-375	361-375
					376-390	376-390	376-390
					391-405	391-405	391-405
					406-420	406-420	406-420
					421-435	421-435	421-435
					436-450	436-450	436-450
					451-465	451-465	451-465
					466-480	466-480	466-480
					481-495	481-495	481-495
					496-510	496-510	496-510
					511-525	511-525	511-525
					526-540	526-540	526-540
					541-555	541-555	541-555
					556-570	556-570	556-570
					571-585	571-585	571-585
					586-600	586-600	586-600
					601-615	601-615	601-615
					616-630	616-630	616-630
					631-645	631-645	631-645
					646-660	646-660	646-660
					661-675	661-675	661-675
					676-690	676-690	676-690
					691-705	691-705	691-705
					706-720	706-720	706-720
					721-735	721-735	721-735
					736-750	736-750	736-750
					751-765	751-765	751-765
					766-780	766-780	766-780
					781-795	781-795	781-795
					796-810	796-810	796-810
					811-825	811-825	811-825
					826-840	826-840	826-840
					841-855	841-855	841-855
					856-870	856-870	856-870
					871-885	871-885	871-885
					886-900	886-900	886-900
					901-915	901-915	901-915
					916-930	916-930	916-930
					931-945	931-945	931-945
					946-960	946-960	946-960
					961-975	961-975	961-975
					976-990	976-990	976-990
					991-1005	991-1005	991-1005
					1006-1020	1006-1020	1006-1020
					1021-1035	1021-1035	1021-1035
					1036-1050	1036-1050	1036-1050
					1051-1065	1051-1065	1051-1065
					1066-1080	1066-1080	1066-1080
					1081-1095	1081-1095	1081-1095
					1096-1110	1096-1110	1096-1110
					1111-1125	1111-1125	1111-1125
					1126-1140	1126-1140	1126-1140
					1141-1155	1141-1155	1141-1155
					1156-1170	1156-1170	1156-1170
					1171-1185	1171-1185	1171-1185
					1186-1200	1186-1200	1186-1200
					1201-1215	1201-1215	1201-1215
					1216-1230	1216-1230	1216-1230
					1231-1245	1231-1245	1231-1245
					1246-1260	1246-1260	1246-1260
					1261-1275	1261-1275	1261-1275
					1276-1290	1276-1290	1276-1290
					1291-1305	1291-1305	1291-1305
					1306-1320	1306-1320	1306-1320
					1321-1335	1321-1335	1321-1335
					1336-1350	1336-1350	1336-1350
					1351-1365	1351-1365	1351-1365
					1366-1380	1366-1380	1366-1380
					1381-1395	1381-1395	1381-1395
					1396-1410	1396-1410	1396-1410
					1411-1425	1411-1425	1411-1425
					1426-1440	1426-1440	1426-1440
					1441-1455	1441-1455	1441-1455
					1456-1470	1456-1470	1456-1470
					1471-1485	1471-1485	1471-1485
					1486-1500	1486-1500	1486-1500
					1501-1515	1501-1515	1501-1515
					1516-1530	1516-1530	1516-1530
					1531-1545	1531-1545	1531-1545
					1546-1560	1546-1560	1546-1560
					1561-1575	1561-1575	1561-1575
					1576-1590	1576-1590	1576-1590
					1591-1605	1591-1605	1591-1605
					1606-1620	1606-1620	1606-1620
					1621-1635	1621-1635	1621-1635
					1636-1650	1636-1650	1636-1650
					1651-1665	1651-1665	1651-1665
					1666-1680	1666-1680	1666-1680
					1681-1695	1681-1695	1681-1695
					1696-1710	1696-1710	1696-1710
					1711-1725	1711-1725	1711-1725
					1726-1740	1726-1740	1726-1740
					1741-1755	1741-1755	1741-1755
					1756-1770	1756-1770	1756-1770
					1771-1785	1771-1785	1771-1785
					1786-1800	1786-1800	1786-1800
					1801-1815	1801-1815	1801-1815
					1816-1830	1816-1830	1816-1830
					1831-1845	1831-1845	1831-1845
					1846-1860	1846-1860	1846-1860
					1861-1875	1861-1875	1861-1875
					1876-1890	1876-1890	1876-1890
					1891-1905	1891-1905	1891-1905
					1906-1920	1906-1920	1906-1920
					1921-1935	1921-1935	1921-1935
					1936-1950	1936-1950	1936-1950
					1951-1965	1951-1965	1951-1965
					1966-1980	1966-1980	1966-1980
					1981-1995	1981-1995	1981-1995
					1996-2010	1996-2010	1996-2010
					2011-2025	2011-2025	2011-2025
					2026-2040	2026-2040	2026-2040
					2041-2055	2041-2055	2041-2055
					2056-2070	2056-2070	2056-2070
					2071-2085	2071-2085	2071-2085
					2086-2100	2086-2100	2086-2100
					2101-2115	2101-2115	2101-2115
					2116-2130	2116-2130	2116-2130
					2131-2145	2131-2145	2131-2145
					2146-2160	2146-2160	2146-2160
					2161-2175	2161-2175	2161-2175
					2176-2190	2176-2190	2176-2190
					2191-2205	2191-2205	2191-2205
					2206-2220	2206-2220	2206-2220
					2221-2235	2221-2235	2221-2235
					2236-2250	2236-2250	2236-2250
					2251-2265	2251-2265	2251-2265
					2266-2280	2266-2280	2266-2280
					2281-2295	2281-2295	2281-2295
					2296-2310	2296-2310	2296-2310
					2311-2325	2311-2325	2311-2325
					2326-2340	2326-2340	2326-2340
					2341-2355	2341-2355	2341-2355
					2356-2370	2356-2370	2356-2370
					2371-2385	2371-2385	2371-2385
					2386-2400	2386-2400	2386-2400
					2401-2415	2401-2415	2401-2415
					2416-2430	2416-2430	2416-2430
					2431-2445	2431-2445	2431-2445
					2446-2460	2446-2460	2446-2460
					2461-2475	2461-2475	2461-2475
					2476-2490	2476-2490	2476-2490
					2491-2505	2491-2505	2491-2505
					2506-2520	2506-2520	2506-2520
					2521-2535	2521-2535	2521-2535
					2536-2550	2536-2550	2536-2550
					2551-2565	2551-2565	2551-2565
					2566-2580	2566-2580	2566-2580
					2581-2595	2581-2595	2581-2595
					2596-2610	2596-2610	2596-2610
					2611-2625	2611-2625	2611-2625
					2626-2640	2626-2640	2626-2640
					2641-2655	2641-2655	2641-2655
					2656-2670	2656-2670	2656-2670
					2671-2685	2671-2685	2671-2685
					2686-2700	2686-2700	2686-2700
					2701-2715	2701-2715	2701-2715
					2716-2730	2716-2730	2716-2730
					2731-2745	2731-2745	2731-2745
					2746-2760	2746-2760	2746-2760
					2761-2775	2761-2775	2761-2775
					2776-2790	2776-2790	2776-2790
					2791-2805	2791-2805	2791-2805
					2806-2820	2806-2820	2806-2820
					2821-2835	2821-2835	2821-2835
					2836-2850	2836-2850	2836-2850
					2851-2865</		

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

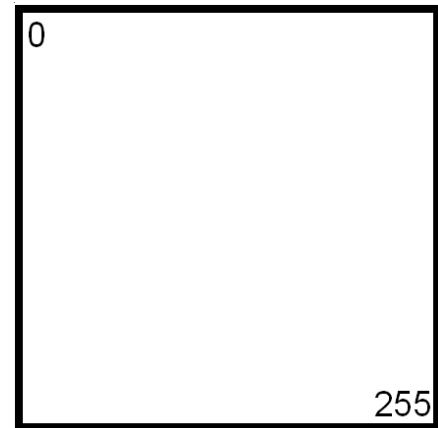
D4	D5	D6	D7	D8	D9	D10
205,206,207,208	209,210,211,212	213,214,215,216	217,218,219,220	221,222,223,224	225,226,227,228	229,230,231,232
			0-15	0-7	0-7	0-7
				8-16	8-16	8-16
		0-31	16-23	16-23	16-23	16-23
			24-31	24-31	24-31	24-31
				32-39	32-39	32-39
		32-63	40-47	40-47	40-47	40-47
			48-55	48-55	48-55	48-55
			56-63	56-63	56-63	56-63
			64-79	64-71	64-71	64-71
		64-95	72-79	72-79	72-79	72-79
			80-87	80-87	80-87	80-87
		86-95	88-95	88-95	88-95	88-95
			96-103	96-103	96-103	96-103
			104-111	104-111	104-111	104-111
		104-111	112-119	112-119	112-119	112-119
			120-127	120-127	120-127	120-127
			128-143	128-135	128-135	128-135
		128-159	136-143	136-143	136-143	136-143
			144-151	144-151	144-151	144-151
		144-159	152-160	152-160	152-160	152-160
			160-175	160-175	160-175	160-175
		160-175	176-183	176-183	176-183	176-183
			184-191	184-191	184-191	184-191
		184-191	192-199	192-199	192-199	192-199
			200-207	200-207	200-207	200-207
		200-223	208-223	208-223	208-223	208-223
			224-231	224-231	224-231	224-231
			232-239	232-239	232-239	232-239
		232-255	240-247	240-247	240-247	240-247
			248-255	248-255	248-255	248-255

Visualizing Subnets Using The Box Method

The box method is the simplest way to visualize the breakdown of subnets and addresses into smaller sizes.

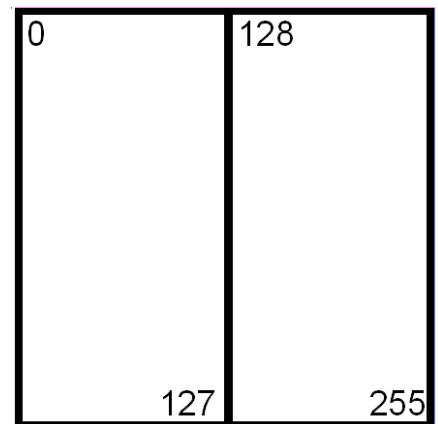
Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



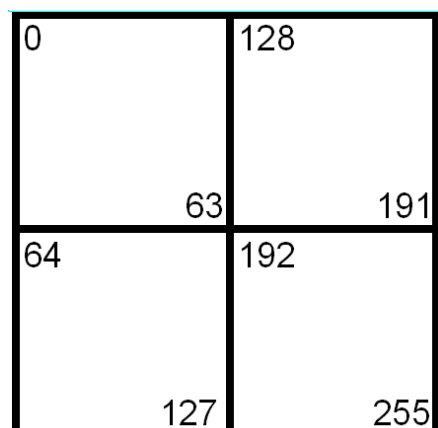
Split the box in half and you get two subnets with 128 addresses,

/25
255.255.255.128
128 Hosts
2 Subnets



Divide the box into quarters and you get four subnets with 64 addresses,

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you get eight subnets with 32 addresses,

/27
255.255.255.224
32 Hosts
8 Subnets

0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

Split the boxes in half again and you get sixteen subnets with sixteen addresses,

/28
255.255.255.240
16 Hosts
16 Subnets

0	32	128	160
15	47	143	175
16	48	144	176
31	63	159	191
64	96	192	224
79	111	207	239
80	112	208	240
95	127	223	255

The next split gives you thirty two subnets with eight addresses,

/29
255.255.255.248
8 Hosts
32 Subnets

0	8	32	40	128	136	160	168
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
71	79	103	111	199	207	321	239
80	88	112	120	208	216	240	248
87	95	119	127	215	223	247	255

The last split gives sixty four subnets with four addresses each,

/30
255.255.255.252
4 Hosts
64 Subnets

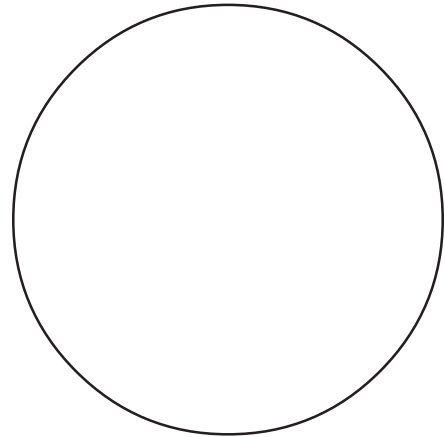
0	8	32	40	128	136	160	168
3	11	35	43	131	139	163	171
4	12	36	44	132	140	164	172
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
19	27	51	59	147	155	179	187
20	28	52	60	148	156	180	188
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
67	75	99	107	195	203	227	235
68	76	100	108	196	204	228	236
71	79	103	111	199	207	321	239
80	88	112	120	208	216	240	248
83	91	115	123	211	219	243	251
84	92	116	124	212	220	244	252
87	95	119	127	215	223	247	255

Visualizing Subnets Using The Circle Method

The circle method is another method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the different sections of the circle you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

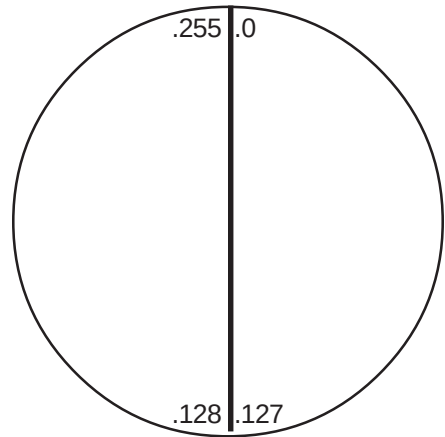
Start with a circle. The whole circle is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



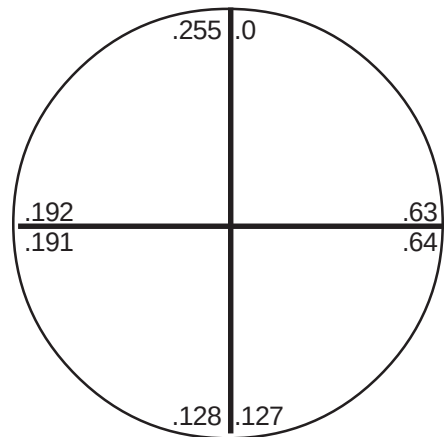
Split the circle in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



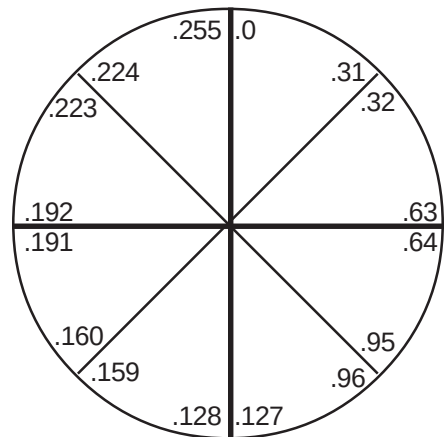
Divide the circle into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



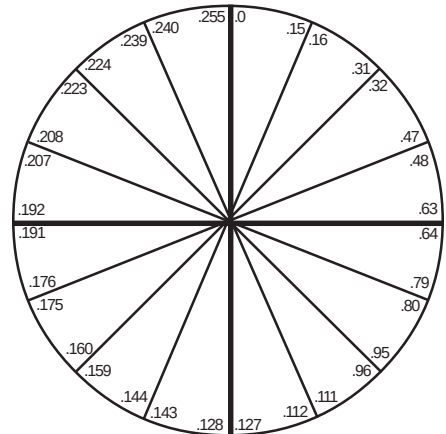
Split each quarter and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets



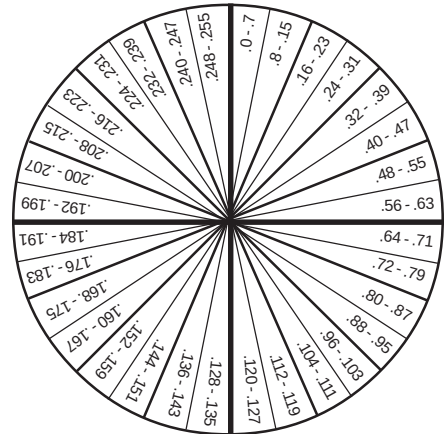
Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets



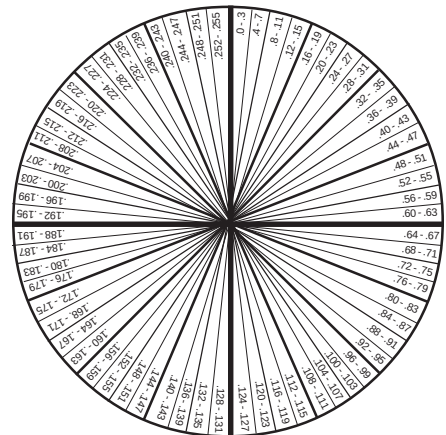
The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets



The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets



Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31	16-23	8-11	4-5
						12-15	6-7
					24-31	16-19	8-9
						20-23	10-11
		32-63	32-47	32-39	40-47	24-27	12-13
						28-31	14-15
				48-63	48-55	32-35	16-17
						36-39	18-19
					56-63	40-43	20-21
						44-47	22-23
		64-127	64-95	64-79	80-87	48-51	24-25
						52-55	26-27
				80-95	88-95	56-59	28-29
						60-63	30-31
		128-191	128-159	128-143	136-143	64-67	32-33
						68-71	34-35
				144-159	152-159	72-75	36-37
						76-79	38-39
		160-191	160-175	160-167	168-171	80-83	40-41
						84-87	42-43
				176-183	184-187	88-91	44-45
						92-95	46-47
		192-255	192-223	192-207	200-207	96-99	48-49
						100-103	50-51
				208-223	216-223	104-107	52-53
						108-111	54-55
		224-255	224-239	224-231	232-239	112-115	56-57
						116-119	58-59
				240-247	248-251	120-123	60-61
						124-127	62-63
		240-255	240-255	240-247	248-251	128-131	64-65
						132-135	66-67
				248-255	252-255	136-139	68-69
						140-143	70-71
						144-147	72-73
						148-151	74-75
						152-155	76-77
						156-159	78-79
						160-163	80-81
						164-167	82-83
						168-171	84-85
						172-175	86-87
						176-179	88-89
						180-183	90-91
						184-187	92-93
						188-191	94-95
						192-195	96-97
						196-199	98-99
						200-203	100-101
						204-207	102-103
						208-211	104-105
						212-215	106-107
						216-219	108-109
						220-223	110-111
						224-227	112-113
						228-231	114-115
						232-235	116-117
						236-239	118-119
						240-243	120-121
						244-247	122-123
						248-251	124-125
						252-255	126-127
							128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					4-7	4-5	2-3
					8-15	8-11	6-7
					16-31	12-15	8-9
						16-19	10-11
						20-23	12-13
						24-27	14-15
			32-63	32-47	28-31	32-35	16-17
					36-39	40-43	18-19
				48-63	44-47	48-51	20-21
					52-55	56-59	22-23
					60-63	64-67	24-25
					64-79	68-71	26-27
						72-75	28-29
						76-79	30-31
			64-127	80-95	84-87	88-91	32-33
					92-95	96-99	34-35
					96-111	100-103	36-37
						104-107	38-39
						108-111	40-41
				112-127	116-119	120-123	42-43
					124-127	128-131	44-45
					128-143	132-135	46-47
						136-139	48-49
						140-143	50-51
			128-191	144-159	148-151	152-155	52-53
					156-159	160-163	54-55
					160-175	164-167	56-57
						168-171	58-59
						172-175	60-61
				176-191	180-183	184-187	62-63
					184-187	188-191	64-65
					192-207	196-199	66-67
						200-203	68-69
						204-207	70-71
			128-255	192-223	208-211	212-215	72-73
					216-219	220-223	74-75
					224-239	228-231	76-77
						232-235	78-79
						236-239	80-81
				240-255	244-247	248-251	82-83
					248-251	252-255	84-85
					240-255	244-247	86-87
						248-251	88-89
						252-255	90-91
						252-255	92-93
						252-255	94-95
						252-255	96-97
						252-255	98-99
						252-255	100-101
						252-255	102-103
						252-255	104-105
						252-255	106-107
						252-255	108-109
						252-255	110-111
						252-255	112-113
						252-255	114-115
						252-255	116-117
						252-255	118-119
						252-255	120-121
						252-255	122-123
						252-255	124-125
						252-255	126-127
						252-255	128-129
						252-255	130-131
						252-255	132-133
						252-255	134-135
						252-255	136-137
						252-255	138-139
						252-255	140-141
						252-255	142-143
						252-255	144-145
						252-255	146-147
						252-255	148-149
						252-255	150-151
						252-255	152-153
						252-255	154-155
						252-255	156-157
						252-255	158-159
						252-255	160-161
						252-255	162-163
						252-255	164-165
						252-255	166-167
						252-255	168-169
						252-255	170-171
						252-255	172-173
						252-255	174-175
						252-255	176-177
						252-255	178-179
						252-255	180-181
						252-255	182-183
						252-255	184-185
						252-255	186-187
						252-255	188-189
						252-255	190-191
						252-255	192-193
						252-255	194-195
						252-255	196-197
						252-255	198-199
						252-255	200-201
						252-255	202-203
						252-255	204-205
						252-255	206-207
						252-255	208-209
						252-255	210-211
						252-255	212-213
						252-255	214-215
						252-255	216-217
						252-255	218-219
						252-255	220-221
						252-255	222-223
						252-255	224-225
						252-255	226-227
						252-255	228-229
						252-255	230-231
						252-255	232-233
						252-255	234-235
						252-255	236-237
						252-255	238-239
						252-255	240-241
						252-255	242-243
						252-255	244-245
						252-255	246-247
						252-255	248-249
						252-255	250-251
						252-255	252-253
						252-255	254-255

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
					4-7	4-7
				16-31	8-15	8-11
					12-15	12-15
		64-127	32-63	32-47	16-23	16-19
					20-23	20-23
				48-63	24-31	24-27
					28-31	28-31
			64-95	64-79	32-39	32-35
					36-39	36-39
				80-95	40-47	40-43
					44-47	44-47
		96-127	96-111	96-103	48-55	48-51
					52-55	52-55
				104-111	56-59	56-59
					60-63	60-63
			112-127	112-119	64-71	64-67
					68-71	68-71
				120-127	72-79	72-75
					76-79	76-79
	128-255	128-191	128-159	128-143	80-87	80-83
					84-87	84-87
				144-159	88-91	88-91
					92-95	92-95
		160-191	160-175	160-167	96-99	96-99
					100-103	100-103
				176-191	104-107	104-107
					108-111	108-111
		192-255	192-223	192-207	112-115	112-115
					116-119	116-119
				208-223	120-123	120-123
					124-127	124-127
		224-255	224-239	224-231	128-131	128-131
					132-135	132-135
				232-239	136-143	136-139
					140-143	140-143
			240-255	240-247	144-151	144-147
					148-151	148-151
				248-255	152-155	152-155
					156-159	156-159
			240-255	240-247	160-163	160-163
					164-167	164-167
				248-255	168-171	168-171
					172-175	172-175
			240-255	240-247	176-179	176-179
					180-183	180-183
				248-255	184-187	184-187
					188-191	188-191
			240-255	240-247	192-195	192-195
					196-199	196-199
				248-255	200-203	200-203
					204-207	204-207
			240-255	240-247	208-211	208-211
					212-215	212-215
				248-255	216-219	216-219
					220-223	220-223
			240-255	240-247	224-227	224-227
					228-231	228-231
				248-255	232-235	232-235
					236-239	236-239
			240-255	240-247	240-243	240-243
					244-247	244-247
				248-255	248-251	248-251
					252-255	252-255

Class A Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/8	0	255.0.0.0	1	16,777,216	16,777,214
/9	1	255.128.0.0	2	8,388,608	8,388,606
/10	2	255.192.0.0	4	4,194,304	4,194,302
/11	3	255.224.0.0	8	2,097,152	2,097,150
/12	4	255.240.0.0	16	1,048,576	1,048,574
/13	5	255.248.0.0	32	524,288	524,286
/14	6	255.252.0.0	64	262,144	262,142
/15	7	255.254.0.0	128	131,072	131,070
/16	8	255.255.0.0	256	65,536	65,534
/17	9	255.255.128.0	512	32,768	32,766
/18	10	255.255.192.0	1,024	16,384	16,382
/19	11	255.255.224.0	2,048	8,192	8,190
/20	12	255.255.240.0	4,096	4,096	4,094
/21	13	255.255.248.0	8,192	2,048	2,046
/22	14	255.255.252.0	16,384	1,024	1,022
/23	15	255.255.254.0	32,768	512	510
/24	16	255.255.255.0	65,536	256	254
/25	17	255.255.255.128	131,072	128	126
/26	18	255.255.255.192	262,144	64	62
/27	19	255.255.255.224	524,288	32	30
/28	20	255.255.255.240	1,048,576	16	14
/29	21	255.255.255.248	2,097,152	8	6
/30	22	255.255.255.252	4,194,304	4	2

Class B Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/16	0	255.255.0.0	1	65,536	65,534
/17	1	255.255.128.0	2	32,768	32,766
/18	2	255.255.192.0	4	16,384	16,382
/19	3	255.255.224.0	8	8,192	8,190
/20	4	255.255.240.0	16	4,096	4,094
/21	5	255.255.248.0	32	2,048	2,046
/22	6	255.255.252.0	64	1,024	1,022
/23	7	255.255.254.0	128	512	510
/24	8	255.255.255.0	256	256	254
/25	9	255.255.255.128	512	128	126
/26	10	255.255.255.192	1,024	64	62
/27	11	255.255.255.224	2,048	32	30
/28	12	255.255.255.240	4,096	16	14
/29	13	255.255.255.248	8,192	8	6
/30	14	255.255.255.252	16,384	4	2

Class C Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/24	0	255.255.255.0	1	256	254
/25	1	255.255.255.128	2	128	126
/26	2	255.255.255.192	4	64	62
/27	3	255.255.255.224	8	32	30
/28	4	255.255.255.240	16	16	14
/29	5	255.255.255.248	32	8	6
/30	6	255.255.255.252	64	4	2

IPv6 Addressing and Subnetting Workbook

Version 1

Global Routing Prefix

Subnet ID

Interface ID

Hexadecimal

Student Name:

2001:0DB9

Types of IPv6 Addresses

Unspecified, Loopback, Embedded IPv4

Unspecified address is an all 0 address and cannot be assigned to an interface. It would be typed as ::. This is only used as a source address to indicate the absence of an actual address.

Loopback Address is all 0's except for the last bit, which is 1. It would be typed as ::1. It operates the same as the IPv4 127.0.0.1 loopback address.

IPv4 Embedded addresses are IPv6 addresses with an IPv4 address embedded in the low-order 32 bits. They are used to transition networks from IPv4 to IPv6.

Address Range:

0000:0000:0000:0000:0000:0000:0000:0000/8 to 00FF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/8

Global Unicast

Global Unicast addresses are used to uniquely identify a specific interface on a host and can be used as a public address on the internet.

Address Range:

2000:0000:0000:0000:0000:0000:0000:0000/3 to 3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/3

Unique local Unicast

Unique local Unicast addresses are roughly the same as IPv4 private addresses.

Address Range:

FC00:0000:0000:0000:0000:0000:0000:0000/7 to FDFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/7

Link-local Unicast

Link-local addresses are unicast addresses that are limited to a point to point connection within a local network. Routers will not forward packets with a link-local address.

Address Range:

FE80:0000:0000:0000:0000:0000:0000:0000/10 to FEBF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/10

Multicast

Multicast addresses are used to send a single packet to multiple destinations simultaneously.

Address Range:

FF00:0000:0000:0000:0000:0000:0000:0000/8 to FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF/8

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Special Thanks to Melvin Baker and Jim Dorsch for taking the time to check this workbook for errors,
and to everyone who has sent in suggestions to improve the series.

A Brief History of TCP/IP Versions

TCP version 1 through TCP version 3 were developed as test versions and not widely used. Contrary to popular belief there was never an IPv1, IPv2, or IPv3. The version numbers were kept intact to avoid confusion when the TCP protocol was split into TCP and IP.

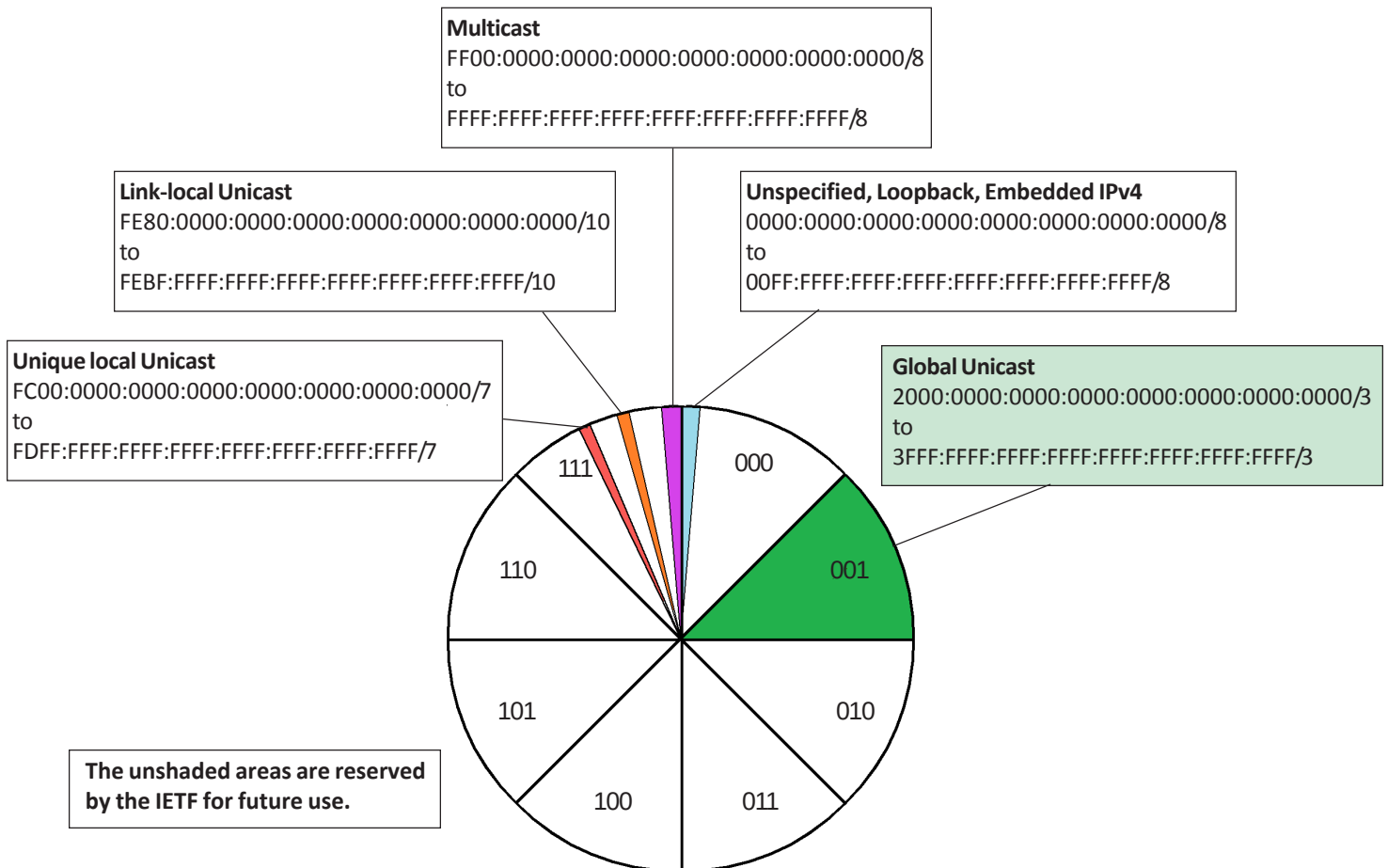
- 1973 - TCP version 1 was developed and documented in RFC 675. At this time IP was part of TCP.
- 1977 - TCP version 2 was developed and documented in March. In August of 1977 it was decided that the TCP protocol was going in the wrong direction.
- 1978 - TCP and IP were split into two separate protocols. Both TCP and IP were part of version 3.
- 1980 - Early development of IPv4 defined in RFC 760.
- 1981 - The current version of IPv4 is defined in RFC 791, 792 and 793. It was the first widely used version of the Internet Protocol.
- 1983 - On January 1, 1983, TCP/IP protocols became the only approved protocol on the ARPANET, replacing the earlier NCP protocol. This was known as flag day.
- 1984 - The number of hosts on the internet breaks 1000.
- 1987 - Hosts on the internet exceeds 10,000.
- 1989 - Host accessing the internet surpasses 100,000.
- 1990 - IPv5 relates to an experimental TCP/IP protocol called the Internet Stream Protocol, Version 2, originally defined in RFC 1190. This protocol was a peer of IPv4 and was designed to work with voice conversations and conferences with delay and bandwidth guarantees. These packets were assigned IP version 5 to differentiate them from "normal" IPv4 packets. This protocol was never introduced to the public, and was always considered experimental. To be sure there would be no confusion, version 5 was skipped over in favor of version 6.
- 1992 - The number of hosts on the internet breaks 1,000,000.
- 1995 - IPv6, introduced as IP Next Generation, was presented in RFC 1883.
- 1997 - The number of hosts using the internet exceeds 19,000,000.
- 1998 - The more fully developed IPv6 obsoletes RFC 1883 with the updated RFC 2460.

IPv4 has been well established for years. IPv6 is still in flux as it undergoes growing pains with changes and adjustments to the rules as it is being implemented.

IPv6

There are 340,282,366,920,938,463,463,374,607,431,768,211,456 possible IPv6 addresses.

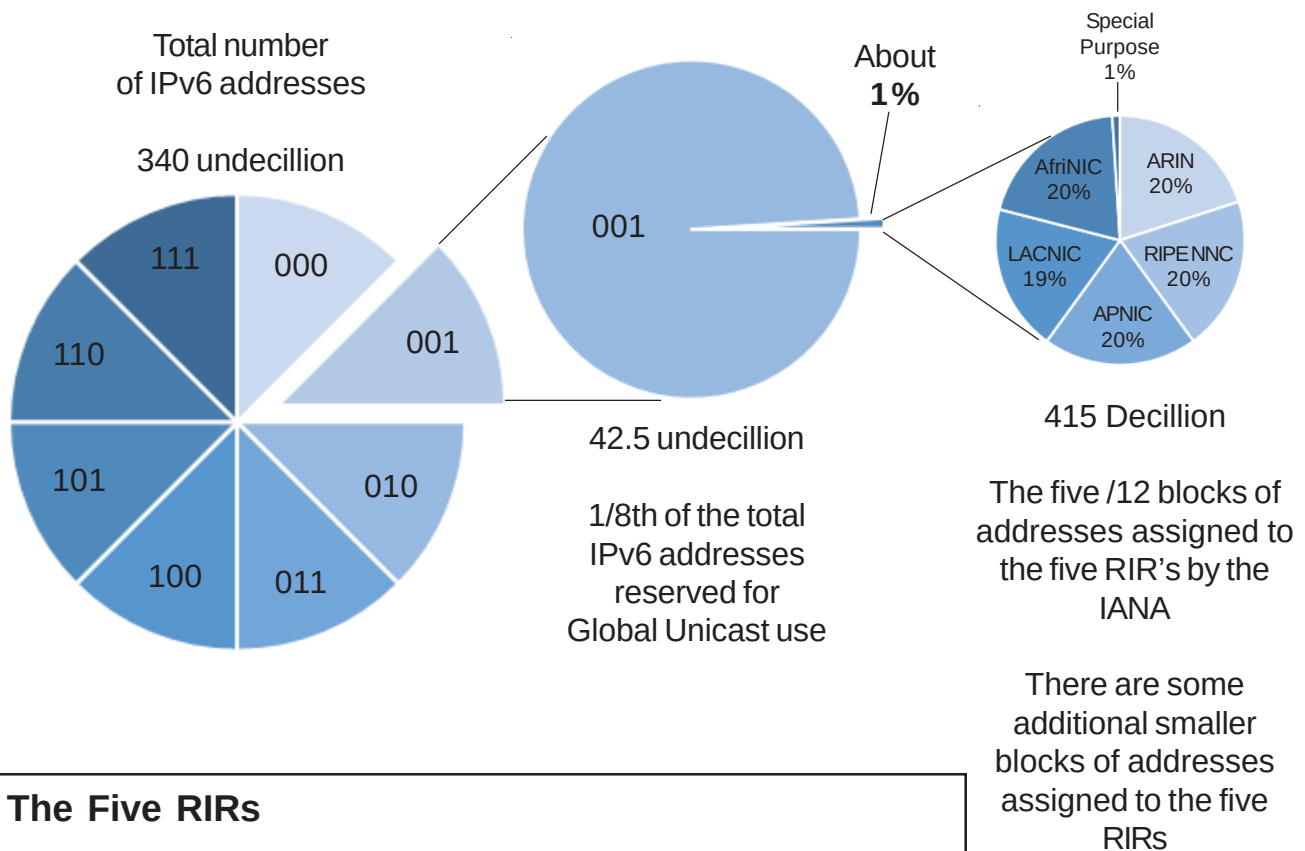
If you want to actually say the number it is three hundred and forty undecillion, two hundred and eighty-two decillion, three hundred and sixty-six nonillion, nine hundred and twenty octillion, nine hundred and thirty-eight septillion, four hundred and sixty-three sextillion, four hundred and sixty-three quintillion, three hundred and seventy-four quadrillion, six hundred and seven trillion, four hundred and thirty-one billion, seven hundred and sixty-eight million, two hundred and eleven thousand, four hundred and fifty-six. (or you can have Windows Narrator say it for you.)



The Internet Assigned Numbers Authority (IANA) divided the available IPv6 addresses into eight equal segments based on the three leading bits of the addresses (000, 001, 010, 011, 100, 101, 110, and 111). Only one eighth of the total available addresses have been reserved for use as global unicast addresses. Four smaller subgroups have been made available for unique local unicast, link-local unicast, multicast, and (unspecified, loopback, embedded IPv4).

IPv6 by the Numbers

340,282,366,920,938,463,374,607,431,768,211,456 Total number of IPv6 Addresses
 42,535,295,865,117,307,932,921,825,928,971,026,432 1/8 or the reserved Global Unicast addresses
 415,383,748,682,786,210,282,439,706,337,607,680 The five /12 ranges assigned to the RIRs
 7,119,157,000 Estimated world population
 58,347,322,398,253,923,924,200,534 Estimated number of IPv6 addresses per person
 (That's over 58 septillion addresses per person and doesn't include the additional smaller blocks of addresses assigned to the five RIRs by the IANA)



The Five RIRs

The **Regional Internet Registry** is an organization that manages the allocation and registration of internet number resources world wide. It has evolved over time to divide the world into five areas, or RIRs.

AfriNIC - African Network Information Centre
ARIN - American Registry for Internet Numbers
APNIC - Asia-Pacific Network Information Centre
LACNIC - Latin America and Caribbean Network Information Centre
RIPE NCC - Réseaux IP Européens Network Coordination Centre

There is a chart in the Reference Section that has all of these listed.

To make IPv6 addresses a little less imposing, two rules were developed to make them easier to work with. Rule 1: Omission of the Leading 0s, and Rule 2: Omission of the all-0 Hextets.

Rule 1: Omission of the Leading 0s

Rule 1 allows you to remove all the leading 0s in each individual hextet.

Sample 1 Unspecified address

Preferred Format: **0000:0000:0000:0000:0000:0000:0000:0000**

Leading 0's removed: 0: 0: 0: 0: 0: 0: 0: 0

or

0:0:0:0:0:0:0:0

Sample 2 - Loopback Address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0001

Leading 0's removed: 0: 0: 0: 0: 0: 0: 0: 1

or

0:0:0:0:0:0:0:1

Sample 3 – Global Unicast Address

Preferred Format: 2000:0000:0000:0000:0000:0000:0000:0001

Leading 0's removed: 2000: 0: 0: 0: 0: 0: 0: 1

or

2000:0:0:0:0:0:0:1

Sample 4 – Global Unicast Address

Preferred Format: 2001:00FE:ACAD:2013:0000:0000:00AA:0271

Leading 0's removed: 2001: FE:ACAD:2013: 0: 0: AA: 271

or

2001: FE:ACAD:2013:0:0:AA:271

Sample 5 – Unique local Unicast Address

Preferred Format: FC80:0000:0000:ACAD:0000:0000:0000:0001

Leading 0's removed: FC80: 0: 0:ACAD: 0: 0: 0: 1

or

FC80:0:0:ACAD:0:0:0:1

Sample 6 – Link-local Address

Preferred Format: FE80:ACAD:0000:0197:0000:0000:0000:FF01

Leading 0's removed: FE80:ACAD: 0: 197: 0: 0: 0:FF01

or

FE80:ACAD:0:197:0:0:0:FF01

Sample 7 – Multicast Address

Preferred Format: FF00:0000:0000:ACAD:0000:0000:FE00:0721

Leading 0's removed: FF00: 0: 0:ACAD: 0: 0:FE00: 721

or

FF00:0:0:ACAD:0:0:FE00:721

Rule 1: Omission of the Leading 0s Problems

Using Rule 1 reduce the IPv6 addresses to their shortened form.

1. 0000:0000:0000:0000:0000:0000:0000:0000

2. 0000:0000:0000:0000:0000:0000:0000:0001

3. 2000:0000:0000:0000:0000:ABCD:0000:0025

4. 3F00:0090:0000:0000:0000:0098:0000:0001

5. 2001:3756:0005:0000:ACAD:0000:0000:0025

6. 3FFF:FF00:0000:0000:ACAD:0000:0000:0127

7. 2001:0000:0000:ABCD:FFFF:0000:0000:0001

8. 3ABC:0001:ACAD:0000:0000:1234:0000:0005

9. FC00:0000:0000:0000:3E00:1275:0000:0034

10. FE95:FC6C:C540:0000:0000:0000:0000:9800

11. FF00:ACAD:0000:0000:1234:0000:0000:0001

Rule 2: Omission of the All-0 Hextets

Rule 2 uses a double colon :: to represent a single contiguous set of all zero hextets. It can only be used once in any IPv6 address.

Sample 1 Unspecified address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0000
Contiguous 0's removed: ::

Sample 2 - Loopback Address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0001
Contiguous 0's removed: ::0001

Sample 3 – Global Unicast Address

Preferred Format: 2000:0000:0000:0000:0000:0000:0000:0001
Contiguous 0's removed: 2000: :0001
or
2000::0001

Sample 4 – Global Unicast Address

Preferred Format: 2001:00FE:ACAD:2013:0000:0000:00AA:0271
Contiguous 0's removed: 2001:00FE:ACAD:2013: :00AA:0271
or
2001:00FE:ACAD:2013::00AA:0271

Sample 5 – Unique local Unicast Address

Preferred Format: FC80:0000:0000:ACAD:0000:0000:0000:0001
Contiguous 0's removed: FC80:0000:0000:ACAD: :0001
or
FC80:0000:0000:ACAD::0001

Sample 6 – Link-local Address

Preferred Format: FE80:ACAD:0000:0197:0000:0000:0000:FF01
Contiguous 0's removed: FE80:ACAD:0000:0197: :FF01
or
FE80:ACAD:0000:0197::FF01

Sample 7 – Multicast Address

Preferred Format: FF00:0000:0000:ACAD:0000:0000:FE00:0721
Contiguous 0's removed: FF00: :ACAD:0000:0000:FE00:0721 (Option #1)
FF00:0000:0000:ACAD: :FE00:0721 (Option #2)
or
FF00::ACAD:0000:0000:FE00:0721 (Option #1)
FF00:0000:0000:ACAD::FE00:0721 (Option #2)

Rule 2: Omission of the All-0 Hextets Problems

Using Rule 2 reduce the IPv6 addresses to their shortened form.

1. 0000:0000:0000:0000:0000:0000:0000:0000

2. 0000:0000:0000:0000:0000:0000:0000:0001

3. 2000:0000:0000:0000:0000:ABCD:0000:0025

4. 3F00:0090:0000:0000:0000:0098:0000:0001

5. 2001:3756:0005:0000:ACAD:0000:0000:0025

6. 3FFF:FF00:0000:0000:ACAD:0025:0000:0127

7. 2001:ACAD:0000:ABCD:FFFF:0000:0000:0001

8. 3ABC:0001:ACAD:0000:0000:1234:0000:0005

9. FC00:0000:0000:0000:3E00:1275:0000:0034

10. FE95:FC6C:C540:0000:0000:0000:0000:9800

11. FF00:ACAD:0000:0000:1234:0000:0000:0001

Combining Rule 1 and Rule 2

To reduce the size of IPv6 address even more you can combine Rule 1 with Rule 2.

Sample 1 Unspecified address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0000
Combined reduction: ::

Sample 2 - Loopback Address

Preferred Format: 0000:0000:0000:0000:0000:0000:0000:0001
Combined reduction: ::1

Sample 3 – Global Unicast Address

Preferred Format: 2000:0000:0000:0000:0000:0000:0000:0001
Combined reduction: 2000: :1
or
2000::1

Sample 4 – Global Unicast Address

Preferred Format: 2001:00FE:ACAD:2013:0000:0000:00AA:0271
Combined reduction: 2001: FE:ACAD:2013: : AA: 271
or
2001:FE:ACAD:2013::AA:271

Sample 5 – Unique local Unicast Address

Preferred Format: FC80:0000:0000:ACAD:0000:0000:0000:0001
Combined reduction: FC80: 0: 0:ACAD: : 1
or
FC80:0:0:ACAD::1

Sample 6 – Link-local Address

Preferred Format: FE80:ACAD:0000:0197:0000:0000:0000:FF01
Combined reduction: FE80:ACAD:0 : 197: :FF01
or
FE80:ACAD:0:197::FF01

Sample 7 – Multicast Address

Preferred Format: FF00:0000:0000:ACAD:0000:0000:FE00:0721
Combined reduction: FF00: :ACAD: 0: 0:FE00: 721 (Option #1)
FF00: 0: 0:ACAD: :FE00: 721 (Option #2)
or
FF00::ACAD:0:0:FE00:721 (Option #1)
FF00:0:0:ACAD::FE00:721 (Option #2)

Combining Rule 1 and Rule 2 Problems

Using Rule 1 and 2 reduce the IPv6 addresses to their shortest form.

1. 0000:0000:0000:0000:0000:0000:0000:0000

2. 0000:0000:0000:0000:0000:0000:0000:0001

3. 2000:0000:0000:0000:0000:ABCD:0000:0025

4. 3F00:0090:0000:0000:0000:0098:0000:0001

5. 2001:3756:0005:0000:ACAD:0000:0000:0025

6. 3FFF:FF00:0000:0000:ACAD:0025:0000:0127

7. 2001:ACAD:0000:ABCD:FFFF:0000:0000:0001

8. 3ABC:0001:ACAD:0000:0000:1234:0000:0005

9. FC00:0000:0000:0000:3E00:1275:0000:0034

10. FE95:FC6C:C540:0000:0000:0000:0000:9800

11. FF00:ACAD:0000:0000:1234:0000:0000:0001

Reverting Reduced Address Problems

The following addresses have been shorted using Rule1 and/or Rule 2. Expand them back to their preferred format.

Sample: FF00:ACAD:ABCD:0:1234::1

FF00:ACAD:ABCD:0000:1234:0000:0000:0001

1. 2000::1

2. ::1

3. 2001:0:0:0:0:ABCD:0:127

4. 3E80:0070::0098:0000:0001

5. 2FFF:38:5:0:ACAD::5

6. 3FFF::ACAD:25:0:100

7. 2002:ACAD:0:1BCD:FFFF::4

8. 3FAA:0025:ACAD::ABCD:0000:0005

9. FFFF::4E00:1235:0:34

10. 3E01:6C:40::9800

IPv6 Address Categories

All IPv6 addresses fall into one of three categories

Unicast -

Unicast addresses identify a unique interface on an IPv6 device.
It is a one to one connection between a source and destination.

Examples of IPv6 Unicast addresses include:

Global Unicast -

Similar to a public IPv4 address
Can be used as a public address on the internet
Globally unique
Can be static or dynamic

Link-Local -

Only used on a local network link to uniquely identify a host
Not routable on the public internet
No router will forward a link-local address
Every IPv6 enabled networked device is required to have a link-local address
Multiple interfaces on the same device can have the same link-local address

Loopback - (::1/128)

Used by a host to ping itself to test the TCP/IP stack
It cannot be assigned to a physical interface

Unspecified Address - (::/128)

Is only used as a source address to indicate the absence of an actual address

Unique Local -

Unique local addresses are roughly the same as IPv4 private addresses

Embedded IPv4 -

Used to transition IPv4 networks into IPv6

Multicast -

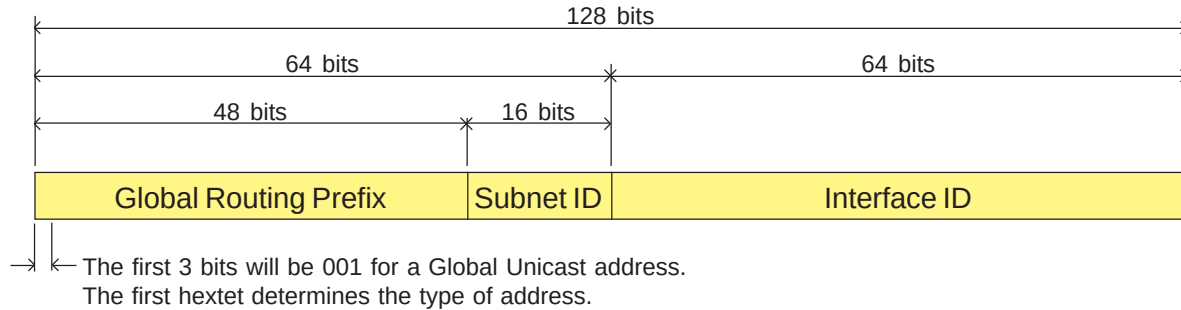
Multicast addresses are used to send a single packet to multiple destinations simultaneously.

Anycast -

Anycast addresses are described as a one-to-nearest or one-to-one-of-many packet delivery. For example, all routers in the same network will be assigned the same Anycast address. A packet sent to that address will only be delivered to the closest router with that address based on routing protocol metrics. Anycast addresses can be pulled from Global Unicast, Site-Local, or Link-Local address ranges. The first address and the last 128 addresses in a /64 Global Unicast range are reserved as the Subnet-Router Anycast Address.

There are no broadcast addresses in IPv6.

Global Unicast IPv6 Address



Global Routing Prefix -

This is assigned by the ISP to a customer or site.

The Global Routing Prefix is determined by the *prefix-length* notation. (example /48 or /64).

This is similar to the network portion of an IPv4 address.

Subnet ID -

This is similar to the subnet portion of an IPv4 address.

The difference is in IPv4 the subnet is borrowed from the host portion of the address.

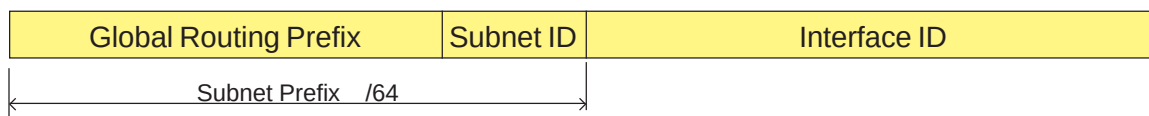
In IPv6 the Subnet ID is a separate field (/48 to /64) and not necessarily part of the Interface ID.

Interface ID -

The Interface ID uniquely identifies a interface on the local subnet.

Subnet Prefix

The Subnet Prefix is the address space used by the Global Routing Prefix and the Subnet ID, and can range from 0 to 128. The preferred Subnet Prefix length is /48 to /64 for customers or sites.



RFC 4291 recommends that the Interface-ID or host portion of your IPv6 address be 64 bits.

A minimum /64 prefix length is required to support Stateless Address Auto-configuration.

IPv6 Prefix Length vs IPv4 CIDR

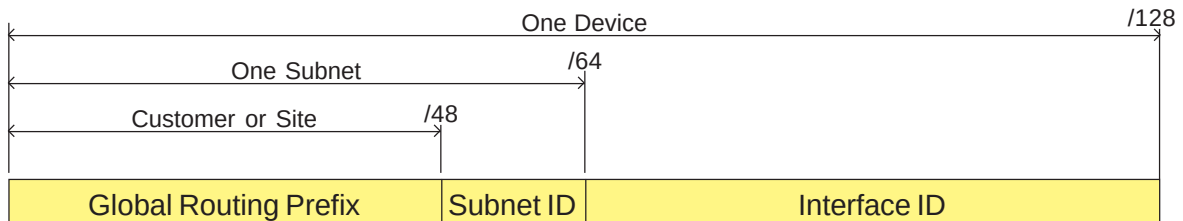
In IPv4 the network portion of the IP address is indicated with a dotted-decimal subnet mask; such as 255.255.255.0. It can also be identified with a CIDR (classless interdomain routing) notation; such as /24.

IPv6 does not use either of these terms to indicate the network portion of an IPv6 address. The network portion of the address is indicated with a **Prefix Length** at the end of the address. While a /48 or /56 looks like a CIDR notation it is not a classless interdomain routing notation. The prefix length indicates the number of nibbles or bits used in the network or subnetwork portion of an IPv6 address.

Global Unicast Prefix Allocations

In 2001, RFC 3177 was written to provide recommendations for how IPv6 Global Unicast addresses should be allocated to customers or Sites.

“In particular, it recommends the assignment of /48 in the general case, /64 when it is known that one and only one subnet is needed and /128 when it is absolutely known that one and only one device is connecting.”

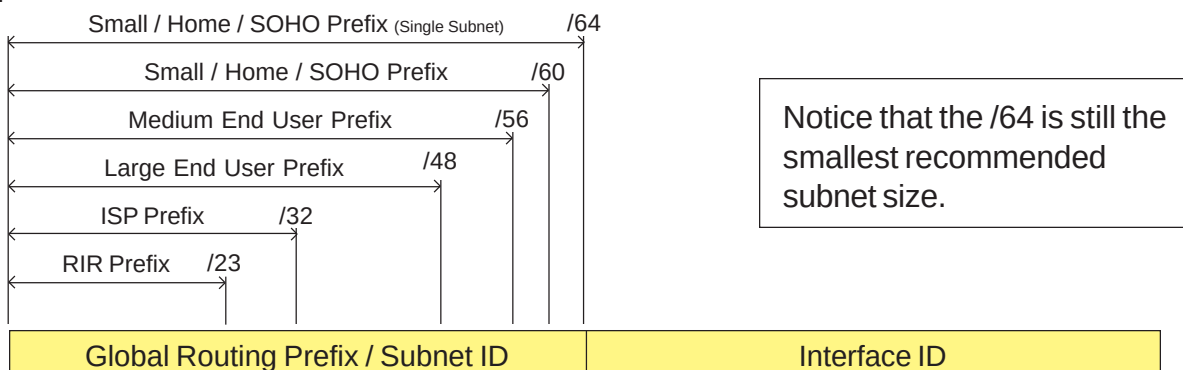


A /48 prefix allows each customer or Site to have 1,208,925,819,614,629,174,706,176 addresses. These can be used as a single subnet, or up to 65,536 subnets with a /64 prefix-length.

The Regional Internet Registries (RIRs) adopted RTC 3177 recommendation in 2002, but began reconsidering the policy in 2005. In March of 2011 RTC 6177 obsoleted RTC 3177 with a new recommendation.

“The exact choice of how much address space to assign end sites is an issue for the operational community.”

This gives local ISPs more options when assigning IPv6 addresses to their customers or Sites.



Typical IPv6 Prefix Assignments:

<u>Subscribers</u>	<u>Prefix- Length</u>	<u># of Subnets at /64</u>	<u>Total number of Possible Addresses</u>
RIR	/23	2,199,023,255,552	40,564,819,207,303,340,847,894,502,572,032
Service Provider (*LIR) -	/32	4,294,967,296	79,228,162,514,264,337,593,543,950,336
Large End User -	/48	65,536	1,208,925,819,614,629,174,706,176
Medium End User -	/56	256	4,722,366,482,869,645,213,696
Small / Home / SOHO -	/60	16	295,147,905,179,352,825,856
Small / Home / SOHO -	/64	1	18,446,744,073,709,551,616

Subnetting on the Nibble Boundary

One of the design themes driving IPv6 was to keep subnetting as simple as possible. The fourth hexet is reserved for subnetting, giving network administrators multiple options for developing a network plan. In order to keep IPv6 addressing as simple as possible it is a **Best Practice** to subnet on the nibble boundary.

Every IPv6 address is comprised of 128 bits, which is represented with 32 hexadecimal numbers.

2001:ACAD:1234:0000:0000:0000:0000:0000/48

Showing this address with 128 binary characters makes it difficult to read and almost impossible for most people to work with.

00100000000000001:1010110010101101:0001001000110100:0000000000000000:
0000000000000000:0000000000000000:0000000000000000:0000000000000000

Each hexadecimal number in an IPv6 address represents 4 bits or a Nibble. An IPv6 address is composed of 32 hexadecimal numbers or 32 Nibbles.

2001:ACAD:1234:0000:0000:0000:0000:0000/48

IPv6 can be subnetted just like IPv4 using individual binary bits. To keep subnetting as simple as possible it is a **Best Practice** to borrow 4 bits, or one Nibble at a time.

Nibble Boundary Subnets

(Subnetting on the Nibble Boundary)

Prefix	# of /64 subnets
/48	65,536
/52	4096
/56	256
/60	16
/64	1

It is a Best Practice to subnet on the Nibble Boundary.

Subnets Based on Individual Binary Bits

(Subnetting within a Nibble)

Prefix subnets	# of /64
/48	65,536
/49	32,768
/50	16,384
/51	8,192
/52	4,096
/53	2,048
/54	1,024
/55	512
/56	256
/57	128
/58	64
/59	32
/60	16
/61	8
/62	4
/63	2
/64	1

Site ID and Sub-Site IDs

Subnetting on the nibble boundary allows you to easily set up proper route aggregation and summarization to use for each subnet or location. It also allows for easier deployment of firewalls based on location or network users. In order to do this you need to assign Site ID's and Sub-Site ID's as necessary. The Site ID is the first address in the subnet you have assigned to a specific location or user group. Sub-Site ID's come into play if you subnet a location or user group into smaller subnets. The first address in each range becomes the Sub-Site ID.

Subnetting on the Nibble boundary gives you these subnetting options.

/48 No Nibbles	/52 1 Nibble	/56 2 Nibbles	/60 3 Nibbles	/64 4 Nibbles
/48 - 1 Subnet				
/52 - 16 Subnets	/52 - 1 Subnet			
/56 - 256 Subnets	/56 - 16 Subnets	/56 - 1 Subnets		
/60 - 4096 Subnets	/60 - 256 Subnets	/60 - 16 Subnets	/60 - 1 Subnets	
/64 - 65,536 Subnets	/64 - 4096 Subnets	/64 - 256 Subnets	/64 - 16 Subnets	/64 - 1 Subnet

As an example, your company has two offices, and within each office there are several groups you want on separate subnets. The groups include: Infrastructure, Management, Marketing, Finance, Research, Warehouse, and Sales.

Your ISP has assigned your company the IPv6 address 2001:ACAD:1234::/48. You will need one Site ID for each office. A /52 Subnet Prefix will give you 16 subnets, or you could use a /56 Subnet Prefix and have 256 subnets. For our purposes we'll use the /56 Subnet Prefix.

Our original IPv6 Range was: 2001:ACAD:1234::/48. Subnetting this address with a /56 Subnet Prefix will take two nibbles from the Subnet ID and give you the following address ranges:

2001:ACAD:1234:**0000**::/56 Save this range for over all infrastructure needs.

2001:ACAD:1234:**0100**::/56 This becomes the Site ID for Office 1.

2001:ACAD:1234:**0200**::/56 This becomes the Site ID for Office 2.

2001:ACAD:1234:**0300**::/56

(Lots of subnets omitted)

2001:ACAD:1234:**FF00**::/56

Office 1 needs subnets for Infrastructure, Management, and Sales. Subnet the Site ID for Office 1 with a /60 Subnet Prefix.

2001:ACAD:1234:01**00**::/60 This becomes the Sub-Site ID for Infrastructure needs.

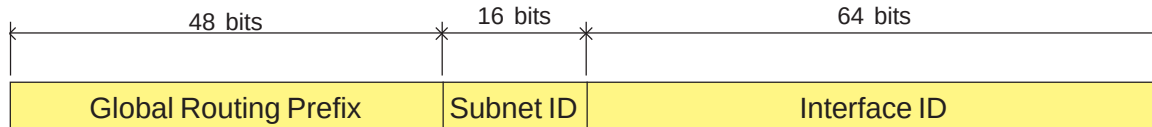
2001:ACAD:1234:01**10**::/60 This becomes the Sub-Site ID for Managment.

2001:ACAD:1234:01**20**::/60 This becomes the Sub-Site ID for Sales.

(Lots of subnets omitted)

2001:ACAD:1234:01**F0**::/60

Subnetting IPv6



Unlike IPv4 which requires you to borrow bits from the host portion, IPv6 has 16 bits or four hexadecimal numbers built into the address specifically allocated for creating subnets. A /48 address will allow you to have a single subnet or up to 65,536 subnets.

Lets say that your ISP has assigned you the 2001:ACAD:1234::/48 IPv6 address.

Global Routing Prefix	Subnet ID	Interface ID
2001:ACAD:1234:	0000	0000:0000:0000:0000

Basic subnetting in IPv6 was designed to make subnetting a very simple process. Start at the /64 bit and start counting up until you've used all the available bits in the subnet ID.

It really is that simple!

2001:ACAD:1234:0000:0000:0000:0000:0000/64	1st subnet
2001:ACAD:1234:0001:0000:0000:0000:0000/64	2nd subnet
2001:ACAD:1234:0002:0000:0000:0000:0000/64	3rd subnet
2001:ACAD:1234:0003:0000:0000:0000:0000/64	4th subnet
2001:ACAD:1234:0004:0000:0000:0000:0000/64	5th subnet
2001:ACAD:1234:0005:0000:0000:0000:0000/64	6th subnet
2001:ACAD:1234:0006:0000:0000:0000:0000/64	7th subnet
2001:ACAD:1234:0007:0000:0000:0000:0000/64	8th subnet
2001:ACAD:1234:0008:0000:0000:0000:0000/64	9th subnet
2001:ACAD:1234:0009:0000:0000:0000:0000/64	10th subnet
2001:ACAD:1234:000A:0000:0000:0000:0000/64	11th subnet
2001:ACAD:1234:000B:0000:0000:0000:0000/64	12th subnet
2001:ACAD:1234:000C:0000:0000:0000:0000/64	13th subnet
2001:ACAD:1234:000D:0000:0000:0000:0000/64	14th subnet
2001:ACAD:1234:000E:0000:0000:0000:0000/64	15th subnet
2001:ACAD:1234:000F:0000:0000:0000:0000/64	16th subnet
2001:ACAD:1234:0010:0000:0000:0000:0000/64	17th subnet
2001:ACAD:1234:0011:0000:0000:0000:0000/64	18th subnet
2001:ACAD:1234:0012:0000:0000:0000:0000/64	19th subnet
2001:ACAD:1234:0013:0000:0000:0000:0000/64	20th subnet

Each subnet contains over 18 quintillion addresses.

(Lots of subnets omitted for space.)

2001:ACAD:1234:FFFC:0000:0000:0000:0000/64	65,533rd subnet
2001:ACAD:1234:FFFD:0000:0000:0000:0000/64	65,534th subnet
2001:ACAD:1234:FFFE:0000:0000:0000:0000/64	65,535th subnet
2001:ACAD:1234:FFFF:0000:0000:0000:0000/64	65,536th subnet

A Medium End user might receive a /56 IPv6 address from their ISP.

Global Routing Prefix	Subnet ID	Interface ID	
2001:ACAD:1234:12	00	0000:0000:0000:0000	
2001:ACAD:1234:12	00	0000:0000:0000:0000/64	1st subnet
2001:ACAD:1234:12	01	0000:0000:0000:0000/64	2nd subnet
2001:ACAD:1234:12	02	0000:0000:0000:0000/64	3rd subnet
2001:ACAD:1234:12	03	0000:0000:0000:0000/64	4th subnet
(Lots of subnets omitted for space.)			
2001:ACAD:1234:12	FD	0000:0000:0000:0000/64	254th subnet
2001:ACAD:1234:12	FE	0000:0000:0000:0000/64	255th subnet
2001:ACAD:1234:12	FF	0000:0000:0000:0000/64	256th subnet

Each subnet contains over 18 quintillion addresses.

A Small End user might receive a /60 IPv6 address from their ISP.

Global Routing Prefix	Subnet ID	Interface ID
2001:ACAD:1234:1230:0000:0000:0000:0000		
2001:ACAD:1234:1230:0000:0000:0000:0000/64	1st subnet	
2001:ACAD:1234:1231:0000:0000:0000:0000/64	2nd subnet	
2001:ACAD:1234:1232:0000:0000:0000:0000/64	3rd subnet	
2001:ACAD:1234:1233:0000:0000:0000:0000/64	4th subnet	
2001:ACAD:1234:1234:0000:0000:0000:0000/64	5th subnet	
2001:ACAD:1234:1235:0000:0000:0000:0000/64	6th subnet	
2001:ACAD:1234:1236:0000:0000:0000:0000/64	7th subnet	
2001:ACAD:1234:1237:0000:0000:0000:0000/64	8th subnet	
2001:ACAD:1234:1238:0000:0000:0000:0000/64	9th subnet	
2001:ACAD:1234:1239:0000:0000:0000:0000/64	10th subnet	
2001:ACAD:1234:123A:0000:0000:0000:0000/64	11th subnet	
2001:ACAD:1234:123B:0000:0000:0000:0000/64	12th subnet	
2001:ACAD:1234:123C:0000:0000:0000:0000/64	13th subnet	
2001:ACAD:1234:123D:0000:0000:0000:0000/64	14th subnet	
2001:ACAD:1234:123E:0000:0000:0000:0000/64	15th subnet	
2001:ACAD:1234:123F:0000:0000:0000:0000/64	16th subnet	

Each subnet contains over 18 quintillion addresses.

A Home or single Site might receive a /64 IPv6 address from their ISP.

Global Routing Prefix	Interface ID
2001:ACAD:1234:1234	0000:0000:0000:0000
2001:ACAD:1234:1234:0000:0000:0000:0000/64	1 subnet

Over 18 quintillion addresses

Common Prefix's and Number of Subnets

Using the Subnet ID, the common Subnet Prefix's available from your ISP are /48, /52, /56, /60, and /64. The ISP could assign a lower Prefix Length, but most business will not need more than 65,536 subnets per Site.

The commonly available /64 subnets are:

/48	65,536 subnets
/52	4096 subnets
/56	256 subnets
/60	16 subnets
/64	1 subnet

Notice that the Subnet ID always changes by four even though we're only using a single hexadecimal character.

Each hexadecimal character equals 4 binary numbers.

(/48, /52, /56, /60, /64)

Each /64 subnet contains over 18 quintillion addresses.

With IPv4 the main concern was using the fewest possible number of addresses through creative subnetting.

The primary concern with IPv6 is making sure you have a prefix length that will cover all the subnets your Site will require and allow for future growth.

Basic Subnetting Problems

Sample Problem

Your ISP has given you the IPv6 address 2001:FE12:A231::/48.

How many /64 subnets are available with this address? 65,536

What are the first six /64 subnets?

2001:FE12:A231::/64

2001:FE12:A231:1::/64

2001:FE12:A231:2::/64

2001:FE12:A231:3::/64

2001:FE12:A231:4::/64

2001:FE12:A231:5::/64

Problem 1

Your ISP has given you the IPv6 address 2000:ACAD:1234:6600::/56.

How many /64 subnets are available with this address?

What are the first four /64 subnets?

What are the last two /64 subnets in this range?

Problem 2

Your ISP has given you the IPv6 address 3FFF:5801:DEAF::/48.

How many /64 subnets are available with this address?

What are the first four /64 subnets?

What are the last two /64 subnets in this range?

Problem 3

Your ISP has given you the IPv6 address 2001:ACAD:5678:1840::/60.

How many /64 subnets are available with this address?

Complete the /64 subnets in this range. (The ISP's Global Routing Prefix is already printed for you.)

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

2001:ACAD:5678:

Problem 4

Your ISP has given you the IPv6 address 3100:6523:AD14:8000::/52.

How many /64 subnets are available with this address?

What are the first four /64 subnets? (The ISP's Global Routing Prefix is already printed for you.)

3100:6523:AD14:

3100:6523:AD14:

3100:6523:AD14:

3100:6523:AD14:

What are the last two /64 subnets in this range?

3100:6523:AD14:

3100:6523:AD14:

Problem 5

Your ISP has given you the IPv6 address 2100:89:4500::/48.

How many /64 subnets are available with this address?

What are the first four /64 subnets? (The ISP's Global Routing Prefix is already printed for you.)

2100:89:4500:

2100:89:4500:

2100:89:4500:

2100:89:4500:

What are the last two /64 subnets in this range?

2100:89:4500:

2100:89:4500:

IPv6 Subnetting Best Practices

IPv6 was designed to make subnetting as simple as possible by using ridiculously large blocks of addresses. Yes, it's wasteful! Most experts agree that IPv6 will last for 100 years and IP will be replaced before we run out of IPv6 address space.

So until the rules and/or the consensus of the experts change, these are the common subnets and best practices you will be working with. Information for subnets can be found in RFC & RTC documents, plus some additional resources listed in the Reference section of this workbook.

The common /64 subnets are:

You can negotiate with your ISP for a larger block

/48	65,536 subnets
/52	4096 subnets
/56	256 subnets
/60	16 subnets
/64	1 subnet

Specific Address Rules and Best Practices:

Point-To-Point Connections -

Best Practice: Use a /64 address range for these two addresses.

In rare cases this approach might present an issue with certain router setups where addresses are bounced back and forth between routers. Reducing the number of addresses in the range can also help avoid neighbor cache exhaustion attacks. (RFC 6164 - section 5)

Acceptable options include:

- /127 -is recommended in RFC 6164. You must disable the router's Subnet-Router Anycast option to avoid issues with the all routers anycast address.
- /126 -is discussed in RFC 2526. The /126 allows the all-zero reserved Anycast address to be avoided.
- /120 - allows all the reserved anycast address to be avoided.
- /112 - allows you to avoid the reserved anycast address, and gives you the entire four-digit hex value after the last colon.

Of all these options the /120 is probably the best choice since it avoids any issues with the Anycast addresses.

It is recommended that you reserve the entire /64 network with each of the above options.

/128 Single Address Subnets -

Best Practice: reserve the first subnet to use for infrastructure needs, such as loopback addresses, etc..

Acceptable option:

Allocate a full /64 range of addresses for each loopback address, but assign it a /128 subnet Prefix.

Anycast addresses -

Best Practice: Don't use the very first address or the last 128 addresses in any /64 network. These can only be assigned to an interface as an Anycast address.

Anycast addresses can be pulled from Global Unicast, Site-Local or Link-Local address ranges. Any address assigned to more than one interface on a subnet becomes an Anycast address. Anycast addresses can only be used by network devices, not a host. No anycast address can be used as the source address of an IPv6 packet.

The first address in every /64 subnet range is reserved for special use. The Interface ID is all zero's. (Example: 2001:ACAD:1234:5678:0000:0000:0000:0000/64) This address is the Subnet-Anycast Address. These addresses are typically used by different protocols such as Mobile IPv6. (RFC 4291)

The last eight bits in every /64 subnet range are reserved for Anycast addresses. These bits are 10000000 to 11111111. This means you can not assign any addresses if the last hextet falls between FF80 and FFFF unless it is a Anycast address. (RFC 2526)

If you try to apply the last 128 addresses to a router without setting them up as an Anycast address you will get the following error message.

```
Router(config-if)#ipv6 address 2001:ACAD:1234:5678:FFFF:FFFF:FFFF:FFFF/64
% 2001:DB8:1:1:FFFF:FFFF:FFFF:FFFF/64 should not be configured on FastEthernet0/0, a reserved anycast
```

Developing an Address Plan

(or IPv6 Subnetting in the Real World)

There is no one right way for developing an IPv6 addressing plan, but the recommended general guidelines include the following:

Step 1: Decide how you are going to divide your network:

- a. by location
- b. by user groups



Administration

Users Include:
Administration
Staff



Academic

Users Include:
Staff
Students



Dormitory

Users Include:
Staff
Students

Subnetting by Location:

To divide by location you would need four subnets. One for each building and one for the overall network infrastructure needs. You also need to hold several extra subnets in reserve for later growth.

Advantages:

This allows you to optimize your routing tables. All the networks within each location will aggregate to a single route.

Subnetting by User Groups:

To subnet the network by user groups you would need four subnets. One for Administration, Staff, and Students, plus one for overall network infrastructure needs. You also need to hold several extra subnets in reserve for later growth.

Advantages:

Subnetting by user groups makes it much easier to implement a security policy. Grouping by usage also helps track addresses for allocation and management.

Best Practice:

Subnetting by either location or user is acceptable. However, with the emphasis on network security, most networks are better served by subnetting user groups. It makes it much easier to maintain a higher level of security.

Step 2: Determine how many primary and secondary subnets your Site will need.

- Create the primary subnets first.
- Then create secondary subnets.

Subnetting by Location:

Primary Subnets: Quantity 4

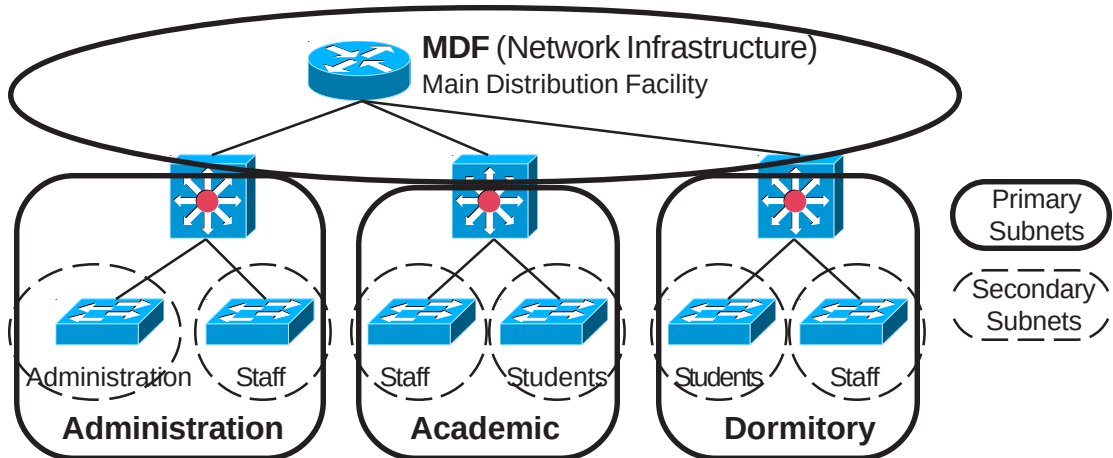
With three buildings you will need four primary subnets. One for each building and one for the overall infrastructure needs.

Secondary Subnets: Quantity 6

Administration will need two secondary subnets: Administration and Staff.

Academic will need two secondary subnets: Staff and Students.

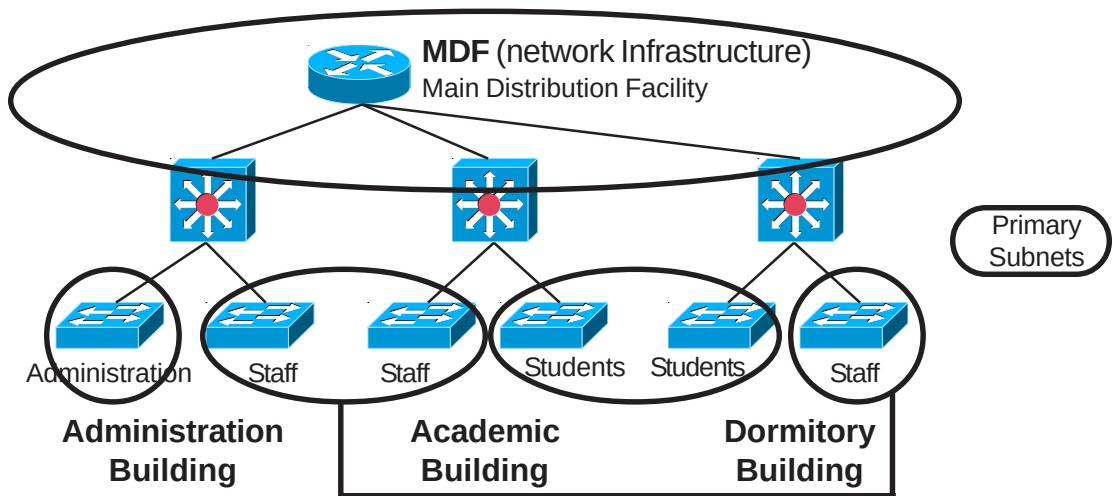
Dormitory will need two secondary subnets: Staff and Students.



Subnetting by User Groups:

Primary Subnets: Quantity 4

With three user groups you will need four subnets. One for each group and one for the overall infrastructure needs.



Step 3: Based on the number of primary and secondary subnets needed assign the address ranges. The ISP has assigned you 2001:ACAD:1234::/48.

Subnetting Options:

/48 No Nibbles	/52 1 Nibble	/56 2 Nibbles	/60 3 Nibbles	/64 4 Nibbles
/48 - 1 Subnet				
/52 - 16 Subnets	/52 - 1 Subnet			
/56 - 256 Subnets	/56 - 16 Subnets	/56 - 1 Subnets		
/60 - 4096 Subnets	/60 - 256 Subnets	/60 - 16 Subnets	/60 - 1 Subnets	
/64 - 65,536 Subnets	/64 - 4096 Subnets	/64 - 256 Subnets	/64 - 16 Subnets	/64 - 1 Subnet

Subnetting by Location:

Primary Subnets: Quantity 4

With three buildings you will need four primary subnets. One for each building and one for the overall infrastructure needs.

Secondary Subnets: Quantity 6

Administration will need two secondary subnets: Administration and Staff.

Academic will need two secondary subnets: Staff and Students.

Dormitory will need two secondary subnets: Staff and Students.

Take the addresses assigned to you by the ISP use one nibble and subnet it into 16 subnets using a /52 Subnet Prefix. This will give you the 4 primary subnets required with several to spare for future growth.

2001:ACAD:1234::/48 becomes:

- ▣ 2001:ACAD:1234::/52 Site ID for over all infrastructure needs.
- ▣ 2001:ACAD:1234:1000::/52 Site ID for the Administration Building.
- ▣ 2001:ACAD:1234:2000::/52 Site ID for the Academic Building.
- ▣ 2001:ACAD:1234:3000::/52 Site ID for the Dormitory.
- ▣ 2001:ACAD:1234:4000::/52
- (Subnets omitted for space.)
- ▣ 2001:ACAD:1234:F000::/52

Site IDs and Sub-Site IDs will be the addresses found in the routing tables.

Take the second, third, and forth ranges and subnet them again by using the next Nibble with a /56 Subnet Prefix. This will create 16 subnets for each location.

Administration Building Site ID 2001:ACAD:1234:1000::/52 becomes:

- ▣ 2001:ACAD:1234:1000::/52 Administration Building Site ID
 - ▣ 2001:ACAD:1234:1000::/56 Sub-Site ID for infrastructure needs.
 - ▣ 2001:ACAD:1234:1100::/56 Sub-Site ID for Administration
 - ▣ 2001:ACAD:1234:1200::/56 Sub-Site ID for Staff

Academic Building Site ID 2001:ACAD:1234:2000::/52 becomes:

- ▣ 2001:ACAD:1234:2000::/52 Academic Building Site ID
 - ▣ 2001:ACAD:1234:2000::/56 Sub-Site ID for infrastructure needs.
 - ▣ 2001:ACAD:1234:2100::/56 Sub-Site ID for Students
 - ▣ 2001:ACAD:1234:2200::/56 Sub-Site ID for Staff

Dormitory Building Site ID 2001:ACAD:1234:3000::/52 becomes:

- ▣ 2001:ACAD:1234:3000::/52 Dormitory Building Site ID
 - ▣ 2001:ACAD:1234:3000::/56 Sub-Site ID for infrastructure needs.
 - ▣ 2001:ACAD:1234:3100::/56 Sub-Site ID for Students
 - ▣ 2001:ACAD:1234:3200::/56 Sub-Site ID for Staff

Subnetting by User Group:

Primary Subnets: Quantity 4

With three user groups you will need four primary subnets. One for each group and one for the overall infrastructure needs. In this example no secondary subnets are required.

Take the address assigned to you by the ISP use one nibble and subnet it into 16 subnets using a /52 Subnet Prefix. This will give you the 4 primary subnets required with several to spare for future growth.

2001:ACAD:1234::/48 becomes:

- ▣ 2001:ACAD:1234::/52 Site ID for over all infrastructure needs.
- ▣ 2001:ACAD:1234:1000::/52 Site ID for the Administration employees.
- ▣ 2001:ACAD:1234:2000::/52 Site ID for the Staff.
- ▣ 2001:ACAD:1234:3000::/52 Site ID for the Students.
- ▣ 2001:ACAD:1234:4000::/52
- (Subnets omitted for space.)
- ▣ 2001:ACAD:1234:F000::/52

IPv6 Subnetting Problems

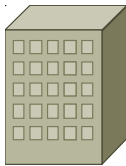
Subnetting on the Nibble Boundary

Sample Problem 1

Using the **minimum number of subnets required** for the primary and secondary sites, design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

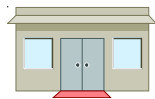
A coffee shop is opening three new stores and a central office/warehouse in your community and needs an IPv6 network plan developed. Each store will need secure network access for three groups: managers, the registers (which will monitor inventory and revenue), and wireless access for guests. The central office/warehouse will need secure network access for several departments: Managers, Finance, Human Relations.

The ISP has given the company 2000:FE23:0054::/48.



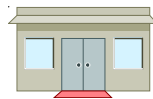
Central Office

Users Include:
Mangers
Finance
Human Relations



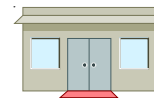
Store 1

Users Include:
Mangers
Registers
Wireless



Store 2

Users Include:
Mangers
Registers
Wireless



Store 3

Users Include:
Mangers
Registers
Wireless

Subnets Based on User Groups

ISP Address: 2000:FE23:0054::/48

- ❑ Infrastructure Site ID: 2000:FE23:0054::/52
- ❑ Managers Site ID: 2000:FE23:0054:1000::/52
- ❑ Finance Site ID: 2000:FE23:0054:2000::/52
- ❑ Human Relations Site ID: 2000:FE23:0054:3000::/52
- ❑ Registers Site ID: 2000:FE23:0054:4000::/52
- ❑ Wireless Site ID: 2000:FE23:0054:5000::/52

Subnets Based on Location

ISP Address: 2000:FE23:0054::/48

- ▣ Infrastructure Site ID: 2000:FE23:0054::/52
- ▣ Central Office Site ID: 2000:FE23:0054:1000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:1000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:1100::/56
- ▣ Finance Sub-Site ID: 2000:FE23:0054:1200::/56
- ▣ Human Relations Sub-Site ID: 2000:FE23:0054:1300::/56
- ▣ Store 1 Site ID: 2000:FE23:0054:2000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:2000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:2100::/56
- ▣ Registers Sub-Site ID: 2000:FE23:0054:2200::/56
- ▣ Wireless Sub-Site ID: 2000:FE23:0054:2300::/56
- ▣ Store 2 Site ID: 2000:FE23:0054:3000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:3000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:3100::/56
- ▣ Registers Sub-Site ID: 2000:FE23:0054:3200::/56
- ▣ Wireless Sub-Site ID: 2000:FE23:0054:3300::/56
- ▣ Store 3 Site ID: 2000:FE23:0054:4000::/52
- ▣ Infrastructure Sub-Site ID: 2000:FE23:0054:4000::/56
- ▣ Managers Sub-Site ID: 2000:FE23:0054:4100::/56
- ▣ Registers Sub-Site ID: 2000:FE23:0054:4200::/56
- ▣ Wireless Sub-Site ID: 2000:FE23:0054:4300::/56

IPv6 Subnetting Problems

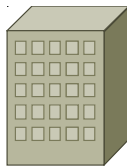
Subnetting on the Nibble Boundary

Sample Problem 2

Using the **minimum number of subnets required** for the primary and secondary sites, design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

A Medical Building is opening in your community and needs an IPv6 network plan developed.

The ISP has given the company 2001:5378:8801::/48.



Medical Building

First Floor Rooms

Patient Check-in
Emergency Room

Users Include:
Administrators
Staff
Guests

Second Floor Rooms

Nurses Station
Ward A

Users Include:
Staff
Guests

Subnets Based on User Groups

ISP Address: 2001:5378:8801::/48

- ▣ Infrastructure Site ID: 2001:5378:8801::/52
- ▣ Administrators Site ID: 2001:5378:8801:1000::/52
- ▣ Staff Site ID: 2001:5378:8801:2000::/52
- ▣ Guests Site ID: 2001:5378:8801:3000::/52

Subnets Based on Location

ISP Address: 2001:5378:8801::/48

- ▣ Infrastructure Site ID: 2001:5378:8801::/52
- ▣ First Floor Site ID: 2001:5378:8801:1000::/52
- ▣ Infrastructure Sub-Site ID: 2001:5378:8801:1000::/56
- ▣ Patient Check-in Sub-Site ID: 2001:5378:8801:1100::/56
- ▣ Administrators Sub-Site ID: 2001:5378:8801:1100::/60
- ▣ Staff Sub-Site ID: 2001:5378:8801:1110::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:1120::/60
- ▣ Emergency Room Sub-Site ID: 2001:5378:8801:1200::/56
- ▣ Administrators Sub-Site ID: 2001:5378:8801:1200::/60
- ▣ Staff Sub-Site ID: 2001:5378:8801:1210::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:1220::/60
- ▣ Second Floor Site ID: 2001:5378:8801:2000::/52
- ▣ Infrastructure Sub-Site ID: 2001:5378:8801:2000::/56
- ▣ Nurses Station Sub-Site ID: 2001:5378:8801:2100::/56
- ▣ Staff Sub-Site ID: 2001:5378:8801:2100::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:2110::/60
- ▣ Ward A Sub-Site ID: 2001:5378:8801:2200::/56
- ▣ Staff Sub-Site ID: 2001:5378:8801:2200::/60
- ▣ Guest Sub-Site ID: 2001:5378:8801:2210::/60

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 1

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

The XYZ Film company is setting up a new office and needs an IPv6 network plan developed. The company will have the following departments:

The ISP has given the company 2001:EE00:2575::/48.

Administration Building

Users Include:
Sales
Accounting
Distribution

Production Building

Users Include:
Casting
Editing

Subnets Based on User Groups

ISP Address: 2001:EE00:2575::/48

☐ Infrastructure Site ID:

☐ Sales Site ID:

☐ Accounting Site ID:

☐ Distribution Site ID:

☐ Casting Site ID:

☐ Editing Site ID:

Subnets Based on Location

ISP Address: 2001:EE00:2575::/48

▣ Infrastructure Site ID:

▣ Administration Site ID:

▣ Infrastructure Sub-Site ID:

▣ Sales Sub-Site ID:

▣ Accounting Sub-Site ID:

▣ Distribution Sub-Site ID:

▣ Production Site ID:

▣ Infrastructure Sub-Site ID:

▣ Casting Sub-Site ID:

▣ Editing Sub-Site ID:

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 2

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

A new medical supply company is opening and needs an IPv6 network plan developed. The company has three buildings and will include the following user groups:

The ISP has given the company 3F01:ABCD:8875::/48.

Building 1

Users Include:
Management
Sales

Building 2

Users Include:
Management
Human Resources

Building 3

Users Include:
Management
Warehouse

Subnets Based on User Groups

ISP Address: 3F01:ABCD:8875::/48

- ▣ Infrastructure Site ID:
- ▣ Management Site ID:
- ▣ Sales Site ID:
- ▣ Human Resources Site ID:
- ▣ Warehouse Site ID:

Subnets Based on Location

ISP Address: 3F01:ABCD:8875::/48

▣ Infrastructure Site ID:

▣ Building 1 Site ID:

▣ Infrastructure Sub-Site ID:

▣ Management Sub-Site ID:

▣ Sales Sub-Site ID:

▣ Building 2 Site ID:

▣ Infrastructure Sub-Site ID:

▣ Management Sub-Site ID:

▣ Human Resources Sub-Site ID:

▣ Building 2 Site ID:

▣ Infrastructure Sub-Site ID:

▣ Management Sub-Site ID:

▣ Warehouse Sub-Site ID:

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 3

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

A paper supply company needs an IPv6 network plan developed. The company has two buildings and will include the following user groups and sub-user groups:

The ISP has given the company 2001:CA21:9000::/48.

Building A

Management Groups

Human Resources

Sales:

Wholesale

Retail

Building B

Production Groups

Warehouse

Shipping:

Domestic

Worldwide

Subnets Based on User Groups

ISP Address: 2001:CA21:9000::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2001:CA21:9000:

☐ Management Groups Site ID: 2001:CA21:9000:

 ☐ HR Sub-Site ID: 2001:CA21:9000:

 ☐ Sales Sub-Site ID: 2001:CA21:9000:

 ☐ Wholesale Sub- Site ID: 2001:CA21:9000:

 ☐ Retail Sub- Site ID: 2001:CA21:9000:

☐ Production Groups Site ID: 2001:CA21:9000:

 ☐ Warehouse Sub-Site ID: 2001:CA21:9000:

 ☐ Shipping Sub-Site ID: 2001:CA21:9000:

 ☐ Domestic Sub-Site ID: 2001:CA21:9000:

 ☐ Worldwide Sub-Site ID: 2001:CA21:9000:

Subnets Based on Location

ISP Address: 2001:CA21:9000::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2001:CA21:9000:

☐ Building A Site ID: 2001:CA21:9000:

☐ Infrastructure Sub-Site ID: 2001:CA21:9000:

☐ HR Sub-Site ID: 2001:CA21:9000:

☐ Sales Sub-Site ID: 2001:CA21:9000:

☐ Wholesale Sub-Site ID: 2001:CA21:9000:

☐ Retail Sub-Site ID: 2001:CA21:9000:

☐ Building B Site ID: 2001:CA21:9000:

☐ Infrastructure Sub-Site ID: 2001:CA21:9000:

☐ Warehouse Sub-Site ID: 2001:CA21:9000:

☐ Shipping Sub-Site ID: 2001:CA21:9000:

☐ Domestic Sub-Site ID: 2001:CA21:9000:

☐ Worldwide Sub-Site ID: 2001:CA21:9000:

IPv6 Subnetting Problems

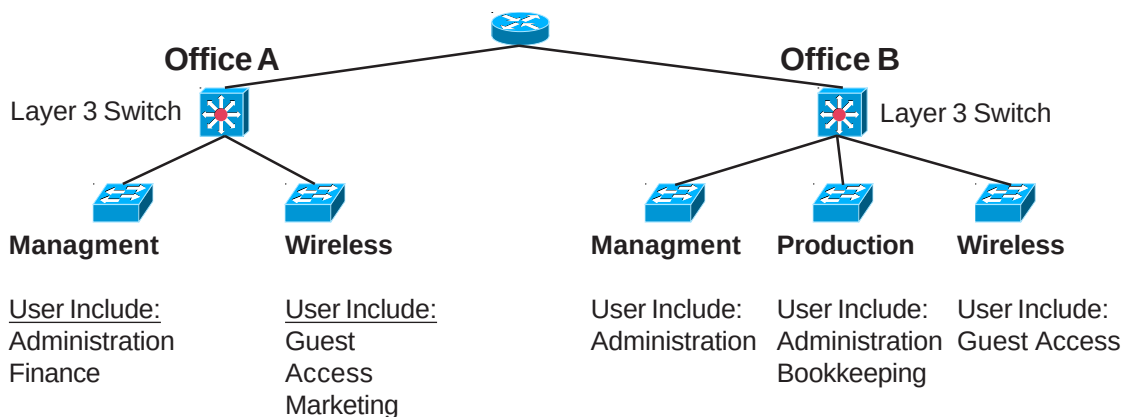
Subnetting on the Nibble Boundary

Problem 4

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

A company is setting up a new server farm and needs an IPv6 network plan developed. The Core Router and the direct connections with layer 3 switches will pull their IPv6 addresses from the Infrastructure Site ID range.

The ISP has given the company 2000:ACAD:1145::/48.



Subnets Based on User Groups

☐ ISP Address: 2000:ACAD:1145::/48 (The ISP's Global Routing Prefix is already printed for you.)

- ☐ Infrastructure Site ID: 2000:ACAD:1145:
- ☐ Administration Site ID: 2000:ACAD:1145:
- ☐ Finance Site ID: 2000:ACAD:1145:
- ☐ Guest Access Site ID: 2000:ACAD:1145:
- ☐ Marketing Site ID: 2000:ACAD:1145:
- ☐ Bookkeeping Site ID: 2000:ACAD:1145:

Subnets Based on Location

ISP Address: 2000:ACAD:1145::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2000:ACAD:1145:

☐ Office A Site ID: 2000:ACAD:1145:

☑ Infrastructure Sub-Site ID: 2000:ACAD:1145:

☑ Management Sub-Site ID: 2000:ACAD:1145:

☑ Administration Sub-Site ID: 2000:ACAD:1145:

☑ Finance Sub-Site ID: 2000:ACAD:1145:

☑ Wireless Access Sub-Site ID: 2000:ACAD:1145:

☑ Guest Access Sub-Site ID: 2000:ACAD:1145:

☑ Marketing Sub-Site ID: 2000:ACAD:1145:

☐ Office B Site ID: 2000:ACAD:1145:

☑ Infrastructure Sub-Site ID: 2000:ACAD:1145:

☑ Management Sub-Site ID: 2000:ACAD:1145:

☑ Administration Sub-Site ID: 2000:ACAD:1145:

☑ Production Sub-Site ID: 2000:ACAD:1145:

☑ Administration Sub-Site ID: 2000:ACAD:1145:

☑ Bookkeeping Sub-Site ID: 2000:ACAD:1145:

☑ Wireless Access Sub-Site ID: 2000:ACAD:1145:

☑ Guest Access Sub-Site ID: 2000:ACAD:1145:

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 5

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for user groups and a second plan for location.

The company has multiple floors in a high rise building and will include the following user groups and sub-user groups:

The ISP has given the company 3F01:AA07:3907::/48.

33rd Floor

Manufacturing Groups

Marketing

Inventory

Shipping

34th Floor

Admin Groups

Human Resources (HR):

Hiring

Benifits

Financial:

Purchasing

Sales

Subnets Based on User Groups

☐ ISP Address: 3F01:AA07:3907::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 3F01:AA07:3907:

☐ Manufacturing Groups Site ID: 3F01:AA07:3907:

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:

☐ Marketing Sub-Site ID: 3F01:AA07:3907:

☐ Inventory Sub-Site ID: 3F01:AA07:3907:

☐ Shipping Sub- Site ID: 3F01:AA07:3907:

☐ Admin Groups Site ID: 3F01:AA07:3907:

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:

☐ HR Sub-Site ID: 3F01:AA07:3907:

☐ Hiring Sub-Site ID: 3F01:AA07:3907:

☐ Benfits Sub-Site ID: 3F01:AA07:3907:

☐ Financial Sub-Site ID: 3F01:AA07:3907:

☐ Purchasing Sub-Site ID: 3F01:AA07:3907:

☐ Sales Sub-Site ID: 3F01:AA07:3907:

Subnets Based on Location

ISP Address: 3F01:AA07:3907::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 3F01:AA07:3907:

☐ 33rd Floor Site ID: 3F01:AA07:3907:

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:

☐ Manufacturing Groups Sub-Site ID: 3F01:AA07:3907:

☐ Marketing Sub-Site ID: 3F01:AA07:3907:

☐ Inventory Sub-Site ID: 3F01:AA07:3907:

☐ Shipping Sub-Site ID: 3F01:AA07:3907:

☐ 34th Floor Site ID: 3F01:AA07:3907:

☐ Infrastructure Sub-Site ID: 3F01:AA07:3907:

☐ Admin Groups Sub-Site ID: 3F01:AA07:3907:

☐ HR Sub-Site ID: 3F01:AA07:3907:

☐ Hiring Sub-Site ID: 3F01:AA07:3907:

☐ Benefits Sub-Site ID: 3F01:AA07:3907:

☐ Financial Sub-Site ID: 3F01:AA07:3907:

☐ Purchasing Sub-Site ID: 3F01:AA07:3907:

☐ Sales Sub-Site ID: 3F01:AA07:3907:

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 6

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

This medium sized company will include the following user groups and sub-user groups:

The ISP has given the company 2001:0:17::/52

Management

Users Include:
Laboratory
Administrators
Staff

Finance

Users Include:
Staff

Marketing Department

Users Include:
Advertising:
Radio
TV
Web
Sales:
Retail
Wholesale

Subnets Based on User Groups

☐ ISP Address: 2001:0:17::/52 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID:	<u>2001:0:17:</u>	
☐ Administrators Site ID:	<u>2001:0:17:</u>	
☐ Staff Site ID:	<u>2001:0:17:</u>	
☐ Advertising Site ID:	<u>2001:0:17:</u>	
☐ Infrastructure Sub-Site ID:	<u>2001:0:17:</u>	
☐ Radio Sub-Site ID:	<u>2001:0:17:</u>	
☐ TV Sub- Site ID:	<u>2001:0:17:</u>	
☐ Web Sub- Site ID:	<u>2001:0:17:</u>	
☐ Sales Site ID:	<u>2001:0:17:</u>	
☐ Infrastructure Sub-Site ID:	<u>2001:0:17:</u>	
☐ Retail Sub-Site ID:	<u>2001:0:17:</u>	
☐ Wholesale Sub-Site ID:	<u>2001:0:17:</u>	

Subnets Based on Location

☐ ISP Address: 2001:0:17::/52 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2001:0:17:

☐ Management Site ID: 2001:0:17:

☐ Infrastructure Sub-Site ID: 2001:0:17:

☐ Administrators Sub-Site ID: 2001:0:17:

☐ Staff Sub-Site ID: 2001:0:17:

☐ Finance Site ID: 2001:0:17:

☐ Infrastructure Sub-Site ID: 2001:0:17:

☐ Staff Sub-Site ID: 2001:0:17:

☐ Marketing Dept Site ID: 2001:0:17:

☐ Infrastructure Sub-Site ID: 2001:0:17:

☐ Advertising Sub-Site ID: 2001:0:17:

☐ Radio Sub-Site ID: 2001:0:17:

☐ TV Sub-Site ID: 2001:0:17:

☐ Web Sub-Site ID: 2001:0:17:

☐ Sales Sub-Site ID: 2001:0:17:

☐ Retail Sub-Site ID: 2001:0:17:

☐ Wholesale Sub-Site ID: 2001:0:17:

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 7

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

This medium sized company is setting up a new IPv6 addressing plan which will include the following user groups and sub-user groups:

The ISP has given the company 3F00:3589:0:5000::/52

Office A

Users Include:
Management
Human Relations (HR):
 Record Keeping
 Insurance

Office B

Users Include:
Management
Finance:
 Sales

Office C

Users Include:
Management
Purchasing:
 Inventory
 Distribution

Subnets Based on User Groups

ISP Address: 3F00:3589:0:5000::/52 (The ISP's Global Routing Prefix is already printed for you.)

☐	Infrastructure Site ID:	3F00:3589:0:	
☐	Management Site ID:	3F00:3589:0:	
☐	HR Site ID:	3F00:3589:0:	
☐	Infrastructure Sub-Site ID:	3F00:3589:0:	
☐	Record Keeping Sub-Site ID:	3F00:3589:0:	
☐	Insurance Site ID:	3F00:3589:0:	
☐	Finance Site ID:	3F00:3589:0:	
☐	Infrastructure Sub-Site ID:	3F00:3589:0:	
☐	Sales Sub-Site ID:	3F00:3589:0:	
☐	Purchasing Sub-Site ID:	3F00:3589:0:	
☐	Inventory Sub-Site ID:	3F00:3589:0:	
☐	Distribution Sub-Site ID:	3F00:3589:0:	

Subnets Based on Location

ISP Address: 3F00:3589:0:5000::/52 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID:	3F00:3589:0:	
☐ Office A Site ID:	3F00:3589:0:	
☑ Infrastructure Sub-Site ID:	3F00:3589:0:	
☑ Management Sub-Site ID:	3F00:3589:0:	
☑ HR Sub- Site ID:	3F00:3589:0:	
☑ Record Keeping Sub-Site ID:	3F00:3589:0:	
☑ Insurance Sub-Site ID:	3F00:3589:0:	
☐ Office B Site ID:	3F00:3589:0:	
☑ Infrastructure Sub-Site ID:	3F00:3589:0:	
☑ Management Sub-Site ID:	3F00:3589:0:	
☑ Finance Sub-Site ID:	3F00:3589:0:	
☑ Sales Sub-Site ID:	3F00:3589:0:	
☐ Office C Site ID:	3F00:3589:0:	
☑ Infrastructure Sub-Site ID:	3F00:3589:0:	
☑ Management Sub-Site ID:	3F00:3589:0:	
☑ Purchasing Sub-Site ID:	3F00:3589:0:	
☑ Inventory Sub-Site ID:	3F00:3589:0:	
☑ Distribution Sub-Site ID:	3F00:3589:0:	

IPv6 Subnetting Problems

Subnetting on the Nibble Boundary

Problem 8

Using the minimum number of subnets required for the primary and secondary sites design two IPv6 address plans that meets the following requirements. Create one plan for location and the second plan for user groups.

A Health Care facility is upgrading their network to IPv6 and will include the following user groups and sub-user groups:

The ISP has given the company 2000:2531:FE00::/48.

Emergency

Users Include:
Nurses/Staff
Laboratory
Obstetrics
Pediactric

Admissions

Users Include:
Nurses/Staff
Records

Patient Wards

Users Include:
Ward A:
Nurses/Staff
Guest WIFI
Ward B:
Nurses/Staff
Guest WIFI

Subnets Based on User Groups

ISP Address: 2000:2531:FE00::/48 (The ISP's Global Routing Prefix is already printed for you.)

Infrastructure Site ID:	<u>2000:2531:FE00:</u>	
Nurses/Staff Site ID:	<u>2000:2531:FE00:</u>	
Laboratory Site ID:	<u>2000:2531:FE00:</u>	
Obstetrics Site ID:	<u>2000:2531:FE00:</u>	
Pediatric Site ID:	<u>2000:2531:FE00:</u>	
Records Site ID:	<u>2000:2531:FE00:</u>	
Guest WIFI Site ID:	<u>2000:2531:FE00:</u>	

Subnets Based on Location

ISP Address: 2000:2531:FE00::/48 (The ISP's Global Routing Prefix is already printed for you.)

☐ Infrastructure Site ID: 2000:2531:FE00:

☐ Emergency Site ID: 2000:2531:FE00:

☐ Infrastructure Sub-Site ID: 2000:2531:FE00:

☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:

☐ Laboratory Sub-Site ID: 2000:2531:FE00:

☐ Obstetrics Sub-Site ID: 2000:2531:FE00:

☐ Pediatric Sub-Site ID: 2000:2531:FE00:

☐ Admissions Site ID: 2000:2531:FE00:

☐ Infrastructure Sub-Site ID: 2000:2531:FE00:

☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:

☐ Records Sub-Site ID: 2000:2531:FE00:

☐ Patient Wards Site ID: 2000:2531:FE00:

☐ Infrastructure Sub-Site ID: 2000:2531:FE00:

☐ Ward A Sub-Site ID: 2000:2531:FE00:

☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:

☐ Guest WIFI Sub-Site ID: 2000:2531:FE00:

☐ Ward B Sub-Site ID: 2000:2531:FE00:

☐ Nurses/Staff Sub-Site ID: 2000:2531:FE00:

☐ Guest WIFI Sub-Site ID: 2000:2531:FE00:

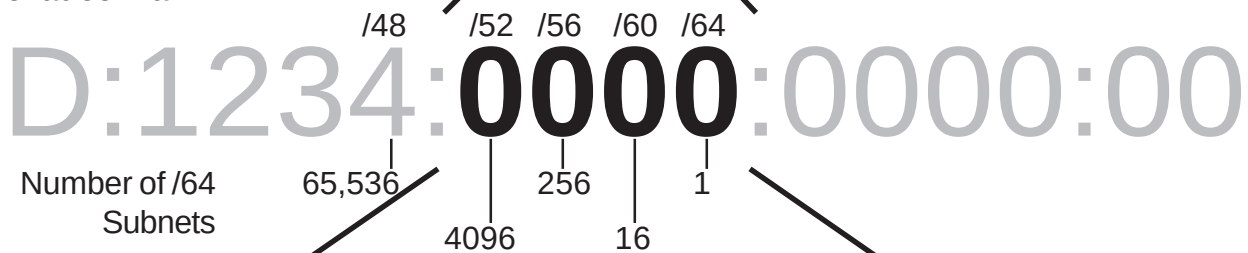
Subnetting Within a Nibble

It is a Best Practice to subnet on the Nibble Boundary. However, subnetting within a Nibble is an acceptable practice. It tends to make subnetting, implementation, and troubleshooting more difficult.

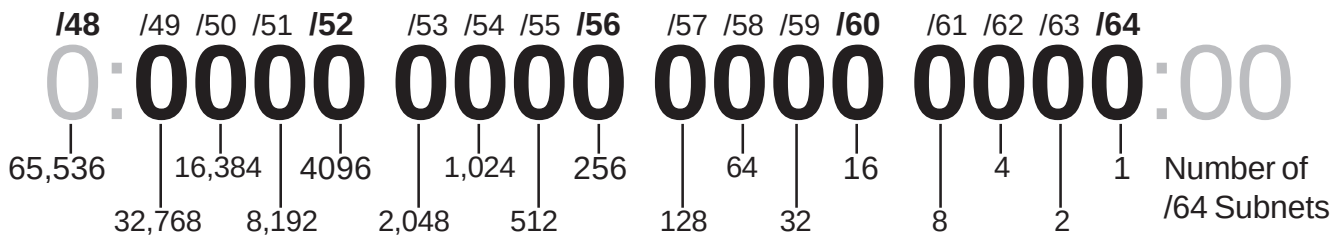
Subnet ID Break Down

2001:ACAD:1234:0000:0000:0000:0000:0000/48

Hexadecimal



Binary

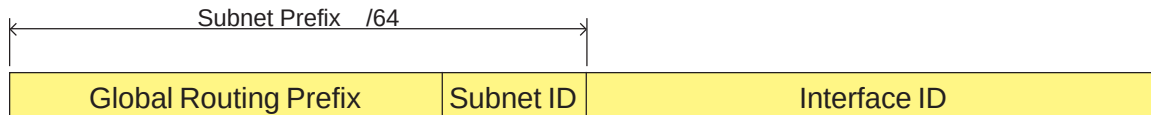


Subnetting Beyond the /64 Boundary

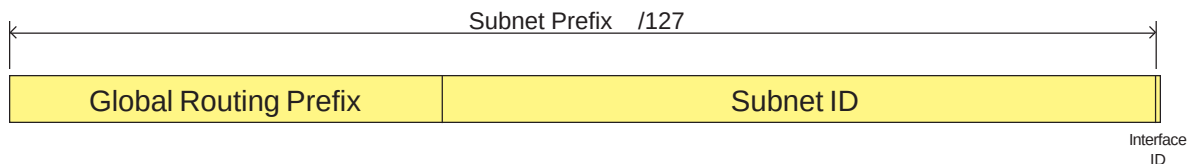
What happens if you need more subnets than the 16 bit Subnet Prefix will allow? Or your IPv4 address conservatism kicks in and you decide that using a /64 Global Routing Prefix with over 18 quintillion addresses for a single point-to-point serial connection is more than you can handle; there are some options.

IPv6 was designed to be very flexible. The Subnet Prefix is the address space used by the Global Routing Prefix and the Subnet ID, and can range from 0 to 128.

Standard /48 Subnet Prefix



Point-To-Point /127 Subnet Prefix



Just as you could borrow host bits in IPv4, you can borrow Interface ID bits in IPv6. This allows you to create more subnets with fewer addresses. Before you get too excited, there are a few rules and best practices you need to take into account.

Borrowing bits from the Interface ID should only be done on network infrastructure links. Loopback addresses, point-to-point links, addresses that are usually statically assigned.

Any subnet that includes end devices needs to stay on a /64 or lower prefix. This would be computers, tablets, smart phones, servers, printers, anything that might be on a subnet that connects to the internet.

According to RFC 5375, a /64 prefix is required to support a number of benefits offered by IPv6; such as:

- Stateless Address Autoconfiguration
- Neighbor Discovery (ND)
- Secure Neighborhood Discovery (SEND)
- privacy extensions
- parts of Mobile IPv6
- PIM-SM with Embedded-RP
- SHIM6 [SHIM6]

Plus a number of other features currently in development, or being proposed, which will rely on a /64 prefix.

The bottom line is that IPv6 was designed to waste an unbelievable amount of addresses and it's OK. So while it is possible to subnet beyond the /64 subnet prefix it is not recommended.

Define the Following IPv6 Terms:

(Use the definitions/explanations from this workbook)

1. Global Routing Prefix -

2. Subnet ID -

3. Interface ID -

4. Subnet Prefix -

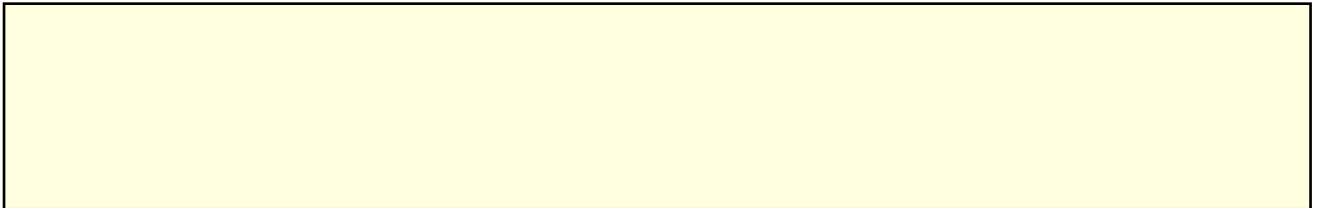
5. Nibble -

6. Unspecified address -

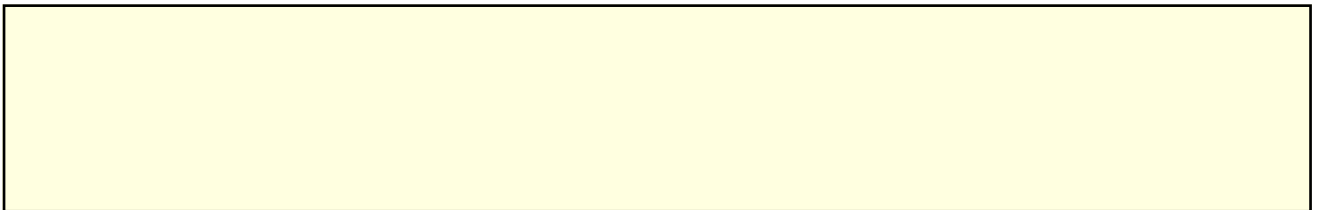
7. Loopback address -



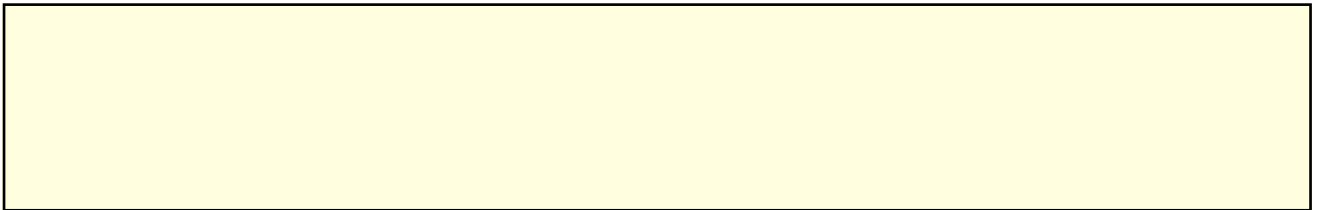
8. Ipv4 Embedded address -



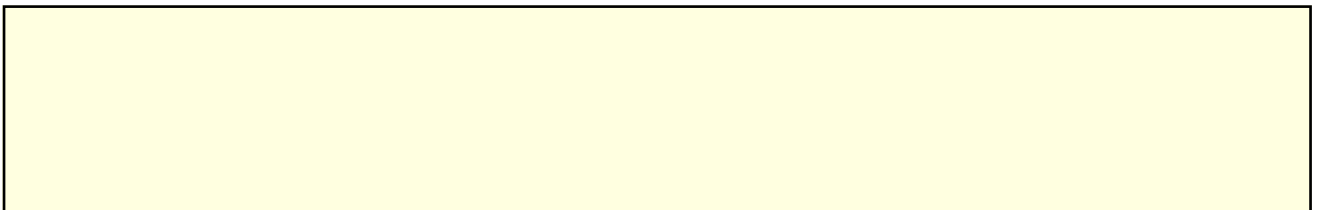
9. Global Unicast address -



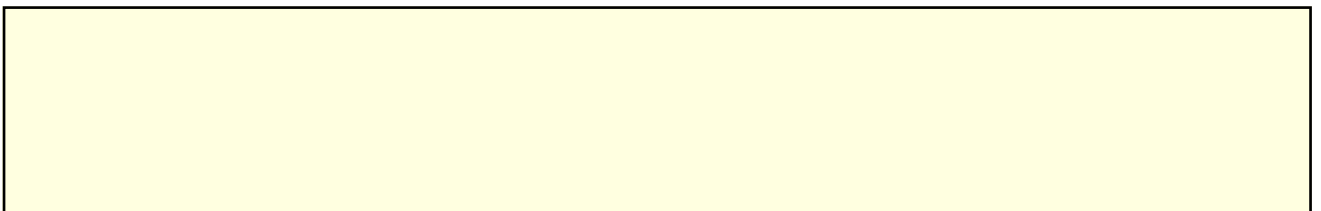
10. Unique Local address -



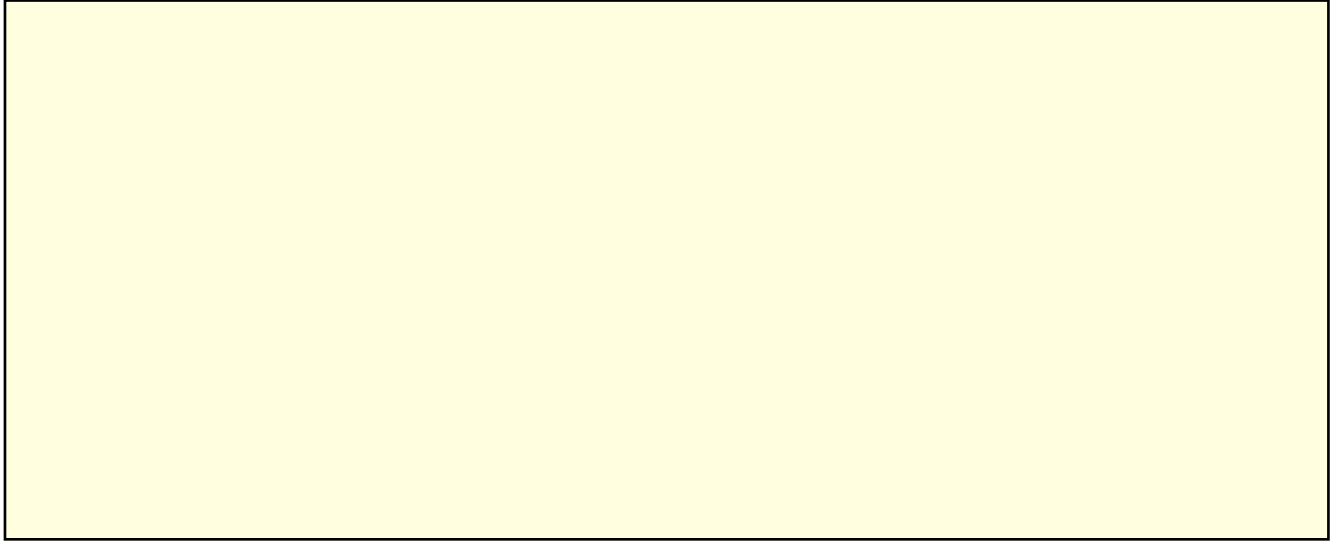
11. Multicast address -



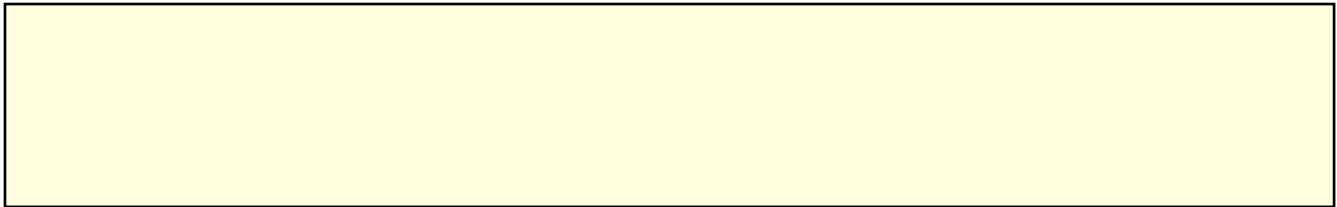
12. Unicast address -



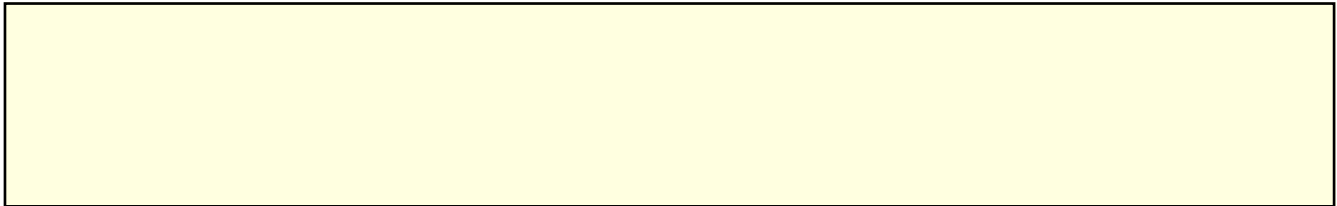
13. Anycast address -



14. Site ID -



15. Sub-Site ID -



16. Prefix Length -



What Type of IPv6 Address is this?

Base on the information supplied on the inside front cover of this workbook identify the following IPv6 addresses as either: Unspecified, Loopback, Embedded IPv4, Global Unicast, Unique local Unicast, Link-local Unicast, or Multicast.

1. 2000:ACAD:1234::/48 Global Unicast
2. 0000:0000:0000:0000:0000:0000:0000:0000 Unspecified
3. FE80:ACAD:1234::/48
4. FDFF:8771:3321::/48
5. FFCD:984:1::/48
6. 3F98::/48
7. ::1
9. 3000::0001/64
10. FEA1:8934:3021:8945:1234:ACAD:FE23:0001/64
11. 00AB:2307:4829::/56
13. FF45:6543:ACAD::/60
14. 2ABC:ACAD:AAAA:0000:0000:0000:0000:00001/64
15. FC12:0000:ACAD:1234:5678:9101:1121:3141/48
16. 2345:FE66:FECD:9999:2365::1/52
17. ::
18. 0000:0000:0000:0000:0000:0000:0000:0001
19. FFF8:0000:00001::0023/64
20. 0023:5935:F441::/48
22. 2001:ABCD:1234:FFFF:ACAD::45/60
23. 3211:FCAB:EEEE::/48
24. FCCC:25:1::/48

Reference Section

IANA IPv6 Global Unicast Address Allocations to the RIRs

Prefix	Designation	Date	Whois	Status
2001:0000::/23	IANA	7/1/1999	whois.iana.org	ALLOCATED
2001:0200::/23	APNIC	7/1/1999	whois.apnic.net	ALLOCATED
2001:0400::/23	ARIN	7/1/1999	whois.arin.net	ALLOCATED
2001:0600::/23	RIPE NCC	7/1/1999	whois.ripe.net	ALLOCATED
2001:0800::/23	RIPE NCC	5/2/2002	whois.ripe.net	ALLOCATED
2001:0a00::/23	RIPE NCC	11/2/2002	whois.ripe.net	ALLOCATED
2001:0c00::/23	APNIC	5/2/2002	whois.apnic.net	ALLOCATED
2001:0e00::/23	APNIC	1/1/2003	whois.apnic.net	ALLOCATED
2001:1200::/23	LACNIC	11/1/2002	whois.lacnic.net	ALLOCATED
2001:1400::/23	RIPE NCC	2/1/2003	whois.ripe.net	ALLOCATED
2001:1600::/23	RIPE NCC	7/1/2003	whois.ripe.net	ALLOCATED
2001:1800::/23	ARIN	4/1/2003	whois.arin.net	ALLOCATED
2001:1a00::/23	RIPE NCC	1/1/2004	whois.ripe.net	ALLOCATED
2001:1c00::/22	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:2000::/20	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:3000::/21	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:3800::/22	RIPE NCC	5/4/2001	whois.ripe.net	ALLOCATED
2001:3c00::/22	IANA			RESERVED
2001:4000::/23	RIPE NCC	6/11/2004	whois.ripe.net	ALLOCATED
2001:4200::/23	AFRINIC	6/1/2004	whois.afrinic.net	ALLOCATED
2001:4400::/23	APNIC	6/11/2004	whois.apnic.net	ALLOCATED
2001:4600::/23	RIPE NCC	8/17/2004	whois.ripe.net	ALLOCATED
2001:4800::/23	ARIN	8/24/2004	whois.arin.net	ALLOCATED
2001:4a00::/23	RIPE NCC	10/15/2004	whois.ripe.net	ALLOCATED
2001:4c00::/23	RIPE NCC	12/17/2004	whois.ripe.net	ALLOCATED
2001:5000::/20	RIPE NCC	9/10/2004	whois.ripe.net	ALLOCATED
2001:8000::/19	APNIC	11/30/2004	whois.apnic.net	ALLOCATED
2001:a000::/20	APNIC	11/30/2004	whois.apnic.net	ALLOCATED
2001:b000::/20	APNIC	3/8/2006	whois.apnic.net	ALLOCATED
2002:0000::/16	6to4	2/1/2001		ALLOCATED
2003:0000::/18	RIPE NCC	1/12/2005	whois.ripe.net	ALLOCATED
2400:0000::/12	APNIC	10/3/2006	whois.apnic.net	ALLOCATED
2600:0000::/12	ARIN	10/3/2006	whois.arin.net	ALLOCATED
2610:0000::/23	ARIN	11/17/2005	whois.arin.net	ALLOCATED
2620:0000::/23	ARIN	9/12/2006	whois.arin.net	ALLOCATED
2800:0000::/12	LACNIC	10/3/2006	whois.lacnic.net	ALLOCATED
2a00:0000::/12	RIPE NCC	10/3/2006	whois.ripe.net	ALLOCATED
2c00:0000::/12	AFRINIC	10/3/2006	whois.afrinic.net	ALLOCATED
2d00:0000::/8	IANA	7/1/1999		RESERVED
2e00:0000::/7	IANA	7/1/1999		RESERVED
3000:0000::/4	IANA	7/1/1999		RESERVED
3ffe::/16	IANA	2008-04		RESERVED
5f00::/8	IANA	2008-04		RESERVED

Prefix-Length	Number of IPs		
/128	1	/68	1,152,921,504,606,840,000
/127	2	/67	2,305,843,009,213,690,000
/126	4	/66	4,611,686,018,427,380,000
/125	8	/65	9,223,372,036,854,770,000
/124	16	/64 - Residential	18,446,744,073,709,500,000
/123	32	/63	36,893,488,147,419,100,000
/122	64	/62	73,786,976,294,838,200,000
/121	128	/61	147,573,952,589,676,000,000
/120	256	/60 - Residential	295,147,905,179,352,000,000
/119	512	/59	590,295,810,358,705,000,000
/118	1,024	/58	1,180,591,620,717,410,000,000
/117	2,048	/57	2,361,183,241,434,820,000,000
/116	4,096	/56 - Medium	4,722,366,482,869,640,000,000
/115	8,192	/55	9,444,732,965,739,290,000,000
/114	16,384	/54	18,889,465,931,478,500,000,000
/113	32,768	/53	37,778,931,862,957,100,000,000
/112	65,536	/52	75,557,863,725,914,300,000,000
/111	131,072	/51	151,115,727,451,828,000,000,000
/110	262,144	/50	302,231,454,903,657,000,000,000
/109	524,288	/49	604,462,909,807,314,000,000,000
/108	1,048,576	/48 - Large	1,208,925,819,614,620,000,000,000
/107	2,097,152	/47	2,417,851,639,229,250,000,000,000
/106	4,194,304	/46	4,835,703,278,458,510,000,000,000
/105	8,388,608	/45	9,671,406,556,917,030,000,000,000
/104	16,777,216	/44	19,342,813,113,834,000,000,000,000
/103	33,554,432	/43	38,685,626,227,668,100,000,000,000
/102	67,108,864	/42	77,371,252,455,336,200,000,000,000
/101	134,217,728	/41	154,742,504,910,672,000,000,000,000
/100	268,435,456	/40	309,485,009,821,345,000,000,000,000
/99	536,870,912	/39	618,970,019,642,690,000,000,000,000
/98	1,073,741,824	/38	1,237,940,039,285,380,000,000,000,000
/97	2,147,483,648	/37	2,475,880,078,570,760,000,000,000,000
/96	4,294,967,296	/36	4,951,760,157,141,520,000,000,000,000
/95	8,589,934,592	/35	9,903,520,314,283,040,000,000,000,000
/94	17,179,869,184	/34	19,807,040,628,566,000,000,000,000,000
/93	34,359,738,368	/33	39,614,081,257,132,100,000,000,000,000
/92	68,719,476,736	/32 - Service LIR	79,228,162,514,264,300,000,000,000,000
/91	137,438,953,472	/31	158,456,325,028,528,000,000,000,000,000
/90	274,877,906,944	/30	316,912,650,057,057,000,000,000,000,000
/89	549,755,813,888	/29	633,825,300,114,114,000,000,000,000,000
/88	1,099,511,627,776	/28	1,267,650,600,228,220,000,000,000,000,000
/87	2,199,023,255,552	/27	2,535,301,200,456,450,000,000,000,000,000
/86	4,398,046,511,104	/26	5,070,602,400,912,910,000,000,000,000,000
/85	8,796,093,022,208	/25	10,141,204,801,825,800,000,000,000,000,000
/84	17,592,186,044,416	/24	20,282,409,603,651,600,000,000,000,000,000
/83	35,184,372,088,832	/23 - ISP	40,564,819,207,303,300,000,000,000,000,000
/82	70,368,744,177,664	/22	81,129,638,414,606,600,000,000,000,000,000
/81	140,737,488,355,328	/21	162,259,276,829,213,000,000,000,000,000,000
/80	281,474,976,710,656	/20	324,518,553,658,426,000,000,000,000,000,000
/79	562,949,953,421,312	/19	649,037,107,316,853,000,000,000,000,000,000
/78	1,125,899,906,842,620	/18	1,298,074,214,633,700,000,000,000,000,000,000
/77	2,251,799,813,685,240	/17	2,596,148,429,267,410,000,000,000,000,000,000
/76	4,503,599,627,370,490	/16	5,192,296,858,534,820,000,000,000,000,000,000
/75	9,007,199,254,740,990	/15	10,384,593,717,069,600,000,000,000,000,000,000
/74	18,014,398,509,481,900	/14	20,769,187,434,139,300,000,000,000,000,000,000
/73	36,028,797,018,963,900	/13	41,538,374,868,278,600,000,000,000,000,000,000
/72	72,057,594,037,927,900	/12	83,076,749,736,557,200,000,000,000,000,000,000
/71	144,115,188,075,855,000	/11	166,153,499,473,114,000,000,000,000,000,000,000
/70	288,230,376,151,711,000	/10	332,306,998,946,228,000,000,000,000,000,000,000
/69	576,460,752,303,423,000	/9	664,613,997,892,457,000,000,000,000,000,000,000
		/8	1,329,227,995,784,910,000,000,000,000,000,000,000

IPv6 Resources

Web Sites:

ARIN IPv6 Wiki

<http://www.getipv6.info/display/IPv6/IPv6+Info+Home>

Cisco Support Community IPv6 Subnetting - Overview and Case Study

<https://supportforums.cisco.com/docs/DOC-17232>

Videos:

IPv6 for CCNAs with Anthony Sequeira

Video Series by the Cisco Learning Network - Parts 1, 2 & 3

<https://learningnetwork.cisco.com/docs/DOC-20357>

PDF Resources:

Preparing An IPv6 Address Plan, Version 2, 18 September 2013

https://www.ripe.net/lir-services/training/material/IPv6-for-LIRs-Training-Course/IPv6_addr_plan4.pdf

Best Current Operational Practices - IPv6 Subnetting (v1)

http://www.ipbcop.org/wp-content/uploads/2012/02/BCOP-IPv6_Subnetting.pdf

6net An IPv6 Deployment Guide by The European 6NET Consortium

<http://www.6net.org/book/deployment-guide.pdf>

IPv6 Addressing At-A-Glance By Cisco

http://www.cisco.com/en/US/technologies/tk648/tk872/technologies_white_paper0900aecd8026003d.pdf

IPv6 Implementation Guide, Cisco IOS Release 15.2M&T

<http://www.cisco.com/en/US/docs/ios-xml/ios/ipv6/configuration/15-2mt/ipv6-15-2mt-book.pdf>

Printed Books:

IPv6 Fundamentals A Straightforward Approach to Understanding IPv6

By Rick Graziani

ISBN-13: 978-1-58714-313-7

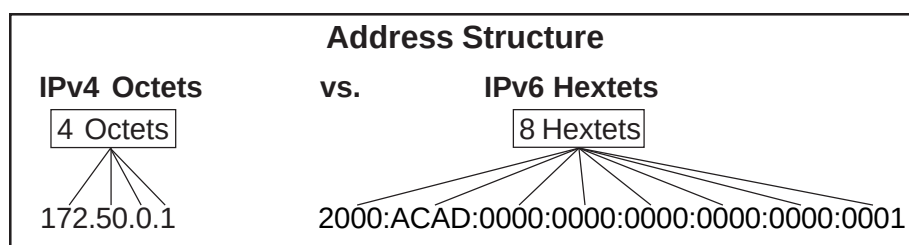
Understanding IPv6 Third Edition

By Joseph Davies

ISBN: 978-0-7356-5914-8

Available subnets within the Subnet ID using the Nibble boundary.

/48 No Nibbles	/52 1 Nibble	/56 2 Nibbles	/60 3 Nibbles	/64 4 Nibbles
/48 - 1 Subnet				
/52 - 16 Subnets	/52 - 1 Subnet			
/56 - 256 Subnets	/56 - 16 Subnets	/56 - 1 Subnets		
/60 - 4096 Subnets	/60 - 256 Subnets	/60 - 16 Subnets	/60 - 1 Subnets	
/64 - 65,536 Subnets	/64 - 4096 Subnets	/64 - 256 Subnets	/64 - 16 Subnets	/64 - 1 Subnet



What is a Site?

A Site = one building
A home, apartment, or house = a Site
A campus with 10 buildings = 10 Sites
A single building with 15 businesses = 15 Sites

Conversion Chart

Decimal (Base 10)	Hexadecimal (Base 16)	Binary (Base 2)
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

VLSM

Variable-Length Subnet Mask

IPv4

Workbook

Version 2.3

192.168.

192.168.10.96

192.168.10.126

172.31.15.0

10.250.1.0

Student Name:

IPv4 Address Classes

Class A	1 – 127	Leading bit pattern	0	00000000.00000000.00000000.00000000 Network . Host . Host . Host
Class B	128 – 191	Leading bit pattern	10	10000000.00000000.00000000.00000000 Network . Network . Host . Host
Class C	192 – 223	Leading bit pattern	110	11000000.00000000.00000000.00000000 Network . Network . Network . Host
Class D	224 – 239	(Reserved for multicast)		
Class E	240 – 255	(Reserved for experimental, used for research)		

Speciality Address Ranges

Loopback -	Only the single 127.0.0.1 address is used, addresses 127.0.0.0 to 127.255.255.255 are reserved. Any address within this block will loop back to the local host.
Link-Local Addresses -	IPv4 addresses in the address block 169.254.0.0 to 169.254.255.255 (169.254.0.0/16) are designated as link-local addresses.
TEST-NET Addresses -	The address block 192.0.2.0 to 192.0.2.255 (192.0.2.0/24) is set aside for teaching and learning purposes.
Experimental Addresses -	The addresses in the block 240.0.0.0 to 255.255.255.254 are listed as reserved for future use (RFC 3330).

Private Address Space

Class A	10.0.0.0 to 10.255.255.255
Class B	172.16.0.0 to 172.31.255.255
Class C	192.168.0.0 to 192.168.255.255

Default Subnet Masks

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

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Special Thanks to Melvin Baker and Jim Dorsch for taking the time to check this workbook for errors, and to everyone who has sent in suggestions to improve the series.

Classful vs. Classless Subnetting

When you're subnetting an IP address for a network you have two options: classful and classless. Classful subnetting is the simplest method. It tends to be the most wasteful because it uses more addresses than are necessary. In classful subnetting you use the same subnet mask for each subnet, and all the subnets have the same number of addresses in them.

Classless addressing allows you to use different subnet masks and create subnets tailored to the number of users in each group. This technique is referred to as VLSM, Variable Length Subnet Masks.

What is VLSM

Variable Length Subnet Masks allow you a much tighter control over your addressing scheme. If you use a class C address with a default subnet mask you end up with one subnet containing 256 addresses. By using VLSM you can adjust the number of subnets and number of addresses depending on the specific needs of your network. The same rules apply to a class A or B addresses.

VLSM is supported by the following protocols: RIP version 2, OSPF, EIGRP, Dual IS-IS, and BGP. You need to configure your router for Variable Length Subnet Masks by setting up one of these protocols. Then configure the subnet masks of the various interfaces in the IP address interface sub-command.

Benefits of VLSM

- Allows efficient use of address space
- Allows the use of multiple subnet mask lengths
- Breaks up an address block into smaller custom blocks
- Allows for route summarization
- Provides more flexibility in network design
- Supports hierarchical enterprise networks

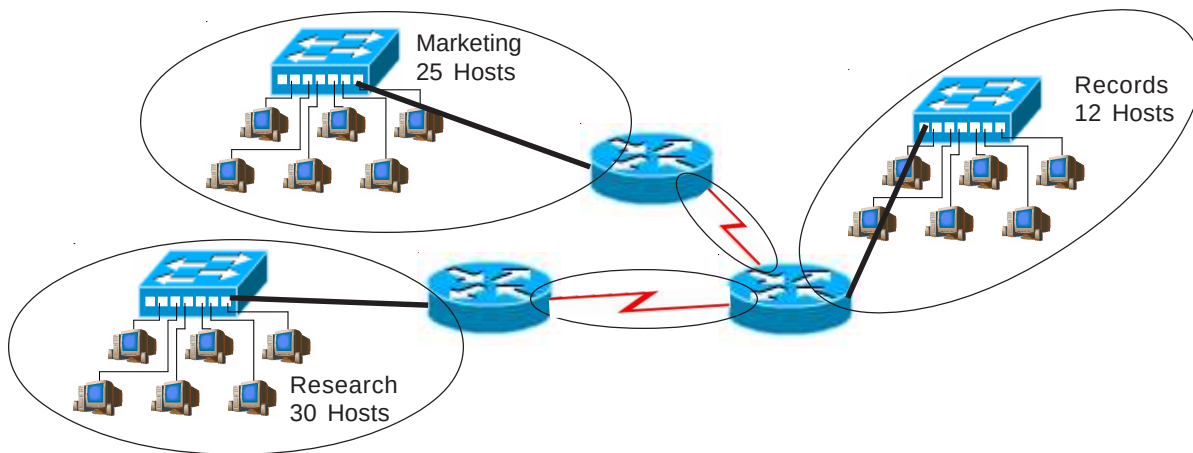
This workbook explores three different methods to figure out sub-subnets: the box method, the circle method, and a VLSM chart.

Classful Subnetting Example

When you're subnetting an IP address for a network you have two options: classful and classless. Classful subnetting is the simplest method. It also tends to be the most wasteful because it uses more addresses than are necessary. In classful subnetting you use the same subnet mask for each subnet, and all the subnets have the same number of addresses in them.

In this example you need five subnets, each one containing 30 hosts. The serial connections only require two address each so you are wasting 28 usable addresses in each of the serial subnet ranges.

IP Address: 192.168.1.0



The Box Method for visualizing subnets

Classful Subnet Ranges

192.168.1.0	to	192.168.1.31	/27
192.168.1.32	to	192.168.1.63	/27
192.168.1.64	to	192.168.1.95	/27
192.168.1.96	to	192.168.1.127	/27
192.168.1.128	to	192.168.1.159	/27
192.168.1.160	to	192.168.1.191	/27
192.168.1.192	to	192.168.1.223	/27
192.168.1.224	to	192.168.1.255	/27

/27
255.255.255.224
32 Hosts
8 Subnets

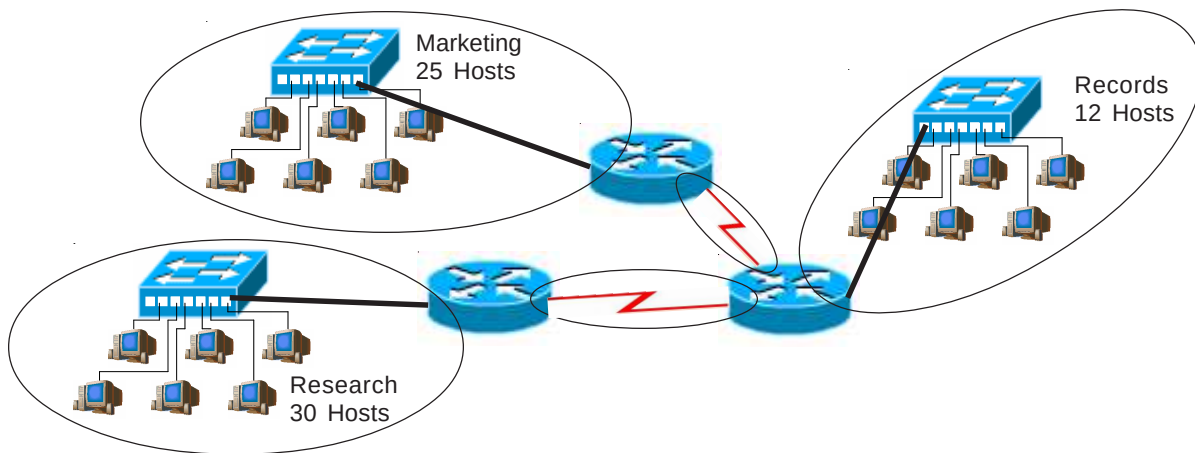
0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

Classless Subnetting Example

Classless addressing allows you to use different subnet masks and create subnets tailored to the number of users in each subnetwork. There are fewer wasted IP addresses using smaller subnets.

In this example you need at total of five subnets, two containing 30 hosts, one containing 12 hosts, and two serial connections that only require two usable addresses each.

IP Address: 192.168.1.0

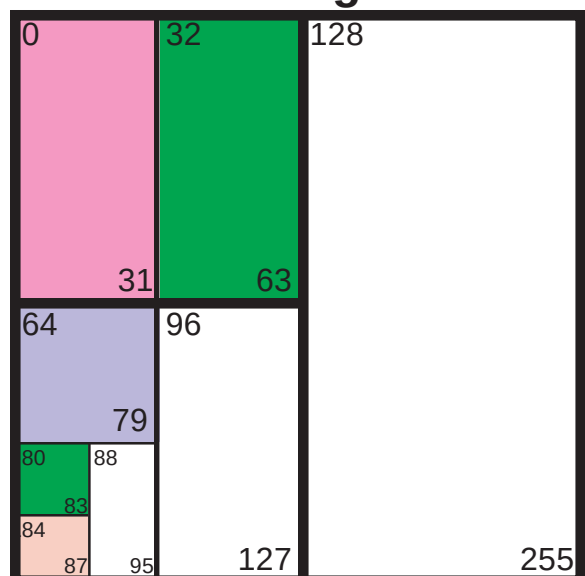


By adjusting the subnet masks you can cut your address usage by almost half in this example. This type of subnetting requires a network protocol which will support it such as: RIP version 2, EIGRP, OSPF, or BGP.

The Box Method for visualizing subnets

Classless Subnet Ranges

192.168.1.0	to	192.168.1.31	/27
192.168.1.32	to	192.168.1.63	/27
192.168.1.64	to	192.168.1.79	/28
192.168.1.80	to	192.168.1.87	/30
192.168.1.88	to	192.168.1.95	/29
192.168.1.96	to	192.168.1.127	/27
192.168.1.128	to	192.168.1.255	/25

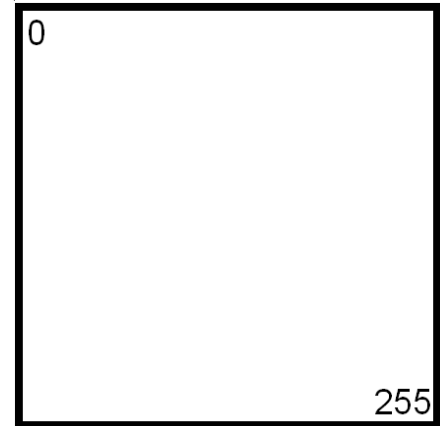


Visualizing Subnets Using The Box Method

The box method is a simple way to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

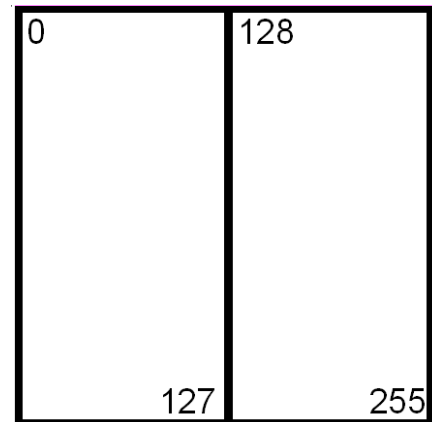
Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



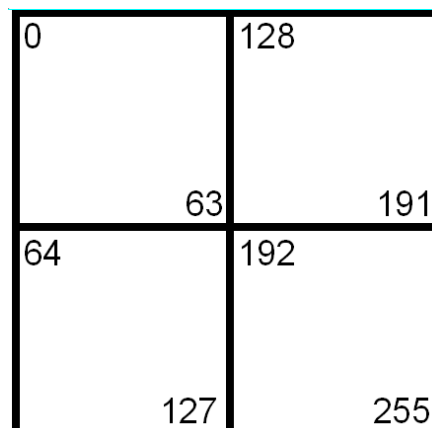
Split the box in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



Divide the box into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets

0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets

0	32	128	160
15	47	143	175
16	48	144	176
31	63	159	191
64	96	192	224
79	111	207	239
80	112	208	240
95	127	223	255

The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets

0	8	32	40	128	136	160	168
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
87	95	119	127	215	223	247	255

The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets

0	8	32	40	128	136	160	168
3	11	35	43	131	139	163	171
4	12	36	44	132	140	164	172
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
19	27	51	59	147	155	179	187
20	28	52	60	148	156	180	188
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
67	75	99	107	195	203	227	235
68	76	100	108	196	204	228	236
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
83	91	115	123	211	219	243	251
84	92	116	124	212	220	244	252
87	95	119	127	215	223	247	255

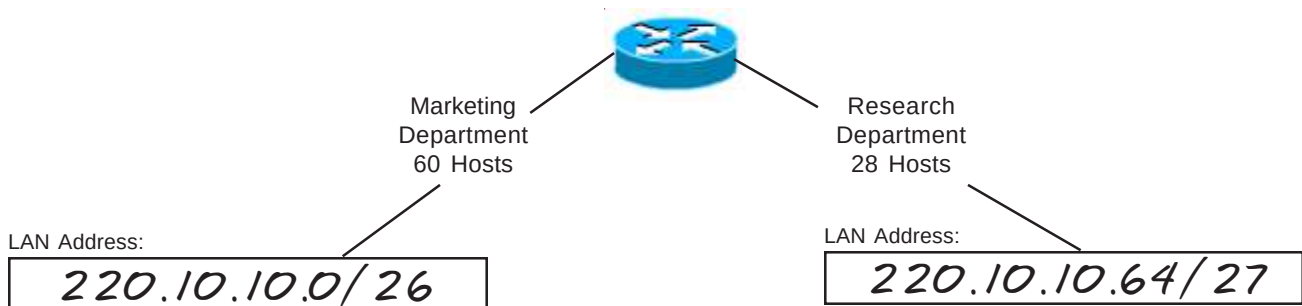
VLSM Addressing

Box Method

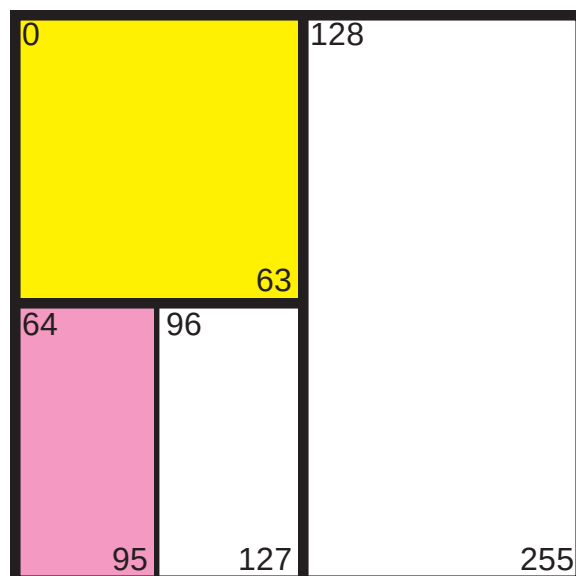
(Sample)

Problem 1

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This business will be using the class C address 220.10.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.



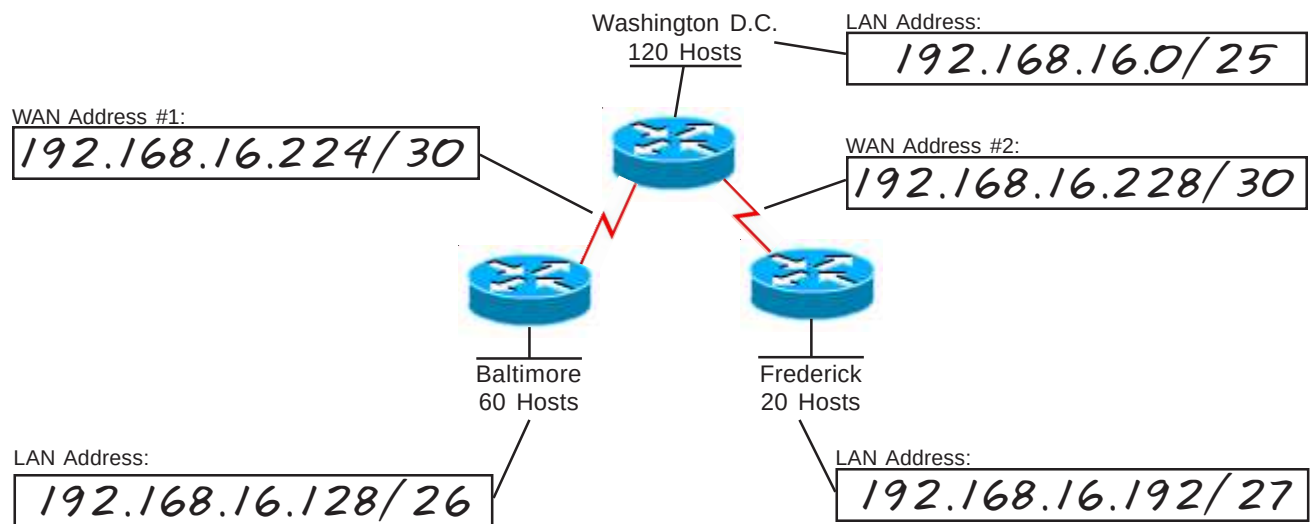
VLSM Addressing

Box Method

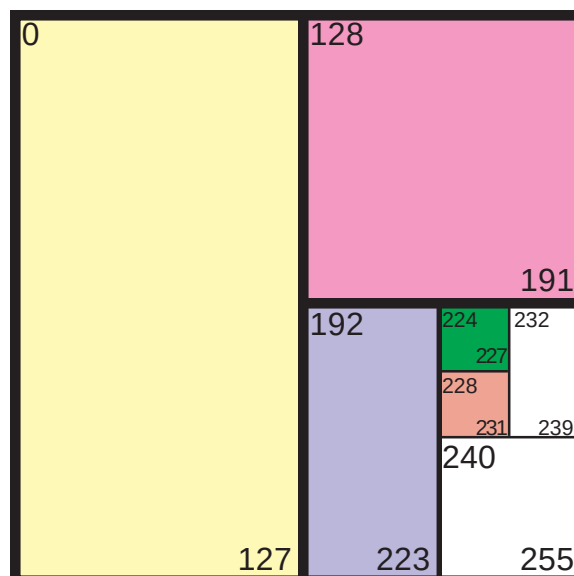
(Sample)

Problem 2

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.16.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

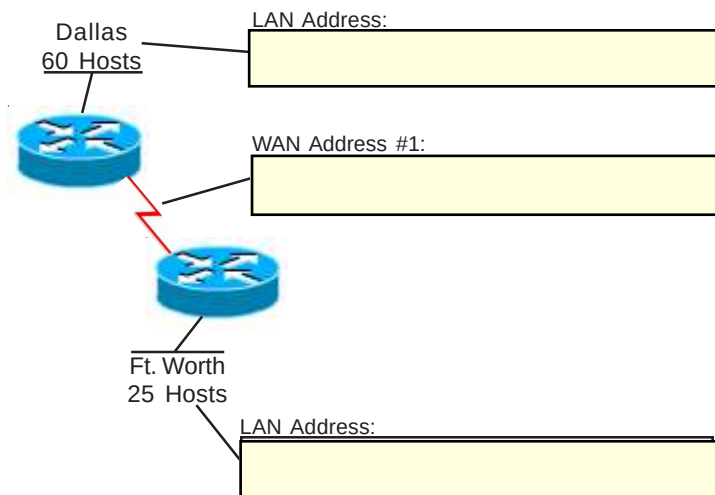


VLSM Addressing

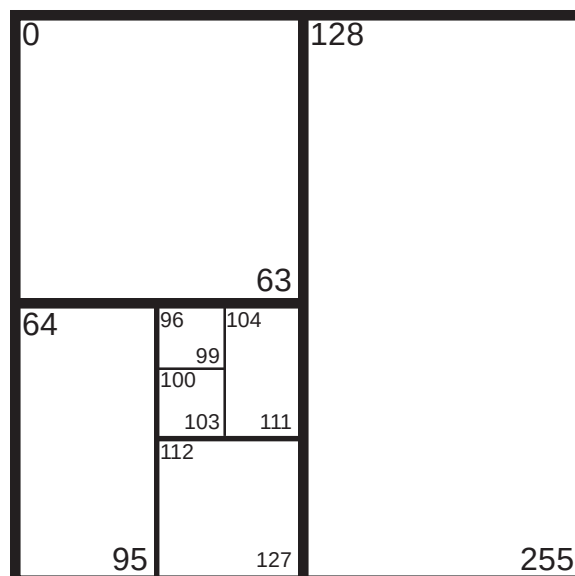
Box Method

Problem 3

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 190.10.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

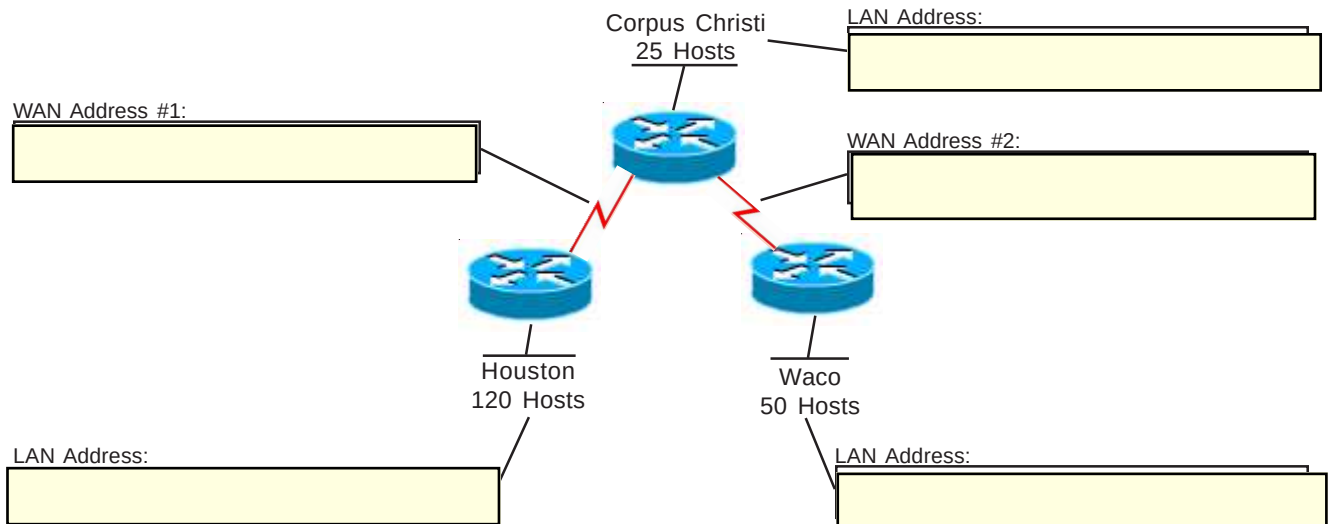


VLSM Addressing

Box Method

Problem 4

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 220.108.38.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

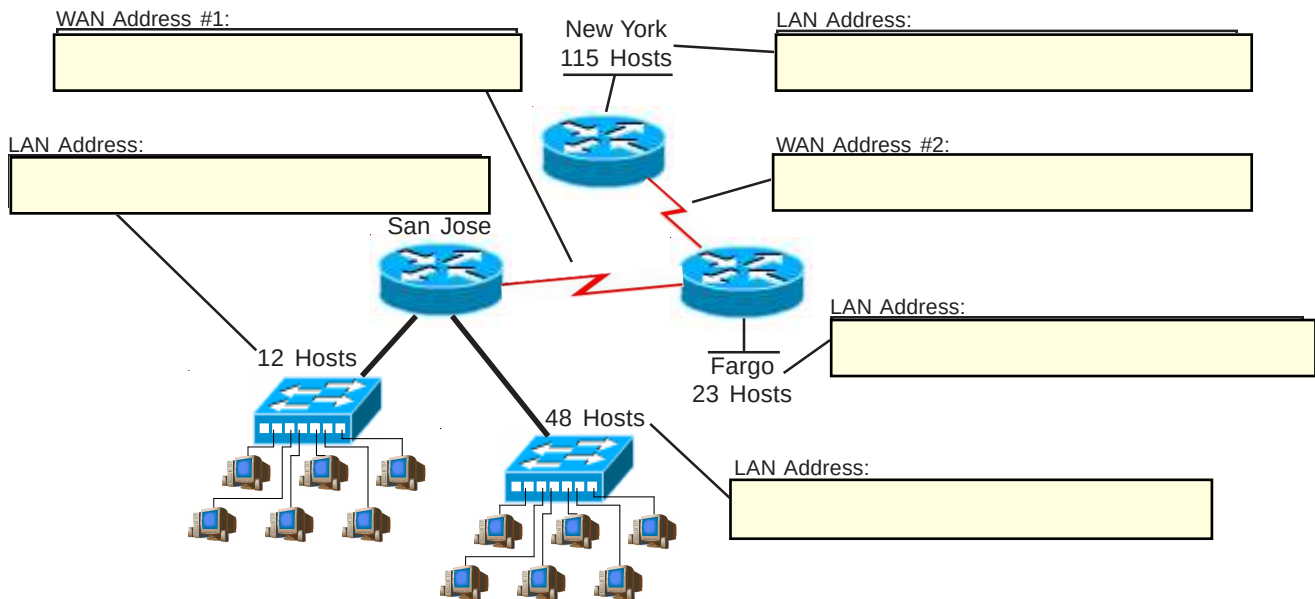
0	8	32	40	128	136	160	168
	3	11	35	43	131	139	163
4	12	36	44	132	140	164	172
	7	15	39	47	135	143	167
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247

VLSM Addressing

Box Method

Problem 5

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

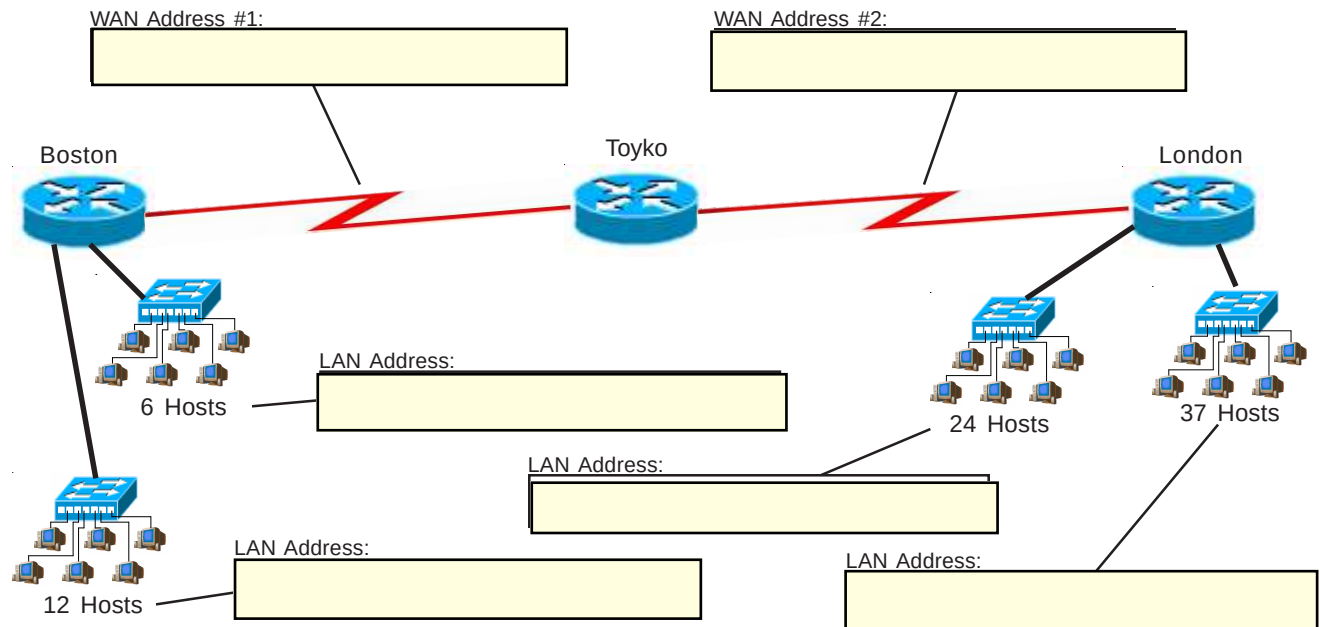
0	8	32	40	128	136	160	168
	3	11	35	43	131	139	171
4	12	36	44	132	140	164	172
	7	15	39	47	135	143	175
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247

VLSM Addressing

Box Method

Problem 6

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 222.10.150.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

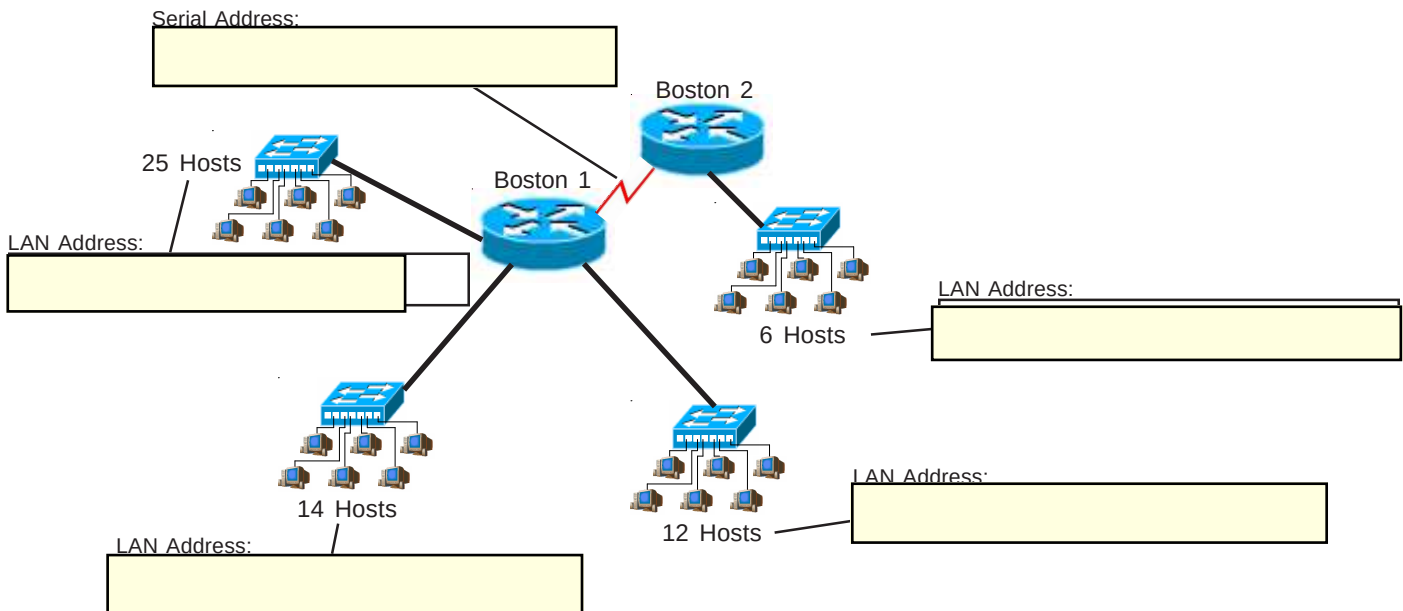
0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

VLSM Addressing

Box Method

Problem 7

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and subnet mask in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 200.150.70.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

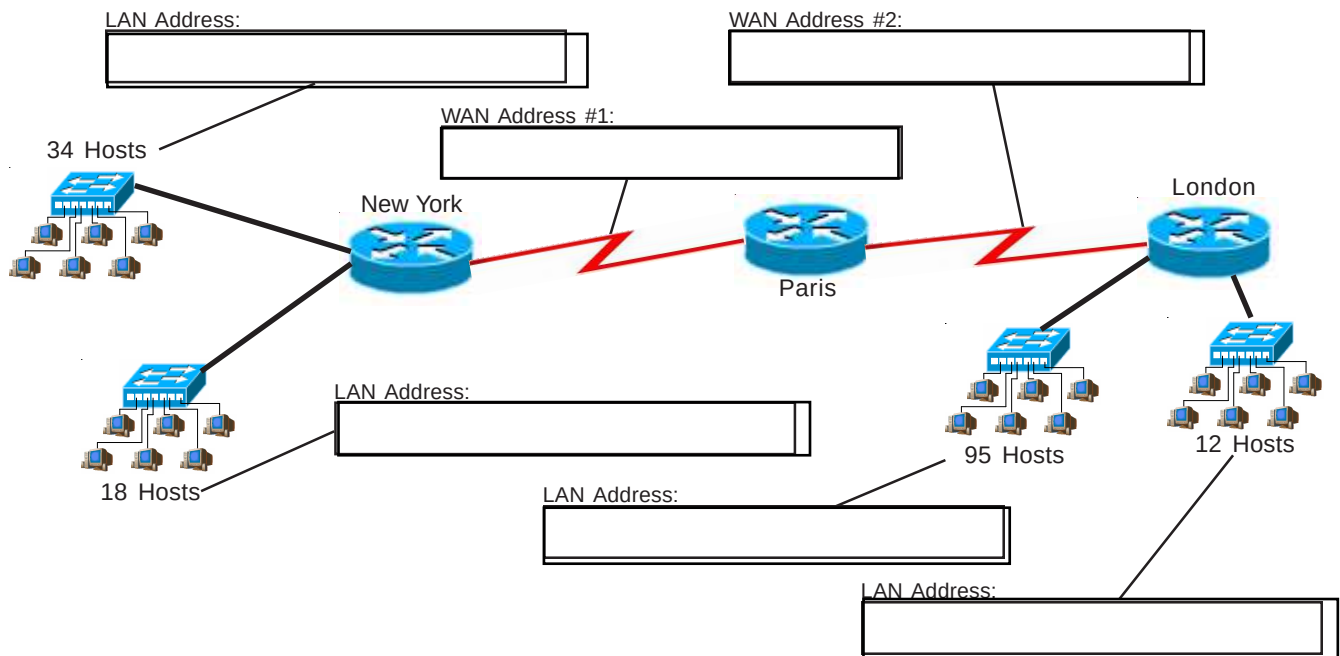
0	128
63	191
64	192
127	255

VLSM Addressing

Box Method

Problem 8

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and subnet mask in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.24.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

0
255

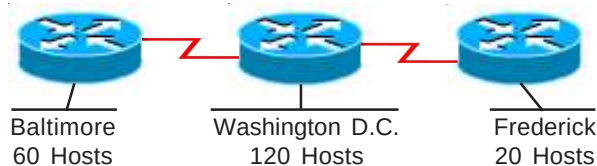
Configuring Subnets Using The Numeric Method

The Numeric method extends the technique from the IP Addressing and Subnetting workbook to allow for classless subnetting. Start by listing the needed hosts per subnets from largest to smallest.

Step 1

In this sample the needed host groups are:

- 120 Washington D. C.
- 60 Baltimore
- 20 Frederick
- 2 Serial #1
- 2 Serial #2



Step 2

Lay out the IP address the same way you would in the IP Addressing and Subnetting workbook showing the binary values of the bits, the number of subnets, and the number of hosts.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
192.10.10.0	0	0	0	0	0	0	0	0	0	

Step 3

Draw a line allowing for 120 hosts. Using the value of the last bit borrowed or magic number, write in your first range. Then add the subnet address to start the second range.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
192.10.10.0	0	0	0	0	0	0	0	0	0	

	Address Ranges		
Hosts	192.168.16.0	to	192.168.16.127
120	192.168.16.128	to	
			CIDR /25

Draw a second line for the 60 hosts. Use the value of the last bit borrowed or magic number to lay out the second largest subnet. Add the subnet address which will begin the third range.

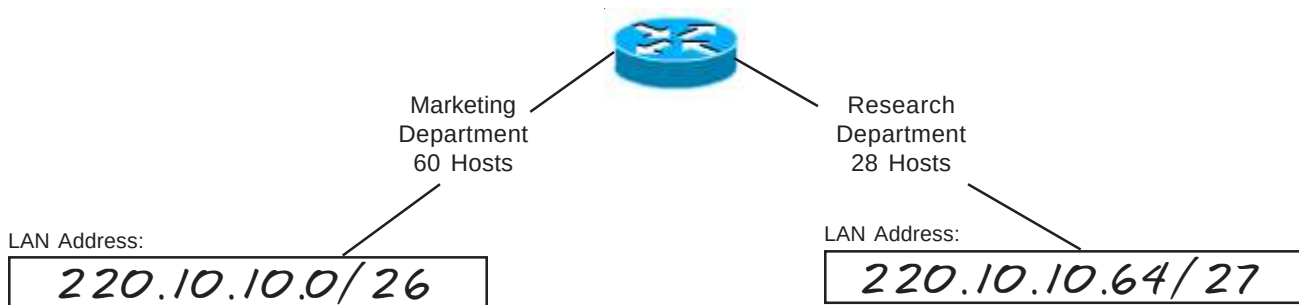
Step 5

Continue the same process until all the subnets are accounted for. The CIDR is the total number of bits borrowed for the network and subnetwork portion of the address.

17

(Sample)

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 220.10.10.0. Start with your largest groups first.

[illegible]

<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>
60	220.10.10.0 to 220.10.10.63	/26
28	220.10.10.64 to 220.10.10.95	/27

VLSM Addressing

Numeric Method

Problem 11

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 200.20.20.0. Start with your largest groups first.

LAN Address:

Marketing
Department
100 Hosts

Marketing
Department
50 Hosts



Department
25 Hosts

LAN Address:

LAN Address:

Show your work for Problem 11 in the space below.

	100 Hosts	50 Hosts	25 Hosts									Number of Hosts
Number of Subnets	2	4	8	16	32	64	128	256				
	128	64	32	16	8	4	2	1				Binary values
200.20.20.0	0	0	0	0	0	0	0	0				

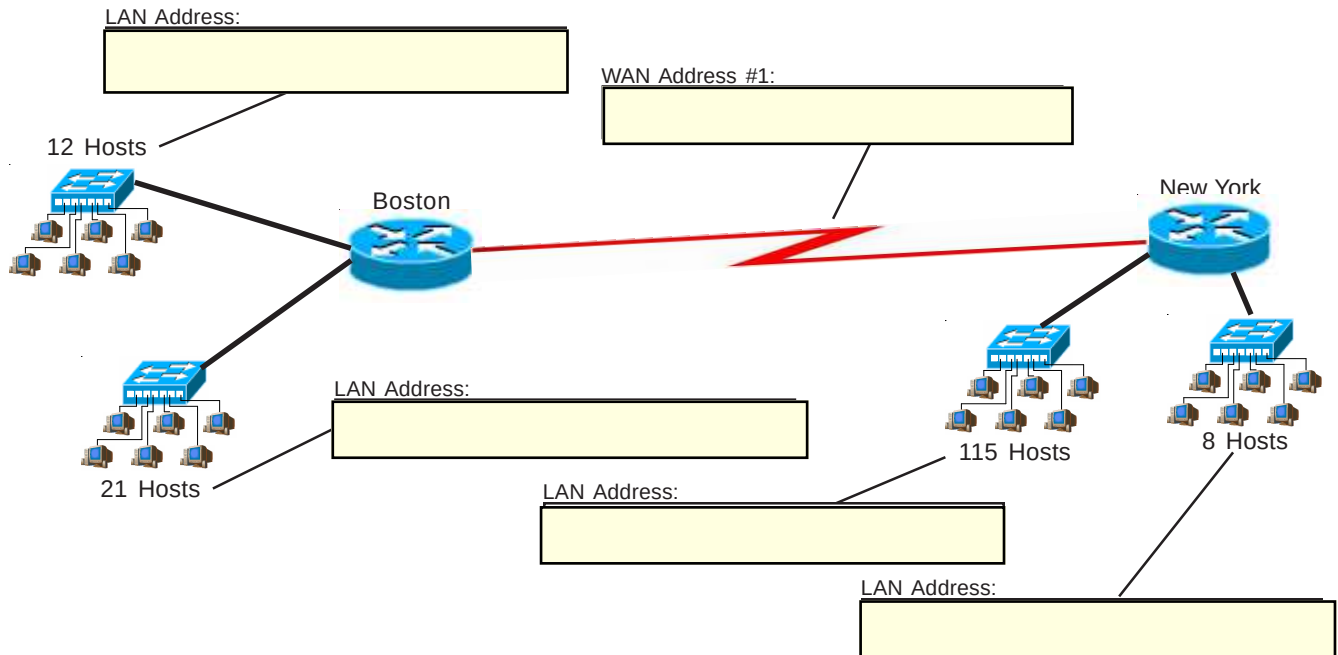
<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>
100		
50		
25		

VLSM Addressing

Numeric Method

Problem 12

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 199.55.70.0. Start with your largest groups first.



Show your work for Problem 12 in the space below.

	<i>Hosts</i>		<i>Hosts</i>		<i>Hosts</i>		<i>Hosts</i>				
	256	128	64	32	16	8	4	2	-	Number of	
Number of Subnets	- 2	4	8	16	32	64	128	256		Hosts	
	128	64	32	16	8	4	2	1	-	Binary values	
199.55.70.0	0	0	0	0	0	0	0	0			

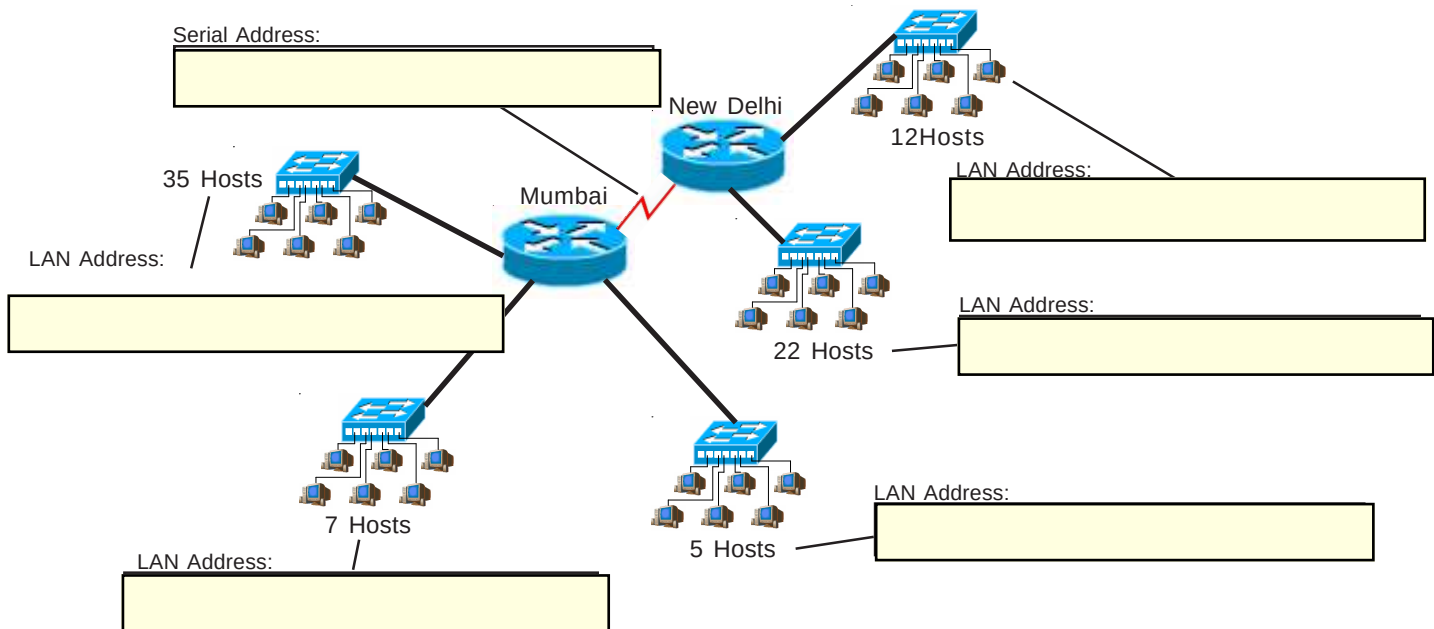
<i>Hosts</i>	<i>Address Ranges</i>	<i>CIDR</i>
115		
21		
12		
8		
2		

VLSM Addressing

Numeric Method

Problem 13

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 200.150.70.0. Start with your largest groups first.



Show your work for Problem 13 in the space below.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets	-	2	4	8	16	32	64	128	256	
	128	64	32	16	8	4	2	1	-	Binary values
200.150.70.0	0	0	0	0	0	0	0	0	0	

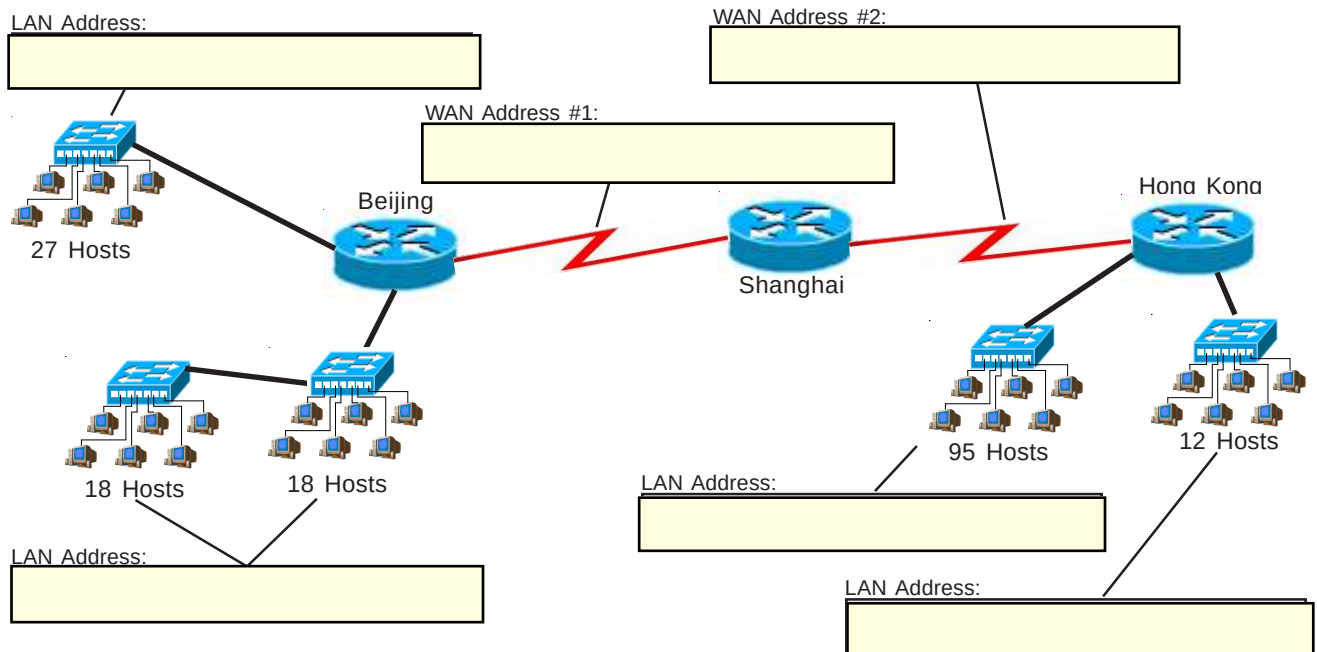
Hosts	Address Ranges	CIDR

VLSM Addressing

Numeric Method

Problem 14

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 210.10.10.0. Start with your largest groups first.



Show your work for Problem 14 in the space below.

210.10.10.0 0 0 0 0 0 0 0 0

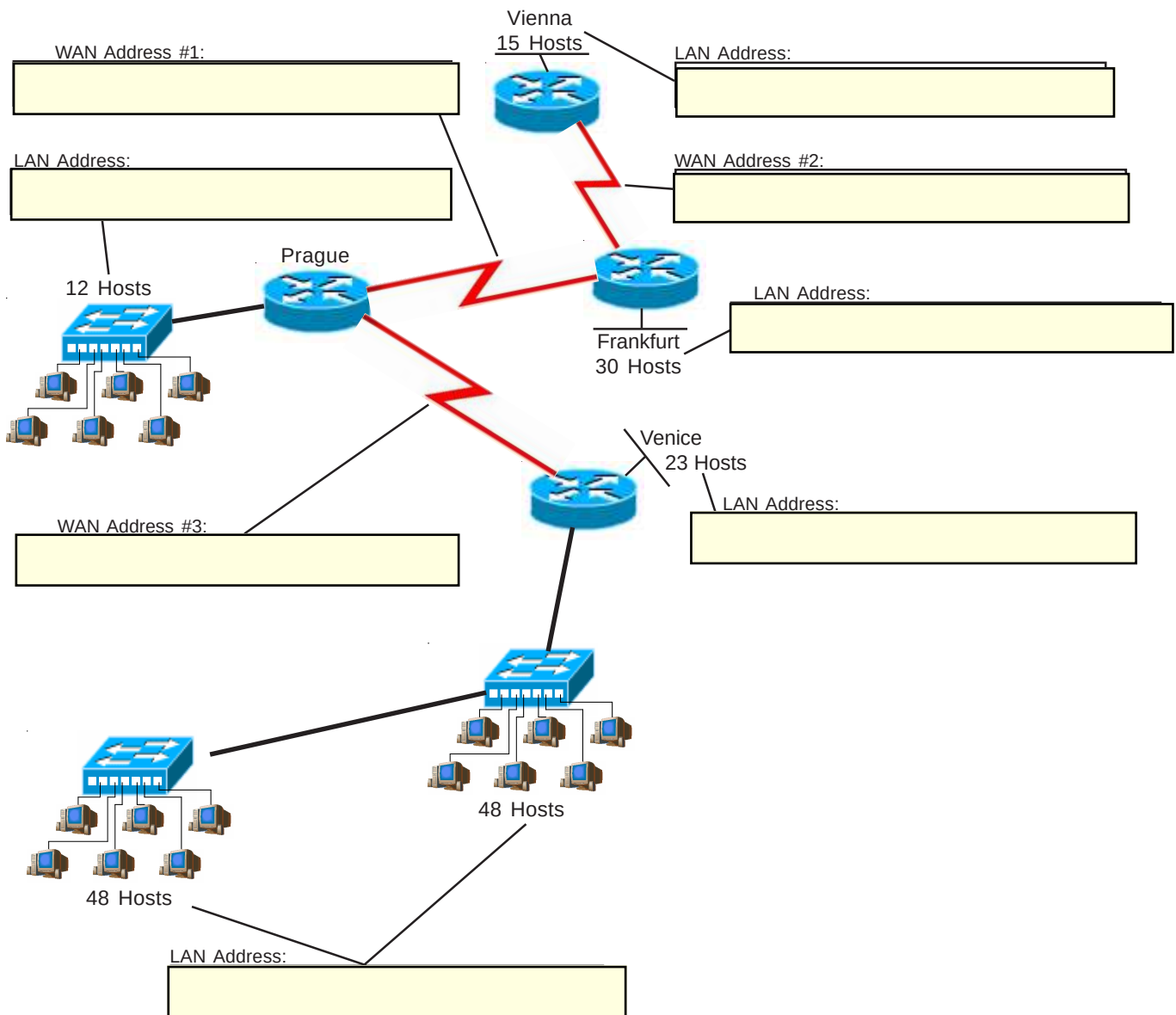
<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>

VLSM Addressing

Numeric Method

Problem 15

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 192.168.150.0. Start with your largest groups first.

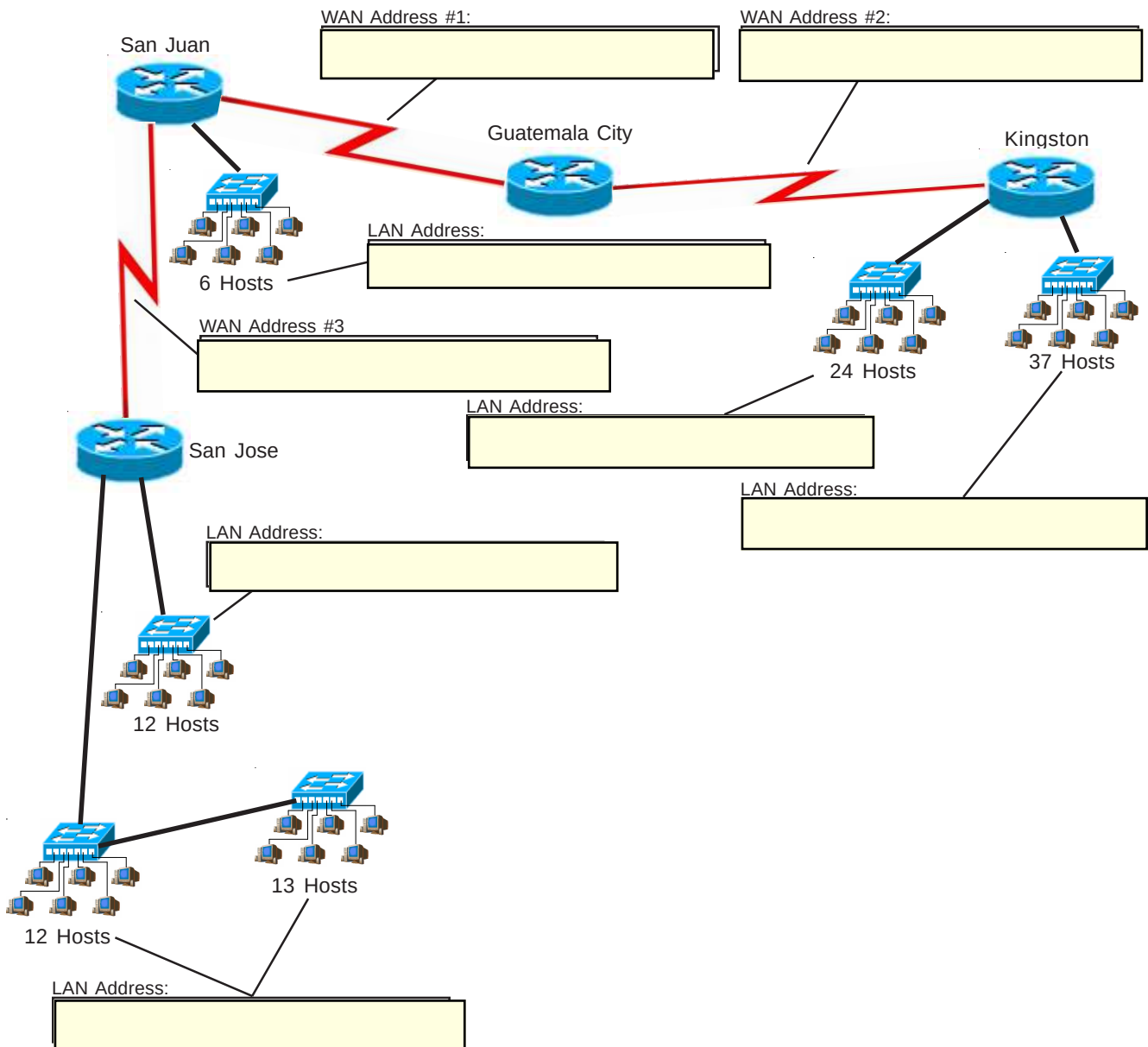


VLSM Addressing

Numeric Method

Problem 16

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Layout and show the subnet address and CIDR in the space below. This business will be using the class C address 195.75.23.0. Start with your largest groups first.



Visualizing Subnets Using a VLSM Chart

The VLSM chart is the third method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You can adjust each sub-subnet to the correct size needed.

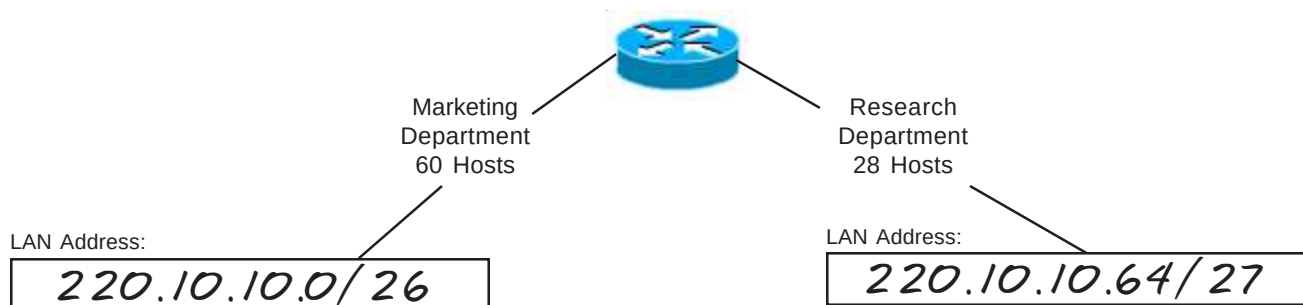
VLSM Addressing

VLSM Chart Method

(Sample)

Problem 17

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This business will be using the class C address 220.10.10.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

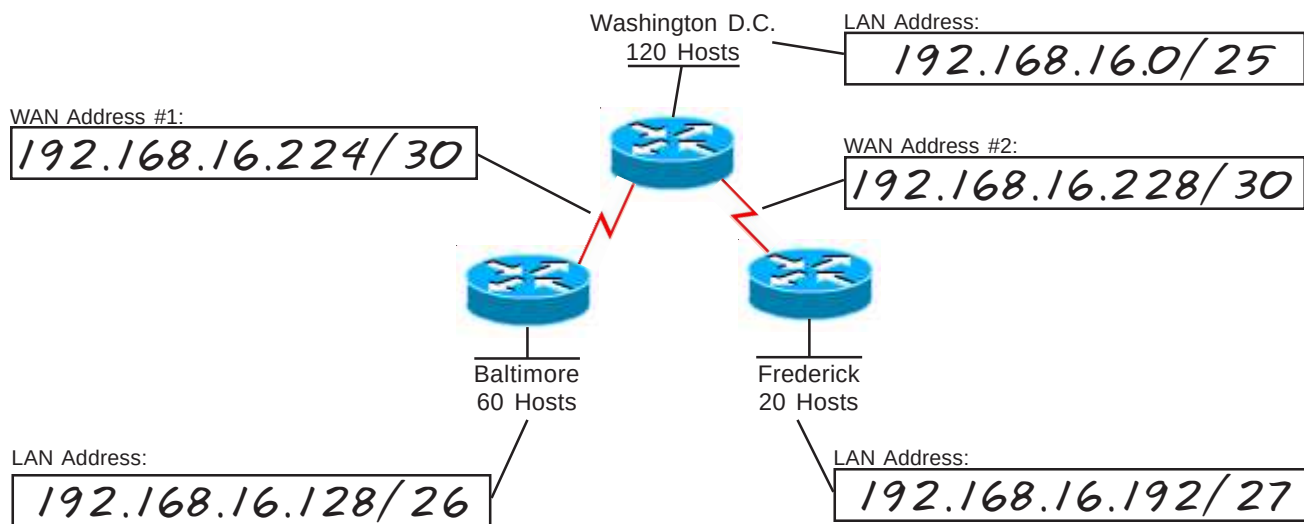
VLSM Addressing

VLSM Chart Method

(Sample)

Problem 18

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 192.168.16.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

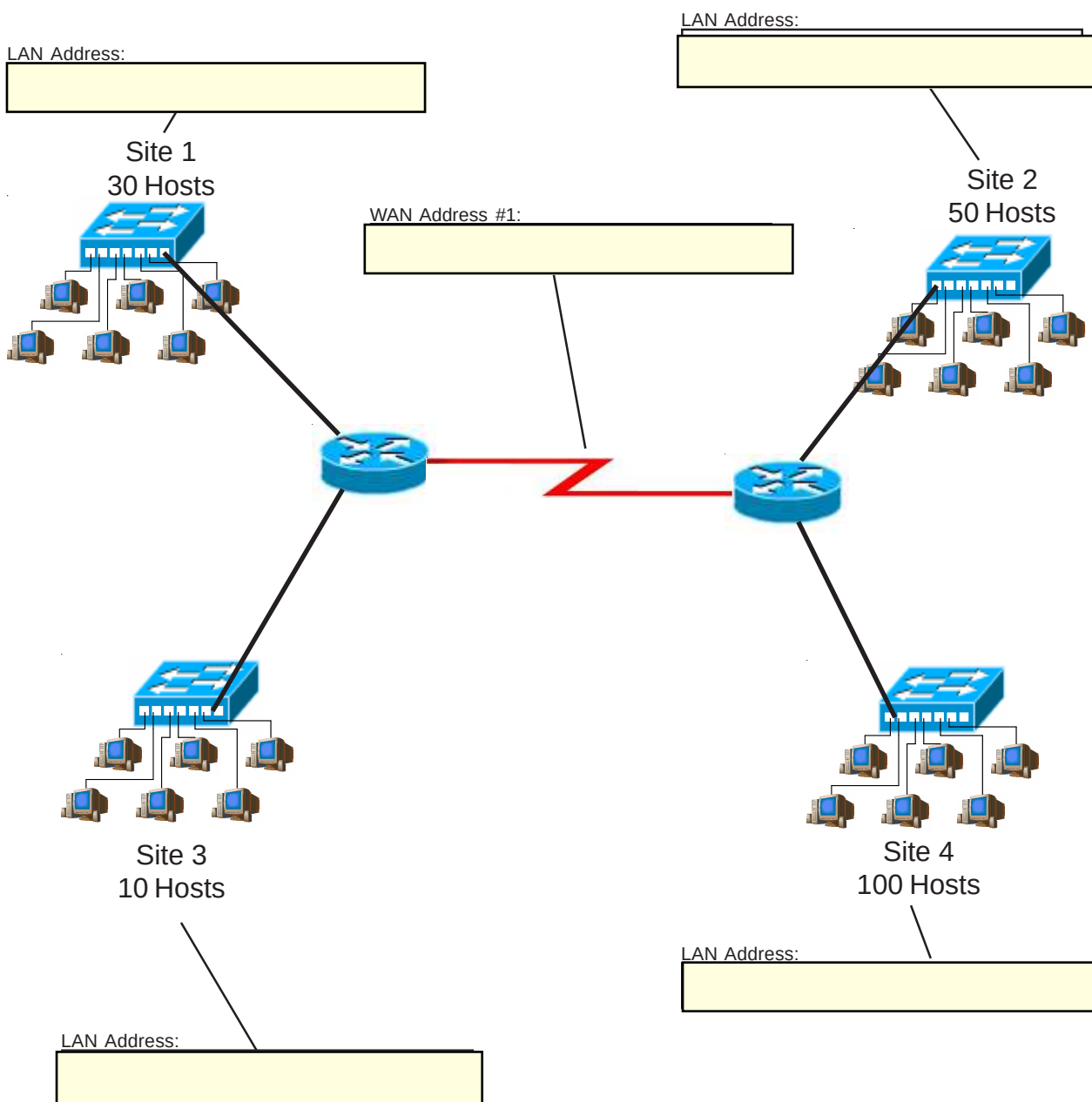
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
		32-63	32-47	32-39	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
		160-191	160-175	160-175	160-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 19

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 199.55.78.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

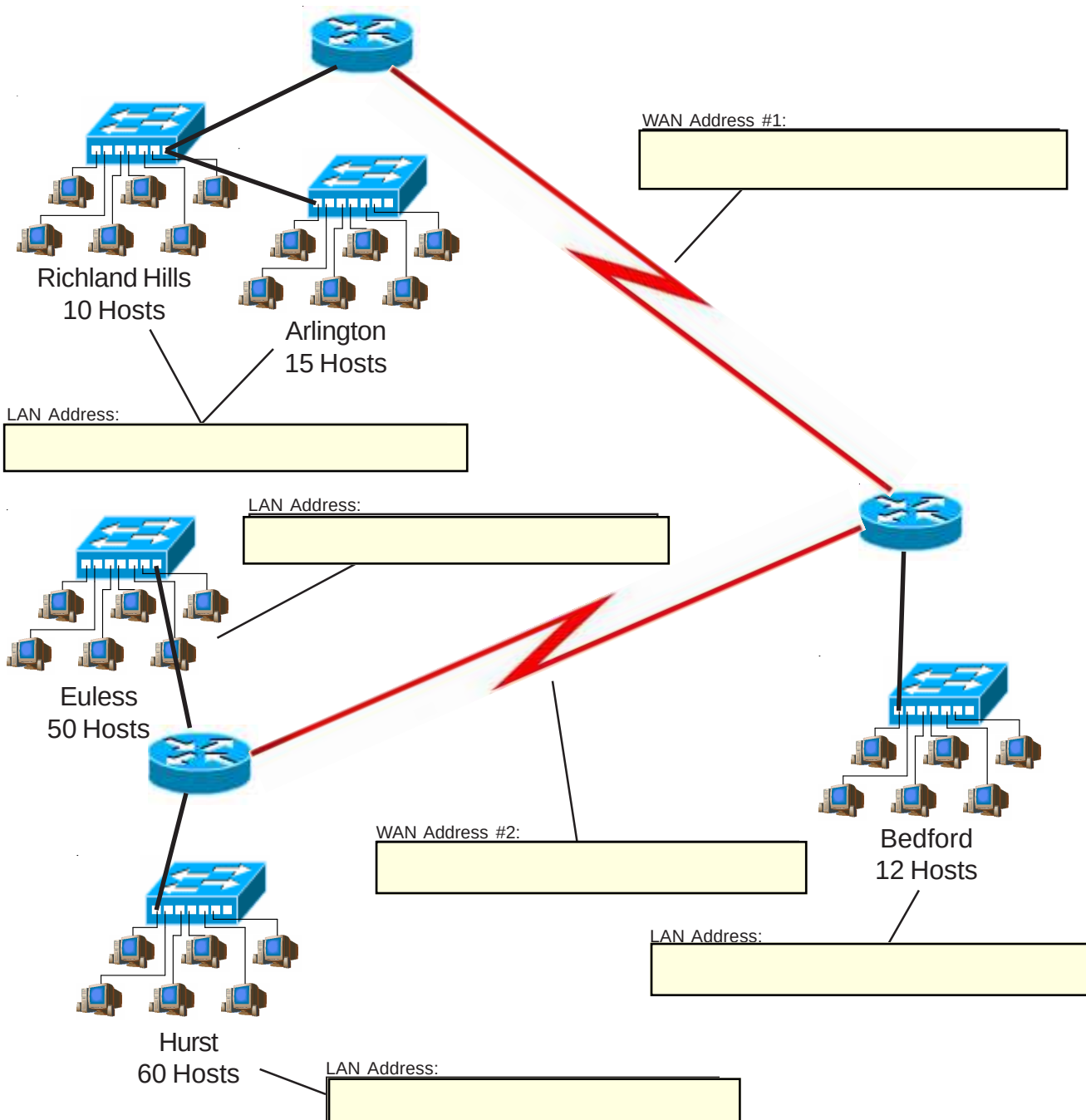
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 20

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 223.150.50.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

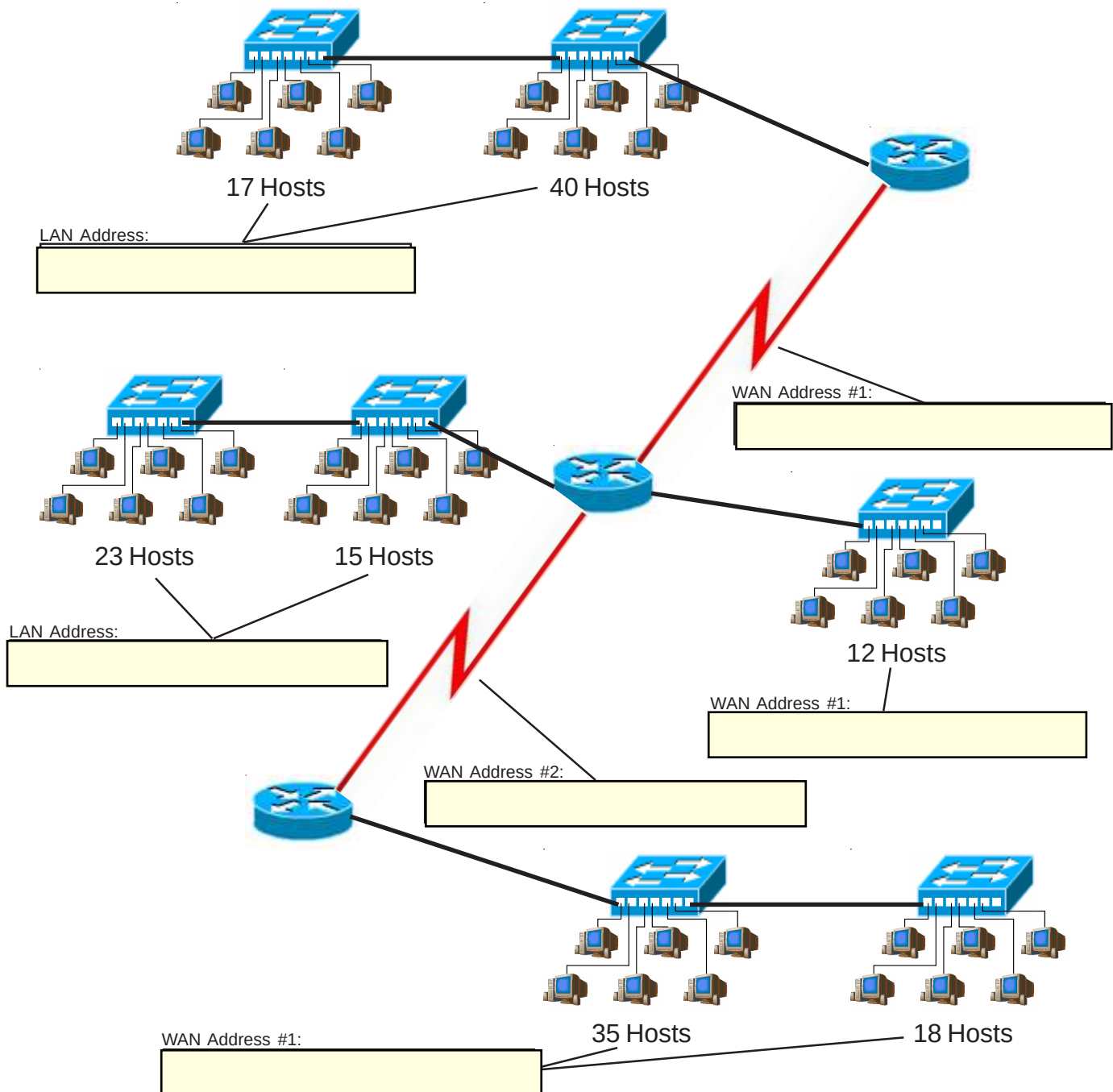
/24	/25	/26	/27	/28	/29	/30					
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts					
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7 8-11 12-15 16-19 20-23 24-27 28-31					
					16-31	8-15	32-35 36-39 40-43 44-47				
						32-47	16-23	48-51 52-55 56-59 60-63			
							48-63	24-31	64-67 68-71 72-75 76-79		
				64-127				64-95	64-79	80-83 84-87 88-91 92-95	
					80-95					96-99 100-103 104-107 108-111	
						96-127			96-111	112-115 116-119 120-123 124-127	
					112-127		128-131 132-135 136-139 140-143 144-147 148-151 152-155 156-159				
			128-255				128-191	128-159	128-143	160-163 164-167 168-171 172-175	
					144-159					176-179 180-183 184-187 188-191	
						160-191				160-175	192-195 196-199 200-203 204-207
											176-191
				192-255					192-223		
					240-247 244-247 248-251 252-255						
										224-255	
					248-255						

VLSM Addressing

VLSM Chart Method

Problem 21

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 222.22.2.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

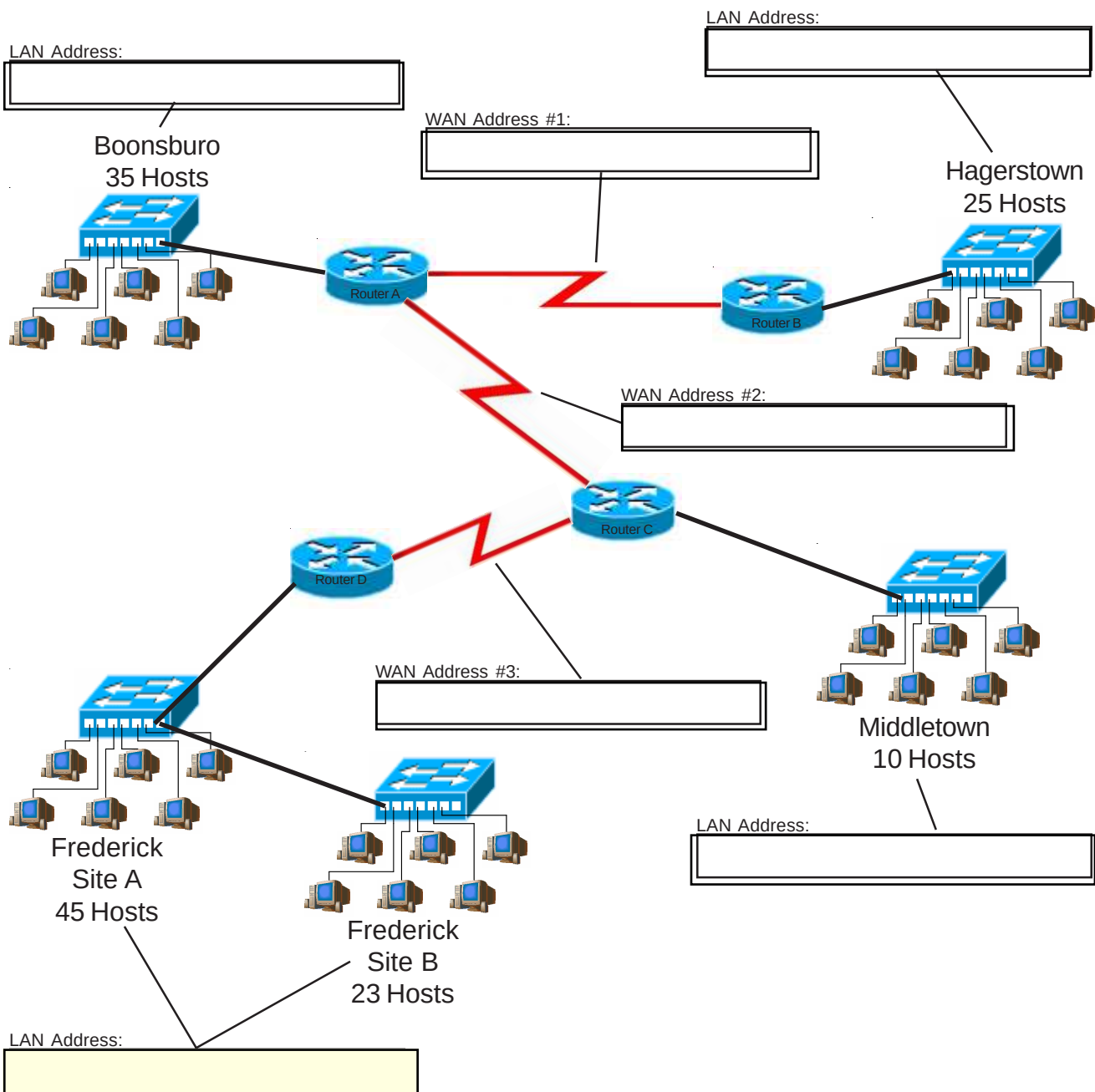
/24	/25	/26	/27	/28	/29	/30						
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts						
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7 8-11 12-15 16-19 20-23 24-27 28-31						
					16-31	8-15	32-35 36-39 40-43 44-47					
						32-47	16-23	48-51 52-55 56-59 60-63				
							48-63	24-31	64-67 68-71 72-75 76-79			
				64-127				64-95	64-79	80-83 84-87 88-91 92-95		
					80-95					96-99 100-103 104-107 108-111		
						96-127			96-111	112-115 116-119 120-123 124-127		
					112-127		128-131 132-135 136-139 140-143 144-147 148-151 152-155 156-159					
			128-255				128-191	128-159	128-143	160-163 164-167 168-171 172-175 176-179 180-183 184-187 188-191		
					144-159					192-195 196-199 200-203 204-207 208-211 212-215 216-219 220-223 224-227		
						160-191				160-175	228-231 232-235 236-239 240-243 244-247 248-251 252-255	
											176-191	240-247 248-255
				192-255					192-223			192-207
					208-223							
										224-255		224-239
					240-255						240-247 248-255	

VLSM Addressing

VLSM Chart Method

Problem 22

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 200.20.2.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

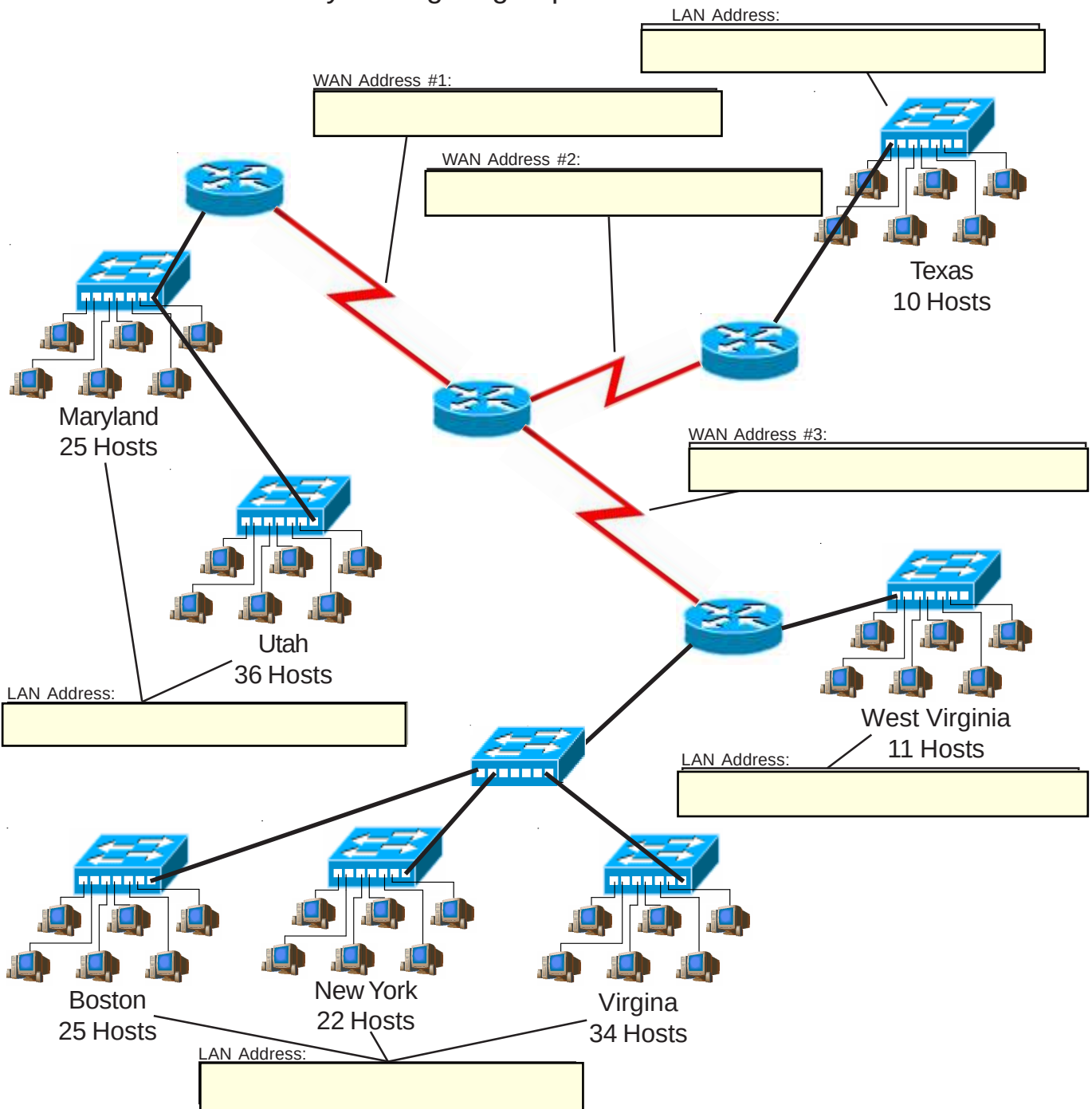
/24	/25	/26	/27	/28	/29	/30					
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts					
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7 8-11 12-15 16-19 20-23 24-27 28-31					
					16-31	8-15	32-35 36-39 40-43 44-47				
						32-47	16-23	48-51 52-55 56-59 60-63			
							48-63	24-31	64-67 68-71 72-75 76-79		
				64-127				64-95	64-79	80-83 84-87 88-91 92-95	
					80-95					96-99 100-103 104-107 108-111	
						96-127			96-111	112-115 116-119 120-123 124-127	
					112-127		128-131 132-135 136-139 140-143 144-147 148-151 152-155 156-159				
			128-255				128-191	128-159	128-143	160-163 164-167 168-171 172-175 176-179 180-183 184-187 188-191	
					144-159					192-195 196-199 200-203 204-207 208-211 212-215 216-219 220-223	
						160-191				160-175	224-227 228-231 232-235 236-239 240-243 244-247 248-251 252-255
											176-191
				192-255					192-223	192-207	
					208-223						240-247 248-251 252-255
						224-255				224-239	240-247 248-251 252-255
					240-255						240-247 248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 23

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 190.150.23.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

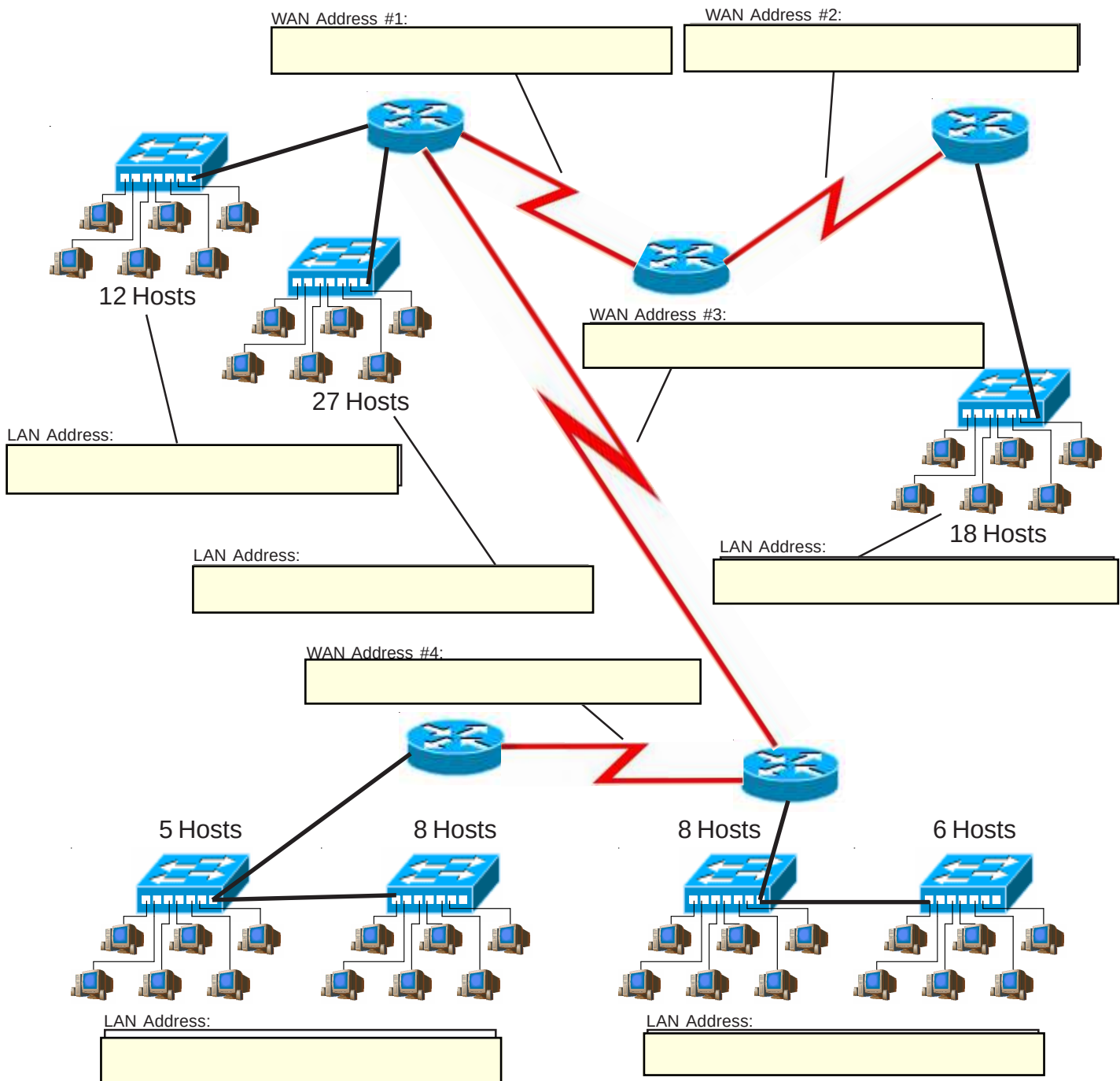
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
					4-7	4-7
				16-31	8-15	8-11
					12-15	12-15
		32-63	32-47	16-23	16-19	16-19
					20-23	20-23
				24-31	24-27	24-27
					28-31	28-31
			48-63	32-39	32-35	32-35
					36-39	36-39
				40-47	40-43	40-43
					44-47	44-47
		64-127	64-95	48-55	48-51	48-51
					52-55	52-55
				56-63	56-59	56-59
					60-63	60-63
				64-71	64-67	64-67
					68-71	68-71
				72-79	72-75	72-75
					76-79	76-79
			80-95	80-87	80-83	80-83
					84-87	84-87
				88-95	88-91	88-91
					92-95	92-95
		96-127	96-111	96-103	96-99	96-99
					100-103	100-103
				104-111	104-107	104-107
					108-111	108-111
			112-127	112-119	112-115	112-115
					116-119	116-119
				120-127	120-123	120-123
					124-127	124-127
	128-255	128-191	128-159	128-143	128-135	128-131
					132-135	132-135
				136-143	136-139	136-139
					140-143	140-143
			144-159	144-151	144-147	144-147
					148-151	148-151
				152-159	152-155	152-155
					156-159	156-159
			160-191	160-175	160-163	160-163
					164-167	164-167
				168-175	168-171	168-171
					172-175	172-175
		192-255	160-191	176-183	176-179	176-179
					180-183	180-183
				184-191	184-187	184-187
					188-191	188-191
			192-223	192-207	192-195	192-195
					196-199	196-199
				200-207	200-203	200-203
					204-207	204-207
			208-223	208-215	208-211	208-211
					212-215	212-215
				216-223	216-219	216-219
					220-223	220-223
		224-255	224-239	224-231	224-227	224-227
					228-231	228-231
				232-239	232-235	232-235
					236-239	236-239
			240-255	240-247	240-243	240-243
					244-247	244-247
				248-255	248-251	248-251
					252-255	252-255

VLSM Addressing

VLSM Chart Method

Problem 24

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 192.168.1.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

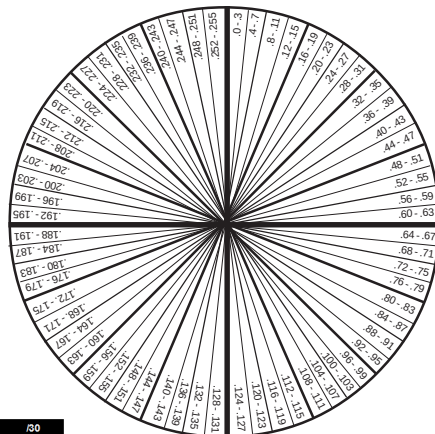
/24		/25		/26		/29		/30	
255.255.255.0 256 Hosts		255.255.255.128 128 Hosts		255.255.255.192 64 Hosts		255.255.255.248 8 Hosts		255.255.255.252 4 Hosts	
0 - 255	0-127	0-63	0-31	0-15	0-7		0-3		
					4-7		4-7		
					8-15		8-11		
					12-15		12-15		
				16-31	16-23		16-19		
					20-23		20-23		
					24-31		24-27		
					28-31		28-31		
			32-63	32-47	32-39		32-35		
					36-39		36-39		
					40-47		40-43		
					44-47		44-47		
				48-63	48-55		48-51		
					52-55		52-55		
					56-63		56-59		
					60-63		60-63		
		64-127	64-95	64-79	64-71		64-67		
					68-71		68-71		
					72-79		72-75		
					76-79		76-79		
				80-95	80-87		80-83		
					84-87		84-87		
					88-95		88-91		
					92-95		92-95		
			96-127	96-111	96-103		96-99		
					100-103		100-103		
					104-111		104-107		
					108-111		108-111		
				112-127	112-119		112-115		
					116-119		116-119		
					120-123		120-123		
					124-127		124-127		
	128-255	128-191	128-159	128-143	128-135		128-131		
					132-135		132-135		
					136-143		136-139		
					140-143		140-143		
				144-159	144-151		144-147		
					148-151		148-151		
					152-155		152-155		
					156-159		156-159		
			160-191	160-175	16-167		160-163		
					164-167		164-167		
					168-175		168-171		
					172-175		172-175		
				176-191	176-183		176-179		
					180-183		180-183		
					184-191		184-187		
					188-191		188-191		
		192-255	192-223	192-207	192-199		192-195		
					196-199		196-199		
					200-207		200-203		
					204-207		204-207		
				208-223	208-215		208-211		
					212-215		212-215		
					216-223		216-219		
					220-223		220-223		
			224-255	224-239	224-231		224-227		
					228-231		228-231		
					232-235		232-235		
					236-239		236-239		
				240-255	240-247		240-243		
					244-247		244-247		
					248-251		248-251		
					252-255		252-255		

Practical VLSM Problems

Use the VLSM method of your choice to complete the following problems.

(Reference pages are in the back of the workbook.)

0	8	32	40	128	136	160	168
	3	11	35	43	131	139	163
4	12	36	44		132	140	164
	7	15	39	47	135	143	167
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247
							255



VLSM Chart 24-30 Bits (4th octet)

/24 256 Hosts	/25 128 Hosts	/26 64 Hosts	/27 32 Hosts	/28 16 Hosts	/29 8 Hosts	/30 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
				16-31	8-15	4-7
				32-47	16-23	8-15
			32-63	48-63	24-31	16-23
				64-79	32-39	24-31
				80-95	40-47	32-39
		64-127	64-95	96-111	48-55	40-47
				112-127	56-63	48-55
				128-143	64-71	56-63
			96-127	144-159	72-79	64-71
				160-175	80-87	72-79
				176-191	88-95	80-87
	128-255	128-191	128-159	192-207	96-103	88-95
				208-223	104-111	96-103
				224-239	112-119	104-111
			160-191	240-255	120-127	112-119
					128-135	120-127
					136-143	128-135
		192-255	192-223	224-239	144-151	136-143
				240-255	152-159	144-151
					160-167	152-159
			224-255		168-175	160-167
					176-183	168-175
					184-191	176-183
					192-199	184-191

VLSM Addressing

(Sample)

Problem 25

You are developing a school network with the class C address 192.168.2.0/24. There will be three computer labs with 30 computers each that need to be on different sub-subnets. Forty eight classrooms with one computer each that will comprise a single sub-subnet. The administrative office and guidance office contain a total of seven computers which will need to be grouped together. Plan for four more mini labs with six computers to each sub-subnetwork. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.2.0	/26	192.168.2.1	192.168.2.62	192.168.2.63
2	192.168.2.64	/27	192.168.2.65	192.168.2.94	192.168.2.95
3	192.168.2.96	/27	192.168.2.97	192.168.2.126	192.168.2.127
4	192.168.2.128	/27	192.168.2.129	192.168.2.158	192.168.2.159
5	192.168.2.160	/28	192.168.2.161	192.168.2.174	192.168.2.175
6	192.168.2.176	/29	192.168.2.177	192.168.2.182	192.168.2.183
7	192.168.2.184	/29	192.168.2.185	192.168.2.190	192.168.2.191
8	192.168.2.192	/29	192.168.2.193	192.168.2.198	192.168.2.199
9	192.168.2.200	/29	192.168.2.201	192.168.2.206	192.168.2.207
10					
11					
12					
13					
14					

VLSM Addressing

(Sample)

Problem 26

You are setting up a small business network with the class C address 220.55.80.0/24. The marketing division will need 12 computers. Research and development needs 27 computers. The reception area will need two computers. Management requires 19 computers. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	220.55.80.0	/27	220.55.80.1	220.55.80.30	220.55.80.31
2	220.55.80.32	/27	220.55.80.	220.55.80.62	220.55.80.63
3	220.55.80.64	/28	220.55.80.65	220.55.80.78	220.55.80.79
4	220.55.80.80	/30	220.55.80.81	220.55.80.82	220.55.80.83
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 27

You are setting up a medium sized network with the class C address 222.37.34.0/24. Marketing needs 29 computers. Research and development needs 110 computers. Bookkeeping will use 12 computers. The reception area will need three computers. Management requires 60 computers. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/x)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 28

A shipping company needs to set up its network across several locations. The Denver office needs six computers. The Waco office needs 22 computers. The Fargo office will need five computers. The WAN links between all three locations need to be included in the solution. Using the IP address 192.168.10.0/24 divide the network using VLSM. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/x)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 29

A new school is being built in the local school district. It will have three computer labs with 28 computers each. There will be 58 classrooms with 2 computers each that need to be on one sub-subnet. The office staff and administrators will need 7 computers. The guidance and attendance office will have 5 computers. The school has been given the address 223.145.75.0/24. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/x)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 30

A local college is setting up a campus wide network. The technology wing will be on its own network address of 192.168.250.0/24. The office wing will include 15 computers. There are 2 labs of 20 computers each, 2 labs of 30 computers each and one lab of 35 computers. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/x)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 31

You are setting up a network for a company in four locations. Location A has 8 computers. Location B has 122 computers. Location C has 4 computers. Location D has 55 computers. There is a WAN connection between all four locations. Complete the information required below using the class C address 192.168.10.0. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/x)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 32

A college dormitory is being remodeled. A new network is being installed. There are 50 dorm rooms with two drops each that will be on one sub-subnet. The offices will have 5 drops. The reception desk will have three drops. A small study hall will include 30 drops. Using the IP address 192.168.12.0/24 complete the information required below using VLSM. Work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/x)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 33

You are setting up a business network with the class C address 219.75.160.0/24. The marketing division will need 19 computers. Research and development needs 40 computers. The reception area will need four computers. Management requires 12 computers. Divide the network using variable length subnet information. On the opposite page draw a detailed map of this network. Include the name and sub-subnet IP addresses for each branch of the network with the subnet mask. One router with four ethernet ports will be used for this network.

Subnet	Subnet Address	Subnet Mask (/x)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

Problem 33 Detailed Map

Draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network, and the subnet mask.



VLSM Addressing

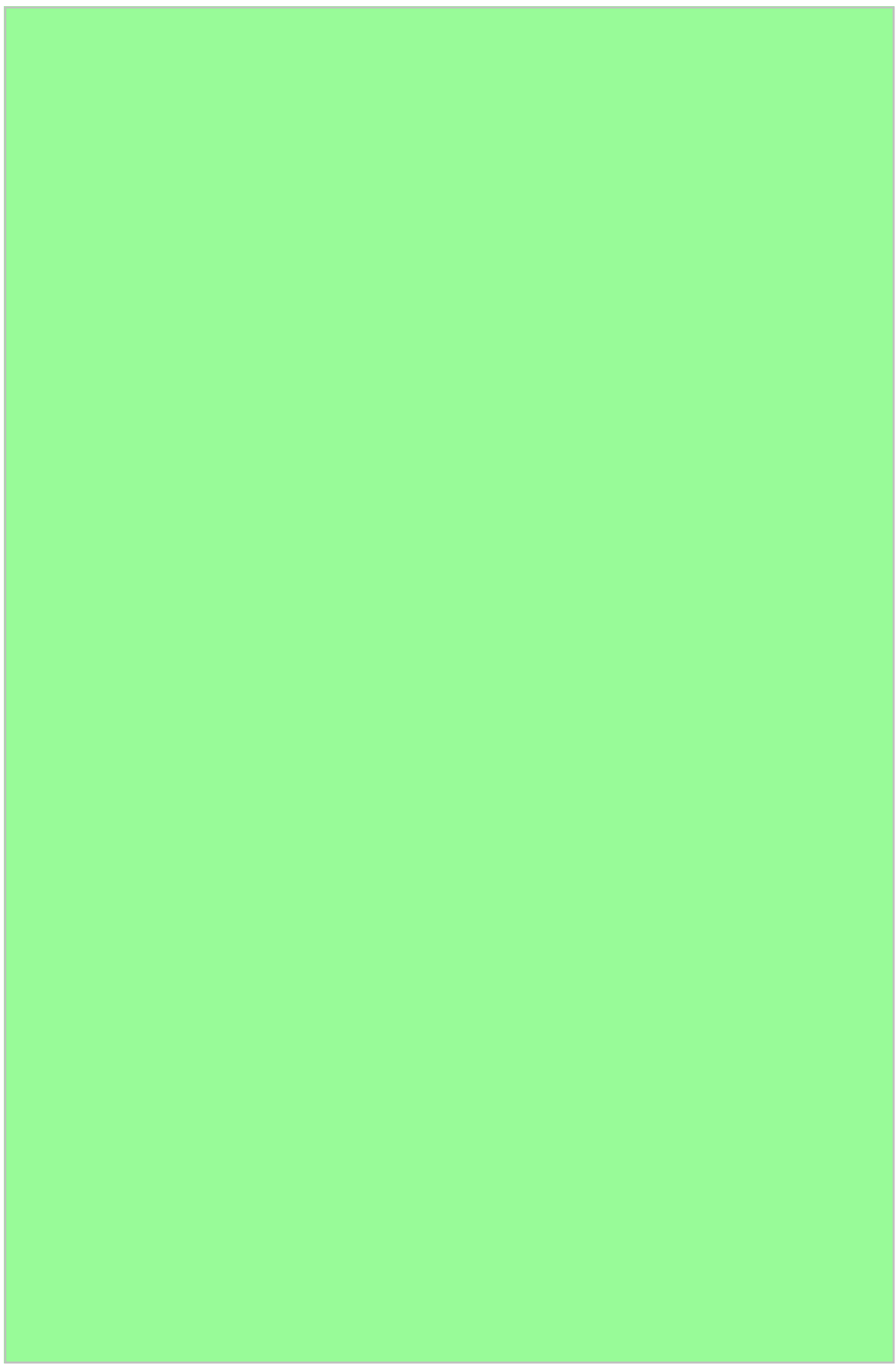
Problem 34

A small company needs to set up its network across several locations. The New York branch office needs 15 computers. The San Jose office needs 66 computers. The Trinidad office will need 18 computers. You will need two WAN links between the routers. Using the IP address 195.20.5.0/24 divide the network using VLSM. On the opposite page draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network. Label the WAN links with the same information. Complete the information required below. Work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/x)	First Usable Host	Last Usable Host	Broadcast Address
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

Problem 34 - Detailed Map

Draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network.



Class A and B VLSM Problems

10.0.0.0

Class A Addresses
VLSM Chart 8-15 Bits (2nd octet)

10	11	12	13	14	15
255.255.0.0 16,777,216 hosts	255.255.0.0 65,536 hosts	255.255.0.0 1,638,400 hosts	255.255.0.0 409,600 hosts	255.255.0.0 102,400 hosts	255.255.0.0 25,600 hosts
0-255	0-127	0-63	0-31	0-15	0-7
				16-31	8-15
			32-47	48-63	24-31
				64-79	32-47
		64-127	64-95	96-111	48-63
				112-127	64-79
			128-159	160-175	80-95
				176-191	96-111
	128-255	128-191	128-159	160-175	160-175
				176-191	176-191
			192-223	224-239	192-207
				240-255	208-223
		192-255	224-239	240-255	224-239
				256-271	240-255
			272-303	304-319	272-287
				320-335	288-303

Class B Addresses
VLSM Chart 16-23 Bits (3rd octet)

16	17	18	19	20	21	22	23
255.255.0.0 65,536 hosts	255.255.0.0 16,384 hosts	255.255.0.0 4,096 hosts	255.255.0.0 1,024 hosts	255.255.0.0 256 hosts	255.255.0.0 64 hosts	255.255.0.0 16 hosts	255.255.0.0 4 hosts
0-255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
				16-31	8-15	4-7	2-3
			32-47	48-63	24-31	12-15	6-7
				64-79	32-47	16-23	8-11
		64-127	64-95	96-111	48-63	24-31	12-15
				112-127	64-79	32-47	16-23
			128-159	160-175	80-95	40-55	20-35
				176-191	96-111	48-63	24-31
	128-255	128-191	128-159	160-175	160-175	160-175	160-175
				176-191	176-191	176-191	176-191
			192-223	224-239	224-239	224-239	224-239
				240-255	240-255	240-255	240-255
		192-255	224-239	240-255	240-255	240-255	240-255
				256-271	256-271	256-271	256-271
			272-303	304-319	304-319	304-319	304-319
				320-335	320-335	320-335	320-335

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

24	25	26	27	28	29	30
255.255.255.0 256 hosts	255.255.255.0 64 hosts	255.255.255.0 16 hosts	255.255.255.0 4 hosts	255.255.255.0 2 hosts	255.255.255.0 1 host	255.255.255.0 1 host
0-255	0-127	0-63	0-31	0-15	0-7	0-3
				16-31	8-15	4-7
			32-47	48-63	24-31	12-15
				64-79	32-47	16-23
		64-127	64-95	96-111	48-63	24-31
				112-127	64-79	32-47
			128-159	160-175	80-95	40-55
				176-191	96-111	48-63
	128-255	128-191	128-159	160-175	160-175	160-175
				176-191	176-191	176-191
			192-223	224-239	224-239	224-239
				240-255	240-255	240-255
		192-255	224-239	240-255	240-255	240-255
				256-271	256-271	256-271
			272-303	304-319	304-319	304-319
				320-335	320-335	320-335

VLSM

with

Class A and B Addresses

We've gone over the practical applications of using VLSM on class C addresses. The same approach works with class A and B addresses. For example an ISP may have a class A address which it needs to subnet between its customers. Each customer may need to take their addresses and subnet them again in order to use them more effectively. The real trick to this is to remember which octet of the IP address you are working with.

Sample Problem 35

Part 1 of 3

Use the **Class A** address chart to break down the address for different ISP customers. At this stage of the problem you are creating subnets using the second octet of the IP address.

ISP Addresses 15.0.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Customer #1	8 million	<i>15.0.0.0 to 15.127.255.255</i>	<i>/9</i>
Customer #2	2 million	<i>15.128.0.0 to 15.159.255.255</i>	<i>/11</i>
Customer #3	2,000,000	<i>15.160.0.0 to 15.191.255.255</i>	<i>/11</i>
Customer #4	1,000,000	<i>15.192.0.0 to 15.207.255.255</i>	<i>/12</i>
Customer #5	500,000	<i>15.208.0.0 to 15.215.255.255</i>	<i>/13</i>
Customer #6	450,000	<i>15.216.0.0 to 15.223.255.255</i>	<i>/13</i>
Customer #7	200,000	<i>15.224.0.0 to 15.227.255.255</i>	<i>/14</i>
Customer #8	130,000	<i>15.228.0.0 to 15.229.255.255</i>	<i>/15</i>
Customer #9	100,000	<i>15.230.0.0 to 15.231.255.255</i>	<i>/15</i>

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
						8-11	4-5
						12-15	6-7
				16-31		16-19	8-9
					16-23	20-23	10-11
					24-31	24-27	12-13
						28-31	14-15
						30-31	16-17
						32-35	18-19
		32-63	32-47	32-39		36-39	20-21
					40-47	44-47	22-23
						48-51	24-25
					48-55	52-55	26-27
			48-63	56-63		60-63	28-29
							30-31
						64-67	32-33
						68-71	34-35
						72-75	36-37
						76-79	38-39
		64-127	64-95	64-79		80-83	40-41
					80-87	84-87	42-43
						88-91	44-45
					88-95	92-95	46-47
			96-111	96-103		96-99	48-49
					104-111	100-103	50-51
						104-107	52-53
						108-111	54-55
			112-127	112-119		112-115	56-57
					120-127	116-119	58-59
						120-123	60-61
						124-127	62-63
	128-255	128-191	128-159	128-143		128-131	64-65
					128-135	132-135	66-67
					136-143	136-139	68-69
						140-143	70-71
			144-159	144-151		144-147	72-73
					148-151	148-149	74-75
					152-159	152-155	76-77
						156-159	78-79
			160-191	160-175		160-163	80-81
					168-175	164-167	82-83
						168-171	84-85
						172-175	86-87
			176-191	176-183		176-179	88-89
					180-183	180-183	90-91
						184-187	92-93
					184-191	188-191	94-95
		192-255	192-223	192-207		192-195	96-97
					200-207	196-199	98-99
						200-203	100-101
						204-207	102-103
			208-223	208-215		208-211	104-105
					216-223	212-215	106-107
						216-219	108-109
						220-223	110-111
			224-255	224-231		224-227	112-113
					232-239	228-231	114-115
						232-235	116-117
					240-247	236-239	118-119
			240-255	240-247		240-243	120-121
					248-255	244-247	122-123
						248-251	124-125
						252-255	126-127
							128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Sample Problem 35

Part 2 of 3

Customer #5 has a total of 524,288 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for their different clients. At this stage of the problem you are creating sub-subnets with the third octet of the IP address.

ISP Addresses 15.208.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Client #1	7,500	<i>15.208.0.0 to 15.208.31.255</i>	<i>/19</i>
Client #2	5,000	<i>15.208.32.0 to 15.208.63.255</i>	<i>/19</i>
Client #3	4,500	<i>15.208.64.0 to 15.208.95.255</i>	<i>/19</i>
Client #4	2,000	<i>15.208.96.0 to 15.208.103.255</i>	<i>/21</i>
Client #5	1,450	<i>15.208.104.0 to 15.208.111.255</i>	<i>/21</i>
Client #6	1,150	<i>15.208.112.0 to 15.208.119.255</i>	<i>/21</i>
Client #7	900	<i>15.208.120.0 to 15.208.123.255</i>	<i>/22</i>
Client #8	750	<i>15.208.124.0 to 15.208.127.255</i>	<i>/22</i>
Client #9	450	<i>15.208.128.0 to 15.208.129.255</i>	<i>/23</i>

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
						8-11	4-5
						12-15	6-7
						16-19	8-9
						20-23	10-11
						24-27	12-13
						28-31	14-15
			32-63	16-31	16-23	32-35	16-17
					24-31	36-39	18-19
						40-43	20-21
						44-47	22-23
						48-51	24-25
						52-55	26-27
						56-59	28-29
						60-63	30-31
		64-127	64-95	32-47	32-39	64-67	32-33
					40-47	68-71	34-35
						72-75	36-37
						76-79	38-39
				48-63	48-55	80-83	40-41
					56-63	84-87	42-43
						88-91	44-45
						92-95	46-47
			96-127	64-79	64-71	96-99	48-49
					72-79	100-103	50-51
						104-107	52-53
						108-111	54-55
				80-95	80-87	112-115	56-57
					88-95	116-119	58-59
						120-123	60-61
						124-127	62-63
	128-255	128-191	128-159	128-143	128-135	128-131	64-65
					136-143	132-135	66-67
						136-139	68-69
						140-143	70-71
				144-159	144-151	148-151	72-73
					152-159	152-155	74-75
						156-159	76-77
						160-163	78-79
			160-191	160-175	160-167	164-167	80-81
					168-175	168-171	82-83
						172-175	84-85
						176-179	86-87
				176-191	176-183	180-183	88-89
					184-191	188-191	90-91
						192-195	92-93
						196-199	94-95
		192-255	192-223	192-207	200-207	204-207	96-97
					208-215	212-215	98-99
					216-223	216-219	100-101
						220-223	102-103
				224-239	224-231	228-231	104-105
					232-239	236-239	106-107
					240-247	244-247	108-109
					248-255	252-255	110-111
			224-255	240-255	240-247	244-247	112-113
					248-251	252-251	114-115
							116-117
							118-119
							120-121
							122-123
							124-125
							126-127

VLSM

with

Class A and B Addresses

Sample Problem 35

Part 3 of 3

Client #8 has a total of 1,024 addresses. Use the **Class C** address chart to break down the sub-subnetwork addresses for their different branch offices. At this stage of the problem you are creating sub-subnets with the fourth octet of the IP address.

ISP Addresses 15.208.124.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Branch #1	100	<i>15.208.124.0 to 15.208.124.127</i>	<i>/ 25</i>
Branch #2	55	<i>15.208.124.128 to 15.208.124.191</i>	<i>/ 26</i>
Branch #3	25	<i>15.208.124.192 to 15.208.124.223</i>	<i>/ 27</i>
Branch #4	6	<i>15.208.124.224 to 15.208.124.231</i>	<i>/ 29</i>
Branch #5	4	<i>15.208.124.232 to 15.208.124.239</i>	<i>/ 29</i>
Branch #6	2	<i>15.208.124.240 to 15.208.124.243</i>	<i>/ 30</i>
Branch #7	2	<i>15.208.124.244 to 15.208.124.247</i>	<i>/ 30</i>
Branch #8	2	<i>15.208.124.248 to 15.208.124.251</i>	<i>/ 30</i>
Branch #9	2	<i>15.208.124.252 to 15.208.124.255</i>	<i>/ 30</i>

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30		
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts		
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3		
					4-7			
				16-31	8-15	8-11		
					12-15			
			32-63	32-47	16-23	16-19		
					20-23			
				48-63	24-31	24-27		
					28-31			
				64-127	64-95	64-79	32-39	32-35
							36-39	
		80-95	40-43					
			44-47					
		96-127	96-111		48-55	48-51		
					52-55			
			112-127		56-59			
					60-63			
			128-255		128-191	128-159	64-71	64-67
							68-71	
		72-75						
		76-79						
		144-159		80-83				
				84-87				
				88-91				
				92-95				
		160-191		96-99				
				100-103				
	104-107							
	108-111							
	112-115							
	116-119							
	192-255	192-223		192-207		120-123		
						124-127		
				208-223		128-131		
						132-135		
				224-255	136-139			
					140-143			
		144-147						
		148-151						
		192-255		192-223	192-207	152-155		
						156-159		
	208-223					160-163		
						164-167		
	224-239				168-171			
					172-175			
					240-247	176-179		
						180-183		
						240-255	244-247	184-187
								188-191
	192-195							
	196-199							
	248-251			200-203				
				204-207				
				208-211				
				212-215				
			216-219					
			220-223					
	248-255		224-227					
			228-231					
	248-255	232-235						
		236-239						
	248-255	240-243						
		244-247						
	248-255	248-251						
		252-255						

VLSM

with

Class A and B Addresses

Problem 36

Part 1 of 3

The company you are working for is using the IP address 110.0.0.0 sub-subneted for multiple offices around the world. Use the **Class A** address chart to break down the sub-subnetwork addresses for the different offices.

At this stage of the problem you are creating sub-subnets with the third octet of the IP address. Remember which octet of the IP address you are working in.

Company Address 110.0.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Moskva	3,050,000		
New York	1,540,000		
St. Petersburg	1,075,000		
London	975,000		
Ekaterinoburg	525,000		
Munchen	450,000		
Napoli	150,000		
Birmingham	130,000		
Rotterdam	95,000		

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					4-7	4-7	2-3
					8-15	8-11	4-5
					16-23	12-15	6-7
					24-31	16-19	8-9
					32-39	20-23	10-11
			32-63	16-31	24-27	28-31	12-13
					32-35	30-31	14-15
				32-47	36-39	40-43	16-17
					40-43	44-47	18-19
				48-63	44-47	48-51	20-21
					48-51	52-55	22-23
		64-127	64-95	64-79	56-59	60-63	24-25
					60-63	64-67	26-27
				80-95	64-67	68-71	28-29
					68-71	72-75	30-31
			96-127	96-111	72-75	76-79	32-33
					76-79	80-83	34-35
				112-127	80-83	84-87	36-37
					84-87	88-91	38-39
			128-191	128-143	88-91	92-95	40-41
					92-95	96-99	42-43
				144-159	96-103	100-103	44-45
					100-103	104-107	46-47
				160-175	104-111	108-111	48-49
					108-111	112-115	50-51
				176-191	112-119	116-119	52-53
					116-119	120-123	54-55
		128-255	128-159	128-143	120-127	124-127	56-57
					124-127	128-131	58-59
				144-159	128-135	132-135	60-61
					132-135	136-139	62-63
			160-191	160-175	136-139	140-143	64-65
					140-143	144-147	66-67
				176-191	144-151	148-151	68-69
					148-151	152-155	70-71
			192-223	192-207	152-155	156-159	72-73
					156-159	160-163	74-75
				208-223	160-163	164-167	76-77
					164-167	168-171	78-79
		192-255	224-239	224-231	168-171	172-175	80-81
					172-175	176-179	82-83
				232-239	176-179	180-183	84-85
					180-183	184-187	86-87
			240-255	240-247	184-187	188-191	88-89
					188-191	192-195	90-91
				248-255	192-199	196-199	92-93
					196-199	200-203	94-95
				252-255	200-207	204-207	96-97
					204-207	208-211	98-99
				256-259	208-215	212-215	100-101
					212-215	216-219	102-103
				260-263	216-223	220-223	104-105
					220-223	224-227	106-107
				264-267	224-231	228-231	108-109
					228-231	232-235	110-111
				268-271	232-239	236-239	112-113
					236-239	240-243	114-115
				272-275	240-247	244-247	116-117
					244-247	248-251	118-119
				276-279	248-251	252-255	120-121
					252-255		122-123
				280-283			124-125
							126-127
				284-287			128-129
							130-131
				288-291			132-133
							134-135
				292-295			136-137
							138-139
				296-299			140-141
							142-143
				300-303			144-145
							146-147
				304-307			148-149
							150-151
				308-311			152-153
							154-155
				312-315			156-157
							158-159
				316-319			160-161
							162-163
				320-323			164-165
							166-167
				324-327			168-169
							170-171
				328-331			172-173
							174-175
				332-335			176-177
							178-179
				336-339			180-181
							182-183
				340-343			184-185
							186-187
				344-347			188-189
							190-191
				348-351			192-193
							194-195
				352-355			196-197
							198-199
				356-359			200-201
							202-203
				360-363			204-205
							206-207
				364-367			208-209
							210-211
				368-371			212-213
							214-215
				372-375			216-217
							218-219
				376-379			220-221
							222-223
				380-383			224-225
							226-227
				384-387			228-229
							230-231
				388-391			232-233
							234-235
				392-395			236-237
							238-239
				396-399			240-241
							242-243
				400-403			244-245
							246-247
				404-407			248-249
							250-251
				408-411			252-253
							254-255

VLSM

with

Class A and B Addresses

Problem 36

Part 2 of 3

London in part 1 of this problem has been given 1,048,576 hosts, with the address range of 110.128.0.0 to 110.143.255.255 /12 (255.240.0.0).

Based on the information below supply the required address ranges and subnet masks for each office. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different areas of the network.

London
Address Range 110.128.0.0 to 110.143.255.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Office #1	6,450		
Office #2	3,780		
Office #3	2,750		
Office #4	2,000		
Office #5	1,000		
Office #6	845		
Office #7	500		
Office #8	450		
Office #9	300		

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16	/17	/18	/19		/23			
255.255.0.0 65,536 Hosts	255.255.128.0 32,768 Hosts	255.255.192.0 16,384 Hosts	255.255.224.0 8,192 Hosts			255.255.254.0 512 Hosts		
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1	
						4-7	2-3	
				16-31	8-15	8-11	4-5	
						12-15	6-7	
					16-23	16-19	8-9	
					20-23	20-21	10-11	
			32-63	24-31	24-27	24-25	12-13	
					28-31	26-27	14-15	
				32-47	32-39	32-35	16-17	
					36-39	36-37	18-19	
					40-47	40-43	20-21	
					44-47	44-45	22-23	
			48-63	48-55	48-51	24-25		
				52-55	52-53	26-27		
				56-59	56-57	28-29		
				60-63	60-61	30-31		
			64-127	64-95	64-79	64-71	64-67	64-65
						68-71	68-69	66-67
						72-75	72-73	68-69
						76-79	74-75	70-71
					80-95	80-87	80-83	80-81
						84-87	84-85	82-83
						88-95	88-91	86-87
						92-95	92-93	88-89
					96-111	96-103	96-99	94-95
						100-103	100-101	96-97
						104-111	104-107	98-99
						108-111	108-109	100-101
						112-115	112-113	102-103
						116-119	114-115	104-105
						120-123	116-117	106-107
						124-127	118-119	108-109
					96-127	112-119	112-111	110-111
						116-119	116-115	112-113
						120-123	120-121	114-115
						124-127	122-123	116-117
		128-129				124-125	118-119	
		132-135				126-127	120-121	
		136-139				128-129	122-123	
		140-143				130-131	124-125	
		128-191		128-143	128-135	128-131	128-129	
					136-143	132-135	130-131	
					140-143	136-139	132-133	
					144-147	140-141	134-135	
					148-151	142-143	136-137	
					152-155	144-145	138-139	
					156-159	146-147	140-141	
					160-163	148-149	142-143	
				144-159	144-151	144-147	144-145	
					148-151	148-149	146-147	
					152-155	152-153	148-149	
					156-159	156-157	150-151	
					160-163	160-161	152-153	
					164-167	164-165	154-155	
					168-171	166-167	156-157	
					172-175	168-169	158-159	
				160-191	160-167	160-163	160-161	
					168-175	164-167	162-163	
					176-179	168-171	164-165	
					180-183	172-173	166-167	
					184-187	176-177	168-169	
					188-191	178-179	170-171	
					192-195	180-181	172-173	
	196-199				182-183	174-175		
	192-255	192-207		192-199	184-185	176-177		
				196-199	186-187	178-179		
				200-203	188-189	180-181		
				204-207	190-191	182-183		
				208-211	192-193	184-185		
				212-215	194-195	186-187		
				216-219	196-197	188-189		
				220-223	198-199	190-191		
		208-223		208-215	200-201	192-193		
				216-223	202-203	194-195		
				224-227	204-205	196-197		
				228-231	206-207	198-199		
				232-235	208-209	200-201		
				236-239	210-211	202-203		
				240-243	212-213	204-205		
			244-247	214-215	206-207			
		224-239	224-231	216-217	208-209			
			232-239	218-219	210-211			
			240-247	220-221	212-213			
			248-255	222-223	214-215			
			252-255	224-225	216-217			
			256-257	226-227	218-219			
			260-261	228-229	220-221			
			262-263	230-231	222-223			
	240-255	240-247	232-233	224-225				
		248-255	234-235	226-227				
		252-255	236-237	228-229				
		256-257	238-239	230-231				
		260-261	240-241	232-233				
		262-263	242-243	234-235				
		264-265	244-245	236-237				
		266-267	246-247	238-239				

VLSM

with

Class A and B Addresses

Problem 36

Part 3 of 3

Office #7 in part 2 of this problem has been given 512 hosts, with the address range of 110.128.80.0 / 23 (255.255.254.0).

Based on the information below supply the required address ranges and subnet masks for each school area. Use the **Class C** address chart to break down the sub-subnetwork addresses for the different areas of the network. **Hint:** Another way to look at this problem is to see that with the third octet range of 80 to 81 you have access to 2 groups of 255 addresses or two Class C VLSM charts.

Office #7
Address Range 110.128.80.0 to 110.128.81.255

Customer Name	Number of Addresses	Address Range	CIDR
1st Floor	125		
2nd Floor	75		
5th Floor	50		
8th Floor	45		
4th Floor	30		
Basement	14		
7th Floor	12		
3rd Floor	6		
6th Floor	4		

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-103	96-99	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

VLSM

with

Class A and B Addresses

Problem 37

Part 1 of 3

The school system you are working for is using the private address of 172.32.0.0 to subnet the entire district. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different schools and offices.

At this stage of the problem you are creating sub-subnets with the third octet of the IP address. Remember which octet of the IP address you are working in.

School System Address 172.32.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
North High	2,400		
South High	2,000		
North Middle	1,200		
South Middle	1,000		
Central Elem.	550		
Southern Elem.	475		
Eastern Elem.	450		
Central Office	400		
Western Elem.	300		

Numeric Method
Show your work for Problem 37 in the space below.

Number of Hosts	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	256	128	64	32	16	8	4	2
Number of Subnets	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536
Binary values	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1

<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>

VLSM

with

Class A and B Addresses

Problem 37

Part 2 of 3

Eastern Elementary has been given 512 hosts, with the address range of 172.32.42.0 / 21 (255.255.248.0).

Based on the information below supply the required address ranges and subnet masks for each school area.

Hint:

Another way to look at this problem is to see that with the third octet range of 42 to 43 you have access to 2 groups of 255 addresses (172.32.42.0 and 172.32.43.0). Think in terms of having two Class C address ranges.

Eastern Elementary School
Address Range 172.32.42.0 to 172.32.43.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Students	250		
Printers	45		
Staff	40		
Network Devices	25		
Administrative	12		

Numeric Method

Show your work for Problem 37 in the space below.

Number of Subnets 256 128 64 32 16 8 4 2 - Number of Hosts
 ~ 2 4 8 16 32 64 128 256
 128 64 32 16 8 4 2 1 - Binary values
172. 32. 42 . 0 0 0 0 0 0 0 0

<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>

Number of Subnets 256 128 64 32 16 8 4 2 - Number of Hosts
 ~ 2 4 8 16 32 64 128 256
 128 64 32 16 8 4 2 1 - Binary values
172. 32. 43 . 0 0 0 0 0 0 0 0

<u>Hosts</u>	<u>Address Ranges</u>	<u>CIDR</u>

VLSM

with

Class A and B Addresses

Problem 37

Part 3 of 3

South High in part 1 of this problem has been given 2,048 hosts, with the address range of 172.32.16.0 / 21 (255.255.248.0).

Based on the information below supply the required address ranges and subnet masks for each school area.

South High School
Address Range 172.32.16.0 to 172.32.23.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Students	1,000		
Network Devices	250		
Printers	200		
Staff	150		
Administrative	50		

VLSM

with

Class A and B Addresses

Problem 38

Part 1 of 4

Use the **Class A** address chart to break down the address for different business customers by country. At this stage of this problem you are creating subnets in the second octet of the IP address.

Addresses 75.0.0.0

Customer Name	Number of Addresses	Address Range	CIDR
United States	6.5 million		
China	4 million		
Japan	1 million		
Germany	500,000		
Russia	455,000		
Australia	450,000		
Brazil	125,000		
Canda	90,000		
Denmark	88,000		

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
					16-31	8-11	4-5
						12-15	6-7
			32-63	32-47	16-23	16-19	8-9
					24-31	20-23	10-11
					48-63	24-27	14-15
						28-31	16-17
				64-79	32-39	32-35	18-19
					40-47	36-39	20-21
					56-63	40-43	22-23
						44-47	24-25
			96-127	96-111	48-55	48-51	26-27
					52-55	52-55	28-29
					56-59	56-59	30-31
					60-63	60-63	32-33
				128-143	64-71	64-67	34-35
					72-79	68-71	36-37
					80-95	72-75	38-39
						76-79	40-41
		128-255	128-191	128-159	80-87	80-83	42-43
					84-87	84-87	44-45
					88-95	88-91	46-47
					96-103	92-95	48-49
						96-99	50-51
				144-159	100-103	100-103	52-53
					104-111	104-107	54-55
					112-127	108-111	56-57
						112-115	58-59
			160-191	160-175	116-119	116-119	60-61
					120-127	120-123	62-63
					176-183	124-127	64-65
						128-131	66-67
				192-207	128-135	132-135	68-69
					136-143	136-139	70-71
					144-151	140-143	72-73
						144-147	74-75
			224-255	224-239	148-151	148-151	76-77
					152-155	152-155	78-79
					156-159	156-159	80-81
					160-167	160-163	82-83
						164-167	84-85
				240-255	168-175	168-171	86-87
					172-175	172-175	88-89
					176-179	176-179	90-91
					248-255	180-183	92-93
						184-187	94-95
					252-255	188-191	96-97
						192-195	98-99
					256-259	196-199	100-101
						200-203	102-103
					260-263	204-207	104-105
						208-211	106-107
					264-267	212-215	108-109
						216-219	110-111
					268-271	220-223	112-113
						224-227	114-115
					272-275	228-231	116-117
						232-235	118-119
					276-279	236-239	120-121
						240-243	122-123
					280-283	244-247	124-125
						248-251	126-127
					284-287	252-255	128-129
						256-259	130-131
					288-291	260-263	132-133
						264-267	134-135
					292-295	268-271	136-137
						272-275	138-139
					296-299	276-279	140-141
						280-283	142-143
					300-303	284-287	144-145
						288-291	146-147
					304-307	292-295	148-149
						296-299	150-151
					308-311	300-303	152-153
						304-307	154-155
					312-315	308-311	156-157
						312-315	158-159
					316-319	316-319	160-161
						320-323	162-163
					320-323	324-327	164-165
						328-331	166-167
					324-327	332-335	168-169
						336-339	170-171
					328-331	340-343	172-173
						344-347	174-175
					332-335	348-351	176-177
						352-355	178-179
					336-339	356-359	180-181
						360-363	182-183
					340-343	364-367	184-185
						368-371	186-187
					344-347	372-375	188-189
						376-379	190-191
					348-351	380-383	192-193
						384-387	194-195
					352-355	388-391	196-197
						392-395	198-199
					356-359	396-399	200-201
						400-403	202-203
					360-363	404-407	204-205
						408-411	206-207
					364-367	412-415	208-209
						416-419	210-211
					368-371	420-423	212-213
						424-427	214-215
					372-375	428-431	216-217
						432-435	218-219
					376-379	436-439	220-221
						440-443	222-223
					380-383	444-447	224-225
						448-451	226-227
					384-387	452-455	228-229
						456-459	230-231
					388-391	460-463	232-233
						464-467	234-235
					392-395	468-471	236-237
						472-475	238-239
					396-399	476-479	240-241
						480-483	242-243
					400-403	484-487	244-245
						488-491	246-247
					404-407	492-495	248-249
						496-499	250-251
					408-411	500-503	252-253
						504-507	254-255

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 2 of 4

The United States customers have a total of 8,388,608 addresses. Use the **Class A** address chart to break down the sub-subnetwork addresses for their different areas. At this stage of this problem you are creating sub-subnets in the second octet of the IP address.

Addresses Range: 75.0.0.0 to 75.127.255.255

Customer Name	Number of Addresses	Address Range	CIDR
Client #1	1,950,000		
Client #2	1,000,000		
Client #3	950,000		
Client #4	700,000		
Client #5	550,000		
Client #6	500,000		
Client #7	450,000		

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8	/9	/10	/11	/12	/13	/14	/15
255.0.0.0 16,777,216 Hosts	255.128.0.0 8,388,608 Hosts	255.192.0.0 4,194,304 Hosts	255.224.0.0 2,097,152 Hosts	255.240.0.0 1,048,576 Hosts	255.248.0.0 524,288 Hosts	255.252.0.0 262,144 Hosts	255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
						8-11	4-5
						12-15	6-7
						16-19	8-9
						20-23	10-11
				16-31	16-23	20-23	12-13
					24-31	24-27	14-15
						28-31	16-17
						32-35	18-19
						36-39	20-21
						40-43	22-23
			32-63	32-47	32-39	36-39	24-25
						40-43	26-27
						44-47	28-29
						48-51	30-31
						52-55	32-33
						56-59	34-35
				48-63	48-55	52-55	36-37
					56-63	60-63	38-39
						64-67	40-41
						68-71	42-43
						72-75	44-45
						76-79	46-47
			64-95	64-79	64-71	68-71	48-49
					72-79	76-79	50-51
						80-83	52-53
						84-87	54-55
						88-91	56-57
						92-95	58-59
				80-95	80-87	84-87	60-61
					88-95	92-95	62-63
						96-99	64-65
						100-103	66-67
						104-107	68-69
						108-111	70-71
			96-127	96-111	96-103	100-103	72-73
					104-111	104-107	74-75
						112-115	76-77
						116-119	78-79
						120-123	80-81
						124-127	82-83
				112-127	112-119	116-119	84-85
					120-127	124-127	86-87
						128-129	88-89
						130-131	90-91
						132-133	92-93
						134-135	94-95
128-255	128-191	128-159	128-143	128-135	128-131	96-97	
				136-143	136-139	98-99	
					140-143	100-101	
					144-147	102-103	
					148-151	104-105	
					152-155	106-107	
			144-159	144-151	144-147	108-109	
				152-159	152-155	110-111	
					156-159	112-113	
					160-163	114-115	
					164-167	116-117	
					168-171	118-119	
		160-191	160-175	160-167	164-167	120-121	
				168-175	168-171	122-123	
					172-175	124-125	
					176-179	126-127	
					180-183	128-129	
					184-187	130-131	
			176-191	176-183	180-183	132-133	
				184-191	184-187	134-135	
					188-191	136-137	
					192-195	138-139	
					196-199	140-141	
					200-203	142-143	
		192-223	192-207	192-199	196-199	144-145	
				200-207	200-203	146-147	
					204-207	148-149	
					208-211	150-151	
					212-215	152-153	
					216-219	154-155	
			208-223	208-215	212-215	156-157	
				216-223	220-223	158-159	
					224-227	160-161	
					228-231	162-163	
					232-235	164-165	
					236-239	166-167	
		224-255	224-239	224-231	228-231	168-169	
				232-239	236-239	170-171	
					240-243	172-173	
					244-247	174-175	
					248-251	176-177	
					252-255	178-179	
			240-255	240-247	244-247	180-181	
				248-255	248-251	182-183	
					252-255	184-185	
						186-187	
						188-189	
						190-191	

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 3 of 4

Client #7 has a total of 524,288 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for their different clients. At this stage of this problem you are creating sub-subnets in the third or forth octet of the IP address.

Hint: Another way to look at this problem is to see that with the second octet range of 104 to 111 you have access to 8 groups of 65,536 addresses or 8 Class B VLSM charts.

ISP Addresses 75.104.0.0 to 75.111.255.255

Customer Name	Number of Addresses	Address Range	CIDR
Office #1	60,000		
Office #2	45,000		
Office #3	30,000		
Office #4	24,000		
Office #5	15,000		
Office #6	10,000		
Office #7	8,000		
Office #8	2,000		
Office #9	1,000		

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
							6-7
							8-9
							10-11
				16-31	16-23	16-19	12-13
						20-23	14-15
					24-31	24-27	16-17
						28-31	18-19
							20-21
							22-23
		32-63	32-47	32-39			24-25
							26-27
							28-29
							30-31
				40-47			32-33
							34-35
				48-63	48-55	48-51	36-37
							38-39
							40-41
							42-43
							44-45
							46-47
							48-49
							50-51
							52-53
							54-55
							56-57
							58-59
		64-127	64-95	64-79	64-71	64-67	60-61
						68-71	62-63
					72-79	72-75	64-65
						76-79	66-67
			80-95	80-87			68-69
							70-71
					88-95	88-91	72-73
							74-75
							76-77
							78-79
							80-81
							82-83
							84-85
							86-87
							88-89
							90-91
							92-93
							94-95
							96-97
							98-99
							100-101
							102-103
							104-105
							106-107
							108-109
							110-111
							112-113
							114-115
							116-117
							118-119
							120-121
							122-123
							124-125
							126-127
							128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255
	128-255	128-191	128-159	128-143	128-135	128-131	128-129
							130-131
					136-143	136-139	132-133
							134-135
			144-159	144-151	144-147	144-143	136-137
							138-139
					148-151	148-145	140-141
							142-143
					152-159	152-155	144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
				224-225			
		226-227					
		228-229					
		230-231					
		232-233					
		234-235					
		236-237					
		238-239					
		240-241					
		242-243					
		244-245					
		246-247					
		248-249					
		250-251					
		252-253					
		254-255					
	128-255	192-255	224-255	240-255	240-247	240-243	240-241
					244-247	242-243	
					248-251	244-245	
					252-255	246-247	

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 4 of 4

Office #7 from part 3 of 4 has a total of 8,192 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different branch offices. At this stage of this problem you are creating sub-subnets in the third octet of the IP address.

Hint: Remember that the range of this problem is between 128 and 159 in the third octet. Your subnetting will start in the middle of the chart not at the top for this range.

ISP Addresses 75.107.128.0 to 75.107.159.255

Customer Name	Number of Addresses	Address Range	CIDR
Branch #1	4,000		
Branch #2	2,000		
Branch #3	1,000		
Branch #4	500		
Branch #5	450		

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
			16-31	16-31	16-23	16-19	8-9
						20-23	10-11
					24-31	24-27	12-13
						28-31	14-15
			32-63	32-47	32-39	32-35	16-17
						36-39	18-19
					40-47	40-43	20-21
						44-47	22-23
				48-63	48-55	48-51	24-25
						52-55	26-27
					56-63	56-59	28-29
						60-63	30-31
		64-127	64-95	64-79	64-71	64-67	32-33
						68-71	34-35
					72-79	72-75	36-37
						76-79	38-39
			80-95	80-87	80-83	80-81	40-41
						84-87	42-43
					88-95	88-91	44-45
						92-95	46-47
			96-127	96-111	96-103	96-99	48-49
						100-103	50-51
					104-111	104-107	52-53
						108-111	54-55
				112-127	112-119	112-115	56-57
						116-119	58-59
					120-127	120-123	60-61
						124-127	62-63
	128-255	128-191	See Hint 128-159	128-143	128-135	128-131	64-65
						132-135	66-67
					136-143	136-139	68-69
						140-143	70-71
				144-159	144-151	144-147	72-73
						148-151	74-75
					152-159	152-155	76-77
						156-159	78-79
			160-191	160-175	160-167	160-163	80-81
						164-167	82-83
					168-175	168-171	84-85
						172-175	86-87
				176-191	176-183	176-179	88-89
						180-183	90-91
					184-191	184-187	92-93
						188-191	94-95
		192-255	192-223	192-207	192-199	192-195	96-97
						196-199	98-99
					200-207	200-203	100-101
						204-207	102-103
				208-223	208-215	208-211	104-105
						212-215	106-107
					216-223	216-219	108-109
						220-223	110-111
			224-255	224-239	224-231	224-227	112-113
						228-231	114-115
					232-239	232-235	116-117
						236-239	118-119
				240-255	240-247	240-243	120-121
						244-247	122-123
					248-255	248-251	124-125
						252-255	126-127

Reference Charts and Support Materials

Class A Addresses
VLSM Chart 8-15 Bits (2nd octet)

/8	/9	/10	/11	/12	/13	/14	/15
208.0.0.0 8,192 hosts	208.128.0.0 4,096 hosts	208.192.0.0 2,048 hosts	208.224.0.0 1,024 hosts	208.256.0.0 512 hosts	208.288.0.0 256 hosts	209.0.0.0 128 hosts	209.64.0.0 64 hosts
0-255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
				8-15	8-7	4-3	2-1
				16-31	16-23	16-19	16-17
			32-63	32-31	32-23	32-19	32-17
				32-39	32-27	32-23	32-19
				32-47	32-35	32-27	32-19
		64-127	64-95	64-63	64-47	64-31	64-15
				64-63	64-47	64-31	64-15
				64-63	64-47	64-31	64-15
			96-127	96-95	96-79	96-63	96-47
				96-95	96-79	96-63	96-47
				96-95	96-79	96-63	96-47
	128-255	128-191	128-159	128-127	128-111	128-95	128-79
				128-127	128-111	128-95	128-79
				128-127	128-111	128-95	128-79
			160-191	160-159	160-143	160-127	160-111
				160-159	160-143	160-127	160-111
				160-159	160-143	160-127	160-111
		192-255	192-223	192-223	192-207	192-191	192-175
				192-223	192-207	192-191	192-175
				192-223	192-207	192-191	192-175
			224-255	224-223	224-207	224-191	224-175
				224-223	224-207	224-191	224-175
				224-223	224-207	224-191	224-175

Class B Addresses
VLSM Chart 16-23 Bits (3rd octet)

/16	/17	/18	/19	/20	/21	/22	/23
208.208.0.0 65,536 hosts	208.208.128.0 32,768 hosts	208.208.192.0 16,384 hosts	208.208.224.0 8,192 hosts	208.208.240.0 4,096 hosts	208.208.248.0 2,048 hosts	208.208.252.0 1,024 hosts	208.208.254.0 512 hosts
0-255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
				8-15	8-7	4-3	2-1
				16-31	16-23	16-19	16-17
		32-63	32-43	32-31	32-23	32-19	32-17
				32-39	32-27	32-23	32-19
				32-47	32-35	32-27	32-19
		64-127	64-95	64-63	64-47	64-31	64-15
				64-63	64-47	64-31	64-15
				64-63	64-47	64-31	64-15
			96-127	96-95	96-79	96-63	96-47
				96-95	96-79	96-63	96-47
				96-95	96-79	96-63	96-47
	128-255	128-191	128-159	128-127	128-111	128-95	128-79
				128-127	128-111	128-95	128-79
				128-127	128-111	128-95	128-79
			160-191	160-159	160-143	160-127	160-111
				160-159	160-143	160-127	160-111
				160-159	160-143	160-127	160-111
		192-255	192-223	192-223	192-207	192-191	192-175
				192-223	192-207	192-191	192-175
				192-223	192-207	192-191	192-175
			224-255	224-223	224-207	224-191	224-175
				224-223	224-207	224-191	224-175
				224-223	224-207	224-191	224-175

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

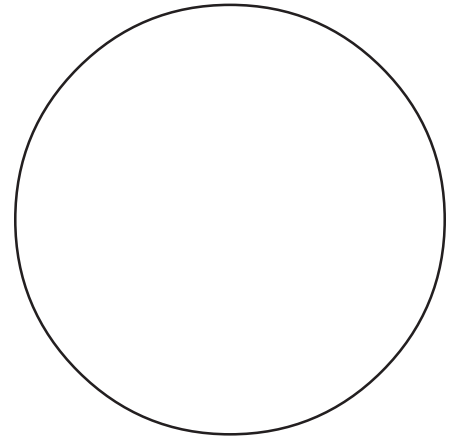
/24	/25	/26	/27	/28	/29	/30
208.208.256.0 256 hosts	208.208.256.128 128 hosts	208.208.256.192 64 hosts	208.208.256.224 32 hosts	208.208.256.240 16 hosts	208.208.256.248 8 hosts	208.208.256.252 4 hosts
0-255	0-127	0-63	0-31	0-15	0-7	0-3
				8-15	8-7	4-3
				16-31	16-23	16-19
		32-63	32-43	32-31	32-23	32-19
				32-39	32-27	32-23
				32-47	32-35	32-27
		64-127	64-95	64-63	64-47	64-31
				64-63	64-47	64-31
				64-63	64-47	64-31
			96-127	96-95	96-79	96-63
				96-95	96-79	96-63
				96-95	96-79	96-63
	128-255	128-191	128-159	128-127	128-111	128-95
				128-127	128-111	128-95
				128-127	128-111	128-95
			160-191	160-159	160-143	160-127
				160-159	160-143	160-127
				160-159	160-143	160-127
		192-255	192-223	192-223	192-207	192-191
				192-223	192-207	192-191
				192-223	192-207	192-191
			224-255	224-223	224-207	224-191
				224-223	224-207	224-191
				224-223	224-207	224-191

Visualizing Subnets Using The Circle Method

The circle method is another method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the different sections of the circle you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

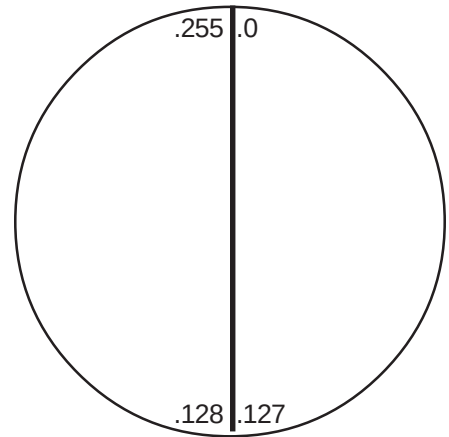
Start with a circle. The whole circle is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



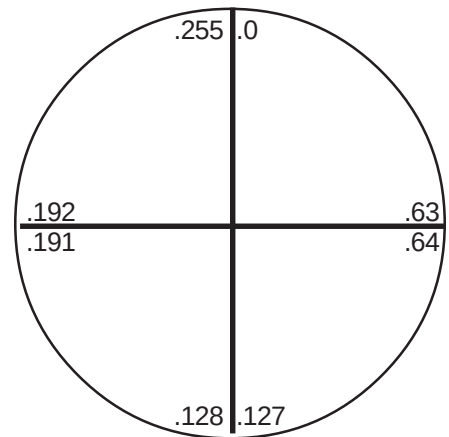
Split the circle in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



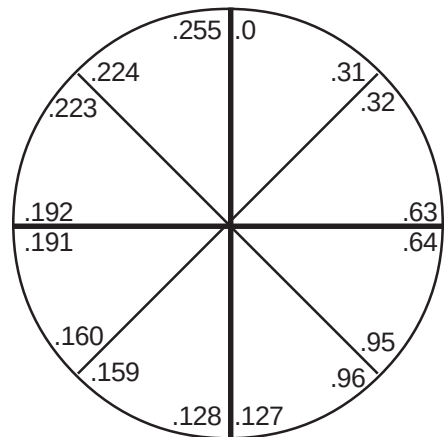
Divide the circle into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



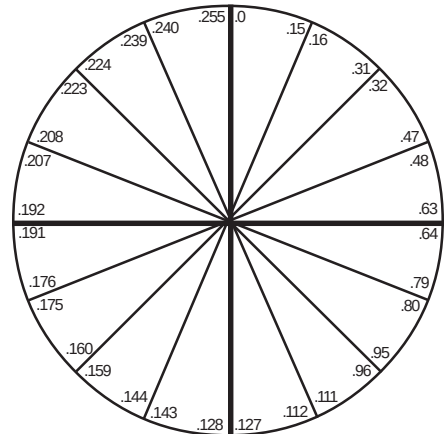
Split each quarter and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets



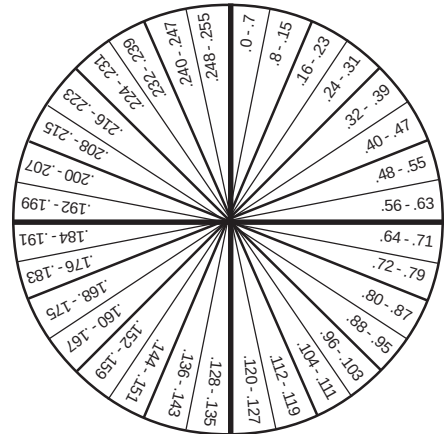
Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets



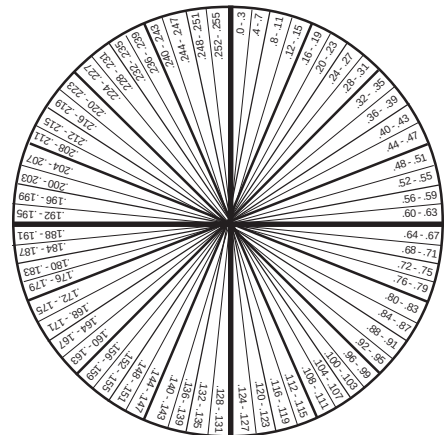
The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets



The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets

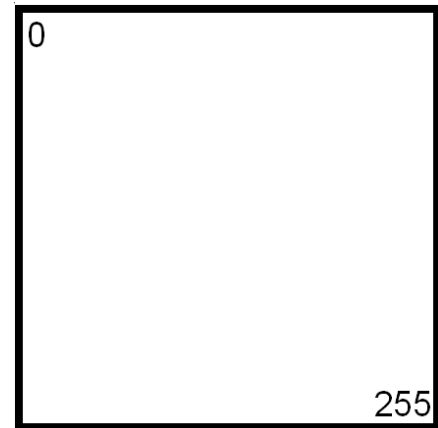


Visualizing Subnets Using The Box Method

The box method is a simple way to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

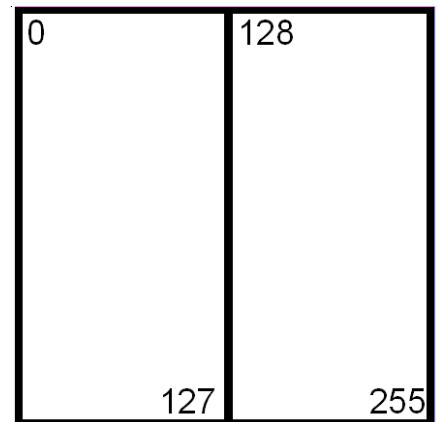
Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



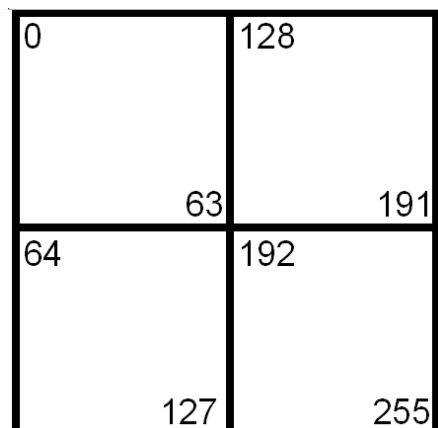
Split the box in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



Divide the box into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets

0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets

0	32	128	160
15	47	143	175
16	48	144	176
31	63	159	191
64	96	192	224
79	111	207	239
80	112	208	240
95	127	223	255

The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets

0	8	32	40	128	136	160	168
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
71	79	103	111	199	207	321	239
80	88	112	120	208	216	240	248
87	95	119	127	215	223	247	255

The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets

0	8	32	40	128	136	160	168
3	11	35	43	131	139	163	171
4	12	36	44	132	140	164	172
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
19	27	51	59	147	155	179	187
20	28	52	60	148	156	180	188
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
67	75	99	107	195	203	227	235
68	76	100	108	196	204	228	236
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
83	91	115	123	211	219	243	251
84	92	116	124	212	220	244	252
87	95	119	127	215	223	247	255

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8	/9	/10	/11	/12	/13	/14	/15			
255.0.0.0 16,777,216 Hosts	255.128.0.0 8,388,608 Hosts	255.192.0.0 4,194,304 Hosts	255.224.0.0 2,097,152 Hosts	255.240.0.0 1,048,576 Hosts	255.248.0.0 524,288 Hosts	255.252.0.0 262,144 Hosts	255.254.0.0 131,072 Hosts			
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1			
					8-15	4-7	2-3			
				16-31	16-23	8-11	4-5			
					24-31	12-15	6-7			
					32-47	32-39	16-19	8-9		
						40-47	20-23	10-11		
			48-63			48-55	24-27	12-13		
						56-63	28-31	14-15		
				64-127		64-95	64-71	32-35	16-17	
							72-79	36-39	18-19	
					80-95		80-87	40-43	20-21	
							88-95	44-47	22-23	
		96-111	96-103				48-51	24-25		
			104-111		52-55		26-27			
			112-127		112-119	56-59	28-29			
					120-127	60-63	30-31			
					128-255	128-191	128-143	128-135	64-67	32-33
								136-143	68-71	34-35
		144-159						144-151	72-75	36-37
								152-159	76-79	38-39
			160-191	160-167				80-83	40-41	
				168-175				84-87	42-43	
				176-191			176-183	88-91	44-45	
							184-191	92-95	46-47	
	192-255	192-223					192-199	96-99	48-49	
							200-207	100-103	50-51	
			208-223				208-215	104-107	52-53	
							216-223	108-111	54-55	
				224-239		224-231	112-115	56-57		
						232-239	116-119	58-59		
		240-255				240-247	120-123	60-61		
						248-255	124-127	62-63		
			128-159			128-135	128-131	64-65	64-65	
							136-139	68-69	66-67	
				144-159			144-147	72-73	68-69	
							152-155	74-75	70-71	
	160-175	160-163					76-77	72-73		
		168-171					78-79	74-75		
		176-191				176-179	80-81	76-77		
						184-187	82-83	78-79		
				192-199	192-195	192-191	84-85	80-81		
						196-199	86-87	82-83		
	200-207					200-203	88-89	84-85		
						204-207	90-91	86-87		
		208-209	208-205			92-93	88-89			
			212-213			94-95	90-91			
			216-217		216-219	96-97	92-93			
					220-223	98-99	94-95			
224-225	224-227				100-101	96-97				
	228-231				102-103	98-99				
	232-233	232-235			104-105	100-101				
		236-239			106-107	102-103				
		240-241	240-243	108-109	104-105					
			244-247	110-111	106-107					
248-249			248-251	112-113	108-109					
			252-255	114-115	110-111					
	256-257		256-259	116-117	112-113					
			260-263	118-119	114-115					
		264-265	264-267	120-121	116-117					
			268-271	122-123	118-119					
272-273			272-275	124-125	120-121					
			276-279	126-127	122-123					
	280-281		280-283	128-129	124-125					
			284-287	130-131	126-127					
		288-289	288-291	132-133	128-129					
			292-295	134-135	130-131					
296-297			296-299	136-137	132-133					
			300-303	138-139	134-135					
	304-305		304-307	140-141	136-137					
			308-311	142-143	138-139					
		312-313	312-315	144-145	140-141					
			316-319	146-147	142-143					
320-321			320-323	148-149	144-145					
			324-327	150-151	146-147					
	328-329		328-331	152-153	148-149					
			332-335	154-155	150-151					
		336-337	336-339	156-157	152-153					
			340-343	158-159	154-155					
344-345			344-347	160-161	156-157					
			348-351	162-163	158-159					
	352-353		352-355	164-165	160-161					
			356-359	166-167	162-163					
		360-361	360-363	168-169	164-165					
			364-367	170-171	166-167					
368-369			368-371	172-173	168-169					
			372-375	174-175	170-171					
	376-377		376-379	176-177	172-173					
			380-383	178-179	174-175					
		384-385	384-387	180-181	176-177					
			388-391	182-183	178-179					
392-393			392-395	184-185	180-181					
			396-399	186-187	182-183					
	400-401		400-403	188-189	184-185					
			404-407	190-191	186-187					
		408-409	408-411	192-193	188-189					
			412-415	194-195	190-191					
416-417			416-419	196-197	192-193					
			420-423	198-199	194-195					
	424-425		424-427	200-201	196-197					
			428-431	202-203	198-199					
		432-433	432-435	204-205	200-201					
			436-439	206-207	202-203					
440-441			440-443	208-209	204-205					
			444-447	210-211	206-207					
	448-449		448-451	212-213	208-209					
			452-455	214-215	210-211					
		456-457	456-459	216-217	212-213					
			460-463	218-219	214-215					
464-465			464-467	220-221	216-217					
			468-471	222-223	218-219					
	472-473		472-475	224-225	220-221					
			476-479	226-227	222-223					
		480-481	480-483	228-229	224-225					
			484-487	230-231	226-227					
488-489			488-491	232-233	228-229					
			492-495	234-235	230-231					
	496-497		496-499	236-237	232-233					
			500-503	238-239	234-235					
		504-505	504-507	240-241	236-237					
			508-511	242-243	238-239					
512-513			512-515	244-245	240-241					
			516-519	246-247	242-243					
	520-521		520-523	248-249	244-245					
			524-527	250-251	246-247					
		528-529	528-531	252-253	248-249					
			532-535	254-255	250-251					
536-537			536-539	256-257	252-253					
			540-543	258-259	254-255					
	544-545		544-547	260-261	256-257					
			548-551	262-263	258-259					
		552-553	552-555	264-265	260-261					
			556-559	266-267	262-263					
560-561			560-563	268-269	264-265					
			564-567	270-271	266-267					
	568-569		568-571	272-273	268-269					
			572-575	274-275	270-271					
		576-577	576-579	276-277	272-273					
			580-583	278-279	274-275					
584-585			584-587	280-281	276-277					
			588-591	282-283	278-279					
	592-593		592-595	284-285	280-281					
			596-599	286-287	282-283					
		600-601	600-603	288-289	284-285					
			604-607	290-291	286-287					
608-609			608-611	292-293	288-289					
			612-615	294-295	290-291					
	616-617		616-619	296-297	292-293					
			620-623	298-299	294-295					
		624-625	624-627	300-301	296-297					
			628-631	302-303	298-299					
632-633			632-635	304-305	300-301					
			636-639	306-307	302-303					
	640-641		640-643	308-309	304-305					
			644-647	310-311	306-307					
		648-649	648-651	312-313	308-309					
			652-655	314-315	310-311					
656-657			656-659	316-317	312-313					
			660-663	318-319	314-315					
	664-665		664-667	320-321	316-317					
			668-671	322-323	318-319					
		672-673	672-675	324-325	320-321					
			676-679	326-327	322-323					
680-681			680-683	328-329	324-325					
			684-687	330-331	326-327					
	688-689		688-691	332-333	328-329					
			692-695	334-335	330-331					
		696-697	696-699	336-337	332-333					
			700-703	338-339	334-335					
704-705			704-707	340-341	336-337					
			708-711	342-343	338-339					
	712-713		712-715	344-345	340-341					
			716-719	346-347	342-343					
		720-721	720-723	348-349	344-345					
			724-727	350-351	346-347					
728-729			728-731	352-353	348-349					
			732-735	354-355	350-351					
	736-737		736-739	356-357	352-353					
			740-743	358-359	354-355					
		744-745	744-747	360-361	356-357					
			748-751	362-363	358-359					
752-753			752-755	364-365	360-361					
			756-759	366-367	362-363					
	760-761		760-763	368-369	364-365					
			764-767	370-371	366-367					
		768-769	768-771	372-373	368-369					
			772-775	374-375	370-371					
776-777			776-779	376-377	372-373					
			780-783	378-379	374-375					
	784-785		784-787	380-381	376-377					
			788-791	382-383	378-379					
		792-793	792-795	384-385	380-381					
			796-799	386-387	382-383					
800-801			800-803	388-389	384-385					
			804-807	390-391	386-387					
	808-809		808-811	392-393	388-389					
			812-815	394-395	390-391					
		816-817	816-819	396-397	392-393					
			820-823	398-399	394-395					
824-825			824-827	400-401	396-397					
			828-831	402-403	398-399					
	832-833		832-835	404-405	400-401					
			836-839	406-407	402-403					
		840-841	840-843	408-409	404-405					
			844-847	410-411	406-407					
848-849			848-851	412-413	408-409					
			852-855	414-415	410-411					
	856-857		856-859	416-417	412-413					
			860-863	418-419	414-415					
		864-865	864-867	420-421	416-417					
			868-871	422-423	418-419					
872-873			872-875	424-425	420-421					
			876-879	426-427	422-423					
	880-881		880-883	428-429	424-425					
			884-887	430-431	426-427					
		888-889	888-891	432-433	428-429					
			892-895	434-435	430-431					
896-897			896-899	436-437	432-433					
			900-903	438-439	434-435					
	904-905		904-907	440-441	436-437					
			908-911	442-443	438-439					
		912-913	912-915	444-445	440-441					
			916-919	446-447	442-443					
920-921			920-923	448-449	444-445					
			924-927	450-451	446-447					
	928-929		928-931	452-453	448-449					
			932-935	454-455	450-451					
		936-937	936-939	456-457	452-453					
			940-943	458-459	454-455					
944-945			944-947	460-461	456-457					
			948-951	462-463	458-459					
	952-953		952-955	464-465	460-461					
			956-959	466-467	462-463					
		960-961	960-963	468-469	464-465					
			964-967	470-471	466-467					
968-969			968-971	472-473	468-469					
			972-975	474-475	470-471					
	976-977		976-979	476-477	472-473					
			980-983	478-479	474-475					
		984-985	984-987	480-481	476-477					
			988-991	482-483	478-479					
992-993			992-995	484-485	480-481					
			996-999	486-487	482-483					
	1000-1001		1000-1003	488-489	484-485					
			1004-1007	490-491	486-487					
		1008-1009	1008-1011	492-493	488-489					
			1012-1015	494-495	490-491					
1016-1017			1016-1019	496-497	492-493					
			1020-1023	498-499	494-495					
	1024-1025		1024-1027	500-501	496-497					
			1028-1031	502-503	498-499					
		1032-1033	1032-1035	504-505	500-501					
			1036-1039	506-507	502-503					
1040-1041			1040-1043	508-509	504-505					
			1044-1047	510-511	506-507					
	1048-1049		1048-1051	512-513	508-509					
			1052-1055	514-515						

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
						16-19	8-9
						20-23	10-11
						24-27	12-13
						28-31	14-15
						32-35	16-17
						36-39	18-19
						40-43	20-21
						44-47	22-23
						48-51	24-25
						52-55	26-27
						56-59	28-29
						60-63	30-31
		64-127	32-63	32-47	64-71	64-67	32-33
						68-71	34-35
					80-87	80-83	36-37
						84-87	38-39
					88-95	88-91	40-41
						92-95	42-43
						96-99	44-45
						100-103	46-47
						104-107	48-49
						108-111	50-51
						112-115	52-53
						116-119	54-55
						120-123	56-57
						124-127	58-59
						128-131	60-61
						132-135	62-63
	128-255	128-191	128-159	128-143	128-135	128-131	64-65
						132-135	66-67
					136-143	136-139	68-69
						140-143	70-71
			144-159	144-151	144-147	144-147	72-73
						148-151	74-75
					152-159	152-155	76-77
						156-159	78-79
						160-163	80-81
						164-167	82-83
						168-171	84-85
						172-175	86-87
						176-179	88-89
						180-183	90-91
						184-187	92-93
						188-191	94-95
						192-195	96-97
		192-255	192-223	192-207	192-199	196-199	98-99
						200-203	100-101
					204-207	204-207	102-103
						208-211	104-105
			208-223	208-215	208-211	212-215	106-107
						216-219	108-109
					220-223	220-223	110-111
						224-227	112-113
						228-231	114-115
						232-235	116-117
						236-239	118-119
						240-243	120-121
						244-247	122-123
						248-251	124-125
						252-255	126-127
			224-255	224-239	224-231	228-231	128-129
						232-235	130-131
				240-255	240-247	244-247	132-133
						248-251	134-135
						252-255	136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

Class C Addresses

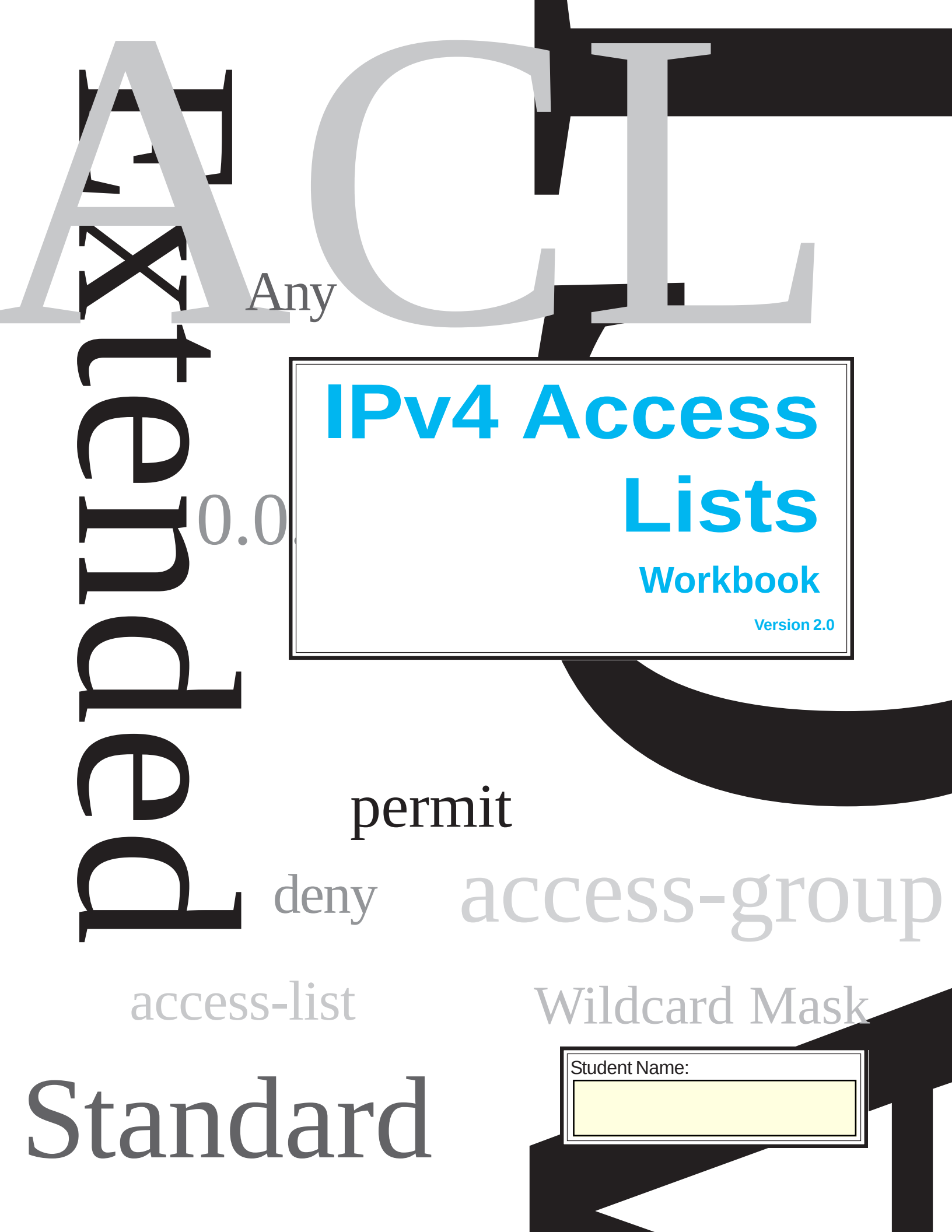
VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
					4-7	4-7
				16-31	8-15	8-11
					12-15	12-15
		64-127	32-63	32-47	16-23	16-19
					20-23	20-23
				48-63	24-31	24-27
					28-31	28-31
			64-95	64-79	32-39	32-35
					36-39	36-39
				80-95	40-47	40-43
					44-47	44-47
		96-127	96-111	96-103	48-55	48-51
					52-55	52-55
				104-111	56-59	56-59
					60-63	60-63
			112-127	112-119	64-67	64-67
					68-71	68-71
				120-127	72-75	72-75
					76-79	76-79
	128-255	128-191	128-159	128-143	80-87	80-83
					84-87	84-87
				144-159	88-91	88-91
					92-95	92-95
			160-191	160-175	96-99	96-99
					100-103	100-103
				176-191	104-107	104-107
					108-111	108-111
		192-255	192-223	192-207	112-115	112-115
					116-119	116-119
				208-223	120-123	120-123
					124-127	124-127
			224-255	224-239	128-131	128-131
					132-135	132-135
				240-255	136-143	136-139
					140-143	140-143
			224-239	224-231	144-147	144-147
					148-151	148-151
				232-239	152-155	152-155
					156-159	156-159
			240-255	240-247	160-163	160-163
					164-167	164-167
				248-255	168-171	168-171
					172-175	172-175
			224-239	224-231	176-179	176-179
					180-183	180-183
				232-239	184-187	184-187
					188-191	188-191
			224-239	224-231	192-195	192-195
					196-199	196-199
				232-239	200-203	200-203
					204-207	204-207
			224-239	224-231	208-211	208-211
					212-215	212-215
				232-239	216-219	216-219
					220-223	220-223
			224-239	224-231	224-227	224-227
					228-231	228-231
				232-239	232-235	232-235
					236-239	236-239
			224-239	224-231	240-243	240-243
					244-247	244-247
				232-239	248-251	248-251
					252-255	252-255

Class A Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/8	0	255.0.0.0	1	16,777,216	16,777,214
/9	1	255.128.0.0	2	8,388,608	8,388,606
/10	2	255.192.0.0	4	4,194,304	4,194,302
/11	3	255.224.0.0	8	2,097,152	2,097,150
/12	4	255.240.0.0	16	1,048,576	1,048,574
/13	5	255.248.0.0	32	524,288	524,286
/14	6	255.252.0.0	64	262,144	262,142
/15	7	255.254.0.0	128	131,072	131,070
/16	8	255.255.0.0	256	65,536	65,534
/17	9	255.255.128.0	512	32,768	32,766
/18	10	255.255.192.0	1,024	16,384	16,382
/19	11	255.255.224.0	2,048	8,192	8,190
/20	12	255.255.240.0	4,096	4,096	4,094
/21	13	255.255.248.0	8,192	2,048	2,046
/22	14	255.255.252.0	16,384	1,024	1,022
/23	15	255.255.254.0	32,768	512	510
/24	16	255.255.255.0	65,536	256	254
/25	17	255.255.255.128	131,072	128	126
/26	18	255.255.255.192	262,144	64	62
/27	19	255.255.255.224	524,288	32	30
/28	20	255.255.255.240	1,048,576	16	14
/29	21	255.255.255.248	2,097,152	8	6
/30	22	255.255.255.252	4,194,304	4	2

Class B Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/16	0	255.255.0.0	1	65,536	65,534
/17	1	255.255.128.0	2	32,768	32,766
/18	2	255.255.192.0	4	16,384	16,382
/19	3	255.255.224.0	8	8,192	8,190
/20	4	255.255.240.0	16	4,096	4,094
/21	5	255.255.248.0	32	2,048	2,046
/22	6	255.255.252.0	64	1,024	1,022
/23	7	255.255.254.0	128	512	510
/24	8	255.255.255.0	256	256	254
/25	9	255.255.255.128	512	128	126
/26	10	255.255.255.192	1,024	64	62
/27	11	255.255.255.224	2,048	32	30
/28	12	255.255.255.240	4,096	16	14
/29	13	255.255.255.248	8,192	8	6
/30	14	255.255.255.252	16,384	4	2

Class C Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/24	0	255.255.255.0	1	256	254
/25	1	255.255.255.128	2	128	126
/26	2	255.255.255.192	4	64	62
/27	3	255.255.255.224	8	32	30
/28	4	255.255.255.240	16	16	14
/29	5	255.255.255.248	32	8	6
/30	6	255.255.255.252	64	4	2



IPv4 Access Lists Workbook

Version 2.0

Standard

Student Name:

Access-List Numbers

IP Standard	1	to	99
IP Extended	100	to	199
Ethernet Type Code	200	to	299
Ethernet Address	700	to	799
DECnet and Extended DECnet	300	to	399
XNS	400	to	499
Extended XNS	500	to	599
Appletalk	600	to	699
48-bit MAC Addresses	700	to	799
IPX Standard	800	to	899
IPX Extended	900	to	999
IPX SAP (service advertisement protocol)	1000	to	1099
IPX SAP SPX	1000	to	1099
Extended 48-bit MAC Addresses	1100	to	1199
IPX NLSP	1200	to	1299
IP Standard, expanded range	1300	to	1999
IP Extended, expanded range	2000	to	2699
SS7 (voice)	2700	to	2999
Standard Vines	1	to	100
Extended Vines	101	to	200
Simple Vines	201	to	300
Transparent bridging (protocol type)	200	to	299
Transparent bridging (vendor type)	700	to	799
Extended Transparent bridging	1100	to	1199
Source-route bridging (protocol type)	200	to	299
Source-route bridging (vendor type)	700	to	799

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for taking the time to check this workbook for errors, and making suggestions for improvements.

Instructors (and anyone else for that matter) please do not post the Instructors version on public websites.
When you do this you are giving everyone else worldwide the answers. Yes, students look for answers this way.
It also discourages others; myself included, from posting high quality materials.

What are Access Control Lists?

ACLs...

...are a sequential list of instructions that tell a router which packets to permit or deny.

General Access Lists Information

Access Lists...

...are read sequentially.

...are set up so that as soon as the packet matches a statement it stops comparing and permits or denies the packet.

...need to be written to take care of the most abundant traffic first.

...must be configured on your router before you can deny packets.

...can be written for all supported routed protocols; but each routed protocol must have a different ACL for each interface.

...must be applied to an interface to work.

How routers use Access Lists

(Outbound Port - Default)

- ❑ The router checks to see if the packet is routable. If it is it looks up the route in its routing table.
- ❑ The router then checks for an ACL on that outbound interface.
- ❑ If there is no ACL the router switches the packet out that interface to its destination.
- ❑ If there is an ACL the router checks the packet against the access list statements sequentially. Then permits or denies each packet as it is matched.
- ❑ If the packet does not match any statement written in the ACL it is denied because there is an implicit “deny any” statement at the end of every ACL.

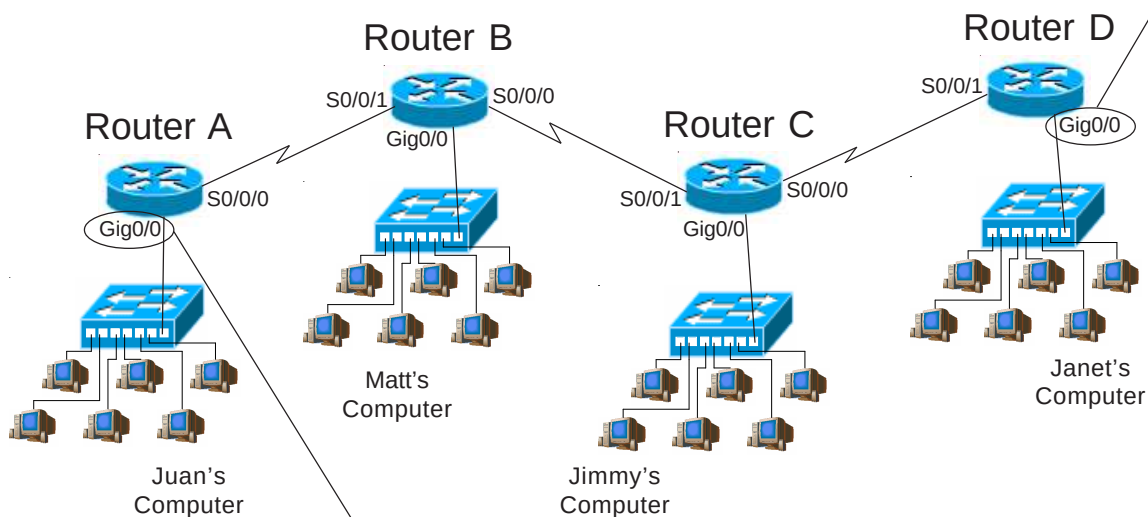
Standard Access Lists

Standard Access Lists...

- ...are numbered from 1 to 99 or 1300 to 1999.
- ...filter (permit or deny) only source addresses.
- ...do not have any destination information so it must be placed as close to the destination as possible.
- ...work at layer 3 of the OSI model.

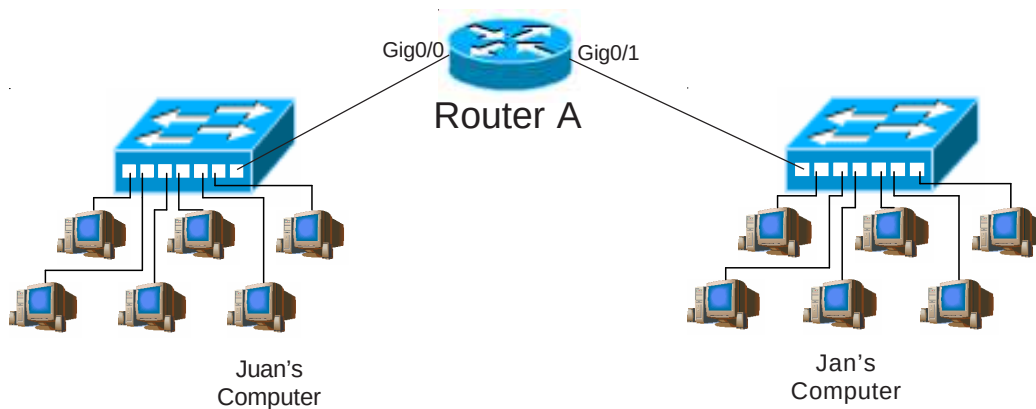
Why standard ACLs are placed close to the destination.

If you want to block traffic from Juan's computer from reaching Janet's computer with a standard access list you would place the ACL close to the destination on Router D, interface Gig0/0. Since it is using only the source address to permit or deny packets the ACL here will not effect packets reaching Routers B, or C.

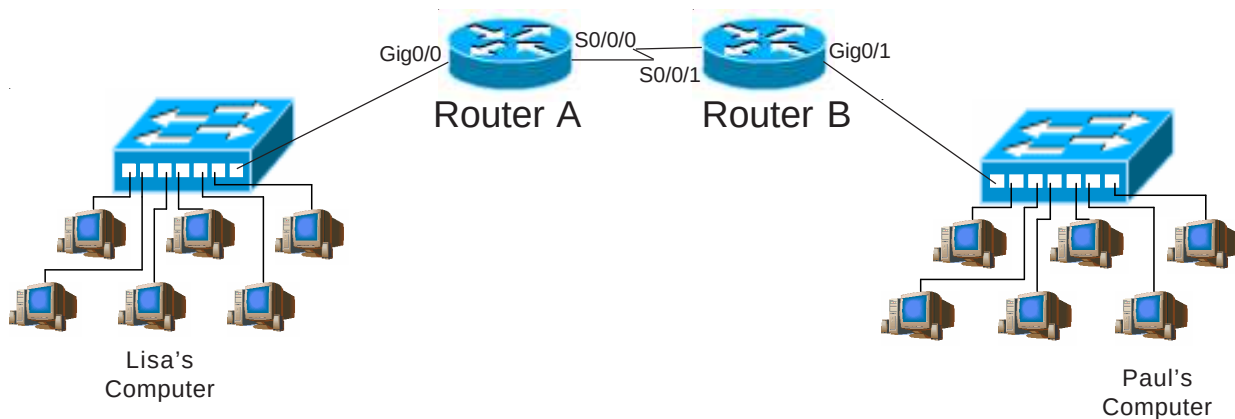


If you place the ACL on router A to block traffic to Router D it will also block all packets going to Routers B, and C; because all the packets will have the same source address.

Standard Access List Placement Sample Problems



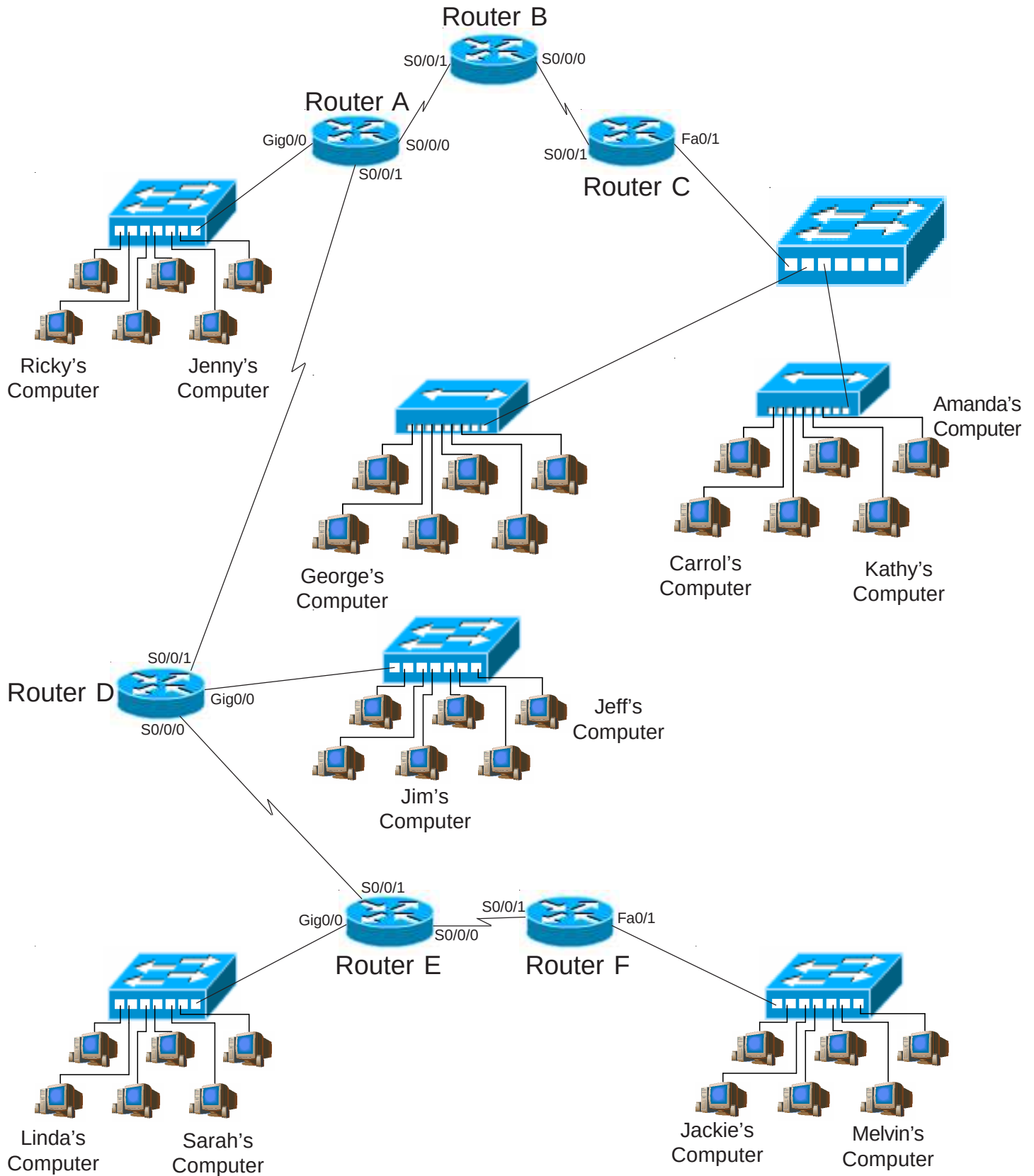
In order to permit packets from Juan's computer to arrive at Jan's computer you would place the standard access list at router interface Gig0/1.



Lisa has been sending unnecessary information to Paul. Where would you place the standard ACL to deny all traffic from Lisa to Paul?
Router Name Router B Interface Gig0/1

Where would you place the standard ACL to deny traffic from Paul to Lisa?
Router Name Router A Interface Gig0/0

Standard Access List Placement



Standard Access List Placement

1. Where would you place a standard access list to permit traffic from Ricky's computer to reach Jeff's computer?

Router Name Router D
Interface Gig0/0

2. Where would you place a standard access list to deny traffic from Melvin's computer from reaching Jenny's computer?

Router Name Router A
Interface Gig0/0

3. Where would you place a standard access list to deny traffic to Carrol's computer from Sarah's computer?

Router Name
Interface

4. Where would you place a standard access list to permit traffic to Ricky's computer from Jeff's computer?

Router Name
Interface

5. Where would you place a standard access list to deny traffic from Amanda's computer from reaching Jeff and Jim's computer?

Router Name
Interface

6. Where would you place a standard access list to permit traffic from Jackie's computer to reach Linda's computer?

Router Name
Interface

7. Where would you place a standard access list to permit traffic from Ricky's computer to reach Carrol and Amanda's computer?

Router Name
Interface

8. Where would you place a standard access list to deny traffic to Jenny's computer from Jackie's computer?

Router Name
Interface

9. Where would you place a standard access list to permit traffic from George's computer to reach Linda and Sarah's computer?

Router Name
Interface

10. Where would you place an ACL to deny traffic from Jeff's computer from reaching George's computer?

Router Name
Interface

11. Where would you place a standard access list to deny traffic to Sarah's computer from Ricky's computer?

Router Name
Interface

12. Where would you place an ACL to deny traffic from Linda's computer from reaching Jackie's computer?

Router Name
Interface

Extended Access Lists

Extended Access Lists...

...are numbered from 100 to 199 or 2000 to 2699.

...filter (permit or deny) based on the:

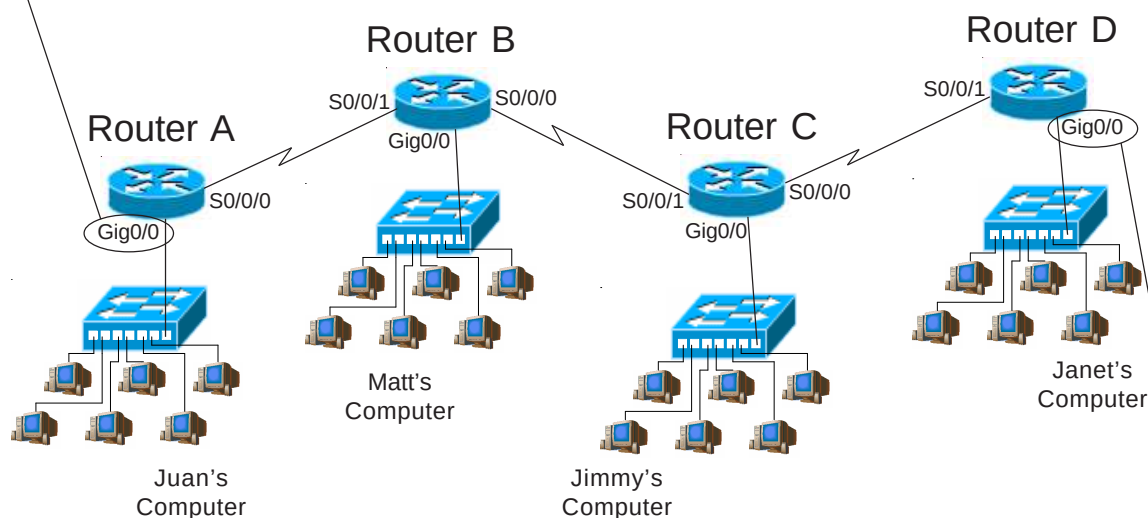
- source address
- destination address
- protocol
- application / port number

... are placed close to the source.

...work at both layer 3 and 4 of the OSI model.

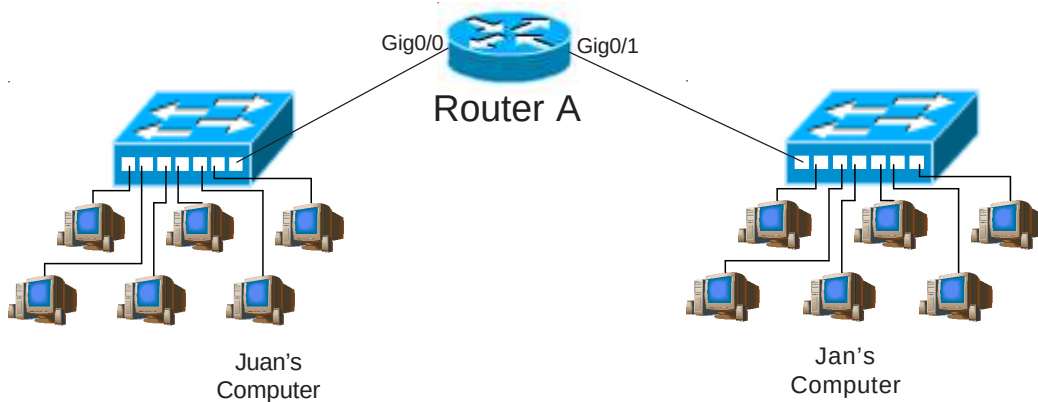
Why extended ACLs are placed close to the source.

If you want to deny traffic from Juan's computer from reaching Janet's computer with an extended access list you would place the ACL **close to the source** on Router A, interface Gig0/0. Since it can permit or deny based on the destination address it can reduce backbone overhead and not affect traffic to Routers B or C.

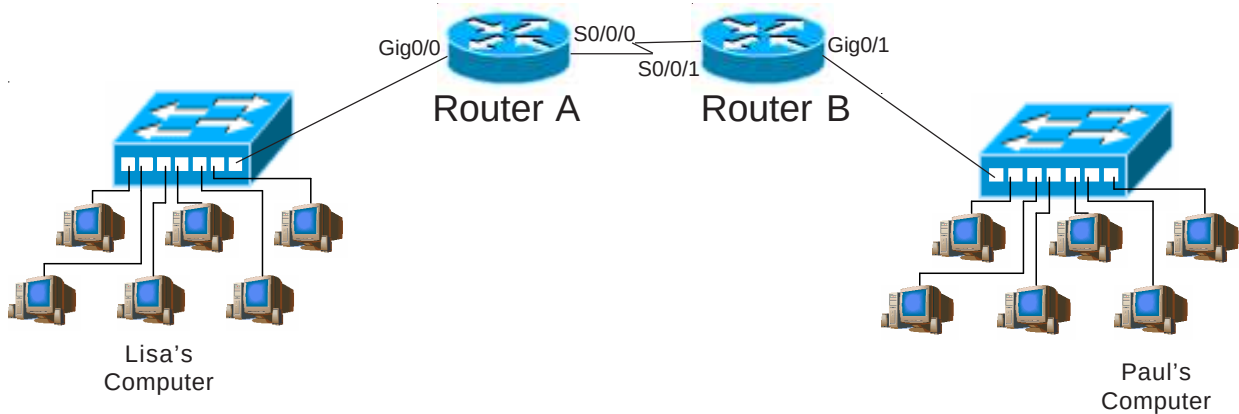


If you place the ACL on Router D to block traffic from Router A, it will work. However, Routers B and C will have to route the packet before it is finally blocked at Router D. This increases the volume of useless network traffic.

Extended Access List Placement Sample Problems



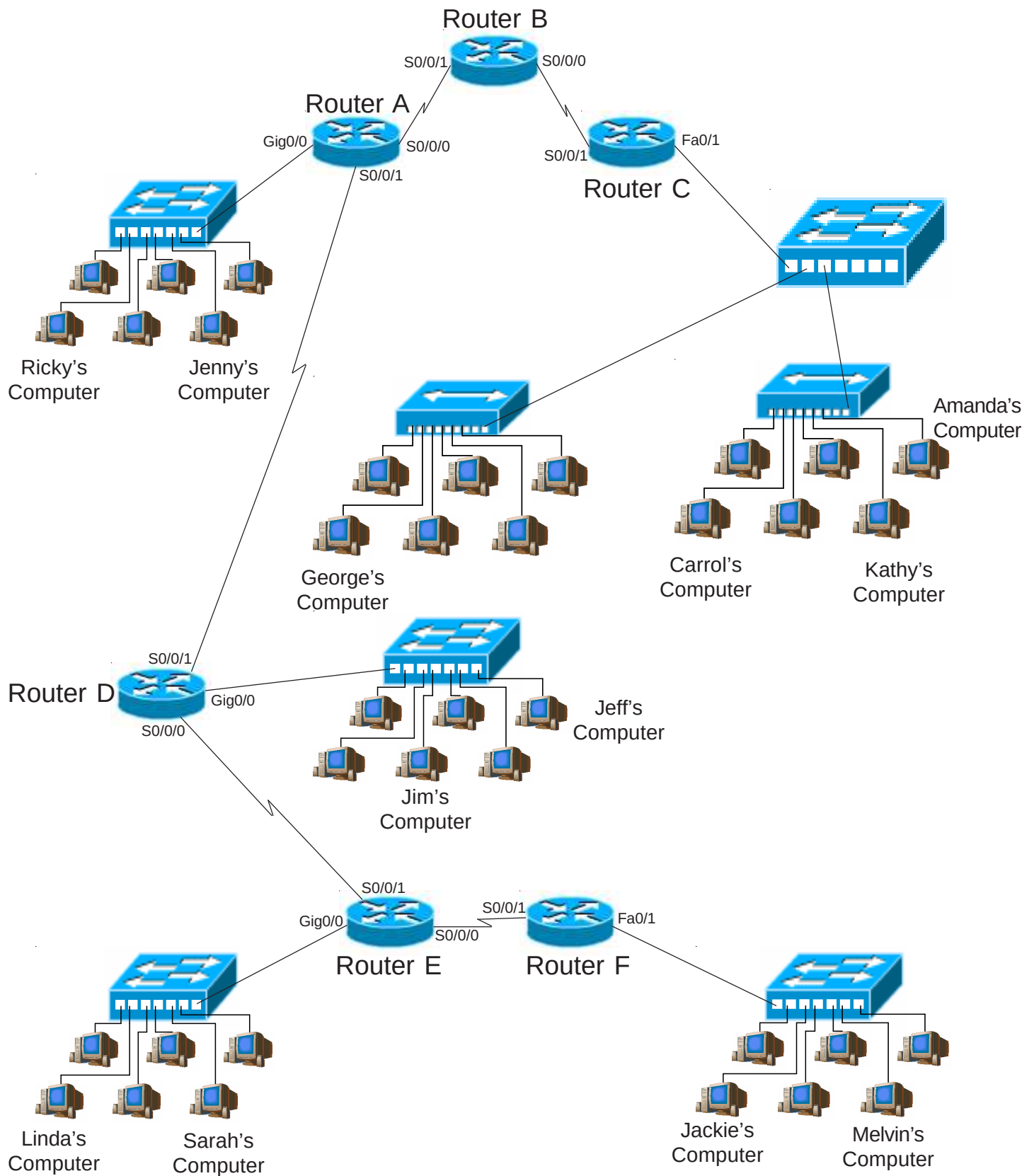
In order to permit packets from Juan's computer to arrive at Jan's computer you would place the extended access list at router interface Gig0/0.



Lisa has been sending unnecessary information to Paul. Where would you place the extended ACL to deny all traffic from Lisa to Paul?
Router Name Router A Interface Gig0/0

Where would you place the extended ACL to deny traffic from Paul to Lisa?
Router Name Router B Interface Gig0/1

Extended Access List Placement



Extended Access List Placement

1. Where would you place an ACL to deny traffic from Jeff's computer from reaching George's computer?

Router Name Router D
Interface Gig0/0

2. Where would you place an extended access list to permit traffic from Jackie's computer to reach Linda's computer?

Router Name Router F
Interface FA0/1

3. Where would you place an extended access list to deny traffic to Carrol's computer from Ricky's computer?

Router Name
Interface

4. Where would you place an extended access list to deny traffic to Sarah's computer from Jackie's computer?

Router Name
Interface

5. Where would you place an extended access list to permit traffic from Carrol's computer to reach Jeff's computer?

Router Name
Interface

6. Where would you place an extended access list to deny traffic from Melvin's computer from reaching Jeff and Jim's computer?

Router Name
Interface

7. Where would you place an extended access list to permit traffic from George's computer to reach Jeff's computer?

Router Name
Interface

8. Where would you place an extended access list to permit traffic from Jim's computer to reach Carrol and Amanda's computer?

Router Name
Interface

9. Where would you place an ACL to deny traffic from Linda's computer from reaching Kathy's computer?

Router Name
Interface

10. Where would you place an extended access list to deny traffic to Jenny's computer from Sarah's computer?

Router Name
Interface

11. Where would you place an extended access list to permit traffic from George's computer to reach Linda and Sarah's computer?

Router Name
Interface

12. Where would you place an extended access list to deny traffic from Linda's computer from reaching Jenny's computer?

Router Name
Interface

Choosing to Filter Incoming or Outgoing Packets

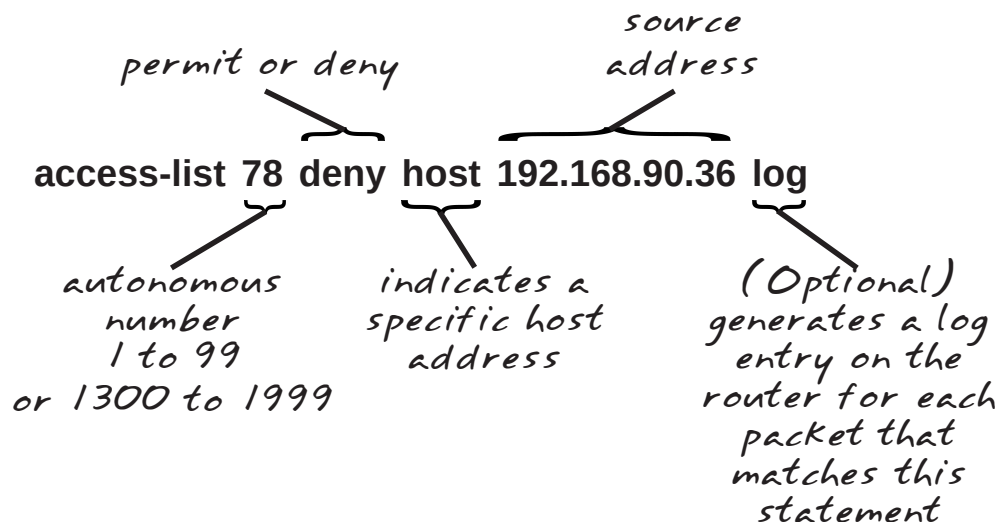
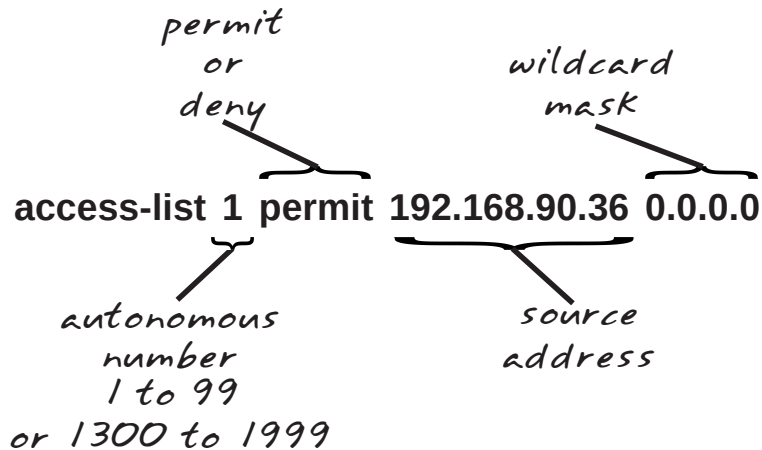
Access Lists on your incoming port...

- ...requires less CPU processing.
- ...filters and denies packets before the router has to make a routing decision.

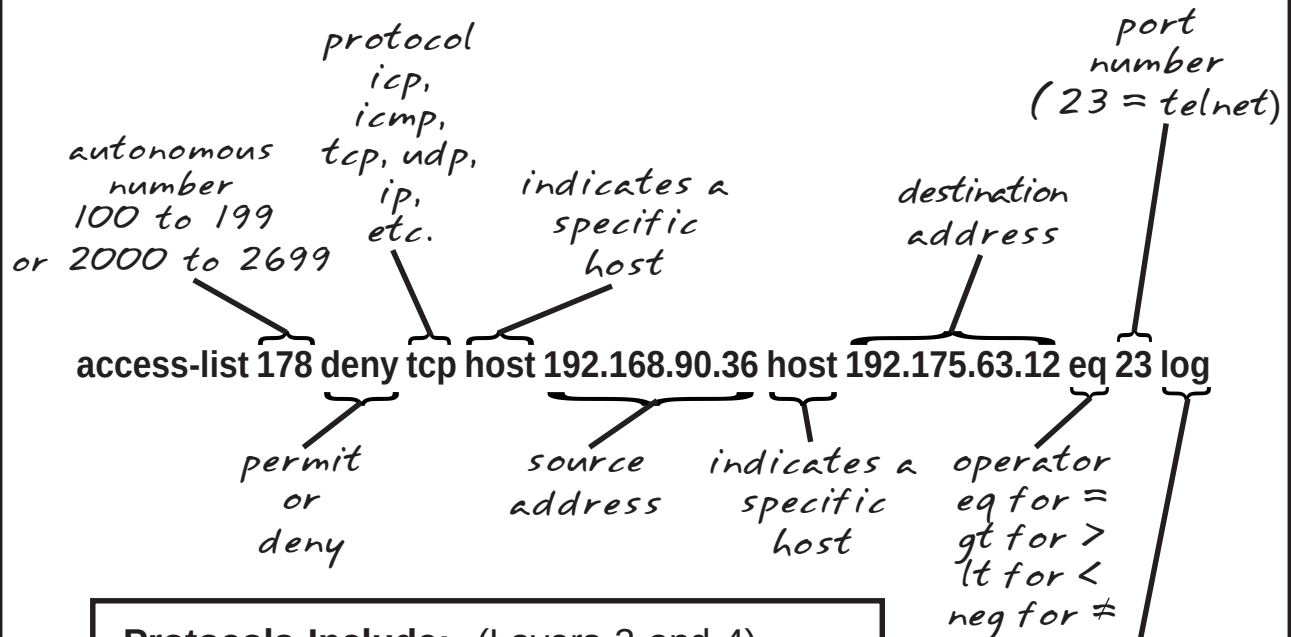
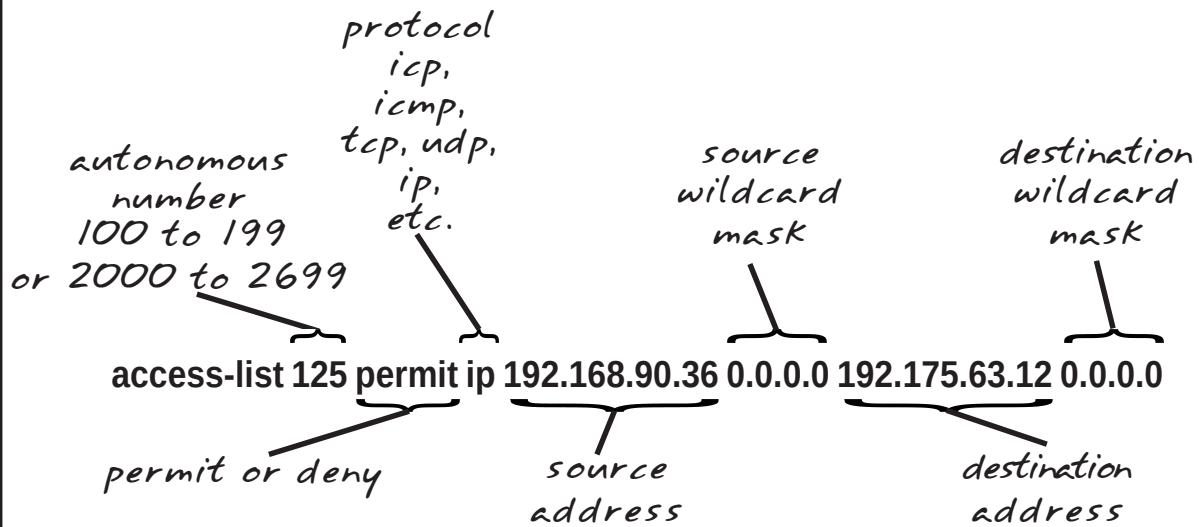
Access Lists on your outgoing port...

- ...are outbound by default unless otherwise specified.
- ...increases the CPU processing time because the routing decision is made and the packet switched to the correct outgoing port before it is tested against the ACL.

Breakdown of a Standard ACL Statement



Breakdown of an Extended ACL Statement



Protocols Include: (Layers 3 and 4)

IP	IGMP	IPINIP
TCP	GRE	OSPF
UDP	IGRP	NOS
ICMP	EIGRP	Integer 0-255

To match any internet protocol use IP.

What are Named Access Control Lists?

Named ACLs...

...are standard or extended ACLs which have an alphanumeric name instead of a number. (ie. 1-99 or 100-199)

Named Access Lists Information

Named Access Lists...

- ...identify ACLs with an intuitive name instead of a number.
- ...eliminate the limits imposed by using numbered ACLs.
(798 for standard and 799 for extended)
- ...names should be typed in all CAPS to make it easier to see.
- ...provide the ability to modify your ACLs without deleting and reloading the revised access list. It will only allow you to add statements to the end of the existing statements.
- ...are not compatible with any IOS prior to Release 11.2.
- ...can not repeat the same name on multiple ACLs.

Applying a Standard Named Access List called "GEORGE"

Write a named standard access list called "GEORGE" on Router A, interface E1 to block Melvin's computer from sending information to Kathy's computer; but will allow all other traffic.

Place the access list at:

Router Name: Router A

Interface: E1

Access-list Name: GEORGE

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# ip access-list standard GEORGE
Router(config-std-nacl)# deny host 72.16.70.35
Router(config-std-nacl)# permit any
Router(config-std-nacl)# interface gig0/1
Router(config-if)# ip access-group george out
Router(config-if)# exit
Router(config)# exit
```

Applying an extended Named Access List called "GRACIE"

Write a named extended access list called "GRACIE" on Router A, Interface E0 called "Gracie" to deny HTTP traffic intended for web server 192.168.207.27, but will permit all other HTTP traffic to reach the only the 192.168.207.0 network. Deny all other IP traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: E0
Access-list Name: GRACIE

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# ip access-list extended GRACIE
Router(config-ext-nacl)# deny tcp any host 192.168.207.27 eq www
Router(config-ext-nacl)# permit tcp any 192.168.207.0 0.0.0.255 eq www
Router(config-ext-nacl)# interface gig0/1
Router(config-if)# ip access-group gracie in
Router(config-if)# exit
Router(config)# exit
```

Choices for Using Wildcard Masks

Wildcard masks are usually set up to do one of four things:

1. Match a specific host.
2. Match an entire subnet.
3. Match a specific range.
4. Match all addresses.

1. Matching a specific host.

For standard access lists:

Access-List 10 permit 192.168.150.50 0.0.0.0

or

Access-List 10 permit 192.168.150.50 (standard ACL's
assume a 0.0.0.0 mask)

or

Access-List 10 permit host 192.168.150.50

For extended access lists:

Access-list 110 deny ip 192.168.150.50 0.0.0.0 any

or

Access-list 110 deny ip host 192.168.150.50 any

2. Matching an entire subnet

Example 1

Address: 192.168.50.0 Subnet Mask: 255.255.255.0

Access-list 25 deny 192.168.50.0 0.0.0.255

Example 2

Address: 172.16.0.0 Subnet Mask: 255.255.0.0

Access-list 12 permit 172.16.0.0 0.0.255.255

Example 3

Address: 10.0.0.0 Subnet Mask: 255.0.0.0

Access-list 125 deny udp 10.0.0.0 0.255.255.255 any

3. Match a specific range

Example 1

Address: 10.250.50.112 Subnet Mask: 255.255.255.224

255.255.255.255
Custom Subnet mask: -255.255.255.224
Wildcard: 0. 0. 0. 31

Access-list 125 permit udp 10.250.50.112 0.0.0.31 any

Example 2

Address Range: 192.168.16.0 to 192.168.16.127

192.168.16.127
-192.168.16. 0
Wildcard: 0. 0. 0.127

Access-list 125 deny ip 192.168.16.0 0.0.0.127 any
(This ACL would block the lower half of the subnet.)

Example 3

Address: 172.250.16.32 to 172.250.31.63

172.250.31. 63
-172.250.16. 32
Wildcard: 0. 0.15. 31

Access-list 125 permit ip 172.250.16.32 0.0.15.31 any

4. Match everyone.

For standard access lists:

Access-List 15 permit any

or

Access-List 15 deny 0.0.0.0 255.255.255.255

For extended access lists:

Access-List 175 permit ip any any

or

Access-List 175 deny tcp 0.0.0.0 255.255.255.255 any

Creating Wildcard Masks

- ❑ Just like a subnet mask the wildcard mask tells the router what part of the address to check or ignore. Zero (0) must match exactly, one (1) will be ignored.
- ❑ The source address can be a single address, a range of addresses, or an entire subnet.
- ❑ As a rule of thumb the wildcard mask is the inverse of the subnet mask.

Example #1:

IP Address and subnet mask: 204.100.100.0 255.255.255.0

IP Address and wildcard mask: 204.100.100.0 0.0.0.255

- ❑ All zero's (or 0.0.0.0) means the address must match exactly.

Example #2:

10.10.150.95 0.0.0.0 (This address must match exactly.)

- ❑ One's will be ignored.

Example #3:

10.10.150.95 0.0.0.255 (Any 10.10.150.0 subnet address will match.
10.10.150.0 to 10.10.150.255)

- ❑ This also works with subnets.

Example #4:

IP Address and subnet mask: 192.170.25.30 255.255.255.224

IP Address and wildcard mask: 192.170.25.30 0.0.0.31
(Subtract the subnet mask from
255.255.255.255 to create the wildcard)

Do the math... $255 - 255 = 0$ (This is the inverse of the subnet mask.)
 $255 - 224 = 31$

Example #5:

IP Address and subnet mask: 172.24.128.0 255.255.128.0

IP Address and wildcard mask: 172.24.128.0 0.0.127.255

Do the math... $255 - 255 = 0$ (This is the inverse of the subnet mask.)
 $255 - 128 = 127$
 $255 - 0 = 255$

Wildcard Mask Problems

1. Create a wildcard mask to match this exact address.
IP Address: 192.168.25.70
Subnet Mask: 255.255.255.0 0 . 0 . 0 . 0
2. Create a wildcard mask to match this range.
IP Address: 210.150.10.0
Subnet Mask: 255.255.255.0 0 . 0 . 0 . 255
3. Create a wildcard mask to match this host.
IP Address: 195.190.10.35
Subnet Mask: 255.255.255.0
4. Create a wildcard mask to match this range.
IP Address: 172.16.0.0
Subnet Mask: 255.255.0.0
5. Create a wildcard mask to match this range.
IP Address: 10.0.0.0
Subnet Mask: 255.0.0.0
6. Create a wildcard mask to match this exact address.
IP Address: 165.100.0.130
Subnet Mask: 255.255.255.192
7. Create a wildcard mask to match this range.
IP Address: 192.10.10.16
Subnet Mask: 255.255.255.224
8. Create a wildcard mask to match this range.
IP Address: 171.50.75.128
Subnet Mask: 255.255.255.192
9. Create a wildcard mask to match this host.
IP Address: 10.250.30.2
Subnet Mask: 255.0.0.0
10. Create a wildcard mask to match this range.
IP Address: 210.150.28.16
Subnet Mask: 255.255.255.240
11. Create a wildcard mask to match this range.
IP Address: 172.18.0.0
Subnet Mask: 255.255.224.0
12. Create a wildcard mask to match this range.
IP Address: 135.35.230.32
Subnet Mask: 255.255.255.248

Basic Wildcard Mask Problems

Based on the given information list the range of source addresses for each ACE statement.

1. access-list 10 permit 192.168.150.50 0.0.0.0

Answer: 192.168.150.50

2. access-list 5 permit any

Answer: Any address

3. access-list 125 deny tcp 195.223.50.0 0.0.0.63 host 172.168.10.1 fragments

Answer: 195.223.50.0 to 195.223.50.63

4. access-list 11 deny 210.10.10.0 0.0.0.255

Answer:

5. access-list 108 deny ip 192.220.10.0 0.0.0.15 172.32.4.0 0.0.0.255

Answer:

6. access-list 171 deny any host 175.18.24.10 fragments

Answer:

7. access-list 105 permit 192.168.15.0 0.0.0.255 any

Answer:

8. access-list 109 permit tcp 172.16.10.0 0.0.0.255 host 192.168.10.1 eq 80

Answer:

9. access-list 111 permit ip any any

Answer:

10. access-list 195 permit udp 172.30.12.0 0.0.0.127 172.50.10.0 0.0.0.255

Answer:

11. **access-list 110 permit ip 192.168.15.0 0.0.0.3 192.168.30.10 0.0.0.0**

Answer:

12. **access-list 120 permit ip 192.168.15.0 0.0.0.7 192.168.30.10 0.0.0.0**

Answer:

13. **access-list 130 permit ip 192.168.15.0 0.0.0.15 192.168.30.10 0.0.0.0**

Answer:

14. **access-list 140 permit ip 192.168.15.0 0.0.0.31 192.168.30.10 0.0.0.0**

Answer:

15. **access-list 150 permit ip 192.168.15.0 0.0.0.63 192.168.30.10 0.0.0.0**

Answer:

16. **access-list 101 Permit ip 192.168.15.0 0.0.0.127 192.168.30.10 0.0.0.0**

Answer:

17. **access-list 185 permit ip 192.168.15.0 0.0.0.255 192.168.30.0 0.0.0.255**

Answer:

18. **access-list 160 deny udp 172.16.0.0 0.0.1.255 172.18.10.18 0.0.0.0 gt 22**

Answer:

19. **access-list 195 permit icmp 172.85.0.0 0.0.15.255 172.50.10.0 0.0.0.255**

Answer:

20. **access-list 10 permit 175.15.120.0 0.0.0.255**

Answer:

21. **access-list 190 permit tcp 192.15.10.0 0.0.0.31 any**

Answer:

22. **access-list 100 permit ip 10.0.0.0 0.255.255.255 172.50.10.0 0.0.0.255**

Answer:

(Slightly More) **Advanced Wildcard Mask Problems**

Not every address range ends with a zero making it simple to determine the address range. Using basic subnetting you can adjust the host portion of the IP address with the wildcard mask to indicate specific ranges.

IP Address: 192 . 100 . 10 . 0
Custom Subnet Mask: 255.255.255.240

Address Ranges: 192.100.10.0 to 192.100.10.15
192.100.10.16 to 192.100.10.31
192.100.10.32 to 192.100.10.47 (Range in the sample below)
192.100.10.48 to 192.100.10.63
192.100.10.64 to 192.100.10.79
192.100.10.80 to 192.100.10.95
192.100.10.96 to 192.100.10.111
192.100.10.112 to 192.100.10.127
192.100.10.128 to 192.100.10.143
192.100.10.144 to 192.100.10.159
192.100.10.160 to 192.100.10.175
192.100.10.176 to 192.100.10.191
192.100.10.192 to 192.100.10.207
192.100.10.208 to 192.100.10.223
192.100.10.224 to 192.100.10.239
192.100.10.240 to 192.100.10.255

Based on this sample to deny or permit the third subnet we would create the correct wildcard mask by inverting the subnet mask from 255.255.255.240 to 0.0.0.15. The third address range would be 192.100.10.32 to 192.100.10.47 .

A standard ACE statement would be written as:

access-list 10 permit 192.100.10.32 0.0.0.15

The wildcard mask indicates that address range 0 to 15 will be permitted. If the source address was 192.100.10.0 it would indicate the first 16 addresses (0 to 15). Since the source address is 192.100.10.32, it indicates that the address range will be from 192.100.10.32 to 192.100.10.47 (32 to 47). The IP address indicates the starting point for the wildcard mask to begin counting up from.

This technique gives you greater control with the addresses you want to permit or deny. It is a **Best Practice** to stay within standard address ranges.

Based on the given information list the range of source addresses for each ACE statement.

1. access-list 10 permit 192.100.10.64 0.0.0.15

Answer: 192.100.10.64 to 192.100.10.79

2. access-list 5 permit 172.16.128.0 0.0.63.255

Answer: 172.16.128.0 to 172.16.191.255

3. access-list 125 deny tcp 192.100.10.208 0.0.0.15 host 192.168.10.1 log

Answer:

4. access-list 11 deny 210.48.72.192 0.0.0.31

Answer:

5. access-list 108 deny ip 192.168.5.184 0.0.0.7 192.64.4.0 0.0.0.255

Answer:

6. access-list 171 deny 172.32.128.0 0.0.15.255 any

Answer:

7. access-list 105 permit 165.50.196.0 0.0.3.255 any

Answer:

8. access-list 109 permit tcp 172.16.128.0 0.0.127.255 host 172.16.20.1

Answer:

9. access-list 111 permit ip 192.168.1.236 0.0.0.3 any

Answer:

10. access-list 195 permit udp 172.32.160.0.0 0.0.31.255 172.100.10.0 0.0.0.255

Answer:

11. access-list 110 permit ip 10.64.0.0 0.63.255.255 172.168.40.10 0.0.0.0

Answer:

Wildcard Mask Problems

Based on the given information list the range of destination addresses for each ACE statement.

1. **access-list 125 deny tcp 195.223.50.0 0.0.0.63 host 172.168.10.1 fragments**

Answer: 172.168.10.1

2. **access-list 115 permit any any**

Answer: Any address

3. **access-list 150 permit ip 192.168.30.10 0.0.0.0 192.168.15.0 0.0.0.63**

Answer: 192.168.15.0 to 192.168.15.63

4. **access-list 120 deny tcp 172.32.4.0 0.0.0.255 192.220.10.0 0.0.0.15**

Answer:

5. **access-list 108 deny ip 192.220.10.0 0.0.0.15 172.32.4.0 0.0.0.255**

Answer:

6. **access-list 101 deny ip 140.130.110.100 0.0.0.0 0.0.0.0 255.255.255.255**

Answer:

7. **access-list 105 permit any 192.168.15.0 0.0.0.255**

Answer:

8. **access-list 120 permit ip 192.168.15.10 0.0.0.0 172.16.40.0 0.0.3.255**

Answer:

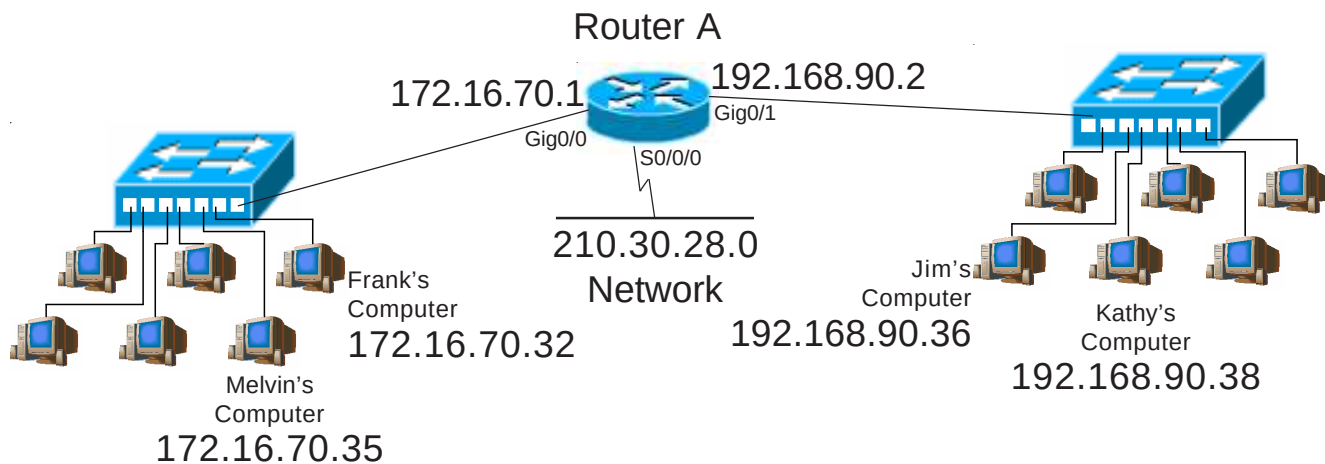
9. **access-list 160 deny udp 172.16.0.0 0.0.1.255 172.18.104.0 0.0.7.255 eq 21**

Answer:

10. **access-list 150 permit ip 192.168.30.0 0.0.0.63 192.168.15.96 0.0.0.31**

Answer:

Writing Standard Access Lists...



Standard Access List Sample #1

Write a standard access list to block Melvin's computer from sending information to Kathy's computer; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: Gig0/1
 Access-list #: 10

Note:

A standard access list blocks only the source address. Melvin will also be blocked from sending information to Jim or anyone else on the 192.168.90.0 network.

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 10 deny 172.16.70.35
                        or
                        access-list 10 deny 172.16.70.35 0.0.0.0
                        or
                        access-list 10 deny host 172.16.70.35
Router(config)# access-list 10 permit 0.0.0.0 255.255.255.255
                        or
                        access-list 10 permit any
Router(config)# interface gig0/1
Router(config-if)# ip access-group 10 out
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

Router# *show configuration*

(This will show which access groups are associated with particular interfaces)

Router# *show access list 10*

(This will show detailed information about this ACL)

Standard Access List Sample #2

Include a remark with each statement of your ACL. Write a standard access list to block Jim's computer from sending information to Frank's computer; but will allow all other traffic from the 192.168.90.0 network. Permit all traffic from the 210.30.28.0 network to reach the 172.16.70.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 28

[Writing and installing an ACL]

```
Router# configure terminal
Router(config)# access-list 28 remark Block Jim from reaching Frank
Router(config)# access-list 28 deny 192.168.90.36
                        or
                    access-list 28 deny 192.168.90.36 0.0.0.0
                        or
                    access-list 28 deny host 192.168.90.36
Router(config)# access-list 28 remark Allow all other traffic
Router(config)# access-list 28 permit 192.168.90.0 0.0.0.255
Router(config)# access-list 28 remark Allow all traffic
Router(config)# access-list 28 permit 210.30.28.0 0.0.0.255
Router(config)# interface Gig0/0
Router(config-if)# ip access-group 28 out
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Remark Command]

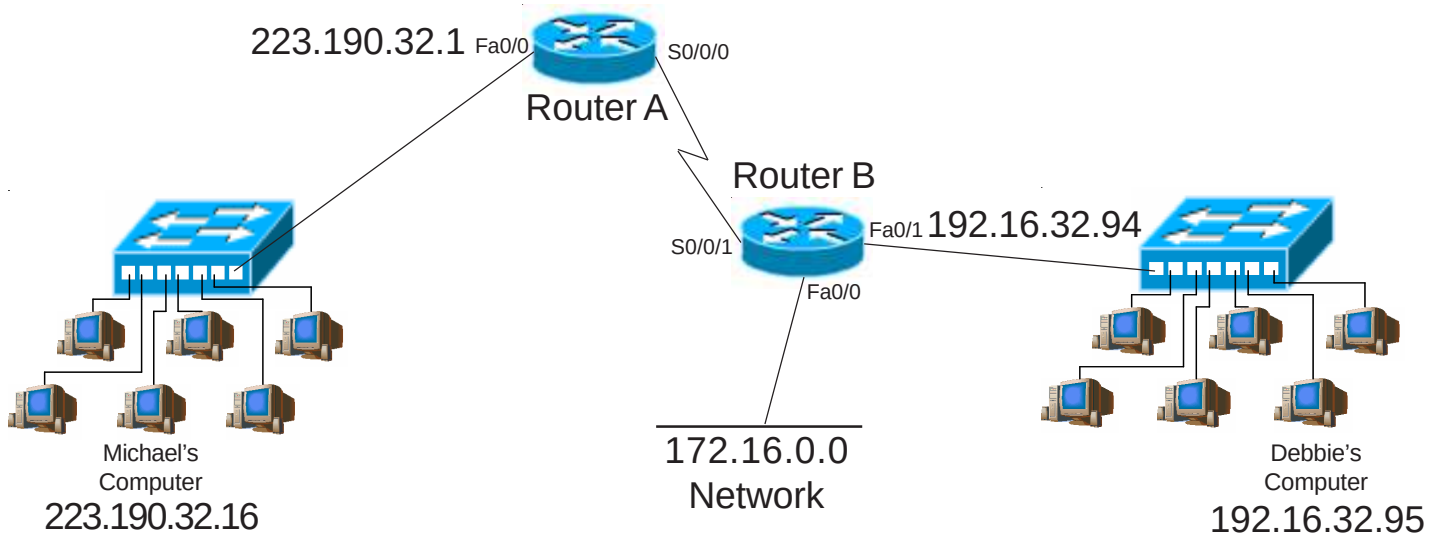
The remark command allows you to place text within the ACL so it can be viewed after it is inserted on the router. It can be viewed using the show run or any command that lists the ACL.

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 28 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 28 out
Router(config-if)# exit
Router(config)# no access-list 28
Router(config)# exit
```



Standard Access List Problem #1

Write a standard access list to block Debbie's computer from receiving information from Michael's computer; but will allow all other traffic. List all the command line options for this problem. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:

Interface:

Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #2

Write a standard access list to permit Debbie's computer to receive information from Michael's computer; but will deny all other traffic from the 223.190.32.0 network. Block all traffic from the 172.16.0.0 network. Permit all other traffic. List all the command line options for this problem. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
Interface:
Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)#

Router(config)#

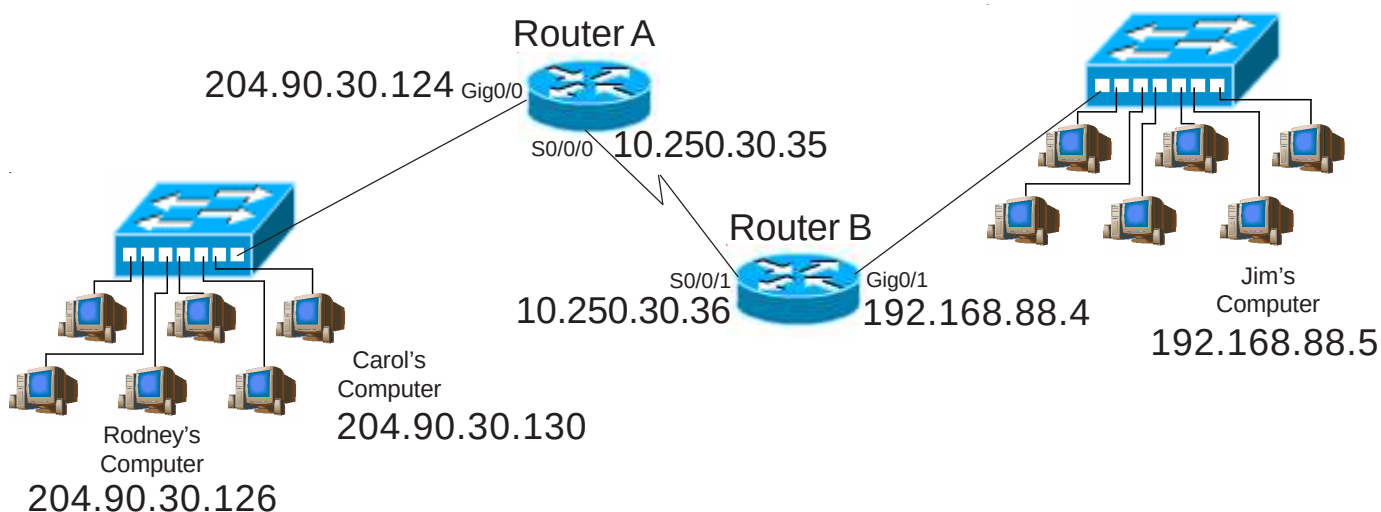
Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*



Standard Access List Problem #3

Write a standard access list to block Rodney and Carol's computer from sending information to Jim's computer; but will allow all other traffic from the 204.90.30.0 network. Block all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #4

Include a remark with each statement of your ACL. Using a minimum number of commands write a standard access list named "Ralph" to block Carol's computer from sending information to Jim's computer; but will permit Jim to receive data from Rodney. Block the upper half of the 204.90.30.0 range from reaching Jim's computer while permitting the lower half of the range. Block all other traffic. For help with blocking the upper half of the range review page 13 or the wildcard mask problems on pages 16 and 17. For help with named ACLs review pages 12 and 13. For help with the remark command review page 23.

Place the access list at:

Router Name:

Interface:

Access-list Name:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

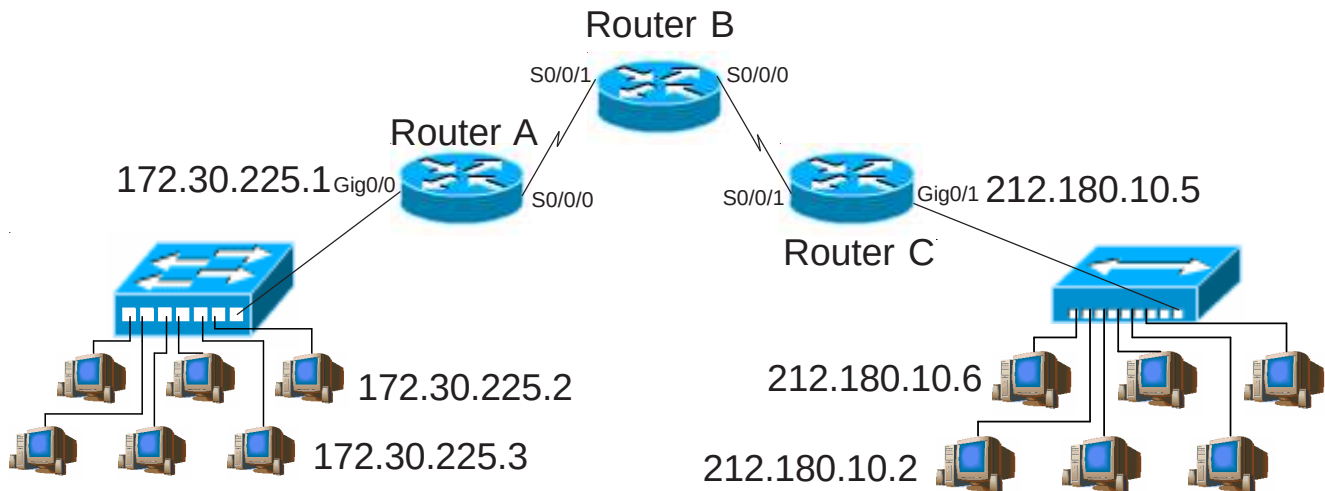
Router(config-std-nacl)#

Router(config-std-nacl)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*



Standard Access List Problem #5

Write a standard access list to block 172.30.225.2 and 172.30.225.3 from sending information to the 212.180.10.0 network; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #6

Add a remark to each statement explaining its purpose. Write a standard access list to block and log 212.180.10.2 from sending information to the 172.30.225.0 network. Permit and log 212.180.10.6 to send data to the 172.30.225.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written. Check the example on page 10 for help with the logging option. For help with the remark command review page 23.

Place the access list at:

Router Name:
Interface:
Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

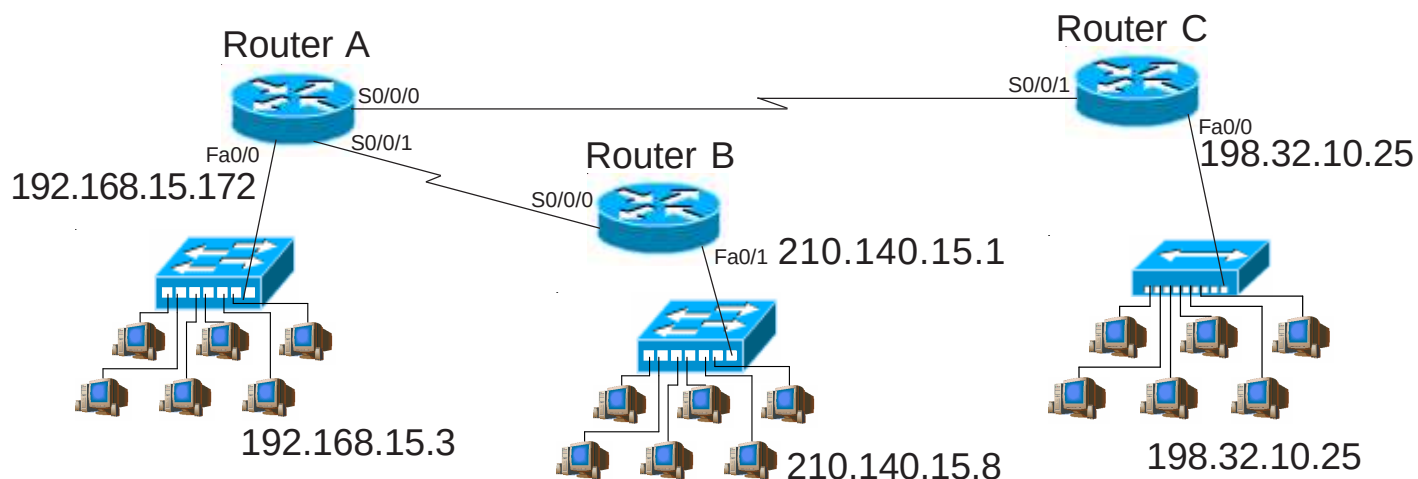
Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*



Standard Access List Problem #7

Write a standard access list to block the addresses 192.168.15.1 to 192.168.15.31 from sending information to the 210.140.15.0 network. Do not permit any traffic from 198.32.10.25 to reach the 210.140.15.0 network. Permit all other traffic. For help with this problem review page 13 or the wildcard mask problems on pages 16 and 17.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #8

Write a standard named access list called "CISCO_LAB_A" to permit traffic from the lower half of the 198.32.10.0 network to reach 192.168.15.0 network; block the upper half of the addresses. Allow host 198.32.10.192 to reach network 192.168.15.0. Permit all other traffic. For help with this problem review page 13 or the wildcard masks problems on pages 16 and 17. For assistance with named ACLs review pages 12 and 13.

Place the access list at:

Router Name:

Interface:

Access-list Name:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config-std-nacl)#

Router(config-std-nacl)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Standard Access List Problem #9

Write a standard access list to block network 192.168.255.0 from receiving information from the following addresses: 10.250.1.1, 10.250.2.1, 10.250.4.1, and the entire 10.250.3.0 255.255.255.0 network. Allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A

Interface: Fa0/0

Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

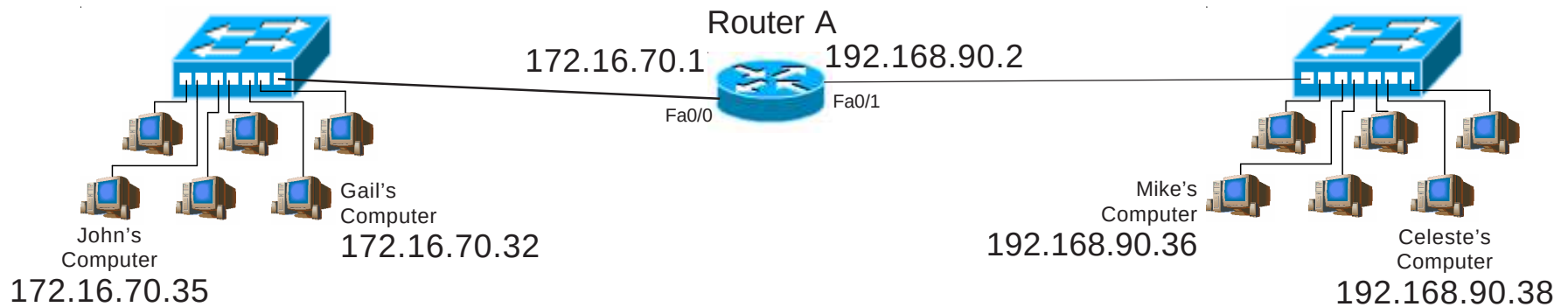
Router(config)# *interface fa0/0*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Writing Extended Access Lists...



Extended Access List Sample #1

Deny/Permit Specific Addresses

Write an extended access list to prevent John's computer from sending information to Mike's computer; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: FA0/0
 Access-list #: 110

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 110 deny ip 172.16.70.35 0.0.0.0 192.168.90.36 0.0.0.0
                        or
                        access-list 110 deny ip host 172.16.70.35 host 192.168.90.36
Router(config)# access-list 110 permit ip any any
                        or
                        access-list 110 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/0
Router(config-if)# ip access-group 110 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

```
Router# show configuration (This will show which access groups
                           are associated with particular interfaces)

Router# show access list 110 (This will show detailed information
                             about this ACL)
```

Extended Access List Sample #2

Deny/Permit Specific Addresses

Write an extended access list to block the 172.16.70.0 network from receiving information from Mike's computer at 192.168.90.36. Block the lower half of the ip addresses from 192.168.90.0 network from reaching Gail's computer at 172.16.70.32. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: FA0/1
Access-list #: 135

[Writing and installing an ACL]

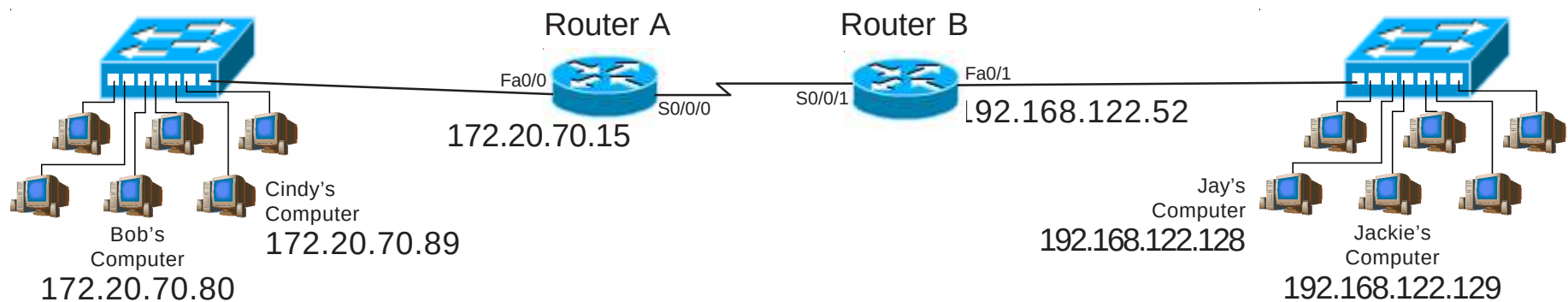
```
Router# configure terminal
Router(config)# access-list 135 deny ip 192.168.90.36 0.0.0.0 172.16.70.0 0.0.0.255
                        or
                        access-list 135 deny ip host 192.168.90.36 172.16.70.0 0.0.0.255
Router(config)# access-list 135 deny ip 192.168.90.0 0.0.0.127 172.16.70.32 0.0.0.0
                        or
                        access-list 135 deny ip 192.168.90.0 0.0.0.127 host 172.16.70.32
Router(config)# access-list 135 permit ip any any
                        or
                        access-list 135 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/1
Router(config-if)# ip access-group 135 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface fa0/1
Router(config-if)# no ip access-group 135 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface fa0/1
Router(config-if)# no ip access-group 135 out
Router(config-if)# exit
Router(config)# no access-list 135
Router(config)# exit
```



Extended Access List Problem #1 Deny/Permit Specific Addresses

Write an extended access list to prevent Jay's computer from receiving information from Cindy's computer. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*
 Router(config-if)# *ip access-group*
 Router(config-if)# *exit*
 Router(config)# *exit*
 Router# *copy run start*

Extended Access List Problem #2

Deny/Permit Specific Addresses

Write an extended access list to block the 172.20.70.0 255.255.255.0 network from receiving information from Jackie's computer at 192.168.122.129. Block the lower half of the ip addresses from 192.168.122.0 network from reaching Cindy's computer at 172.20.70.89. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

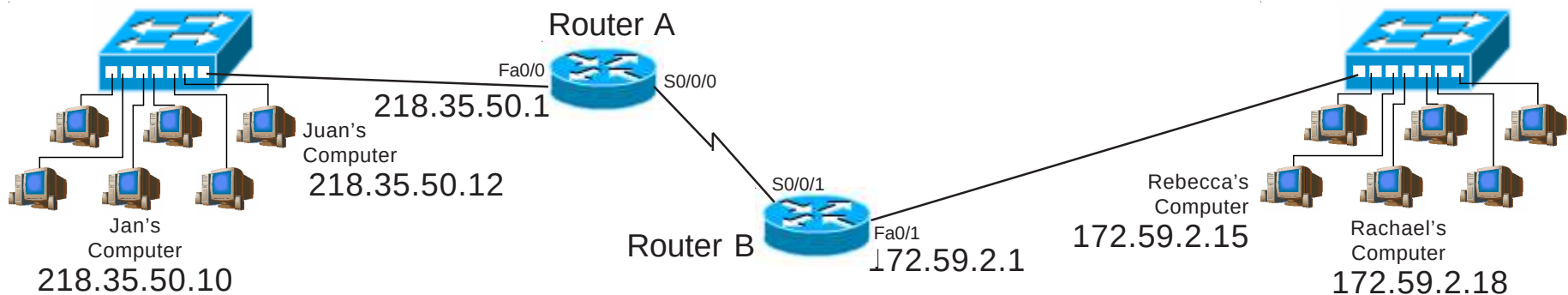
Router Name:
Interface:
Access-list #:

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)#

Router(config)# *interface*
Router(config-if)# *ip access-group*
Router(config-if)# *exit*
Router(config)# *exit*
Router# *copy run start*



Extended Access List Problem #3 Deny/Permit Specific Addresses

Write a named extended access list called "LAB_166" to permit Jan's computer at 218.35.50.10 to receive packets from Rachael's computer at 172.59.2.18; but not Rebecca's computer at 172.59.2.15. Deny all other packets. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list Name:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config-ext-nacl)#

Router(config-ext-nacl)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #4

Deny/Permit Specific Addresses

Write an extended access list to allow Juan's computer at 218.35.50.12 to send information to Rebecca's computer at 172.59.2.15; but not Rachael's computer at 172.59.2.18. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:

Interface:

Access-list #:

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)#

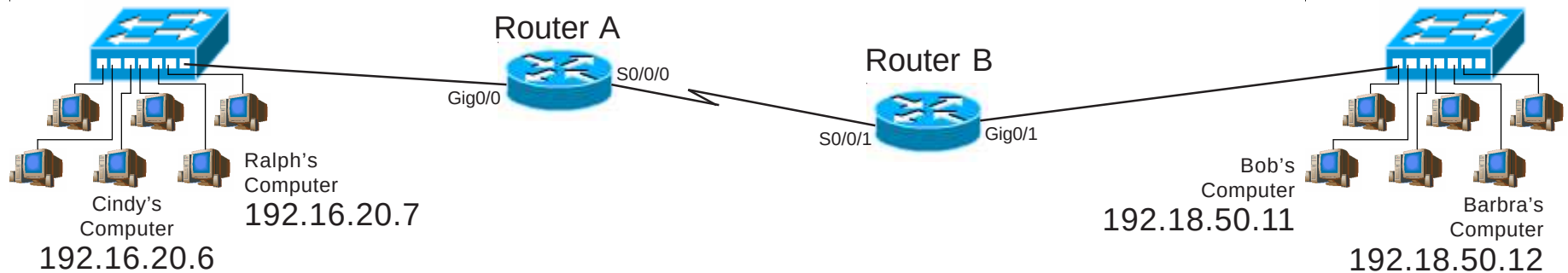
Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Sample #3

Deny/Permit Entire Ranges

Write an extended access list to permit the 192.16.20.0 network to receive packets from the 192.18.50.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Gig0/1
 Access-list #: 111

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 111 permit ip 192.18.50.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config)# access-list 111 deny ip any any
or
Router(config)# access-list 111 deny ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface gig0/1
Router(config-if)# ip access-group 111 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

```
Router# show configuration (This will show which access groups are associated
                           with particular interfaces)

Router# show access list 111 (This will show detailed information about this ACL)
```

Extended Access List Sample #4

Deny/Permit Entire Ranges

Add a remark to each statement. Write an extended access list to block the 192.18.50.0 network from receiving information from the 192.16.20.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 188

[Remark Command]

The remark command allows you to place text within the ACL so it can be viewed after it is inserted on the router. It can be viewed using the show run or any command that lists the ACEs.

[Writing and installing an ACL]

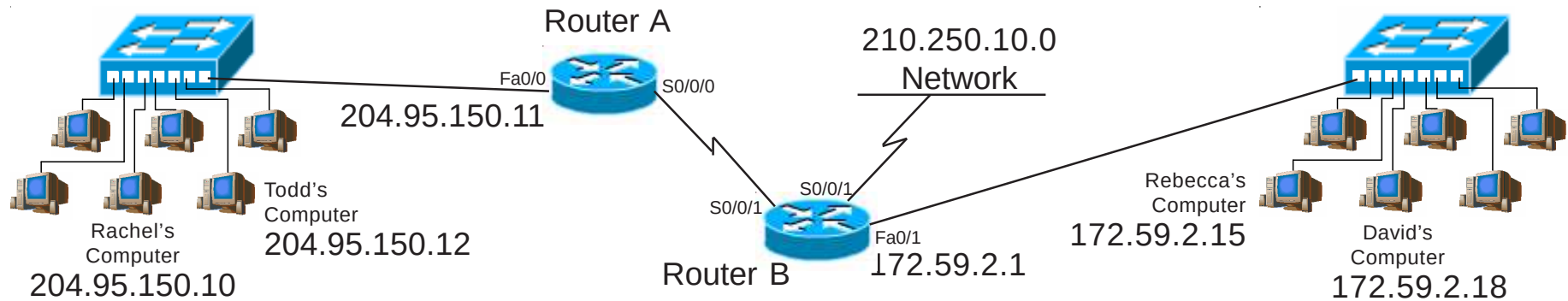
```
Router# configure terminal
Router(config)# access-list 188 remark block all traffic from the Science lab
Router(config)# access-list 188 deny ip 192.16.20.0 0.0.0.255 192.18.50.0 0.0.0.255
Router(config)# access-list 188 remark allow everyone else unrestricted access
Router(config)# access-list 188 permit ip any any
                        or
                        access-list 188 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface gig0/0
Router(config-if)# ip access-group 188 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 188 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface gig0/0
Router(config-if)# no ip access-group 188 out
Router(config-if)# exit
Router(config)# no access-list 188
Router(config)# exit
```



Extended Access List Problem #5 Deny/Permit Entire Ranges

Include a remark with each statement of your ACL. Write an extended access list to permit network 204.95.150.0 to send packets to network 172.59.0.0, but not to the 210.250.10.0 network. Permit all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*
 Router(config-if)# *ip access-group*
 Router(config-if)# *exit*
 Router(config)# *exit*

Extended Access List Problem #6

Deny/Permit Entire Ranges

Write an extended access list to allow Rachel's computer at 204.95.150.10 to receive information from the 172.59.2.0 255.255.255.0 network. Deny all other hosts on the 204.95.150.0 network access from the 172.59.2.0 255.255.255.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:

Interface:

Access-list #:

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)#

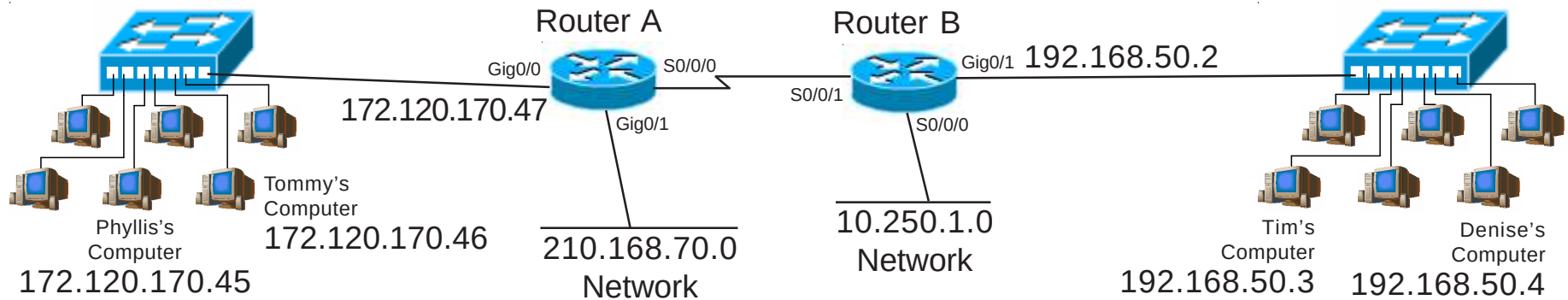
Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #7 Deny/Permit Entire Ranges

Write a named extended access list called "Godzilla" to prevent the 172.120.0.0 network from sending information to the 210.168.70.0, and 10.250.1.0 255.255.255.0 networks; but will permit traffic to the 192.168.50.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list Name:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config-ext-nacl)#

Router(config-ext-nacl)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #8

Deny/Permit Entire Ranges

Assuming default subnet masks write an extended access list to permit Tim at 192.168.50.3 to receive data from the 172.120.0.0 network. Allow the 192.168.50.0 network to receive information from Phyllis's computer at 172.120.170.45. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:

Interface:

Access-list #:

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)#

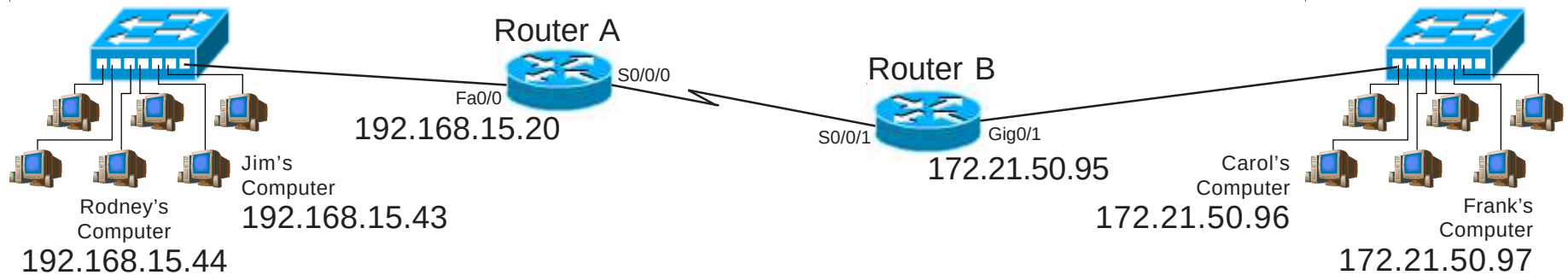
Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Sample #5

Deny/Permit a Range of Addresses

Write an extended access list to deny the first 15 usable addresses of the 192.168.15.0 network from reaching the 172.21.0.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
 Interface: Fa0/0
 Access-list #: 185

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 185 deny ip 192.168.15.0 0.0.0.15 172.21.50.0 0.0.255.255
Router(config)# access-list 185 permit ip any any
or
Router(config)# access-list 185 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/0
Router(config-if)# ip access-group 185 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

```
Router# show configuration (This will show which access groups are associated with particular interfaces)

Router# show access list 185 (This will show detailed information about this ACL)
```

Extended Access List Sample #6

Deny/Permit a Range of Addresses

Write an extended access list which will allow the lower half of 192.168.15.0 network access to the 172.21.50.0 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Fa0/0
Access-list #: 121

[Writing and installing an ACL]

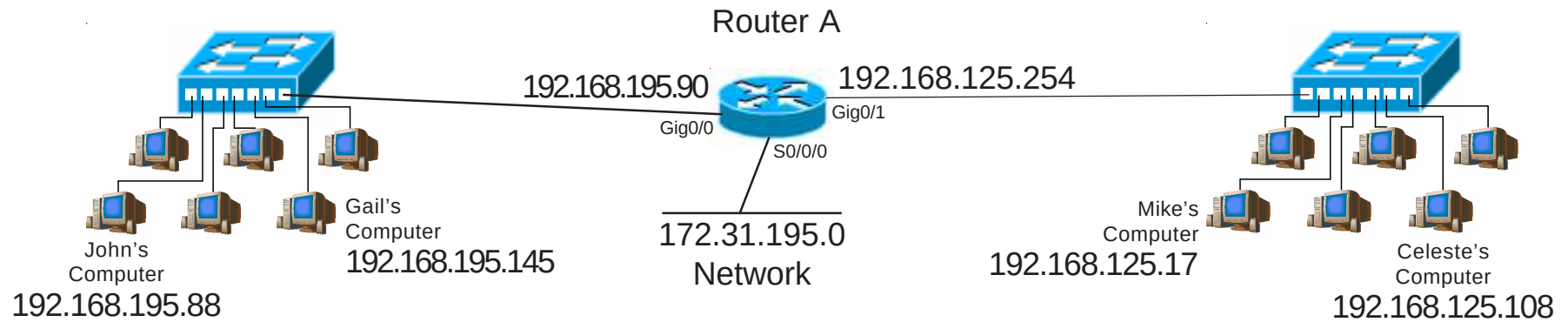
```
Router# configure terminal
Router(config)# access-list 121 permit ip 192.168.15.0 0.0.0.127 172.21.50.0 0.0.0.255
Router(config)# access-list 121 deny ip any any
                        or
                        access-list 121 deny ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface fa0/0
Router(config-if)# ip access-group 121 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface fa0/0
Router(config-if)# no ip access-group 121 in
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface fa0/0
Router(config-if)# no ip access-group 121 in
Router(config-if)# exit
Router(config)# no access-list 121
Router(config)# exit
```



Extended Access List Problem #9 Deny/Permit a Range of Addresses

Write an extended access list to prevent the first 31 usable addresses in the 192.168.125.0 network from reaching the 192.168.195.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*
 Router(config-if)# *ip access-group*
 Router(config-if)# *exit*

Extended Access List Problem #10 Deny/Permit a Range of Addresses

Include a remark with each statement of your ACL. Write a named extended access list called "Media_Center" to permit the range of addresses from 172.31.195.1 through 172.31.195.7 to send data to the 192.168.125.0 network. Deny all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:

Interface:

Access-list Name:

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)#

Router(config-ext-nacl)#

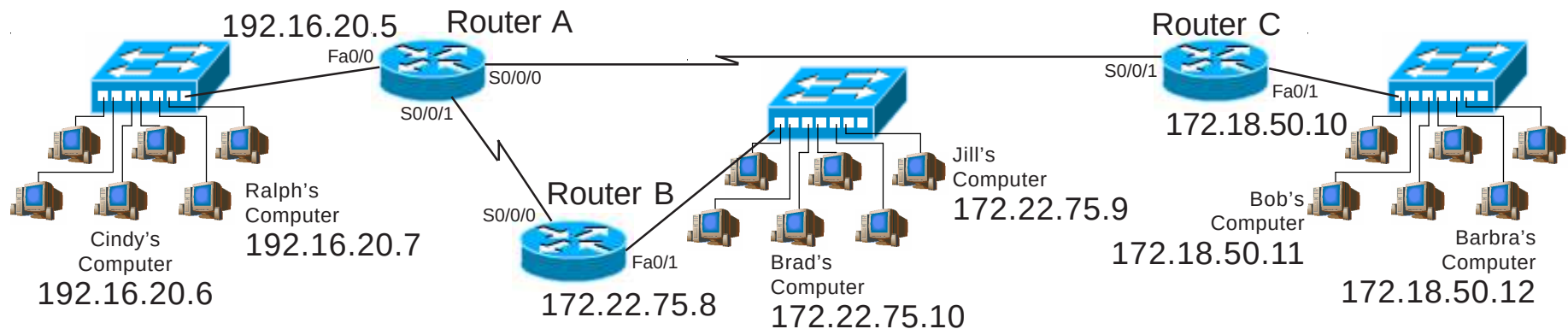
Router(config-ext-nacl)# *interface*

Router(config-if)# *ip access-group* *in or out* (circle one)

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #11. Deny/Permit a Range of Addresses

Write an extended access list to permit the first 3 usable addresses in the 192.16.20.0 network to reach the 172.22.75.0 network. Deny the addresses from 192.16.20.4 through 192.16.20.31 from reaching the 172.22.75.0 network. Permit all other traffic. Keep in mind that there are multiple ways this ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Extended Access List Problem #12 Deny/Permit a Range of Addresses

Write an extended access list to deny the addresses from 172.22.75.8 through 172.22.75.127 from sending data to the 172.18.50.0 network. Deny the first half of the addresses from the 172.22.75.0 network from reaching the 192.16.20.0 network. Permit all other traffic. Keep in mind that there are multiple ways this ACL can be written.

Place the access list at:

Router Name:

Interface:

Access-list #:

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)#

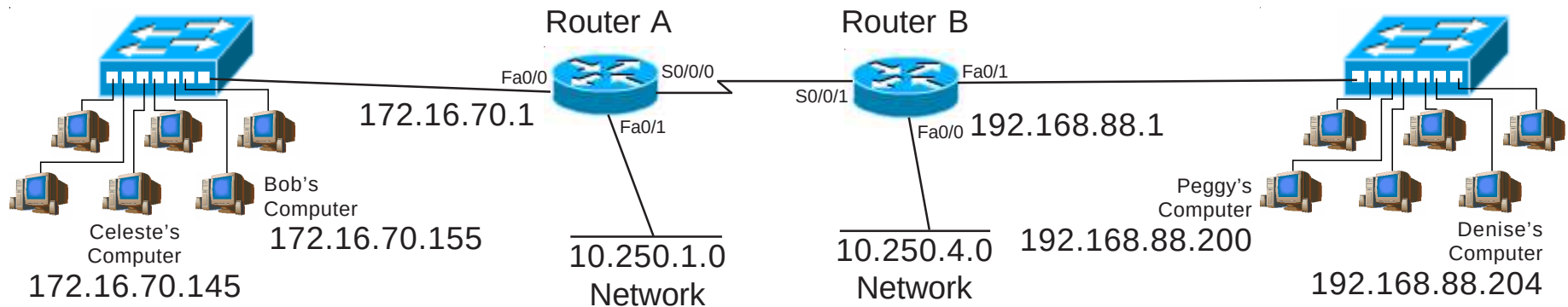
Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #13 Deny/Permit a Range of Addresses

Include a remark with each statement of your ACL. Write an extended access list to permit the first 63 usable addresses in the 192.168.88.0 network to reach the lower half of the addresses in the 172.16.70.0 network; but not the upper half. Deny all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Extended Access List Problem #14 Deny/Permit a Range of Addresses

Write an extended access list to deny the addresses from 10.250.1.0 through 10.250.1.63 from sending data to Denise's computer. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
Interface:
Access-list #:

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)#

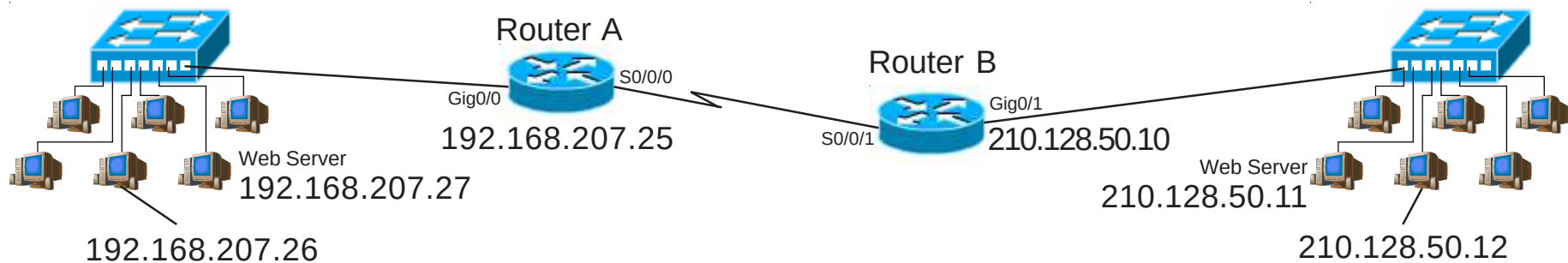
Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Sample #7

Deny/Permit Port Numbers

Write an extended access list to deny HTTP traffic intended for web server 192.168.207.27 from all other networks, but will permit all other HTTP traffic to reach the 192.168.207.0 network. Deny all other IP traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Gig0/1
 Access-list #: 198

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 198 deny tcp any 192.168.207.27 0.0.0.0 eq www
or
access-list 198 deny tcp any host 192.168.207.27 eq www
Router(config)# access-list 198 permit tcp any 192.168.207.0 0.0.0.255 eq www
Router(config)# interface gig0/1
Router(config-if)# ip access-group 198 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

```
Router# show configuration (This will show which access groups are associated
with particular interfaces)

Router# show access list 198 (This will show detailed information about this ACL)
```

Extended Access List Sample #8

Deny/Permit Port Numbers

Write an extended access list on Router B to deny pings between hosts on the 210.128.50.0 and the 192.168.207.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
Interface: Gig0/1
Access-list #: 134

[Writing and installing an ACL]

```
Router# configure terminal
Router(config)# access-list 134 deny icmp 210.128.50.0 0.0.0.255 192.168.207.0 0.0.0.255
Router(config)# access-list 134 permit ip any any
Router(config)# interface gig0/1
Router(config-if)# ip access-group 134 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

Hint:

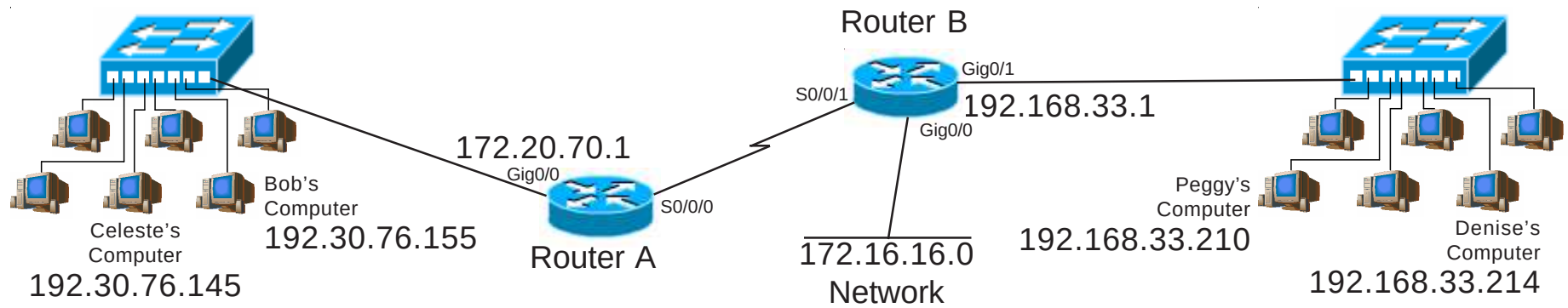
It's OK to use multiple protocols in the same ACL.

[Disabling ACL's]

```
Router# configure terminal
Router(config)# interface Gig0/1
Router(config-if)# no ip access-group 134 out
Router(config-if)# exit
Router(config)# exit
```

[Removing an ACL]

```
Router# configure terminal
Router(config)# interface Gig0/1
Router(config-if)# no ip access-group 134 out
Router(config-if)# exit
Router(config)# no access-list 134
Router(config)# exit
```



Extended Access List Sample #9

Deny/Permit Port Numbers

Write an Extended access list to permit Denise's computer to use TFTP with Bob's computer. Deny all other traffic from the 192.168.33.0 network to the 192.30.76.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B
 Interface: Gig0/1
 Access-list #: 145

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 145 permit udp 192.168.33.214 0.0.0.0 192.30.76.155 0.0.0.0 eq tftp
or
access-list 145 permit udp host 192.168.33.214 host 192.30.76.155 eq tftp
Router(config)# interface Gig0/1
Router(config-if)# ip access-group 145 in
Router(config-if)# exit
Router(config)# exit
```

[Viewing information about existing ACL's]

```
Router# show configuration (This will show which access groups are associated with particular interfaces)

Router# show access list 45 (This will show detailed information about this ACL)
```

Extended Access List Sample #10

Deny/Permit Port Numbers

Write an extended access list to deny FTP traffic from ip addresses 192.30.76.0 through 192.30.76.13 to any destination. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router A
Interface: Gig0/0
Access-list #: 155

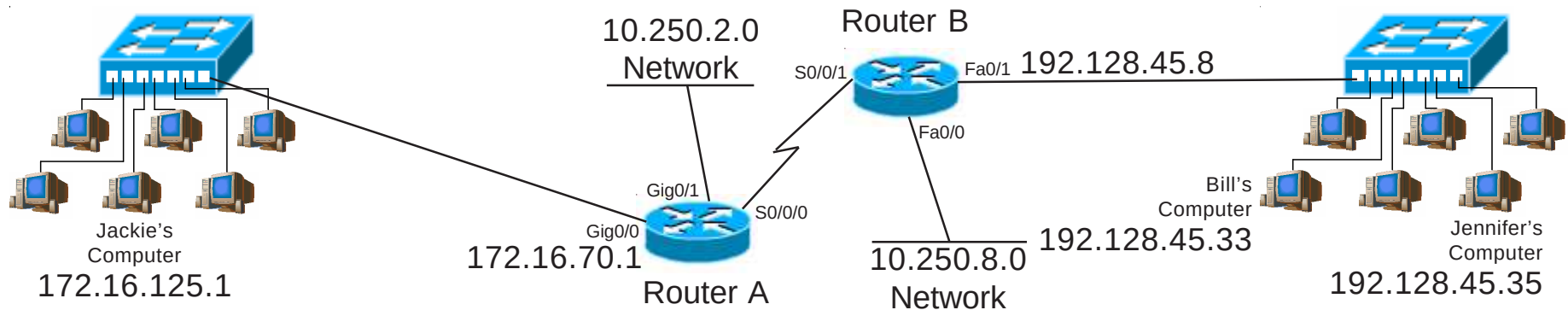
[Writing and installing an ACL]

```
Router# configure terminal
Router(config)# access-list 155 deny tcp 192.30.76.0 0.0.0.8 any eq ftp (Blocks 0 to 7)
Router(config)# access-list 155 deny tcp 192.30.76.8 0.0.0.4 any eq ftp (Blocks 8 to 11)
Router(config)# access-list 155 deny tcp 192.30.76.12 0.0.0.1 any eq ftp (Blocks 12 to 13)
Router(config)# access-list 155 permit ip any any
                        or
                        access-list 155 permit ip 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255
Router(config)# interface gig0/0
Router(config-if)# ip access-group 155 in
Router(config-if)# exit
Router(config)# exit
Router# copy run start
```

Hint:
It's OK to use multiple protocols
in the same ACL.

The first three TCP statements could be shortened to:

```
access-list 155 permit tcp 192.30.76.14 0.0.0.1 any eq ftp (Permits 14 and 15)
access-list 155 deny tcp 192.30.76.8 0.0.0.15 any eq ftp (Blocks 0 to 15)
```



Extended Access List Problem #15 Deny/Permit a Port Numbers

Write an extended access list to permit ICMP traffic from the 192.128.45.0 network to reach the 172.16.125.0 255.255.255.0 and 10.250.2.0 255.255.255.0 networks. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Extended Access List Problem #16 Deny/Permit a Port Numbers

Write a named extended access list called "PEGGYS_LAB" to deny telnet from 10.250.8.0 through 10.250.8.127 from reaching the 192.128.45.0 network. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
Interface:
Access-list Name:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config-std-nacl)#

Router(config-std-nacl)#

Router(config-ext-nacl)# *interface*

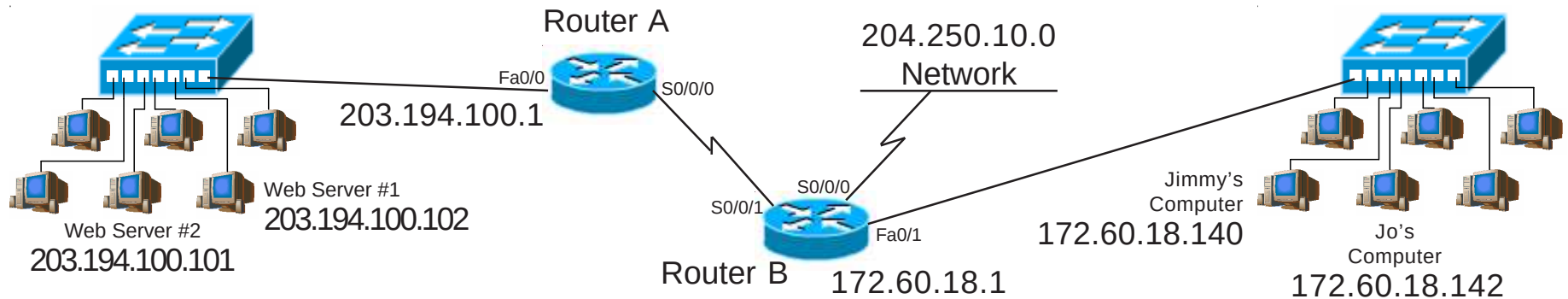
Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*

Router# *copy run start*



Extended Access List Problem #17 Deny/Permit Port Numbers

Write an access list to deny Jimmy's computer from sending ftp packets to Web Server 1, but permit ftp to Web Server #2. Permit all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Extended Access List Problem #18

Deny/Permit Port Numbers

Write an extended access list to deny all HTTP traffic intended for the web server at 203.194.100.102 from the 172.66.0.0 network. Permit all other HTTP traffic from the 204.250.10.0 and 172.60.0.0 networks to any other web servers. Deny all other IP traffic to the 203.194.100.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
Interface:
Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

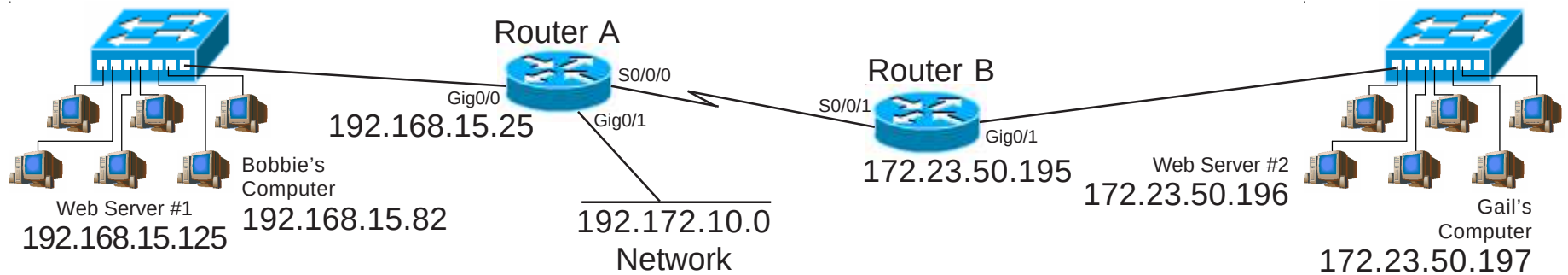
Router(config)# *interface*

Router(config-if)# *ip access-group*

Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*



Extended Access List Problem #19

Deny/Permit Port Numbers

Include a remark with each statement of your ACL. Write an extended access list to permit TFTP traffic from all hosts on the 192.168.15.0 network. Deny all other traffic. For help with the remark command review page 41. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)# *interface*
 Router(config-if)# *ip access-group*
 Router(config-if)# *exit*
 Router(config)# *exit*

Extended Access List Problem #20

Deny/Permit Port Numbers

Write an extended access list that permits web traffic from web server #2 at 172.23.50.196 to reach everyone on the 192.168.15.0 network. Deny all other IP traffic going to the 192.172.10.0, and 192.168.15.0 networks from the 172.25.50.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:

Interface:

Access-list #:

[Writing and installing an ACL]

Router# *configure terminal*

Router(config)#

Router(config)# *interface*

Router(config-if)# *ip access-group*

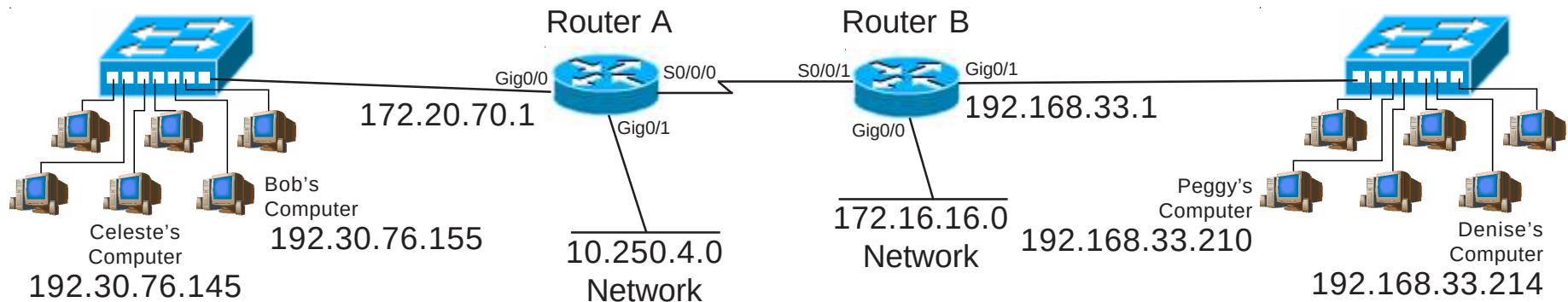
Router(config-if)# *exit*

Router(config)# *exit*

Router# *copy run start*

Writing Access Lists to Restrict Telnet Access...

Restricting access to telnet can be a very usefull option. Telnet is considered a very insecure protocol because it sends passwords through the network in clear-text. By switching from the *access-group* command to the *access-class* command you can increase your security by allowing only those users through that you want to use telnet. The *access-class* command also allows you to apply this access list to the vty connections.



Standard Access List Sample #11 Deny/Permit Telnet

Write a standard access list to permit Denise's and Bob's computers to telnet into Router B. Deny all other telnet traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name: Router B

Interface: line VTY 0 4

Access-list #: 45

(using line VTY 0 4 instead of an interface like E1 allows you to apply this access list to all VTY lines with one statement)

[Writing and installing an ACL]

```
Router# configure terminal (or config t)
Router(config)# access-list 45 permit 192.168.33.214 0.0.0.0
```

or

```
access-list 45 permit host 192.168.33.214
Router(config)# access-list 45 permit 192.30.76.155 0.0.0.0
```

or

```
access-list 45 permit host 192.30.76.155
```

```
Router(config)# line vty 0 4
```

```
Router(config-line)# access-class 45 in
```

```
Router(config-line)# exit
```

```
Router(config)# exit
```

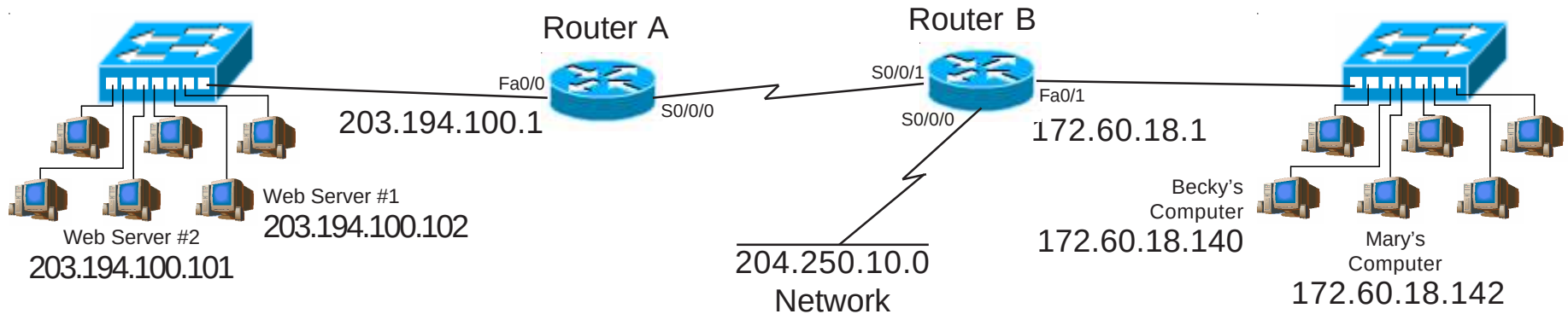
[Viewing information about existing ACL's]

```
Router# show configuration
```

(This will show which access groups are associated with particular interfaces)

```
Router# show access list 45
```

(This will show detailed information about this ACL)



Access List Problem #21 Deny/Permit Telnet

Write a standard access list to permit Becky and Mary's computer to telnet into Router B. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:
 Interface:
 Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)#
 Router(config-line)# *access-class*
 Router(config-line)# *exit*
 Router(config)# *exit*

Access List Problem #22

Deny/Permit Telnet

Write a standard access list to permit which will permit Web Server #1 to telnet into Router A. Log the telnet attempts. Deny all other telnet access. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:

Interface:

Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

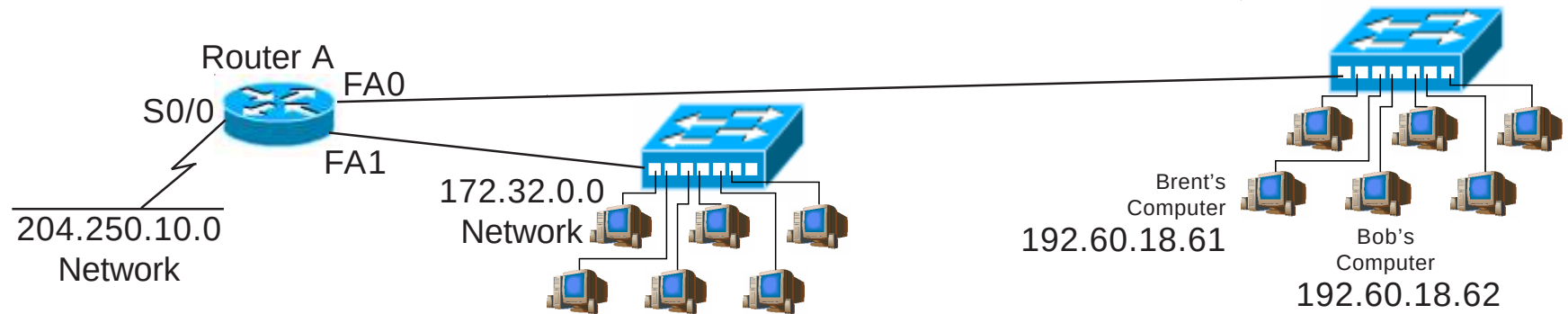
Router(config)#

Router(config)#

Router(config-line)# *access-class*

Router(config-line)# *exit*

Router(config)# *exit*



Access List Problem #23

Deny/Permit Telnet

Write a standard access list to deny Brent and Bob's computer telnet access into Router A. Permit all other telnet traffic from the 192.60.18.0 network. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written.

Place the access list at:

Router Name:

Interface:

Access-list #:

[Writing and installing an ACL]

Router# *configure terminal (or config t)*

Router(config)#

Router(config)#

Router(config-line)# *access-class*

Router(config-line)# *exit*

Router(config)# *exit*

Reference

Port Numbers

Port numbers are now assigned by the ICANN (Internet Corporation for Assigned Names and Numbers). Commonly used TCP and UDP applications are assigned a port number; such as: HTTP - 80, POP3 - 110, FTP - 20. When an application communicates with another application on another node on the internet, it specifies that application in each data transmission by using its port number. You can also type the name (ie. Telnet) instead of the port number (ie. 23). Port numbers range from 0 to 65536 and are divided into three ranges:

Well Known Ports	0 to 1,023
Registered Ports	1,024 to 49,151
Dynamic and/or Private Ports	49,152 to 65,535

Below is a short list of some commonly used ports. For a complete list of port numbers go to <http://www.iana.org/assignments/port-numbers>.

Some commonly used port numbers:

0	Reserved	
1	TCPMUX	(TCP Port Service Multiplexer)
5	RJE	(Remote Job Entry)
7	ECHO	
9	DISCARD	
11	SYSTAT	(Active users)
13	DAYTIME	
17	QUOTE	(Quote of the day)
18	MSP	(Message Send Protocol)
19	CHARGEN	(Character generator)
20	FTP-DATA	(File Transfer Protocol - Data)
21	FTP	(File Transfer Protocol - Control)
22	SSH	(Remote Login Protocol)
23	Telnet	(Terminal Connection)
25	SMTP	(Simple Mail Transfer Protocol)
29	MSG ICP	
37	TIME	
39	RLP	(Resource Location Protocol)
42	NAMESERV	(Host Name Server)

43	NICNAME	(Who Is)
49	LOGIN	(Login Host Protocol)
53	DNS	(Domain Name Server)
67	BOOTP	(Bootstrap Protocol Server)
68	BOOTPS	(Bootstrap Protocol Client)
69	TFTP	(Trivial File Transfer Protocol)
70	GOPHER	(Gopher Services)
75		(Any Private Dial-out Service)
79	FINGER	
80	HTTP	(Hypertext Transfer Protocol)
95	SUPDUP	(SUPDUP Protocol)
101	HOSTNAME	(NIC Host Name Server)
108	SNAGAS	(SNA Gateway Access Server)
109	POP2	(Post Office Protocol - Version 2)
110	POP3	(Post Office Protocol - Version 3)
113	AUTH	(Authentication Service)
115	SFTP	(Simple File Transfer Protocol)
117	UUCP-PATH	(UUCP Path Service)
118	SQLSERV	(SQL Services)
119	NNTP	(Newsgroup)
123	NTP	(Network Time Protocol)
137	NetBIOS-NS	(NetBIOS Name Service)
139	NetBIOS-SSN	(NetBIOS Session Service)
143	IMAP	(Interim Mail Access Protocol)
150	SQL-NET	(NetBIOS Session Service)
156	SQLSRV	(SQL Service)
161	SNMP	(Simple Network Management Protocol)
179	BGP	(Border Gateway Protocol)
190	GACP	(Gateway Access Control Protocol)
194	IRC	(Internet Relay Chat)
197	DLS	(Directory Location Service)
389	LDAP	(Lightweight Directory Access Protocol)
396	NETWARE-IP	(Novell Netware over IP)
443	HTTPS	(HTTP MCom)
444	SNPP	(Simple Network Paging Protocol)
445	Microsoft-DS	
458	Apple QuickTime	
546	DHCP Client	
547	DHCP Server	
563	SNEWS	
569	MSN	

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31	16-23	8-11	4-5
					24-31	12-15	6-7
						16-19	8-9
						20-23	10-11
						24-27	12-13
						28-31	14-15
		32-63	32-47	32-39	32-35	32-35	16-17
					36-39	36-39	18-19
				40-47	40-43	40-43	20-21
					44-47	44-47	22-23
					48-51	48-51	24-25
					52-55	52-55	26-27
			48-63	56-63	56-59	56-59	28-29
					60-63	60-63	30-31
				64-79	64-71	64-67	32-33
					72-79	68-71	34-35
					80-87	80-83	36-37
					88-95	88-91	38-39
		64-127	64-95	96-103	96-99	96-99	40-41
					100-103	100-103	42-43
				104-111	104-107	104-107	44-45
					108-111	108-111	46-47
					112-119	112-115	48-49
					120-127	116-119	50-51
			96-127	128-135	128-135	128-135	52-53
					136-143	136-139	54-55
				144-159	144-151	144-147	56-57
					152-159	148-151	58-59
					160-167	160-163	60-61
					168-175	164-167	62-63
		128-191	128-159	176-183	176-179	176-179	64-65
					184-191	180-183	66-67
				184-191	188-191	184-187	68-69
					192-199	188-191	70-71
					200-207	192-195	72-73
					208-215	196-199	74-75
		128-255	160-191	216-223	216-219	216-219	76-77
					224-231	224-227	78-79
				224-239	232-235	228-231	80-81
					240-247	232-235	82-83
					248-255	236-239	84-85
						240-243	86-87
		192-255	192-223	248-255	252-255	244-247	88-89
						248-251	90-91
						252-255	92-93
							94-95
							96-97
							98-99
			224-255	240-255	240-243	240-243	100-101
					244-247	244-247	102-103
					248-251	248-251	104-105
						252-255	106-107
							108-109
							110-111

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					4-7	4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
			16-31	16-31	16-23	16-19	8-9
					20-23	20-21	10-11
					24-31	24-27	12-13
						28-31	14-15
			32-63	32-47	32-39	32-35	16-17
					36-39	36-37	18-19
					40-47	40-43	20-21
						44-47	22-23
				48-63	48-55	48-51	24-25
					52-55	52-53	26-27
					56-63	56-59	28-29
						60-63	30-31
		64-127	64-95	64-79	64-71	64-67	32-33
					72-79	68-71	34-35
						72-75	36-37
						76-79	38-39
				80-95	80-87	80-83	40-41
					88-95	84-87	42-43
						88-91	44-45
						92-95	46-47
			96-127	96-111	96-103	96-99	48-49
						100-103	50-51
					104-111	104-107	52-53
						108-111	54-55
				112-127	112-119	112-115	56-57
					120-127	116-119	58-59
						120-123	60-61
						124-127	62-63
	128-255	128-191	128-159	128-143	128-135	128-131	64-65
					136-143	132-135	66-67
						136-139	68-69
						140-143	70-71
				144-159	144-151	144-147	72-73
					152-159	148-151	74-75
						152-155	76-77
						156-159	78-79
			160-191	160-175	160-167	160-163	80-81
					168-175	164-167	82-83
						168-171	84-85
						172-175	86-87
				176-191	176-183	176-179	88-89
						180-183	90-91
						184-187	92-93
					184-191	188-191	94-95
		192-255	192-223	192-207	192-199	192-195	96-97
						196-199	98-99
					200-207	200-203	100-101
						204-207	102-103
				208-223	208-215	208-211	104-105
					216-223	212-215	106-107
						216-219	108-109
						220-223	110-111
			224-255	224-239	224-231	224-227	112-113
					232-239	228-231	114-115
						232-235	116-117
						236-239	118-119
				240-255	240-247	240-243	120-121
					248-255	244-247	122-123
						248-251	124-125
						252-255	126-127

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255